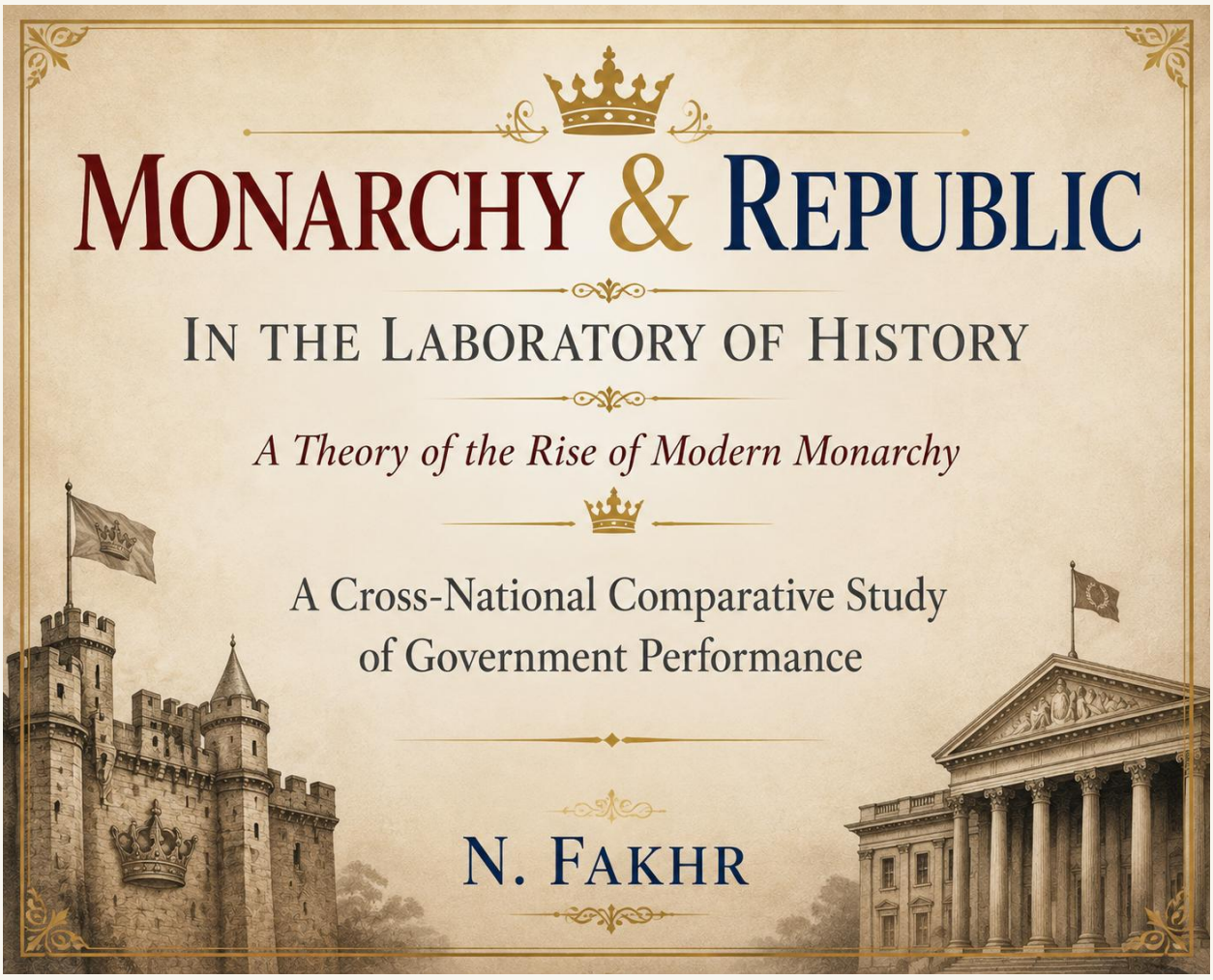


MONARCHY AND REPUBLIC IN THE LABORATORY OF HISTORY

A Theory of the Rise of Modern Monarchy

A Cross-National Comparative Study of Government Performance

Third Edition (Revised and Expanded)

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Author's Note

The first edition of this book was published in the winter of 2025. The text you are now reading is the third edition, substantially revised and expanded from the earlier versions. Its structure, arguments, methodology, statistical analyses, and writing have all undergone extensive review and refinement.

This English edition was translated from the original Persian by the author. The translation was undertaken with the aim of preserving as faithfully as possible the arguments, terminology, methodological choices, and intended meaning of the Persian text.

I hope this edition brings the book one step closer to its central aim: evaluating systems of government on the basis of empirical evidence.

Methodologically, this study is a cross-national comparative quantitative study based on secondary data analysis, using data from multiple reputable international databases to compare the performance of contemporary monarchies and republics across a range of indicators.

N. Fakhr
Autumn 2026

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Preface

About 2,300 years ago, the Greek astronomer Aristarchus, working without a telescope and relying only on observation, measurement, and the systematic comparison of data, reached a conclusion that challenged the prevailing belief of his time: the Sun did not revolve around the Earth; rather, the Earth revolved around the Sun.

Yet his idea failed to change the widely accepted view. What people regarded as “common sense” seemed to tell them otherwise. Every day, they saw the Sun rise on one side of the sky and set on the other. It therefore **seemed obvious** that the Sun must be moving around the Earth. Moreover, many believed that human beings were the highest form of creation and that their home should therefore occupy the center of the universe.

The geocentric view of the universe was not merely a belief held by ordinary people. Many of the greatest scholars and astronomers of their time, from Ptolemy to Al-Biruni, accepted it as well.

In those days, truth was generally sought in one of two ways: by turning to sacred texts or by relying on pure reason and logic. Systematic observation, measurement, and empirical testing had not yet acquired the place they hold today. For this reason, even some eighteen centuries later, when Copernicus examined a large body of astronomical measurements and once again arrived at a heliocentric model, it took many years for this view to gradually replace the prevailing belief.

But the scientific method slowly demonstrated its power. In its simplest form, it can be described in three steps. First, observations lead to a hypothesis intended to explain a phenomenon. Next, a way is designed to test that hypothesis. Finally, through careful measurement and the systematic collection of data, the hypothesis is tested against reality. If a hypothesis repeatedly survives independent tests and succeeds in explaining a broad range of observations, it may gradually develop into a scientific theory—not an eternal truth, but the best available explanation until a better one emerges.

Scientific testing, however, does not always take place in a laboratory under perfectly controlled conditions. For many of the most important scientific questions, such control is either impossible or ethically unacceptable.

Suppose, for example, that we want to know whether smoking increases the risk of lung cancer or heart disease. We cannot randomly select thousands of healthy people, force half of them to smoke for twenty or thirty years, forbid the other half from smoking, and then wait to see which group develops more disease. Such an experiment might seem ideal from a methodological standpoint, but it would clearly be ethically unacceptable.

Science, however, does not come to a halt in such situations. Instead of creating experimental groups, researchers study groups that have **formed naturally in the real world**. Millions of people choose to smoke, while millions of others never do. Scientists follow these individuals for years, sometimes decades, record their health outcomes, and use statistical methods to account, as far as possible, for factors such as age, sex, income, occupation, diet, and other variables that may influence the results.

No single study of this kind can establish a causal relationship with absolute certainty. But when hundreds of independent studies, conducted in different countries, involving millions of people and covering different periods, repeatedly point in the same direction; when the risk of disease rises as smoking increases; and when laboratory findings support the same conclusion, the accumulated evidence gradually produces a clear picture of reality.

Today, few people doubt that smoking is one of the major risk factors for lung cancer and many other diseases—not because a single experiment proved everything, but because **a vast body of independent evidence, accumulated over decades, has consistently pointed to the same conclusion**.

Science, therefore, is not merely the art of conducting controlled experiments. It is also the art of carefully drawing knowledge from a reality that often cannot be recreated in a laboratory.

Today, many examples of this kind of research can be found across different fields of science. During the COVID-19 pandemic, for instance, many of the most important questions were answered not in laboratories, but within human societies themselves. Using data from millions of people in different countries, researchers studied vaccine effectiveness, the role of age and underlying health conditions, patterns of virus transmission, and dozens of other factors. Each individual study had its limitations, but when the findings of hundreds of independent studies were considered together, a reasonably clear picture of reality emerged. This was another example of how accumulated evidence can provide more reliable knowledge than any single observation or experience.

The social and political sciences, although they face even greater complexities, follow much the same logic. Countries cannot normally be treated like subjects in a laboratory experiment, randomly divided into two groups, and subjected to different systems of government for several decades so that the results can later be compared. Such an experiment would be neither practically possible nor ethically acceptable.

History, however, has conducted something resembling such an experiment—though certainly not a perfectly controlled one—on an enormous scale.

Today, dozens of countries are monarchies, while more than a hundred are republics. Over decades, and in some cases centuries, these countries have followed different paths, and the results of their performance have been recorded in thousands of economic, social, political, and governance indicators. Although far from perfect, this body of evidence provides us with the

largest natural experiment available for an empirical study of how different systems of government perform.

But this raises a more basic question: why conduct such a study at all? Are the strengths and weaknesses of different systems of government not already known?

In much of contemporary political thought, the republic is presented as a symbol of modernity, freedom, and popular sovereignty, while monarchy is often regarded as a relic of the past. This view, rooted partly in historical experiences such as the American and French Revolutions, has become so deeply embedded in modern political discourse that many of us assume we already know the answer before looking at the empirical evidence.

Yet the history of science has repeatedly shown that even humanity's most self-evident beliefs have not always matched reality. There was a time when it seemed obvious that the Sun revolved around the Earth. Disease was once thought to arise from polluted air or evil spirits. Even a few decades ago, many people did not accept that smoking could cause lung cancer. In the early twentieth century, a large proportion of the world's population regarded the superiority of communism over other systems of government as self-evident. In each of these cases, it was empirical evidence—not philosophical argument or human intuition—that challenged deeply rooted beliefs.

For this reason, the central question of this book is not **which system is more appealing from a philosophical or moral point of view**. Instead, it asks: **if we look only at the empirical record of countries, do we find a meaningful pattern in the performance of monarchies and republics?**

Such a question cannot be answered by quoting philosophers or pointing to a handful of famous examples. It can only be addressed through a systematic examination of the data the world has accumulated over decades.

An important feature of this study is that the data used here were not originally collected to answer this question. International institutions, national statistical agencies, universities, and research organizations have spent years measuring indicators such as income per capita, life expectancy, crime rates, quality of governance, education, freedom, and dozens of other variables. They were not trying to demonstrate that monarchy is better than republic, or vice versa. Their purpose was simply to measure these indicators.

In this sense, the process through which these data were collected can, to a considerable extent, be regarded as "blind" and independent with respect to the hypothesis examined in this book. The data were produced before this study was conceived and independently of it. The likelihood that those who collected the data consciously or unconsciously shaped them to support or reject this book's hypothesis is therefore considerably lower. This independence makes these datasets particularly valuable for putting the book's central question to an empirical test.

This does not mean, of course, that all data are free from error or limitation. Any statistical indicator may suffer from measurement problems or differences in how countries collect and report information. But such limitations are different from bias introduced by an effort to prove a particular hypothesis. Moreover, when the results of dozens of independent indicators—collected by different institutions for different purposes—are considered together, they can provide a more reliable picture of reality.

Yet the independence of the data from the hypothesis is not, by itself, enough to produce a valid conclusion. Countries differ in geography, natural resources, culture, history, population, level of development, and countless other factors. For this reason, this book does not rely on a single country, a single indicator, or a handful of well-known examples. Its method is based on the principle of **accumulating evidence**: each indicator reveals only one part of the picture, and only when similar patterns repeatedly emerge across independent areas can we speak with greater confidence about whether a relationship exists.

This book examines indicators related to five fundamental goals of government: **welfare, security, order, justice, and liberty**. In each area, the performance of monarchies and republics will first be compared globally, then at the continental level, and finally within the Middle East. At the end, the findings from these independent analyses will be brought together to determine whether the evidence as a whole presents a consistent picture of the performance of the two systems of government.

This book was not written to prove a belief already held. If the data support conventional assumptions, that conclusion should be accepted. If they present a different picture, we should have the courage to reconsider our beliefs.

Nor was this book written to persuade readers to think as the author does. Its purpose is to encourage them, before reaching a judgment, to ask the same question Aristarchus asked: **What do the data tell us?**

Perhaps, at the end of this investigation, no meaningful pattern will emerge, and the conclusion will be that the form of government, by itself, does not play a decisive role in the fate of nations. But if a stable and recurring pattern does emerge from the mass of data, that pattern deserves to be considered with an open mind and without prejudice.

No study, however extensive, can have the final word on a subject as complex as government and national development. But every study can take us one step closer to the truth—especially when its conclusions are based not on assumptions, but on evidence.

These days, in my country, Iran, the question of which form of government is better and more effective is the subject of intense debate. But this book is written neither for supporters of monarchy nor for supporters of republic. Its intended audience is anyone willing to re-examine their beliefs—whatever those beliefs may be—in the light of empirical evidence, and to allow the data to speak before judgment is passed.

In the pages ahead, neither you nor I will be the final judge. The data will.

N. Fakhr
Autumn 2026

Where Do We Begin?

If we want to judge the performance of two systems of government, we must first answer a simple but fundamental question: **What criteria should we use to evaluate a government?**

Is the mere fact that a government is called a "republic" or a "monarchy" enough to judge one superior to the other? Or should we look instead at the results each system has produced in the lives of its citizens?

The answer is not as simple as it may first appear. Before we can measure how well governments perform, we need to know **what we expect a government to do.**

If we believe that the sole purpose of government is to provide security, we may reach a very different conclusion from someone who considers liberty its most important responsibility. Likewise, if we place justice above welfare, or welfare above social order, our criteria for judging governments will change accordingly.

Before examining the data, therefore, we need to establish the criteria by which governments will be judged.

This brings us to an important point. Empirical science can tell us which governments have been more successful at increasing welfare, reducing crime, or improving public health. But science alone cannot determine **which of these goals is morally or philosophically more important.**

In other words, science can help answer questions about "what is," but the question of "what ought to be" belongs to the realm of political philosophy.

To determine the criteria for evaluating governments, then, we must first take a brief look at political thought.

The history of political thought is filled with disagreement about the purpose of government. Some philosophers have regarded order and security as its most important responsibilities. Others have placed individual liberty at the center of their thinking. Some have considered justice the most fundamental duty of government, while others have viewed the welfare and flourishing of citizens as the most important measure of its success.

These disagreements can be so profound that, at first glance, it may seem impossible to find a common standard for evaluating governments. A closer look, however, reveals a different picture.

Despite the many differences among schools of political philosophy, most of them share considerable common ground regarding a set of fundamental functions of government. Their main disagreement is usually about the **priority** given to these goals, rather than about **whether**

the goals themselves matter. One theory, for example, may place liberty above justice, while another may give justice priority over welfare. Yet it is difficult to find a major political theory that regards security, liberty, order, justice, or welfare as entirely unimportant.

Rather than adopting any one school of political philosophy as the foundation of this study, we can therefore focus on the ground they share and select criteria that appear, directly or indirectly, in most major theories of government.

On this basis, this book evaluates government performance in five main areas:

- **Welfare:** the government's ability to create suitable conditions for a good standard of living, meet basic needs, and improve the economic well-being of its citizens.
- **Security:** the government's ability to protect the lives, property, and safety of society against internal and external threats.
- **Order:** the government's ability to establish effective laws, enforce them properly, maintain stable public institutions, and create a predictable environment for social life and economic activity.
- **Justice:** the extent to which citizens are equal before the law, laws are applied fairly, and people are able to exercise and defend their rights.
- **Liberty:** the extent to which the government respects the individual, civil, and political liberties of its citizens.

These five areas do not, of course, cover every possible dimension of government, and other criteria could certainly be proposed. They do, however, have two important advantages. First, in one form or another, they appear throughout much of the tradition of political philosophy. Second, reasonably reliable and comparable international indicators are available for measuring each of them. This makes an empirical comparison of different systems of government possible.

Beginning with the next chapter, each of these goals will be examined separately. We will see how monarchies and republics have performed in each area and whether the global data reveal any meaningful differences or patterns between them. No conclusion, however, will be based on a single indicator or a single area. Each indicator captures only one part of reality, and **a final conclusion can be reached only when evidence from different areas is considered together**.

Just as we cannot judge a person's overall health from the result of a single medical test, we cannot evaluate a system of government on the basis of a single indicator. A reliable picture emerges only when the results of several independent indicators are examined together. When those results point in the same direction, our confidence that we are observing a genuine pattern increases. When they differ, the disagreement may reveal more complex aspects of government performance.

Accordingly, no conclusion in this book will rest on a single indicator or isolated observation. Each conclusion will emerge from the combined weight of multiple independent indicators and empirical evidence.

The final judgment on monarchy and republic will therefore be postponed until all five areas have been examined. The next chapter begins this journey with the first goal: **the welfare of citizens**.

Welfare Report

GDP per Capita

Why GDP per Capita?

Welfare is multidimensional, and no single indicator can measure all of its aspects on its own. Income, purchasing power, the distribution of wealth, the quality of education and healthcare, economic security, and many other factors all contribute to the well-being of citizens. For this reason, this book uses several complementary indicators to assess welfare, each shedding light on a different part of the picture.

In this chapter, we begin our examination of welfare with **GDP per capita**.

GDP per capita shows the average value of goods and services produced by a country's economy per person over the course of a year. In general, the higher this figure is, the greater the economic resources available to a society to support the well-being of its citizens.

Of course, this indicator alone cannot capture every dimension of welfare. GDP per capita, for example, does not directly tell us how income is distributed, how unequal a society is, what citizens can actually afford with their income, or many other aspects of quality of life. It should therefore not be treated as a complete measure of welfare.

For this reason, the conclusions of this book do not rely on this indicator alone. In the following sections, we will also examine **GDP per capita based on purchasing power parity (PPP)** and the **Gini coefficient** to develop a more complete picture of citizens' welfare. GDP per capita adjusted for purchasing power parity largely accounts for differences in price levels between countries, while the Gini coefficient provides a measure of how income is distributed within a society. Each of these indicators therefore captures a different dimension of welfare, and together they provide a broader basis for comparing the performance of governments.

Nevertheless, GDP per capita is strongly associated with many important aspects of well-being, including the quality of nutrition and housing, access to education and healthcare, and life expectancy. For this reason, economists and international institutions have long used it as one of the key indicators for comparing levels of welfare across countries.

It therefore provides a useful starting point for our examination of welfare. We can then add complementary indicators to build a more comprehensive picture of government performance.

Global Comparison of GDP per Capita: Monarchies and Republics

Now that we have introduced GDP per capita as our first measure of welfare, we can compare monarchies and republics using this indicator.

The data in this section are drawn primarily from the World Bank database for 2023 (Appendix RDS-A01). In the few cases where 2023 data were unavailable for a country, the most recently published figure was used in order to include as many countries as possible in the analysis.

We will first examine the distribution of the data and the descriptive statistics for the two groups. We will then use a statistical test to determine whether the observed difference between their mean GDP per capita values is statistically significant.

One of the first questions readers may ask is whether any apparent advantage of monarchies is simply driven by a handful of oil-rich countries with exceptionally high incomes. This is an entirely reasonable question and one that should not be overlooked. Later in this chapter, we will therefore examine oil-rich countries separately to better understand how much this factor may be influencing the results.

Before turning to the statistical tests, however, let us first look at the distribution of GDP per capita across the two groups.

Descriptive Statistics for GDP per Capita

The kernel density distribution provides an overall picture of the difference between the two groups. For a more precise comparison, however, we also need to examine the main descriptive statistics. The table below shows the number of countries in each group, along with the mean, median, standard deviation, minimum, and maximum GDP per capita.

Statistic	Monarchies	Republics
Number of countries	28	150
Mean GDP per capita	$50,800	$12,988
Median	$34,654	$5,446
Sample standard deviation	$60,342	$19,547
Minimum	$921	$250
Maximum	$256,800	$106,819

The values in the table were calculated from data for 28 monarchies and 150 republics.

The first striking difference between the two groups is their mean. Average GDP per capita is approximately $50,800 in monarchies, compared with $12,988 in republics. Thus, **the average**

GDP per capita of monarchies is about 3.91 times that of republics. The absolute difference between the two means is approximately $37,812 per person.

The mean alone, however, is not enough to describe these data, since a small number of countries with exceptionally high values can pull the average upward. This makes the comparison of medians particularly important. The median GDP per capita is approximately $34,654 among monarchies and $5,446 among republics. In other words, **the median GDP per capita of monarchies is about 6.36 times that of republics**.

This suggests that the observed difference between the two groups is not simply the result of a few exceptionally high values at the upper end of the distribution. Even when we use the median, which is less sensitive to extreme values, a substantial gap between the two groups remains.

In both groups, the mean is higher than the median. This is consistent with the shape of the distribution and indicates that GDP per capita in both groups is skewed toward higher values. Among monarchies, countries such as Monaco, Liechtenstein, and Luxembourg are found at the upper end of the distribution, while among republics, Ireland, Switzerland, Singapore, and the United States occupy the higher end.

The standard deviation is also considerably larger among monarchies. The sample standard deviation is approximately $60,342 for monarchies, compared with $19,547 for republics. This indicates that monarchies form a much more diverse group in terms of GDP per capita: they include countries with values below $1,000 as well as countries with values above $200,000.

The range of the data tells a similar story. Among monarchies, GDP per capita ranges from approximately $921 in Lesotho to about $256,800 in Monaco. Among republics, it ranges from approximately $250 in Burundi to about $106,819 in Ireland.

Overall, the descriptive statistics reveal three main features of the data. First, both the mean and the median GDP per capita are substantially higher among monarchies than among republics. Second, GDP per capita varies more widely among monarchies. Third, the distributions of both groups are asymmetric and influenced by very high values at the upper end.

The descriptive statistics therefore reveal a substantial difference between the two groups, but they do not yet tell us how statistically reliable that difference is, given the variability of the data and the unequal number of countries in the two groups. To answer that question, the next section examines the difference between the means using **Welch's t-test**.

Statistical Inference

So far, the distribution graph and descriptive statistics have given us an initial picture of the difference between monarchies and republics. One question, however, remains: does the observed difference reflect a genuine difference between the two groups, or could it simply be the result of natural variation in the data?

This is the question addressed by inferential statistics. While descriptive statistics summarize and describe the available data, inferential statistics help us determine whether the patterns we observe are also statistically reliable.

Because the same method will be used repeatedly in later chapters, it is useful to explain its basic principles here once. Readers who are not interested in the methodological details may skip this section and go directly to the test results, since each result will be interpreted in plain language. Those who would like a more detailed understanding of the methodology can refer to the **Research Methodology** appendix, which provides a fuller discussion of the research design, the theoretical basis of the statistical tests, the selection of indicators, and the limitations of the study.

In this study, differences between the means of two groups are tested using **Welch's t-test**. This test is designed to compare the means of two independent groups and is particularly appropriate when the groups differ in size or when the variability of their data is unequal.

Both conditions apply in the present study. The number of monarchies and republics is not the same, and, as we saw in the descriptive statistics, the variability in GDP per capita also differs substantially between the two groups. Welch's t-test was therefore chosen to assess whether the difference between their means is statistically significant.

The logic of the test is straightforward. We begin by assuming that there is no real difference between the means of the two groups and that the observed difference is simply the result of random variation in the data. This assumption is known as the **null hypothesis (H_0)**.

The **alternative hypothesis (H_1)**, by contrast, states that the two group means are different and that the observed difference cannot be attributed to chance alone.

The test then calculates how likely it would be, if the null hypothesis were true, to observe a difference as large as the one found in the data—or an even larger one. This probability is expressed by the **p-value**.

Following common practice in the social and empirical sciences, this study uses a significance level of **0.05**. If the p-value is below 0.05, the null hypothesis is rejected and the observed difference is considered statistically significant.

Statistical significance, however, does not by itself establish a causal relationship. The test tells us only that the observed difference is unlikely to be explained by chance alone. Understanding why the difference exists, and what other factors may contribute to it, requires further analysis.

The same statistical approach will be applied throughout the following chapters, where appropriate to the nature of each indicator. This explanation therefore provides the methodological foundation for the statistical analyses used throughout the book.

Now that the basis of the inferential method is clear, we can examine the test results for GDP per capita in monarchies and republics.

Statistical Significance Test for the Difference in GDP per Capita

Having examined the distribution of the data, calculated the descriptive statistics, and introduced the method of statistical inference, we can now ask whether the observed difference in GDP per capita between monarchies and republics is also statistically significant.

For this purpose, **Welch's t-test** was applied to data from 28 monarchies and 150 republics. As explained in the previous section, this test is appropriate for comparing the means of two independent groups when their sample sizes and levels of variability differ. The results are presented below.

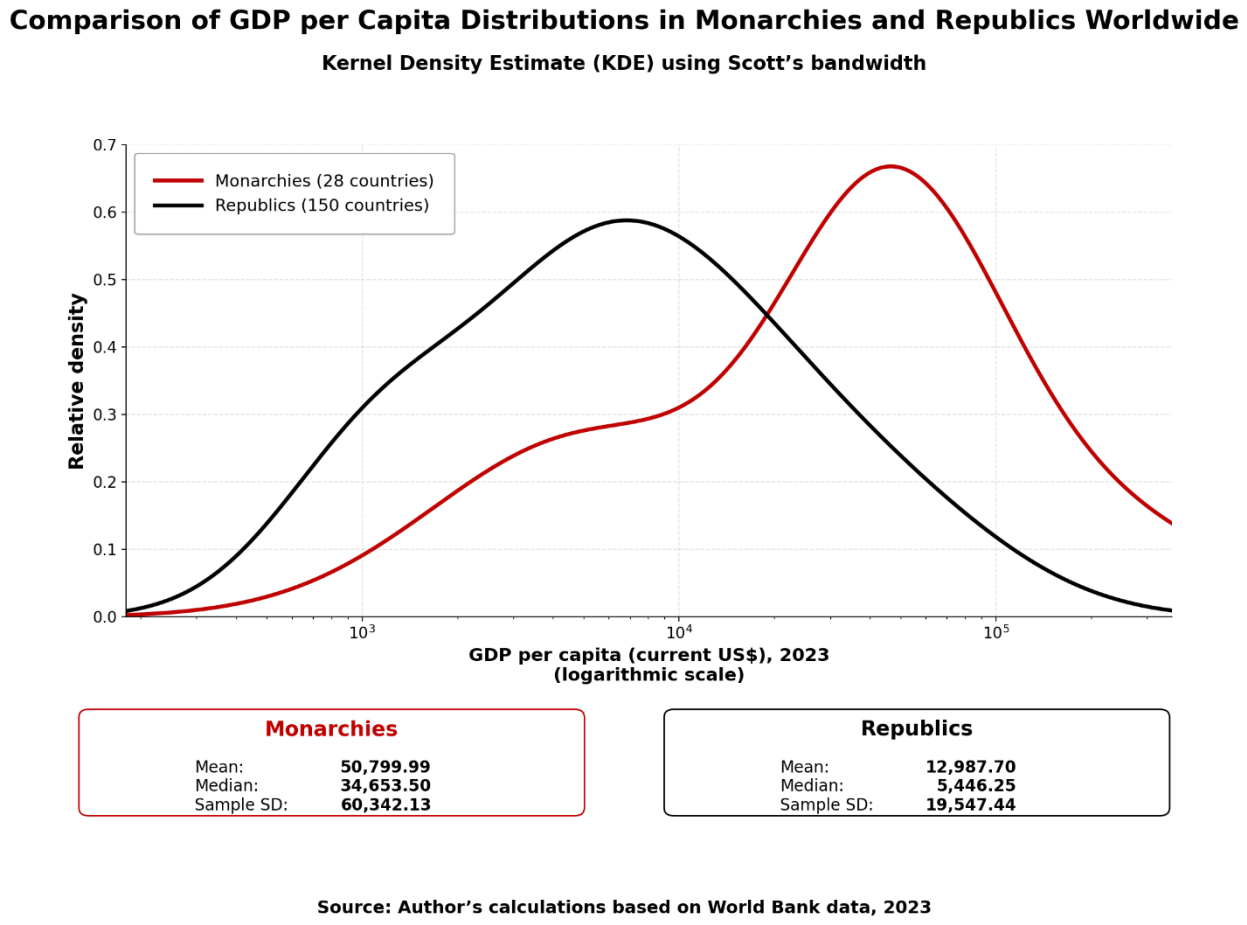
Statistic	Value
Number of monarchies	28
Number of republics	150
Mean for monarchies	$50,800
Mean for republics	$12,988
Difference in means	$37,812
t-statistic	3.284
Welch's degrees of freedom	28.07
p-value	0.00275
95% confidence interval for the mean difference	$14,228 to $61,397
Effect size (Hedges' g)	1.27

The test yields a p-value of approximately 0.00275, well below the significance level of 0.05. We therefore reject the null hypothesis that the two groups have equal mean GDP per capita. In other words, the observed difference between monarchies and republics is unlikely to be explained by random variation alone and is **statistically significant**.

The effect size, which measures the magnitude of the difference, is approximately 1.27 (Hedges' g = 1.27). This indicates that the difference between the two groups is not only statistically significant but also large in practical terms. In addition, the 95% confidence interval for the difference between the means ranges from approximately $14,228 to $61,397. Because the entire interval lies above zero, it further supports the finding that mean GDP per capita is higher among monarchies than among republics.

Statistical significance, however, **does not by itself explain the cause of this difference**. One important question is whether oil-rich countries with exceptionally high incomes are driving up

the average among monarchies. In the next section, we therefore repeat the comparison after excluding oil-rich countries to assess how much this factor may be contributing to the result.



The chart shows that the distribution of GDP per capita among monarchies is, overall, shifted toward higher values than the distribution among republics. Most republics are concentrated at lower levels of GDP per capita, while the monarchy curve shows greater density at higher levels.

At the same time, the considerable overlap between the two curves indicates that the groups are not completely separate. Both include countries with a wide range of income levels. The broad and asymmetric shape of the distributions also reflects the substantial variation within each group.

Overall, the chart is descriptively consistent with the summary statistics: both the mean and the median GDP per capita are substantially higher among monarchies than among republics. The chart alone, however, is not sufficient to establish statistical significance or a causal relationship. These questions must be examined using descriptive statistics and inferential tests.

Excluding Oil-Rich Countries from the Sample

One question that may arise when interpreting the results of this chapter is whether the higher average GDP per capita among monarchies is simply driven by a handful of oil-rich countries with exceptionally high incomes. To address this possibility, the statistical analysis was repeated after excluding the world's major oil powers.

Before conducting this analysis, however, we first need to define what is meant by an "oil-rich country." In this study, countries are classified as oil-rich on the basis of their **proven oil reserves**, rather than their level of oil production or exports.

The reason for this choice is that oil reserves can largely be regarded as a natural and geological endowment. Unlike the way those resources are managed and exploited, their existence is not directly the result of the quality of governance, economic policies, or government performance. Oil production and exports, by contrast, may be influenced by factors such as technology, investment, international sanctions, OPEC quotas, extraction capacity, economic decisions, and other government policies. Since this study examines the relationship between political systems and national performance, using oil production or exports to identify oil-rich countries could introduce part of the effect of government performance into the control variable itself, potentially biasing the analysis. Proven oil reserves therefore provide a more appropriate measure for controlling for the natural endowment of oil.

The next step is to determine a threshold that separates the world's major oil powers from other countries. This threshold should not be entirely arbitrary, since doing so could raise the concern that it was chosen specifically to strengthen a particular result. For this reason, the threshold used in this study was determined from the actual distribution of the data.

To do this, countries were ranked in descending order by their proven oil reserves (Appendix RDS-O01), and the differences between successive values were examined. The results showed a noticeable drop after countries with approximately 25 billion barrels or more in proven reserves. Beyond this point, countries with medium and smaller reserves begin to appear. This clear **natural break** in the distribution provides a reasonable boundary for distinguishing the world's major oil powers from other countries.

Accordingly, this study defines countries with at least **25 billion barrels of proven oil reserves** as major oil powers. This definition is not based on a purely subjective cutoff, but on the structure of the observed data distribution, giving the threshold a methodological basis.

The 25-billion-barrel threshold does not, of course, imply that countries below it have no significant oil resources. Many countries with smaller reserves are still important oil producers. The purpose of this analysis, however, is not to exclude every country that possesses oil, but rather to remove those countries whose exceptionally large reserves place them among the world's major oil powers and give them an unusual natural advantage. The threshold should

therefore be understood as a methodological tool for controlling for this natural advantage, rather than as an absolute dividing line between oil and non-oil countries.

Using this criterion, fourteen countries were classified as major oil powers:

1. Venezuela
2. Saudi Arabia
3. Iran
4. Canada
5. Iraq
6. Kuwait
7. United Arab Emirates
8. Russia
9. United States
10. Libya
11. Nigeria
12. Kazakhstan
13. China
14. Qatar

Of these fourteen countries, thirteen were included in the main sample comparing monarchies and republics and were therefore excluded from the robustness analysis: **four monarchies and nine republics**. Canada, as a Commonwealth realm, was included in the separate analysis of Commonwealth countries and was not part of the main sample of 28 monarchies and 150 republics. Its exclusion therefore did not affect the size of the main sample.

All statistical calculations—including the descriptive statistics and Welch's t-test—were then repeated using the remaining countries. This allows us to examine whether the observed difference between monarchies and republics persists after controlling for the possible influence of exceptionally large oil reserves.

It is important to emphasize that this analysis is a **robustness test**, not a replacement for the main analysis. Its purpose is to assess how stable the original findings remain after removing one of the most important potential sources of bias.

Descriptive Statistics After Excluding Oil-Rich Countries

After the major oil powers were excluded, mean GDP per capita was $50,921 among monarchies and $12,787 among republics. Thus, **even after excluding these countries, the average GDP per capita of monarchies remained approximately 3.98 times that of republics**.

The medians also show a substantial difference. Median GDP per capita was $33,192 among monarchies and $5,320 among republics. In other words, **the median GDP per capita of monarchies was approximately 6.24 times that of republics**.

Because the median is less sensitive than the mean to exceptionally high values, this finding suggests that the observed difference cannot be attributed simply to the presence of a few extremely wealthy countries. The gap is also clearly visible at the center of the two distributions.

The sample standard deviation was approximately $64,926 among monarchies and $19,220 among republics. The greater dispersion among monarchies indicates that, even after the major oil powers are excluded, this group still includes a diverse range of low-income, middle-income, and very wealthy countries.

Statistical Significance Test After Excluding Oil-Rich Countries

To determine whether the observed difference between monarchies and republics persists after excluding the major oil powers, **Welch's t-test** was applied. As explained earlier, this test is appropriate for comparing the means of two independent groups with unequal sample sizes and variances.

The hypotheses were defined as follows:

- **Null hypothesis (H_0):** The mean GDP per capita of monarchies and republics is equal.
- **Alternative hypothesis (H_1):** The mean GDP per capita of the two groups is different.

As in the main analysis, the significance level was set at **0.05**.

Test Results

Statistic	Value
Number of monarchies	**24**
Number of republics	**141**
Mean for monarchies	**$50,921**
Mean for republics	**$12,787**
Difference in means	**$38,134**
t-statistic	**2.856**
Welch's degrees of freedom	**23.69**
p-value	**0.0088**
95% confidence interval	**$10,559 to $65,709**
Effect size (Hedges' g)	**1.26**

Since the p-value (≈ 0.0088) is below the significance level of 0.05, the null hypothesis is rejected. **Thus, even after excluding the major oil powers, the difference in mean GDP per capita between monarchies and republics remains statistically significant.**

The effect size, Hedges' g ≈ 1.26, is also considered large by conventional standards. The observed difference is therefore not only statistically significant but also substantial in magnitude.

The 95% confidence interval for the difference between the means ranges from approximately $10,559 to $65,709. Since the entire interval lies above zero, the data are not consistent with equal group means. More precisely, under the statistical method used here, the estimated interval contains only positive values for the difference between the two means.

Comparison with the Initial Analysis

Statistic	Initial Analysis	After Excluding Oil-Rich Countries
Number of monarchies	**28**	**24**
Number of republics	**150**	**141**
Mean for monarchies	**$50,800**	**$50,921**
Mean for republics	**$12,988**	**$12,787**
Difference in means	**$37,812**	**$38,134**
t-statistic	**3.284**	**2.856**
Welch's degrees of freedom	**28.07**	**23.69**
p-value	**0.00275**	**0.0088**
Hedges' g	**1.27**	**1.26**

This comparison shows that excluding the major oil powers reduced the t-statistic and increased the p-value. This change, however, was not caused by a reduction in the difference between the two group means. On the contrary, the mean difference increased slightly, from approximately $37,812 in the initial analysis to about $38,134 after the oil-rich countries were excluded. The smaller sample size and the resulting increase in the standard error of the mean difference caused the t-statistic to fall and the p-value to rise.

The group means themselves also reveal an interesting pattern. After the major oil powers were excluded, the mean for monarchies increased slightly from $50,800 to $50,921, while the mean for republics declined from $12,988 to $12,787. As a result, the ratio between the two means increased from approximately 3.91 in the initial analysis to about 3.98 after the exclusion of the major oil powers.

The effect size changed only slightly, from 1.27 to 1.26, and remained within the range generally considered large. Overall, excluding the major oil powers did not alter the main finding. In fact, the raw difference between the two group means became slightly larger. Despite the reduction in statistical power resulting from the smaller sample, the difference in mean GDP per capita between monarchies and republics remained statistically significant and substantial in magnitude.

Conclusion

This analysis can be regarded as a robustness test of the chapter's main findings. If the higher average GDP per capita among monarchies were driven solely by a handful of major oil-rich countries, we would expect the difference between the two groups to shrink substantially—or lose statistical significance—once those countries were excluded. The results show that this did not happen.

After excluding four monarchies and nine republics with exceptionally large oil reserves, the average GDP per capita of monarchies remained nearly four times that of republics. The median for monarchies was also more than six times that of republics, and Welch's t-test continued to show a statistically significant difference. The effect size also remained large.

Notably, after these countries were excluded, the difference in mean GDP per capita between the two groups did not decrease; it increased slightly. Thus, in this dataset, the presence of major oil powers does not explain the higher average GDP per capita observed among monarchies compared with republics.

This finding, of course, **does not by itself prove** that the type of political system causes the observed difference. Many other historical, geographical, institutional, and economic factors may also contribute to it. Nevertheless, the robustness test indicates that exceptionally large oil reserves, on their own, are not sufficient to explain the observed gap between monarchies and republics.

Excluding Oil-Rich Countries and Very Wealthy Microstates

The previous section showed that excluding countries with exceptionally large oil reserves does not eliminate the observed difference between monarchies and republics. However, another possible objection remains: a few very small and exceptionally wealthy monarchies, such as Monaco, Liechtenstein, and Luxembourg, may disproportionately raise the average GDP per capita of monarchies.

Because of their small size, very low populations, and distinctive economic structures, these countries are unusual cases in the global economy. Comparing them directly with more typical countries could therefore produce an exaggerated picture of the economic position of monarchies.

To examine this possibility, a second robustness test was conducted. In this analysis, in addition to the major oil powers, four wealthy monarchical microstates—Monaco, Liechtenstein, Luxembourg, and Andorra—were excluded from the sample. The purpose was to determine whether the difference between the two groups would persist after these exceptional cases were removed, or whether much of the observed gap was driven by a handful of very small and wealthy countries.

After these exclusions, the sample consisted of **20 monarchies and 141 republics**.

Descriptive Statistics After Excluding Oil-Rich Countries and Wealthy Microstates

The descriptive statistics are presented below.

Descriptive Statistic	Monarchies	Republics
Number of countries	20	141
Mean GDP per capita	$28,924	$12,787
Median GDP per capita	$25,159	$5,320
Sample standard deviation	$26,955	$19,220
Minimum	$921	$250
Maximum	$90,984	$106,819

After excluding the oil-rich countries and the four wealthy microstates, the mean GDP per capita of monarchies fell substantially. This decline is expected, since several countries with exceptionally high per capita incomes were removed from the monarchy group. Nevertheless, **the average GDP per capita of monarchies remained approximately 2.26 times that of republics**.

The difference between the medians also remains striking. Median GDP per capita is approximately $25,159 among monarchies and $5,320 among republics. In other words, **the median GDP per capita of monarchies is approximately 4.73 times that of republics**.

The median is particularly important here because, unlike the mean, it is much less affected by a small number of extremely high values. The substantial gap that remains between the two medians therefore suggests that the observed difference cannot be attributed simply to a handful of exceptionally wealthy countries.

The sample standard deviation is approximately $26,955 among monarchies and $19,220 among republics. Compared with the previous analysis, removing the four exceptionally wealthy microstates greatly reduces the dispersion within the monarchy group. Even so, GDP per capita remains somewhat more dispersed among monarchies than among republics.

Statistical Significance Test

To determine whether the difference between the two group means remained statistically significant after these exclusions, **Welch's t-test** was applied once again. This test is appropriate for comparing two groups with unequal sample sizes and different levels of variability.

Statistic	Value
Difference between group means	$16,137
Welch's t-statistic	2.586
Welch's adjusted degrees of freedom (df)	21.82
p-value	0.0169
95% confidence interval for the mean difference	$3,188 to $29,086
Standardized effect size (Hedges' g)	0.79
Effect size interpretation	Moderate to large; very close to the conventional threshold for a large effect

The p-value indicates how likely it would be, if there were actually no difference between the two group means, to observe a difference this large or larger simply as a result of random variation in the data. Throughout this study, p-values below 0.05 are treated as statistically significant evidence against the hypothesis of equal means.

Since the p-value (≈ 0.0169) is below the significance level of 0.05, the hypothesis of equal group means is rejected. **Thus, even after excluding countries with exceptionally large oil reserves and the four very wealthy microstates, the difference in mean GDP per capita between monarchies and republics remains statistically significant.**

The 95% confidence interval for the difference between the means ranges from approximately $3,188 to $29,086. Since the entire interval lies above zero, this result is also consistent with a statistically significant difference between the two group means at the 5% level.

The effect size, Hedges' g, is approximately 0.79. Effect size provides a standardized measure of how substantial the difference between the groups is, independent of the original unit of measurement. Under commonly used approximate guidelines, values around 0.2 are considered small, around 0.5 moderate, and around 0.8 large. The observed value can therefore be described as a **moderate-to-large effect, very close to the conventional threshold for a large effect**.

These results show that excluding the exceptional cases substantially reduces the magnitude of the difference between the two groups, but does not eliminate it.

Conclusion of the Second Robustness Test

This analysis represents the most stringent robustness test conducted in this chapter. In the first step, countries whose economic position could be strongly influenced by exceptionally large oil reserves were excluded. Four monarchical microstates were then removed because their very small populations and distinctive economic structures could disproportionately raise the average GDP per capita of monarchies.

After these restrictions were applied, mean GDP per capita among monarchies fell from approximately $50,800 in the full sample to about $28,924. This decline shows that the very wealthy microstates do indeed make a substantial contribution to the higher initial average among monarchies. Nevertheless, the mean for monarchies remained more than 2.2 times that of republics, and the difference between the two groups remained statistically significant.

The median shows a similar pattern. Even after these exceptional cases were removed, median GDP per capita among monarchies remained approximately 4.7 times that of republics. This is particularly important because the median is much less sensitive to extremely high values at the upper end of the distribution.

The observed advantage of monarchies on this indicator therefore cannot be attributed solely to a handful of oil-rich countries or exceptionally wealthy microstates. This finding does not, however, mean that the type of government causes the economic difference between the two groups. It shows only that, in the available data, the observed statistical relationship persists even after two important groups of exceptional countries are excluded.

Identifying the causes of this difference would require examining other variables, including institutional quality, historical background, initial levels of development, culture, geography, economic structure, political stability, and public policy.

Summary of the Robustness Tests

To assess the robustness of the main finding, GDP per capita was compared between monarchies and republics in three stages.

The first analysis included all countries in the original sample. In the second, countries with exceptionally large oil reserves were excluded to determine whether oil wealth was the main factor behind the observed difference. In the third and more stringent analysis, four very wealthy monarchical microstates were also excluded in addition to the major oil powers.

The results of all three analyses are summarized below.

Analysis	Monarchies	Republics	Mean: Monarchies	Mean: Republics	Mean Difference	Welch's t	p-value	Hedges' g	Interpretation
Full sample	**28**	**150**	**$50,800**	**$12,988**	**$37,812**	**3.284**	**0.00275**	**1.27**	**Significant difference with a very large effect size**

Analysis	Monarchies	Republics	Mean: Monarchies	Mean: Republics	Mean Difference	Welch's t	p-value	Hedges' g	Interpretation
Excluding major oil powers	24	141	$50,921	$12,787	$38,134	2.856	0.0088	1.26	Significant difference with a very large effect size
Excluding major oil powers and very wealthy microstates	20	141	$28,924	$12,787	$16,137	2.586	0.0169	0.79	Significant difference with a moderate-to-large effect size, close to large

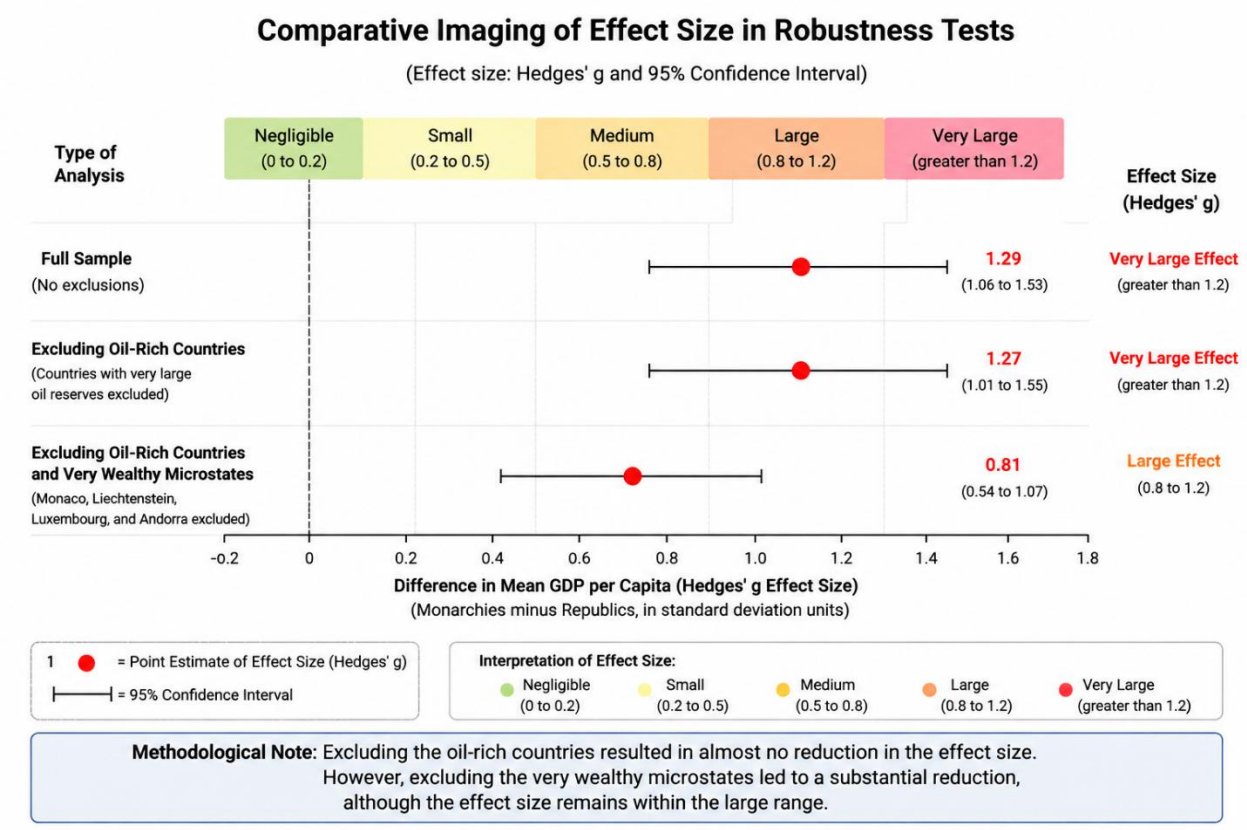
In this table, the p-value indicates how inconsistent the observed evidence is with the assumption that the two group means are equal. In all three analyses, the p-value is below the 0.05 threshold. The difference between the mean GDP per capita of monarchies and republics therefore remains statistically significant at every stage.

The effect size provides a standardized measure of the magnitude of the difference. In the full sample, the effect size was approximately 1.27. After the major oil powers were excluded, it remained almost unchanged at 1.26. This stability is consistent with the previous analysis and indicates that removing the major oil powers had little effect on the relative magnitude of the difference between the two groups.

In the third analysis, after the four very wealthy microstates were also excluded, the effect size fell to approximately 0.79. This decline indicates that these exceptionally wealthy microstates make a substantial contribution to the magnitude of the initial difference between the two group means. Nevertheless, the resulting effect remains moderate to large, very close to the conventional threshold for a large effect, and the difference remains statistically significant.

The two robustness tests therefore reveal different patterns regarding the source of the observed gap. Excluding the major oil powers produced virtually no reduction in the difference between the means; in fact, the raw mean difference increased slightly. By contrast, removing the very wealthy monarchical microstates substantially reduced the gap. The data therefore suggest that wealthy microstates contribute to the magnitude of the initial difference between the two groups, but their presence is not sufficient to explain the entire gap.

In other words, after approximately controlling for the effect of exceptionally large oil reserves and then removing the smallest and wealthiest monarchies, mean GDP per capita among monarchies remains more than twice that of republics, their median remains nearly five times as high, and the difference between the group means remains statistically significant. The main finding of this chapter therefore cannot be attributed solely to these two groups of exceptional cases.



Global Comparison of GDP per Capita: Constitutional and Non-Constitutional Monarchies

In the previous section, we found that monarchies, on average, have higher GDP per capita than republics, and that the difference is statistically significant. Monarchies, however, are not a completely homogeneous group. In some, the monarch plays a largely symbolic role and

political power rests with elected institutions. In others, the monarch continues to exercise substantial executive and political authority.

This raises an important question:

Is the economic advantage observed among monarchies shared by different types of monarchy, or is it mainly associated with one of them?

To examine this question, monarchies were divided into two groups:

- **Constitutional monarchies:** 15 countries
- **Semi-constitutional and absolute monarchies:** 13 countries

Each of these groups was then compared separately with the 150 republics.

The analysis proceeds in four stages. First, we examine the distribution of the data. Next, we present the descriptive statistics for each group. We then use Welch's t-test to assess the differences between the group means. Finally, the results are interpreted in plain language.

This comparison allows us to determine whether the pattern observed in the overall comparison between monarchies and republics is also present in both types of monarchy, or whether the overall result is driven mainly by the performance of one of these two groups.

Descriptive Statistics

To compare the distributions of GDP per capita more closely, the main descriptive statistics were calculated for all three groups. In this section, the term **"non-constitutional monarchies"** refers collectively to semi-constitutional and absolute monarchies. All monetary values are expressed in current US dollars.

Descriptive Statistic	Republics	Constitutional Monarchies	Semi-Constitutional and Absolute Monarchies
Number of countries	**150**	**15**	**13**
Mean GDP per capita	**$12,988**	**$43,884**	**$58,780**
Median	**$5,446**	**$46,812**	**$32,891**
Sample standard deviation	**$19,547**	**$36,970**	**$80,432**
Minimum	**$250**	**$921**	**$3,756**
First quartile (Q1)	**$1,974**	**$9,298**	**$5,652**
Third quartile (Q3)	**$14,193**	**$59,380**	**$49,851**

Descriptive Statistic	Republics	Constitutional Monarchies	Semi-Constitutional and Absolute Monarchies
Maximum	**$106,819**	**$133,231**	**$256,800**
Coefficient of variation	**1.51**	**0.84**	**1.37**

Source: Author's calculations based on the research dataset.

Interpretation of the Descriptive Statistics

The descriptive statistics show that mean GDP per capita is higher in both types of monarchy than in republics.

The mean is approximately **3.38 times higher in constitutional monarchies** and **4.53 times higher in semi-constitutional and absolute monarchies** than in republics.

The mean, however, can be influenced by a small number of exceptionally wealthy countries. The median therefore provides an important additional perspective. Median GDP per capita in constitutional monarchies is approximately **8.60 times** the median in republics, while in semi-constitutional and absolute monarchies it is approximately **6.04 times** the median in republics.

Thus, the higher GDP per capita observed among monarchies is not limited to a few exceptionally high values. Even when we use the median, which is less sensitive to extreme values, a substantial gap remains between republics and both types of monarchy.

A comparison of the first quartile shows the same pattern, although less strongly. The first quartile is approximately $9,298 in constitutional monarchies and $5,652 in semi-constitutional and absolute monarchies, compared with approximately $1,974 in republics.

Put simply, even in the lower part of the distribution, GDP per capita values are higher in both groups of monarchies than in republics. As with the other descriptive statistics, however, this comparison describes only the distribution of the observed data and does not by itself provide a basis for statistical inference.

There is, however, considerable variation within the group of semi-constitutional and absolute monarchies. The mean for this group is approximately $58,780, while the median is about $32,891 and the maximum reaches approximately $256,800. The large gap between the mean and median, together with the very high standard deviation, indicates that a few exceptionally wealthy countries strongly stretch the upper end of this group's distribution.

The coefficient of variation provides another perspective on these differences. Constitutional monarchies, with a coefficient of variation of approximately 0.84, are the most homogeneous of the three groups in relative terms. The corresponding figure is approximately 1.37 for semi-

constitutional and absolute monarchies and 1.51 for republics. Relative to their respective means, therefore, republics show the greatest dispersion, while constitutional monarchies show the least.

Overall, the descriptive statistics show that both types of monarchy occupy higher levels than republics in terms of the mean, median, and distributional quartiles. To determine how statistically reliable the differences between the means are, however, inferential tests are required.

Inferential Statistics

To compare mean GDP per capita across the groups, **Welch's t-test** was used.

Welch's test is appropriate when two independent groups have unequal sample sizes and equal variances cannot be assumed. In this study, the number of republics is much larger than the number of countries in either monarchy group, and the variability of the data also differs substantially across the groups. Welch's test is therefore more appropriate than a t-test that assumes equal variances.

Statistical Hypotheses

For each comparison, the hypotheses were defined as follows:

Null hypothesis (H_0): The mean GDP per capita of the two groups is equal.

Alternative hypothesis (H_1): The mean GDP per capita of the two groups is different.

In symbolic form:

$\mathbf{H_0: \mu_m = \mu_r}$

$\mathbf{H_1: \mu_m \neq \mu_r}$

where:

- μ_m is the mean GDP per capita of the monarchy group being examined.
- μ_r is the mean GDP per capita of the republics.

A **two-tailed test** was used. This tests whether the two group means differ without assuming in advance which group has the higher mean.

Welch's t-test Results

Comparison	Mean Difference	Welch's t	Welch's df	Two-tailed p-value	95% CI for Mean Difference	Hedges' g	Result
Constitutional monarchies vs. republics	**$30,897**	**3.192**	**14.79**	**0.0061**	**$10,243 to $51,550**	**1.42**	**Statistically significant; very large effect**
Semi-constitutional and absolute monarchies vs. republics	**$45,792**	**2.047**	**12.12**	**0.0629**	**−$2,882 to $94,466**	**1.58**	**Not statistically significant in the two-tailed test; very large effect**

Explanation of the statistics

The Welch t-statistic expresses the difference between the two group means relative to the standard error of that difference. The standard error itself depends on both the variability of the data and the number of observations in each group. The larger the absolute value of the t-statistic, the stronger the statistical evidence against the hypothesis of equal means.

Welch's degrees of freedom are calculated from the sample sizes and variances of the two groups and are used to determine the reference distribution of the test and its corresponding p-value.

The p-value indicates how compatible a difference this large or larger—in either direction—would be with random variation if the null hypothesis were true. In this study, a p-value below 0.05 is used as the threshold for statistical significance.

The 95% confidence interval provides a range of values for the difference between the two group means that are compatible with the observed data. If this interval does not include zero, the two-tailed test is statistically significant at the 5% level.

Hedges' g expresses the magnitude of the difference between the two groups in standardized terms. As an approximate guideline, values around 0.2 are generally considered small, around 0.5 moderate, and around 0.8 or higher large.

Interpretation of the Test Results

For constitutional monarchies compared with republics, the difference in mean GDP per capita is approximately $30,897. Welch's test yields **t = 3.192** and **p ≈ 0.0061**. Since the p-value is well

below 0.05, the hypothesis of equal means is rejected. The difference in mean GDP per capita between constitutional monarchies and republics is therefore statistically significant.

The 95% confidence interval for this difference ranges from approximately $10,243 to $51,550 and lies entirely above zero. In addition, the effect size is **Hedges' g ≈ 1.42**, indicating a very large difference between the two groups.

The result for semi-constitutional and absolute monarchies requires a more careful interpretation. Their mean GDP per capita is approximately $45,792 higher than that of republics, and the effect size is also very large (**Hedges' g ≈ 1.58**). However, the two-tailed Welch test yields **p ≈ 0.0629**, slightly above the predetermined 0.05 threshold. Under the criterion adopted in this study, the hypothesis of equal means is therefore not rejected in this comparison.

The 95% confidence interval for the mean difference ranges from approximately −$2,882 to $94,466 and includes zero, which is consistent with the significance-test result.

At first glance, a very large effect size alongside a statistically non-significant result may appear contradictory, but there is no contradiction. Effect size measures the magnitude of the observed difference, whereas the p-value depends not only on the size of that difference but also on sample size and variability. The semi-constitutional and absolute monarchy group contains only 13 countries and has a standard deviation of more than $80,000. As a result, the estimated mean for this group carries considerable uncertainty, producing a wide confidence interval.

The results therefore show that the advantage observed in the overall comparison between monarchies and republics appears in both monarchy subgroups in the descriptive statistics and in the magnitude of the effect, but the inferential evidence is not equally strong for the two groups. Among constitutional monarchies, the difference is both large and statistically significant. Among semi-constitutional and absolute monarchies, the observed difference and effect size are even larger, but because of the small number of countries and the very high variability of the data, the two-tailed test does not reach statistical significance at the 0.05 level.

The available data therefore do not support the conclusion that the statistical advantage observed in the full monarchy sample has been established with the same level of confidence for both types of monarchy. What can be stated more precisely is that **constitutional monarchies show a clear and statistically significant difference from republics, while semi-constitutional and absolute monarchies show a very large observed difference but also substantial statistical uncertainty because of their small sample size and high variability.**

Global Comparison of GDP per Capita: Commonwealth Realms and Republics

In addition to the independent monarchies examined in the previous sections, there is another group of countries whose distinctive legal and historical characteristics make them worth examining separately. These countries are known as the **Commonwealth realms**. Each is an

independent state with its own constitution, parliament, and government, while recognizing the British monarch as its head of state. They should therefore not be regarded as colonies or dependencies of the United Kingdom.

From a constitutional perspective, the Commonwealth realms are forms of constitutional monarchy. However, because they share the same monarch and have a distinct historical and institutional background, they are treated as a separate group in this study.

This distinction is important for two reasons. First, it allows the economic performance of the Commonwealth realms to be assessed without combining them with other monarchies. Second, it allows us to examine whether the pattern observed among other monarchies also appears in this distinctive group.

In this section, **14 Commonwealth realms** are compared with **150 republics**.

As in the previous sections, the comparison proceeds in four stages:

1. Examining the distribution of the data using a **Kernel Density Estimate (KDE)** plot
2. Presenting the **descriptive statistics**
3. Applying **Welch's t-test**
4. **Interpreting the results**

Descriptive Statistics

The table below presents the main descriptive statistics for the two groups. All monetary values are expressed in current US dollars.

Descriptive Statistic	Republics	Commonwealth Realms
Number of countries	150	14
Mean GDP per capita	$12,988	$22,329
Median	$5,446	$12,516
Sample standard deviation	$19,547	$20,987
Coefficient of variation	1.51	0.94

Source: Author's calculations based on World Bank data.

Interpretation of the Descriptive Statistics

The descriptive statistics show that **mean GDP per capita in the Commonwealth realms is approximately 1.72 times that of the republics**.

The difference is not limited to the mean. **Median GDP per capita in the Commonwealth realms is also approximately 2.30 times that of the republics**. This suggests that the descriptive difference is not driven solely by a small number of very wealthy countries, since the median is also substantially higher among the Commonwealth realms.

The coefficient of variation also indicates that the Commonwealth realms are more homogeneous in relative terms. It is approximately 0.94 for the Commonwealth realms, compared with 1.51 for the republics. Thus, GDP per capita shows greater dispersion relative to the group mean among republics.

Overall, the descriptive statistics indicate that the Commonwealth realms have higher GDP per capita than the republics in this dataset. To determine how statistically reliable the difference between the two group means is, however, inferential testing is required.

Inferential Statistics

To compare mean GDP per capita between the two groups, **Welch's t-test** was used.

This test is appropriate for comparing two independent groups with unequal sample sizes and different levels of variability, both of which apply to the present comparison.

Statistical Hypotheses

Null hypothesis (H_0):
The mean GDP per capita of the Commonwealth realms and republics is equal.

Alternative hypothesis (H_1):
The mean GDP per capita of the Commonwealth realms and republics is different.

In symbolic form:

H_0: $\mu c = \mu r$

H_1: $\mu c \neq \mu r$

As in the previous comparisons, a **two-tailed test** was used. The direction of the difference was therefore not assumed in advance; the test simply examines whether the two group means differ.

Welch's t-test Results

Statistic	Value
Difference between group means	$9,341
Welch's t-statistic	1.602
Adjusted degrees of freedom (df)	15.18
Two-tailed p-value	0.1298
Effect size (Hedges' g)	0.47
Effect size interpretation	Close to moderate

Interpretation of the Results

Mean GDP per capita in the Commonwealth realms is approximately $9,341 higher than in the republics. However, Welch's test yields **$t = 1.602$** and **$p \approx 0.1298$**.

Since the p-value is above the 0.05 significance level, the hypothesis of equal means is not rejected in the two-tailed test. Therefore, **the available data do not provide sufficient evidence to conclude that the difference in mean GDP per capita between the Commonwealth realms and republics is statistically significant**.

This result does not establish that the two groups are equal. In the observed data, the mean for the Commonwealth realms is more than $9,000 higher than that of the republics. However, the Commonwealth group contains only 14 countries, and the data also show considerable variability. As a result, the uncertainty surrounding the estimate is too large for the observed difference to reach statistical significance at the 5% level.

The effect size is **Hedges' $g \approx 0.47$**. Under commonly used approximate guidelines, values around 0.2 are considered small, around 0.5 moderate, and around 0.8 or higher large. The observed standardized difference can therefore be described as **close to a moderate effect**.

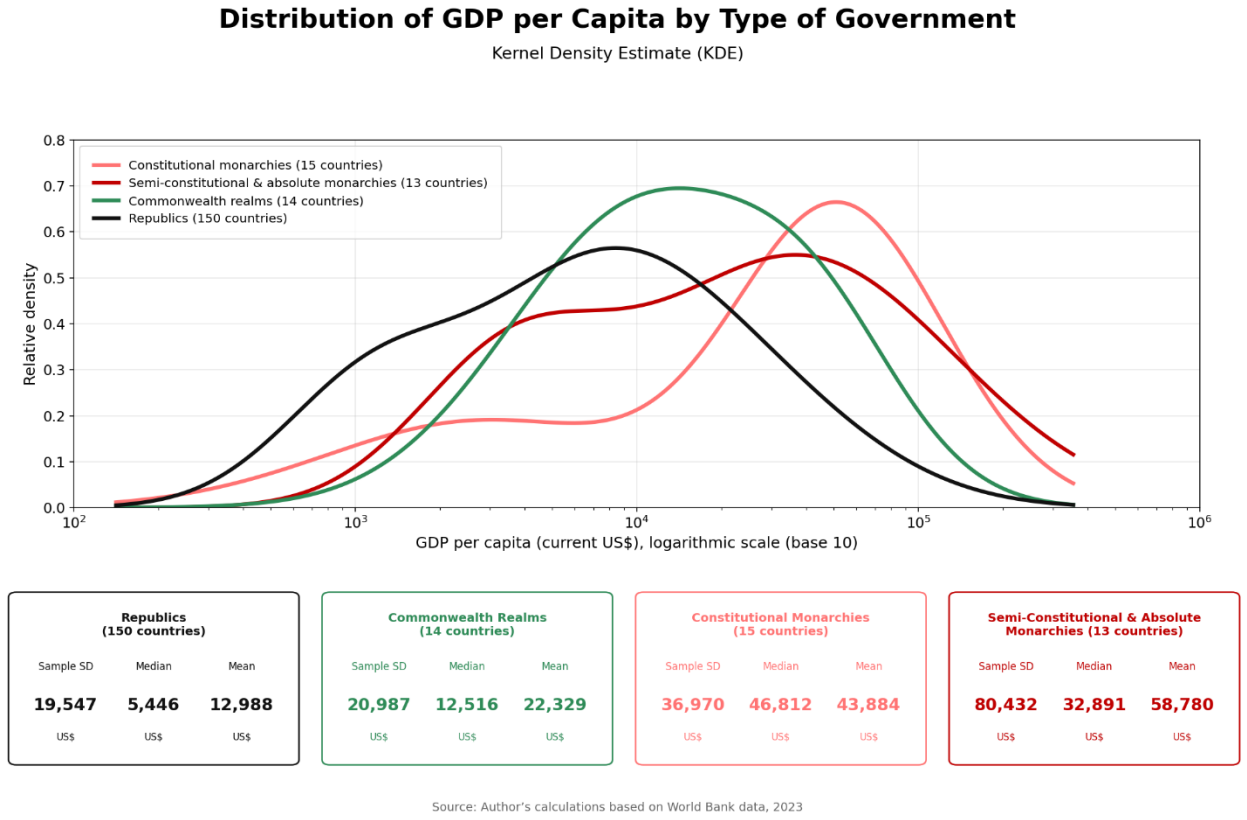
This result illustrates the distinction between descriptive statistics, effect size, and statistical significance. The descriptive statistics show that the Commonwealth realms in this sample have higher mean and median GDP per capita, and the effect size is not negligible. However, given the limited number of countries in this group, the statistical evidence is insufficient to reject the hypothesis of equal means.

Summary

In the available data, the Commonwealth realms have, on average, a higher GDP per capita than the republics. Their mean GDP per capita is approximately 1.72 times that of the republics, while their median is approximately 2.30 times as high. Relative variability is also lower among the Commonwealth realms.

However, the two-tailed Welch's t-test indicates that this difference is not statistically significant at the conventional 5% level ($p \approx 0.1298$). The effect size is approximately 0.47, which is close to a moderate effect.

Therefore, the most appropriate conclusion is that the Commonwealth realms show a notable descriptive advantage over the republics in this sample, but the available evidence is insufficient to generalize this difference as statistically significant.



The chart shows that the distribution of GDP per capita in all three monarchy-related groups is, overall, shifted toward higher values than in the republics. **Constitutional monarchies**, with a median of about $46,812, show the greatest concentration at higher income levels, and their distribution is also relatively more compact. **Semi-constitutional and absolute monarchies** have the highest mean, but their broader curve indicates much greater variation, with both middle-income countries and a few exceptionally wealthy countries within the group.

The Commonwealth realms occupy an intermediate position between the other monarchy groups and the republics. Both their mean and median are higher than those of the republics, although the gap is smaller than for the other two monarchy groups. By contrast, **the republics** are more heavily concentrated at lower levels of GDP per capita and have the lowest mean and median of all four groups.

Overall, the chart reveals a fairly clear descriptive pattern: constitutional monarchies are most concentrated at higher income levels; semi-constitutional and absolute monarchies have a very high mean but much greater dispersion; the Commonwealth realms occupy an intermediate position; and republics are concentrated mainly in the lower part of the distribution. However, the overlap between the curves shows that the groups are not completely separate, and the chart alone cannot establish statistical significance or a causal relationship.

Comparison of GDP per Capita Between Monarchies and Republics Across Continents and in the Middle East

In the previous sections, we compared the economic performance of monarchies and republics at the global level using **GDP per capita**. Although such a comparison provides an overall picture of the differences between the two forms of government, some of the observed differences may be influenced by the wide variation among countries in their history, culture, geography, economic structure, or regional conditions.

One way to assess the robustness of the findings is to repeat the same analysis using samples that are more homogeneous in certain respects. Countries within the same continent generally share more similarities than countries across the world as a whole, particularly in terms of historical background, geographic setting, economic interactions, cultural characteristics, and, to some extent, political institutions. Comparing monarchies and republics within each continent can therefore help determine whether the pattern observed at the global level also appears in settings where countries share relatively similar conditions.

In addition to the continental comparisons, the **Middle East** is examined separately. Despite its internal diversity, the region shares a number of historical, climatic, cultural, religious, and geopolitical characteristics. It therefore provides another useful setting for examining the relationship between the type of government and countries' economic performance.

For each region, we first examine the distribution of the data and the descriptive statistics. The difference between monarchies and republics is then evaluated using **Welch's t-test**, while the effect size is reported using **Hedges' g**. This allows us to assess not only whether the observed difference is statistically significant, but also how large that difference is.

For Europe, an additional **robustness check** is conducted. In this analysis, republics with a history of communist rule are excluded from the sample to determine whether the observed pattern persists after this change in sample composition. If the results remain similar under this more restricted comparison, we can be more confident that the findings are not simply a product of the particular composition of the original sample, but reflect a more robust pattern.

Comparison of GDP per Capita Between Monarchies and Republics in Asia

With substantial numbers of both monarchies and republics, Asia is one of the most suitable continents for comparing the performance of these two forms of government. Although Asian countries differ considerably in their levels of development, natural resources, and economic structures, they share more historical, geographical, and cultural characteristics with one another than countries across the world as a whole. Examining Asia can therefore help determine whether the pattern observed in the global analysis also appears within a regional setting.

Descriptive Statistics

Among Asian countries, the mean **GDP per capita** is **$27,339** for monarchies and **$9,750** for republics. In other words, the average GDP per capita of Asian monarchies is approximately **2.8 times that of the continent's republics**.

However, as emphasized in the previous sections, the mean alone cannot provide a complete picture of the data, since it may be influenced by a small number of countries with exceptionally high or low incomes. For this reason, the median, sample standard deviation, and the overall shape of the distribution are also examined alongside the mean.

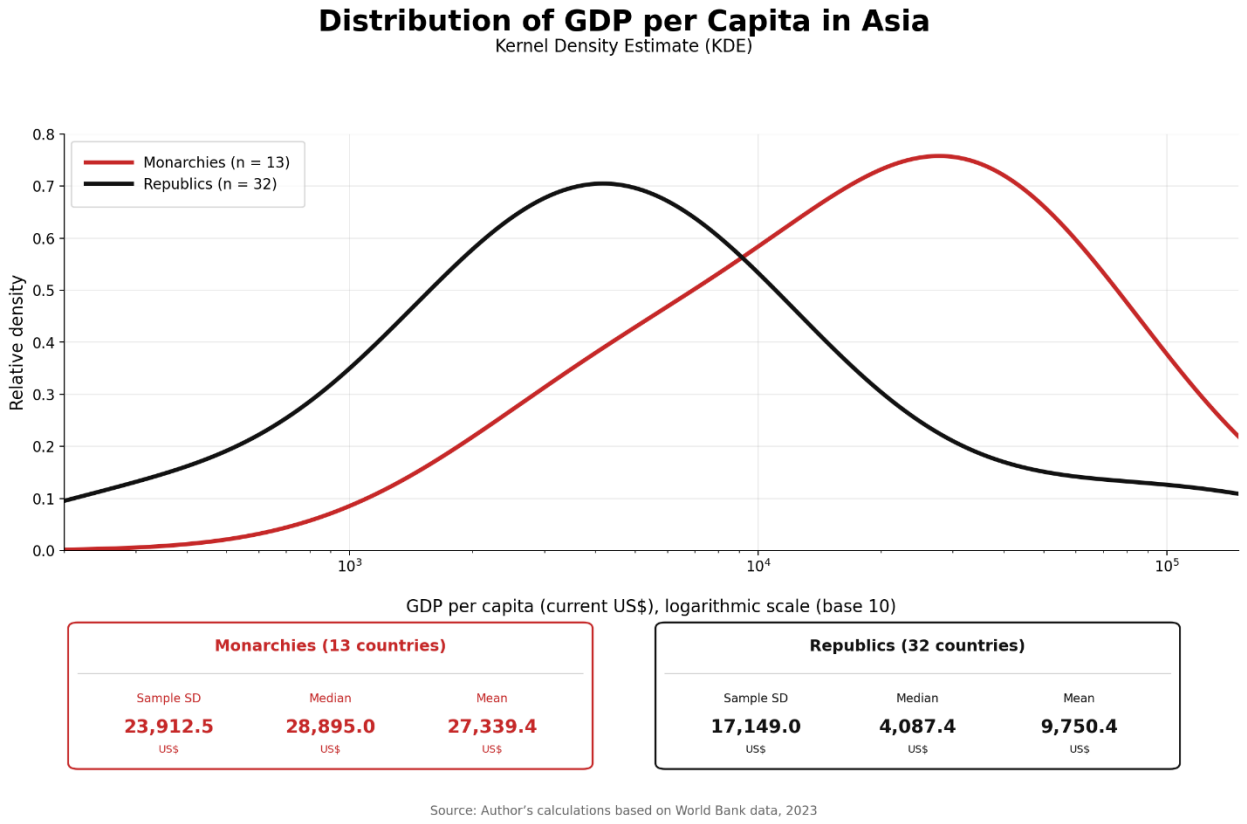
Inferential Statistics

To assess whether the observed difference between Asian monarchies and republics could simply be attributed to sampling variation, **Welch's t-test** was used.

The results show that the difference in mean GDP per capita between the two groups **is statistically significant** ($p < 0.05$). The effect size, measured by **Hedges' g**, is also in the **large** range, indicating that the difference is not only statistically significant but also substantial in magnitude.

Summary

The data for Asia therefore show the same general pattern observed in the global comparison. In this sample, Asian monarchies have, on average, a higher GDP per capita than Asian republics, and the statistical test indicates that this difference cannot readily be attributed to random sampling variation.



The chart shows a clear descriptive shift in the distribution of GDP per capita in Asia. **Monarchies are concentrated at substantially higher income levels than republics**, with both their mean ($27,339) and median ($28,895) well above those of republics ($9,750 and $4,087, respectively).

The distributions nevertheless overlap, indicating that the two groups are not completely separate and that countries with different income levels exist in both. The republics also show a pronounced right-skewed distribution, with most countries concentrated at relatively low income levels and a smaller number extending into much higher levels.

Overall, the chart is consistent with the descriptive statistics: **Asian monarchies tend to occupy the higher end of the GDP-per-capita distribution, while Asian republics are more heavily concentrated at lower levels**. The statistical significance of this difference, however, must be assessed through the inferential tests presented separately.

Comparison of GDP per Capita between Monarchies and Republics in Europe

Europe provides one of the most suitable settings for comparing the economic performance of monarchies and republics. Many countries on the continent share considerable similarities in their levels of development, legal systems, economic structures, human capital, and degree of

integration into the global economy. Comparing the two forms of government within Europe can therefore reduce some of the heterogeneity present in the global comparison and provide a more focused regional test of the observed pattern.

European countries, however, have not followed identical historical trajectories. A substantial number of the continent's republics spent a significant part of the twentieth century under communist rule. The institutional and economic legacies of this historical experience may still be reflected in their current levels of development. For this reason, the European analysis is conducted in two stages. First, all European monarchies and republics are compared. Then, as a **robustness check**, republics with a history of communist rule are excluded from the sample to assess whether the main result remains stable under this alternative sample specification.

Descriptive Statistics

The main European sample consists of **11 monarchies** and **34 republics**. Mean GDP per capita is **$92,065.8** among European monarchies and **$31,206.2** among the continent's republics. Thus, **the mean for monarchies is approximately 2.95 times that of republics**.

Median GDP per capita is also substantially higher among monarchies: **$62,536.7**, compared with **$23,827.3** among republics. The higher median indicates that the difference between the two groups is not merely the result of one or a few exceptionally wealthy countries; it is also evident in the central part of the distribution.

The sample standard deviation is **$66,495.6** for monarchies and **$25,038.3** for republics. The greater dispersion among monarchies indicates that these countries are not homogeneous in terms of GDP per capita, with some very high-income monarchies lying far above the other members of the group. For this reason, examining the full shape of the distribution is particularly important alongside the mean and median.

Inferential Statistics

To compare the means of the two groups, **Welch's t-test** was used. Unlike the classical Student's t-test, Welch's test does not assume equal variances and is therefore more appropriate for the European data, where dispersion is clearly greater among monarchies than among republics.

The test result is as follows:

$$t \approx 2.968$$
$$p \approx 0.0129$$

Since the **p-value is below the 0.05 significance level**, the null hypothesis of equal group means is rejected. More precisely, in the sample examined, **the difference in mean GDP per capita between European monarchies and republics is statistically significant**.

The effect size based on **Hedges' g** is approximately:

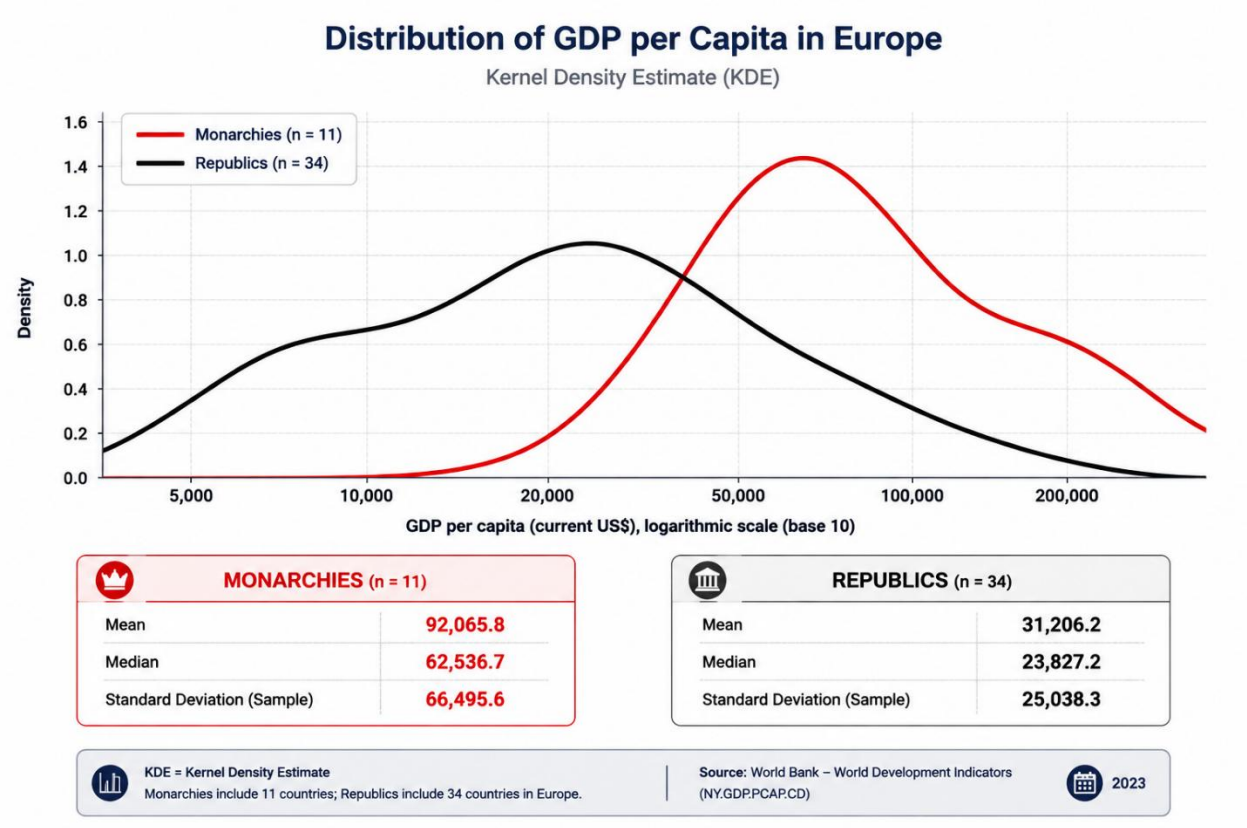
$$g \approx 1.539$$

This value indicates a **very large effect size**. The observed difference is therefore not only statistically significant but also substantial in magnitude.

Interpretation

The results of the main European analysis are consistent with the pattern observed in the global comparison and in the analysis of Asia. European monarchies have higher GDP per capita than the continent's republics in terms of both the mean and the median. The distribution plot likewise shows that a substantial portion of the monarchy distribution is concentrated at higher income levels.

This result, however, **does not by itself establish that the form of government causes the observed economic difference**. Differences in the historical trajectories of European countries—including the experience of communist rule in many republics—may account for part of the observed gap. Therefore, to assess the robustness of the result, the comparison should be repeated using an alternative specification of the republic sample.



The chart shows a clear descriptive difference between the two groups. **The distribution of GDP per capita among European monarchies is shifted substantially toward higher income levels than that of European republics.** Monarchies have both a much higher mean ($92,066) and median ($62,537) than republics ($31,206 and $23,827, respectively).

The wider spread of the monarchy curve also indicates greater variation within this group, partly reflecting the presence of several exceptionally high-income monarchies. Nevertheless, the large difference in medians suggests that the observed gap is not driven solely by a few very wealthy countries.

Overall, **European monarchies are concentrated at considerably higher levels of GDP per capita than European republics**, although the overlap between the distributions shows that the two groups are not completely separate. As with the previous comparisons, the chart is descriptive and does not by itself establish either statistical significance or causality.

Robustness Check for Europe

In the next stage, European republics with a history of communist rule are excluded from the sample, and the remaining republics are compared with the monarchies. This analysis should not be interpreted as **fully controlling for the effects of communism**. Rather, its purpose is to

address a narrower question: whether the main result remains robust when the sample is defined differently.

Robustness Check: Excluding Former Communist Republics in Europe

One possible objection is that the observed difference between European monarchies and republics may reflect historical differences between Western and Eastern Europe rather than differences in form of government. Many European republics were under communist rule until only a few decades ago, and the legacy of those systems may still affect their current levels of economic development. It could therefore be argued that including these countries lowers the average GDP per capita of republics and exaggerates the observed gap between the two groups.

To examine this possibility, a robustness check is conducted in which republics with a history of communist rule are excluded from the sample and the statistical analysis is repeated. An important methodological point, however, should be kept in mind. Excluding these countries does not necessarily bring us closer to a more accurate estimate of the underlying difference between monarchies and republics. On the contrary, one could argue that **the tendency of some republics to develop communist regimes may itself have been one of the possible historical outcomes associated with republican systems. If so, excluding these countries would mean removing part of the historical experience of republics from the analysis.**

This robustness check should therefore not be interpreted as an attempt to estimate the "true effect" of the form of government. Instead, it addresses a more limited counterfactual question: how large would the economic difference between European monarchies and republics be if the European republics that experienced communist rule were excluded from the comparison? Excluding former communist republics does not necessarily provide a "truer" picture of the monarchy–republic comparison. Accordingly, the findings of this section do not replace the main results of the study; they simply show how robust the original result is to this particular objection.

Descriptive Statistics

In this robustness check, **21 European republics with a history of communist rule** were excluded from the sample, leaving **13 republics with no such history**. The same **11 European monarchies** from the previous analysis were retained unchanged. The purpose of this comparison is to examine how much of the economic difference between the two forms of government remains after excluding the possible influence of the communist legacy.

The descriptive statistics are presented in the table below.

Group	Number of Countries	Mean	Median	Sample Standard Deviation
Monarchies	11	$92,065.77	$62,536.70	$66,495.55

Group	Number of Countries	Mean	Median	Sample Standard Deviation
Republics with no history of communist rule	13	$54,320.30	$52,745.80	$25,577.50

As the table shows, excluding former communist republics substantially increases the mean GDP per capita of the republic group. Nevertheless, the mean for monarchies remains considerably higher than that of republics with no history of communist rule. The medians of the two groups also become much closer than in the original analysis, suggesting that **part of the initial gap was associated with the lower economic performance of former communist republics, although the difference does not disappear entirely**.

Inferential Statistics

To determine whether the remaining difference is statistically significant, **Welch's t-test** was again applied. Given the unequal variances and different sample sizes of the two groups, this test remains the more appropriate method for comparing their means.

The results are:

- **Welch's t = 1.775**
- **Degrees of freedom = 11.84**
- **p = 0.100**
- **Hedges' g = 0.749**

In this analysis, the **p-value** is greater than the conventional 0.05 significance level. Therefore, **within this subsample, there is insufficient statistical evidence to reject the null hypothesis of equal means at the 5% significance level**.

The effect size, however, remains substantial. **Hedges' g = 0.749** falls in the **medium-to-large range**. In other words, although the observed difference is no longer statistically significant at the 0.05 level, its estimated magnitude remains notable.

The smaller sample resulting from the exclusion of 21 republics also increases statistical uncertainty and reduces the power of the test. Thus, the absence of statistical significance should not be interpreted as evidence that the two groups are economically equivalent.

Interpretation

This robustness check shows that **a substantial part of the original gap between European monarchies and republics becomes smaller when former communist republics are excluded, but the gap does not disappear**. The mean GDP per capita of monarchies remains considerably higher, while the difference between the medians becomes much narrower. Thus,

the descriptive advantage of European monarchies in the full sample cannot be attributed solely to the inclusion of former communist republics.

At the same time, as explained at the beginning of this section, this analysis should not be interpreted as an estimate of the "net effect" of the form of government. Excluding former communist republics removes from the analysis part of the historical path followed by some European republics. The robustness check therefore addresses a narrower counterfactual question: **how does the monarchy–republic comparison change when European republics that experienced communist rule are excluded?**

Accordingly, the main findings of the study remain based on the complete sample. This robustness check provides a more qualified conclusion: **the descriptive economic gap remains after former communist republics are excluded, and the effect size remains substantial, but the difference is no longer statistically significant at the 0.05 level.**

Comparison of GDP per Capita between Monarchies and Republics in Africa, the Americas, and Oceania

After examining Asia and Europe, we now turn to three other regions of the world: Africa, the Americas, and Oceania. Unlike the two previous continents, these regions contain very few independent monarchies, making comparisons with high statistical power difficult. Africa has only three independent monarchies; the Americas have none, so the only possible comparison is between republics and Commonwealth realms; and Oceania has only one independent monarchy, alongside several Commonwealth realms and republics.

Because the aim of this book is to identify reliable patterns in global data, the main emphasis is placed on comparisons with sufficiently large samples. The discussion of these three regions is therefore kept relatively brief, reporting only the most important descriptive statistics and, where appropriate, statistical test results.

This brevity does not imply that these regions are unimportant; it simply reflects the limitations of the available samples. The next section, by contrast, will focus on the Middle East, where both monarchies and republics are well represented and where countries share many historical, cultural, geographical, and, to a considerable extent, natural-resource characteristics. The comparison between the two forms of government in the Middle East may therefore provide one of the fairest and most informative tests in this study.

Africa

Africa differs considerably from Asia and Europe in the distribution of its political systems. The continent has only three monarchies, compared with fifty republics. This substantial imbalance limits the statistical power of the comparison and requires greater caution in interpreting the results. Nevertheless, examining Africa remains useful because it allows us to see whether the pattern observed in other regions of the world is also present on this continent.

Descriptive Statistics

Among the three African monarchies, mean GDP per capita is **$2,782**, compared with **$2,649** among the fifty African republics. The difference between the two means is very small, although the median is higher among monarchies. Income dispersion is also considerably greater among African republics, indicating substantial economic heterogeneity within this group.

Group	Number of Countries	Mean	Median	Sample Standard Deviation
Monarchies	3	$2,782	$3,672	$1,651
Republics	50	$2,649	$1,590	$3,208

Inferential Statistics

The results of **Welch's t-test** likewise indicate that the observed difference between the two groups is not statistically significant (**$p \approx 0.91$**), while the effect size is essentially negligible (**Hedges' $g \approx 0.04$**). Thus, the available data provide **no evidence of a statistically significant difference in mean GDP per capita between African monarchies and republics**.

The Americas

There are no independent monarchies in the Americas. However, several countries in the region are Commonwealth realms: independent states that recognize the British monarch as their head of state and are constitutionally classified as monarchies. Since these countries have already been examined separately, this section provides only a brief regional comparison of their economic performance with that of republics in the Americas.

Given the small number of Commonwealth realms, the findings in this section should be regarded primarily as descriptive rather than as having the same inferential weight as the comparisons for Asia, Europe, or the Middle East.

Descriptive Statistics

The sample for the Americas includes **9 Commonwealth realms** and **25 republics**. Mean GDP per capita is **$20,202** among the Commonwealth realms, compared with **$13,923** among the republics. Median GDP per capita is also higher among the Commonwealth realms (**$13,980**) than among the republics (**$10,044**). The standard deviations of the two groups are very similar, indicating broadly comparable levels of income dispersion.

Group	Number of Countries	Mean	Median	Sample Standard Deviation
Commonwealth realms	9	$20,202	$13,980	$15,322
Republics	25	$13,923	$10,044	$15,540

Interpretation

Although both mean and median GDP per capita are higher among the Commonwealth realms than among the republics of the Americas, the small number of Commonwealth realms means that this finding should be treated primarily as a **descriptive observation rather than a firm statistical conclusion**. Nevertheless, the direction of the difference is consistent with the broader pattern observed in the global comparison.

Oceania

Oceania is also highly unbalanced in terms of forms of government. The region contains only **one independent monarchy, Tonga**, together with **five Commonwealth realms** and **eight republics** included in this study. Because a single independent monarchy provides no basis for meaningful statistical inference, the analysis here focuses on the descriptive comparison between Commonwealth realms and republics, while Tonga is reported separately for completeness.

Descriptive Statistics

Tonga, the only independent monarchy in Oceania, has a GDP per capita of **$4,682**. By comparison, mean GDP per capita among the region's **five Commonwealth realms** is **$24,780**, while the mean among its **eight republics** is **$6,606**.

Although median GDP per capita among the Commonwealth realms (**$5,465**) is only slightly higher than among the republics (**$5,004**), the mean for the Commonwealth group is substantially higher because of the presence of the two large, high-income economies of Australia and New Zealand. This also explains the much larger standard deviation among the Commonwealth realms.

Group	Number of Countries	Mean	Median	Sample Standard Deviation
Independent monarchy (Tonga)	1	$4,682	$4,682	—
Commonwealth realms	5	$24,780	$5,465	$29,647
Republics	8	$6,606	$5,004	$4,432

Interpretation

With only one independent monarchy, Oceania does not provide a sufficient basis for a direct statistical comparison between monarchies and republics. The descriptive comparison between Commonwealth realms and republics shows a substantially higher mean GDP per capita for the former, but this difference is strongly influenced by Australia and New Zealand. The much smaller difference between the two medians illustrates this clearly.

Accordingly, the results for Oceania should also be treated **as descriptive evidence rather than as a firm statistical conclusion**.

The Middle East: The Fairest Test for Comparing Monarchy and Republic

So far, the results of this study have shown that, both globally and in Asia and Europe, countries with monarchical systems have, on average, performed better economically than republics. Yet an important question remains: are these differences actually related to the form of government, or do they mainly reflect the very different historical, cultural, geographical, and economic conditions found across regions of the world?

The Middle East provides a particularly useful setting in which to examine this question. Compared with many other regions, the countries of the Middle East share significant similarities in historical background, culture, religion, geography, climate, and, to a considerable extent, natural resources. Many of these countries were, until less than a century ago, either part of the Ottoman Empire or under the influence of colonial powers, and many began the process of modern state-building during roughly the same historical period. Moreover, a large share of the world's oil and gas reserves is concentrated in this region, and both Middle Eastern monarchies and republics—although to varying degrees—have benefited from these resources.

From a methodological perspective, these similarities are particularly important. In comparative research, the more similar the units being compared are in their background conditions, the better the opportunity to examine the role of the main variable of interest. In this book, that variable is the **form of government**. Therefore, if a meaningful difference between monarchies and republics is still observed in a region where countries share many historical, cultural, geographical, and economic characteristics, it becomes more difficult to explain that difference solely in terms of environmental or historical factors.

In other words, global comparisons inevitably involve many potential confounding factors, since the countries being compared come from very different continents, cultures, climates, and historical paths. In the Middle East, however, a substantial number of these background conditions are shared across the two groups. The region can therefore be viewed as a useful **natural laboratory** for comparing monarchy and republic—one in which some potential confounding factors are reduced, allowing the relationship between the form of government and economic performance to be examined under more comparable conditions.

Unlike Africa, the Americas, and Oceania, where the number of monarchies is very small, the Middle East contains substantial representation of both forms of government. On one side are the monarchies of **Saudi Arabia, Bahrain, Jordan, Kuwait, Oman, Qatar, and the United Arab Emirates**. On the other are republics such as **Iran, Iraq, Syria, Lebanon, Yemen, and Israel**. This relatively balanced representation, combined with the broad regional similarities among these countries, gives this comparison particular analytical value.

For this reason, the analysis in this section is more than simply another regional comparison. It represents one of the most important tests of the book's central hypothesis. If the pattern observed at the global level also appears in the Middle East, such a finding would strengthen the hypothesis that **the form of government may be one of the factors associated with differences in countries' performance**. As with any observational study, however, these results **cannot by themselves establish a causal relationship**. They can nevertheless provide important empirical evidence relevant to that hypothesis.

It is now time to put this natural laboratory to the test—a region in which monarchies and republics have existed side by side for decades within a broadly shared historical and regional setting, allowing their records to provide one of the clearest empirical tests of the relationship between form of government and economic performance.

Comparison of GDP per Capita between Monarchies and Republics in the Middle East

Descriptive Statistics for the Middle East

Having explained the importance of the Middle East as a natural laboratory for comparing monarchy and republic, we can now begin by examining the descriptive statistics for the two groups.

This study includes **7 monarchies** and **10 republics** in the Middle East. The results reveal a striking difference in their economic performance.

Mean **GDP per capita** is **$37,678** among Middle Eastern monarchies, compared with **$12,115** among the region's republics. In other words, **average GDP per capita in Middle Eastern monarchies is more than three times that of the region's republics**.

The median shows the same pattern. Median GDP per capita is **$29,084** among the monarchies and **$4,008** among the republics. Because the median is less sensitive than the mean to extremely high values, the fact that both measures point in the same direction suggests that the observed difference is not simply the result of one or two exceptionally wealthy countries, but is also visible in the broader distribution of the data.

The standard deviation is higher among the monarchies (**$24,419**) than among the republics (**$16,500**). This indicates that although the region's monarchies generally have higher income levels, there is also considerable variation within the group. Jordan's GDP per capita, for example, is far below that of Qatar or the United Arab Emirates. Despite this variation, however, the overall level of GDP per capita remains substantially higher among the monarchies.

Group	Number of Countries	Mean	Median	Sample Standard Deviation
Monarchies	7	**$37,678**	**$29,084**	**$24,419**

Group	Number of Countries	Mean	Median	Sample Standard Deviation
Republics	10	**$12,115**	**$4,008**	**$16,500**

Statistical Test

To determine whether the difference between the two group means can reasonably be distinguished from random variation, **Welch's t-test** was applied. The two-sided test produced the following results:

t = 2.272, df = 10.44, p = 0.045

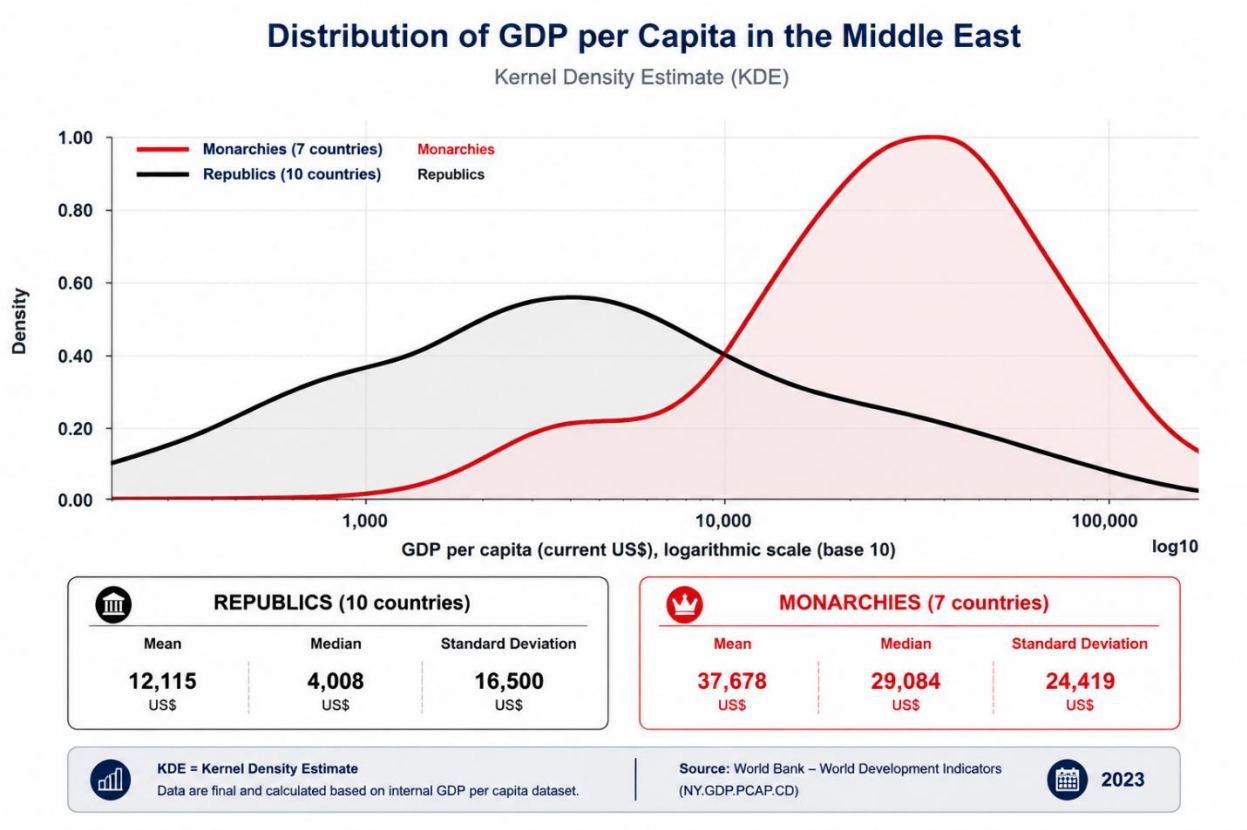
The difference in mean GDP per capita between Middle Eastern monarchies and republics is therefore **statistically significant at the 5% level**.

The estimated effect size, **Hedges' g = 1.124**, also indicates a **large effect**. Thus, the difference between the two groups is not only statistically detectable but also substantial in magnitude.

Interpretation

These results are important for several reasons. First, the difference between the group means is substantial. Second, the fact that both the mean and the median point in the same direction suggests that the gap is not simply driven by a few exceptionally wealthy countries. Third, the difference appears within a region whose countries share many more historical, cultural, and geographical characteristics than the countries included in a global comparison.

For this reason, the Middle Eastern findings are particularly informative. Although descriptive statistics **alone** cannot establish a definitive conclusion—and the small number of countries calls for appropriate caution—they provide an initial indication that the same broad pattern previously observed globally and in Asia and Europe is also present within this **natural laboratory**.



The distribution shows a clear shift toward higher GDP per capita among **Middle Eastern monarchies** compared with the region's **republics**. Monarchies have both a substantially higher mean (**$37,678 vs. $12,115**) and median (**$29,084 vs. $4,008**), indicating that the difference is not driven solely by one or two exceptionally wealthy countries. The republican distribution is concentrated much more heavily at lower income levels, whereas the monarchy distribution peaks considerably farther to the right.

Although the two distributions overlap, the overall pattern indicates a pronounced descriptive economic advantage for monarchies in the region. This visual evidence is consistent with the Welch's t-test reported in the analysis (**$p \approx 0.045$**) and the large effect size (**Hedges' $g \approx 1.12$**), while not by itself establishing a causal relationship between regime type and economic performance.

Purchasing Power per Capita

In the previous section, we found that **GDP per capita** is significantly higher in the world's monarchies than in republics. Nominal income, however, does not always provide a complete picture of living standards, because the prices of goods and services vary across countries. A country may have a lower per capita income, yet because of its lower cost of living, its citizens may be able to purchase more goods and services with that income.

For example, although Afghanistan's GDP per capita is only **$413**, that amount of income in Afghanistan has **purchasing power equivalent to approximately $2,203 in the United States**. To account for such differences, economists use **GDP per capita based on Purchasing Power Parity (PPP)**. This measure adjusts for differences in price levels across countries and is therefore generally more suitable for comparing living standards internationally. For brevity, throughout this chapter we will refer to this indicator as **"Purchasing Power per Capita."** Its unit of measurement is the **international dollar**.

Comparing this indicator also serves as a robustness check on the findings of the previous chapter. If the difference observed between monarchies and republics persists after adjusting for differences in price levels, we can be more confident that the pattern is not merely a consequence of differences in currency values or living costs, but remains visible in a measure that more closely reflects the actual purchasing power of citizens.

The data used in this chapter were obtained from the **World Bank** (Appendix RDS-B01). Data were unavailable for a number of countries, and those countries were therefore excluded from the calculations.

As in the previous chapter, the analysis begins with a global comparison between monarchies and republics. Next, to examine the role of the monarchical institution more closely, the performance of the **Commonwealth realms** is compared with that of republics. Constitutional monarchies and non-constitutional monarchies are then examined separately to determine whether the observed differences are associated simply with the presence or absence of a monarchy, or whether the form of monarchy may also be relevant.

The same statistical framework used in the previous chapter is retained here. Descriptive statistics and distribution plots are presented first, followed by the results of statistical tests. Finally, **robustness checks** are used to assess the stability of the findings.

Global Comparison of Purchasing Power per Capita: Monarchies and Republics

Data source: World Bank
Indicator: GDP per capita, PPP (current international $)
Data year: 2023

To examine differences in economic performance between the two forms of government, purchasing power per capita was compared across the world's **monarchies** and **republics**. By adjusting for differences in price levels across countries, this indicator provides a more comparable measure of the actual purchasing power of citizens.

In addition to descriptive statistics, two independent statistical tests were conducted. **Welch's t-test** was used to assess differences in group means, while the **Mann–Whitney U test** was used to examine differences in the overall distributions.

1. Descriptive Statistics

Group	Countries (n)	Mean	Median	Population SD
Monarchies	**26**	**55,784.75**	**58,224.00**	**38,018.66**
Republics	**148**	**23,435.44**	**15,320.35**	**24,756.62**

Interpretation of the Descriptive Statistics

Based on the 2023 data, mean purchasing power per capita was approximately **55,785 international dollars** among monarchies, compared with approximately **23,435 international dollars** among republics.

In other words, **the mean for monarchies was approximately 2.4 times that of republics**.

The difference between the medians is also substantial. The median was approximately **58,224 international dollars** for monarchies and **15,320 international dollars** for republics. Because the median is less sensitive than the mean to extremely high or low values, this difference indicates that the observed gap is not merely the result of a small number of exceptionally high-income countries.

Nevertheless, both groups exhibit considerable dispersion. Inferential statistical tests were therefore conducted to determine whether the observed differences are statistically significant.

2. Welch's t-test for the Difference in Means

Welch's t-test was used to compare the means of the two groups because, unlike the conventional Student's t-test, it does not require the assumption of equal variances.

The hypotheses were defined as follows:

Null hypothesis (H_0):
Mean purchasing power per capita is equal in the two groups.

Alternative hypothesis (H_1):
Mean purchasing power per capita differs between the two groups.

Results of Welch's t-test

Statistic	Value
Mean — Monarchies	55,784.75
Mean — Republics	23,435.44
Mean difference	32,349.31
t-statistic	4.109
Approximate degrees of freedom	28.710
p-value	0.000302

Interpretation of Welch's t-test

The resulting p-value (**p = 0.000302**) is far below the conventional significance level of 0.05.

The null hypothesis is therefore rejected, indicating that **the observed difference in mean purchasing power per capita between monarchies and republics is statistically significant**.

In other words, under the null hypothesis of equal population means, an observed difference of this magnitude would be highly unlikely to arise from sampling variation alone.

3. Mann–Whitney U Test

Because economic data are often non-normally distributed and may be influenced by countries with exceptionally high or low values, the non-parametric **Mann–Whitney U test** was also conducted.

Rather than directly comparing group means, this test assesses whether observations in one group generally tend to rank higher than observations in the other.

The hypotheses were defined as follows:

Null hypothesis (H_0):
There is no systematic difference between the distributions of purchasing power per capita in the two groups.

Alternative hypothesis (H_1):
The distributions of purchasing power per capita differ between the two groups.

Results of the Mann–Whitney U Test

Statistic	Value
U-statistic	2,943

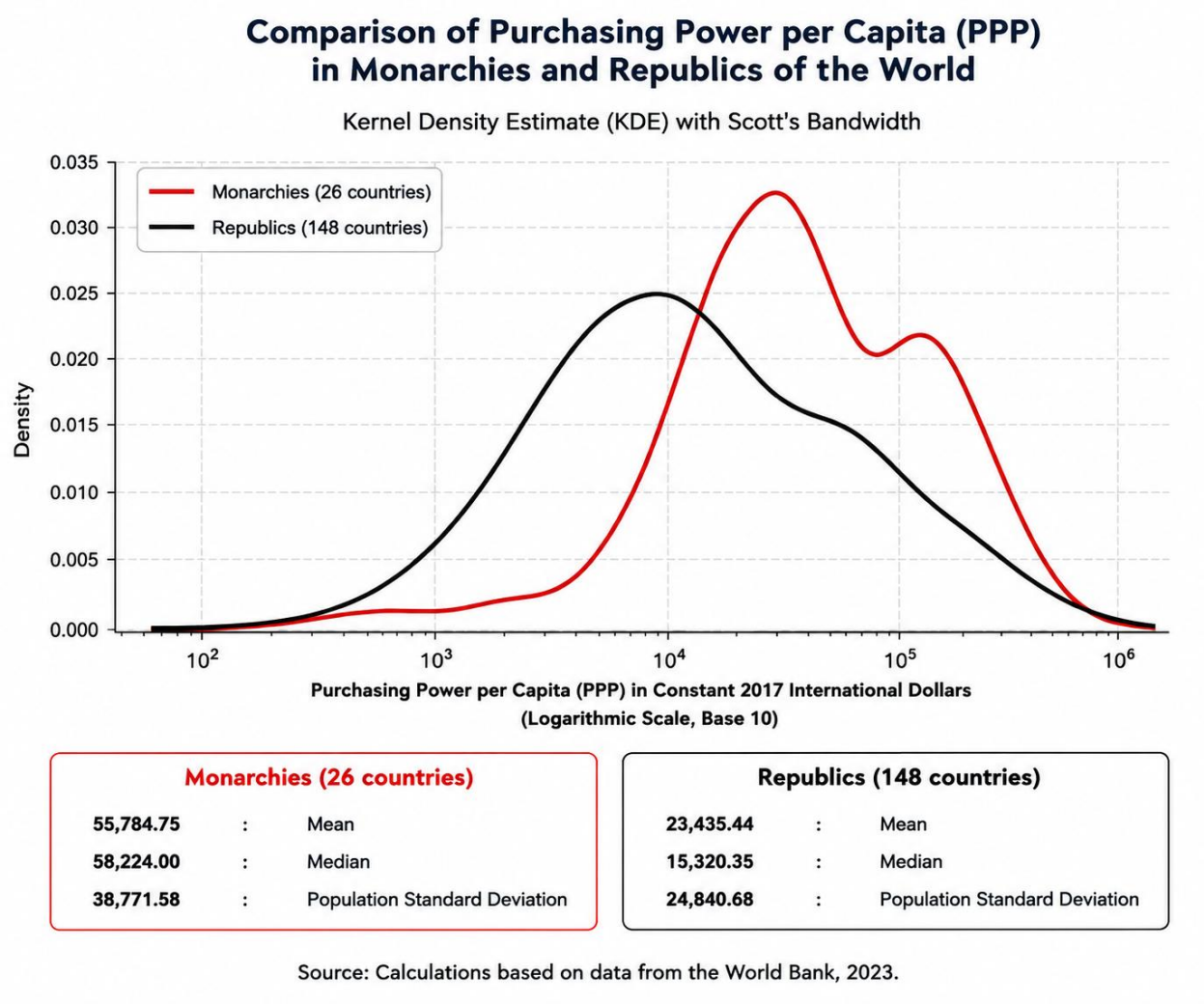
Statistic	Value
p-value	0.000017

Interpretation of the Mann–Whitney U Test

The p-value (**p = 0.000017**) is well below the 0.05 significance threshold.

The null hypothesis is therefore rejected. The test provides strong evidence that **the distributions of purchasing power per capita differ significantly between monarchies and republics**.

This result is consistent with Welch's t-test. Taken together, the two tests indicate that the observed difference is not confined to the group means: the broader distribution of purchasing power per capita also differs significantly between monarchies and republics.



The kernel density estimate (KDE) shows that the distribution of **purchasing power per capita** among the world's monarchies is noticeably shifted toward higher values. The monarchy curve (red) is more heavily concentrated at higher income levels, whereas the republic curve (black) is concentrated primarily at lower levels. This difference is also evident in the descriptive statistics: mean purchasing power per capita is **55,784.75 international dollars** in monarchies, compared with **23,435.44 international dollars** in republics. Likewise, the median for monarchies (**58,224**) is substantially higher than that for republics (**15,320.35**).

This pattern indicates that, in the 2023 data, monarchies tend to occupy higher levels of economic well-being, while republics display greater dispersion and a substantial proportion are concentrated at lower levels of purchasing power per capita. As with the other indicators examined in this study, however, this descriptive pattern **does not by itself establish a causal relationship** between regime type and economic well-being. The statistical significance of the difference between the two groups must therefore be evaluated using the inferential tests reported in the analysis.

Comparison of Purchasing Power per Capita between Constitutional and Non-Constitutional Monarchies Worldwide

Following the overall comparison between monarchies and republics, the next step is to examine differences **within the monarchy group itself**. This comparison investigates whether the structure of monarchy—specifically, the extent to which the monarch's power is legally constrained—is associated with observable differences in economic well-being.

For this purpose, monarchies were divided into two groups:

1. **Constitutional monarchies**
2. **Non-constitutional monarchies**

In addition to descriptive statistics, two independent statistical tests were conducted:

- **Welch's t-test** to compare group means
- **Mann–Whitney U test** to compare the distributions

1. Descriptive Statistics

Group	Countries (n)	Mean	Median	Population SD
Constitutional monarchies	**15**	**58,806.33**	**60,786.70**	**38,519.64**
Non-constitutional monarchies	**11**	**51,664.40**	**53,083.40**	**36,928.37**

Interpretation of the Descriptive Statistics

Based on the 2023 data, mean purchasing power per capita was approximately **58,806 international dollars** in constitutional monarchies and **51,664 international dollars** in non-constitutional monarchies.

Thus, in this dataset, constitutional monarchies had an average purchasing power per capita approximately **7,142 international dollars higher** than that of non-constitutional monarchies.

The medians show a similar difference:

- **Constitutional monarchies:** 60,787 international dollars
- **Non-constitutional monarchies:** 53,083 international dollars

The fact that both the mean and median point in the same direction suggests that the observed difference is not simply driven by a small number of countries with exceptionally high or low values.

However, dispersion is substantial in both groups, and their standard deviations are relatively similar. Inferential tests were therefore conducted to determine whether the observed difference is statistically significant.

2. Welch's t-test for the Difference in Means

The hypotheses were defined as follows:

Null hypothesis (H_0):
Mean purchasing power per capita is equal in the two types of monarchy.

Alternative hypothesis (H_1):
Mean purchasing power per capita differs between the two types of monarchy.

Results of Welch's t-test

Statistic	Value
Mean — Constitutional monarchies	58,806.33
Mean — Non-constitutional monarchies	51,664.40
Mean difference	7,141.93
t-statistic	0.459
Approximate degrees of freedom	22.064
p-value	0.651

Interpretation of Welch's t-test

The resulting p-value (**p = 0.651**) is substantially greater than the conventional significance level of 0.05.

The null hypothesis is therefore **not rejected**.

In other words, although constitutional monarchies have a higher mean purchasing power per capita in this dataset, the observed difference is not statistically significant. The available evidence therefore does not support the conclusion that the two groups have different population means.

3. Mann–Whitney U Test

Because economic data may deviate from a normal distribution, the non-parametric **Mann–Whitney U test** was also conducted.

The hypotheses were defined as follows:

Null hypothesis (H_0):
There is no systematic difference between the distributions of purchasing power per capita in the two groups.

Alternative hypothesis (H_1):
The distributions of purchasing power per capita differ between the two groups.

Results of the Mann–Whitney U Test

Statistic	Value
U-statistic	89
p-value	0.755

Interpretation of the Mann–Whitney U Test

The resulting p-value (**p = 0.755**) is also substantially greater than the 0.05 significance threshold.

The null hypothesis is therefore **not rejected**.

Consistent with Welch's t-test, the Mann–Whitney U test provides no statistically significant evidence of a difference between the two groups.

Conclusion

In the 2023 purchasing-power-per-capita data, constitutional monarchies have a higher mean than non-constitutional monarchies:

- **Constitutional monarchies:** 58,806 international dollars
- **Non-constitutional monarchies:** 51,664 international dollars

However, neither statistical test indicates a statistically significant difference:

- **Welch's t-test:** $p = 0.651$
- **Mann–Whitney U test:** $p = 0.755$

Therefore, **in this dataset, the distinction between constitutional and non-constitutional monarchy is not associated with a statistically significant difference in purchasing power per capita**.

Comparison of Purchasing Power per Capita between Commonwealth Realms and Republics Worldwide

Following the overall comparison between monarchies and republics, a supplementary question arises: **Do the Commonwealth realms, which have a historical and constitutional connection to the British monarchy, differ significantly from the world's republics in terms of economic well-being?**

To address this question, purchasing power per capita was compared between the two groups. In addition to descriptive statistics, two independent statistical tests were conducted:

- **Welch's t-test** to compare group means
- **Mann–Whitney U test** to compare the distributions

1. Descriptive Statistics

Group	Countries (n)	Mean	Median	Population SD
Commonwealth realms	**14**	**28,704.38**	**23,045.05**	**21,526.94**
Republics	**148**	**23,435.44**	**15,320.35**	**24,756.62**

Interpretation of the Descriptive Statistics

Based on the 2023 data, mean purchasing power per capita was approximately **28,704 international dollars** in the Commonwealth realms and **23,435 international dollars** in the world's republics.

Thus, the mean for the Commonwealth realms was approximately **5,269 international dollars higher** than that of the republics.

The medians show a similar pattern:

- **Commonwealth realms:** 23,045 international dollars
- **Republics:** 15,320 international dollars

This difference indicates that the Commonwealth realms also have higher values around the center of the distribution.

However, the number of Commonwealth realms in this comparison is much smaller than the number of republics (**14 versus 148 countries**), and there is substantial dispersion in both groups. Inferential tests were therefore conducted to determine whether the observed difference is statistically reliable.

2. Welch's t-test for the Difference in Means

Welch's t-test evaluates whether the difference between the two group means is statistically significant after accounting for sample size and variability.

The hypotheses were defined as follows:

Null hypothesis (H_0):
Mean purchasing power per capita is equal in the Commonwealth realms and republics.

Alternative hypothesis (H_1):
Mean purchasing power per capita differs between the two groups.

Results of Welch's t-test

Statistic	Value
Mean — Commonwealth realms	**28,704.38**
Mean — Republics	**23,435.44**
Mean difference	**5,268.94**
t-statistic	**0.835**
Approximate degrees of freedom	**16.199**
p-value	**0.416**

Interpretation of Welch's t-test

The resulting p-value (**p = 0.416**) is greater than the conventional significance level of 0.05.

The null hypothesis is therefore **not rejected**.

In other words, **although mean purchasing power per capita is higher among the Commonwealth realms, the difference is not statistically significant in this sample**. The available evidence is therefore insufficient to conclude that the two groups have different population means.

3. Mann–Whitney U Test

Because economic data may have asymmetric or non-normal distributions, the **Mann–Whitney U test** was also conducted to determine whether the overall distributions of the two groups differ.

The hypotheses were defined as follows:

Null hypothesis (H_0):
There is no systematic difference between the distributions of purchasing power per capita in the two groups.

Alternative hypothesis (H_1):
The distributions of purchasing power per capita differ between the two groups.

Results of the Mann–Whitney U Test

Statistic	Value
U-statistic	**1,244**
p-value	**0.216**

Interpretation of the Mann–Whitney U Test

The resulting p-value (**p = 0.216**) is greater than the 0.05 significance threshold.

The null hypothesis is therefore **not rejected**.

Consistent with Welch's t-test, the Mann–Whitney U test provides no statistically significant evidence of a difference between the Commonwealth realms and republics in this dataset.

Conclusion

In the 2023 purchasing-power-per-capita data:

Group	Mean Purchasing Power per Capita
Commonwealth realms	**28,704 international dollars**
Republics	**23,435 international dollars**

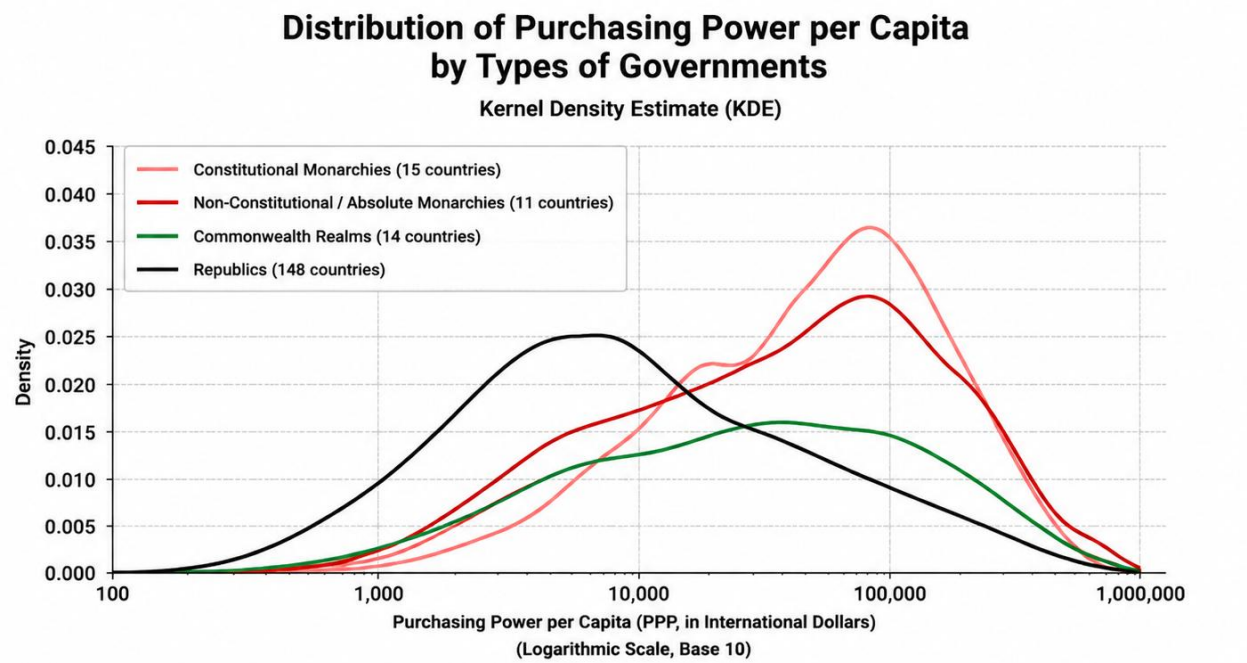
The Commonwealth realms have higher mean and median purchasing power per capita than the world's republics. However:

- **Welch's t-test:** $p = 0.416$
- **Mann–Whitney U test:** $p = 0.216$

Neither test identifies the observed difference as statistically significant.

Therefore, **in this dataset, membership in the Commonwealth-realm group is not, by itself, associated with a statistically demonstrable difference in purchasing power per capita relative to the world's republics**.

This result differs from the overall comparison between monarchies and republics. **In the global comparison, the advantage observed for monarchies was statistically significant and accompanied by a large effect size; when the analysis is restricted to the Commonwealth realms, however, the gap between the two groups becomes smaller and is no longer statistically significant.** The small number of Commonwealth realms and the substantial variation in economic development among them may partly explain this greater statistical uncertainty.



Constitutional Monarchies (15 countries)		
58,806.33	:	Mean
60,786.70	:	Median
39,871.61	:	Population Standard Deviation

Non-Constitutional / Absolute Monarchies (11 countries)		
51,664.40	:	Mean
53,083.40	:	Median
38,730.80	:	Population Standard Deviation

Commonwealth Realms (14 countries)		
28,704.38	:	Mean
23,045.05	:	Median
22,339.56	:	Population Standard Deviation

Republics (148 countries)		
23,435.44	:	Mean
15,320.35	:	Median
24,840.68	:	Population Standard Deviation

Source: Calculations based on data from the World Bank, 2023.

The distribution of purchasing power per capita across the four groups shows that the curves for **constitutional monarchies** and **non-constitutional monarchies** are shifted toward the higher end of the income scale, whereas the distributions of **republics** and **Commonwealth realms** are more concentrated at lower levels.

Constitutional monarchies have the highest mean purchasing power per capita among the four groups, at approximately **58.8 thousand international dollars**, followed by semi-constitutional and absolute monarchies at approximately **51.7 thousand**. By comparison, the mean is about **28.7 thousand** for the Commonwealth realms and **23.4 thousand** for republics.

The positions of the curves also indicate that a substantial proportion of monarchies are located in the higher ranges of the distribution. Nevertheless, the considerable dispersion within all four groups means that differences in means **alone** are insufficient for drawing firm conclusions; statistical tests are required to determine whether the observed differences are statistically significant.

Comparison of Purchasing Power per Capita between Monarchies and Republics across Continents and in the Middle East

Following the global analysis, a regional comparison can help determine whether the pattern observed worldwide is also replicated within individual geographic regions. **Asia**, with its considerable economic and political diversity, is one of the most important regions for such a comparison.

Comparison of Purchasing Power per Capita between Monarchies and Republics in Asia

1. Descriptive Statistics

Group	Countries (n)	Mean	Median	Population SD
Asian Monarchies	**13**	**51,615.02**	**52,726.20**	**33,162.82**
Asian Republics	**31**	**21,911.26**	**14,664.60**	**26,357.24**

Interpretation of the Descriptive Statistics

The 2023 data show that mean purchasing power per capita was approximately **51.6 thousand international dollars** in Asian monarchies, compared with approximately **21.9 thousand international dollars** in Asian republics.

Thus, in this sample, **mean purchasing power per capita in Asian monarchies was approximately 2.4 times that of Asian republics**.

The medians reveal an even more pronounced difference:

- **Monarchies:** 52,726 international dollars
- **Republics:** 14,665 international dollars

This indicates that the observed difference is not merely driven by a small number of countries with exceptionally high values; it is also evident at the center of the distributions.

Nevertheless, both groups exhibit substantial dispersion. Inferential statistical tests were therefore conducted to determine whether the observed difference is statistically significant.

2. Welch's t-test for the Difference in Means

The hypotheses were defined as follows:

Null hypothesis (H_0):
Mean purchasing power per capita is equal in Asian monarchies and Asian republics.

Alternative hypothesis (H_1):
Mean purchasing power per capita differs between the two groups.

Results of Welch's t-test

Statistic	Value
Mean — Asian monarchies	**51,615.02**
Mean — Asian republics	**21,911.26**
Mean difference	**29,703.76**
t-statistic	**2.772**
Approximate degrees of freedom	**18.361**
p-value	**0.0124**

Interpretation of Welch's t-test

The p-value (**p = 0.0124**) is below the conventional significance level of 0.05. The null hypothesis is therefore rejected.

This result indicates that **the difference in mean purchasing power per capita between Asian monarchies and republics is statistically significant in this dataset**.

In other words, under the null hypothesis of equal population means, an observed difference at least this extreme would be relatively unlikely.

3. Mann–Whitney U Test

Because economic data may depart substantially from a normal distribution, the **Mann–Whitney U test** was also conducted to examine whether the two groups differ more generally in the distribution of purchasing power per capita.

The hypotheses were defined as follows:

Null hypothesis (H_0):
There is no systematic difference between the distributions of purchasing power per capita in the two groups.

Alternative hypothesis (H_1):
The distributions of purchasing power per capita differ between the two groups.

Results of the Mann–Whitney U Test

Statistic	Value
U-statistic	**319**
p-value	**0.00262**

Interpretation of the Mann–Whitney U Test

The p-value (**p = 0.00262**) is below the 0.05 significance threshold. The null hypothesis is therefore rejected.

This test also indicates a statistically significant difference between the distributions of purchasing power per capita in Asian monarchies and republics.

The agreement between the two independent tests strengthens the statistical evidence: Welch's test identifies a significant difference in the means, while the Mann–Whitney test indicates a broader systematic difference in the distributions.

Conclusion

The 2023 data for Asia show:

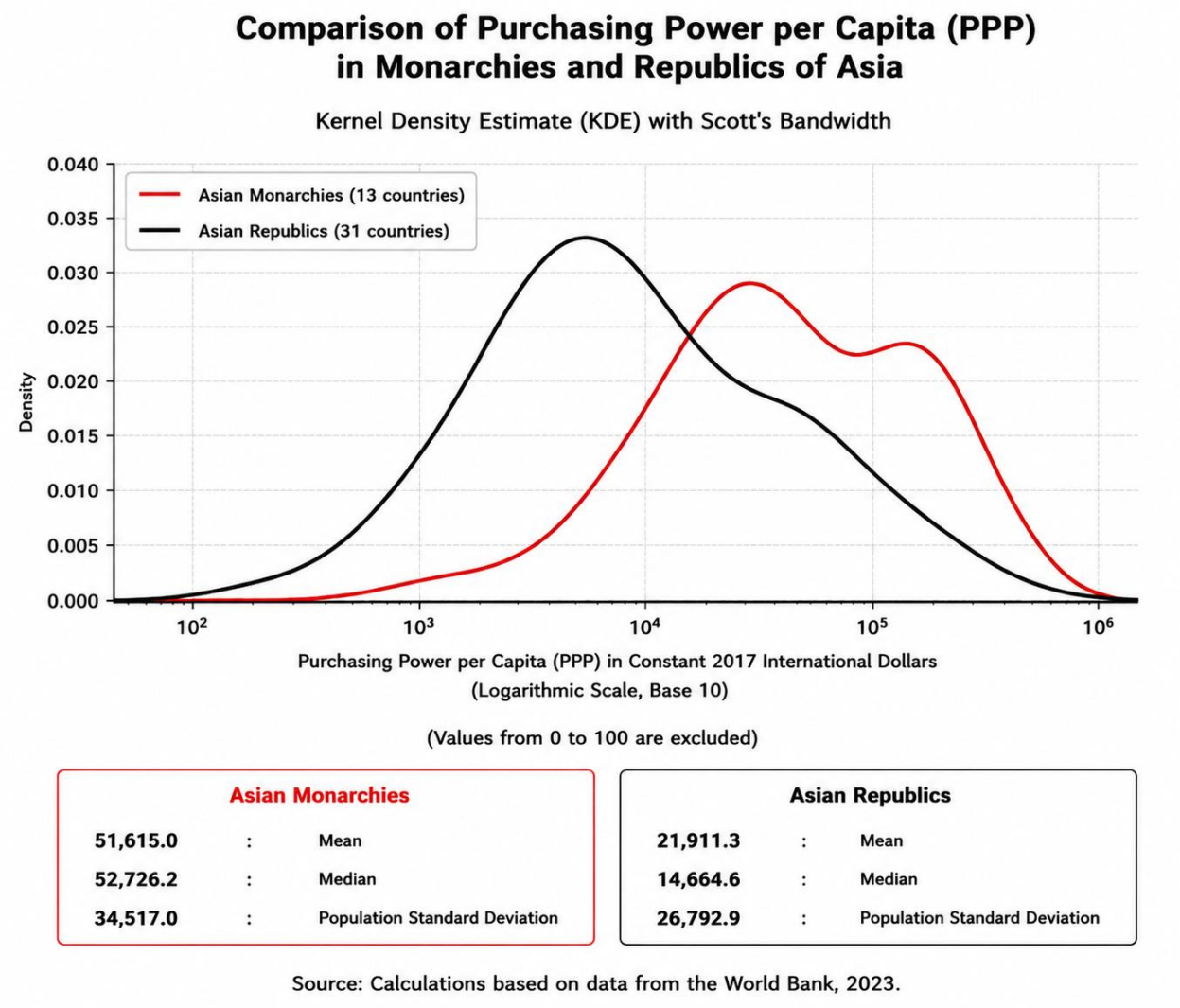
Group	Mean Purchasing Power per Capita
Monarchies	**51,615 international dollars**
Republics	**21,911 international dollars**

Both statistical tests identify a significant difference:

- **Welch's t-test:** p = 0.0124
- **Mann–Whitney U test:** p = 0.00262

Thus, among the Asian countries included in this analysis, monarchies had substantially higher purchasing power per capita than republics, and the observed difference is statistically significant.

This finding is consistent with the global comparison. Its persistence within Asia is particularly noteworthy because the analysis restricts the comparison to a single geographic region, although substantial historical, cultural, and economic heterogeneity remains within Asia itself. As throughout this study, however, these findings establish a **statistical association rather than a causal relationship**.



The distribution of purchasing power per capita across Asian countries shows that the curve for **monarchies is shifted toward higher values** relative to that of republics. Mean purchasing power per capita is approximately **51.6 thousand international dollars** in Asian monarchies, compared with approximately **21.9 thousand international dollars** in Asian republics.

The medians also differ substantially. Half of the monarchies have purchasing power per capita above approximately **52.7 thousand international dollars**, whereas the corresponding median for republics is approximately **14.7 thousand international dollars**.

Despite considerable dispersion within both groups, the distribution of monarchies is more concentrated at higher levels of purchasing power per capita. The statistical tests presented in this section assess whether this observed difference is also **statistically significant**.

Comparison of Purchasing Power per Capita between Monarchies and Republics in Europe

Europe is one of the most suitable regions for comparing systems of government, as many countries on the continent are relatively similar in terms of economic development, historical background, and cultural ties. An intra-regional comparison can therefore reduce, to some extent, the influence of the substantial differences that exist across world regions.

In this section, purchasing power per capita in European countries is compared between monarchies and republics.

1. Descriptive Statistics

Group	Number of Countries (n)	Mean	Median	Population SD
European Monarchies	8	62,184.91	62,463.20	25,763.48
European Republics	34	46,712.56	45,024.35	25,894.72

Interpretation of Descriptive Statistics

According to the 2023 data, mean purchasing power per capita was approximately **62.2 thousand international dollars** in European monarchies, compared with approximately **46.7 thousand international dollars** in European republics.

Thus, in this sample, the mean purchasing power per capita of European monarchies was approximately **15.5 thousand international dollars higher** than that of European republics.

The medians point in the same direction:

- Monarchies: **62,463 international dollars**
- Republics: **45,024 international dollars**

The consistency between the differences in the means and medians suggests that the observed gap is not driven solely by a small number of exceptionally high-income countries, but is also evident in the central part of the distributions.

However, dispersion is quite similar in the two groups. Inferential statistical tests were therefore conducted to determine whether the observed difference is statistically significant.

2. Welch's Test for the Difference in Means

Hypotheses

Null hypothesis (H_0):
The mean purchasing power per capita is equal in European monarchies and European republics.

Alternative hypothesis (H_1):
The mean purchasing power per capita differs between the two groups.

Results of Welch's t-test

Statistic	Value
Mean — European Monarchies	62,184.91
Mean — European Republics	46,712.56
Difference in Means	15,472.35
t-statistic	1.482
Approximate Degrees of Freedom	11.674
p-value	0.165

Interpretation of Welch's t-test

The p-value is greater than the conventional **5% significance level**. The null hypothesis is therefore not rejected.

Although mean purchasing power per capita is higher among European monarchies, **given the sample sizes and dispersion of the data, the difference is not statistically significant in this dataset**.

In other words, the available data do not provide sufficient statistical evidence to conclude that the two groups have different mean levels of purchasing power per capita.

3. Mann–Whitney U Test

The non-parametric **Mann–Whitney U test** was also conducted to examine whether the overall distributions of the two groups differ.

Hypotheses

Null hypothesis (H_0):
The distributions of purchasing power per capita do not differ significantly between the two groups.

Alternative hypothesis (H_1):
The distributions of purchasing power per capita differ between the two groups.

Results of the Mann–Whitney U Test

Statistic	Value
U statistic	82
p-value	0.132

Interpretation of the Mann–Whitney U Test

The p-value is greater than the **5% significance level**. The null hypothesis is therefore not rejected.

Like Welch's test, the Mann–Whitney U test does not provide sufficient statistical evidence to confirm a difference between the distributions of purchasing power per capita in European monarchies and republics in this dataset.

Conclusion

The 2023 European data show the following:

Group	Mean Purchasing Power per Capita
Monarchies	62,185 international dollars
Republics	46,713 international dollars

European monarchies have a higher mean purchasing power per capita, but neither statistical test reaches the conventional threshold for statistical significance:

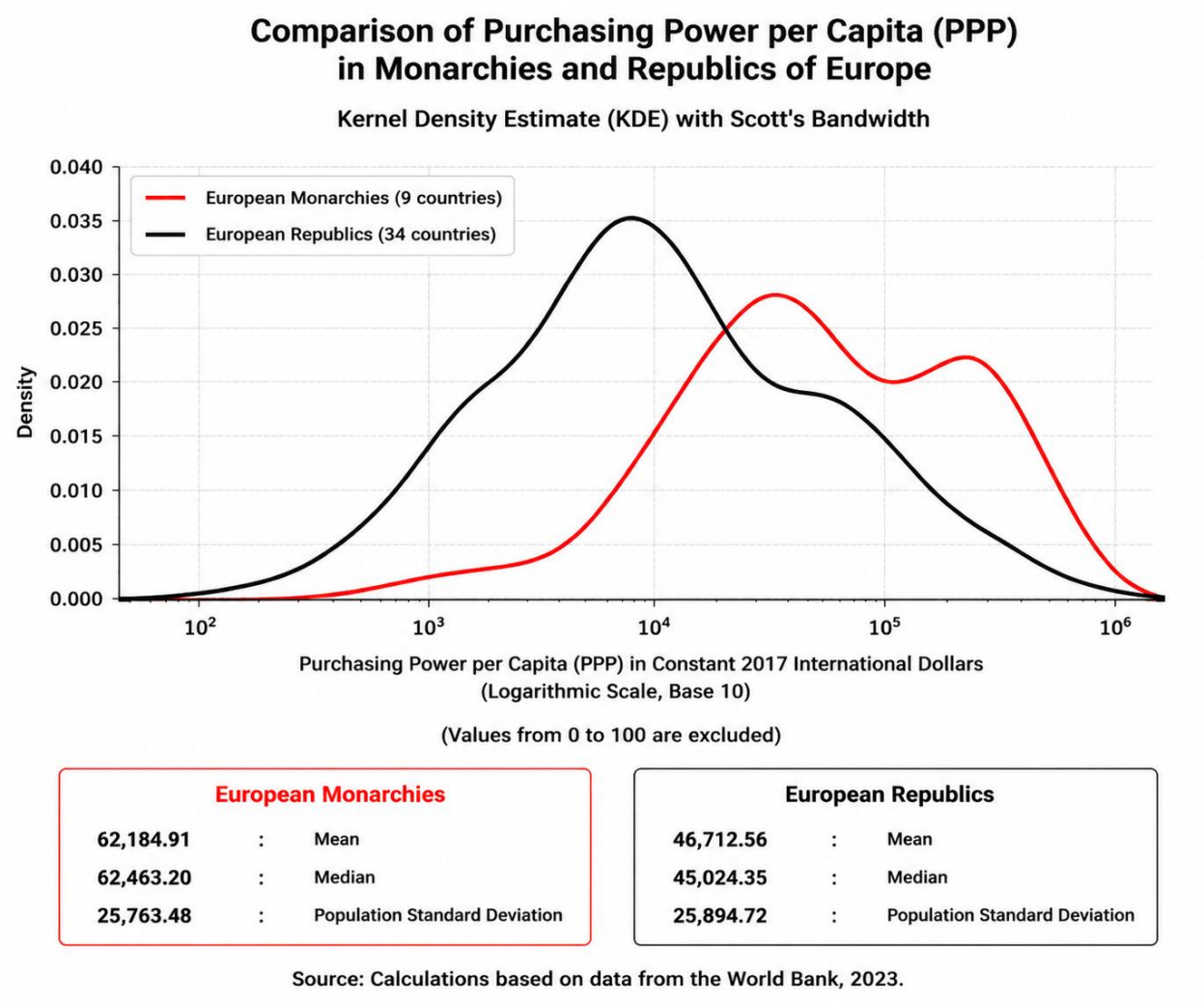
- **Welch's t-test:** p = 0.165
- **Mann–Whitney U test:** p = 0.132

Thus, although European monarchies have a higher average purchasing power per capita in the sample examined, the available statistical evidence is insufficient to conclude that the difference between the two groups is statistically significant.

Overall, **when republics with a history of communist rule are excluded from the comparison, the numerical advantage of European monarchies remains, but the gap between the two groups narrows substantially and is not statistically significant. This suggests that part of the difference observed between monarchies and the full set of European republics is associated with the lower purchasing power per capita of countries with a communist past. Nevertheless, even when monarchies are compared only with republics without a history of communist rule, both the mean and median remain higher among the monarchies.**

One further limitation should be considered when interpreting these results. World Bank purchasing-power-per-capita data were unavailable for two European monarchies, Liechtenstein and Monaco, and these countries were therefore excluded from the

calculations. Given that both are among the wealthiest countries in Europe, their inclusion, had comparable data been available, would likely have raised the mean purchasing power per capita of the monarchy group and potentially widened the observed gap relative to European republics without a history of communist rule. However, because actual comparable data for these two countries were unavailable in the dataset used, the magnitude of this potential effect cannot be determined.



The distribution of purchasing power per capita across European countries shows that the curve for monarchies is shifted toward higher values relative to that for republics. Mean purchasing power per capita is approximately **83.1 thousand international dollars** in European monarchies, compared with approximately **50.9 thousand international dollars** in European republics.

The medians also differ substantially, at approximately **72.6 thousand international dollars** for European monarchies and **48.1 thousand international dollars** for European republics. This

indicates that the distribution of monarchies is concentrated at higher levels of purchasing power per capita.

Although the two curves overlap to some extent, the overall rightward shift of the monarchy distribution is clearly visible. The statistical tests presented below assess whether this observed difference is also statistically significant.

Comparison of Purchasing Power per Capita between European Monarchies and Republics with No History of Communist Rule

Descriptive Statistics

Indicator	European Monarchies	Republics with No History of Communist Rule
Number of Countries	9	13
Mean PPP per Capita	**83,056 international dollars**	**72,239 international dollars**
Median	**72,630 international dollars**	**67,101 international dollars**
Sample Standard Deviation	**29,959 international dollars**	**22,860 international dollars**

Mean purchasing power per capita is **10,818 international dollars higher** among the monarchies, approximately **15% higher** than among European republics with no history of communist rule. The median is also approximately **5,529 international dollars higher** among the monarchies.

Welch's t-test

The two-sided Welch's t-test produced the following results:

t = 0.914, df = 14.21, p = 0.376

Thus, the difference between the two group means is **not statistically significant** at the 5% level.

The effect size is **Hedges' g = 0.401**, indicating a **small-to-moderate effect** in the direction of higher PPP per capita among the monarchies.

Mann–Whitney U Test

To ensure that the conclusion did not depend solely on a comparison of means, the non-parametric **Mann–Whitney U test** was also conducted:

U = 72, p = 0.385

Conclusion

This test likewise indicates that the difference between the two groups is **not statistically significant** at the 5% level.

Importantly, despite their different statistical foundations, both tests lead to a consistent conclusion: after excluding republics with a history of communist rule, the monarchies still have higher **mean and median** PPP per capita, but the available statistical evidence is insufficient to establish a statistically significant difference.

Another limitation should be considered when interpreting this result. In the World Bank dataset used in this study, purchasing power per capita data were unavailable for **Liechtenstein and Monaco**, and these two monarchies were therefore excluded from the calculations. Since both are among Europe's wealthiest countries, it is possible that, had comparable data been available, the mean for the monarchy group would have been higher and the gap between the two groups wider. However, because the required data were unavailable from the source used in this study, the magnitude and actual impact of their inclusion cannot be calculated.

Overall, this robustness check suggests that **the communist historical legacy accounts for part, but not all, of the observed difference in Europe**. Once this historical factor is removed from the comparison, the descriptive advantage of the monarchies remains, although the available evidence is insufficient to establish a statistically significant difference between the two groups.

Comparison of Purchasing Power per Capita between Monarchies and Republics in Africa, the Americas, and Oceania

Africa

Africa differs considerably from Asia and Europe in terms of the distribution of political systems. The continent has only three monarchies, compared with fifty republics. This substantial imbalance limits the statistical power of the comparison and requires greater caution in interpreting the results. Nevertheless, examining Africa remains valuable because it allows us to assess whether the pattern observed in the global comparisons is also present on this continent.

Descriptive Statistics

Among the three African monarchies, mean **purchasing power per capita** is **8,031 international dollars**, compared with **7,154 international dollars** among the fifty African republics. The difference between the two means is relatively small, although the medians show

a somewhat larger gap. Income dispersion, meanwhile, is considerably greater among the African republics, indicating greater economic heterogeneity within this group.

Group	Number of Countries	Mean	Median	Standard Deviation
Monarchies	3	8,031	9,903	4,516
Republics	50	7,154	4,107	7,119

Interpretation

Welch's test indicates that the observed difference is not statistically significant (**p ≈ 0.777**), while the effect size is also very small (**Hedges' g ≈ 0.12**). Thus, unlike in some other regions of the world, the African data do not indicate that type of government is associated with a substantial difference in **purchasing power per capita**. In this region, therefore, economic performance cannot be explained simply by whether a country is a monarchy or a republic.

The Americas

There are no independent monarchies in the Americas. However, several countries on the continent are Commonwealth realms and, in constitutional terms, are classified as constitutional monarchies. Since these countries were examined separately earlier, this section provides only a comparison of their economic performance with that of the republics of the Americas.

Descriptive Statistics

This study includes **9 Commonwealth realms** and **25 republics** in the Americas. Mean **purchasing power per capita** is **29,052 international dollars** in the Commonwealth realms, compared with **25,081 international dollars** in the republics. The median is also higher among the Commonwealth realms (**26,254 international dollars**) than among the republics (**21,282 international dollars**). The standard deviations of the two groups are quite similar, suggesting broadly comparable levels of dispersion.

Group	Number of Countries	Mean	Median	Standard Deviation
Commonwealth Realms	9	29,052	26,254	16,282
Republics	25	25,081	21,282	16,454

Interpretation

Although both the mean and median **purchasing power per capita** are higher among the Commonwealth realms than among the republics of the Americas, Welch's test indicates that the difference is not statistically significant (**p ≈ 0.542**), and the effect size is small (**Hedges' g ≈ 0.24**). The regional data therefore do not provide sufficient evidence to attribute the observed difference to type of government. Nevertheless, the direction of the difference is consistent with the broader pattern observed in the global comparison and is therefore not inconsistent with the wider findings of this study.

Oceania

Oceania also has an uneven distribution of political systems. There is only one independent monarchy in the region, **Tonga**. In addition, the study includes **five Commonwealth realms** and **eight republics**. Because a single independent monarchy does not permit a meaningful statistical comparison between monarchies and republics, the analysis focuses primarily on the Commonwealth realms and republics, while Tonga is reported separately to provide a complete picture.

Descriptive Statistics

Tonga, the only independent monarchy in Oceania, has a **purchasing power per capita** of **2,599 international dollars**. By comparison, mean purchasing power per capita among the five Commonwealth realms is **29,122 international dollars**, compared with **9,233 international dollars** among the eight republics.

Although the median among the Commonwealth realms (**7,338 international dollars**) is slightly lower than that of the republics (**7,864 international dollars**), the presence of the two large economies of Australia and New Zealand raises the group mean substantially. This also explains the much larger standard deviation among the Commonwealth realms.

Group	Number of Countries	Mean	Median	Standard Deviation
Independent Monarchy (Tonga)	1	2,599	2,599	—
Commonwealth Realms	5	29,122	7,338	33,507
Republics	8	9,233	7,864	5,528

Interpretation

Although mean **purchasing power per capita** is substantially higher among the Commonwealth realms than among the republics of Oceania, much of this difference is driven by the two large economies of Australia and New Zealand. Welch's test indicates that the observed difference is not statistically significant (**$p \approx 0.274$**), although the effect size (**Hedges' $g \approx 0.72$**) is larger than those observed in Africa and the Americas. The Oceania data alone are therefore insufficient to support a conclusion about the role of government type and should be interpreted alongside evidence from other regions.

Conclusion

Africa, the Americas, and Oceania all face limitations related to small or highly unbalanced sample sizes. Consequently, none of these regions alone provides a sufficiently strong basis for a definitive judgment about the relationship between type of government and purchasing power per capita. Nevertheless, examining these regions remains informative

because none produces evidence that directly contradicts the broader pattern observed in the global comparisons.

These three regions therefore do not constitute the primary empirical basis of the argument developed in this study. Rather, their results complement the evidence from other regions and contribute to a more comprehensive picture of the relationship between type of government and countries' economic performance.

Comparison of Purchasing Power per Capita between Monarchies and Republics in the Middle East

Descriptive Statistics for the Middle East

Given the importance of the Middle East as a natural laboratory for comparing monarchies and republics, as discussed earlier, we can now examine the results for purchasing power per capita.

This study includes **7 monarchies** and **9 republics** in the Middle East. The results show that the difference between the two groups remains substantial even after adjusting for differences in price levels and the purchasing power of money across countries.

Mean **purchasing power per capita** is **64,201 international dollars** in the Middle Eastern monarchies, compared with **25,685 international dollars** in the region's republics. In other words, mean purchasing power per capita in the Middle Eastern monarchies is approximately **2.5 times** that of the republics.

The median confirms the same pattern. Median purchasing power per capita is **64,224 international dollars** among the monarchies and **18,540 international dollars** among the republics. Because the median is less affected by extremely high values than the mean, the consistency between these two measures indicates that the observed difference is not simply the result of one or two exceptionally wealthy countries but is also reflected in the broader distribution of the data.

The standard deviation is higher among the monarchies (**36,167 international dollars**) than among the republics (**21,452 international dollars**). This indicates that considerable variation exists within the region's monarchies as well. Nevertheless, this dispersion does not prevent their overall level of **purchasing power per capita** from remaining substantially higher than that of the Middle Eastern republics.

Group	Number of Countries	Mean	Median	Standard Deviation
Monarchies	7	64,201	64,224	36,167
Republics	9	25,685	18,540	21,452

Interpretation

1. Magnitude of the Descriptive Difference

The first notable finding is the magnitude of the difference between the two groups. **Mean purchasing power per capita in the Middle Eastern monarchies is approximately 2.5 times that of the region's republics, and the median confirms the same pattern.** The consistency between the mean and median suggests that the observed advantage cannot be attributed solely to a few exceptionally wealthy countries; rather, the difference is also visible across the broader distribution.

The descriptive statistics therefore provide an initial indication that the pattern previously observed globally and in some other regions of the world is also present in the Middle East.

2. Statistical Significance

Although descriptive statistics are informative, they are not sufficient on their own to establish whether the observed difference could plausibly be due to sampling variability. Welch's t-test was therefore used to assess the statistical significance of the difference.

The results indicate that **the difference between Middle Eastern monarchies and republics is statistically significant** (**$p \approx 0.033$**). The null hypothesis of equal group means is therefore rejected, and the available data support the presence of a difference in economic performance between the monarchies and republics in this regional sample.

3. Effect Size

Statistical significance alone does not indicate whether the magnitude of an observed difference is substantively important. For this reason, the effect size was also calculated.

Hedges' $g \approx 1.27$, which is conventionally regarded as a **large effect size**. This indicates that the difference between the two groups is not only statistically detectable but also substantial in standardized terms.

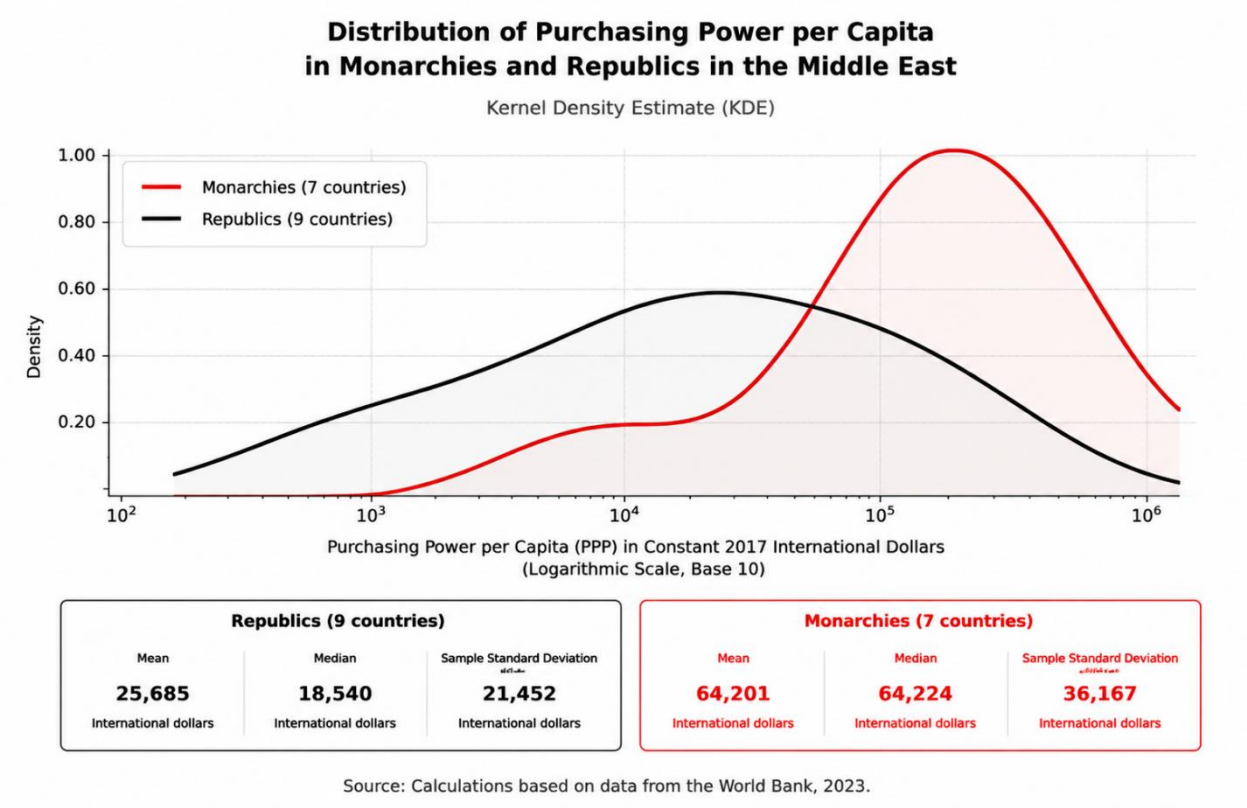
4. Conclusion

Overall, the Middle Eastern data present a consistent picture across all three levels of analysis: descriptive statistics, inferential testing, and effect size. The region's monarchies have substantially higher mean and median purchasing power per capita, the difference is statistically significant, and its standardized magnitude is large.

The importance of this finding becomes more apparent when considering that the comparison is conducted within a region whose countries share, to a greater extent than in many global comparisons, important historical, cultural, geographical, and economic characteristics. This

greater regional comparability reduces some—though not all—of the contextual heterogeneity that complicates global comparisons. The Middle East can therefore be regarded as one of the more informative regional tests of the central hypothesis examined in this study.

As with all observational research, however, these findings **are not sufficient on their own to establish a causal relationship** between type of government and economic performance. Nevertheless, the convergence of the descriptive statistics, statistical significance, and large effect size, together with the greater contextual similarity of the countries being compared, provides substantial empirical support for the hypothesis that **type of government may be one of the factors associated with differences in countries' economic performance.**



The distribution of purchasing power per capita in Middle Eastern countries shows a pronounced rightward shift for monarchies relative to republics. Average purchasing power per capita is approximately **$64.2 thousand** in Middle Eastern monarchies, compared with about **$25.7 thousand** in the region's republics.

The medians also differ substantially: approximately **$64.2 thousand** for monarchies versus **$18.5 thousand** for republics. This indicates that the higher values observed among monarchies are not merely the result of a few exceptionally wealthy countries; rather, the central part of their distribution is also concentrated at considerably higher levels of purchasing power.

Despite substantial dispersion within both groups, the overall separation between the two distributions is clearly visible. Indeed, the visual contrast in the Middle East appears stronger than in Europe and even Asia. Nevertheless, the apparent separation of the curves cannot by itself establish statistical significance, making the inferential tests reported in the analysis essential for determining whether the observed difference is statistically supported.

Gini Index

So far, the economic indicators examined in this study have shown that monarchies have, on average, performed better than republics. However, this raises an important question: **Is greater wealth alone sufficient to evaluate the success of a system of government?** If the wealth generated in a country is concentrated largely in the hands of a small segment of society, while the majority of citizens receive little benefit from it, can such economic performance still be regarded as successful?

To address this question, we must examine not only how much wealth is produced, but also how that wealth is distributed. One of the most widely used measures of economic inequality is the **Gini index**. It measures inequality in the distribution of income or, in some countries, consumption. A lower Gini value indicates a more equal distribution, whereas a higher value indicates greater inequality. In the World Bank's scale, a value of 0 represents perfect equality and a value of 100 represents perfect inequality.

This direction should be kept in mind throughout the analyses that follow. **Unlike GDP per capita and purchasing power per capita, a higher value of the Gini index does not represent better performance; in terms of economic equality, lower values indicate more equal outcomes.**

At the same time, a low Gini index does not necessarily imply better overall economic performance, nor is a high Gini index by itself sufficient grounds for a negative assessment of an economy. The relationship between economic growth and inequality is complex and can vary across countries and periods. Rapid economic transformation, for example, may be accompanied by changes in inequality as different groups benefit from new economic opportunities at different rates. The Gini index should therefore be interpreted alongside other measures of welfare and development rather than in isolation. The World Bank likewise cautions that the Gini measures relative inequality and that differences in underlying income and consumption surveys can affect cross-country comparability.

Accordingly, the purpose of this section is not simply to determine which form of government has the lower Gini index. Rather, it is to examine **whether the higher levels of economic welfare observed in the countries of the two groups are also associated with a more equal distribution of income or consumption**. This comparison brings us one step closer to evaluating another fundamental objective of government: **economic justice**.

The data used in this chapter were obtained from the **World Bank** and refer to **2024 or the nearest available year for each country** (Appendix RDS-C01). Countries for which no data were available were excluded from the analysis. The results presented in this chapter are therefore based on the most recent available observations for countries with sufficient data.

Global Comparison of the Gini Index: Monarchies and Republics

Indicator: Gini Index
Data year: 2024 or the nearest available year
Source: World Bank

The Gini Index is one of the most widely used measures of inequality in the distribution of income or consumption. Its value typically ranges from 0 to 100: lower values indicate a more equal distribution and less inequality, while higher values indicate greater inequality. Therefore, unlike indicators such as GDP per capita and purchasing power per capita, **a lower value is preferable from the standpoint of economic equality**.

In this section, the Gini Index is compared between the world's monarchies and republics. In addition to descriptive statistics, two statistical tests are used: **Welch's t-test** to compare group means and the **Mann–Whitney U test** to compare the rank distributions of the data. **Hedges' g** is also calculated to measure the standardized magnitude of the difference between the two groups.

1. Descriptive Statistics

Group	Countries (n)	Mean	Median	Sample Standard Deviation
Monarchies	19	33.35	32.40	7.34
Republics	143	36.61	35.90	7.04

Interpretation of the Descriptive Statistics

The mean Gini Index is **33.35** among monarchies and **36.61** among republics. Since a lower Gini value indicates less inequality, **monarchies in this dataset show, on average, a lower level of inequality**.

The medians point in the same direction: **32.40** for monarchies and **35.90** for republics. Thus, both the mean and the central value of the distribution are lower in the monarchy group.

The sample standard deviations are also very similar: **7.34** among monarchies and **7.04** among republics. The observed difference between the groups is therefore not accompanied by a substantial difference in overall dispersion.

Descriptive statistics alone, however, cannot determine whether the observed difference is statistically significant. Inferential tests are therefore required.

2. Welch's t-test for the Difference in Means

The hypotheses were defined as follows:

Null hypothesis (H_0): The mean Gini Index is equal in monarchies and republics.

Alternative hypothesis (H_1): The mean Gini Index differs between the two groups.

Results of Welch's t-test

Statistic	Value
Mean — Monarchies	**33.35**
Mean — Republics	**36.61**
Mean difference	**−3.25**
t-statistic	**−1.824**
Welch's adjusted degrees of freedom	**22.63**
Two-tailed p-value	**0.081**
95% confidence interval for the mean difference	**−6.95 to 0.44**
Hedges' g	**−0.46**
Effect size	**Small to moderate; close to moderate**

Interpretation of Welch's t-test

The p-value (**$p \approx 0.081$**) is above the pre-specified significance level of 0.05. The null hypothesis is therefore not rejected at the 5% level.

In other words, although the mean Gini Index is approximately **3.25 points lower** among monarchies than among republics, Welch's test does not provide sufficient evidence to conclude that the **group means** differ significantly at the conventional 5% level.

The 95% confidence interval for the mean difference extends from approximately **−6.95 to 0.44**. Because this interval includes zero, it is consistent with the non-significant result of the Welch test.

This does not establish that the two group means are equal. It simply means that, given the available data and the chosen significance threshold, there is insufficient statistical evidence to reject the hypothesis of equal means.

Effect Size

To assess the magnitude of the observed difference, **Hedges' g** was also calculated. Its value is approximately **−0.46**. The negative sign indicates that the mean Gini Index is lower among monarchies than among republics; it should not be interpreted as a negative effect in a value-laden sense.

In absolute terms, **$|g| \approx 0.46$**, indicating a **small-to-moderate effect, close to moderate**. Thus, even though Welch's test does not reach statistical significance at the 5% level, the standardized magnitude of the observed difference is not negligible.

This distinction is important: the **p-value** reflects the strength of statistical evidence against the null hypothesis, whereas the **effect size** describes the magnitude of the observed difference.

3. Mann–Whitney U Test

To examine the difference between the two groups from another perspective, the non-parametric **Mann–Whitney U test** was also conducted. This test is based on the ranks of the observations and does not require the data to follow a normal distribution.

The hypotheses were defined as follows:

Null hypothesis (H_0): There is no significant difference in the rank distribution of Gini values between the two groups.

Alternative hypothesis (H_1): The rank distributions of Gini values differ between the two groups.

Results of the Mann–Whitney U Test

Statistic	Value
U-statistic	922.5
Two-tailed p-value	0.023

Interpretation of the Mann–Whitney U Test

The p-value (**$p \approx 0.023$**) is below the 0.05 significance level. The null hypothesis is therefore rejected.

This result indicates that **the Gini values of the two groups differ significantly in their rank distributions**. Since both the mean and median Gini Index are lower among monarchies, the observed direction of the difference is toward lower Gini values in the monarchy group.

At first glance, the Mann–Whitney result may appear inconsistent with the Welch result, but there is no contradiction. Welch's test specifically evaluates differences in **means**, whereas the Mann–Whitney test is based on the **ranks of the observations** and captures a different aspect of the difference between the groups. It is therefore possible for one test to reach statistical significance while the other does not.

Without additional assumptions about the shapes of the two distributions, the Mann–Whitney test should also not be interpreted simply as a test of differences in medians.

Which Test Is More Appropriate for These Data?

In this study, **Welch's t-test** and the **Mann–Whitney U test** examine different aspects of the comparison between monarchies and republics and should not be treated as substitutes for one another.

Welch's test specifically addresses whether the **mean Gini Index** differs between the two groups. It is appropriate for groups with unequal sample sizes and unequal variances. The difference in sample size—**19 monarchies versus 143 republics**—is therefore not, by itself, a reason to regard Welch's test as inappropriate.

The Mann–Whitney test, by contrast, is based on ranks and examines whether values in one group tend systematically to occupy higher or lower ranks than those in the other. Because it does not depend on an assumption of normality, it provides complementary information about the difference between the groups.

The most appropriate approach in this study is therefore to **report both tests**, rather than selecting one according to which produces a statistically significant result.

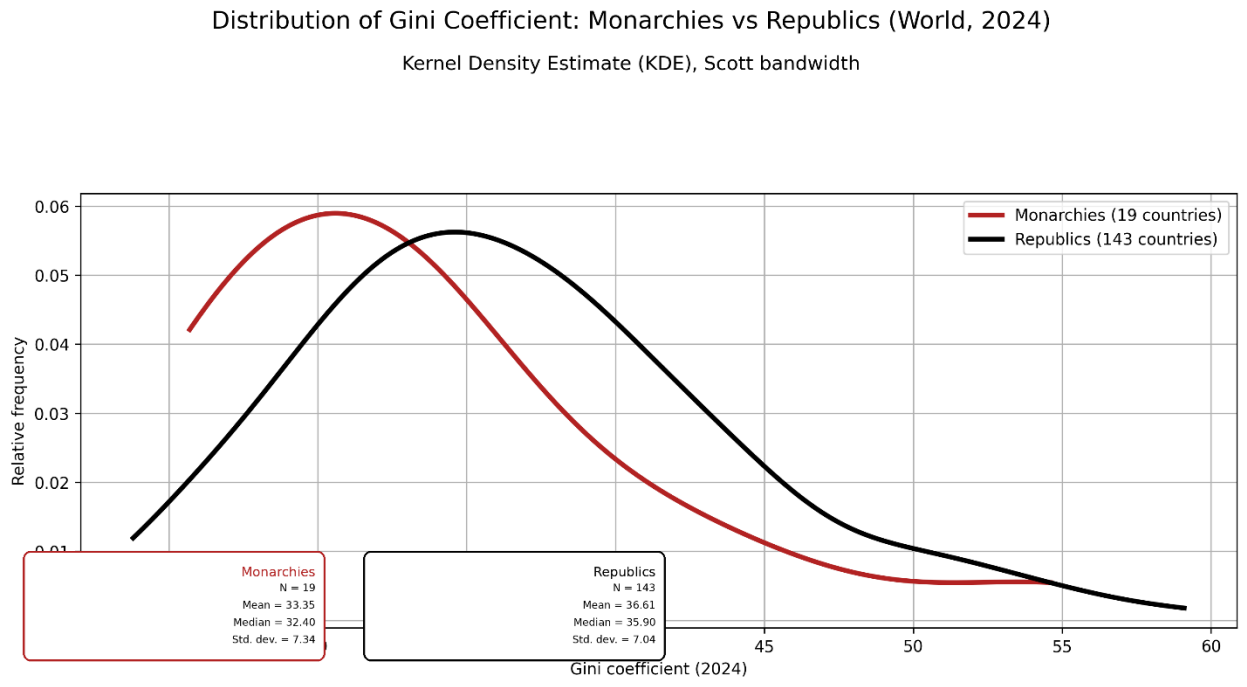
Conclusion

In the available data, the mean Gini Index is **33.35** among monarchies and **36.61** among republics, while the corresponding medians are **32.40** and **35.90**. Both descriptive measures therefore point toward lower inequality in the monarchy group.

The standardized effect size is approximately **0.46** in absolute magnitude, indicating a small-to-moderate difference, close to moderate.

However, the two inferential tests do not produce identical results. Welch's t-test, which evaluates differences in means, does not reach statistical significance at the 5% level ($\mathbf{p \approx 0.081}$), whereas the Mann–Whitney U test indicates a statistically significant rank-based difference between the two groups ($\mathbf{p \approx 0.023}$).

The most cautious conclusion, therefore, is that monarchies in the available data have lower Gini values than republics in both mean and median terms, and the standardized magnitude of the difference is not negligible. The Mann–Whitney test provides statistically significant evidence of a rank-based difference between the groups, while the evidence for a difference in means does not reach the conventional 5% significance threshold in Welch's test.



The distribution of the Gini Index across countries shows that the curve for monarchies is shifted slightly toward lower values relative to republics. Since a lower Gini Index indicates lower income inequality, this shift suggests that, in this dataset, **monarchies tend to exhibit lower levels of income inequality on average**.

The mean Gini Index is **33.35 for monarchies** and **36.61 for republics**. The corresponding medians are **32.40** and **35.90**, respectively. Thus, the observed difference is not confined to a few particular observations but is also visible in the central tendency of the distributions.

Nevertheless, the two curves overlap substantially, indicating considerable variation within both groups. The distributional pattern alone therefore cannot establish either statistical significance or a causal relationship between government type and inequality. **The statistical tests reported in the accompanying analysis are required to assess the significance of the observed differences.**

Comparison of the Gini Index: Constitutional vs. Semi-Constitutional and Absolute Monarchies Worldwide

Following the overall comparison between monarchies and republics, a comparison within the monarchy group allows us to examine whether differences in monarchical structure are associated with differences in the level of income inequality.

For this purpose, monarchies were divided into two groups:

1. **Constitutional monarchies**
2. **Semi-constitutional and absolute monarchies**

Two statistical tests were used to examine differences between the groups:

- **Welch's t-test** to compare the means
- **Mann–Whitney U test** to compare the distributions

1. Descriptive Statistics

Group	Countries (n)	Mean	Median	Population SD
Constitutional monarchies	**13**	**32.50**	**32.40**	**5.24**
Semi-constitutional and absolute monarchies	**6**	**35.20**	**31.80**	**9.86**

Interpretation of the Descriptive Statistics

The mean Gini Index is **32.50** among constitutional monarchies and **35.20** among semi-constitutional and absolute monarchies.

Since a lower Gini Index indicates lower inequality, the descriptive statistics suggest that, in this sample, **constitutional monarchies have a lower average Gini Index**, indicating somewhat lower income inequality.

However, the difference between the two groups is relatively modest, and dispersion is substantially greater among semi-constitutional and absolute monarchies. Statistical tests are therefore required to determine whether the observed difference is sufficiently robust to be distinguished from sampling variation.

2. Welch's t-test for the Difference in Means

The hypotheses were defined as follows:

Null hypothesis (H_0): The mean Gini Index is equal in the two types of monarchy.

Alternative hypothesis (H_1): The mean Gini Index differs between the two types of monarchy.

Results of Welch's t-test

Statistic	Value
Mean — Constitutional monarchies	**32.50**
Mean — Semi-constitutional and absolute monarchies	**35.20**
Mean difference	**−2.70**
t-statistic	**−0.579**

Statistic	Value
Approximate degrees of freedom	**6.21**
p-value	**0.583**

Interpretation of Welch's t-test

The p-value (**p = 0.583**) is substantially greater than the conventional 5% significance level. The null hypothesis is therefore not rejected.

Although the mean Gini Index is lower among constitutional monarchies, **the observed difference is not statistically significant in this dataset**.

In other words, the available data do not provide sufficient statistical evidence to conclude that the two types of monarchy differ in their mean level of inequality.

3. Mann–Whitney U Test

To examine the distributions without assuming normality, the **Mann–Whitney U test** was also conducted.

The hypotheses were defined as follows:

Null hypothesis (H_0): There is no significant difference in the rank distribution of Gini values between the two groups.

Alternative hypothesis (H_1): The rank distributions of Gini values differ between the two groups.

Results of the Mann–Whitney U Test

Statistic	Value
U-statistic	**36**
p-value	**0.831**

Interpretation of the Mann–Whitney U Test

The p-value (**p = 0.831**) is substantially greater than the 5% significance level. The null hypothesis is therefore not rejected.

This test likewise provides **no statistically significant evidence of a rank-based difference in Gini values between the two types of monarchy**.

Conclusion

For the Gini Index data used in this study:

Group	Mean Gini Index
Constitutional monarchies	**32.50**
Semi-constitutional and absolute monarchies	**35.20**

Constitutional monarchies have a lower mean Gini Index in this sample. However, neither statistical test identifies a significant difference:

- **Welch's t-test:** $p = 0.583$
- **Mann–Whitney U test:** $p = 0.831$

Therefore, in the available Gini Index data, there is no statistically significant evidence that inequality differs between constitutional monarchies and semi-constitutional or absolute monarchies. Although the mean Gini Index is lower among constitutional monarchies, the observed difference is not sufficiently robust to distinguish the two groups statistically.

Comparison of the Gini Index: Commonwealth Realms and Republics Worldwide

Following the overall comparison between monarchies and republics, it is also useful to examine the **Commonwealth Realms** separately. These countries share a historical and constitutional connection to the British monarchy, while differing considerably in their economic and social structures.

In this section, the Gini Index of the Commonwealth Realms is compared with that of the world's republics to determine whether the two groups differ in their levels of income inequality.

1. Descriptive Statistics

Group	Countries (n)	Mean	Median	Population SD
Commonwealth Realms	**9**	**38.97**	**39.90**	**4.28**
Republics	**143**	**36.61**	**35.90**	**7.02**

Interpretation of the Descriptive Statistics

The mean Gini Index is **38.97** among the Commonwealth Realms and **36.61** among the world's republics.

Since a lower Gini Index indicates lower inequality, the descriptive statistics suggest that, in this dataset, the Commonwealth Realms have a somewhat higher average Gini Index than republics, indicating somewhat greater inequality.

The medians point in the same direction:

- **Commonwealth Realms:** 39.90
- **Republics:** 35.90

Thus, unlike the overall comparison between monarchies and republics, the Commonwealth Realms in this comparison are **not associated with lower Gini values**.

However, the number of Commonwealth Realms for which data are available is small. Statistical tests are therefore required to determine whether the observed difference is sufficiently robust to distinguish the two groups statistically.

2. Welch's t-test for the Difference in Means

The hypotheses were defined as follows:

Null hypothesis (H_0): The mean Gini Index is equal in the Commonwealth Realms and republics.

Alternative hypothesis (H_1): The mean Gini Index differs between the two groups.

Results of Welch's t-test

Statistic	Value
Mean — Commonwealth Realms	**38.97**
Mean — Republics	**36.61**
Mean difference	**2.36**
t-statistic	**1.550**
Approximate degrees of freedom	**11.04**
p-value	**0.149**

Interpretation of Welch's t-test

The p-value (**p = 0.149**) is greater than the conventional 5% significance level. The null hypothesis is therefore not rejected.

Although the mean Gini Index is higher among the Commonwealth Realms, **the difference in means is not statistically significant**.

In other words, the available data do not provide sufficient statistical evidence to conclude that mean inequality differs between the Commonwealth Realms and the world's republics.

3. Mann–Whitney U Test

To examine the difference between the two groups without assuming normality, the **Mann–Whitney U test** was also conducted.

The hypotheses were defined as follows:

Null hypothesis (H_0): There is no significant difference in the rank distribution of Gini values between the two groups.

Alternative hypothesis (H_1): The rank distributions of Gini values differ between the two groups.

Results of the Mann–Whitney U Test

Statistic	Value
U-statistic	830
p-value	0.146

Interpretation of the Mann–Whitney U Test

The p-value (**p = 0.146**) is above the 5% significance level. The null hypothesis is therefore not rejected.

This test likewise provides **no statistically significant evidence of a rank-based difference in Gini values between the Commonwealth Realms and the world's republics**.

Conclusion

For the Gini Index data used in this study:

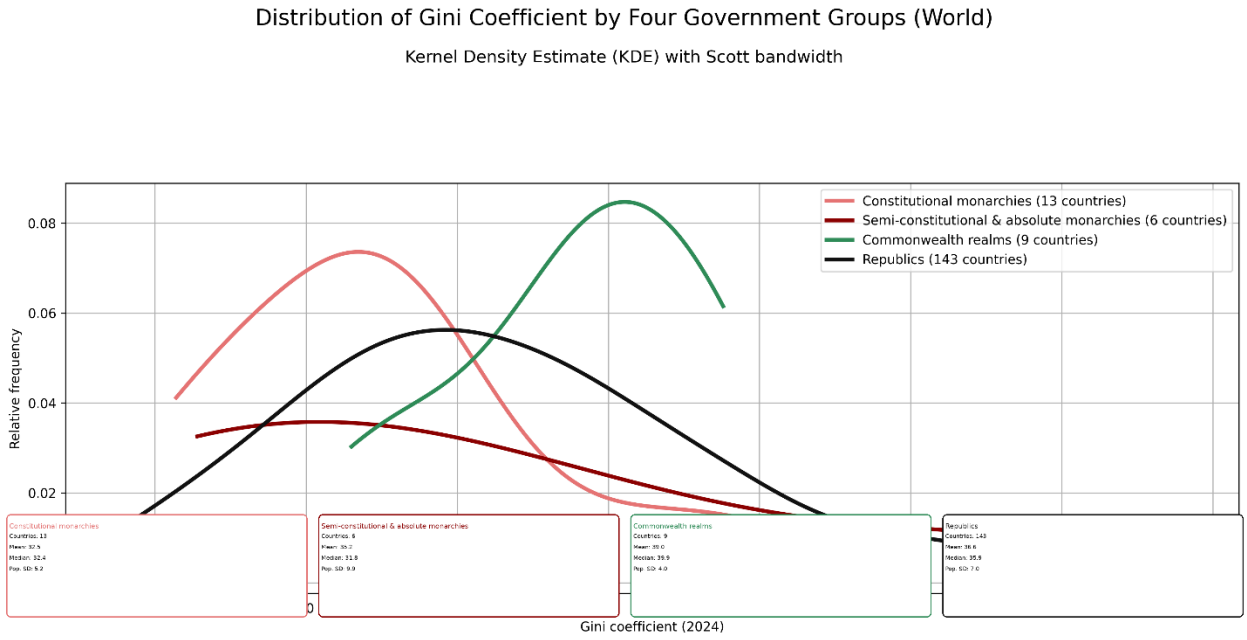
Group	Mean Gini Index
Commonwealth Realms	38.97
Republics	36.61

The Commonwealth Realms have a higher mean Gini Index in this sample, indicating greater inequality on average. However, neither statistical test identifies a significant difference:

- **Welch's t-test:** p = 0.149
- **Mann–Whitney U test:** p = 0.146

Therefore, the available Gini Index data provide no statistically significant evidence of a difference in inequality between the Commonwealth Realms and the world's republics.

Although both the mean and median Gini values are higher among the Commonwealth Realms, the observed difference cannot be statistically distinguished from sampling variation in this dataset.



The distribution of the Gini Index across the four government groups shows that patterns of inequality are not identical across them. Since lower Gini values indicate lower inequality, **constitutional monarchies**, with a mean of **32.5**, have the lowest average among the four groups, whereas the **Commonwealth Realms**, with a mean of **39.0**, have the highest.

Semi-constitutional and absolute monarchies, with a mean of **35.2**, and **republics**, with a mean of **36.6**, occupy intermediate positions. The location of the curves also indicates that the distribution for constitutional monarchies is more concentrated at lower Gini values, whereas the distribution for the Commonwealth Realms is shifted toward higher values.

Despite these observed differences, the substantial overlap among the four curves indicates that **statistical comparisons are necessary to determine whether the differences between the groups are significant**. The shapes of the distributions alone do not provide sufficient grounds for definitive conclusions.

Comparison of the Gini Index: Monarchies and Republics Across Continents and in the Middle East

Comparison of the Gini Index: Monarchies and Republics in Asia

In this section, the Gini Index of Asian countries is compared between two groups:

- **Asian monarchies**
- **Asian republics**

Two statistical tests were conducted to examine differences between the groups:

1. **Welch's t-test** to compare the means
2. **Mann–Whitney U test** to compare the distributions

1. Descriptive Statistics

Group	Countries (n)	Mean	Median	Population SD
Asian monarchies	**7**	**33.87**	**33.50**	**3.42**
Asian republics	**31**	**34.26**	**34.50**	**4.62**

Interpretation of the Descriptive Statistics

The mean Gini Index is **33.87** among Asian monarchies and **34.26** among Asian republics.

Since a lower Gini Index indicates lower inequality, the descriptive statistics suggest that Asian monarchies have, on average, a slightly lower level of inequality.

However, the difference between the means is very small:

33.87 − 34.26 = −0.39

Thus, the observed difference is less than one Gini point.

The medians also differ only slightly:

- **Monarchies:** 33.50
- **Republics:** 34.50

At the descriptive level, therefore, the two groups exhibit very similar patterns. Statistical tests are required to determine whether the small observed difference is statistically meaningful.

2. Welch's t-test for the Difference in Means

The hypotheses were defined as follows:

Null hypothesis (H_0): The mean Gini Index is equal in Asian monarchies and republics.

Alternative hypothesis (H_1): The mean Gini Index differs between the two groups.

Results of Welch's t-test

Statistic	Value
Mean — Asian monarchies	**33.87**
Mean — Asian republics	**34.26**
Mean difference	**−0.39**
t-statistic	**−0.210**
Approximate degrees of freedom	**12.06**
p-value	**0.837**

Interpretation of Welch's t-test

The p-value (**p = 0.837**) is substantially greater than the conventional 5% significance level. The null hypothesis is therefore not rejected.

This result indicates that **the difference in the mean Gini Index between Asian monarchies and republics is not statistically significant**.

In other words, the available data provide no statistical evidence that the two groups differ in their mean level of inequality.

3. Mann–Whitney U Test

To examine the overall distributions without assuming normality, the **Mann–Whitney U test** was also conducted.

The hypotheses were defined as follows:

Null hypothesis (H_0): There is no significant difference in the rank distribution of Gini values between the two groups.

Alternative hypothesis (H_1): The rank distributions of Gini values differ between the two groups.

Results of the Mann–Whitney U Test

Statistic	Value
U-statistic	103
p-value	0.872

Interpretation of the Mann–Whitney U Test

The p-value (**p = 0.872**) is substantially greater than the 5% significance level. The null hypothesis is therefore not rejected.

This test likewise provides **no statistically significant evidence of a rank-based difference in Gini values between Asian monarchies and republics**.

Conclusion

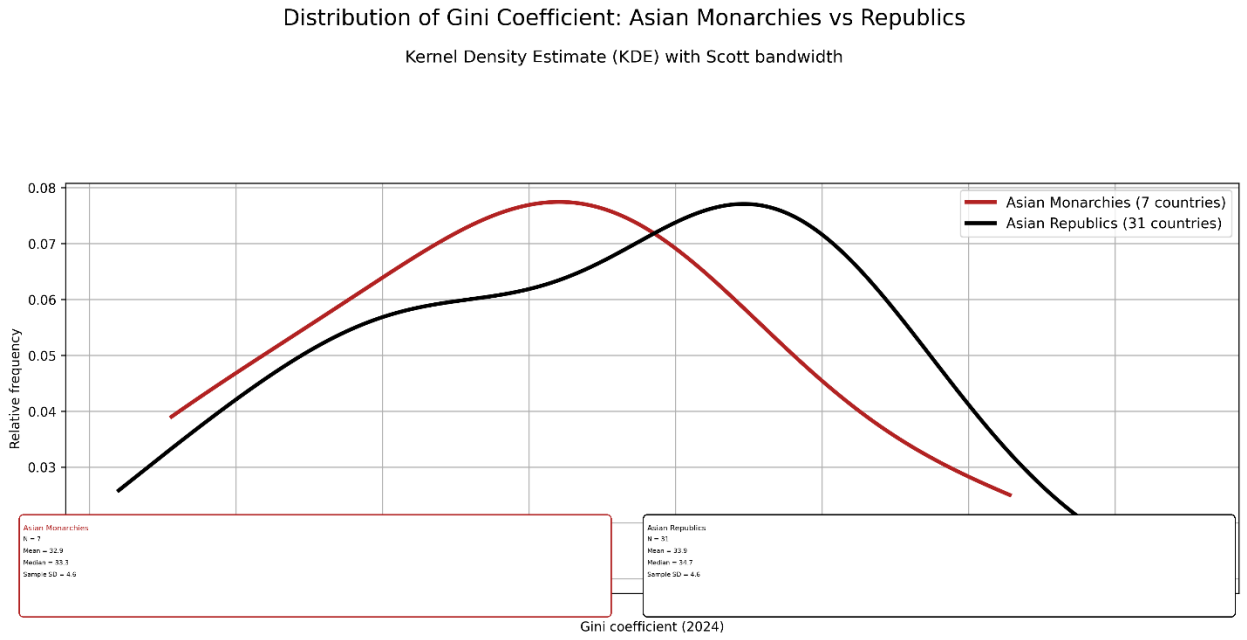
For the Gini Index data used in this study for Asia:

Group	Mean Gini Index
Monarchies	33.87
Republics	34.26

Asian monarchies have a slightly lower mean Gini Index, but the difference is very small and neither statistical test identifies it as significant:

- **Welch's t-test:** p = 0.837
- **Mann–Whitney U test:** p = 0.872

Therefore, among the Asian countries examined, no statistically significant difference in the Gini Index is observed between monarchies and republics. Although the mean Gini Index is slightly lower among monarchies, the difference is too small to be statistically distinguished between the two groups in the available data.



The distribution of the Gini Index across Asian countries shows that monarchies and republics exhibit **very similar patterns**, with substantial overlap between their curves.

The mean Gini Index is **32.9 for Asian monarchies** and **33.9 for Asian republics**, while the corresponding medians are **33.3** and **34.7**, respectively. Since a lower Gini Index indicates lower inequality, the descriptive statistics suggest that Asian monarchies have slightly lower Gini values.

However, the close proximity and substantial overlap of the two distributions indicate that the observed difference is limited. **Statistical tests are therefore necessary to determine whether this small difference is statistically meaningful or simply reflects sampling variation.**

Comparison of the Gini Index: Monarchies and Republics in Europe

Europe is one of the most suitable regions for an intra-regional comparison of government systems because its countries are relatively similar in terms of development levels, historical background, and economic structures. In this section, the Gini Index of European countries is compared between monarchies and republics.

1. Descriptive Statistics

Group	Countries (n)	Mean	Median	Population SD
European monarchies	8	29.70	29.60	2.98
European republics	33	31.11	31.80	3.86

Interpretation of the Descriptive Statistics

The mean Gini Index is **29.70** among European monarchies and **31.11** among European republics.

Since a lower Gini Index indicates lower inequality, the descriptive statistics suggest that European monarchies in this sample have, on average, lower Gini values.

The medians point in the same direction:

- **Monarchies:** 29.60
- **Republics:** 31.80

Thus, at the descriptive level, European monarchies exhibit somewhat lower inequality. However, the difference between the two groups is relatively modest, and statistical tests are required to determine whether it is statistically significant.

2. Welch's t-test for the Difference in Means

The hypotheses were defined as follows:

Null hypothesis (H_0): The mean Gini Index is equal in European monarchies and republics.

Alternative hypothesis (H_1): The mean Gini Index differs between the two groups.

Results of Welch's t-test

Statistic	Value
Mean — European monarchies	**29.70**
Mean — European republics	**31.11**
Mean difference	**−1.41**
t-statistic	**−1.067**
Approximate degrees of freedom	**12.72**
p-value	**0.306**

Interpretation of Welch's t-test

The p-value (**p = 0.306**) is greater than the conventional 5% significance level. The null hypothesis is therefore not rejected.

Although the mean Gini Index is lower among European monarchies, **the difference in means is not statistically significant**.

In other words, the available data do not provide sufficient statistical evidence to conclude that European monarchies and republics differ in their mean level of inequality.

3. Mann–Whitney U Test

To examine the overall distributions without assuming normality, the **Mann–Whitney U test** was also conducted.

The hypotheses were defined as follows:

Null hypothesis (H_0): There is no significant difference in the rank distribution of Gini values between the two groups.

Alternative hypothesis (H_1): The rank distributions of Gini values differ between the two groups.

Results of the Mann–Whitney U Test

Statistic	Value
U-statistic	99
p-value	0.285

Interpretation of the Mann–Whitney U Test

The p-value (**p = 0.285**) is greater than the 5% significance level. The null hypothesis is therefore not rejected.

This test likewise provides **no statistically significant evidence of a rank-based difference in Gini values between European monarchies and republics**.

Conclusion

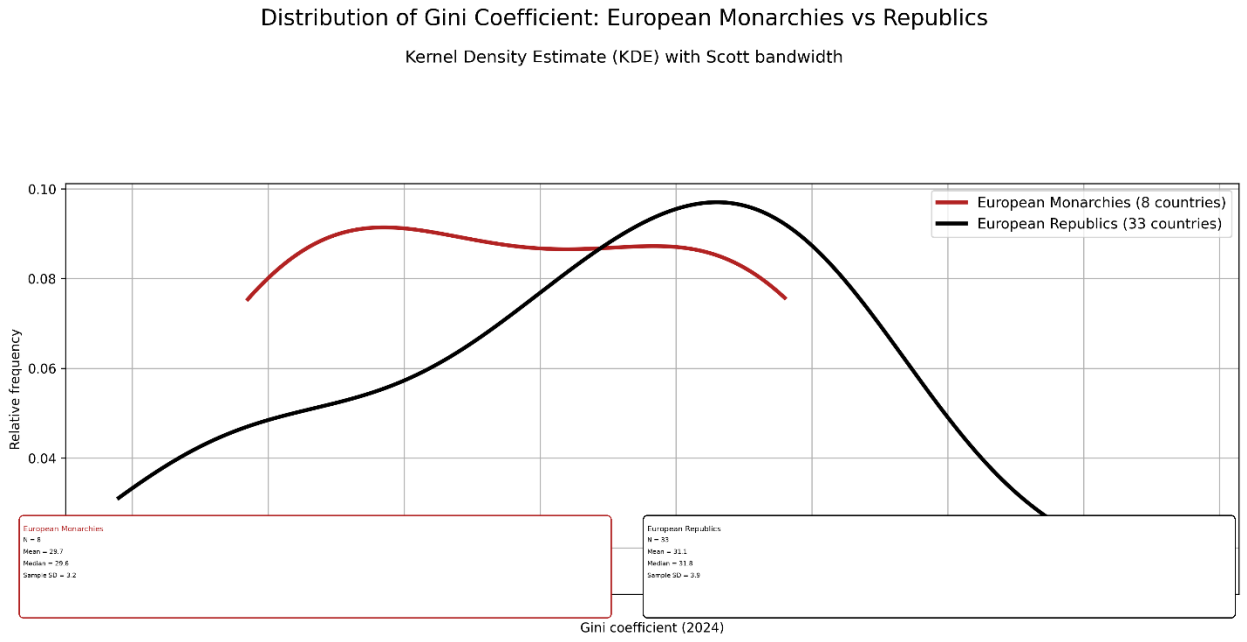
For the available European Gini Index data:

Group	Mean Gini Index
Monarchies	29.70
Republics	31.11

European monarchies have a lower mean Gini Index and therefore exhibit lower inequality at the descriptive level. However, neither statistical test identifies the difference as significant:

- **Welch's t-test:** p = 0.306
- **Mann–Whitney U test:** p = 0.285

Therefore, among the European countries for which data are available, monarchies have a lower average Gini Index than republics, but the difference is not statistically significant. The available evidence is therefore insufficient to establish a systematic difference in inequality between the two groups.



The distribution of the Gini Index across European countries shows that the curve for monarchies is slightly shifted toward lower values. Since a lower Gini Index indicates lower income inequality, this pattern suggests that European monarchies in this dataset tend, on average, to exhibit lower levels of inequality.

The mean Gini Index is **29.7 for European monarchies** and **31.1 for European republics**. The medians are **29.6** and **31.8**, respectively. Thus, the observed difference is not confined to a few particular observations but is also visible in the central values of the two distributions.

Despite this shift, the two curves overlap substantially, indicating that the difference between the groups is relatively modest. Whether the observed difference is statistically significant should therefore be assessed using the statistical tests presented in the analysis.

Comparison of the Gini Index Between European Monarchies and European Republics That Have Never Been Communist

To examine whether a history of communist rule might affect the results of the comparison between European monarchies and republics, European republics with a history of communist government were excluded from the sample in this robustness test. This left **8 European monarchies** to be compared with **12 European republics that had never experienced communist rule**.

Descriptive Statistics

Form of Government	Number of Countries	Mean	Median	Sample SD
European Monarchies	8	29.70	29.60	3.19
European Republics (No History of Communist Rule)	12	32.06	32.60	2.59

After excluding republics with a history of communist rule, the difference between the two groups became **larger than in the comparison of Europe as a whole**. The mean Gini Index of republics with no history of communist rule is approximately **2.36 points higher** than that of the monarchies. The medians point in the same direction: **32.60 for the republics compared with 29.60 for the monarchies**.

Since a lower Gini Index indicates lower inequality, the descriptive statistics suggest that, within this subsample, European monarchies exhibit a more equal distribution of income or consumption.

Statistical Significance Test

Welch's t-test was used to compare the means of the two groups. The result was:

$t = -1.74$, $p = 0.105$

Therefore, at the 0.05 significance level, the null hypothesis of equal means cannot be rejected, and **there is insufficient statistical evidence to conclude that the mean Gini Index differs significantly between the two groups**.

Nevertheless, the p-value is lower than in the previous comparison that included all European republics (**$p \approx 0.31$**). This change is consistent with the larger descriptive difference between the two groups, although it should not, by itself, be interpreted as independent evidence of a genuine difference.

Effect Size

The standardized effect size in this comparison is:

Hedges' $g \approx -0.80$

The negative sign indicates that the mean Gini Index is lower among the monarchies and should not be interpreted as having any independent normative meaning. The absolute magnitude of the effect, approximately **0.80, falls within the range conventionally regarded as a large effect**.

This distinction is important because it indicates that, although Welch's test does not reach statistical significance at the 5 percent level, **the observed standardized difference is not small**. Given the limited number of countries in the two groups, however, the estimated effect size should also be interpreted with caution.

Conclusion

This robustness test produces a noteworthy result. If the lower Gini Index observed among European monarchies in the initial comparison were primarily attributable to the presence of republics with a communist past, one would expect the difference between the two groups to decrease after those countries were excluded. **In the available data, however, the opposite occurs: after excluding republics with a history of communist rule, the difference in the mean Gini Index between monarchies and republics becomes larger.**

Therefore, **the descriptively lower Gini Index of European monarchies cannot be attributed to the inclusion of post-communist republics in the comparison group**. Within this subsample, both the mean and median Gini Index remain lower among monarchies, and the estimated effect size is also large.

However, Welch's test does not reach statistical significance at the conventional 5 percent level (**p = 0.105**). Accordingly, this result should be interpreted as **noteworthy descriptive evidence and a useful robustness check, rather than conclusive statistical evidence of a difference between the two groups**.

Comparison of the Gini Index Between Monarchies and Republics in Africa, the Americas, and Oceania

Africa

In Africa, the difference in the mean Gini Index between monarchies and republics is small, and Welch's test likewise does not indicate a statistically significant difference between the two groups. The effect size is also small. Therefore, based on the available data, **there is insufficient statistical evidence of a systematic difference in inequality between African monarchies and republics**.

The Americas

In the Americas, the comparison between Commonwealth realms and republics produces a similar result. Although the mean Gini Index differs slightly between the two groups, the difference is not statistically significant, and the effect size remains small. Therefore, **the available data provide insufficient evidence of a systematic difference in inequality between Commonwealth realms and republics in the Americas**.

Oceania

In Oceania, the comparison of Commonwealth realms and monarchies with available data against republics likewise reveals no statistically significant difference. In addition, the limited number of countries with available data in the groups being compared increases the uncertainty of the estimates. Accordingly, **the available data do not provide sufficient evidence to establish a systematic difference in inequality across forms of government in this region**.

Comparison of the Gini Index Between Monarchies and Republics in the Middle East

Descriptive Statistics

In this section, **3 Middle Eastern monarchies** and **10 Middle Eastern republics** for which Gini Index data were available from the World Bank were compared.

Form of Government	Number of Countries	Mean	Median	Sample SD
Monarchies	3	31.73	33.70	4.67
Republics	10	34.30	35.70	5.18

Descriptively, the mean Gini Index in Middle Eastern monarchies is approximately **2.6 points lower** than in the republics. Since lower values of this index indicate lower inequality, the observed difference points toward lower inequality among the monarchies. The median is also lower among the monarchies, while the dispersion of the data is relatively similar in the two groups.

Interpretation of the Results

1. Magnitude of the Descriptive Difference

The corrected data show that the mean Gini Index of Middle Eastern monarchies is not higher, but lower than that of the region's republics. The difference between the two groups is approximately **2.6 points**, indicating that the monarchies with available data have, on average, lower Gini Index values.

The medians point in the same direction: **33.70 for monarchies compared with 35.70 for republics**. However, only three monarchies in the region have data included in this analysis. Estimates for this group are therefore subject to considerable uncertainty and should be interpreted with caution.

2. Statistical Significance Test

Welch's test showed that the difference between the group means is not statistically significant:

t = −0.814, p = 0.466

Therefore, the null hypothesis of equal means cannot be rejected at the 0.05 significance level. In other words, **although the monarchies with available data have a lower mean Gini Index, the available statistical evidence is insufficient to conclude that the mean Gini Index differs significantly between the two groups**.

This result does not establish that the two groups are equal. It simply indicates that, given the available data, the null hypothesis of equal means cannot be rejected at the conventional significance level.

3. Effect Size

The standardized effect size in this comparison is approximately:

Hedges' g ≈ −0.50

The negative sign indicates that the mean Gini Index is lower among the monarchies and carries no independent normative meaning. The absolute magnitude of the effect, approximately **0.50, falls within the range conventionally regarded as a medium effect**.

Thus, although Welch's test does not identify a statistically significant difference, the observed standardized difference is not negligible. However, because only three monarchies are included in this comparison, **the effect-size estimate itself is subject to considerable uncertainty** and should be interpreted cautiously.

Conclusion

In the Middle East, monarchies with available data have both a lower mean and a lower median Gini Index than republics, while the observed effect size is in the medium range. Nevertheless, Welch's test does not find the difference between the group means to be statistically significant (p = 0.466), and the very limited number of monarchies with available data increases the uncertainty surrounding the estimate.

Therefore, unlike the much clearer findings for **GDP per capita and GDP per capita at purchasing power parity**, the Gini Index results for the Middle East do not provide sufficient statistical evidence of a systematic difference between the two forms of government. At the same time, the direction of the descriptive statistics is also inconsistent with the proposition that monarchies exhibit greater inequality: in the available data, both the mean and median Gini Index are lower among the monarchies.

This final point is particularly relevant to the broader analysis: **the Gini data do not indicate that the economic advantage observed for monarchies in the preceding indicators has been accompanied by greater income inequality.**

Security Report

So far, we have seen that, across the indicators examined for the most important goal of government—**welfare**—monarchies have, overall, performed better than republics. We now turn to the second goal of government, **security**, to examine which of the two "vaccines"—monarchy or republic—has in practice been associated with greater protection against violence, terrorism, and political instability.

Political Stability and Absence of Violence/Terrorism

The World Bank publishes a set of measures known as the **Worldwide Governance Indicators (WGI)**, which assess different dimensions of governance. One of these is **Political Stability and Absence of Violence/Terrorism**, which, for brevity, will hereafter be referred to as the **"Absence of Violence" indicator**.

This indicator does not measure actual incidents of violence or terrorism alone. Rather, it **captures perceptions of the likelihood of political instability and/or politically motivated violence, including terrorism**. It therefore provides a broader picture of political stability and security than a simple count of violent or terrorist incidents.

This study uses the indicator's **Percentile Rank**. The percentile rank represents each country's relative position compared with the other countries assessed and is expressed on a scale from **0 to 100**. Higher values indicate better performance in terms of political stability and the absence of violence and terrorism. In other words, the higher a country's percentile rank, the more favorable its relative position on this indicator.

The use of percentile ranks also involves an important interpretive consideration: **the numerical distance between two percentile ranks should not be interpreted directly as an absolute difference in the level of security**. These values primarily represent countries' relative positions within the global distribution.

The data used in this chapter are for **2023** (Appendix RDS-D01).

Global Comparison of Monarchies and Republics on the "Political Stability and Absence of Violence/Terrorism" Indicator

1. Descriptive Statistics

Group	Number of Countries (n)	Mean	Median	Sample SD
Monarchies	28	65.08	67.30	22.74
Republics	152	41.61	39.10	27.45

Interpretation of the Descriptive Statistics

The mean score on the **Political Stability and Absence of Violence/Terrorism** indicator is **65.08** for monarchies and **41.61** for republics.

Because a higher percentile rank on this indicator represents a more favorable position in terms of political stability and the absence of violence and terrorism, the descriptive statistics show that monarchies in this dataset have, on average, substantially higher scores than republics.

The medians point in the same direction:

- Monarchies: **67.30**
- Republics: **39.10**

Thus, the observed difference is not driven solely by a few countries with exceptionally high scores; it is also evident in the central part of the distributions.

Despite considerable dispersion within both groups, descriptive statistics alone are insufficient to determine whether the observed difference is statistically significant. Inferential tests were therefore conducted to examine the difference more rigorously.

2. Welch's t-test for the Difference in Means

The hypotheses were defined as follows:

Null hypothesis (H_0): The mean score on the Political Stability and Absence of Violence/Terrorism indicator is equal in monarchies and republics.

Alternative hypothesis (H_1): The mean score differs between the two groups.

Welch's t-test Results

Statistic	Value
Mean — Monarchies	65.08
Mean — Republics	41.61
Mean Difference	**23.47**
t-statistic	**4.849**
Welch-adjusted degrees of freedom	**42.90**
Two-tailed p-value	**0.0000166**
95% CI for the Mean Difference	**13.71 to 33.23**

Interpretation of Welch's t-test

The p-value (**p ≈ 0.0000166**) is far below the conventional significance level of 0.05. The null hypothesis is therefore rejected.

The result indicates that **the difference in the mean scores of monarchies and republics is statistically significant**. On average, monarchies score approximately **23.47 percentile-rank points** higher than republics.

The 95% confidence interval for the mean difference extends from approximately **13.71 to 33.23 points**. Because this interval does not include zero, it is consistent with the conclusion that the difference between the two group means is statistically significant.

It is important to note that the indicator is reported as a **percentile rank**. The difference of 23.47 points should therefore not be interpreted as meaning that monarchies are "23.47 percent safer" than republics. Rather, it represents the difference between the average positions of the two groups on the indicator's percentile-rank scale.

3. Mann–Whitney U Test

To examine the difference between the two groups from another perspective, the non-parametric **Mann–Whitney U test** was also conducted. This test is based on the ranks of the observations and does not depend on an assumption of normally distributed data.

The hypotheses were defined as follows:

Null hypothesis (H_0): There is no statistically significant difference in the rank distributions of the indicator between the two groups.

Alternative hypothesis (H_1): The rank distributions of the indicator differ between the two groups.

Mann–Whitney U Test Results

Statistic	Value
U statistic	**3,152**
Two-tailed p-value	**0.0000535**

Interpretation of the Mann–Whitney U Test

The p-value (**p ≈ 0.0000535**) is far below the 0.05 significance level. The null hypothesis is therefore rejected.

This test likewise indicates that **the indicator values differ significantly between the two groups in terms of their ranks**. Given that both the mean and median are substantially higher among monarchies than among republics, the observed difference is in the direction of higher indicator values for monarchies.

The consistency between the two tests is important. Welch's t-test identifies a statistically significant difference in the means, while the Mann–Whitney U test provides significant evidence of a difference in the rank distributions of the two groups. The result is therefore not dependent on the choice of a single statistical method.

4. Effect Size

To assess the standardized magnitude of the observed difference between the two groups, **Hedges' g** was calculated:

Hedges' g ≈ 0.87

By conventional benchmarks, this represents a **large effect size**. Thus, the observed difference between monarchies and republics is not only statistically significant but also substantial in standardized magnitude.

The positive sign indicates that monarchies have a higher mean score than republics. Since higher values on this indicator represent a more favorable position in terms of political stability and the absence of violence and terrorism, the direction of the effect size is fully consistent with the descriptive statistics.

Summary

In the 2023 data, the mean score on the Political Stability and Absence of Violence/Terrorism indicator is **65.08** for monarchies and **41.61** for republics. The corresponding medians are **67.30** and **39.10**, respectively. Both descriptive measures therefore point toward better performance among monarchies.

Both inferential tests also identify a statistically significant difference between the two groups:

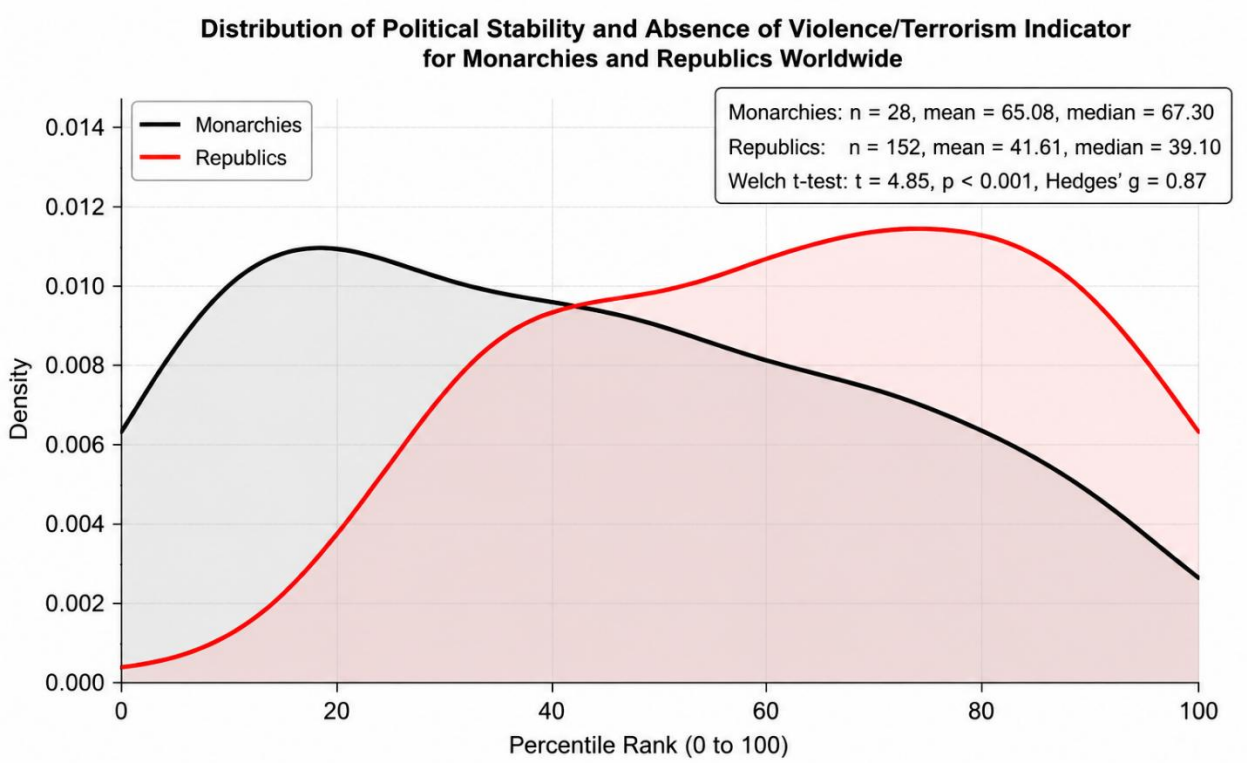
Test	Result
Welch's t-test	**p = 0.0000166 → Significant**
Mann–Whitney U	**p = 0.0000535 → Significant**

In addition, **Hedges' g ≈ 0.87** indicates a large effect size, while the 95% confidence interval for the mean difference, **13.71 to 33.23 percentile-rank points**, lies entirely above zero.

Conclusion

The 2023 data show that monarchies worldwide performed better, on average, than republics on the Political Stability and Absence of Violence/Terrorism indicator. The higher mean and median among monarchies, the very strong statistical significance found by both tests, and the large effect size all point in the same direction. Thus, the observed difference is not only substantial descriptively but is also supported by strong statistical evidence of a difference between the two groups.

These findings, however, do not by themselves establish that regime type is the direct cause of the observed difference. A variety of factors—including levels of development, geographical conditions, social structures, natural resources, and countries' geopolitical environments—may also influence political stability. Moreover, some factors, such as wars or foreign interventions, may themselves be partly influenced by the quality of governance and by governments' domestic and foreign policies. Consequently, as with any observational study, fully disentangling cause and effect remains inherently difficult.



Comparison of Political Stability and Absence of Violence/Terrorism Indicator

The distribution of the **Political Stability and Absence of Violence/Terrorism** indicator across countries worldwide shows that the distribution for monarchies is shifted toward higher values relative to that of republics. Since higher values on this indicator represent greater political

stability and a lower likelihood of violence and instability, this shift indicates that monarchies in this dataset exhibit, on average, higher levels of political stability.

The mean indicator score is **65.08** for monarchies and **41.61** for republics. The corresponding medians are **67.30** and **39.10**, respectively. Thus, the observed difference is not driven solely by a few countries with exceptionally high scores; it is also evident in the central part of the distributions.

The partial overlap between the two curves reflects considerable variation among countries within both groups. Nevertheless, the greater concentration of the monarchy distribution at higher indicator values reveals a distinct overall pattern between the two groups. The statistical tests presented below assess whether this observed difference is also statistically significant.

Comparison of the "Political Stability and Absence of Violence/Terrorism" Indicator between Constitutional and Non-Constitutional Monarchies Worldwide

Descriptive Statistics

Indicator	Constitutional Monarchies	Semi-Constitutional and Absolute Monarchies
Number of countries	15	13
Mean	63.36	67.05
Median	62.10	70.10
Sample SD	19.34	26.82

Statistical Significance Test (Welch's t-test)

- **t = −0.412**
- **p = 0.684**

Therefore, **the difference between the two group means is not statistically significant**.

Effect Size

- **Hedges' g = −0.16**

This value indicates a **very small effect size**.

Magnitude of the Descriptive Difference

The mean score on the Political Stability and Absence of Violence/Terrorism indicator is **63.36** for constitutional monarchies and **67.05** for semi-constitutional and absolute monarchies. Thus,

descriptively, the latter group scores approximately **3.69 percentile-rank points** higher. The corresponding medians are **62.10** for constitutional monarchies and **70.10** for semi-constitutional and absolute monarchies.

Nevertheless, there is considerable dispersion within both groups, particularly among semi-constitutional and absolute monarchies, where the sample standard deviation is **26.82**, compared with **19.34** among constitutional monarchies.

Statistical Significance

To assess whether the difference between the two group means extends beyond random sampling variation, Welch's t-test was conducted. The test indicates that the difference is **not statistically significant** (**t = −0.412, p = 0.684**).

Effect Size

The estimated **Hedges' g is −0.16**, corresponding to a **very small effect size**. The negative sign simply reflects the direction of the comparison: the mean score for constitutional monarchies is slightly lower than that for semi-constitutional and absolute monarchies. The magnitude of the effect indicates that the practical difference between the two types of monarchy is small.

Conclusion

Based on the 2023 data, **there is no statistical evidence that constitutional monarchies and semi-constitutional and absolute monarchies differ in their performance on the Political Stability and Absence of Violence/Terrorism indicator.** The difference between the group means is small, statistically non-significant, and accompanied by a very small effect size. Thus, for this indicator, the distinction between these two types of monarchy is not associated with a substantial difference in security-related performance.

Comparison of the "Political Stability and Absence of Violence/Terrorism" Indicator between Commonwealth Realms and Republics Worldwide

Descriptive Statistics

Based on the classification used in the current dataset, the Commonwealth group includes **14 countries**, while the global republic group includes **152 countries**.

Indicator	Commonwealth Realms	Republics Worldwide
Number of countries	14	152
Mean	75.84	41.61
Median	80.60	39.10

Indicator	Commonwealth Realms	Republics Worldwide
Sample SD	18.17	27.46

The mean score on the Political Stability and Absence of Violence/Terrorism indicator is **75.84** for the Commonwealth realms and **41.61** for republics worldwide. Thus, the Commonwealth realms have a mean score approximately **34.23 percentile-rank points** higher than that of the republics.

The median is also substantially higher among the Commonwealth realms: **80.60**, compared with **39.10** for republics. The fact that both the mean and median point in the same direction indicates that the descriptive advantage of the Commonwealth realms is not driven solely by a few exceptional countries, but is also evident in the central part of the distribution.

Scores are also more dispersed among republics. The sample standard deviation is **27.46** for republics, compared with **18.17** for the Commonwealth realms. Thus, in addition to having a lower mean, the republic group displays greater variation in performance on this indicator.

Statistical Significance

To assess the statistical significance of the difference between the two group means, **Welch's t-test** was used. The results are:

- **t = 6.408**
- **Approximate df = 18.97**
- **p = 0.0000038**
- **p < 0.001**

The difference between the Commonwealth realms and republics worldwide is therefore **highly statistically significant**.

The 95% confidence interval for the difference between the group means is:

23.05 to 45.41 percentile-rank points

Because the entire confidence interval lies above zero, the data provide strong statistical evidence that the mean score of the Commonwealth realms is higher than that of republics worldwide.

It is important to note that these values are **percentile ranks**. Thus, the 34.23-point difference should not be interpreted as meaning that the Commonwealth realms are "34.23 percent more stable" or "34.23 percent safer." Rather, it represents the difference between the average positions of the two groups on the indicator's percentile-rank scale.

Effect Size

The estimated effect size is:

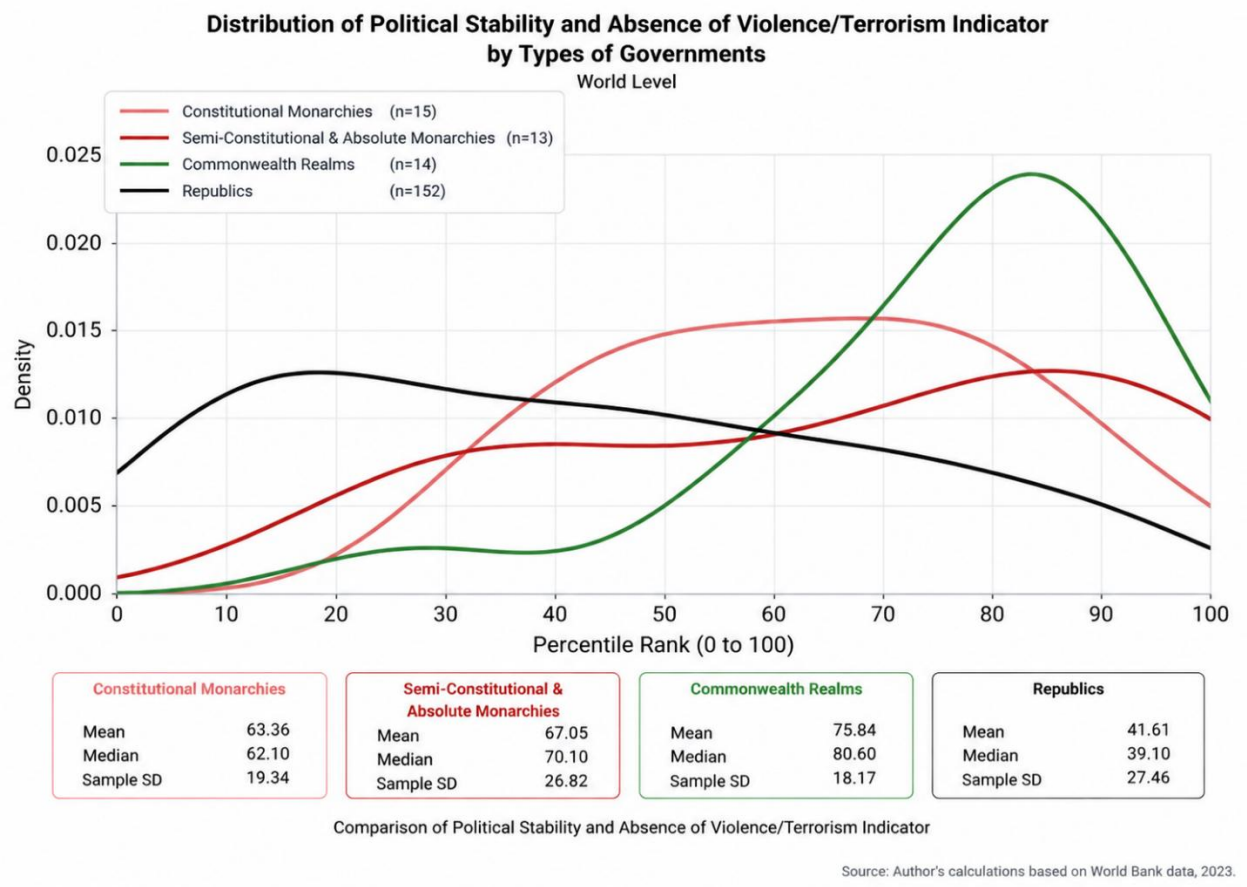
Hedges' g = 1.27

This represents a **large effect size**. Thus, the observed difference is not only statistically significant but also substantial in standardized terms. In other words, the separation between the two groups is large relative to the variability observed within them.

Conclusion

The 2023 data show that the Commonwealth realms, on average, performed substantially better than republics worldwide on the Political Stability and Absence of Violence/Terrorism indicator. Their mean score was approximately 34.23 percentile-rank points higher; the difference was highly statistically significant, and the standardized effect size was large.

In this comparison, membership in the Commonwealth-realm group is therefore **associated with** higher levels of political stability and absence of violence and terrorism. However, because this is an observational comparison, the finding **does not by itself establish a causal relationship** between constitutional structure and political stability. Other factors—including levels of development, geography, social structure, and the geopolitical environment—may also contribute to the observed difference.



The figure shows the distribution of the **Political Stability and Absence of Violence/Terrorism** indicator across four groups of political systems: constitutional monarchies, semi-constitutional and absolute monarchies, Commonwealth realms, and republics. Unlike bar charts, which display only the mean of each group, the kernel density estimation (KDE) plot illustrates how country scores within each group are distributed across the full range of the indicator.

All three monarchy-related groups—constitutional monarchies, semi-constitutional and absolute monarchies, and Commonwealth realms—are shifted toward higher scores relative to republics. By contrast, the distribution for republics is more concentrated in the low-to-middle range of the indicator and also exhibits greater dispersion.

Among the three monarchy-related groups, the Commonwealth realms show the strongest concentration at the upper end of the scale, indicating relatively high levels of political stability among most countries in this group. Constitutional monarchies and semi-constitutional and absolute monarchies display comparatively similar distributions, with substantial overlap between their curves. This pattern is consistent with the statistical results reported in this chapter, which indicate that the difference between these two groups is not statistically significant.

Overall, the figure suggests that the observed difference between monarchies and republics is not driven solely by a small number of exceptionally high- or low-performing countries. Rather, the overall distribution of the data also shows a greater concentration of monarchy-related groups at higher levels of political stability and absence of violence. Nevertheless, the overlap among the curves indicates that political system is only one of the factors associated with variation in this indicator; historical, economic, social, and geopolitical factors may also contribute to differences in country performance.

Comparison of the “Political Stability and Absence of Violence/Terrorism” Indicator between Monarchies and Republics across Continents and in the Middle East

Comparison of the “Political Stability and Absence of Violence/Terrorism” Indicator between Monarchies and Republics in Asia

Indicator: Political Stability and Absence of Violence/Terrorism
Source: World Bank data

Higher values on this indicator represent a more favorable position in terms of political stability and a lower likelihood of politically motivated violence and terrorism.

The comparison includes **13 Asian monarchies** and **33 Asian republics**. To assess the difference between the two groups, two statistical tests were used: **Welch’s t-test** to compare group means and the **Mann–Whitney U test** to compare the rank distributions of the observations.

1. Descriptive Statistics

Group	n	Mean	Median	Sample SD
Asian Monarchies	13	60.38	58.80	21.66
Asian Republics	33	30.72	26.10	24.01

Interpretation of Descriptive Statistics

The mean score on the Political Stability and Absence of Violence/Terrorism indicator is **60.38** for Asian monarchies and **30.72** for Asian republics.

Because higher values indicate a more favorable position in terms of political stability and absence of violence and terrorism, the descriptive statistics show that Asian monarchies, on average, have substantially higher scores than Asian republics.

The medians point in the same direction:

- Asian monarchies: **58.80**
- Asian republics: **26.10**

Thus, the observed difference is not driven solely by a few countries with exceptionally high scores; it is also evident in the central part of the distributions. The substantial difference in both means and medians provides descriptive evidence of a broad separation between the two groups.

2. Welch's t-test for the Difference in Means

The hypotheses were defined as follows:

Null hypothesis (H_0): The mean Political Stability and Absence of Violence/Terrorism score is equal in the two groups.

Alternative hypothesis (H_1): The mean score differs between the two groups.

Welch's t-test Results

Statistical Measure	Value
Mean — Asian Monarchies	60.38
Mean — Asian Republics	30.72
Mean Difference	**29.66**
t-statistic	**4.21**
Approximate df	**27.5**
Two-sided p-value	**0.00024**

Interpretation of Welch's t-test

The probability value (**p = 0.00024**) is far below the conventional 5% significance level. The null hypothesis is therefore rejected.

The difference in mean Political Stability and Absence of Violence/Terrorism scores between Asian monarchies and republics is statistically significant.

The mean score of Asian monarchies is approximately **29.66 percentile-rank points** higher than that of Asian republics. As with the global comparison, this difference should not be interpreted as meaning that monarchies are "29.66 percent more stable." Rather, it represents the difference between the average positions of the two groups on the indicator's percentile-rank scale.

3. Mann–Whitney U Test

To examine the difference between the two groups from a rank-based perspective without assuming normally distributed data, the non-parametric **Mann–Whitney U test** was also performed.

The hypotheses were:

Null hypothesis (H_0): The rank distributions of the indicator do not differ significantly between the two groups.

Alternative hypothesis (H_1): The rank distributions differ between the two groups.

Mann–Whitney U Test Results

Statistical Measure	Value
U-statistic	**361**
Two-sided p-value	**0.00052**

Interpretation of the Mann–Whitney U Test

The probability value (**p = 0.00052**) is well below the 5% significance level. The null hypothesis is therefore rejected.

The Mann–Whitney U test also provides statistically significant evidence that the indicator values differ between Asian monarchies and republics in their rank distributions.

The agreement between Welch's t-test and the Mann–Whitney U test strengthens the result: the observed difference is evident both in the comparison of group means and in the rank-based distribution of country scores. Thus, the finding is not dependent on a single statistical method.

Conclusion

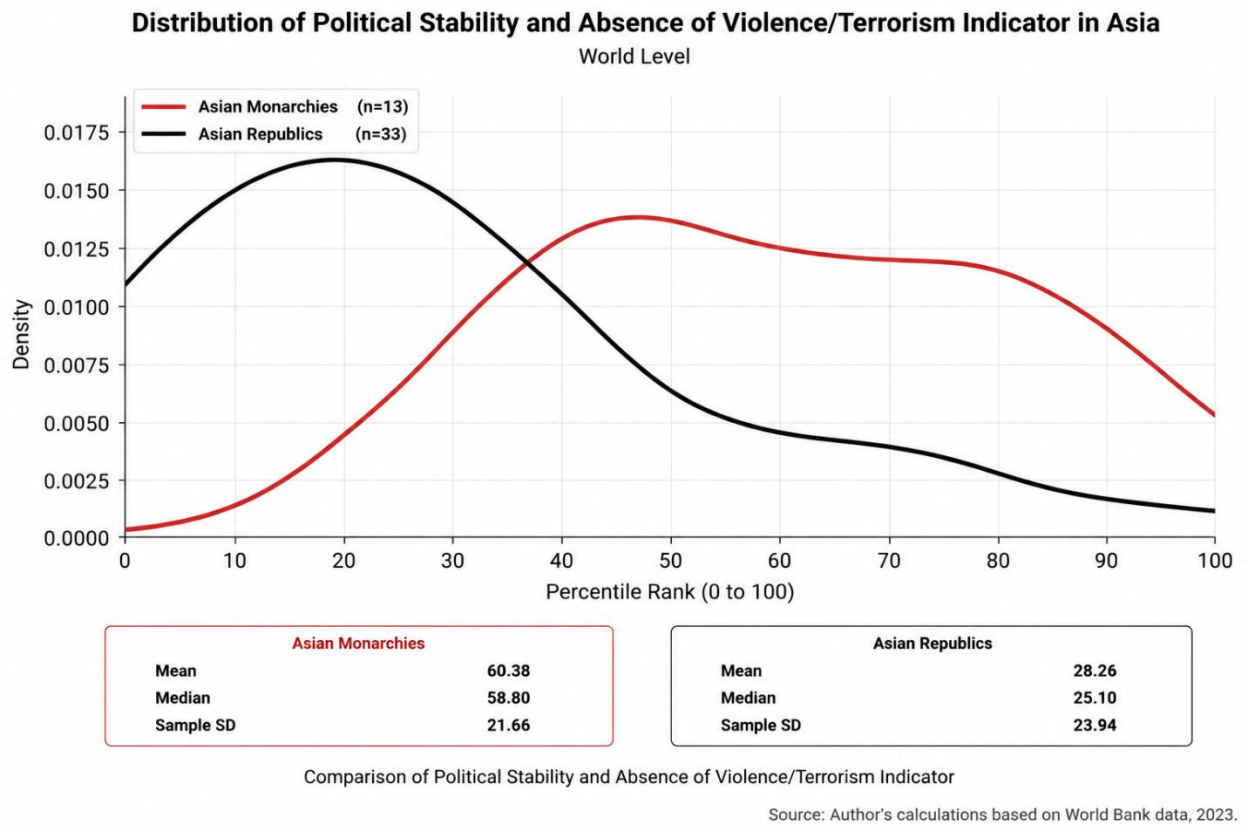
Among the Asian countries included in the dataset, monarchies have a mean Political Stability and Absence of Violence/Terrorism score of **60.38**, compared with **30.72** for republics. The corresponding medians are **58.80** and **26.10**, respectively.

Both statistical tests indicate a highly significant difference:

Test	Result
Welch's t-test	**p = 0.00024 → statistically significant**
Mann–Whitney U	**p = 0.00052 → statistically significant**

Among the Asian countries for which data are available, monarchies have substantially higher average scores on the Political Stability and Absence of Violence/Terrorism indicator than republics, and the difference is statistically significant under both Welch's t-test and the Mann–Whitney U test. The consistency of the means, medians, and two inferential tests provides strong evidence of a systematic difference between the two groups in this dataset.

As elsewhere in this study, however, this observational comparison should not by itself be interpreted as establishing a causal effect of political system on political stability. Other historical, economic, social, geographic, and geopolitical differences among Asian countries may also contribute to the observed pattern.



Overall, the Asian data are consistent with the pattern observed in the global comparison: **monarchies in the region have, on average, shown greater political stability and a lower perceived risk of politically motivated violence, terrorism, and instability than republics**. However, this finding is based on an observational comparison and **does not by itself establish that political system is the direct cause of the observed difference**. Historical differences, levels of development, social structures, wars, foreign interventions, and countries' geopolitical circumstances may also have contributed to this pattern.

Comparison of the "Political Stability and Absence of Violence/Terrorism" Indicator between Monarchies and Republics in Europe

Indicator: Political Stability and Absence of Violence/Terrorism
Source: World Bank data

Higher values on this indicator represent a more favorable position in terms of political stability and a lower likelihood of politically motivated violence and terrorism.

Two statistical tests were used:

1. **Welch's t-test** to compare the group means.
2. **Mann–Whitney U test** to compare the rank distributions of the observations.

1. *Descriptive Statistics*

Group	n	Mean	Median	Sample SD
European Monarchies	11	77.36	76.90	15.73
European Republics	34	59.39	64.70	21.66

Interpretation of Descriptive Statistics

The mean Political Stability and Absence of Violence/Terrorism score is **77.36** for European monarchies and **59.39** for European republics.

Because higher values indicate a more favorable position in terms of political stability and absence of violence and terrorism, the descriptive statistics show that European monarchies have, on average, substantially higher scores than European republics.

The medians point in the same direction:

- European monarchies: **76.90**
- European republics: **64.70**

Thus, the observed difference is not driven solely by a few countries with exceptionally high scores; it is also evident in the central part of the distributions.

2. Welch's t-test for the Difference in Means

The hypotheses were defined as follows:

Null hypothesis (H_0): The mean Political Stability and Absence of Violence/Terrorism score is equal in European monarchies and republics.

Alternative hypothesis (H_1): The mean score differs between the two groups.

Welch's t-test Results

Statistical Measure	Value
Mean — European Monarchies	77.36
Mean — European Republics	59.39
Mean Difference	**17.98**
t-statistic	**2.984**
Approximate df	**23.37**
Two-sided p-value	**0.0066**

Interpretation of Welch's t-test

The probability value (**p = 0.0066**) is below the conventional 5% significance level. The null hypothesis is therefore rejected.

The results indicate that **the difference in mean Political Stability and Absence of Violence/Terrorism scores between European monarchies and republics is statistically significant**.

In this dataset, European monarchies have a mean score approximately **17.98 percentile-rank points** higher than European republics. Because the indicator is expressed as a percentile rank, this difference should not be interpreted as meaning that monarchies are "17.98 percent more stable." Rather, it represents the difference between the average positions of the two groups on the indicator's percentile-rank scale.

3. Mann–Whitney U Test

To examine the difference between the two groups from a rank-based perspective without assuming normally distributed data, the **Mann–Whitney U test** was also performed.

The hypotheses were:

Null hypothesis (H_0): The rank distributions of the indicator do not differ significantly between the two groups.

Alternative hypothesis (H_1): The rank distributions differ between the two groups.

Mann–Whitney U Test Results

Statistical Measure	Value
U-statistic	**276.5**
Two-sided p-value	**0.0187**

Interpretation of the Mann–Whitney U Test

The probability value (**p = 0.0187**) is below the 5% significance level. The null hypothesis is therefore rejected.

The Mann–Whitney U test also provides statistically significant evidence that the indicator values differ between European monarchies and republics in their rank distributions.

The agreement between the two tests strengthens the result. Welch's t-test identifies a statistically significant difference in the group means, while the Mann–Whitney U test provides

significant evidence of a difference in the rank distributions. Thus, the finding is not dependent on a single statistical method.

Conclusion

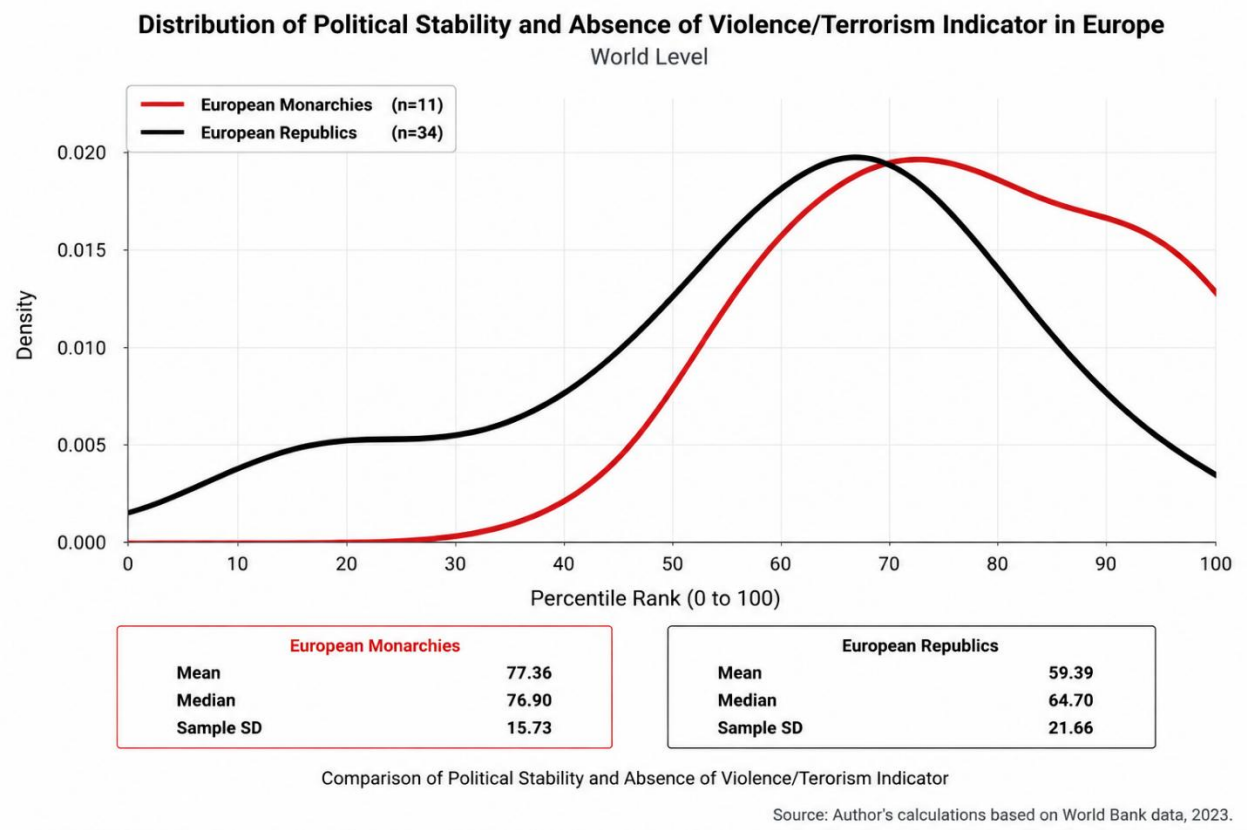
Among the European countries included in the dataset, monarchies have a mean Political Stability and Absence of Violence/Terrorism score of **77.36**, compared with **59.39** for republics. The corresponding medians are **76.90** and **64.70**, respectively.

Both statistical tests indicate a significant difference:

Test	Result
Welch's t-test	**p = 0.0066 → statistically significant**
Mann–Whitney U	**p = 0.0187 → statistically significant**

Among the European countries for which data are available, monarchies have, on average, substantially higher scores on the Political Stability and Absence of Violence/Terrorism indicator than republics, and the difference is statistically significant under both Welch's t-test and the Mann–Whitney U test. The consistency of the means, medians, and two inferential tests provides evidence of a systematic difference between the two groups in this dataset.

As with the other regional comparisons, however, this is an observational finding and **does not by itself establish that political system is the direct cause of the difference**. Historical trajectories, levels of development, social and institutional structures, geopolitical conditions, wars, and other country-specific factors may also contribute to the observed pattern.



The kernel density estimate (KDE) plot reflects the same overall pattern. The distribution for European monarchies is concentrated primarily at the higher end of the index, with much of the density clustered around scores between 70 and 90. By contrast, the distribution for European republics is more dispersed and, alongside countries with very high levels of stability, also includes a number of Eastern European and post-communist republics with relatively lower scores. This greater heterogeneity helps explain the higher dispersion observed among European republics compared with monarchies.

Overall, the European data, consistent with the global and Asian results, indicate that monarchies have, on average, higher scores for political stability and absence of violence and terrorism. As in the other sections of this study, however, this finding is based on an observational comparison and **does not, by itself, establish a causal relationship between the form of government and political stability**. Factors such as historical legacies, the experience of communist rule, levels of development, institutional structures, and geopolitical conditions may also contribute to the observed pattern.

Robustness Test for Europe: Excluding Republics with a History of Communist Rule

Indicator	European Monarchies	Republics with No History of Communist Rule
Number of countries	11	13

Indicator	European Monarchies	Republics with No History of Communist Rule
Mean Political Stability and Absence of Violence score	77.36	72.85
Median	76.90	71.10
Sample standard deviation	15.73	13.45

The mean Political Stability and Absence of Violence score is therefore 4.52 points, or approximately 6.2%, higher among the monarchies. The median is also about 5.8 points higher among the monarchies.

Welch's t-test

Using a two-tailed Welch's t-test:

t = 0.749, df = 19.87, p = 0.463

Thus, the difference between the group means is not statistically significant at the 5% level.

The effect size was:

Hedges' g = 0.300

indicating a **small effect in favor of the monarchies**.

Interpretation of the Robustness Test

As in the preceding comparisons, the potential influence of Europe's communist legacy can also be examined. After excluding republics that had experienced communist rule, the 11 European monarchies were compared with 13 republics that had never been under communist rule.

The results show that the monarchies retain their descriptive advantage after this exclusion, although the gap between the two groups narrows. The mean Political Stability and Absence of Violence score is 77.36 for the monarchies and 72.85 for the republics with no history of communist rule. The corresponding medians are 76.90 and 71.10. Thus, whether measured by the mean or the median, the monarchies continue to record higher scores.

However, **Welch's t-test indicates that the difference between the two group means is not statistically significant** ($t = 0.749$, $p = 0.463$). Hedges' g is 0.300, indicating a small standardized effect in favor of the monarchies.

Thus, excluding republics with a history of communist rule does not reverse the direction of the difference, but it substantially reduces its magnitude. European monarchies still have

higher mean and median Political Stability and Absence of Violence scores, but this restricted comparison does not provide sufficient statistical evidence to reject the null hypothesis of equal group means.

This restricted comparison, however, should not replace the primary European comparison. From a methodological standpoint, excluding European republics with a history of communist rule from the overall analysis is not warranted, because these countries constitute a real part of the historical experience of republican government in Europe. Removing them could introduce a form of **selection bias by excluding unfavorable cases from the historical record of republics**.

Within the analytical framework of this book, the tendency of some republics to develop into communist or one-party regimes, together with the institutional legacies left by those regimes, is not treated simply as an external factor wholly independent of the form of government. Rather, it is considered part of the historical trajectory—and potentially part of the consequences—of the republican experience itself. **Republics with a history of communist rule should therefore remain in the primary comparison, while their exclusion is used here only as a robustness test.**

Comparison of the Political Stability and Absence of Violence/Terrorism Indicator in Africa, the Americas, and Oceania

Africa

Group	Number of Countries	Mean	Median	Sample SD
Monarchies	3	32.07	30.80	3.01
Republics	50	28.41	22.05	22.26

In Africa, the mean Political Stability and Absence of Violence/Terrorism score of the continent's three monarchies is slightly higher than that of its republics. However, the difference between the two groups is smaller than in the other continents, and the very limited number of African monarchies requires particular caution in interpreting the results.

The dispersion among republics is also substantially greater than among monarchies, indicating that African republics are considerably more heterogeneous in terms of political stability.

The Americas

Group	Number of Countries	Mean	Median	Sample SD
Commonwealth Realms	9	78.21	80.60	10.35
Republics	26	46.41	45.00	23.03

In the Americas, the Commonwealth realms score, on average, more than **31 percentile-rank points** higher than the republics. In addition to their higher mean, the Commonwealth realms

display less dispersion, with most of them concentrated among countries with relatively high levels of political stability, whereas the republics exhibit a much wider range of performance.

Although the difference between the Commonwealth realms and republics in the Americas is striking, it should not be attributed directly or entirely to the form of government. The composition of the sample in this region is distinctive: a substantial proportion of the Commonwealth realms are small, sparsely populated Caribbean island states. Owing to their particular geographical circumstances, their distance from many of the principal centers of regional and international conflict, and their more limited exposure to certain security threats, these countries may naturally obtain higher scores on the Political Stability and Absence of Violence/Terrorism indicator.

Geographical characteristics and sample composition should therefore be regarded as potential confounding factors when interpreting these results, and the observed difference should not be attributed entirely to the form of government.

Oceania

Group	Number of Countries	Mean	Median	Sample SD
Monarchies	1	90.00	90.00	—
Commonwealth Realms	5	71.56	79.60	28.69
Republics	8	83.06	89.60	13.24

In Oceania, there is only one independent monarchy, **Tonga**, and consequently no meaningful statistical inference can be made about the continent's independent monarchies as a group. Moreover, unlike the pattern observed in the other continents, **the mean score of Oceania's republics is higher than that of its Commonwealth realms**. This difference is largely influenced by the very low score of Papua New Guinea within the Commonwealth group, which substantially lowers the group mean.

Given the small number of countries involved, the results for Oceania should be interpreted with particular caution and should not be generalized to the same extent as the global, Asian, or European findings.

Oceania also exhibits a phenomenon similar to that observed in the Americas, although in the opposite direction. A substantial proportion of the republics in this region are small and relatively remote island states whose particular geographical circumstances leave them less exposed to many of the security threats commonly encountered in other parts of the world. The higher average score of Oceania's republics therefore cannot, by itself, be attributed to their form of government.

Here too, **sample composition and geographical characteristics, alongside other political, economic, and social factors, are likely to contribute to the observed pattern.**

Comparison of the Political Stability and Absence of Violence/Terrorism Indicator in the Middle East

Group	Number of Countries	Mean	Median	Sample SD
Monarchies	7	55.53	58.80	19.69
Republics	10	12.75	8.80	17.18

The Middle East exhibits one of the clearest differences between the two forms of government. The mean score for the region's monarchies is **55.53**, compared with only **12.75** for its republics, corresponding to a mean difference of approximately **42.78 percentile-rank points**. The medians reinforce the same pattern: **58.80** for the monarchies compared with **8.80** for the republics. This substantial difference indicates that the higher scores of the monarchies are not merely the result of one or two exceptionally high-scoring countries, but are also evident in the central tendency of the two distributions.

Statistical Significance

Welch's two-tailed t-test indicates that the difference between the two group means is statistically significant:

- **t = 4.643**
- **df ≈ 11.85**
- **p ≈ 0.00059**
- **95% confidence interval for the mean difference: 22.68 to 62.88 percentile-rank points**
- **Hedges' g ≈ 2.23**

The p-value is well below the conventional 5% significance threshold, and the entire 95% confidence interval lies above zero. Thus, the null hypothesis of equal group means is rejected.

Effect Size

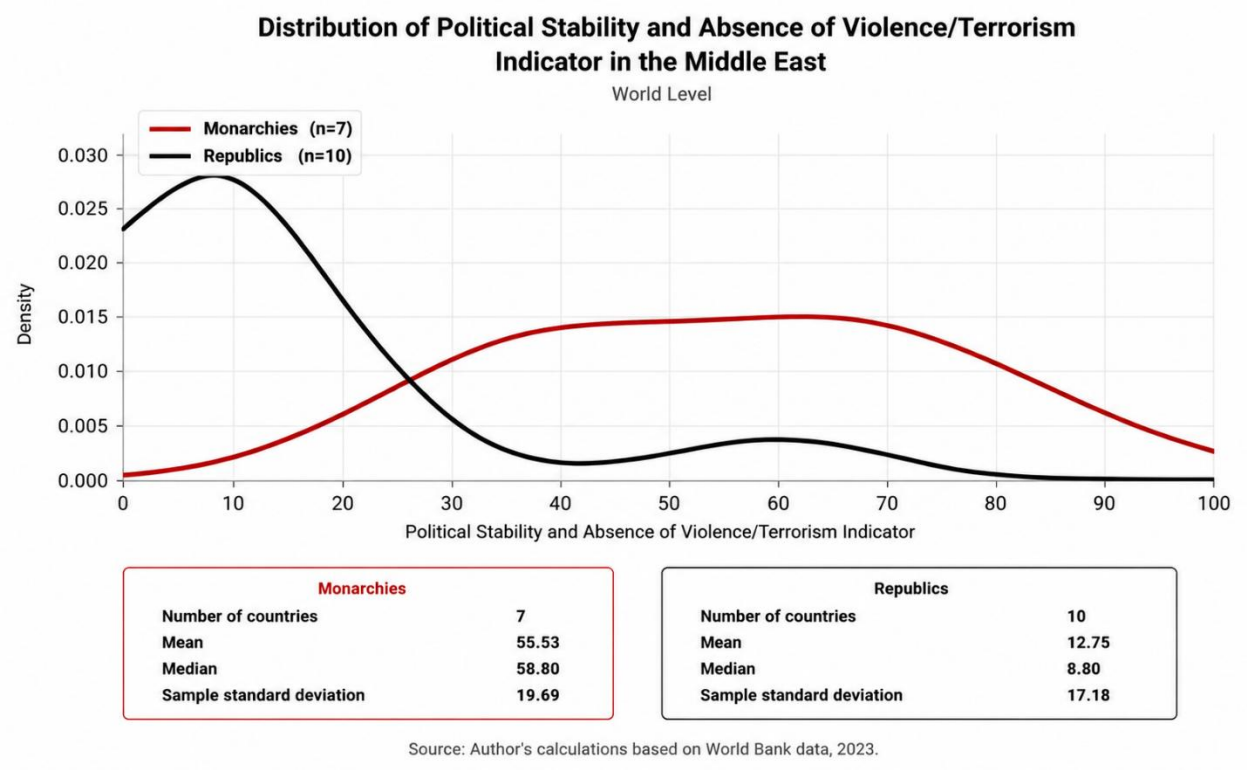
The estimated **Hedges' g of approximately 2.23 represents an exceptionally large standardized effect**. This indicates that the difference between the two groups is not only statistically significant but also very large in magnitude.

As elsewhere in this analysis, however, the difference of 42.78 points should be interpreted as a difference in **percentile-rank scores**, not as evidence that monarchies are "42.78 percent more stable" than republics.

Conclusion

Among the Middle Eastern countries included in the analysis, monarchies recorded substantially higher Political Stability and Absence of Violence/Terrorism scores than republics. The difference is evident in both the mean and the median, is statistically significant under Welch's t-test, and is accompanied by an exceptionally large effect size.

The magnitude of the difference makes the Middle East particularly noteworthy within the regional comparisons. Nevertheless, these results remain observational and **do not, by themselves, establish that the form of government is the direct cause of the observed difference**. Wars, foreign intervention, geopolitical rivalry, economic structures, institutional development, and other historical and social factors may also contribute to the exceptionally large gap observed between the two groups.



Monarchies	
Number of countries	7
Mean	55.53
Median	58.80
Sample standard deviation	19.69

Republics	
Number of countries	10
Mean	12.75
Median	8.80
Sample standard deviation	17.18

Source: Author's calculations based on World Bank data, 2023.

The distribution curve for the republics is concentrated predominantly in the lower range of the indicator, whereas the distribution for the monarchies is clearly shifted toward higher scores.

However, this finding should not be interpreted as proof of a simple causal relationship between monarchy and political stability. The Middle East is a region in which civil wars, foreign interventions, ethnic and sectarian conflicts, regional rivalries, state institutional collapse, and differences in access to economic resources simultaneously affect national stability. Many of the

region’s republics have experienced war, revolution, sanctions, occupation, or political breakdown in recent decades, but disentangling the effect of regime type from these factors is difficult within an observational comparison.

Accordingly, the present finding indicates a **strong empirical association**, rather than definitive proof that regime type alone caused the observed difference.

The Record on Order

We now turn to the third objective of government: maintaining **order**. One of the most important instruments available to governments for establishing and preserving order is the enactment and enforcement of laws and regulations. The mere existence of laws, however, does not in itself indicate that the resulting political and economic order is desirable. All governments establish regulations, but the orientation and quality of those regulations can vary considerably.

In some countries, laws and regulations are designed to facilitate economic activity, investment, competition, and private-sector development. In others, poorly designed regulations may contribute to greater monopoly power, rent-seeking, economic uncertainty, and preferential treatment for government-affiliated firms or groups with political influence. The central issue, therefore, is not simply whether a legal and regulatory order exists, but how effectively that order structures economic life and enables citizens and firms to engage in free and productive activity.

Nor does the concept of order, as understood by philosophers as one of the objectives of government, refer merely to preserving those in power or preventing disruption of the status quo. An order that can legitimately be regarded as an objective of government is one that makes social and economic relations predictable, clarifies rights and obligations, and provides the conditions for cooperation and constructive activity among citizens.

Regulatory Quality

As part of its Worldwide Governance Indicators, the World Bank publishes an indicator entitled **"Regulatory Quality"**. This indicator reflects perceptions of the government's ability to formulate and implement sound policies and regulations that permit and promote private-sector development. Like the other Worldwide Governance Indicators, this measure draws on a combination of data from surveys of citizens and firms and assessments by experts. It should therefore not be interpreted as a direct assessment of the text of laws themselves or of the benefits received by each individual citizen.

For each country, the World Bank reports both an estimated governance score and a **percentile rank**. The interpretation of percentile ranks was explained in the preceding section. In this section, the percentile ranks of monarchies and republics on the Regulatory Quality indicator are compared.

The data used in this analysis are for **2023** (Appendix RDS-E01).

Comparison of the “Regulatory Quality” Indicator Between Monarchies and Republics Worldwide

1. Descriptive Statistics

Group	Number of Countries (n)	Mean	Median	Sample SD
Monarchies	28	72.61	80.90	23.10
Republics	152	41.79	39.85	27.12

Interpretation of Descriptive Statistics

The mean Regulatory Quality score is **72.61** for monarchies and **41.79** for republics.

Because a higher value on this indicator represents better regulatory quality, the descriptive statistics show that, in this dataset, monarchies have substantially higher average scores than republics.

The medians point in the same direction:

- Monarchies: **80.90**
- Republics: **39.85**

Thus, the observed difference is not driven solely by a few countries with exceptionally high scores; it is also evident in the central tendency of the two distributions.

Despite the relatively substantial dispersion within both groups, descriptive statistics alone are insufficient to determine whether the observed difference is statistically significant. Inferential tests were therefore conducted.

2. Welch’s t-test for the Difference in Means

The hypotheses were defined as follows:

Null hypothesis (H_0): The mean Regulatory Quality score is equal in monarchies and republics.

Alternative hypothesis (H_1): The mean Regulatory Quality score differs between the two groups.

Results of Welch’s t-test

Statistic	Value
Mean — Monarchies	72.61
Mean — Republics	41.79
Difference in means	**30.82**

Statistic	Value
t-statistic	**6.304**
Approximate degrees of freedom	**41.96**
Two-tailed p-value	**0.000000145**

Interpretation of Welch's t-test

The p-value (**p ≈ 0.000000145**) is far below the conventional 5 percent significance level. The null hypothesis is therefore rejected.

The result indicates that **the difference in mean Regulatory Quality scores between monarchies and republics is highly statistically significant**.

In this dataset, monarchies have, on average, substantially higher Regulatory Quality scores than republics.

Because the indicator is reported as a **percentile rank**, the 30.82-point difference should not be interpreted as meaning that monarchies have "30.82 percent better regulations." Rather, it represents the difference between the average percentile-rank positions of the two groups on the Regulatory Quality indicator.

3. Mann–Whitney U Test

To examine the difference between the two groups from another perspective, the nonparametric **Mann–Whitney U test** was also conducted. This test is based on the ranking of observations and does not depend on an assumption of normally distributed data.

The hypotheses were:

Null hypothesis (H_0): The ranked distributions of Regulatory Quality scores do not differ significantly between the two groups.

Alternative hypothesis (H_1): The ranked distributions of Regulatory Quality scores differ between the two groups.

Results of the Mann–Whitney U Test

Statistic	Value
U statistic	**3,412**
Two-tailed p-value	**0.000000407**

Interpretation of the Mann–Whitney U Test

The p-value (**p ≈ 0.000000407**) is far below the 5 percent significance level. The null hypothesis is therefore rejected.

This test likewise indicates that **Regulatory Quality scores differ significantly between monarchies and republics in terms of their ranked distributions**. Given that both the mean and median are markedly higher among monarchies, the direction of the observed difference is toward higher Regulatory Quality scores in the monarchy group.

The agreement between the two tests is important. Welch's t-test identifies a statistically significant difference in means, while the Mann–Whitney U test provides significant evidence of a difference in the ranked distributions. Thus, the finding is not dependent on the choice of a single statistical method.

Summary

For the 2023 Regulatory Quality data:

Group	Mean Score
Monarchies	**72.61**
Republics	**41.79**

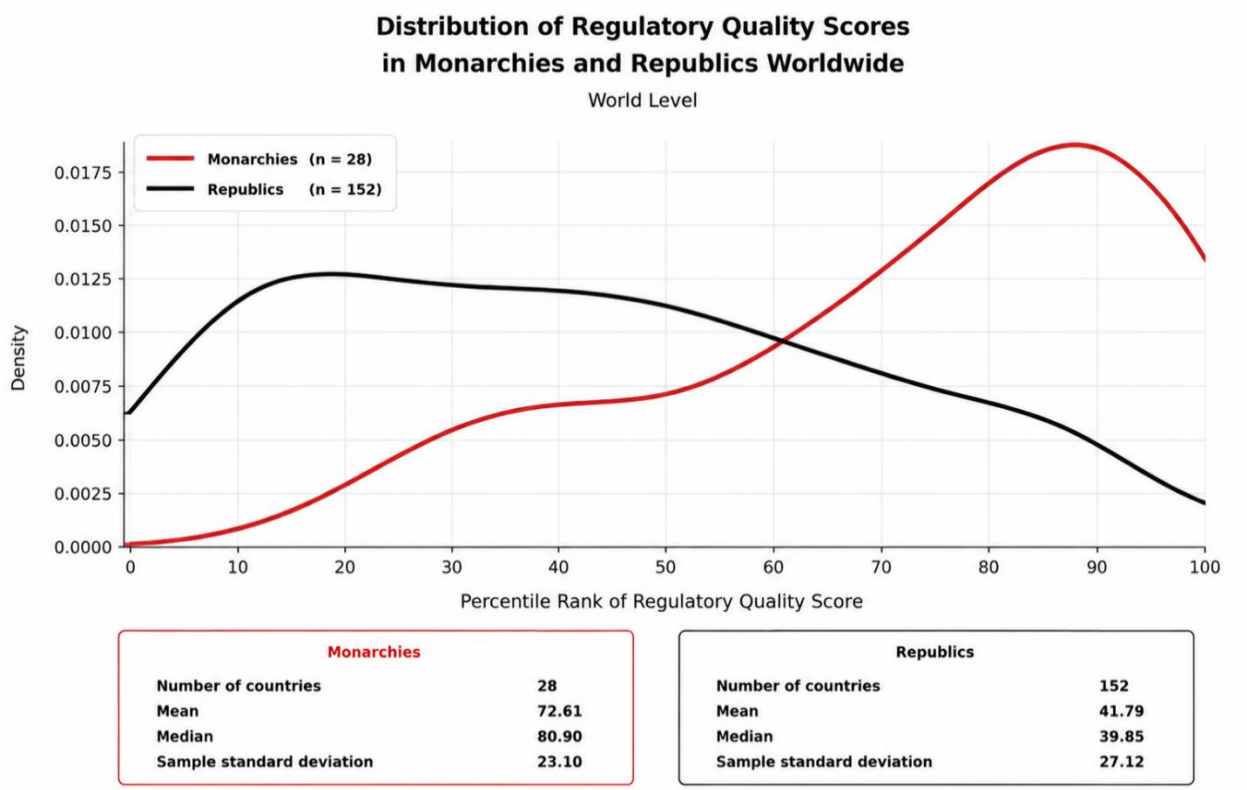
Both inferential tests identify a statistically significant difference:

Test	Result
Welch's t-test	**p = 0.000000145 → Significant**
Mann–Whitney U	**p = 0.000000407 → Significant**

Conclusion

Among countries with available data in 2023, monarchies had substantially higher average Regulatory Quality scores than republics. The difference is highly statistically significant under both Welch's t-test and the Mann–Whitney U test, while the substantially higher median among monarchies indicates that the pattern is also evident in the central part of the distributions rather than being driven solely by a small number of exceptionally high-performing countries.

As with the preceding comparisons, however, these results establish an **empirical association rather than a causal effect**. They do not, by themselves, demonstrate that regime type directly causes differences in regulatory quality. Other historical, economic, institutional, and regional factors may also contribute to the observed pattern.



The distribution of Regulatory Quality scores across countries worldwide shows that the curve for monarchies is shifted toward higher values relative to that for republics. Since higher values on this indicator represent better regulatory quality, greater government capacity to formulate and implement sound policies, and a more favorable environment for economic activity, this pattern indicates that monarchies in this dataset perform better on average.

The mean Regulatory Quality score is **72.61** for monarchies and **41.79** for republics. The corresponding medians are **80.90** and **39.85**, respectively. Thus, the observed difference is not driven solely by a few countries with exceptionally high scores; it is also clearly visible in the central tendency of the two distributions.

Although the two curves overlap to some extent, the greater concentration of monarchies at the higher end of the indicator and of republics at lower values reveals a substantial difference in the overall distributional patterns of the two groups. The statistical tests presented below determine whether this observed difference is also statistically significant.

Comparison of the "Regulatory Quality" Indicator Between Constitutional and Non-Constitutional Monarchies Worldwide

To examine whether the type of monarchy is associated with regulatory quality, monarchies were divided into two groups: **constitutional monarchies** and **semi-constitutional and absolute monarchies**. Their performance was then compared using the World Bank's 2023 Regulatory Quality indicator.

The results show that **constitutional monarchies**, with a mean score of **78.21** and a median of **89.60**, performed better than **semi-constitutional and absolute monarchies**, which had a mean of **66.15** and a median of **68.90**. The standard deviations of the two groups are also very similar—**22.90** and **22.47**, respectively—indicating approximately comparable levels of dispersion.

Nevertheless, **Welch's t-test** did not find the difference between the two group means to be statistically significant (**$t = 1.40$, $p > 0.05$**). At the same time, the effect size, **Hedges' $g = 0.52$**, indicates a **moderate effect**. Thus, although the observed difference is not negligible in standardized terms, the limited number of countries in the two groups means that the available statistical evidence is insufficient to draw a firm conclusion that one type of monarchy outperforms the other.

Overall, the findings indicate that both groups of monarchies perform relatively well on the Regulatory Quality indicator. Although constitutional monarchies have a higher average score, this difference is not statistically significant at the conventional level. Based on the available data, therefore, it cannot be concluded that the type of monarchy, by itself, is associated with a statistically significant difference in regulatory quality.

Comparison of the "Regulatory Quality" Indicator Between Commonwealth Realms and Republics Worldwide

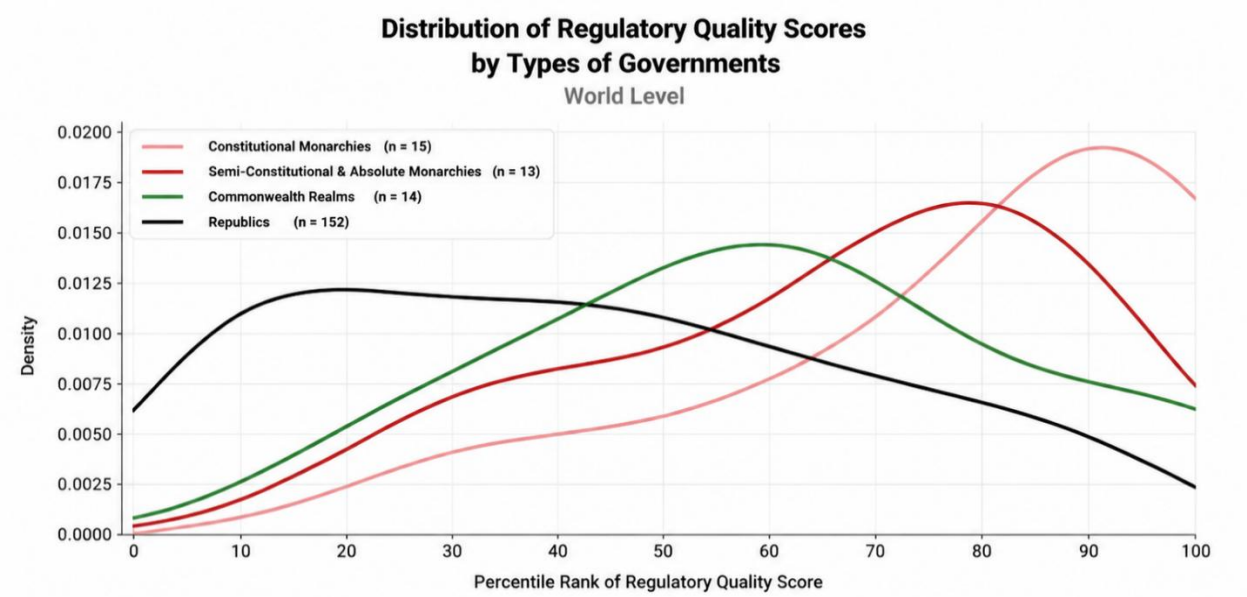
To assess the position of the **Commonwealth realms** on the Regulatory Quality indicator, their performance was compared with that of republics worldwide using the World Bank's 2023 Regulatory Quality data.

The results show that the **Commonwealth realms**, with a mean score of **61.21** and a median of **62.05**, performed better than **republics worldwide**, which had a mean of **41.79** and a median of **39.85**. The standard deviation is **24.63** for the Commonwealth realms and **27.12** for republics, indicating relatively similar levels of dispersion in the two groups.

Welch's t-test indicates that the difference between the two group means is **statistically significant** (**$t = 2.798$, $p = 0.0129$**). The estimated difference in means is **19.42 points**, with a **95% confidence interval from 4.71 to 34.13 points**. The effect size, **Hedges' $g = 0.72$**, indicates a **moderate-to-large effect**.

On this basis, the findings show that the Commonwealth realms, on average, performed better than republics worldwide on the Regulatory Quality indicator. The difference is not only statistically significant, but its effect size also suggests that the observed gap is substantial in practical terms.

As emphasized throughout this study, however, this is an observational comparison. It reflects the historical performance of the two groups of countries and, **by itself**, is insufficient to establish a causal relationship between regime type and regulatory quality.



The chart above shows the distribution of the **Regulatory Quality** indicator across four groups of countries: **constitutional monarchies, semi-constitutional and absolute monarchies, Commonwealth realms, and republics worldwide**. Each curve represents the density distribution of countries across different values of the indicator. Thus, a curve shifted further to the right indicates a greater concentration of countries in that group at higher percentile ranks of regulatory quality.

As the chart shows, the distribution of **constitutional monarchies** is more heavily concentrated at the higher end of the indicator than those of the other groups. This pattern is consistent with the group's mean of **78.21** and median of **89.60**, indicating that, descriptively, constitutional monarchies have the highest central values among the four groups examined.

Semi-constitutional and absolute monarchies are also concentrated largely at relatively high values of the indicator. This group has a mean of **66.15** and a median of **68.90**, both substantially higher than the corresponding values for republics worldwide. However, as the preceding statistical test showed, the difference between this group and constitutional monarchies is not statistically significant. The chart therefore does not provide a basis for concluding that either of these two types of monarchy definitively outperforms the other.

The distribution of the **Commonwealth realms** is likewise more concentrated in the middle and upper ranges of the indicator than that of republics worldwide. Their mean of **61.21** and median of **62.05**, compared with a mean of **41.79** and median of **39.85** for republics, indicate descriptively better performance among the Commonwealth realms. As shown by the Welch test, this difference is also statistically significant.

By contrast, **republics worldwide** display a broader distribution across the full range of the indicator, with a larger share of their distribution concentrated at low and intermediate values. Their mean of **41.79** and median of **39.85** are also lower than the corresponding values for each of the other three groups.

Overall, the chart shows that, **in this dataset, all three monarchy-related groups are more concentrated at higher levels of regulatory quality than republics worldwide**. However, the magnitude of the differences among the monarchy groups themselves varies, and the shape of the distributions alone is not sufficient to determine whether these differences are statistically significant. The chart should therefore be interpreted alongside the results of the statistical tests for each comparison

Comparison of the “Regulatory Quality” Indicator between Monarchies and Republics across Continents and the Middle East

Comparison of the “Regulatory Quality” Indicator between Monarchies and Republics in Asia

1. Descriptive Statistics

Group	Number of Countries (n)	Mean	Median	Sample SD
Asian Monarchies	13	67.80	68.90	17.46
Asian Republics	33	35.70	33.00	26.11

Interpretation of Descriptive Statistics

The mean Regulatory Quality score is **67.80** for Asian monarchies and **35.70** for Asian republics. The median is **68.90** for monarchies and **33.00** for republics.

Thus, based on the available data, **Asian monarchies not only have a higher mean, but the central value of their distribution is also substantially higher.** The fact that the mean and median point in the same direction indicates that the observed difference is not merely driven by a few countries with exceptionally high values, but is also evident in the central part of the distribution.

The sample standard deviation is also lower among monarchies (**17.46**) than among republics (**26.11**), indicating greater dispersion in Regulatory Quality scores among Asian republics.

2. Welch’s t-test for the Difference in Means

The hypotheses were defined as follows:

Null hypothesis (H_0): The mean Regulatory Quality score is equal for Asian monarchies and republics.

Alternative hypothesis (H_1): The mean Regulatory Quality score differs between the two groups.

Results of Welch’s t-test

Measure	Value
Mean — Monarchies	67.80
Mean — Republics	35.70
Difference in Means	**32.10**
t-statistic	**4.833**

Measure	Value
Approximate Degrees of Freedom	**32.89**
Two-tailed p-value	**0.00003**

Interpretation of Welch's t-test

The probability value (**p = 0.00003**) is far below the conventional 5 percent significance level. The null hypothesis is therefore rejected.

The result indicates that the difference in mean Regulatory Quality scores between Asian monarchies and republics is statistically significant. The mean percentile rank of Asian monarchies on this indicator is approximately **32.10 points** higher than that of Asian republics.

3. Mann–Whitney U Test

To examine the difference between the two groups from another perspective and without relying on the assumption of normally distributed data, the nonparametric **Mann–Whitney U test** was also conducted.

Results of the Mann–Whitney U Test

Measure	Value
U-statistic	**359**
Two-tailed p-value	**0.00044**

Interpretation of the Mann–Whitney U Test

The probability value (**p = 0.00044**) is below the 5 percent significance level. The null hypothesis is therefore rejected.

This test likewise indicates a statistically significant rank-based difference in Regulatory Quality scores between Asian monarchies and republics. Given the substantially higher mean and median among monarchies, the direction of the observed difference is toward higher indicator values in the monarchy group.

The consistency of the Welch and Mann–Whitney results is important. Welch's test identifies a statistically significant difference in means, while the Mann–Whitney test also provides significant evidence of a rank-based difference between the two groups. The observed result is therefore not dependent on the choice of a single statistical method.

Summary

Group	Mean Regulatory Quality Score
Asian Monarchies	**67.80**
Asian Republics	**35.70**

Both statistical tests confirm a significant difference between the two groups:

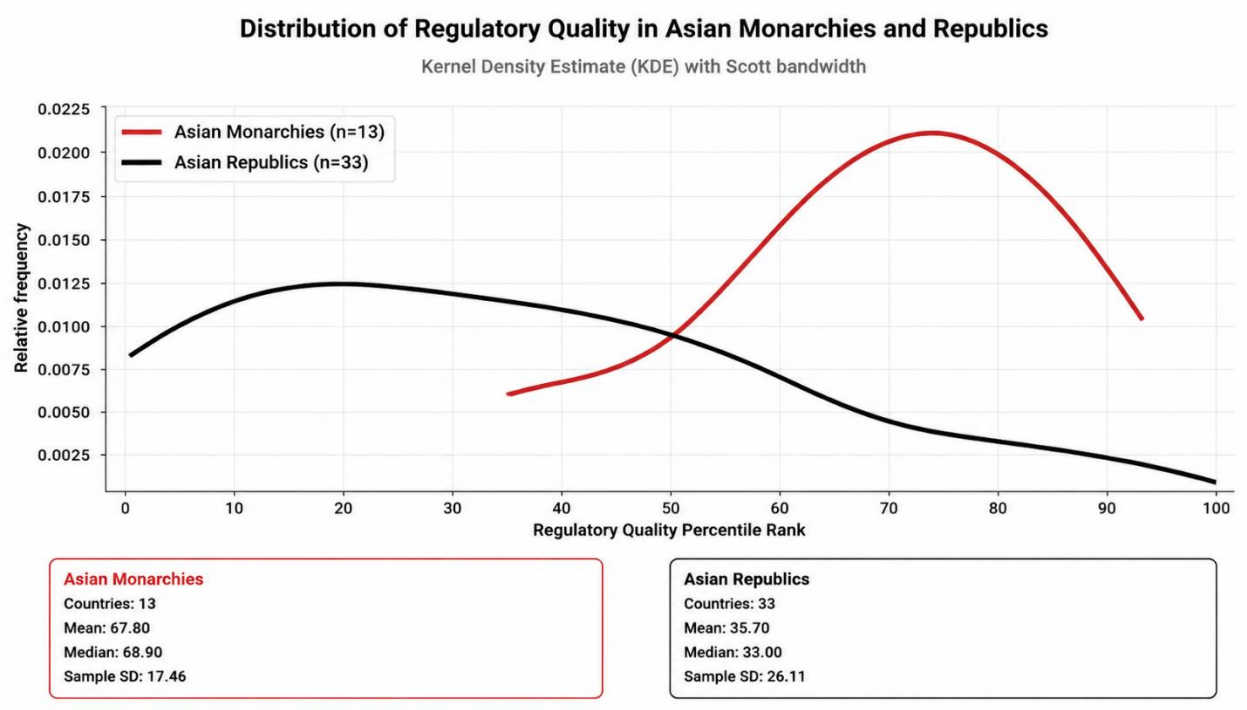
Test	Result
Welch's t-test	**p = 0.00003 → Significant**
Mann–Whitney U	**p = 0.00044 → Significant**

Conclusion

Among Asian countries with available data, monarchies have, on average, higher percentile ranks on the Regulatory Quality indicator than republics, and the difference between the two groups is statistically significant in both tests.

Overall, the within-continent comparison for Asia is consistent with the pattern observed in the global comparison. On average, the monarchies of Asia rank higher on the Regulatory Quality indicator, while their scores also show less dispersion than those of Asian republics. Based on the 2023 data, **Asian monarchies have a clearly better record on this indicator—which reflects perceptions of governments' ability to formulate and implement policies and regulations that support and promote private-sector activity and development—than the republics of the continent.**

However, this finding is based on an observational comparison and **does not, by itself, establish a causal relationship between the form of government and regulatory quality**. Historical, economic, and institutional factors, as well as differences in levels of development, may also contribute to the observed difference between the two groups.



The distribution of the Regulatory Quality indicator across Asian countries shows that the curve for monarchies is shifted toward higher values relative to that for republics. Since higher values on this indicator represent a more favorable position in terms of regulatory quality, this pattern indicates that, on average, Asian monarchies performed better on this indicator in the dataset examined.

The mean Regulatory Quality score is **67.80** for Asian monarchies and **35.70** for Asian republics. The corresponding medians are **68.90** and **33.00**, respectively. The fact that both the mean and median point in the same direction indicates that the observed difference is not merely driven by a few countries with exceptionally high values, but is also evident in the central part of the distributions.

Although the two curves overlap to some extent, indicating within-group variation in both categories, the distribution of monarchies is more concentrated at higher values of the indicator, whereas that of republics is more concentrated at lower values. The greater dispersion among republics is also consistent with their higher sample standard deviation (**26.11**, compared with **17.46** for monarchies).

The statistical tests presented below determine whether this observed difference is also statistically significant.

Comparison of the “Regulatory Quality” Indicator Between Monarchies and Republics in Europe

1. Descriptive Statistics

Group	Number of Countries (n)	Mean	Median	Sample SD
European Monarchies	11	91.80	93.40	7.00
European Republics	34	70.07	73.35	21.38

Interpretation of the Descriptive Statistics

The mean Regulatory Quality percentile rank is **91.80** for European monarchies and **70.07** for European republics.

The medians show the same pattern:

- Monarchies: **93.40**
- Republics: **73.35**

These figures indicate that **European monarchies not only have a higher mean, but the center of their distribution is also located at a substantially higher level**. In addition, the much lower standard deviation among monarchies indicates that the countries in this group are clustered more closely around similarly high levels of regulatory quality, whereas European republics display considerably greater variation.

2. Welch’s t-test for the Difference in Means

The hypotheses were defined as follows:

Null hypothesis (H_0): The mean Regulatory Quality percentile rank is equal in European monarchies and republics.

Alternative hypothesis (H_1): The mean Regulatory Quality percentile rank differs between the two groups.

Results of Welch’s t-test

Statistic	Value
Mean — Monarchies	91.80
Mean — Republics	70.07
Difference in Means	**21.73**
t-statistic	**5.137**
Approximate Degrees of Freedom	**42.94**

Statistic	Value
Two-sided p-value	**0.0000065**

Interpretation of Welch's t-test

The p-value (**p ≈ 0.0000065**) is far below the conventional 5% significance level. The null hypothesis is therefore rejected.

The result indicates that **the difference in mean Regulatory Quality percentile rank between European monarchies and republics is statistically significant**. In this dataset, European monarchies have an average percentile rank approximately **21.73 points higher** than European republics.

3. Mann–Whitney U Test

To examine the difference between the two groups from a nonparametric perspective, the **Mann–Whitney U test** was also conducted. This test is based on the ranks of the observations and does not require the assumption of normally distributed data.

Results of the Mann–Whitney U Test

Statistic	Value
U-statistic	**324**
Two-sided p-value	**0.000312**

Interpretation of the Mann–Whitney U Test

The p-value (**p ≈ 0.000312**) is well below the 5% significance level. The null hypothesis is therefore rejected.

This test also provides strong evidence that the Regulatory Quality values of European monarchies and republics differ significantly in rank. Given that both the mean and median are substantially higher among monarchies, the direction of the observed difference is toward higher Regulatory Quality percentile ranks among European monarchies.

The consistency of the two tests is important. Welch's t-test identifies a statistically significant difference in means, while the Mann–Whitney U test provides significant evidence of a difference in the rank distributions. Thus, the finding is not dependent on a single statistical method.

Summary

Group	Mean Regulatory Quality Percentile Rank
European Monarchies	**91.80**

Group	Mean Regulatory Quality Percentile Rank
European Republics	**70.07**

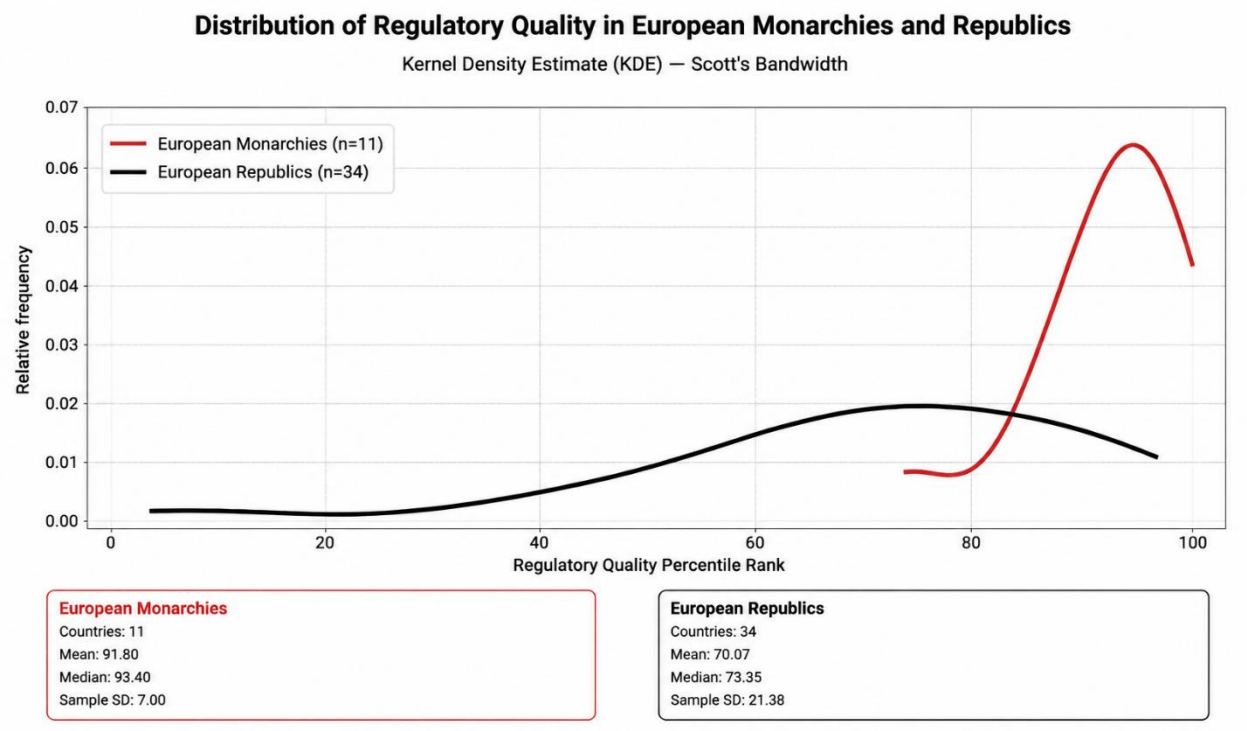
The results of both statistical tests are significant:

Test	Result
Welch's t-test	**p = 0.0000065 → Significant**
Mann–Whitney U	**p = 0.000312 → Significant**

Conclusion

Among European countries with available data, monarchies have, on average, substantially higher Regulatory Quality percentile ranks than republics. Both statistical tests provide strong evidence of a difference between the two groups. The higher median and markedly lower dispersion among monarchies further indicate that their higher performance is not driven merely by a small number of exceptionally high-scoring countries, but is reflected more broadly across the group.

As with the other comparisons in this study, however, this is an **observational comparison**. The results describe the empirical record of the two groups and, **by themselves**, do not establish a causal relationship between the form of government and regulatory quality.



The distribution of the Regulatory Quality indicator across European countries shows that the curve for monarchies is noticeably shifted toward higher percentile ranks. Since higher values on this indicator represent better regulatory quality and a greater perceived ability of governments to formulate and implement policies and regulations that support private-sector development, this pattern indicates that European monarchies in this dataset tend to occupy substantially higher levels of regulatory quality.

The mean Regulatory Quality percentile rank is 91.80 for European monarchies and 70.07 for European republics. The corresponding medians are 93.40 and 73.35, respectively. This difference indicates that the higher performance observed among monarchies is not driven merely by a few exceptionally high-performing countries, but is also evident at the center of the distribution.

The greater concentration of the monarchy curve at the upper end of the indicator, together with its lower dispersion, also suggests that countries in this group display a relatively more consistent pattern of regulatory quality. By contrast, the curve for republics spans a much wider range of values, indicating greater variation in performance among countries in this group.

The statistical tests presented below determine whether this observed difference is also statistically significant.

Comparison of the "Regulatory Quality" Indicator Between European Monarchies and European Republics That Have Never Been Communist

One question is whether the observed advantage of European monarchies is simply due to the inclusion, among the republics, of Eastern European countries with a history of communist rule. To examine this possibility, an additional comparison was conducted in which European monarchies were compared only with **European republics that have never experienced communist rule**. This substantially reduces the potential influence of the institutional legacy of communist systems on the comparison.

This analysis includes **11 monarchies** and **13 non-communist European republics**. The results show that the mean **Regulatory Quality percentile rank** is **91.80** for European monarchies, compared with **83.97** for non-communist European republics. European monarchies therefore score, on average, approximately **7.83 percentile-rank points higher** than this group of republics.

The medians show the same pattern. The median is **93.40** among European monarchies and **85.40** among non-communist republics. Although the gap between the two groups is smaller than in the previous comparison with all European republics, monarchies still occupy the higher position.

European monarchies also display less dispersion. Their sample standard deviation is **7.00**, compared with **9.69** for non-communist European republics. This indicates that Regulatory

Quality scores among European monarchies are not only higher on average but also more tightly clustered.

Welch's t-test indicates that the difference in means is statistically significant at the 5% level (t = 2.29, df = 21.53, p = 0.032). Thus, even after excluding republics with a history of communist rule, the data continue to provide statistical evidence that European monarchies have a higher mean Regulatory Quality percentile rank than non-communist European republics.

This comparison reveals an important point. In the previous analysis, one might have suspected that the lower average among European republics was largely attributable to the inclusion of Eastern European countries with a communist legacy. Once these countries are excluded, the gap becomes smaller, but **does not disappear entirely**. In other words, removing republics with a communist past reduces part of the previously observed difference, yet European monarchies continue to outperform European republics without a history of communist rule on the Regulatory Quality indicator.

This supplementary analysis therefore suggests that the observed advantage of European monarchies cannot be attributed solely to the legacy of communist rule among some European republics. Even when compared with republics that have never experienced communist rule, European monarchies still recorded higher Regulatory Quality percentile ranks in the 2023 data. As emphasized throughout this study, however, this remains an observational comparison and **by itself** is insufficient to establish a causal relationship between form of government and regulatory quality.

This restricted comparison should not, however, replace the main analysis. From a methodological perspective, excluding European republics with a history of communist rule from the overall comparison is not justified, because these countries constitute a genuine part of the historical experience of republics in Europe. Excluding them could introduce a form of **selection bias** by removing unfavorable cases from the record of republics. Within the analytical framework of this book, the historical tendency of some republics toward communism, the establishment of one-party rule, and the institutional legacies left by those systems are not treated simply as external factors wholly independent of the form of government; rather, they are considered part of the historical trajectories and potential outcomes of the republican experience itself. Therefore, republics with a history of communist rule should remain included in the primary comparison.

Comparison of the “Regulatory Quality” Indicator Between Monarchies and Republics in Africa, the Americas, and Oceania

Africa

In Africa, **three monarchies** were compared with **50 republics**. The results show that the mean **Regulatory Quality percentile rank** is **34.90** for African monarchies and **27.94** for African republics. The corresponding medians are **29.70** and **24.75**, respectively.

Group	Number of Countries	Mean	Median	Sample SD
African Monarchies	3	34.90	29.70	14.25
African Republics	50	27.94	24.75	18.32

In Africa as well, monarchies have a higher mean than republics, although the difference between the two groups is considerably smaller than in Asia and Europe. Moreover, because there are only three monarchies on the continent, the results of this comparison should be interpreted with greater caution.

The Americas

In the Americas, where there are no independent monarchies, **nine Commonwealth realms** were compared with **26 republics**. The mean Regulatory Quality percentile rank is **62.73** for the Commonwealth realms and **43.35** for the republics of the Americas. The corresponding medians are **62.30** and **43.85**, respectively, reinforcing the overall descriptive advantage of the Commonwealth realms.

Group	Number of Countries	Mean	Median	Sample SD
Commonwealth Realms	9	62.73	62.30	15.11
Republics of the Americas	26	43.35	43.85	23.04

Thus, in the Americas as well, the Commonwealth realms have, on average, higher Regulatory Quality percentile ranks than the republics of the region.

Oceania

Oceania has only **one independent monarchy (Tonga)**. Therefore, in addition to reporting its score, **five Commonwealth realms** were compared with the **nine republics** of the region. The mean Regulatory Quality percentile rank is **58.48** for the Commonwealth realms and **44.71** for the republics. The corresponding medians are **44.80** and **42.90**, respectively.

Group	Number of Countries	Mean	Median	Sample SD
Independent Monarchy	1	37.30	37.30	—
Commonwealth Realms	5	58.48	44.80	33.98
Republics of Oceania	9	44.71	42.90	9.98

Because there is only one independent monarchy in Oceania, a statistical comparison involving that group alone is not possible. Nevertheless, the comparison between the Commonwealth realms and the republics of Oceania shows the same descriptive direction observed in the other regions: the mean Regulatory Quality percentile rank is higher among the Commonwealth realms than among the republics.

Overall, the results for Africa, the Americas, and Oceania are consistent in direction with the global, Asian, and European comparisons. In all three regions, monarchies or Commonwealth realms have, on average, higher Regulatory Quality percentile ranks than republics. However, because of the small number of countries in some groups—particularly in Africa and Oceania—these regional results should be interpreted with greater caution.

Comparison of the "Regulatory Quality" Indicator Between Monarchies and Republics in the Middle East

Indicator	Monarchies	Republics
Number of Countries	7	10
Mean	72.24	30.56
Median	68.90	20.75
Sample SD	9.97	30.29

The difference between the two groups is striking. The mean Regulatory Quality percentile rank in Middle Eastern monarchies is **41.68 points higher** than in republics and is approximately **2.36 times as high**. The median among monarchies is also more than three times that of republics.

Another notable feature is the substantial difference in dispersion between the two groups. The sample standard deviation is only **9.97** among monarchies, compared with **30.29** among republics. Thus, the monarchies of the region not only have a substantially higher mean percentile rank, but their scores are also much more tightly clustered and homogeneous.

Welch's t-test

A two-sided Welch's t-test produced the following result:

t = 4.049, df = 11.58, p = 0.00173

Thus, **the difference in means between the two groups is highly statistically significant**. The null hypothesis of equal means is rejected even at the 1% significance level.

The Hedges' g effect size is:

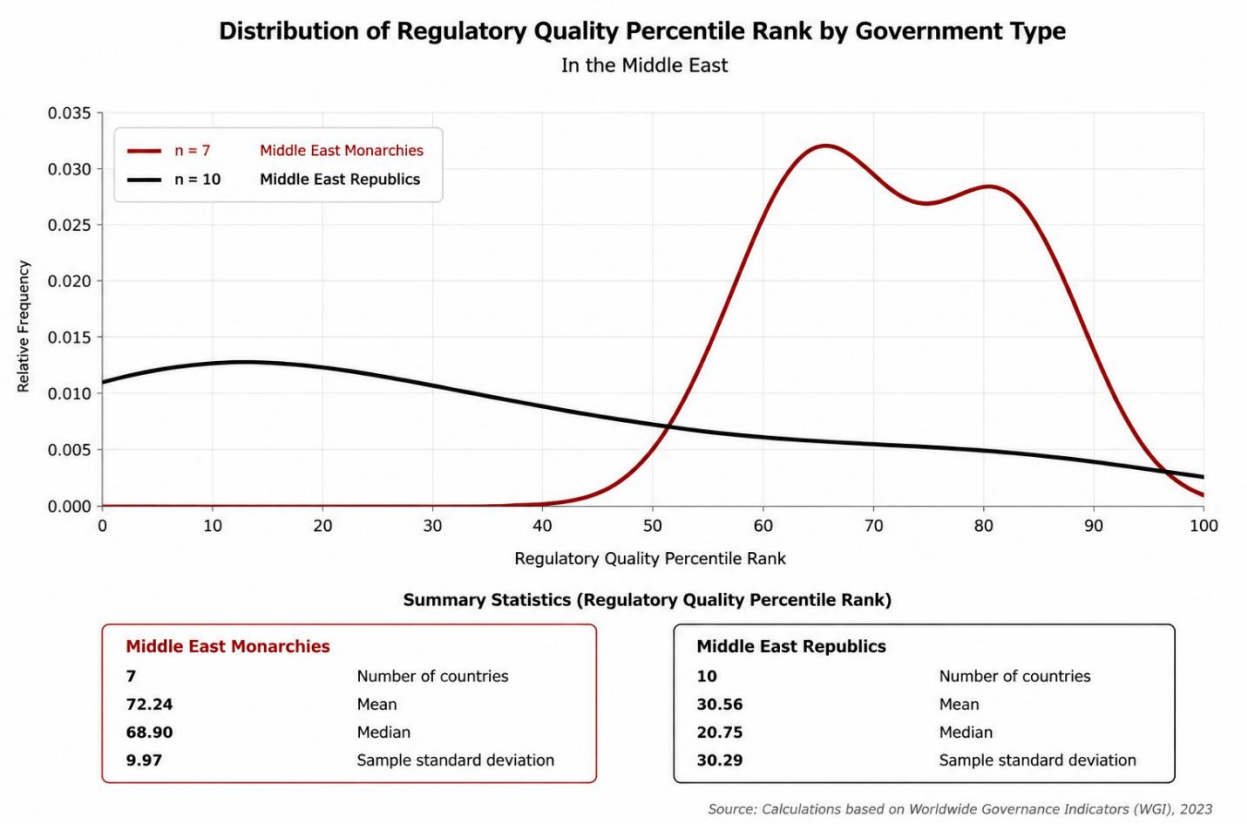
g = 1.628

This represents a **very large effect in favor of the monarchies**.

Summary

To assess the statistical significance of the difference in means, **Welch's t-test** was used. **The test indicates a highly statistically significant difference between the two groups (t = 4.049, df = 11.58, p = 0.0017**). The estimated effect size, **Hedges' g = 1.63**, is **very large**.

Thus, in the Middle East, the observed association between form of government and Regulatory Quality is not merely a small difference in average scores. Monarchies rank substantially higher than republics on this indicator, the difference is highly statistically significant, and the effect size is very large. At the same time, the much lower dispersion of scores among monarchies indicates that their relatively high performance is also notably consistent across the countries in this group.



The distribution plot shows that the curve for **Middle Eastern monarchies** is shifted markedly to the right of that for republics. This shift indicates that, overall, the monarchies in the region achieved higher percentile ranks on the **Regulatory Quality** indicator. The greater concentration of the monarchy curve also suggests that performance among these countries is relatively more consistent, whereas the wider dispersion of the republic curve reflects substantial variation in regulatory quality across Middle Eastern republics.

Despite this variation, the overlap between the two curves is limited, with most of the distribution for monarchies concentrated at higher values than that of republics. This pattern is consistent with the statistical results presented in this section: the mean percentile rank for Middle Eastern monarchies (**72.24**) is substantially higher than that for the region's republics (**30.56**). The distribution plot therefore visually reinforces the findings of the statistical tests: **in the 2023 data, Middle Eastern monarchies, on average, performed substantially better than the region's republics on the Regulatory Quality indicator.**

From an Empirical Pattern to a New Hypothesis: “New or Modern Monarchy”

Thus far, we have seen that monarchies have outperformed republics, on average, not only in the indicators examined under **welfare** and **security**, but also in **Regulatory Quality**, where they have achieved substantially higher rankings. More precisely, monarchies have, on average, performed better in formulating and implementing policies and regulations that support private-sector activity and economic development.

This latest finding, at least at first glance, appears inconsistent with what is commonly expected of republics. In republics, executive officials and members of legislative bodies are elected, directly or indirectly, by the people. One might therefore expect policies and regulations, more than under any other form of government, to be designed in the public interest and to improve citizens’ lives. Historically, by contrast, monarchies were often associated with extensive privileges for royal families, aristocrats, landowners, religious authorities, and supporters of the court, while the broader population in many past monarchical societies enjoyed far fewer rights and privileges.

This brings us to a fundamental question:

When, and through what transformations, did monarchies that once looked like that come to look like this?

In the scientific method, observations can lead us to formulate an explanatory hypothesis. We must then derive testable implications from that hypothesis and examine them against independent data. A hypothesis remains viable so long as new evidence does not falsify it; and the more demanding and diverse the tests it survives, the greater our confidence in its explanatory power.

The data examined so far are still insufficient to establish a causal relationship between the form of government and national performance. Yet the recurring pattern in the results makes it reasonable to formulate a new hypothesis. This hypothesis seeks to explain why many contemporary monarchies, contrary to the historical image of monarchy, have performed favorably in several important areas of governance.

To understand this transformation, we must first return to the economic and political structure of the premodern world.

In an era before industry, technology, and trade existed on anything like their present scale, a large share of human production and wealth depended on land and what could be produced from it. Land was not merely an economic asset; it was also the principal source of income, taxation, labor, and military power. Usually, control of land also entailed control over the peasants or laborers who worked it. The more land a person controlled, the greater his wealth and influence

generally were. Under such conditions, one way for a monarch to increase power and wealth was to expand the realm and acquire new territories.

Monarchs could not administer all these territories directly. In many preindustrial societies, therefore, portions of land—or rights to the income generated by them—were granted to military commanders, princes, aristocrats, and landowners. Such grants could be temporary or hereditary, and in return the holder was expected to pay taxes or tribute, provide military forces, and support the monarch in times of war.

This system took different forms in different parts of the world, and its various manifestations should not be treated as identical. Yet across many of them—from parts of Europe to the Middle East and East Asia—the central government depended on landed classes to administer territory, collect taxes, and supply soldiers. This dependence compelled monarchs to grant them extensive legal, economic, and political privileges.

As a result, laws were often structured in favor of aristocrats and landowners. In some societies, peasants lacked freedom to move, change occupations, own property, or choose where to live, and were legally or economically tied to the land they worked. Many important governmental and military offices were also, either formally or in practice, reserved for aristocrats and the upper classes.

Alongside landowners, religious institutions also played an important role in legitimizing royal power. In many ancient monarchies, religious authorities portrayed the monarch’s rule as divinely sanctioned and obedience to the monarch as a religious duty. In return, the state recognized privileges for religious institutions, including tax exemptions, land ownership, judicial autonomy, or exemption from certain public obligations.

In this way, stratified societies emerged in which different social classes possessed different legal rights and obligations. The law did not necessarily apply equally to everyone; family status, land ownership, religious affiliation, and proximity to the royal court could determine the extent of an individual’s rights.

Monarchs, of course, were not always entirely satisfied with this arrangement. The same aristocrats and landowners who supplied the state with military and financial resources could become rivals to the monarch when central authority weakened. They sometimes fought one another, defied the government, or even attempted to seize the throne. For this reason, many monarchs sought to reduce the power of local aristocracies, centralize military and taxation systems, and bring the administration of the country more directly under the control of the central government.

These efforts were not always sustainable. Weak communications, limited administrative capacity, and the difficulty of controlling vast territories often forced governments to become dependent once again on local powers. In many periods, sharing power with landowners was the easiest—or the only practicable—way to govern the realm.

Within such a structure, the monarch was often a conqueror, or the heir of a conqueror, who regarded the country as the property of his dynasty. Its territory, resources, and population were primarily instruments for maintaining power, securing wealth, perpetuating the dynasty, and expanding the realm. The people were not yet “equal citizens,” but subjects who supplied the government with labor, taxes, and soldiers.

With the growth of industry, technology, trade, and urbanization, however, this structure gradually changed. Land did not cease to have economic importance, but it lost its monopoly as the principal source of wealth and power. Capital, industry, knowledge, technology, trade, finance, and skilled human resources became new sources of wealth creation. Government revenue, too, no longer came primarily from land and agricultural production; economic activity in cities, companies, industries, and expanding markets became increasingly important sources of taxation.

The organization of armies also changed. Central governments became capable of maintaining standing armies and more systematic administrative and taxation systems, reducing their dependence on the military forces of local aristocrats. Advances in communications, education, bureaucracy, and technology likewise made more direct administration and control of the country possible.

Alongside these transformations, the source of monarchical legitimacy also changed. The doctrine of the divine right of kings, at least in many contemporary monarchies, gave way to constitutions, formal rules of succession, and the political and social acceptance of the institution of monarchy. The country was no longer a prize of war acquired by a monarch, but a political unit with recognized borders, public institutions, and a population endowed with citizenship rights.

In the modern international system, territorial conquest has also lost much of the legitimacy and economic utility it once possessed. National borders have become more firmly established, while the costs of war and occupation have risen dramatically. Consequently, the power and prestige of governments have come to depend more than before on economic development, political stability, the quality of public administration, and the welfare of the population.

As a result, **the contemporary monarch does not necessarily have the same sources of power, the same dependencies, or the same objectives as the monarch of the past**. To preserve his position, he need not necessarily secure the support of privileged landed classes and religious institutions in the way monarchs once did, nor does the survival of the monarchy necessarily depend on conquering new territories. In some monarchies, royal succession has acquired such legal and social stability that the monarch’s position, compared with that of elected politicians or personalist dictators, is less exposed to constant competition for control of power.

A political leader generally faces two fundamental concerns: first, preserving his own power and political position, together with those of his supporters; and second, increasing the strength,

wealth, stability, and prestige of the country. The relative importance of these two objectives is not the same in every political system or for every ruler.

An elected politician must think about the next election, preserve a governing coalition, satisfy influential constituencies, and defeat partisan rivals. A personalist dictator may likewise devote a large share of the state’s capacity to controlling opponents, preventing coups, and maintaining the loyalty of military and security forces. A monarch whose position and succession are relatively securely established in law and political convention may be less preoccupied than either of these with short-term competition to retain office.

This does not mean that monarchs have no concern with preserving power, or that monarchies are immune to political crises, succession disputes, social unrest, or the risk of removal. Nor can we assume that all monarchs are necessarily benevolent or competent. The more limited claim is that **the stability of the monarchical office may, under certain conditions, reduce the importance of short-term competition for power and lengthen the government’s time horizon.**

Under such conditions, increasing the wealth, stability, and prestige of the country may become one of the principal sources of prestige for the monarch and the royal family themselves. The monarch can no longer behave like a conqueror seeking to extract the greatest possible benefit from his spoils before the opportunity disappears. His success becomes more closely tied than before to the success of the country with which the name and reputation of his dynasty are associated.

It is here that the analogy between the modern monarch and a parent becomes meaningful. A parent, under ordinary circumstances, does not regard the home and family as temporary assets to be exploited as quickly as possible. Parents see their own future as intertwined with that of their children and generally seek to leave them something more prosperous, secure, and stable. A monarch who sees the continuity of his dynasty as bound up with the continuity and prosperity of the country may likewise view the country through such a long-term, intergenerational horizon.

This analogy does not equate citizens with children, nor does it justify depriving them of political agency. Rather, the point is that the relationship of a modern monarch to the country may be more long-term and intergenerational than the relationship of a conqueror to conquered territory. Parents, too, do not always act correctly, justly, or without authoritarian tendencies; what matters in this analogy is the alignment of their long-term interests with the future of the family.

The ancient monarch was often a conqueror, or the heir of a conqueror, who sought to use the territory, resources, and people under his rule to preserve power and expand the dominance of his dynasty. In some respects, such behavior resembled that of a personalist dictator who places the country’s resources primarily at the service of his own political survival.

The modern monarch, provided that his position and succession possess sufficient institutional stability, may face the problem of day-to-day political survival less intensely and may therefore

be able to devote a greater share of his attention to the stability and prosperity of the country. From this perspective, the distinction between such a monarch and a dictator should not be sought merely in the degree of authoritarianism or in claims of benevolence, but rather in the structure of succession, the degree to which power is institutionalized, the time horizon of decision-making, and the way the ruler’s interests are linked to the country’s future.

Both monarchs and parents may at times act in an authoritarian manner; but within this hypothesis, what matters is the prevailing direction of their decisions. A personalist dictator may desire national prosperity insofar as it helps preserve his own power, whereas a modern monarch may see the survival of his dynasty’s prestige and position as dependent on the country’s long-term prosperity. This distinction, of course, is not yet an established finding; it is part of a hypothesis that must be tested in the remainder of the study.

In the premodern world, the sources of power of many monarchs were broadly similar: conquest, land ownership, aristocratic support, military power, and religious legitimacy. It is natural that similarities in the sources of power should also produce similarities in governmental objectives and behavior. If, in the contemporary world, we likewise observe recurring patterns of behavior among monarchies that differ from those of ancient monarchies, we may consider the possibility that changes in the sources and structure of monarchical power have also produced changes in its behavior.

For this reason, applying the same name to ancient and contemporary monarchies may itself create a cognitive error. The shared label can give the impression that we are dealing with a fixed and unchanged phenomenon, **even though the economic structure, source of legitimacy, rules of succession, relationship between government and society, and legal status of the population have all changed.**

Accordingly, from this point onward I distinguish between two concepts:

Ancient Monarchy is a system in which power rests primarily on conquest, territorial ownership, the support of privileged classes, personal military force, and traditional or religious legitimacy, and in which the country is regarded, to a significant extent, as the property of the ruling dynasty.

New or Modern Monarchy is a system in which the monarchical office is embedded within the framework of the modern state and its institutions, succession follows relatively well-defined rules, the distinction between public property and the property of the royal family is largely recognized, and the prestige of the monarchy is more closely tied than in the past to the stability, welfare, and success of the country.

This definition does not imply that all contemporary monarchies are equally modern or that they fully possess all of these characteristics. There are substantial differences among them, and some remain closer than others to the model of ancient monarchy. “Modern monarchy,” therefore, is

used here as an **analytical concept that may exist in varying degrees**, rather than as a label to be applied indiscriminately to every existing monarchy.

Thus far in this great experiment of contemporary history, the indicators examined have not been inconsistent with this hypothesis. Monarchies have, on average, performed better in welfare, security, and regulatory quality. At this point, however, a fundamental principle must be emphasized: **a hypothesis that has encountered only supportive evidence has not yet undergone its most demanding test.**

The scientific value of this hypothesis will become clearer when we expose it to indicators that may count against it. We must examine whether monarchies display the same pattern in government effectiveness, rule of law, control of corruption, political stability, quality of public services, and other dimensions of governance. We must also ask whether different results emerge in indicators where republics should, in theory, possess an advantage.

Subsequent data may strengthen the hypothesis; they may show that it is valid only under certain conditions and for certain types of monarchy; or we may encounter indicators that weaken or even falsify it. None of these possibilities should be excluded in advance.

The time has therefore not yet come for a final verdict. What we have seen so far is not the end of the experiment, but the beginning of its more demanding stage. We now have a hypothesis that must be confronted, without reservation, with new data. If the new monarchy is genuinely a different phenomenon from ancient monarchy, that difference should also reveal itself across a broader set of independent indicators and in diverse geographical and institutional comparisons.

So we continue the experiment—not in order to confirm the hypothesis, but to see whether the data that follow are capable of falsifying it. The procedure will be as follows:

Testing Roadmap for the “New or Modern Monarchy” Hypothesis

Stage One: The Welfare Record

Hypothesis prediction: If modern monarchy is more successful, it should generate greater economic welfare for its citizens.

Test	Indicator
1	GDP per Capita
2	Purchasing Power per Capita
3	Gini Coefficient

⬇

Stage Two: The Security Record

Hypothesis prediction: A government that regards itself as responsible for protecting its citizens should provide a safer environment.

Test	Indicator
4	Political Stability and Absence of Violence/Terrorism

⬇

Stage Three: The Order Record

Hypothesis prediction: If this pattern holds, both the quality of laws and regulations and the effectiveness of their implementation should be higher.

Test	Indicator
5	Regulatory Quality
6	Rule of Law
7	Government Effectiveness
8	Inflation

⬇

Stage Four: The Justice Record

Hypothesis prediction: A government that regards itself as responsible to its people should exhibit lower levels of corruption.

Test	Indicator
9	Control of Corruption
10	Corruption Perceptions

⬇

Stage Five: The Freedom Record

Hypothesis prediction: If this pattern genuinely works to the benefit of citizens, its effects should also be visible in individual freedoms and quality of life.

Test	Indicator
11	Property Rights
12	Human Development
13	Voice and Accountability

⬇

Stage Six: A Composite Indicator

Hypothesis prediction: If the advantages predicted by the hypothesis are not confined to a single dimension of governance, they should also be reflected in a broader measure that captures the combined outcome of multiple aspects of national performance.

Test	Indicator
14	Passport Power

⬇

Final Assessment

Has the “New or Modern Monarchy” hypothesis successfully passed the combined evidence from all fourteen independent tests?

Status of the “New or Modern Monarchy” Hypothesis After the First Five Indicators

Status	Indicator	Result
✓	**GDP per Capita**	**Consistent with the hypothesis**
✓	**Purchasing Power per Capita**	**Consistent with the hypothesis**
◐	**Gini Coefficient**	**Mixed evidence; difference favors the hypothesis**
✓	**Political Stability and Absence of Violence**	**Consistent with the hypothesis**
✓	**Regulatory Quality**	**Consistent with the hypothesis**

Rule of Law

In assessing the record of **"Order,"** we have so far seen that monarchies, on average, performed better than republics on the **Regulatory Quality** indicator. However, having sound laws and regulations is not sufficient, by itself, to establish order. Law demonstrates its real value when it is enforced in practice, when rights and obligations are reliable, and when citizens, businesses, and government institutions can have confidence in the mechanisms through which the law is enforced.

For this reason, the next step is to examine the **Rule of Law** indicator.

The World Bank also publishes this indicator as part of the **Worldwide Governance Indicators (WGI)** and reports it in the form of a **percentile rank** (Appendix RDS-F01). As explained earlier, the percentile rank indicates a country's relative position compared with the other countries assessed; thus, a higher value represents a more favorable position. The data used in this section are from **2023**.

The **Rule of Law** indicator reflects perceptions of the extent to which agents have confidence in and abide by the rules of society. Among other factors, it incorporates the quality of contract enforcement, property rights, the police and the courts, as well as the likelihood of crime and violence. Thus, the indicator does not merely measure the existence of laws; rather, it provides a picture of **how reliably rules and legal institutions operate and are enforced in practice**.

If the **"New or Modern Monarchy" hypothesis** introduced earlier can explain part of the variation in government performance, we would expect its effects not to end with the formulation of laws and regulations, but to be observable, at least to some extent, in the rule and enforcement of law as well. This, however, is merely a **testable prediction**, and the data will determine whether it is supported. We must therefore examine whether the Rule of Law indicator also reveals a pattern consistent with the hypothesis or instead poses a challenge to it.

Global Comparison of the "Rule of Law" Indicator Between Monarchies and Republics

Indicator: Rule of Law
Source: World Bank
Data year: 2023

For this indicator, a higher percentile rank represents a more favorable position in terms of the rule of law.

1. Descriptive Statistics

Group	n	Mean	Median	Sample SD
Monarchies	28	73.72	78.55	21.55
Republics	152	41.85	39.60	27.49

Interpretation of the Descriptive Statistics

The mean percentile rank for the Rule of Law indicator is **73.72** for monarchies and **41.85** for republics, representing a difference of approximately **31.87 percentile-rank points** between the two groups.

The medians point in the same direction:

- Monarchies: **78.55**
- Republics: **39.60**

The observed difference, therefore, is not confined to the means. The substantially higher median among monarchies indicates that the pattern is not merely the result of a few monarchies with exceptionally high rankings, but is also evident in the central part of the distribution.

The sample standard deviation is also higher among republics (**27.49**) than among monarchies (**21.55**), indicating greater dispersion in Rule of Law scores among republics. Descriptive statistics alone, however, cannot establish whether the difference between the groups is statistically significant. Inferential tests are therefore required.

2. Welch's t-Test for the Difference in Means

The hypotheses were defined as follows:

Null hypothesis (H_0): The mean percentile rank for the Rule of Law indicator is equal in monarchies and republics.

Alternative hypothesis (H_1): The mean percentile rank for the Rule of Law indicator differs between the two groups.

Welch's Test Results

Statistic	Value
Mean — Monarchies	73.72
Mean — Republics	41.85
Difference in means	**31.87**
t-statistic	**6.86**

Statistic	Value
Approximate degrees of freedom	**45.0**
Two-sided p-value	**< 0.001**

Interpretation of Welch's Test

The p-value is far below the conventional 5% significance level. The null hypothesis is therefore rejected.

The result indicates that **the difference in the mean percentile rank for the Rule of Law indicator between monarchies and republics worldwide is statistically significant**. In the 2023 data, the mean for monarchies is approximately **31.87 percentile-rank points** higher than that for republics.

It is important to note that the data used here are expressed as **percentile ranks**. Accordingly, the 31.87-point difference should not be interpreted as meaning that the rule of law is "31.87 percent better" in monarchies. Rather, it represents the difference between the average positions of the two groups on the indicator's percentile-rank scale.

3. Mann–Whitney U Test

To examine the difference between the two groups from another perspective, without relying on an assumption of normally distributed data, the nonparametric **Mann–Whitney U test** was also performed. This test is based on the ranking of observations.

The hypotheses were defined as follows:

Null hypothesis (H_0): Rule of Law values in the two groups do not differ significantly in terms of their ranks.

Alternative hypothesis (H_1): Rule of Law values in the two groups differ significantly in terms of their ranks.

Mann–Whitney U Test Results

Statistic	Value
U-statistic	**3,437**
Two-sided p-value	**< 0.001**

Interpretation of the Mann–Whitney U Test

The result of this test likewise indicates that **Rule of Law values in monarchies and republics differ significantly in terms of their ranks**.

Given that both the mean and median are substantially higher among monarchies, the direction of the observed difference is toward higher Rule of Law values in the monarchy group.

The convergence of the two tests is important. Welch's test identifies a statistically significant difference in the means, while the Mann–Whitney test provides significant evidence of a rank-based difference between the groups. The result, therefore, does not depend on the choice of a single statistical method.

Summary

For the 2023 data:

Group	Mean Rule of Law Percentile Rank
Monarchies	**73.72**
Republics	**41.85**

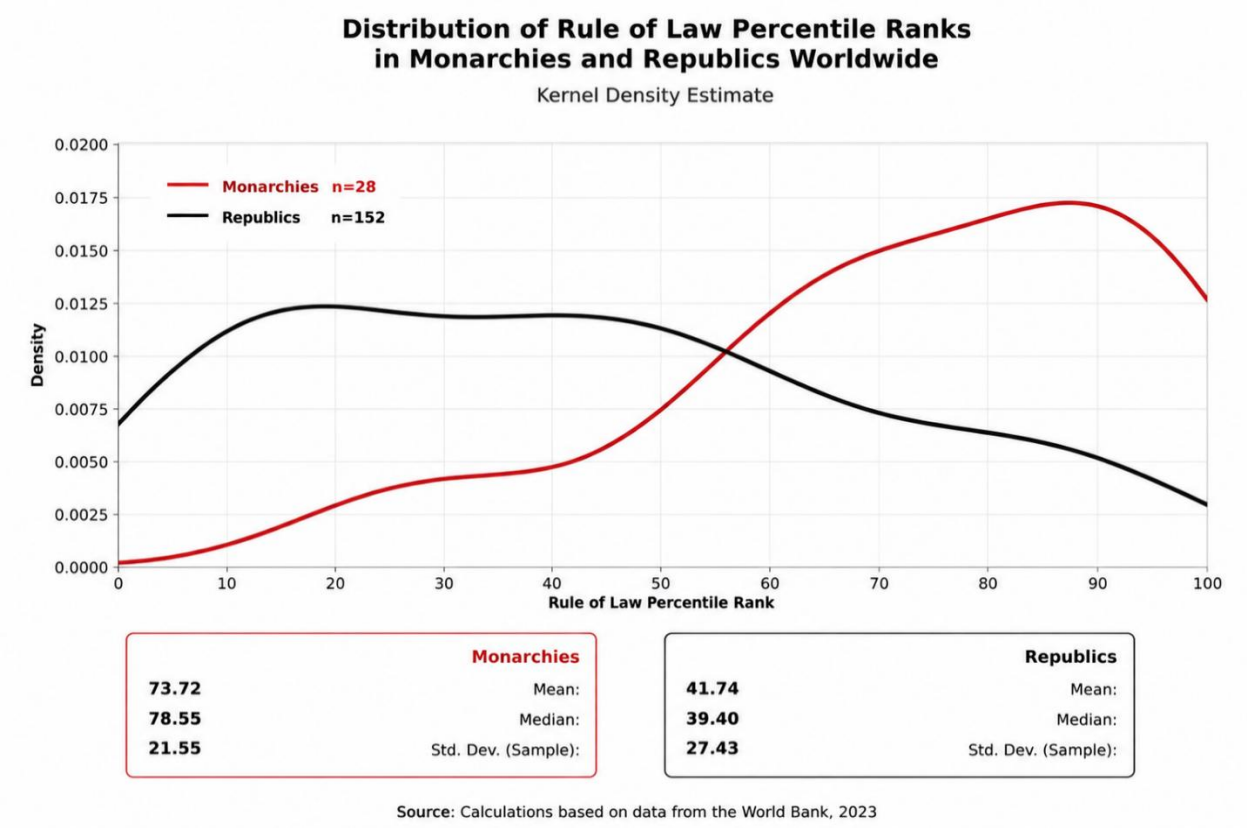
The statistical tests produced the following results:

Test	Result
Welch's t-test	**$p < 0.001$ → Significant**
Mann–Whitney U	**$p < 0.001$ → Significant**

Conclusion

Among countries with available data in 2023, monarchies had, on average, a higher percentile rank on the Rule of Law indicator than republics. The difference between the group means was approximately 31.87 percentile-rank points, and both statistical tests indicated a significant difference between the two groups. The substantially higher median among monarchies further suggests that the observed difference is not merely driven by a small number of exceptionally high-performing countries, but is also evident in the central part of the distribution.

This finding is consistent with the prediction of the **"New or Modern Monarchy" hypothesis** and indicates that the pattern previously observed for **Regulatory Quality** is repeated in the **Rule of Law** indicator. Nevertheless, as with the other comparisons in this study, this finding is based on observational data and **does not, by itself, establish a causal relationship between the form of government and the rule of law**. Levels of development, institutional history, social structures, political stability, and historical and geographical factors may also contribute to differences in countries' performance.



The distribution curve for monarchies (red) is noticeably shifted toward higher Rule of Law scores compared with the curve for republics (black). The greater concentration of the red curve in the higher-score range, together with the limited overlap between the two curves, indicates that monarchies generally performed better on this indicator in the 2023 data. Nevertheless, the two distributions still overlap to some extent, and exceptions can be observed in both groups. This pattern is also consistent with the results of the statistical tests presented in this section.

Comparison of the Rule of Law Index between Constitutional and Non-Constitutional Monarchies Worldwide

Among monarchies, the mean Rule of Law score is **78.21** for constitutional monarchies and **66.15** for semi-constitutional and absolute monarchies. The median is also higher among constitutional monarchies, at **89.60**, compared with **68.90** for the other group. These descriptive statistics indicate that constitutional monarchies, on average, achieved higher Rule of Law scores.

However, **Welch's t-test showed that the difference was not statistically significant**. Therefore, based on the available data, there is insufficient statistical evidence to conclude that the type of monarchy, among monarchies themselves, is associated with a significant difference in Rule of Law performance.

Comparison of the Rule of Law Index between Commonwealth Realms and Republics Worldwide

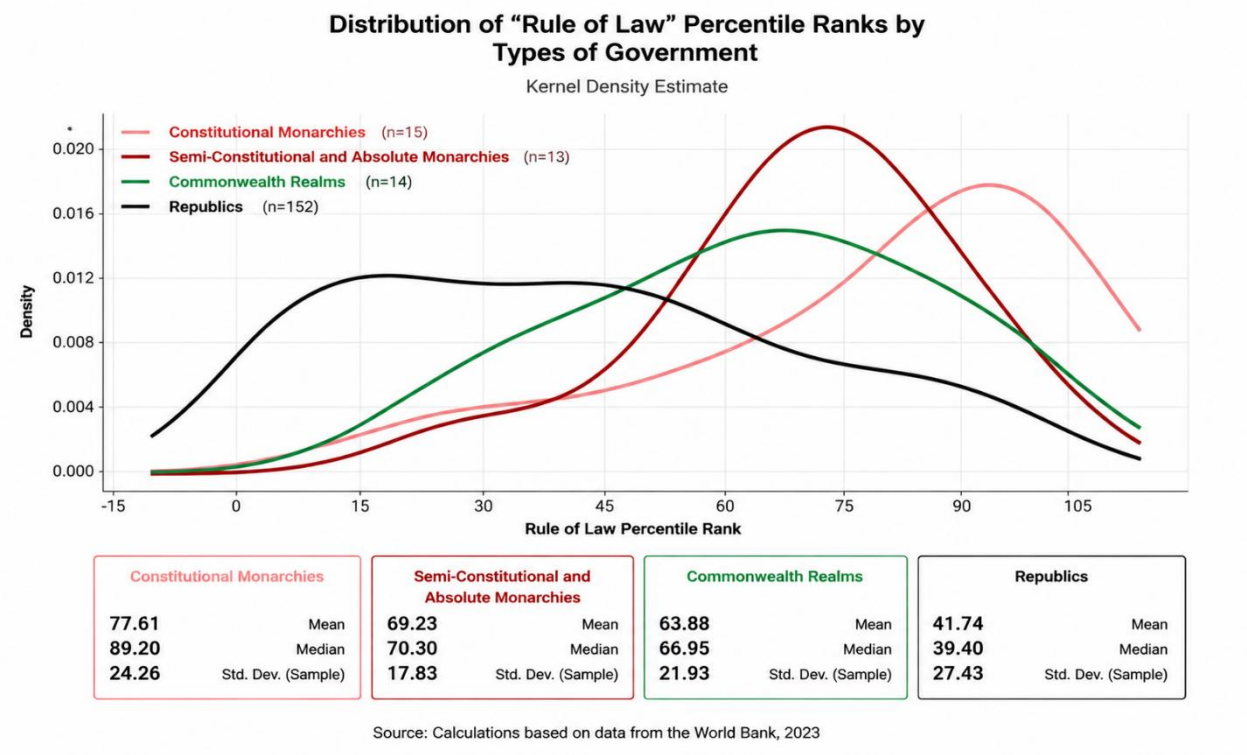
For the Rule of Law indicator, the **14 Commonwealth realms** have a mean of **63.88**, a median of **66.95**, and a standard deviation of **21.93**. In comparison, the **152 republics worldwide** have a mean of **41.74**, a median of **39.40**, and a standard deviation of **27.43**.

Thus, **the mean score of the Commonwealth realms is approximately 22.14 points higher than that of the republics. Welch's t-test indicates that this difference is statistically significant**:

t = 3.53, df = 16.99, p = 0.0026

The **95% confidence interval** for the mean difference ranges from **8.91 to 35.36 points**. Since this interval does not include zero, it is consistent with the result of the significance test. The effect size, measured by **Hedges' g, is 0.82**, indicating a relatively large difference between the two groups.

Overall, **the 2023 data show that the Commonwealth realms performed better than the world's republics on the Rule of Law indicator not only descriptively, but also by a statistically significant margin.**



The chart shows that **the distribution of scores for constitutional monarchies is shifted more strongly toward the higher end of the Rule of Law indicator than that of the other groups**. With a mean of **77.61** and a median of **89.20**, this group ranks highest among the four. **They are followed by semi-constitutional and absolute monarchies**, with a mean of **69.23**, and **Commonwealth realms**, with a mean of **63.88**. In contrast, **the distribution for republics is more concentrated in the middle and lower ranges of the indicator**, with a mean of only **41.74**.

At the same time, the overlap among the curves indicates that there is no clear-cut boundary separating the different types of government, and countries with varying levels of performance can be found within each group. Nevertheless, the overall rightward shift of the three monarchy-related distributions relative to that of the republics indicates that, in the 2023 data, these countries generally achieved higher Rule of Law scores. The lower dispersion among semi-constitutional and absolute monarchies also indicates that the scores within this group were more closely clustered around their mean, whereas republics displayed the greatest heterogeneity among the four groups.

Comparison of the Rule of Law Indicator Across Continents and the Middle East

Comparison of the Rule of Law Indicator in Asian Monarchies and Republics

Indicator: Rule of Law

A higher value on this indicator represents stronger rule of law, more effective enforcement of laws, and better-performing legal institutions.

Data Check

Group	Number of Countries with Data
Asian Monarchies	13
Asian Republics	33

1. Descriptive Statistics

Group	Number (n)	Mean	Median	Sample SD
Asian Monarchies	13	66.73	67.50	16.57
Asian Republics	33	33.21	32.50	24.83

Interpretation of Descriptive Statistics

The mean Rule of Law score is **66.73** for Asian monarchies and **33.21** for Asian republics.

The medians are:

- **Monarchies:** 67.50
- **Republics:** 32.50

This difference shows that, in this dataset, **Asian monarchies not only have a higher mean, but the central value of their distribution is also approximately twice that of the republics**. Thus, the observed difference is not merely driven by a few exceptionally high-performing countries; it is also evident in the central part of the distributions.

2. Welch's t-test for the Difference in Means

Hypotheses

Null hypothesis (H_0):
The mean Rule of Law score is equal in Asian monarchies and republics.

Alternative hypothesis (H_1):
The mean Rule of Law score differs between the two groups.

Welch's t-test Results

Item	Value
Mean — Monarchies	66.73
Mean — Republics	33.21
Difference in Means	33.52
t-statistic	5.314
Approximate degrees of freedom	32.96
p-value	0.0000073

Interpretation of Welch's t-test

The p-value is far below the **5% significance level**.

Therefore, the null hypothesis is rejected.

This result indicates that the difference in the mean Rule of Law scores between Asian monarchies and republics is statistically significant.

3. Mann–Whitney U Test

To examine whether the overall distributions of the two groups differ without relying on the assumption of normality, a **Mann–Whitney U test** was also conducted.

Mann–Whitney U Test Results

Item Value

U-statistic 379

p-value 0.000063

Interpretation of the Mann–Whitney U Test

The p-value is below the **5% significance level**.

Therefore, the null hypothesis is rejected.

This test also indicates a statistically significant difference between the distributions of Rule of Law scores in Asian monarchies and republics.

The agreement between the results of these two independent tests indicates that the observed difference is not limited to the group means but is also evident in the overall pattern of the distributions.

Summary

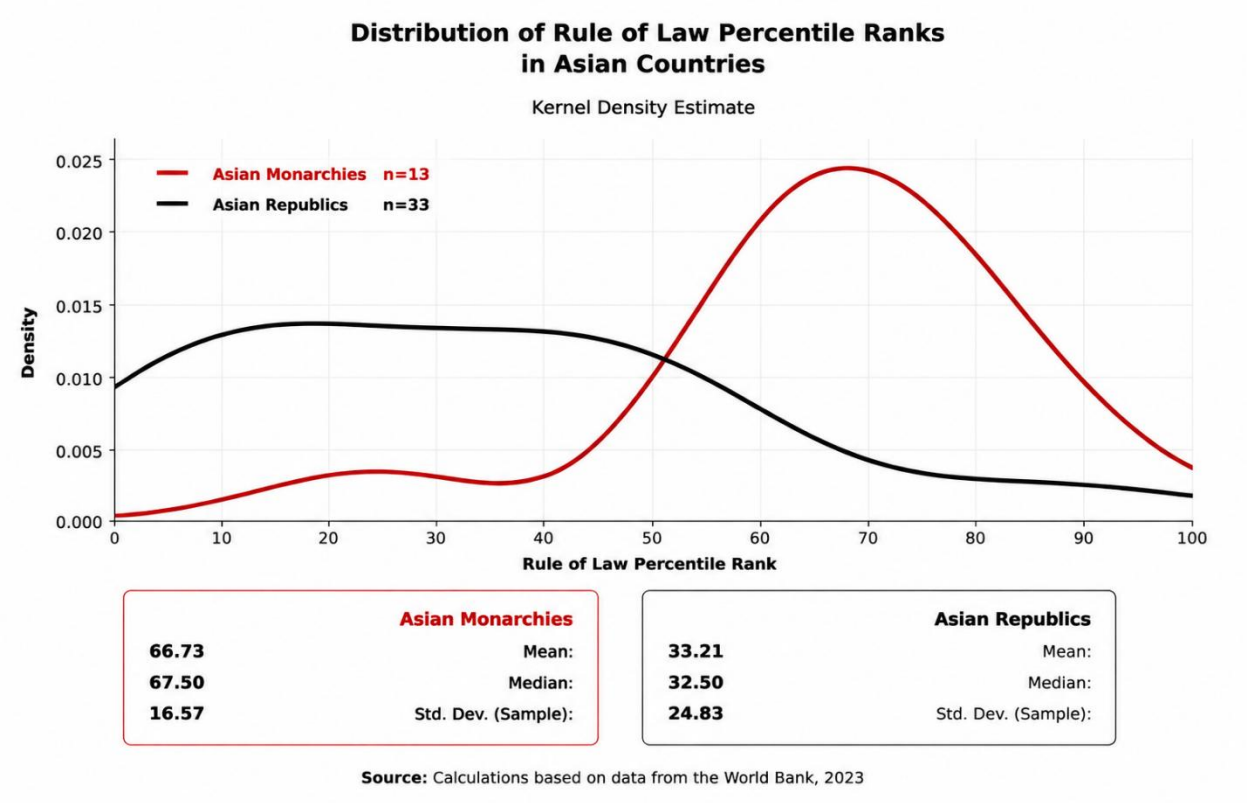
Group	Mean Rule of Law Score
Asian Monarchies	66.73
Asian Republics	33.21

Test results:

Test	Result
Welch's t-test	p = 0.0000073 → Statistically significant
Mann–Whitney U	p = 0.000063 → Statistically significant

Final conclusion:

Among Asian countries with available data, monarchies have, on average, higher Rule of Law scores than republics, and the difference is statistically significant according to both statistical tests.



The distribution curve for **Asian monarchies** is noticeably shifted toward higher Rule of Law scores, while the curve for **Asian republics** is more concentrated in the lower and middle ranges of the index. The greater dispersion of the republics' curve also indicates more heterogeneous performance within this group.

Overall, the figure confirms the **descriptive advantage of Asian monarchies**, a pattern that is also consistent with the statistically significant result of **Welch's t-test** and the **large effect size**.

Comparison of the Rule of Law Indicator Between Monarchies and Republics in Europe

For this indicator, higher values represent stronger rule of law, more effective enforcement of laws, and better-performing legal institutions.

Data Check — Europe

Group	Number of Countries with Data
European Monarchies	11
European Republics	34

1. Descriptive Statistics

Group	Number (n)	Mean	Median	Sample SD
European Monarchies	11	91.80	93.40	7.00
European Republics	34	66.23	67.00	25.04

Interpretation of Descriptive Statistics

The mean Rule of Law score is **91.80** for European monarchies and **66.23** for European republics.

The medians are:

- **Monarchies:** 93.40
- **Republics:** 67.00

This difference indicates that **European monarchies occupy a higher position in terms of the Rule of Law indicator both in their mean and at the center of their distribution**.

The lower dispersion among monarchies also indicates that countries in this group have relatively more similar Rule of Law performance, whereas European republics display a much wider range of outcomes.

2. Welch's t-test for the Difference in Means

The hypotheses were defined as follows:

Null hypothesis (H_0):
The mean Rule of Law score is equal in European monarchies and republics.

Alternative hypothesis (H_1):
The mean Rule of Law score differs between the two groups.

Welch's t-test Results

Item	Value
Mean — Monarchies	91.80
Mean — Republics	66.23
Difference in Means	**25.57**
t-statistic	**5.12**
Approximate degrees of freedom	**38.9**
p-value	**0.000008**

Interpretation of Welch's t-test

The p-value is far below the conventional **5% significance level**.

Therefore, the null hypothesis is rejected.

This result indicates that **the difference in mean Rule of Law scores between European monarchies and republics is statistically significant**.

3. Mann–Whitney U Test

To examine the difference between the two groups without relying on the assumption of normally distributed data, a **Mann–Whitney U test** was also conducted.

Mann–Whitney U Test Results

Item	Value
U-statistic	**280**
p-value	**0.0008**

Interpretation of the Mann–Whitney U Test

The p-value is below the **5% significance level**.

Therefore, the null hypothesis is rejected.

This test also indicates that **Rule of Law values in European monarchies and republics differ significantly in terms of their ranks**.

The consistency of the two independent tests is important: Welch's t-test identifies a statistically significant difference in means, while the Mann–Whitney U test provides significant evidence of a rank-based difference between the groups. Thus, the observed result is not limited to the difference in means alone.

Summary

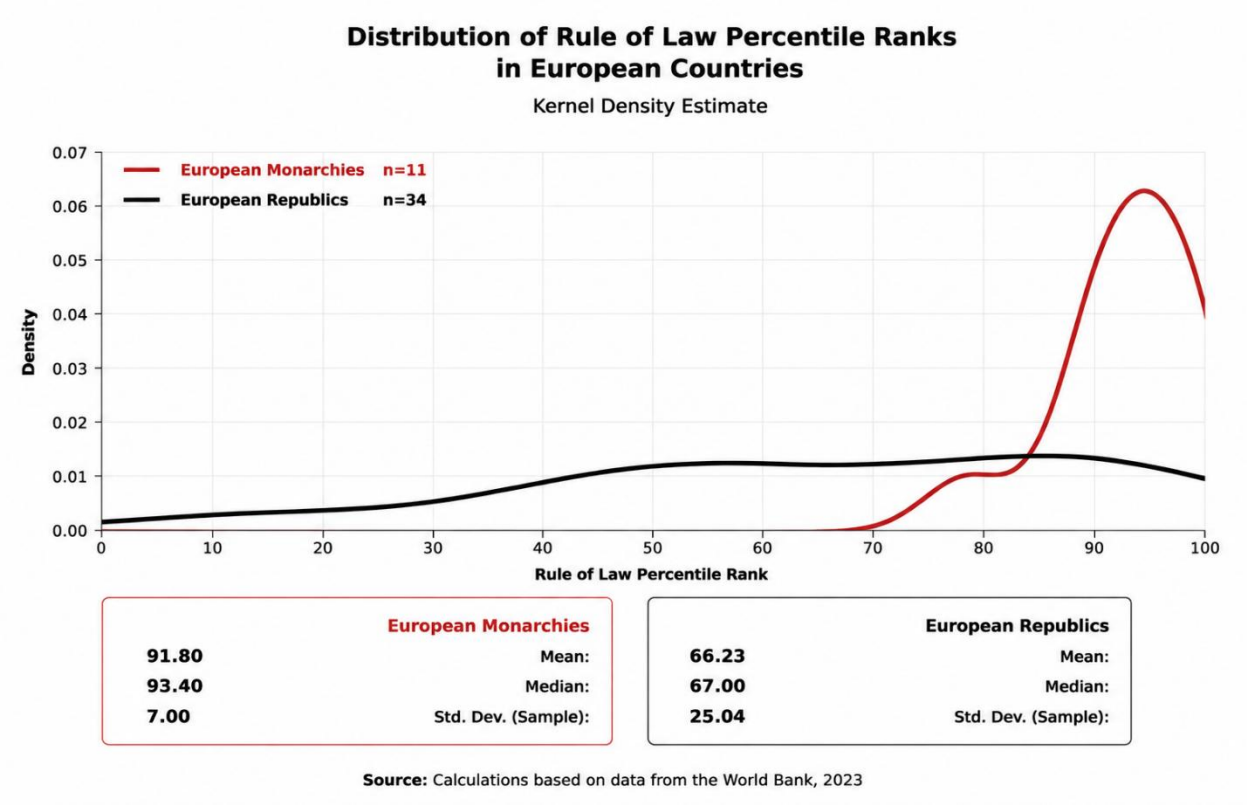
Group	Mean Rule of Law Score
European Monarchies	**91.80**
European Republics	**66.23**

The statistical tests produced the following results:

Test	Result
Welch's t-test	**p = 0.000008 → Statistically significant**
Mann–Whitney U	**p = 0.0008 → Statistically significant**

Final Conclusion

Among European countries with available data, monarchies have, on average, higher Rule of Law scores than republics, and the difference is statistically significant according to both statistical tests.



The distribution curve for **European monarchies** is concentrated entirely in the upper range of the Rule of Law indicator. Compared with the curve for European republics, it is both **shifted to the right and considerably less dispersed**. In contrast, the distribution for republics covers a much wider range, extending from countries with very high scores to those with relatively low scores.

This pattern indicates that European monarchies not only have a **higher average level of Rule of Law**, but also display **greater consistency and homogeneity** in their performance on this indicator. The figure is therefore consistent with the results of **Welch's t-test and the large effect size**, and confirms the descriptive advantage of European monarchies over the republics of the continent.

Europe Robustness Test: Rule of Law

To examine the role of the **communist legacy** in the observed difference between European monarchies and republics, republics that had previously experienced communist rule were excluded from this comparison. In this robustness test, **11 European monarchies** were compared with **13 European republics that had never experienced communist rule**.

Indicator	European Monarchies	Republics with No Communist History
Number of countries	**11**	**13**
Mean	**92.62**	**84.95**
Median	**93.40**	**91.50**
Sample standard deviation	**6.08**	**14.47**

The mean Rule of Law score is **7.67 points higher** in European monarchies. The scores are also considerably less dispersed among monarchies: the standard deviation among the republics is more than twice that of the monarchies.

Welch's t-test

The two-tailed Welch's t-test yielded:

t = 1.739, df = 16.65, p = 0.1005

Thus, at the conventional 5% significance level, **the difference between the two group means is not statistically significant**.

The Hedges' g effect size is:

g = 0.647

indicating a **moderate-to-large effect in favor of the monarchies**.

Thus, excluding republics with a history of communist rule reduces the gap between European monarchies and republics, but does not eliminate it. Monarchies still have a higher mean, and the magnitude of the difference remains practically notable. However, given the number of countries available for this comparison, the evidence is insufficient to establish statistical significance in a two-tailed test.

One statistical point is also worth noting. If the book's directional hypothesis—**that Rule of Law is higher in monarchies**—had been specified in advance, the **one-tailed p-value would be approximately 0.05025**. This is extremely close to, but still slightly above, the 5% threshold. It is therefore preferable not to describe the result as statistically significant even under the one-tailed test.

Nevertheless, this restricted comparison should not replace the main comparison. From a methodological standpoint, excluding European republics with a communist past from the overall analysis is not warranted, because these countries constitute a real part of the historical experience of republican government in Europe. Removing them could introduce a form of **selection bias** by excluding unfavorable cases from the historical record of republics. Within the analytical framework of this book, the movement of some republics toward communism, the

establishment of one-party rule, and the institutional legacies left by those experiences are not treated simply as external factors wholly independent of the form of government; rather, they are considered part of the historical trajectories and possible outcomes of the republican experience. Republics with a communist past should therefore remain in the main comparison.

Comparing non-communist republics with monarchies is useful only as a **supplementary robustness check or sensitivity analysis**. It shows how much the observed gap changes when countries with that particular historical background are excluded. The results of this supplementary test indicate that part of the large difference between European monarchies and republics is associated with the lower performance of republics with a communist past, but the **descriptive advantage of monarchies remains even after those countries are excluded**.

At the same time, this supplementary analysis does not, by itself, establish that republican government directly caused countries to become communist. Rather, it shows that removing such historical trajectories from an assessment of political systems would provide an incomplete picture of their actual historical experience.

Comparison of the Rule of Law Indicator in Monarchies and Republics in Africa, the Americas, and Oceania

Continent	Country Group	Number	Mean	Median	Sample SD
Africa	Monarchies	3	36.97	34.00	10.47
Africa	Republics	50	27.71	24.75	19.58
Americas	Commonwealth Realms	9	61.26	66.00	17.53
Americas	Republics	26	40.35	38.45	24.62
Oceania	Monarchy and Commonwealth Realms	6	68.33	74.55	26.91
Oceania	Republics	9	62.48	68.40	18.35

In **Africa**, the mean score for monarchies is approximately **9.25 points higher** than that of republics. The median is also higher among monarchies, while their lower dispersion suggests relatively more homogeneous performance, although the number of African monarchies is very small.

In the **Americas**, the Commonwealth realms occupy a clearly higher position. Their mean is approximately **20.90 points higher**, and their median **27.55 points higher**, than those of the republics. The republics also exhibit greater dispersion.

In **Oceania**, the monarchy and Commonwealth realms still have higher mean and median scores, although the gap between the two groups is smaller. The difference in means is approximately **5.86 points**, while the difference in medians is **6.15 points**. In this region, however, dispersion is greater among the monarchy and Commonwealth group than among the republics.

Overall, in all three regions, both the mean and median Rule of Law scores are higher for monarchies or Commonwealth realms than for republics. The magnitude of this descriptive advantage, however, is greatest in the Americas and smallest in Oceania.

Comparison of the Rule of Law Indicator in Middle Eastern Monarchies and Republics

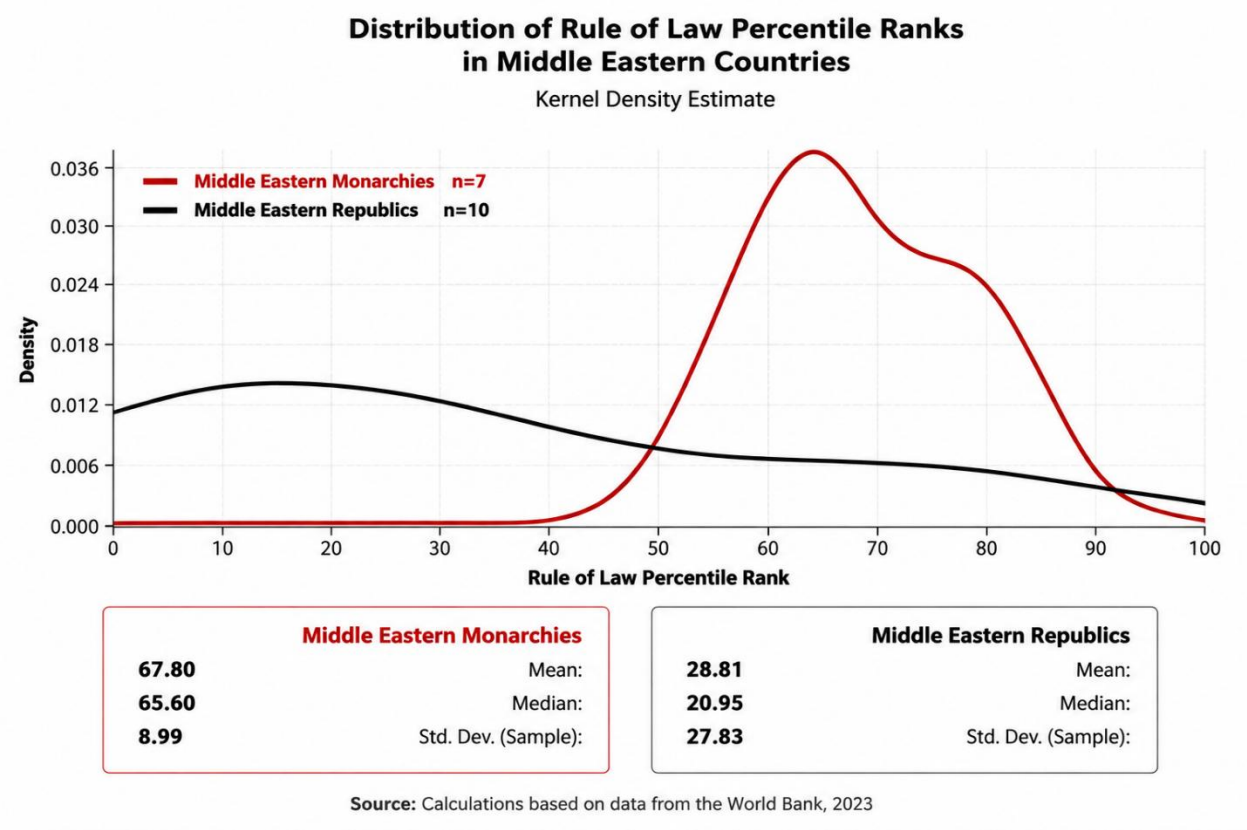
Country Group	Number	Mean	Median	Sample SD
Middle Eastern Monarchies	7	**67.80**	**65.60**	**8.99**
Middle Eastern Republics	10	**28.81**	**20.95**	**27.83**

The mean Rule of Law score in Middle Eastern monarchies is **38.99 points higher** than in the region's republics. The median is also approximately **44.65 points higher** among monarchies. In addition, the substantially lower standard deviation among monarchies indicates considerably more homogeneous performance, whereas Middle Eastern republics display much greater variation.

Welch's t-test produced the following results:

- **t = 4.133**
- **df = 11.502**
- **p = 0.0015**
- **Mean difference = 38.99 points**
- **95% confidence interval for the mean difference: 18.33 to 59.65**
- **Hedges' g = 1.66**

Thus, **the difference between the two groups is highly statistically significant**, and the very large effect size indicates that the gap between Middle Eastern monarchies and republics is substantial rather than marginal. Within this regional comparison, **the monarchies not only have markedly higher mean and median Rule of Law scores, but also display considerably greater consistency and homogeneity in their performance**. Nevertheless, this comparison describes the observed difference in performance between the two groups and **does not, by itself, establish a causal relationship**.



The distribution curve for **Middle Eastern monarchies** is clearly concentrated in the higher ranges of the Rule of Law Index, whereas the curve for **republics** is spread mainly across lower scores and exhibits greater dispersion. This pattern is consistent with the substantial difference between the mean scores of the two groups and the **very large effect size** obtained in the statistical analysis.

Status of the "New or Modern Monarchy" Hypothesis After the Sixth Indicator

Summary:
The **"New or Modern Monarchy" hypothesis** has so far been evaluated through six independent tests. Across all six, the findings have been consistent with the predictions of the hypothesis, and no result has contradicted it. However, since additional indicators will be examined in the following sections, a final assessment of the hypothesis will be deferred until all tests in the book have been completed.

Test	Indicator	Result
✓	GDP per Capita	Consistent with the hypothesis
✓	Purchasing Power per Capita	Consistent with the hypothesis

Test	Indicator	Result
◐	Gini Coefficient	Mixed evidence; difference favors the hypothesis
✓	Political Stability and Absence of Violence	Consistent with the hypothesis
✓	Regulatory Quality	Consistent with the hypothesis
✓	Rule of Law	Consistent with the hypothesis

Government Effectiveness

We now turn to an indicator that may capture the **"protective" dimension of government** more clearly than almost any other. If the **"New or Modern Monarchy" hypothesis** is correct, governments with greater institutional stability would be expected to maintain higher-quality public services even during periods of crisis and to prevent political and economic turmoil from excessively disrupting the lives of citizens, **particularly the most vulnerable groups**.

To assess this dimension, the World Bank provides an indicator known as **Government Effectiveness**. This indicator evaluates the quality of public services, the effectiveness of the civil service, the quality of policy implementation, the relative independence of public services from political pressures, and the credibility of the government's commitment to implementing its policies. Put simply, the indicator reflects the extent to which a government is capable of continuing to provide essential services to its citizens without serious disruption, even under difficult circumstances.

From the perspective of the **"New or Modern Monarchy" hypothesis**, this indicator is particularly important. Just as responsible parents seek, as far as possible, to shield their children from the pressures and difficulties of life, a government that does not see its role merely as the exercise of power, but rather regards itself as a guardian of the welfare and security of its citizens, would be expected to make its greatest effort during times of crisis to preserve the quality of public services and protect the most vulnerable segments of society.

Like the other World Bank governance indicators examined in this book, Government Effectiveness is reported as a **percentile rank**. Thus, a country with a percentile rank of 80 has performed better in government effectiveness than approximately 80 percent of the countries in the world.

We can now examine the average percentile rank of **Government Effectiveness** among the world's monarchies and republics and determine whether the results are consistent with the predictions of the **"New or Modern Monarchy" hypothesis**.

The data used in this section are from the World Bank for 2023 (Appendix RDS-G01).

Comparison of the "Government Effectiveness" Indicator Between Monarchies and Republics Worldwide

Indicator: Government Effectiveness
Data year: 2023
Source: World Bank data

For this indicator, a higher value represents better government performance in terms of the quality of public services, the quality of the civil service, policy implementation, and the effectiveness of the executive system.

Data Check

Group	Countries with Available Data
Monarchies	28
Republics	151

1. Descriptive Statistics

Group	n	Mean	Median	Sample SD
Monarchies	28	74.63	80.20	24.02
Republics	151	42.03	40.35	26.93

Interpretation of Descriptive Statistics

The mean Government Effectiveness score is **74.63 for monarchies** and **42.03 for republics**.

The medians are:

- Monarchies: **80.20**
- Republics: **40.35**

These differences indicate that, in this dataset, **monarchies exhibit higher levels of government effectiveness both in terms of the mean and the central tendency of the distribution**.

The substantial gap between the medians also indicates that the observed difference is not merely driven by a few exceptionally high-performing countries, but is also evident in the main body of the distributions.

2. Welch's Test for the Difference in Means

Hypotheses

Null hypothesis (H_0):
The mean Government Effectiveness score is equal in monarchies and republics.

Alternative hypothesis (H_1):
The mean Government Effectiveness score differs between the two groups.

Welch's Test Results

Item	Value
Mean — Monarchies	74.63
Mean — Republics	42.03
Difference in means	32.60
t-statistic	6.44
Approximate degrees of freedom	39.8
p-value	< 0.001

Interpretation of Welch's Test

The p-value is far below the conventional **5% significance level**. The null hypothesis is therefore rejected.

This result indicates that the difference in mean Government Effectiveness between monarchies and republics worldwide is statistically significant.

In other words, **within this dataset, monarchies have a higher average Government Effectiveness score than republics**.

3. Mann–Whitney U Test

To examine differences in the overall distributions without relying on an assumption of normality, the **Mann–Whitney U test** was also conducted.

Mann–Whitney U Test Results

Item	Value
U statistic	approximately 3,420
p-value	< 0.001

Interpretation of the Mann–Whitney U Test

This test likewise indicates that **the distributions of Government Effectiveness scores differ significantly between monarchies and republics**.

The agreement between the results of these two independent statistical tests suggests that the observed difference is not limited to a difference in means, but is also evident in the overall pattern of the distributions.

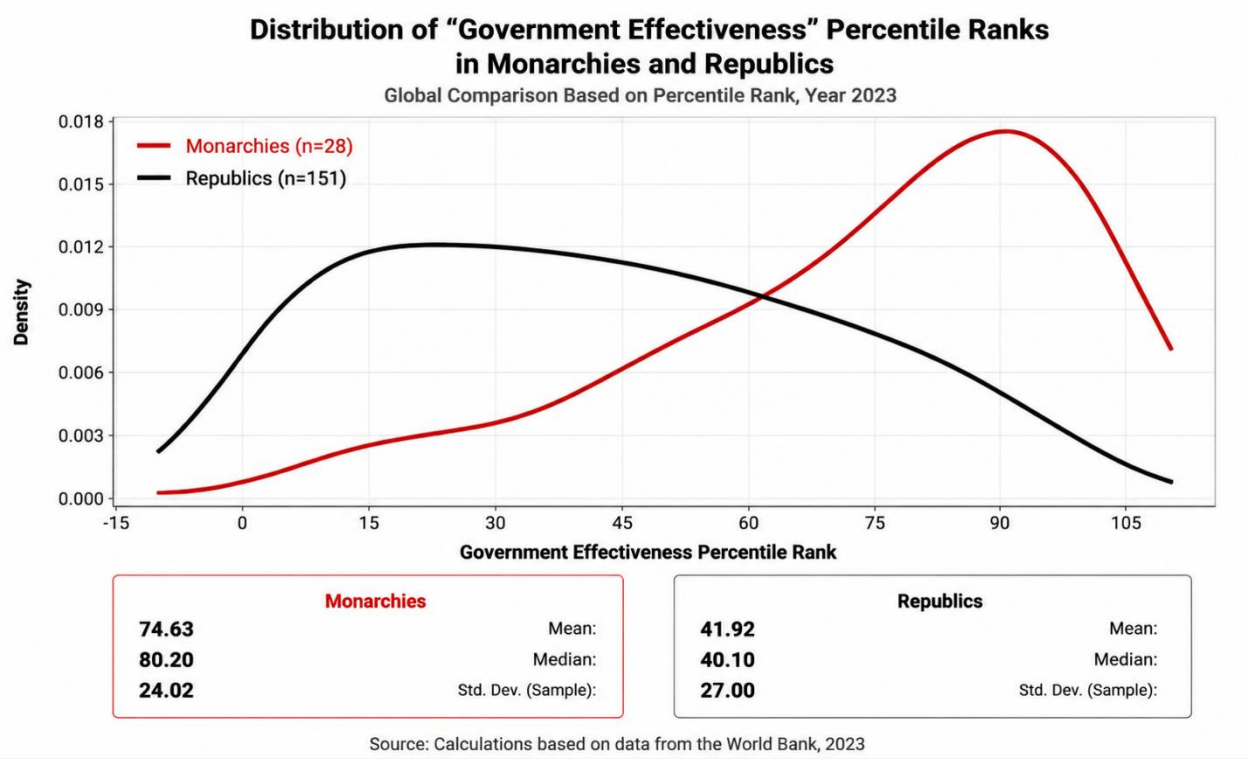
Summary

Group	Mean Government Effectiveness
Monarchies	74.63
Republics	42.03

Test	Result
Welch's t-test	Significant difference (**p < 0.001**)
Mann–Whitney U	Significant difference (**p < 0.001**)

Final Result

Among countries with available data in 2023, monarchies have, on average, higher Government Effectiveness scores than republics, and the difference is statistically significant in both statistical tests.



The distribution curve for monarchies is noticeably shifted toward the right side of the chart, indicating that these countries are concentrated at higher percentile ranks on the **Government Effectiveness** indicator. In contrast, the distribution for republics is concentrated more heavily in the left half of the chart and also shows greater dispersion.

The limited overlap between the two curves indicates that, although both groups include countries with relatively weak or strong performance, monarchies as a whole tend to occupy

substantially higher levels of government effectiveness. This visual pattern is consistent with the statistical test results and suggests that the observed difference between the two groups is not driven merely by a few exceptional countries, but is also evident in the overall distribution of the data.

Comparison of the "Government Effectiveness" Indicator Between Constitutional and Non-Constitutional Monarchies Worldwide

To examine differences in performance among types of monarchy, monarchies were divided into two groups: **constitutional monarchies** and **semi-constitutional and absolute monarchies**. Descriptive statistics for the two groups, based on their 2023 percentile ranks on the Government Effectiveness indicator, are presented below:

Type of Monarchy	Number of Countries	Mean Percentile Rank	Median	Sample SD
Constitutional monarchies	15	78.14	84.40	24.36
Semi-constitutional and absolute monarchies	13	70.58	74.50	23.93

The mean percentile rank for Government Effectiveness is **78.14** among constitutional monarchies and **70.58** among semi-constitutional and absolute monarchies. Thus, the mean for the first group is approximately **7.56 percentile points** higher than that of the second group. The median is also higher among constitutional monarchies: **84.40** compared with **74.50**. The degree of dispersion is very similar in the two groups, with a standard deviation of approximately 24 percentile points in each.

Test of the Difference Between Means

To determine whether the observed difference between the two groups is statistically significant, **Welch's t-test** was used.

Statistical Measure	Value
Difference between group means	7.56
Welch's t-statistic	0.826
Approximate degrees of freedom	25.56
p-value	0.416
95% CI for the difference between means	−11.26 to 26.37
Hedges' g	0.30

The p-value is **0.416**, which is greater than the 0.05 significance level. Therefore, the null hypothesis of equal means cannot be rejected, and **the observed difference between constitutional monarchies and semi-constitutional and absolute monarchies is not statistically significant**.

The 95% confidence interval extends from approximately **11.26 percentile points in favor of the second group** to **26.37 percentile points in favor of the first group**, and includes zero. Thus, the available data do not provide sufficient evidence to establish a consistent difference between the mean scores of the two groups.

Hedges' g is **0.30**, indicating a small effect size. This suggests that **although constitutional monarchies have a numerical advantage in both the mean and the median, the difference is relatively small compared with the within-group variation**.

Summary

On the Government Effectiveness indicator, constitutional monarchies have higher mean and median scores than semi-constitutional and absolute monarchies. However, the statistical test indicates that the difference is not significant, and the effect size is small. On the basis of these data, therefore, **we cannot conclude that either type of monarchy is systematically more effective than the other**.

At the same time, this result indicates that the relatively strong performance of monarchies as a whole compared with republics is not confined to only one type of monarchy, since both groups of monarchies have relatively high mean percentile ranks on the Government Effectiveness indicator. This comparison, however, concerns the observed performance of the two groups and **does not by itself establish a causal relationship**.

Comparison of the "Government Effectiveness" Indicator Between Commonwealth Realms and Republics Worldwide

To examine the performance of Commonwealth realms, the percentile ranks on the Government Effectiveness indicator for **14 Commonwealth realms** were compared with those of **republics worldwide**. Descriptive statistics for the two groups are presented below.

Type of Government	Number of Countries	Mean Percentile Rank	Median	Sample SD
Commonwealth realms	14	58.24	58.75	23.48
Republics worldwide	402	48.59	48.10	29.34

The **mean percentile rank for Government Effectiveness among Commonwealth realms** is **58.24**, compared with **48.59** among republics worldwide. Thus, the mean for Commonwealth

realms is approximately **9.65 percentile points higher than that of republics**. The median is also higher among Commonwealth realms (**58.75**) than among republics (**48.10**). The standard deviation is somewhat lower among Commonwealth realms, suggesting slightly less variation in performance within this group.

Test of the Difference Between Means

To assess the statistical significance of this difference, **Welch's t-test** was used.

Statistical Measure	Value
Difference between group means	9.65
Welch's t-statistic	1.497
Approximate degrees of freedom	14.45
p-value	0.156
95% CI for the difference between means	−4.14 to 23.43
Hedges' g	0.33

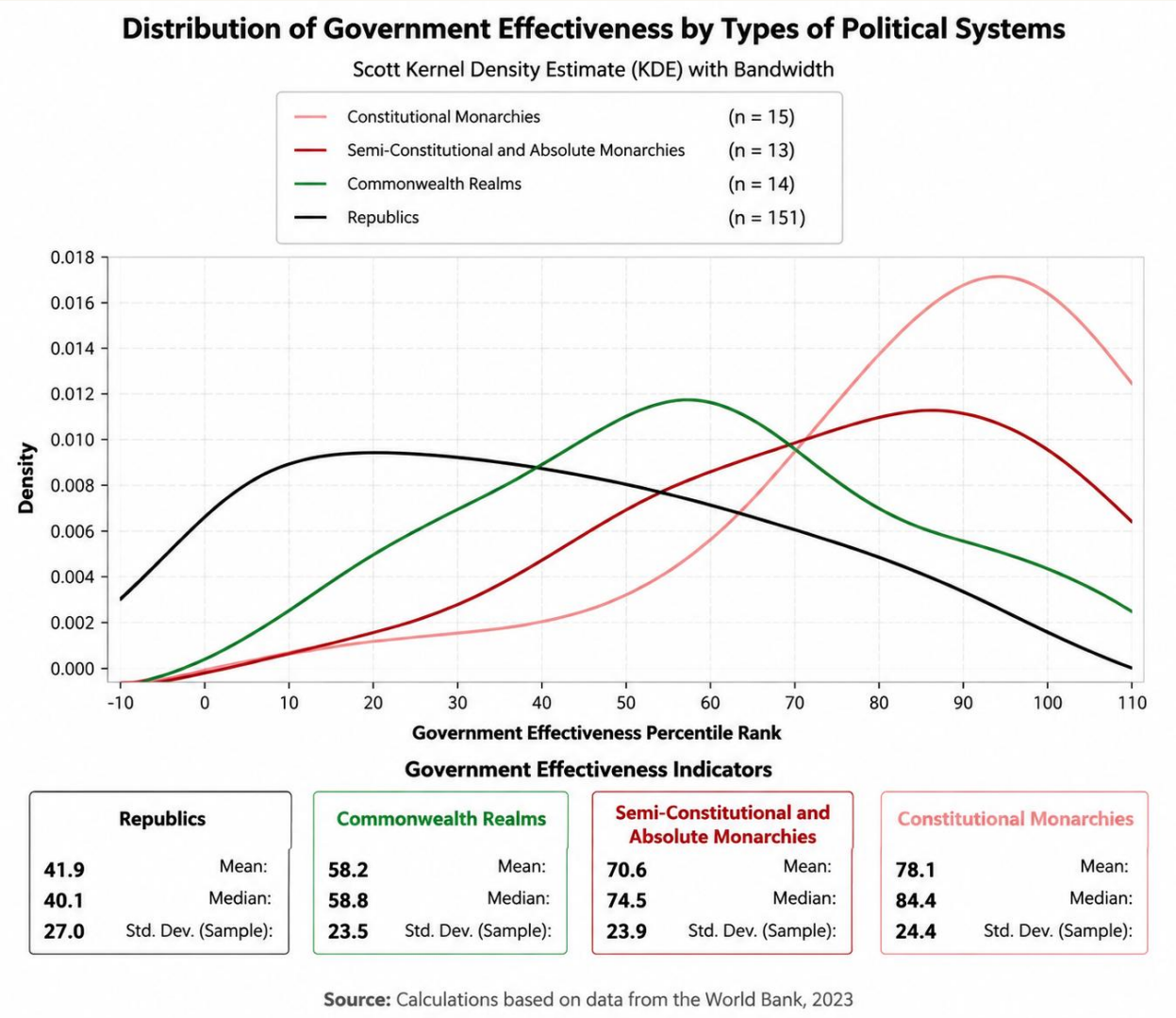
The p-value (**p = 0.156**) is greater than the 0.05 significance level. Therefore, the null hypothesis of equal means cannot be rejected, and the observed difference between Commonwealth realms and republics worldwide is not statistically significant.

The 95% confidence interval extends from approximately **4.14 percentile points in favor of republics** to **23.43 percentile points in favor of Commonwealth realms** and includes zero. The available data therefore do not provide sufficient evidence to establish a consistent difference between the two groups.

Hedges' g is **0.33**, which falls within the range of a **small effect**. Thus, although Commonwealth realms perform better on average than republics worldwide, the magnitude of the difference is relatively limited.

Summary

Commonwealth realms show higher mean and median Government Effectiveness scores than republics worldwide, but the statistical test indicates that this difference is not significant at the 5% level, and the effect size is small. On the basis of the available data, therefore, we cannot conclude that Commonwealth realms systematically outperform republics worldwide on this indicator. This comparison concerns the observed performance of the two groups and does not, by itself, establish a causal relationship.



The kernel density estimate shows that the distribution of the **Government Effectiveness** indicator for constitutional monarchies is generally concentrated at higher levels of the index, with the greatest density of observations occurring among high effectiveness scores. Semi-constitutional and absolute monarchies, although somewhat more dispersed, are also concentrated primarily in the upper half of the distribution.

In contrast, the distribution of republics is broader, with a substantial portion concentrated at lower levels of the indicator. Commonwealth realms are also concentrated mainly within the relatively high range of government effectiveness, and their distribution is closer to that of constitutional monarchies than to that of republics.

Overall, the shapes of the distributions indicate that high Government Effectiveness scores are more concentrated among the monarchy groups than among republics. However, any final

judgment regarding the statistical significance of these differences depends on the results of the statistical tests presented in the following analysis.

Comparison of the "Government Effectiveness" Indicator Between Monarchies and Republics Across Continents and in the Middle East

Comparison of Government Effectiveness Between Monarchies and Republics in Asia

To examine the performance of the two systems of government at the continental level, the percentile ranks on the Government Effectiveness indicator for **13 Asian monarchies** were compared with those of **33 Asian republics**. The data are from 2023.

Descriptive Statistics

Type of Government	Number of Countries	Mean Percentile Rank	Median	Sample SD
Asian monarchies	13	72.68	74.50	17.50
Asian republics	33	37.78	34.00	28.48

The mean percentile rank for Government Effectiveness is **72.68** among Asian monarchies and **37.78** among Asian republics. Thus, the mean for monarchies is approximately **34.90 percentile points** higher than that for republics.

The median shows the same pattern: half of the Asian monarchies have a percentile rank above **74.50**, whereas the median for Asian republics is only **34.00**. In addition, the standard deviation is substantially higher among republics. Thus, the performance of Asian republics is not only lower on average but also considerably more dispersed.

Test of the Difference Between Means

To determine whether the observed difference is merely the result of sampling variation or is supported statistically, **Welch's t-test** was used.

Statistical Measure	Value
Difference between group means	34.90
Welch's t-statistic	5.031
Approximate degrees of freedom	35.58
p-value	0.000014
95% CI for the difference between means	20.82 to 48.98
Hedges' g	1.32

The resulting p-value is approximately **0.000014**, far below the 0.05 significance level. The null hypothesis of equal means is therefore rejected, indicating that **the difference in Government Effectiveness between Asian monarchies and republics is highly statistically significant**.

The 95% confidence interval indicates that the mean for Asian monarchies is estimated to be between approximately **20.82 and 48.98 percentile points** higher than the mean for Asian republics. Because the entire interval favors monarchies and does not include zero, the result provides strong statistical evidence of a difference between the two groups.

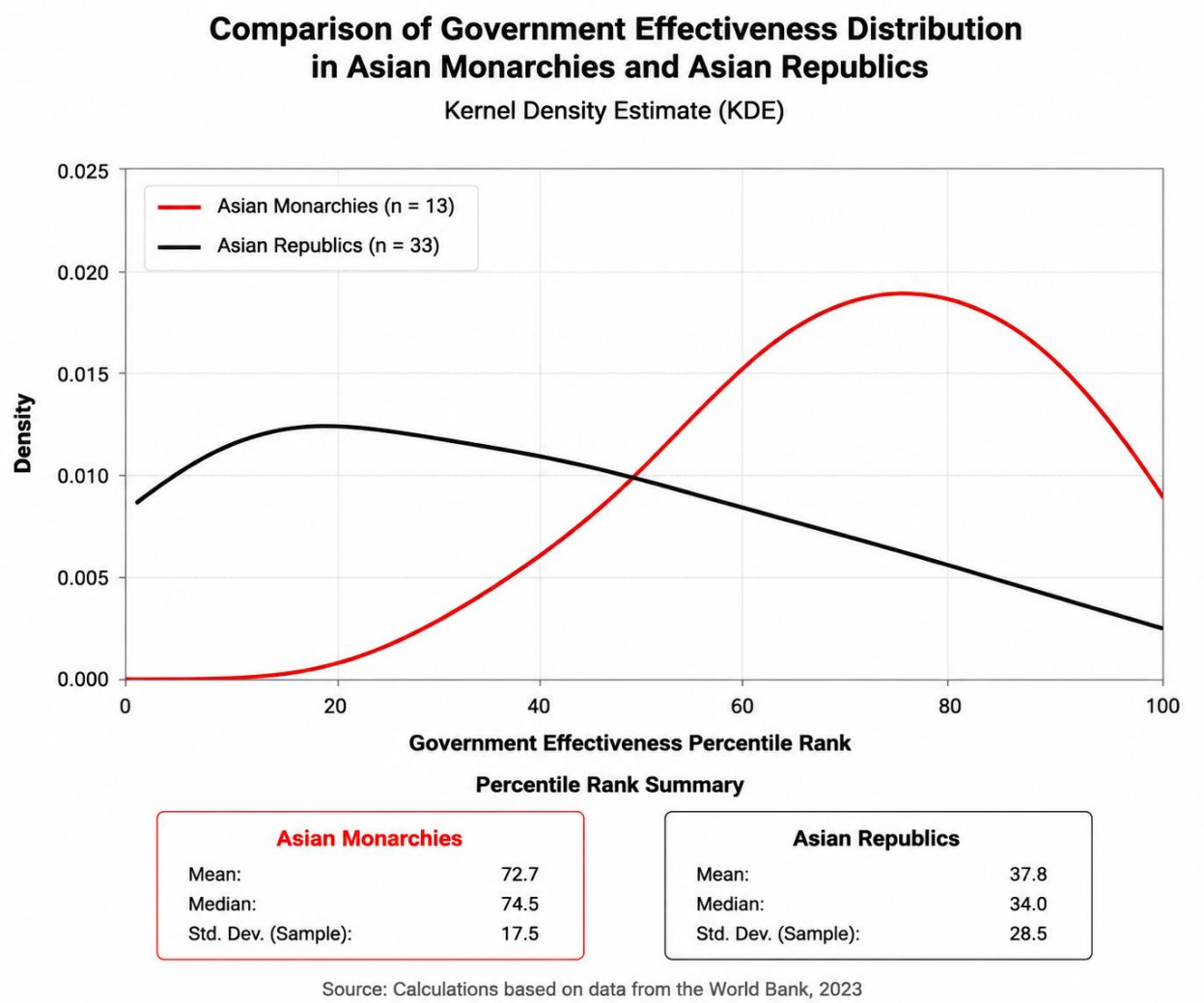
Hedges' g is **1.32**, indicating a **very large effect size**. This means that **the difference between the two groups is not only statistically significant but also substantial in practical terms**.

Summary

The Asian data present a clear picture of the difference in performance between the two groups. **Asian monarchies have substantially higher percentile ranks on the Government Effectiveness indicator in terms of both the mean and the median.** Their performance is also less dispersed, whereas Asian republics display a much wider range, extending from very weak to very strong performance.

Overall, **the evidence from this comparison is consistent with the "New or Modern Monarchy" hypothesis**: in Asia, countries that have retained a monarchical system have recorded substantially stronger performance in Government Effectiveness than republics. **The very large effect size further indicates that the observed difference is substantial in performance terms, rather than merely statistically detectable.**

This comparison, however, concerns the observed difference in performance between the two groups and **does not, by itself, establish a causal relationship**.



The kernel density estimate (KDE) plot shows that the distribution of the **Government Effectiveness** indicator for Asian monarchies is noticeably shifted toward higher percentile ranks, with the greatest concentration of this group occurring in the upper half of the distribution. In contrast, the distribution of Asian republics is more dispersed, with most observations concentrated in the lower and middle percentile ranks. The limited overlap between the two curves also suggests that a substantial proportion of Asian monarchies rank higher in government effectiveness than most Asian republics. However, the final assessment of the statistical significance of these differences is based on the results of the statistical tests presented in this chapter.

Comparison of Government Effectiveness in European Monarchies and Republics

To compare the performance of the two forms of government in Europe, the percentile ranks of the **Government Effectiveness** indicator for **11 European monarchies** were compared with those of **34 European republics**. The data are for 2023.

Descriptive Statistics

Form of Government	Number of Countries	Mean Percentile Rank	Median	Sample Standard Deviation
European Monarchies	11	92.11	95.80	7.85
European Republics	34	66.45	68.40	22.42

The mean percentile rank for Government Effectiveness is **92.11** among European monarchies and **66.45** among the republics of the continent. Thus, the mean for the monarchies is approximately **25.66 percentile points** higher than that of the republics.

The median confirms the same pattern. The median for European monarchies is **95.80**, meaning that at least half of these countries rank within approximately the **top five percent of countries worldwide** in terms of government effectiveness. By comparison, the median for European republics is **68.40**.

Another important difference concerns dispersion. The standard deviation among the monarchies is only **7.85**, compared with **22.42** among the republics. European monarchies therefore form a much more homogeneous group in addition to having a substantially higher mean. European republics span a much wider range, from republics with very high government effectiveness to countries with relatively weak performance.

Test of the Difference Between Means

To determine whether the difference between the two groups is statistically significant, **Welch's t-test** was used.

Statistical Measure	Value
Difference between group means	25.66
Welch's t-statistic	5.682
Approximate degrees of freedom	42.57
p-value	0.0000011
95% CI for the mean difference	16.55 to 34.77
Hedges' g	1.26

The p-value is approximately **0.0000011**, far below both the 0.05 and 0.001 significance levels. The null hypothesis of equal means is therefore rejected, and **the observed difference between European monarchies and republics is highly statistically significant**.

The 95% confidence interval indicates that the mean percentile rank of European monarchies is approximately **16.55 to 34.77 percentile points** higher than that of European republics. The

entire interval lies above zero, clearly indicating that the direction of the difference favors the monarchies.

Hedges' g is **1.26**, representing a **very large effect size**. Thus, **the difference between the two groups is not only statistically significant but also substantial in practical and performance terms**.

Interpretation of the Results

Europe is an especially important case because it includes several of the world's most successful republics, such as Switzerland, Finland, Ireland, Iceland, and Germany. Despite the presence of these highly successful republics, **the overall mean for European monarchies remains substantially higher**.

Part of the difference between the two group means is influenced by the greater heterogeneity among European republics. The republican group includes both Western and Northern European countries with very high government effectiveness and some republics in Eastern and Southeastern Europe with considerably lower percentile ranks. By contrast, **nearly all European monarchies are concentrated in the upper part of the global distribution**.

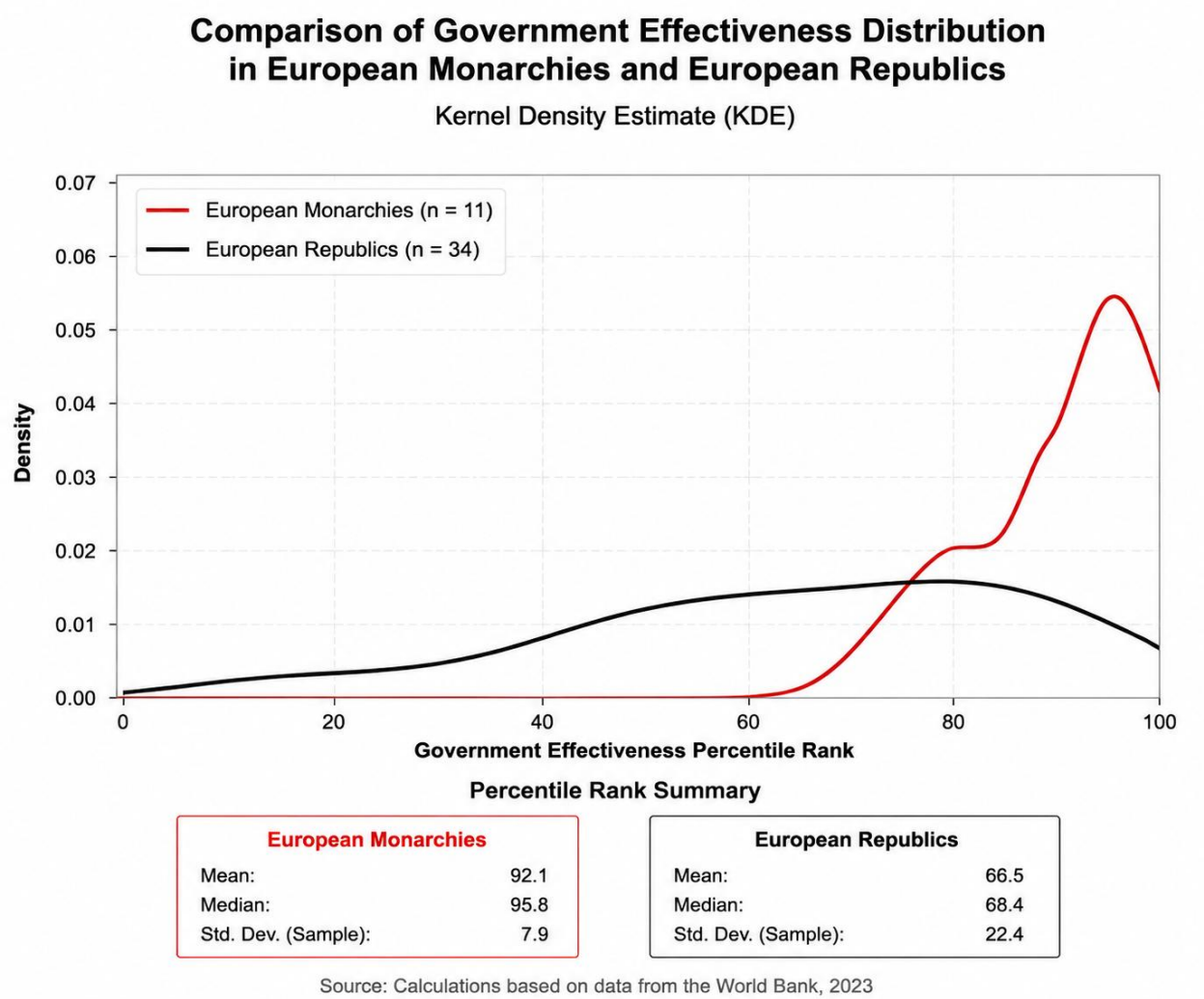
Importantly, the advantage of the monarchies is not driven solely by a few exceptional cases. Even the lower values among European monarchies are relatively high, and most of these countries have percentile ranks close to or above 90. This concentration at high levels produces both a very high mean and median and a remarkably low degree of dispersion among the monarchies.

Conclusion

The comparison of European countries shows that monarchies have recorded substantially stronger performance in Government Effectiveness than the republics of the continent as a whole. Monarchies have higher means and medians, less variation in performance, a highly statistically significant difference in means, and a very large effect size.

This finding is consistent with the "New or Modern Monarchy" hypothesis. European monarchies display very high and stable levels of government effectiveness not only relative to the global average, but even within a continent that includes many of the world's most successful republics.

This comparison, however, concerns the **observed difference in performance** between the two groups and **does not, by itself, establish a causal relationship**.



The kernel density estimate (KDE) plot shows that the distribution of the **Government Effectiveness** indicator for European monarchies is clearly concentrated at very high percentile ranks, with the greatest density occurring near the upper end of the indicator. In contrast, the distribution of European republics is more dispersed; although part of the distribution also lies at high percentile ranks, a substantial concentration is observed at intermediate levels. The relatively limited overlap between the two curves indicates that, overall, European monarchies have performed better than European republics in terms of government effectiveness. However, the final assessment of the statistical significance of this difference is based on the results of the statistical tests presented in this chapter.

Comparison of Government Effectiveness in European Monarchies and European Republics with No History of Communist Rule

Form of Government	Number of Countries	Mean	Median	Standard Deviation
European Monarchies	11	92.11	95.80	7.85

Form of Government	Number of Countries	Mean	Median	Standard Deviation
European Republics with No History of Communist Rule	13	83.05	85.40	12.73

The mean Government Effectiveness percentile rank of European monarchies is **9.06 percentile points higher**. Welch's t-test indicates that this difference is just within the conventional threshold of statistical significance:

Statistical Measure	Value
Welch's t-statistic	2.131
Degrees of freedom	20.30
p-value	0.046
95% confidence interval	0.20 to 17.91
Hedges' g	0.81

Thus, **even after excluding republics with a history of communist rule, European monarchies continue to have a higher mean level of government effectiveness. The difference is statistically significant at the 5% level, and the effect size is large**. However, the p-value is close to the significance threshold, and the result should therefore be interpreted in light of the relatively small sample sizes of the two groups. This comparison, by itself, does not establish a causal relationship.

Nevertheless, this restricted comparison should not replace the main comparison. From a methodological perspective, excluding European republics with a history of communist rule from the overall analysis is not justified, because these countries constitute a real part of the historical experience of republican government in Europe. Excluding them could introduce a form of **selection bias** by removing unfavorable cases from the performance record of republics. Within the analytical framework of this book, the movement of some republics toward communism, the establishment of one-party rule, and the institutional legacies that followed are not treated merely as external factors independent of the form of government, but as part of the historical trajectories and potential consequences of the republican experience itself. Republics with a history of communist rule should therefore remain in the main comparison.

Comparison of Government Effectiveness in Monarchies and Republics in Africa, the Americas, and Oceania

Because the number of independent monarchies in Africa, the Americas, and Oceania is limited, monarchies and Commonwealth realms, where applicable, are treated as a single group in this section and compared with republics on the same continent. Given the small sample sizes in some groups, these comparisons are primarily descriptive.

Africa

Descriptive Statistics

Form of Government	Number of Countries	Mean	Median	Standard Deviation
Monarchies	3	**28.60**	**19.30**	**18.59**
Republics	41	**26.89**	**21.20**	**20.35**

In Africa, the mean Government Effectiveness score of the continent's three monarchies is slightly higher than that of the republics, but the difference between the two groups is very small. Their medians are also close, and there is no substantial difference in the dispersion of their performance. Thus, unlike in Asia and Europe, **Africa shows no clear advantage for either form of government**.

The Americas

Descriptive Statistics

Form of Government	Number of Countries	Mean	Median	Standard Deviation
Commonwealth Realms	9	**60.70**	**61.80**	**14.77**
Republics	24	**41.08**	**37.95**	**20.45**

In the Americas, Commonwealth realms perform, on average, approximately **20 percentile points** better than republics. In addition to their higher mean, their median is substantially higher, while their lower standard deviation indicates more consistent performance. This finding is consistent with the pattern observed in Asia and Europe.

Oceania

Descriptive Statistics

Form of Government	Number of Countries	Mean	Median	Standard Deviation
Monarchies and Commonwealth Realms	6	**52.47**	**40.60**	**32.64**
Republics	8	**55.60**	**57.05**	**13.28**

In Oceania, republics have slightly higher means and medians than the group comprising monarchies and Commonwealth realms. However, performance within the monarchy/Commonwealth group is highly dispersed: countries such as Australia and New Zealand rank among the world's highest performers, whereas some other Commonwealth realms and the monarchy of Tonga have considerably lower percentile ranks. As a result, the pattern in Oceania appears to reflect the substantial heterogeneity within this group more than the form of government itself.

Conclusion

The comparison across these three continents shows that the pattern observed in Asia and Europe is not reproduced with equal strength throughout the world. In Africa, there is little difference between the two groups; in the Americas, Commonwealth realms perform substantially better and more consistently than republics; and in Oceania, the small number of countries and the considerable heterogeneity within the monarchy/Commonwealth group prevent a clear pattern from emerging.

These comparisons are therefore primarily descriptive and should be interpreted with caution. As elsewhere in this book, they show observed differences in group performance and **do not, by themselves, establish a causal relationship**.

Comparison of Government Effectiveness in Middle Eastern Monarchies and Republics

To compare the performance of the two forms of government in the Middle East, the region's **seven monarchies** were compared with **ten Middle Eastern republics** using World Bank data for 2023.

Descriptive Statistics

Form of Government	Number of Countries	Mean Percentile Rank	Median	Standard Deviation
Middle Eastern Monarchies	7	**73.20**	**74.50**	**14.95**
Middle Eastern Republics	10	**28.64**	**12.30**	**30.77**

The mean Government Effectiveness percentile rank is **73.20** for Middle Eastern monarchies and **28.64** for the region's republics. Thus, the mean for monarchies is approximately **44.56 percentile points higher**.

The median confirms the same pattern. The median for monarchies is **74.50**, compared with only **12.30** for republics. Moreover, the standard deviation among the republics is more than twice that of the monarchies, indicating considerably greater heterogeneity in the performance of Middle Eastern republics.

Test of the Difference Between Means

To assess the statistical significance of the difference between the two groups, **Welch's t-test** was used.

Statistical Measure	Value
Difference between group means	**44.56**
Welch's t-statistic	**3.960**
Approximate degrees of freedom	**13.75**
p-value	**0.00147**
95% CI for the mean difference	**20.38 to 68.74**
Hedges' g	**1.65**

The p-value (**p = 0.00147**) is far below both the 0.05 and 0.01 significance levels. The null hypothesis of equal means is therefore rejected, and **the observed difference is highly statistically significant**.

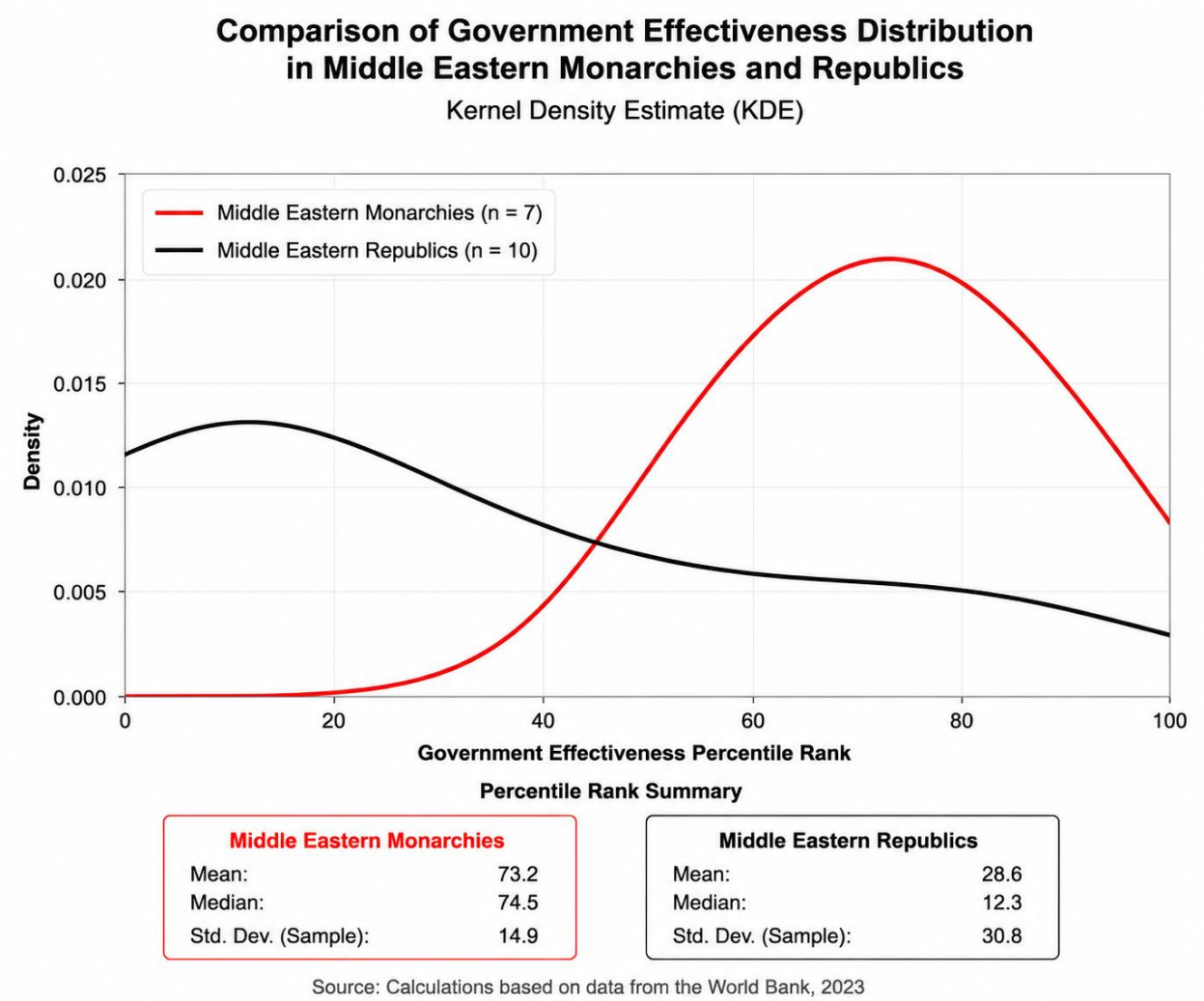
The 95% confidence interval indicates that the mean Government Effectiveness percentile rank of Middle Eastern monarchies is between **20.38 and 68.74 percentile points higher** than that of the region's republics. Because the entire interval lies above zero, the direction of the difference clearly favors the monarchies.

Hedges' g is **1.65**, indicating a **very large effect size**. Thus, **the difference between the two groups is not only statistically significant but also substantial in practical terms**.

Conclusion

The results show that Middle Eastern monarchies have performed substantially better than the region's republics on the Government Effectiveness indicator. Their mean and median percentile ranks are considerably higher, their performance is less dispersed, and the statistical test indicates a highly significant difference between the two groups. This finding is consistent with the "New or Modern Monarchy" hypothesis in the Middle East as well.

As in the other sections of this book, however, this comparison identifies an **observed difference in performance** between the two groups and **does not, by itself, establish a causal relationship**.



The kernel density estimate shows that the distribution of the Government Effectiveness Index for **Middle Eastern monarchies is concentrated predominantly at higher percentile ranks**, whereas the region's republics are more heavily concentrated in the lower and middle ranges. The substantial separation and limited overlap between the two curves indicate a marked difference in the distribution of this indicator across the two forms of government. Overall, **Middle Eastern monarchies perform better than the region's republics on the Government Effectiveness Index**.

Status of the "New or Modern Monarchy" Hypothesis After the Seventh Indicator

Now that the **Government Effectiveness** indicator has been examined using the same methodology, and its results also favor monarchies, the status of the hypothesis can be updated as follows:

Test	Indicator	Result
✓	GDP per Capita	Consistent with the hypothesis
✓	GDP per Capita, PPP	Consistent with the hypothesis
◐	Gini Coefficient	**Mixed evidence; difference favors the hypothesis**
✓	Political Stability and Absence of Violence	Consistent with the hypothesis
✓	Regulatory Quality	Consistent with the hypothesis
✓	Rule of Law	Consistent with the hypothesis
✓	Government Effectiveness	Consistent with the hypothesis

Inflation

One of the most important responsibilities of government is to maintain order in the economy. Stable rules, fiscal discipline, a reasonable degree of monetary-policy independence, and restraint from short-term populist decisions all contribute to price stability. Inflation can therefore be regarded as one of the most important outcomes of economic order in governance.

Conversely, when governments resort to undisciplined monetary and fiscal policies to finance budget deficits, cover political expenditures, or pursue short-term objectives, the usual result is higher inflation and reduced economic stability. The ability to control inflation can therefore serve as an important indicator of the quality of economic governance.

If the "New or Modern Monarchy" hypothesis is correct, and if this system can promote greater economic discipline by encouraging a longer-term decision-making horizon and reducing short-term political incentives, monarchies would be expected, on average, to experience lower inflation than republics. Conversely, if contemporary monarchies remain vulnerable to the autocracy and mismanagement associated with many historical monarchies, their average inflation rate would be expected to exceed that of republics.

To increase the **robustness** of the results and reduce the influence of **short-term fluctuations**, this analysis uses each country's average inflation rate over the five-year period from **2019 to 2023**, rather than the inflation rate for a single year. A multi-year average reduces the disproportionate influence of temporary shocks, such as economic crises or short-lived price surges, and provides a more accurate picture of the longer-term performance of different systems of government in maintaining price stability. Accordingly, throughout the remainder of this section, "inflation" refers to the **five-year average inflation rate for 2019–2023** (Appendix RDS-H01).

Global Comparison of Inflation in Monarchies and Republics

Descriptive Statistics

Form of Government	Number of Countries	Five-Year Mean Inflation (%)	Standard Deviation
Monarchies	25	**2.77**	**1.45**
Republics	148	**66.85**	**494.19**

The five-year mean inflation rate is **2.77%** for monarchies and **66.85%** for republics. At first glance, this difference appears enormous. However, the exceptionally large standard deviation among republics indicates that their mean is heavily influenced by a small number of countries experiencing extremely high inflation. As with many distributions containing extreme outliers, the mean alone therefore does not provide a complete picture of the difference between the two groups.

Welch's t-test

The two-tailed Welch's t-test produced:

t = −1.577, df = 147.02, p = 0.1168

Thus, **the difference between the group means is not statistically significant at the 5% level according to Welch's test**. The principal reason is the exceptionally high dispersion among republics.

Mann–Whitney U Test

To compare the two distributions without relying on the assumption of normally distributed data, a **Mann–Whitney U test** was also conducted:

U = 704, p < 0.000001

This result is highly statistically significant and indicates that inflation values among monarchies are systematically lower than those among republics.

If the directional hypothesis—that inflation is lower in monarchies than in republics—had been specified in advance, the corresponding one-tailed value would be approximately:

p = 0.00000038

Interpretation

To reduce the influence of fluctuations specific to any single year, this analysis compares the **average inflation rate over the five-year period from 2019 to 2023**. The indicator could be calculated for **25 monarchies** and **148 republics**.

The five-year mean inflation rate is only **2.77%** among monarchies, compared with **66.85%** among republics. This very large difference in means, however, must be interpreted cautiously because the republic group contains several countries with extremely high inflation rates that pull the mean sharply upward.

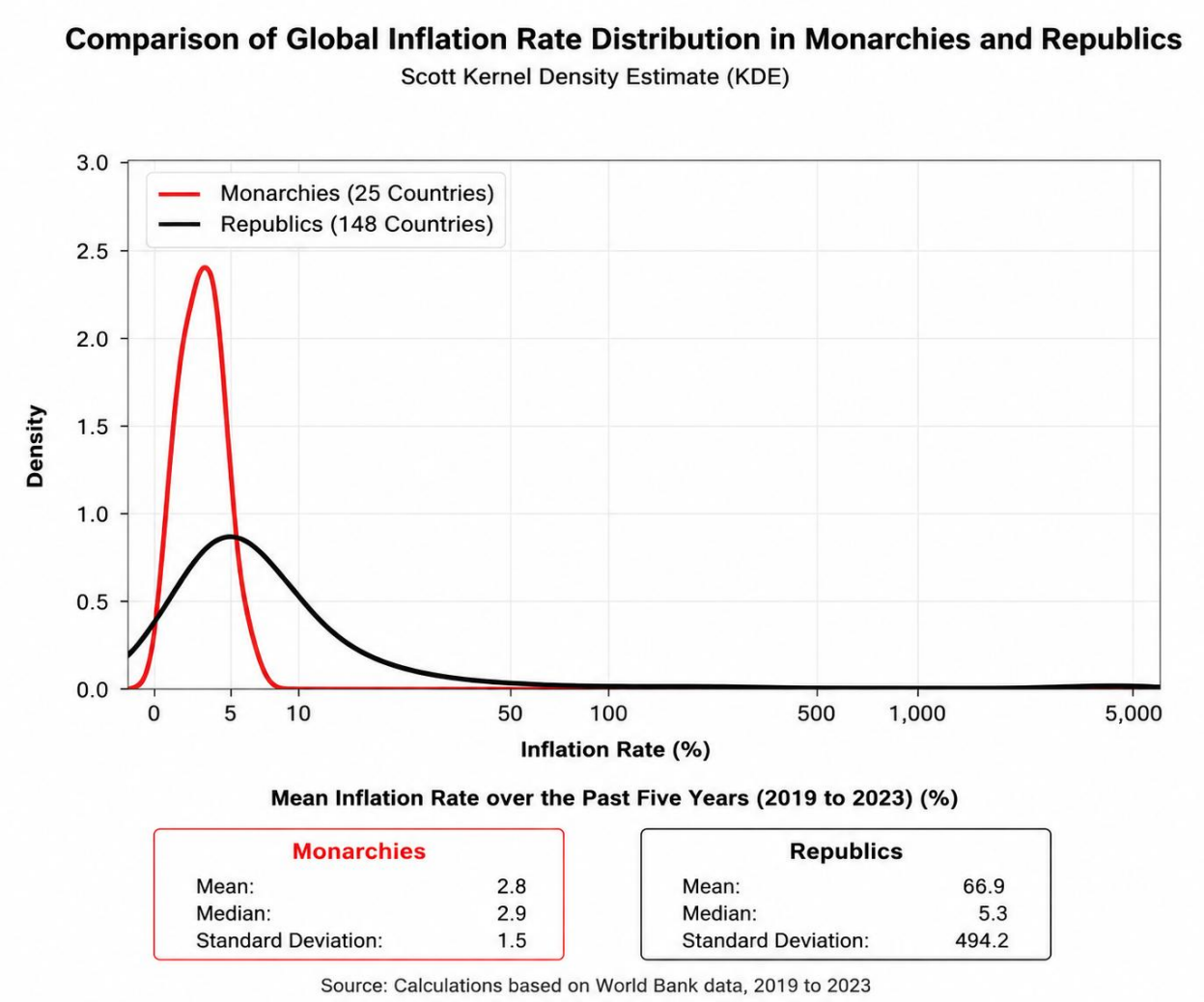
For this reason, the medians provide a more representative picture of the center of the two distributions. The five-year median inflation rate is **2.99%** among monarchies and **5.33%** among republics. Thus, even after reducing the influence of extreme observations, the median republic experienced an inflation rate approximately **1.8 times** that of the median monarchy.

Because of the exceptionally high dispersion among republics, Welch's t-test does not find the difference in group means statistically significant at the 5% level (**t = −1.577, p = 0.1168**). This

result illustrates why comparisons based solely on means can be misleading when distributions are highly skewed and contain extreme outliers.

For this reason, the **Mann–Whitney U test** was also applied. Because this test is based on the ranks of observations, it is substantially less sensitive to extreme values. Its result indicates that the distribution of inflation among monarchies is significantly lower than that among republics (**U = 704, p < 0.000001**).

Taken together, the descriptive evidence and the nonparametric test indicate that monarchies typically experienced lower and more stable inflation than republics during the period examined. For this indicator, given the extreme skewness of the data and the presence of exceptionally high inflation in some republics, the Mann–Whitney test provides a more informative picture of the typical difference between the two groups than a comparison based solely on their means.



نمودار نشان می‌دهد که میانگین تورم پنج‌ساله در پادشاهی‌های جهان عمدتاً در سطوح پایین و در محدودهٔ نسبتاً فشرده‌ای متمرکز است، در حالی که توزیع جمهوری‌ها پراکندگی بسیار بیشتری دارد و به سمت نرخ‌های بسیار بالا کشیده شده است. بنابراین، پادشاهی‌ها نه‌تنها تورم متوسط کمتری داشته‌اند، بلکه از ثبات بیشتری نیز برخوردار بوده‌اند؛ در مقابل، وجود چند مورد تورم شدید و ابرتورم در میان جمهوری‌ها سبب افزایش چشمگیر میانگین و انحراف معیار این گروه شده است.

Comparison of Inflation in Constitutional Monarchies and Semi-Constitutional and Absolute Monarchies

This comparison examines the average inflation rate over the period **2019–2023** across the two types of monarchy. Among countries for which World Bank data were available, **14 countries** were classified as constitutional monarchies and **11 countries** as semi-constitutional and absolute monarchies.

Descriptive Statistics

Type of Monarchy	Number of Countries	Five-Year Mean Inflation (%)	Median (%)	Standard Deviation
Constitutional Monarchies	14	**3.18**	**3.07**	**1.29**
Semi-Constitutional and Absolute Monarchies	11	**2.25**	**1.84**	**1.54**

The five-year mean inflation rate was **3.18%** in constitutional monarchies and **2.25%** in semi-constitutional and absolute monarchies. Thus, the latter group experienced inflation that was, on average, approximately **0.93 percentage points lower**.

The medians show a similar pattern: median inflation was **3.07%** in constitutional monarchies and **1.84%** in semi-constitutional and absolute monarchies. The consistency between the means and medians suggests that the observed difference is not simply the result of one or more unusually high values.

Test of the Difference in Means

Welch’s t-test was used to determine whether the difference in mean inflation between the two groups was statistically significant.

Statistical Measure	Value
Mean difference: Constitutional − Semi-Constitutional and Absolute	**0.93**
Welch’s t	**1.60**
Approximate degrees of freedom	**19.42**
Two-tailed p-value	**0.125**
95% CI for the mean difference	**−0.28 to 2.14**

Statistical Measure	Value
Hedges' g	**0.64**

Mean inflation in constitutional monarchies was approximately **0.93 percentage points higher** than in semi-constitutional and absolute monarchies. However, Welch's test indicates that this difference is not statistically significant at the 95% confidence level (**p = 0.125**). The confidence interval for the difference in means also includes zero; therefore, the available data do not allow us to conclude with confidence that the true mean inflation rates of the two types of monarchy differ.

Hedges' **g = 0.64**, which in magnitude suggests a moderate effect. However, given the relatively small number of countries in each group and the resulting uncertainty of the estimate, this difference is not statistically confirmed. In other words, the data suggest lower inflation among semi-constitutional and absolute monarchies, but the evidence is insufficient for a definitive conclusion.

Conclusion

During the period examined, semi-constitutional and absolute monarchies had lower inflation than constitutional monarchies in terms of both the mean and the median. However, the difference between the group means did not reach statistical significance. Therefore, at this stage, neither type of monarchy can be considered conclusively superior in controlling inflation, although the observed difference favors semi-constitutional and absolute monarchies. This is an observational comparison of country performance and, by itself, does not establish a causal relationship between the type of monarchy and inflation.

Comparison of Inflation in Commonwealth Realms and Republics Worldwide

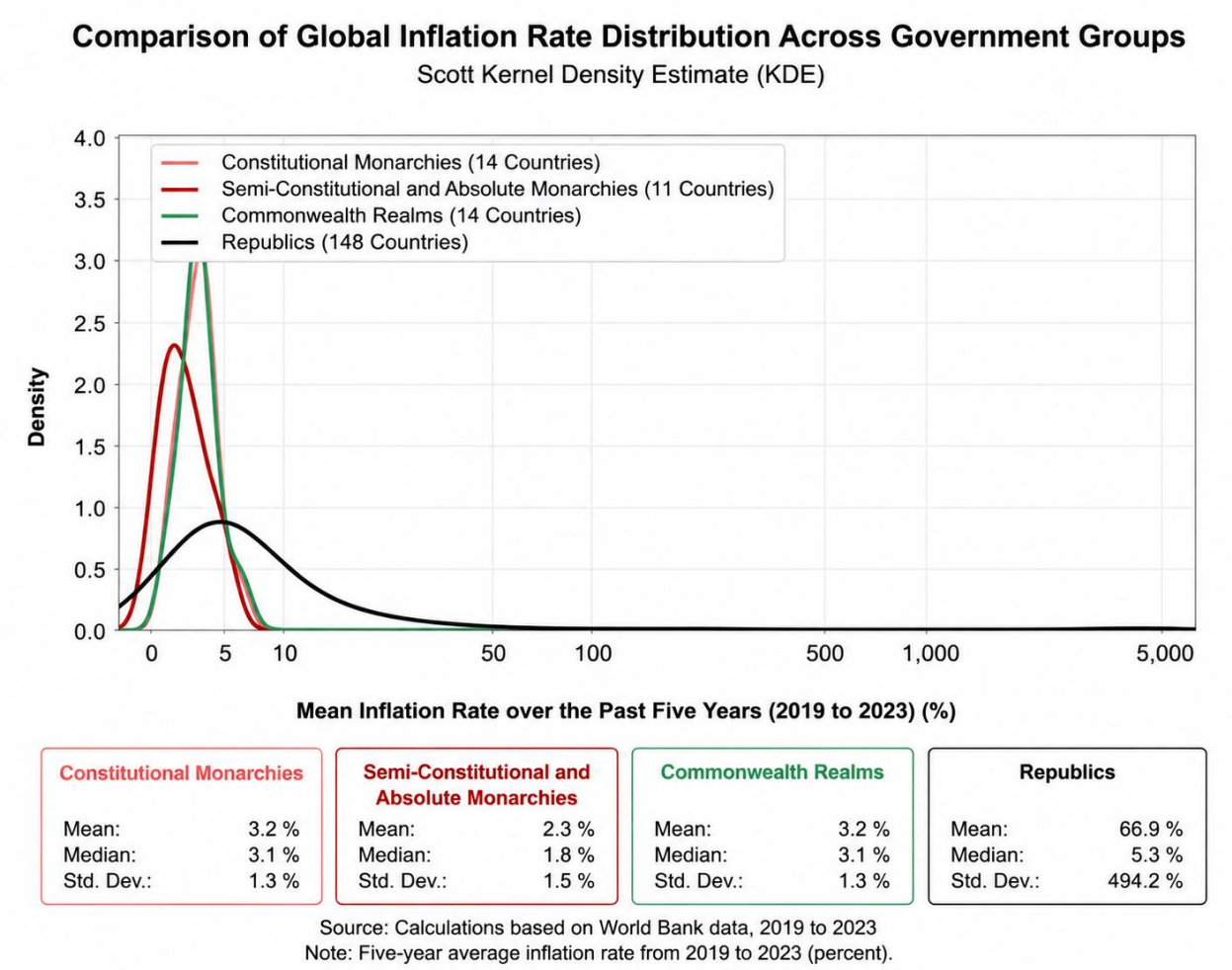
This comparison examines average inflation rates over the period **2019–2023** in Commonwealth realms and republics worldwide. The available data cover **14 Commonwealth realms** and **148 republics**.

Descriptive Statistics

Form of Government	Number of Countries	Five-Year Mean Inflation (%)	Median (%)	Standard Deviation
Commonwealth Realms	14	**3.23**	**3.14**	**1.33**
Republics	148	**66.85**	**5.33**	**494.19**

The five-year mean inflation rate was **3.23%** in Commonwealth realms and **66.85%** in republics. On this basis, average inflation among republics was approximately **63.62 percentage points higher** than among Commonwealth realms.

However, the exceptionally large standard deviation among republics indicates that the group mean is heavily influenced by a small number of republics experiencing extremely high inflation or hyperinflation.



The kernel density estimate shows that all three groups—**constitutional monarchies, semi-constitutional and absolute monarchies, and Commonwealth realms**—are concentrated within the range of low inflation rates, with substantial overlap among their distributions. Among these three groups, **semi-constitutional and absolute monarchies are slightly shifted toward lower inflation rates**, while constitutional monarchies and Commonwealth realms display very similar distributional patterns.

In contrast, the distribution for **republics is considerably more dispersed** and, owing to the presence of countries with extremely high inflation, exhibits a long right tail extending toward much larger values. Overall, this pattern suggests that inflation outcomes across the three monarchy-related groups have been **more concentrated and stable**, whereas republics have experienced **substantially greater variation in inflation rates**.

Comparison of the Inflation Indicator between Monarchies and Republics Across Continents and in the Middle East

Comparison of Inflation between Monarchies and Republics in Asia

This comparison examines the **average annual inflation rate over the 2019–2023 period** in Asian monarchies and republics. Available data covered **13 monarchies** and **31 republics** in Asia.

Descriptive Statistics

Type of Government	Number of Countries	Five-Year Average Inflation (%)	Median (%)	Standard Deviation
Asian monarchies	13	**1.85**	**1.64**	**1.23**
Asian republics	31	**13.08**	**6.80**	**23.61**

The five-year average inflation rate was **1.85%** in Asian monarchies and **13.08%** in Asian republics. Thus, average inflation in Asian republics was approximately **11.23 percentage points higher** than in Asian monarchies.

The **medians show the same pattern**. Median inflation was **1.64%** among monarchies and **6.80%** among republics. Thus, even after reducing the influence of extremely high values, the typical Asian republic experienced substantially higher inflation than the typical Asian monarchy.

Test of the Difference in Means

A **Welch's t-test** was used to determine whether the difference in mean inflation between the two groups was statistically significant.

Statistical Measure	Value
Difference in means (monarchies minus republics)	**−11.23**
Welch's t-statistic	**−2.64**
Approximate degrees of freedom	**30.38**
Two-tailed p-value	**0.013**
95% confidence interval	**−19.91 to −2.54**
Hedges' g	**−0.55**

Welch's test indicates that the **difference in mean inflation between the two groups is statistically significant** (**p = 0.013**). Moreover, the entire 95% confidence interval lies below

zero, supporting the finding that average inflation was lower among Asian monarchies than among the continent's republics.

Hedges' g is **−0.55**, indicating an **effect of moderate magnitude**. In other words, the observed difference is not only statistically significant but also meaningful in terms of its size.

Robust Test of Distributional Differences

Because several republics had exceptionally high inflation rates, the nonparametric **Mann–Whitney U test** was also conducted.

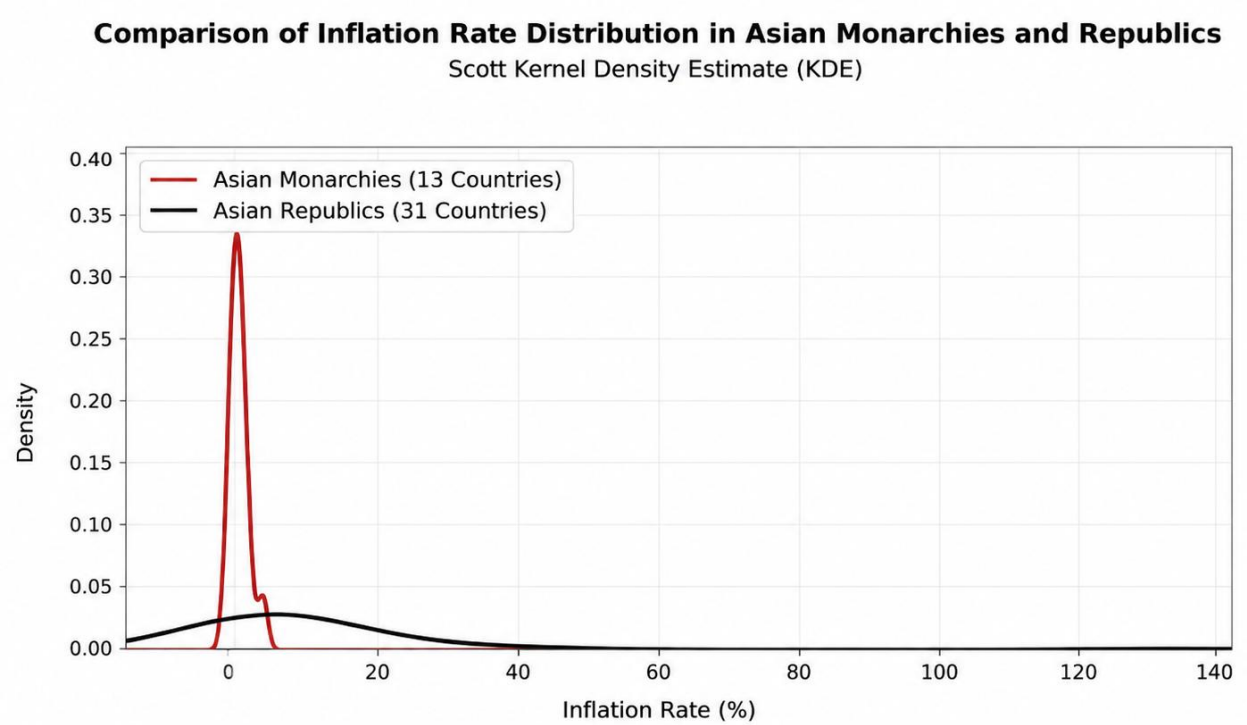
Statistical Measure	Value
Mann–Whitney U statistic	**35**
Two-tailed p-value	**0.00002**
Rank-biserial correlation effect size	**0.83**

The Mann–Whitney test also shows **very strong statistical evidence of a difference between the two groups (p < 0.001)**. The rank-based effect size of **0.83** indicates a **very large difference between their inflation distributions**, with inflation values in Asian monarchies generally falling below those observed in Asian republics in pairwise comparisons.

Conclusion

During the 2019–2023 period, Asian monarchies had lower inflation than Asian republics in terms of both the mean and the median. Unlike the global comparison, in which extreme hyperinflation episodes prevented the Welch test from reaching statistical significance, the difference in means within Asia was statistically significant (p = 0.013). In addition, the Mann–Whitney test revealed a highly significant difference in the overall distributions (p < 0.001).

Overall, the Asian data examined in this study indicate that **Asian monarchies performed better than Asian republics in maintaining price stability and controlling inflation**. As in the other analyses in this book, this finding represents an **observed association and does not, by itself, establish a causal relationship between the type of government and the inflation rate**.



The kernel density estimate shows that the distribution of the **five-year average inflation rate among Asian monarchies** is concentrated predominantly at low rates and within a relatively narrow range. In contrast, while Asian republics are also concentrated around low and moderate inflation rates, their distribution is considerably more dispersed, with a long tail extending toward very high inflation rates. This pattern indicates that **Asian monarchies generally experienced lower and more stable inflation**, whereas several cases of very high inflation among republics substantially increased the dispersion of inflation outcomes within that group.

Comparison of the Inflation Indicator between Monarchies and Republics in Europe

The available data cover **9 European monarchies** and **34 European republics**.

Descriptive Statistics

Type of Government	Number of Countries	Five-Year Average Inflation (%)	Median (%)	Standard Deviation
European monarchies	9	**3.50**	**3.62**	**0.54**
European republics	34	**5.29**	**4.70**	**2.43**

The five-year average inflation rate was **3.50%** in European monarchies and **5.29%** in European republics. Thus, average inflation in the republics was approximately **1.79 percentage points higher** than in the monarchies.

The **medians point in the same direction**: median inflation was **3.62%** among monarchies and **4.70%** among republics, meaning that the median inflation rate in European republics was approximately **1.08 percentage points higher**.

Another notable difference concerns the dispersion of the data. The standard deviation among monarchies was only **0.54**, compared with **2.43** among republics. Thus, **European monarchies not only had lower inflation on average, but their inflation outcomes were also considerably more homogeneous and stable**.

Test of the Difference in Means

A **Welch's t-test** was used to determine whether the difference in mean inflation between the two groups was statistically significant.

Statistical Measure	Value
Difference in means (monarchies minus republics)	**−1.79**
Welch's t-statistic	**−3.94**
Approximate degrees of freedom	**40.64**
Two-tailed p-value	**0.00031**
95% confidence interval	**−2.71 to −0.87**
Hedges' g	**−0.80**

Welch's test indicates that the **difference in mean inflation between the two groups is highly statistically significant** (**$p < 0.001$**). The entire 95% confidence interval also lies below zero, providing strong evidence that average inflation was lower among European monarchies than among the continent's republics.

Hedges' g is **−0.80**, which is at the threshold of a **large effect**. Thus, the observed difference is not only statistically significant but also substantial in terms of its magnitude.

Robust Mann–Whitney Test

To ensure that the result was not dependent solely on a comparison of means, the nonparametric **Mann–Whitney U test** was also conducted.

Statistical Measure	Value
U statistic	**87**
Exact two-tailed p-value	**0.049**
Rank-biserial correlation	**0.43**

The exact Mann–Whitney test reaches statistical significance at the conventional 5% threshold (**p = 0.049**). However, this value is very close to the significance threshold, and the asymptotic approximation yields a value of approximately **0.051**. The evidence from this test should therefore be regarded as **borderline**.

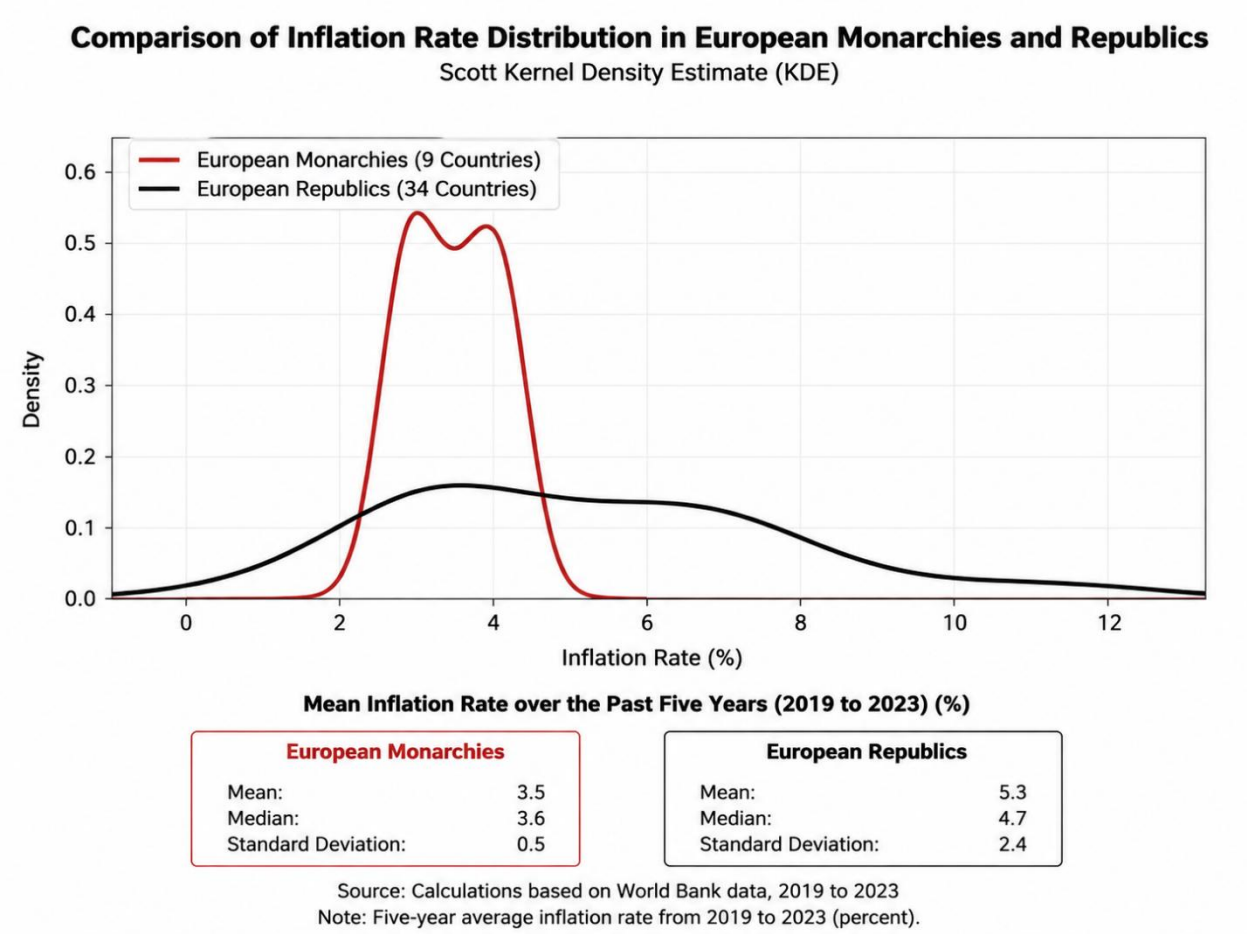
The rank-based effect size of **0.43** indicates a **moderate difference** between the two groups, with inflation values in European monarchies generally occupying lower ranks than those in European republics.

Conclusion

During the 2019–2023 period, European monarchies had lower inflation than European republics in terms of both the mean and the median. Average inflation was 3.50% among monarchies and 5.29% among republics, and Welch's test provides very strong statistical evidence of a difference between the two groups.

Monarchies also exhibited substantially less dispersion, indicating that their inflation outcomes during this period were not only lower but also more stable and consistent across countries.

The exact Mann–Whitney test also places the difference between the two groups just within the conventional threshold of statistical significance, although its evidence is weaker and more borderline than that provided by Welch's test. Overall, the European data examined in this study indicate that **European monarchies recorded a better inflation-control performance than European republics over this five-year period**.



The kernel density estimate shows that the distribution of the **five-year average inflation rate** in both European monarchies and republics is concentrated primarily at relatively low inflation rates. However, the distribution for monarchies is more tightly concentrated within a narrower range. In contrast, European republics exhibit greater dispersion, with some countries experiencing higher inflation rates, causing the distribution to extend further toward larger values.

This pattern indicates that **European monarchies generally experienced greater stability in inflation rates**, although the difference between the two groups is considerably smaller than that observed in the comparison for Asia.

Comparison of Inflation in European Monarchies and European Republics That Have Never Been Communist

This comparison examines the **five-year average inflation rate from 2019 to 2023**. The data include **9 European monarchies** and **13 European republics that have never experienced communist rule**.

Descriptive Statistics

Form of Government	Number of Countries	Five-Year Average Inflation (%)	Median (%)	Standard Deviation
European Monarchies	9	**3.50**	**3.62**	**0.54**
European Republics (Never Communist)	13	**3.17**	**3.02**	**1.03**

The five-year average inflation rate was **3.50%** in European monarchies and **3.17%** in European republics that have never experienced communist rule. Thus, in this comparison, **the average inflation rate in monarchies was approximately 0.33 percentage points higher than in the non-communist republics**.

The medians point in the same direction: 3.62% in the monarchies compared with 3.02% in the non-communist republics.

At the same time, the standard deviation among monarchies (**0.54**) was approximately half that of the non-communist republics (**1.03**). Thus, although the average inflation rates of the two groups were very similar, inflation outcomes among the monarchies were more uniform and less dispersed.

Test of the Difference in Means

Welch's **t-test** was used to determine whether the difference in mean inflation between the two groups was statistically significant.

Statistical Measure	Value
Difference in means (Monarchies − Republics)	**+0.33**
Welch's t-statistic	**0.98**
Approximate degrees of freedom	**18.95**
Two-tailed p-value	**0.339**
95% confidence interval	**−0.38 to +1.04**
Hedges' g	**0.37**

Welch's test indicates that the difference in mean inflation between the two groups is **not statistically significant** (**p = 0.339**). The 95% confidence interval also includes zero. Therefore, the data in this sample do not provide sufficient evidence of a genuine difference in mean inflation between the two groups.

Hedges' g is **0.37**, indicating a **small-to-moderate effect size**.

Mann–Whitney U Test

The non-parametric **Mann–Whitney U test** was also used to compare the distributions of the two groups.

Statistical Measure	Value
U statistic	**79**
Two-tailed p-value	**0.182**
Rank-biserial correlation	**−0.35**

The Mann–Whitney test likewise does not indicate a statistically significant difference between the two groups (**p = 0.182**). Thus, neither the comparison of means nor the comparison of the overall distributions provides sufficient statistical evidence that either group outperformed the other.

Conclusion

After excluding European republics with a history of communist rule, the difference observed in the previous comparison effectively disappears. **The average inflation rates of European monarchies (3.50%) and European republics that have never experienced communist rule (3.17%) are very similar, and neither statistical test finds a significant difference between them.**

Therefore, among European countries without a history of communist rule, the data examined in this study do not indicate that the form of government—monarchy or republic—was associated with a statistically significant difference in inflation performance during 2019–2023. Nevertheless, **the monarchies exhibited less dispersion in inflation rates, indicating more uniform inflation outcomes across the group than among the non-communist republics.**

Comparison of Inflation in Monarchies and Republics in Africa, the Americas, and Oceania

Continent and Form of Government	Number of Countries	Mean Inflation	Median	Standard Deviation
Africa — Monarchies	2	**4.59%**	4.59%	2.23
Africa — Republics	49	**85.05%**	5.74%	507.97
Americas — Commonwealth Realms	9	**2.92%**	2.67%	1.51
Americas — Republics	25	**204.37%**	5.23%	973.71
Oceania — Monarchy of Tonga	1	**4.50%**	4.50%	—

Continent and Form of Government	Number of Countries	Mean Inflation	Median	Standard Deviation
Oceania — Commonwealth Realms	5	**3.80%**	3.67%	0.72
Oceania — Republics	9	**3.58%**	3.47%	1.25

The extremely high mean inflation rates among the republics of Africa and the Americas are heavily influenced by cases of extreme inflation. For this reason, **the medians in these two continents provide a more informative picture of the typical country than the means**.

As with the previous indicators, the very small number of monarchies in Africa, the Americas, and Oceania makes inferential comparisons between forms of government of limited value. The analysis of these three continents is therefore restricted to descriptive statistics. In the Americas, where there are no independent monarchies, the Commonwealth realms are compared with the republics. In Oceania, the region's only independent monarchy, Tonga, is considered alongside the Commonwealth realms.

Africa

The only two monarchies represented in the data, **Lesotho and Morocco**, had a five-year average inflation rate of **4.59%**. Among the 49 African republics, the corresponding mean was **85.05%**. However, the republican mean is heavily influenced by several cases of extremely high inflation, particularly Sudan, making the medians more informative for comparing the typical experience of the two groups.

Median inflation was **4.59%** among the monarchies and **5.74%** among the republics. Thus, even when the influence of extreme values is reduced by focusing on the median, the two African monarchies recorded lower typical inflation than the republics of the continent. With only two monarchies, however, this difference cannot support broad statistical conclusions.

The Americas

The Commonwealth realms had a five-year average inflation rate of only **2.92%**, compared with **204.37%** among the continent's 25 republics. This enormous difference is driven primarily by Venezuela's exceptionally high inflation and, to a lesser extent, by several other republics. The mean alone is therefore not an appropriate measure of the typical difference between the two groups.

The medians, however, point in the same direction: median inflation was **2.67%** among the Commonwealth realms and **5.23%** among the republics. Thus, even without relying on extreme inflation cases, the median republic experienced almost **twice the inflation** of the median Commonwealth realm. The standard deviation among the Commonwealth realms was also only **1.51**, indicating much greater uniformity in their inflation outcomes.

Oceania

Oceania presents a different picture, and the differences are much smaller. Tonga, the region's only independent monarchy, had an average inflation rate of **4.50%**. Among the five Commonwealth realms, average inflation was **3.80%**, compared with **3.58%** among the region's nine republics.

The medians are similarly close: **3.67%** among the Commonwealth realms and **3.47%** among the republics. Thus, unlike in Africa and the Americas, the descriptive advantage in terms of lower inflation lies slightly with the republics, although the difference relative to the Commonwealth realms is very small. At the same time, the standard deviation was **0.72** among the Commonwealth realms, compared with **1.25** among the republics, indicating less dispersion in inflation outcomes among the Commonwealth realms.

Overall Assessment

Taken together, these three continents do not present a uniform pattern. **In Africa, the monarchies recorded lower typical inflation than the republics, while in the Americas the Commonwealth realms did so; in Oceania, by contrast, the republics had slightly lower inflation than the Commonwealth realms.**

Given the very small number of monarchies in these regions, however, these findings are **descriptive rather than inferential** and should not be assigned the same statistical weight as the global, Asian, European, or Middle Eastern comparisons.

Comparison of the Inflation Indicator in Middle Eastern Monarchies and Republics

This comparison examines the **five-year average inflation rate from 2019 to 2023**. The available data cover **7 monarchies** and **9 republics** in the Middle East.

Descriptive Statistics

Form of Government	Number of Countries	Five-Year Mean Inflation (%)	Median (%)	Standard Deviation
Middle Eastern Monarchies	7	**1.35**	**1.45**	**0.81**
Middle Eastern Republics	9	**28.02**	**14.51**	**41.16**

The five-year average inflation rate was **1.35%** in Middle Eastern monarchies and **28.02%** in Middle Eastern republics. Thus, average inflation among the republics was approximately **26.66 percentage points higher** than among the monarchies.

The medians confirm the same pattern. Median inflation was **1.45%** among the monarchies and **14.51%** among the republics. Thus, even when the influence of extremely high values is reduced by focusing on the median, the typical Middle Eastern republic experienced substantially higher inflation than the typical monarchy in the region.

The standard deviation among the republics (**41.16**) was also dramatically higher than among the monarchies (**0.81**), indicating far greater dispersion in inflation rates across the region's republics.

Test of the Difference in Means

Welch's **t-test** was used to determine whether the difference between the two group means was statistically significant.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**−26.66**
Welch's t-statistic	**−1.94**
Approximate degrees of freedom	**8.01**
Two-tailed p-value	**0.088**
95% confidence interval	**−58.30 to +4.98**
Hedges' g	**−0.81**

Although average inflation was substantially higher among Middle Eastern republics, Welch's test did not find the difference statistically significant at the conventional 5% level (p = 0.088). The main reason is the presence of several republics with very high inflation rates, which greatly increases the dispersion of the data.

Nevertheless, Hedges' g was −0.81, indicating a large effect. The observed difference is therefore substantial in magnitude, even though the small sample sizes and very high variability among the republics prevent the difference in means from reaching conventional statistical significance.

Mann–Whitney U Test

To reduce sensitivity to extreme values, the nonparametric **Mann–Whitney U test** was also conducted.

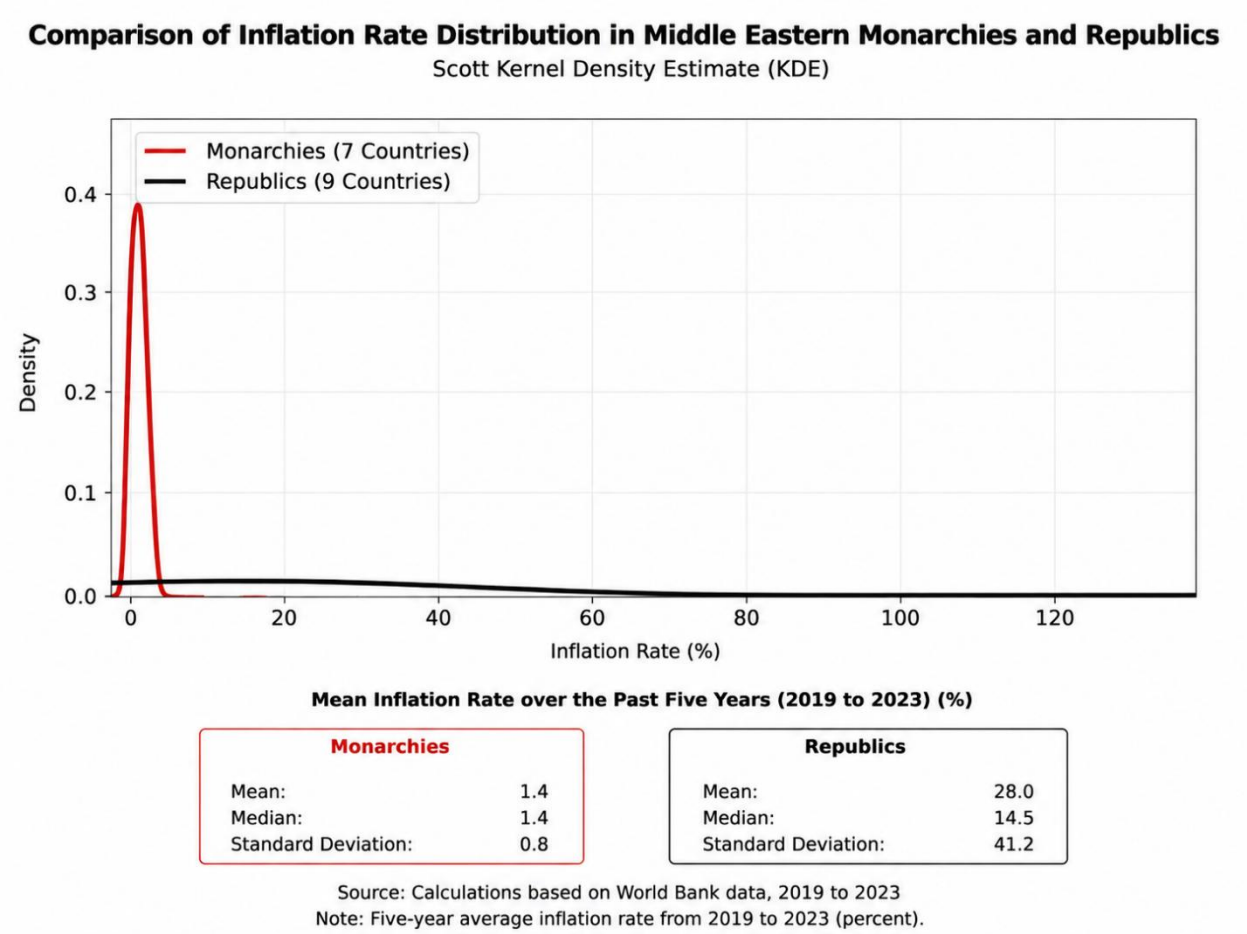
Statistical Measure	Value
U statistic	**3**
Two-tailed p-value	**0.0012**
Rank-biserial correlation	**0.90**

The Mann–Whitney test provides very strong evidence of a difference between the two groups (**p = 0.0012**). The rank-biserial effect size of **0.90** also indicates a very large separation between their inflation distributions, with Middle Eastern monarchies generally occupying substantially lower inflation ranks than the region's republics.

Conclusion

During the 2019–2023 period, **Middle Eastern monarchies had substantially lower inflation than Middle Eastern republics in terms of both the mean and the median**. Although Welch's test did not find the difference in means statistically significant at the 5% level—largely because of several extremely high inflation observations among the republics and the resulting high variance—the large effect size and the Mann–Whitney test both point to a pronounced difference between the two groups.

Overall, the Middle Eastern data in this study indicate that, during 2019–2023, the region's monarchies recorded a better performance than its republics in maintaining price stability and containing inflation. As elsewhere in this book, however, this finding describes an **observed association** and does **not, by itself, establish a causal relationship** between the form of government and the inflation rate.



The kernel density estimate (KDE) shows that the distribution of the five-year average inflation rate among Middle Eastern monarchies is concentrated entirely at low inflation rates, with very little dispersion. In contrast, the republics in the region exhibit substantially greater dispersion and include several countries with very high inflation rates, extending the tail of the distribution toward much higher values. This pattern indicates that, overall, Middle Eastern monarchies experienced lower and more stable inflation, whereas the region's republics were characterized by greater inflation volatility.

Status of the "New or Modern Monarchy" Hypothesis After the Eighth Indicator

Based on the findings obtained so far in this book, the inflation chapter also provides overall support for the hypothesis. Although no statistically significant difference was found in some subgroup comparisons—such as European republics with no history of communist rule—the overall evidence from the global comparison, Asia, and the Middle East was favorable to the hypothesis. The status of the hypothesis can therefore be updated as follows:

Test	Indicator	Result
✓	GDP per Capita	Consistent with the hypothesis

Test	Indicator	Result
✓	Purchasing Power per Capita	Consistent with the hypothesis
◐	Gini Coefficient	**Mixed evidence; difference favors the hypothesis**
✓	Political Stability and Absence of Violence	Consistent with the hypothesis
✓	Regulatory Quality	Consistent with the hypothesis
✓	Rule of Law	Consistent with the hypothesis
✓	Government Effectiveness	Consistent with the hypothesis
✓	Inflation	Consistent with the hypothesis

The Record of Justice

The fourth major objective of government is **the establishment of justice**. Justice is one of the most fundamental expectations people have of government. Citizens may agree or disagree with particular laws, but what most strongly creates a sense of injustice is the discriminatory enforcement of those same laws. If the law is enforced against some groups but ignored for others, or if those with power and wealth can use their influence to evade the law, public trust in government is undermined and justice is effectively eroded.

One of the most important responsibilities of any government, therefore, is to prevent the abuse of public power and ensure the equal application of the law to all citizens. The lower the level of administrative and political corruption, the fewer the opportunities to circumvent laws, enforce them selectively, or grant unfair privileges. For this reason, a government's success in controlling corruption can be regarded as one of the most important indicators of justice in governance.

Control of Corruption

To evaluate government performance in this area, this study uses the World Bank's **Control of Corruption** indicator. This indicator measures the extent to which public power is protected from being exercised for private gain and the extent to which governments have succeeded in controlling administrative and political corruption. The World Bank reports this indicator as a **Percentile Rank**, with higher percentile ranks indicating better performance in controlling corruption. The data used in this chapter are for **2023**. (Appendix RDS-I01)

Global Comparison of the Control of Corruption Indicator in Monarchies and Republics

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Sample Standard Deviation
Monarchies	28	**72.01**	**85.10**	**25.93**
Republics	152	**41.83**	**38.95**	**27.26**

The mean percentile rank for Control of Corruption is **72.01** among monarchies and **41.83** among republics. Thus, **monarchies performed, on average, about 30.18 percentile points better than republics in controlling corruption**.

The medians point in the same direction. The median is **85.10** for monarchies and **38.95** for republics, meaning that the median monarchy ranks more than **46 percentile points** above the median republic in controlling corruption. **This difference indicates that the observed gap is not confined to a comparison of means; it is also clearly visible in the central positions of the two distributions.**

The dispersion of the data is relatively similar in the two groups. The standard deviation is **25.93** among monarchies and **27.26** among republics. Thus, the substantial differences in the means and medians are observed despite broadly similar levels of within-group dispersion.

Test of the Difference in Means

To determine whether the difference between the two group means is statistically significant, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**30.18**
Welch's t	**5.61**
Approximate degrees of freedom	**38.83**
Two-tailed p-value	**0.0000018**
95% confidence interval for the mean difference	**19.30 to 41.05**
Hedges' g	**1.11**

Welch's test shows that **the difference in mean Control of Corruption scores between the two groups is highly statistically significant** (**p < 0.001**). The entire 95% confidence interval also lies above zero. The data in this study therefore show that monarchies performed substantially better than republics on the Control of Corruption indicator.

Hedges' g is **1.11**, indicating a **large effect size**. The observed difference is therefore not only statistically significant but also substantial in magnitude.

Mann–Whitney Test

To compare the two groups without relying solely on differences in means, the nonparametric **Mann–Whitney U test** was also conducted.

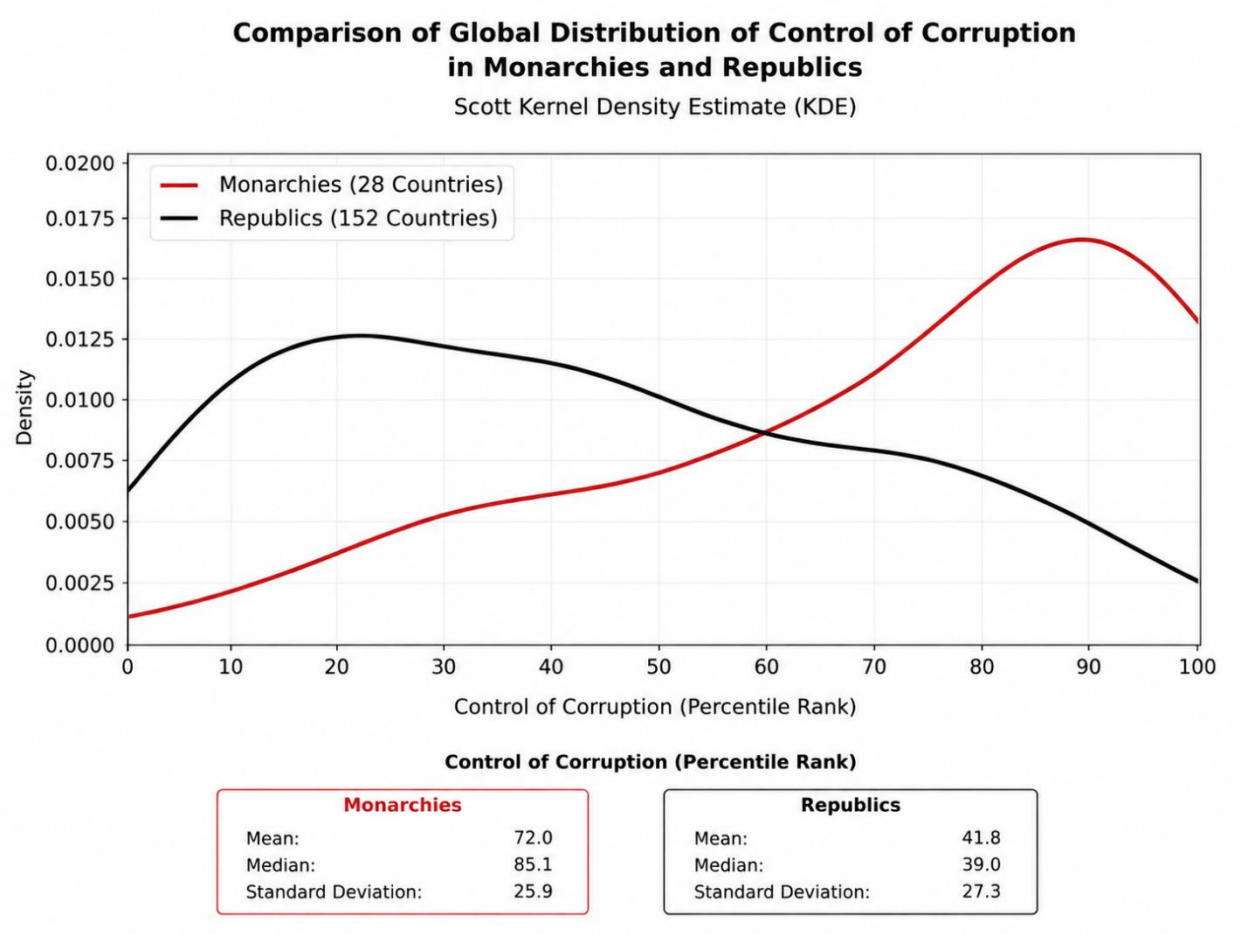
Statistical Measure	Value
Mann–Whitney U	**3349**
Two-tailed p-value	**0.0000015**
Rank-biserial correlation	**0.57**

The Mann–Whitney test likewise provides very strong evidence of a difference between the two groups (**p < 0.001**). The rank-biserial correlation of **0.57** also indicates a **large effect**, showing that Control of Corruption ranks are generally and substantially higher among monarchies than among republics.

Conclusion

In 2023, the world's monarchies performed substantially better than republics on the Control of Corruption indicator. Both the mean and the median were considerably higher among monarchies, and both Welch's t-test and the Mann–Whitney U test provide very strong statistical evidence of a difference between the two groups. The large effect sizes further indicate that the observed difference is not merely statistically significant but also substantial in magnitude.

Overall, the global data examined in this study show that monarchies have a better record than republics in controlling corruption and limiting the abuse of public power for private gain. As with the other analyses in this book, however, this finding represents an **observed association** and does not, by itself, establish a causal relationship between the form of government and the level of corruption control.



The kernel density estimate shows that the distribution of the **Control of Corruption** indicator among the world's monarchies is concentrated primarily at higher percentile ranks, with a large proportion of these countries located at the upper end of the indicator. In contrast, the distribution of republics is more

widely dispersed and is concentrated mainly at lower percentile ranks, although a number of republics have also achieved very high levels of corruption control. This pattern indicates that, overall, monarchies have performed better in controlling corruption, while republics exhibit greater heterogeneity in their performance on this indicator.

Comparison of the Control of Corruption Indicator in Constitutional and Non-Constitutional Monarchies Worldwide

Descriptive Statistics

Type of Monarchy	Number of Countries	Mean	Median	Sample Standard Deviation
Constitutional Monarchies	15	**74.46**	**89.20**	**28.69**
Semi-Constitutional and Absolute Monarchies	13	**69.18**	**75.00**	**23.17**

The mean percentile rank for Control of Corruption is **74.46** among constitutional monarchies and **69.18** among semi-constitutional and absolute monarchies. Thus, constitutional monarchies have a mean approximately **5.28 percentile points** higher than that of the second group.

The difference between the medians is larger than the difference between the means. The median Control of Corruption rank is **89.20** among constitutional monarchies and **75.00** among semi-constitutional and absolute monarchies. In other words, the median country among constitutional monarchies ranks approximately **14.20 percentile points** higher.

However, there is substantial dispersion within both groups. The standard deviation is **28.69** among constitutional monarchies and **23.17** among semi-constitutional and absolute monarchies. The greater dispersion in the first group mainly indicates that, alongside several very high-performing constitutional monarchies, the group also contains some countries with substantially weaker performance.

Test of the Difference in Means

To determine whether the difference between the two group means is statistically significant, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Constitutional − Semi-Constitutional and Absolute)	**5.28**
Welch's t	**0.54**
Approximate degrees of freedom	**25.89**
Two-tailed p-value	**0.595**
95% confidence interval for the mean difference	**−14.88 to 25.43**

Statistical Measure	Value
Hedges' g	**0.19**

Welch's test shows that **the difference in mean Control of Corruption between the two groups is not statistically significant** (**p = 0.595**). The 95% confidence interval also includes zero and is relatively wide. The available data therefore do not provide sufficient evidence to conclude that either type of monarchy has a higher population mean than the other.

Hedges' g is **0.19**, indicating a **small effect**. Thus, **the observed difference in means is not only statistically non-significant but also limited in magnitude**.

Mann–Whitney Test

Because the difference between the group medians is larger than the difference between their means, and because the data exhibit substantial dispersion, the nonparametric **Mann–Whitney U test** was also conducted.

Statistical Measure	Value
Mann–Whitney U	**124.5**
Exact two-tailed p-value	**0.235**
Rank-biserial correlation	**0.28**

The Mann–Whitney test likewise does not indicate a statistically significant difference between the distributions of the two groups (**p = 0.235**). The rank-based effect size of **0.28** suggests a small-to-moderate difference in favor of constitutional monarchies, but the statistical evidence is insufficient to establish this difference conclusively.

Conclusion

In 2023, constitutional monarchies performed better than semi-constitutional and absolute monarchies in terms of both the mean and, particularly, the median Control of Corruption rank. The respective means were **74.46** and **69.18**, while the medians were **89.20** and **75.00**.

Despite this descriptive advantage, neither Welch's test nor the Mann–Whitney test indicates a statistically significant difference between the two groups. The effect size for the difference in means is also small. Based on this sample, therefore, **there is insufficient evidence to conclude that either constitutional monarchies or semi-constitutional and absolute monarchies systematically outperform the other in controlling corruption**.

Overall, the clearest difference on this indicator appears between monarchies and republics, rather than between the two types of monarchy.

Comparison of the Control of Corruption Indicator in Commonwealth Realms and Republics Worldwide

This section compares the performance of **Commonwealth realms** with that of **republics worldwide** on the Control of Corruption indicator. As in the preceding analyses, the data are based on the World Bank's **2023 Control of Corruption percentile ranks**; higher values therefore indicate better performance in controlling corruption.

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Sample Standard Deviation
Commonwealth Realms	14	**68.94**	**67.70**	**20.75**
Republics Worldwide	152	**41.83**	**38.95**	**27.26**

The mean Control of Corruption percentile rank is **68.94** among Commonwealth realms and **41.83** among republics worldwide. Thus, **Commonwealth realms performed, on average, approximately 27.11 percentile points better than republics in controlling corruption**.

The medians show the same pattern. The median Control of Corruption rank is **67.70** among Commonwealth realms and **38.95** among republics worldwide. Thus, the median Commonwealth realm ranks approximately **28.75 percentile points** above the median republic. **This indicates that the observed difference is not confined to the comparison of means; it is also clearly visible in the central positions of the two distributions.**

The standard deviation is also lower among Commonwealth realms (**20.75**) than among republics (**27.26**), indicating that performance among Commonwealth realms is less dispersed as well as higher on average.

Test of the Difference in Means

To determine whether the difference between the two group means is statistically significant, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Commonwealth Realms − Republics)	**27.11**
Welch's t	**4.54**
Approximate degrees of freedom	**17.42**
Two-tailed p-value	**0.00027**
95% confidence interval for the mean difference	**14.53 to 39.68**
Hedges' g	**1.01**

Welch's test shows that **the difference in mean Control of Corruption between Commonwealth realms and republics worldwide is highly statistically significant** (**$p < 0.001$**). The entire 95% confidence interval lies above zero, providing further evidence that the mean is higher among Commonwealth realms.

Hedges' g is **1.01**, indicating a **large effect**. Thus, **the observed difference is not only statistically significant but also substantial in magnitude**.

Mann–Whitney Test

To compare the distributions of the two groups without relying solely on differences in means, the nonparametric **Mann–Whitney U test** was also conducted.

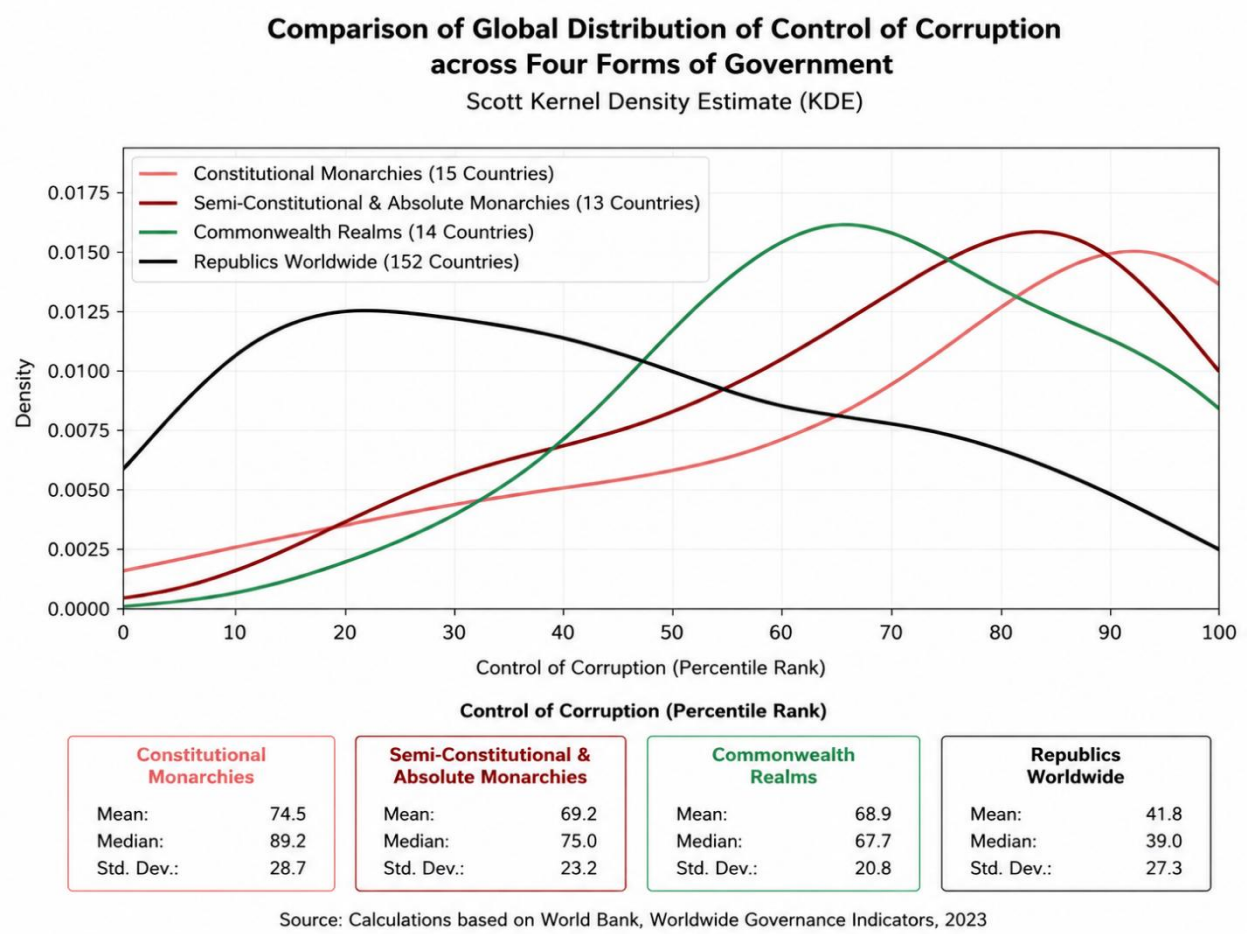
Statistical Measure	Value
Mann–Whitney U	**1654**
Two-tailed p-value	**0.00061**
Rank-biserial correlation	**0.55**

The Mann–Whitney test also provides very strong evidence of a difference between the two groups (**$p < 0.001$**). The rank-biserial correlation of **0.55** indicates a **large effect**, showing that Control of Corruption ranks are generally higher among Commonwealth realms than among republics worldwide.

Conclusion

In 2023, Commonwealth realms performed substantially better than republics worldwide on the Control of Corruption indicator. **Both the mean and the median were considerably higher among Commonwealth realms, and both Welch's t-test and the Mann–Whitney U test provide very strong statistical evidence of a difference between the two groups. The large effect sizes further indicate that the observed difference is substantial in magnitude.**

Overall, the data examined in this study show that Commonwealth realms have a better record than republics worldwide in controlling corruption and limiting the abuse of public power for private gain. As with the other analyses in this book, however, this finding represents an **observed association** and does not, by itself, establish a causal relationship between the form of government and the level of corruption control.



The kernel density estimate (KDE) shows that **constitutional monarchies are the most concentrated at the higher percentile ranks of the Control of Corruption indicator**, with their distribution clustered toward the upper end of the scale. **Semi-constitutional and absolute monarchies, as well as the Commonwealth realms, are also concentrated predominantly at relatively high levels of corruption control**, although their distributions are more dispersed than that of constitutional monarchies. In contrast, **the distribution for republics is broader and is concentrated mainly at lower percentile ranks**, indicating greater heterogeneity in their performance on corruption control.

Comparison of the “Control of Corruption” Indicator between Monarchies and Republics across Continents and the Middle East

Comparison of the “Control of Corruption” Indicator between Monarchies and Republics in Asia

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Asian Monarchies	13	**66.75**	**66.50**	**23.85**
Asian Republics	33	**30.90**	**26.40**	**24.85**

The mean percentile rank for Control of Corruption is **66.75** among Asian monarchies and **30.90** among Asian republics. Thus, **Asian monarchies outperform Asian republics by an average of approximately 35.84 percentile points in controlling corruption**.

The medians confirm the same pattern. The median Control of Corruption percentile rank is **66.50** for monarchies and **26.40** for republics. In other words, the median country among Asian monarchies ranks approximately **40.10 percentile points** higher than the median country among Asian republics. **This substantial difference indicates that the advantage of monarchies is not driven merely by a few exceptional countries, but is also evident across the broader distribution of the data.**

The dispersion of the data is quite similar in the two groups. The standard deviation is **23.85** among monarchies and **24.85** among republics, suggesting that the observed difference primarily reflects a difference in the overall level of performance rather than a difference in the degree of dispersion.

Test of the Difference in Means

To assess whether the difference between the two group means is statistically significant, **Welch’s t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**35.84**
Welch’s t-statistic	**4.54**
Approximate degrees of freedom	**22.90**
Two-tailed p-value	**0.00015**
95% confidence interval	**19.49 to 52.20**
Hedges’ g	**1.43**

Welch's test indicates that **the difference in mean Control of Corruption between Asian monarchies and republics is highly statistically significant** (**$p < 0.001$**). The 95% confidence interval also lies entirely above zero, further supporting the higher average performance of monarchies.

Hedges' g is **1.43**, indicating a **very large effect size**. Thus, **the observed difference is not only statistically significant but also substantial in magnitude**.

Mann–Whitney U Test

To compare the overall distributions of the two groups without relying on assumptions of normality, the nonparametric **Mann–Whitney U test** was also conducted.

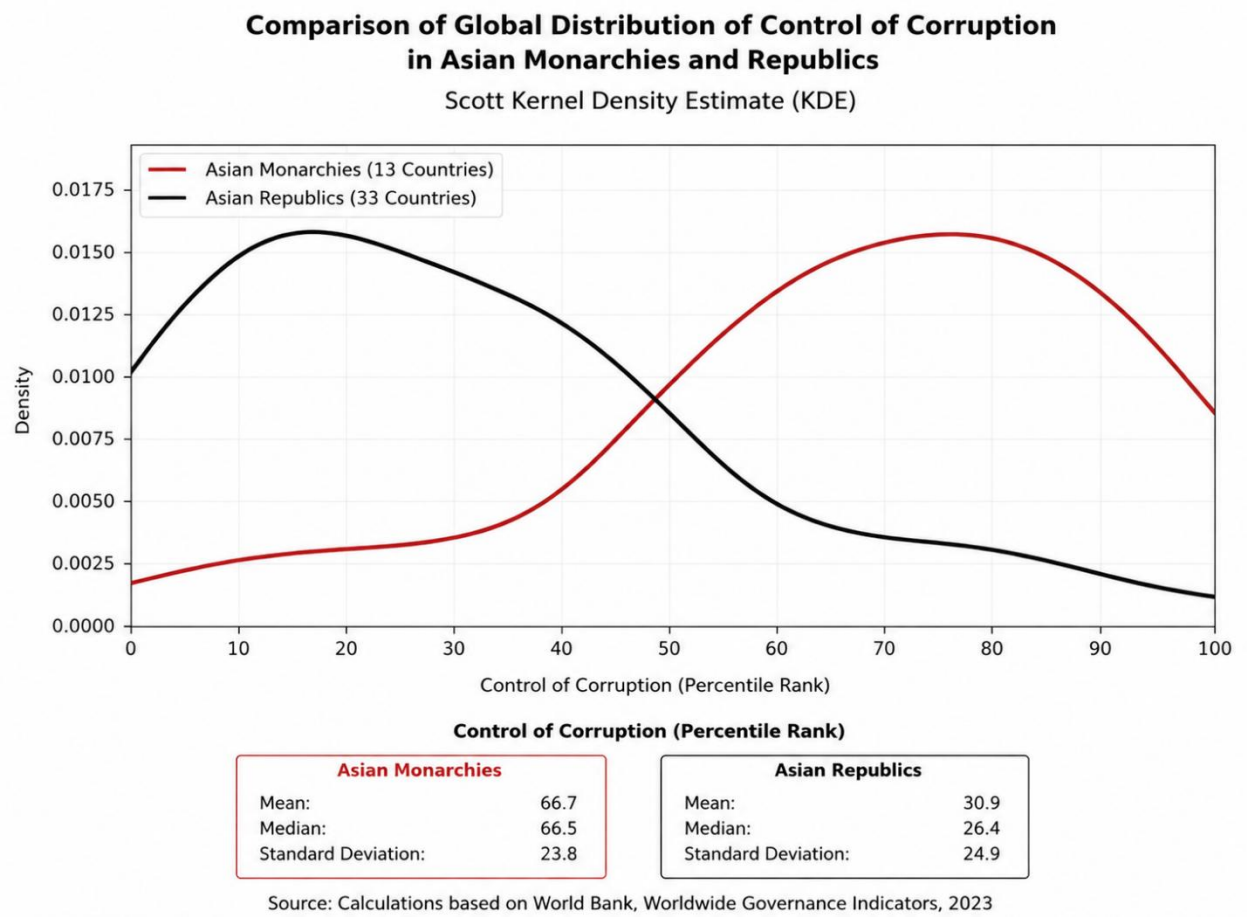
Statistical Measure	Value
Mann–Whitney U statistic	**362**
Two-tailed p-value	**0.00034**
Rank-biserial correlation	**0.69**

The Mann–Whitney test likewise provides very strong evidence of a difference between the two groups (**$p < 0.001$**). The rank-biserial correlation of **0.69** indicates a **large effect**, showing that Control of Corruption percentile ranks are generally and substantially higher among Asian monarchies than among Asian republics.

Conclusion

In 2023, Asian monarchies performed substantially better than Asian republics on the Control of Corruption indicator. **Both the mean and the median are considerably higher among monarchies, and both Welch's t-test and the Mann–Whitney U test confirm the difference with very strong statistical evidence. Moreover, the large effect sizes indicate that the difference is not only statistically significant but also substantial in practical terms.**

Overall, the Asian data in this study indicate that the continent's monarchies have a better record than its republics in controlling corruption and limiting the misuse of public power. As in other sections of this book, however, this finding represents an **observational association** and does not, **by itself**, establish a causal relationship between the form of government and the level of corruption control.



The kernel density estimate (KDE) shows that **the distribution of Asian monarchies is clearly shifted toward higher values of the Control of Corruption indicator**, with the greatest concentration in the upper half of the scale. In contrast, **Asian republics are concentrated primarily at lower levels of the indicator and display greater dispersion**, indicating greater heterogeneity in their performance on corruption control.

Comparison of the “Control of Corruption” Indicator between Monarchies and Republics in Europe

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
European Monarchies	11	91.98	95.30	8.09
European Republics	34	63.87	62.00	22.52

The mean percentile rank for Control of Corruption is **91.98** among European monarchies and **63.87** among European republics. Thus, **European monarchies outperform European republics by an average of approximately 28.11 percentile points in controlling corruption**.

The medians confirm the same pattern. The median Control of Corruption percentile rank is **95.30** for European monarchies and **62.00** for European republics. In other words, the median country among European monarchies ranks approximately **33.30 percentile points** higher than the median country among European republics. **This difference indicates that the advantage of monarchies is not confined to a few exceptional countries, but is also evident across the broader distribution of the data.**

The standard deviation among European monarchies (**8.09**) is also substantially lower than among European republics (**22.52**). This indicates that European monarchies not only have higher levels of corruption control but also display considerably less variation in their performance.

Test of the Difference in Means

To assess whether the difference between the two group means is statistically significant, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**28.11**
Welch's t-statistic	**6.15**
Approximate degrees of freedom	**42.34**
Two-tailed p-value	**0.00000023**
95% confidence interval	**18.89 to 37.33**
Hedges' g	**1.37**

Welch's test indicates that the difference in mean Control of Corruption between European monarchies and republics is highly statistically significant (**$p < 0.001$**). The 95% confidence interval lies entirely above zero, further supporting the higher average performance of monarchies.

Hedges' g is **1.37**, indicating a **very large effect size**. Thus, the observed difference is not only statistically significant but also substantial in magnitude.

Mann–Whitney U Test

To compare the overall distributions of the two groups, the nonparametric **Mann–Whitney U test** was also conducted.

Statistical Measure	Value
Mann–Whitney U statistic	**327**
Two-tailed p-value	**0.000229**
Rank-biserial correlation	**0.75**

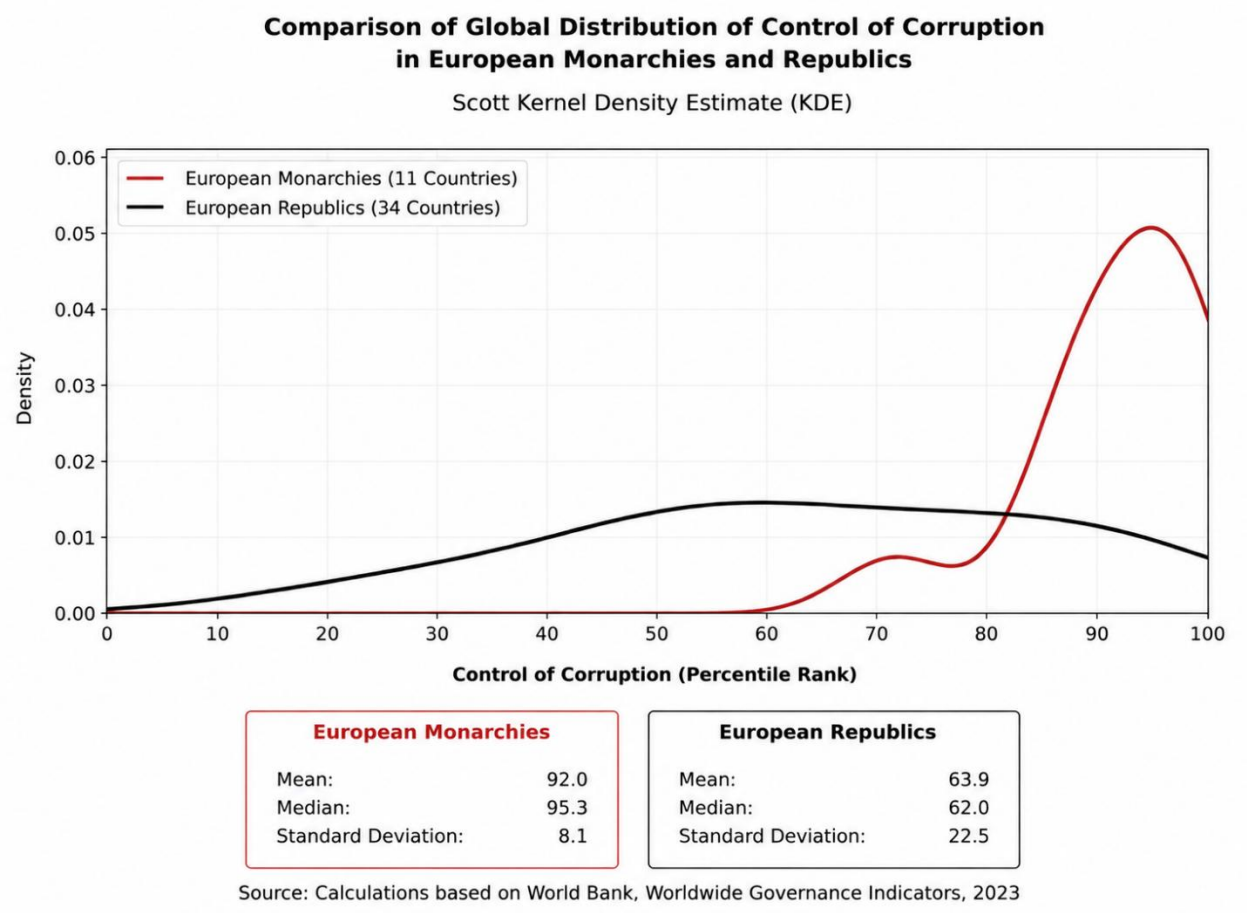
The Mann–Whitney test likewise provides very strong evidence of a difference between the two groups (**$p < 0.001$**). The rank-biserial correlation of approximately **0.75** indicates a **very large effect**, showing a pronounced advantage for European monarchies across the distribution of this indicator.

Conclusion

In 2023, **European monarchies performed substantially better than European republics on the Control of Corruption indicator. Both the mean and the median are considerably higher among monarchies, and both Welch's t-test and the Mann–Whitney U test confirm the difference with very strong statistical evidence. Moreover, the very large effect sizes indicate that the difference is not only statistically significant but also substantial in practical terms.**

The much lower standard deviation among European monarchies also indicates that these countries combine higher levels of corruption control with substantially less variation in performance.

Overall, the European data in this study indicate that European monarchies have a better record than European republics in controlling corruption and limiting the misuse of public power. As in other sections of this book, however, this finding represents an **observational association** and does not, **by itself**, establish a causal relationship between the form of government and the level of corruption control.



The kernel density estimate (KDE) plot shows that the distribution of European monarchies is concentrated almost entirely at very high levels of the Control of Corruption indicator, with relatively little dispersion, suggesting fairly consistent performance across these countries. In contrast, European republics not only have lower values on average but also exhibit a much wider distribution, ranging from republics with very strong performance to countries with relatively weak performance in controlling corruption.

Comparison of the "Control of Corruption" Indicator between European Monarchies and European Republics with No History of Communist Rule

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
European Monarchies	**11**	**91.98**	**95.30**	**8.09**
European Republics with No History of Communist Rule	**13**	**80.95**	**84.00**	**14.98**

The mean percentile rank for Control of Corruption is **91.98** among European monarchies and **80.95** among European republics with no history of communist rule. Thus, **European monarchies perform, on average, approximately 11.04 percentile points better in controlling corruption**.

The medians point in the same direction. The median is **95.30** among European monarchies and **84.00** among the non-communist republics. In other words, the median monarchy ranks approximately **11.30 percentile points** higher than the median republic in this restricted comparison.

The standard deviation is also considerably lower among European monarchies (**8.09**) than among the non-communist republics (**14.98**), indicating that the monarchies combine higher performance with greater consistency.

Test of the Difference in Means

To assess whether the difference between the two group means is statistically significant, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**11.04**
Welch's t-statistic	**2.290**
Approximate degrees of freedom	**18.99**
Two-tailed p-value	**0.0336**
95% confidence interval	**0.95 to 21.12**
Hedges' g	**0.864**

Welch's test indicates that the difference in mean Control of Corruption between the two groups is statistically significant (**p = 0.0336**). The 95% confidence interval lies entirely above zero, indicating higher average performance among European monarchies.

Hedges' g is **0.864**, indicating a **large effect size**. Thus, **the observed difference is not only statistically significant but also substantial in magnitude**.

Mann–Whitney U Test

To compare the overall distributions of the two groups, the nonparametric **Mann–Whitney U test** was also conducted.

Statistical Measure	Value
Mann–Whitney U statistic	**105**

Statistical Measure	Value
Two-tailed p-value	**0.0557**
Rank-biserial correlation	**0.469**

Unlike Welch's test, the Mann–Whitney test does not reach statistical significance at the conventional 5% level (**p = 0.0557**), although the result is very close to the threshold. The rank-biserial correlation of **0.469** indicates a relatively large effect and a general tendency for monarchies to occupy higher ranks.

Conclusion

Once European republics with a history of communist rule are excluded, the gap between monarchies and republics becomes substantially smaller than in the comparison with all European republics. Nevertheless, European monarchies continue to have higher mean and median Control of Corruption scores.

Welch's test finds this difference statistically significant, whereas the Mann–Whitney U test falls just short of the 5% significance threshold. The observed difference nevertheless remains substantial, with a relatively large effect size.

The results therefore suggest that part of the overall difference between European monarchies and republics is associated with the weaker performance of republics with a communist historical legacy. Even after these countries are removed, however, European monarchies still record better Control of Corruption performance than European republics without such a history.

This comparison is particularly noteworthy because the European republics included here, unlike many Eastern European republics, have no history of communist rule, and many have had competitive elections, democratic institutions, and a free press for decades or even more than a century. Since freedom of the press is commonly regarded in political-science literature as an important mechanism for monitoring public power and restraining corruption, one might expect these republics to perform at least as well as, or perhaps better than, European monarchies. Yet the data examined here show that European monarchies still have higher average Control of Corruption scores, with the difference reaching statistical significance under Welch's test.

Comparison of the "Control of Corruption" Indicator in Monarchies and Republics in Africa, the Americas, and Oceania

To complete the regional comparisons, descriptive statistics for Control of Corruption in monarchies or Commonwealth realms in Africa, the Americas, and Oceania are compared with republics in the same regions. Because the number of monarchies and Commonwealth realms is small in some of these regions, this section is limited to descriptive statistics.

Africa

Type of Government	Number of Countries	Mean	Median	Standard Deviation
African Monarchies	3	**30.50**	**33.00**	**4.77**
African Republics	50	**30.93**	**24.30**	**22.88**

In Africa, the mean Control of Corruption score is almost identical in the two groups, with republics having a slightly higher mean. However, the median is clearly higher among monarchies, while their much lower standard deviation indicates more uniform performance. Overall, the descriptive statistics do not indicate a clear advantage for either form of government in this region.

The Americas

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Commonwealth Realms in the Americas	9	**68.97**	**67.00**	**15.50**
Republics in the Americas	26	**39.66**	**36.20**	**26.64**

In the Americas, the Commonwealth realms perform substantially better than republics in terms of both the mean and the median. Their lower standard deviation also indicates that their performance in controlling corruption is not only higher but more consistent.

Oceania

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Monarchy and Commonwealth Realms in Oceania	6	**64.97**	**61.35**	**28.78**
Republics in Oceania	9	**65.51**	**66.00**	**9.81**

In Oceania, republics have slightly higher means and medians than the monarchy/Commonwealth group. However, the difference in means is very small. The substantially higher standard deviation in the monarchy/Commonwealth group indicates much greater variation in performance. Overall, the descriptive statistics for Oceania do not reveal a pronounced difference between the two groups.

Comparison of the “Control of Corruption” Indicator between Monarchies and Republics in the Middle East

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Middle Eastern Monarchies	**7**	**70.13**	**66.50**	**11.96**
Middle Eastern Republics	**10**	**25.94**	**17.65**	**26.31**

The mean percentile rank for Control of Corruption is **70.13** among Middle Eastern monarchies and **25.94** among the region’s republics. Thus, **the mean for monarchies is approximately 44.19 percentile points higher**.

The medians also show a striking difference: 66.50 among monarchies and **17.65** among republics. **This indicates that the observed advantage is not driven merely by one or two exceptional monarchies, but is also clearly visible in the central positions of the two distributions.**

The standard deviation among the republics is more than twice that of the monarchies. This indicates substantially greater heterogeneity among Middle Eastern republics, whereas the monarchies are clustered at a higher and relatively more consistent level of performance.

Test of the Difference in Means

To assess the statistical significance of the difference between the two group means, **Welch’s t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**44.19**
Welch’s t-statistic	**4.67**
Approximate degrees of freedom	**13.35**
Two-tailed p-value	**0.00041**
95% confidence interval	**23.79 to 64.59**
Hedges’ g	**1.93**

Welch’s test indicates that the difference in mean Control of Corruption between Middle Eastern monarchies and republics is highly statistically significant. The p-value is below 0.001, and the entire 95% confidence interval lies above zero.

Hedges’ g is **1.93**, indicating a **very large effect size**. Thus, **the difference between the two groups is not only statistically significant but also exceptionally large in magnitude**.

Mann–Whitney U Test

To avoid relying solely on the assumptions of a parametric test, the distributions of the two groups were also compared using the nonparametric **Mann–Whitney U test**.

Statistical Measure	Value
Mann–Whitney U statistic	**62**
Two-tailed p-value	**0.00679**
Rank-biserial correlation	**0.77**

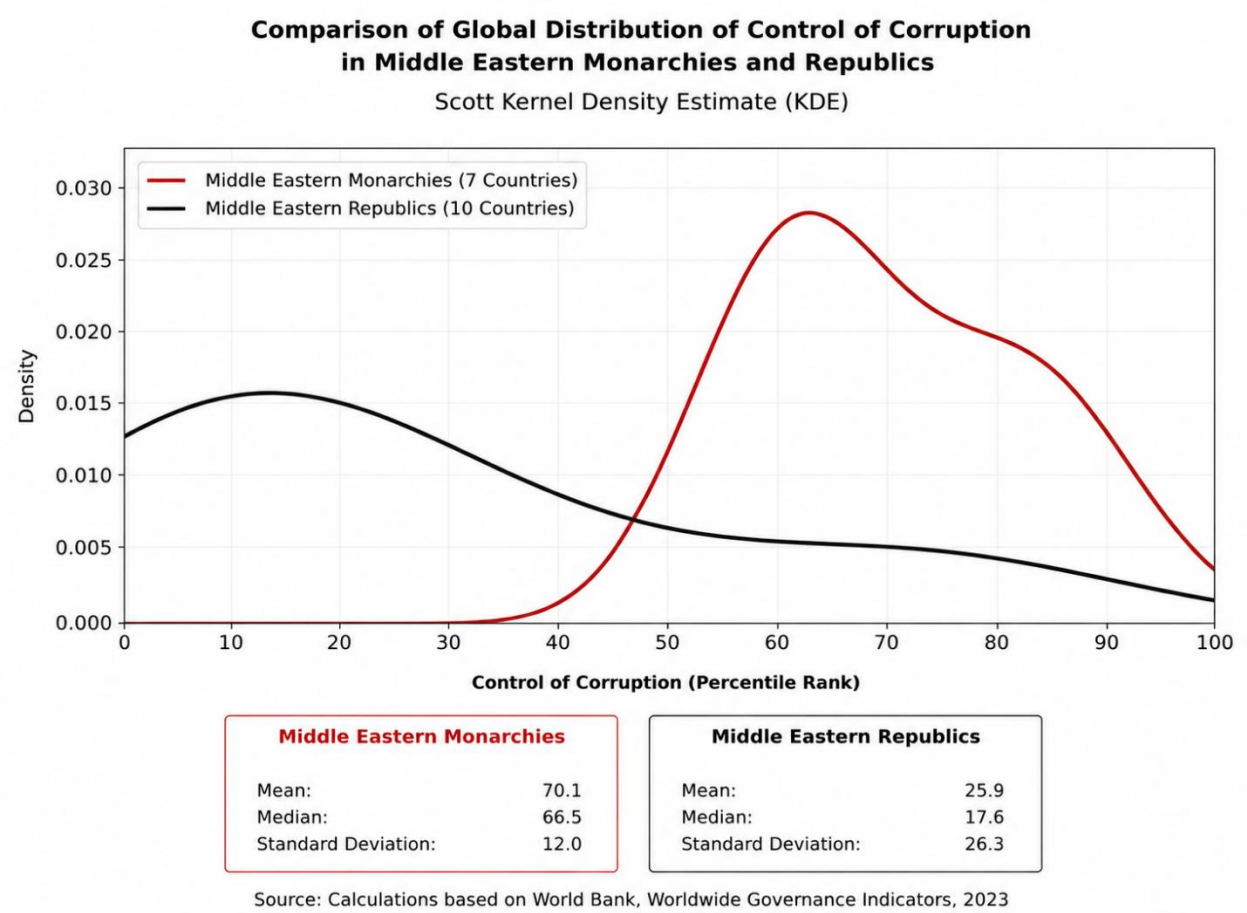
The Mann–Whitney test also indicates a statistically significant difference between the two groups. The rank-biserial correlation of **0.77** indicates a **very large effect size** and a pronounced tendency for Middle Eastern monarchies to occupy higher Control of Corruption ranks.

Conclusion

Middle Eastern monarchies perform substantially better than the region's republics on the Control of Corruption indicator. Their mean is more than **44 percentile points** higher, while their median is nearly **49 percentile points** higher. **Both Welch's t-test and the Mann–Whitney U test support the existence of a difference, and the effect sizes are very large.**

The much lower standard deviation among monarchies also indicates that their stronger performance is not confined to a few exceptional cases, but is more consistent across the group. In contrast, Middle Eastern republics combine much lower mean and median scores with substantially greater dispersion.

Overall, the 2023 data indicate that Middle Eastern monarchies have a considerably better record than the region's republics in controlling corruption and limiting the abuse of public power. This finding represents an **observational association** and does not, **by itself**, establish a causal relationship between the form of government and corruption control.



The kernel density estimate (KDE) shows that the distribution of Middle Eastern monarchies is clearly concentrated at higher levels of the Control of Corruption indicator, with the greatest density occurring in the upper half of the range. In contrast, Middle Eastern republics are predominantly concentrated at lower levels of the indicator and also display considerably greater dispersion, reflecting substantial variation in their performance in controlling corruption. The clear separation between the two curves indicates a pronounced difference in the distributional patterns of the two groups.

Status of the “New or Modern Monarchy” Hypothesis After the Ninth Indicator

With the addition of the **Control of Corruption** chapter, the table is updated as follows:

Test	Indicator	Result
✓	**GDP per Capita**	**Consistent with the hypothesis**
✓	**Purchasing Power per Capita**	**Consistent with the hypothesis**

Test	Indicator	Result
◐	**Gini Coefficient**	**Mixed evidence; difference favors the hypothesis**
✓	**Political Stability and Absence of Violence**	**Consistent with the hypothesis**
✓	**Regulatory Quality**	**Consistent with the hypothesis**
✓	**Rule of Law**	**Consistent with the hypothesis**
✓	**Government Effectiveness**	**Consistent with the hypothesis**
✓	**Inflation**	**Consistent with the hypothesis**
✓	**Control of Corruption**	**Consistent with the hypothesis**

At this stage, the **“New or Modern Monarchy” hypothesis has been evaluated across nine independent indicators**, and the results of all nine are consistent with its predictions. This does not mean that the hypothesis has been conclusively proven. Rather, it means that none of the indicators examined so far has provided evidence sufficient to reject it, while the empirical evidence accumulated thus far continues to support it.

Corruption Perceptions

In the previous section, we examined countries' performance using the World Bank's **Control of Corruption** indicator. That indicator assesses the extent to which governments are able to limit the abuse of public power and control corruption.

We now examine the same issue from a different perspective. The **Corruption Perceptions Index (CPI)**, published annually by **Transparency International**, assesses the perceived level of public-sector corruption based on evaluations by experts and businesspeople. The index reflects perceptions of corruption in the public sector and the abuse of public power for private gain, and is one of the most widely recognized international measures of public-sector corruption.

If the **"New or Modern Monarchy" hypothesis** is correct, modern monarchies would be expected, contrary to the conventional image of historical monarchies, to be associated with lower levels of corruption and abuse of public power and to perform better in this respect than republics in their respective regions.

The data used in this section are from the **2023 Corruption Perceptions Index** (Appendix RDS-J01). A higher score on this index indicates **lower perceived corruption** and therefore better performance in preventing the abuse of public power.

Global Comparison of the "Corruption Perceptions" Indicator between Monarchies and Republics

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Monarchies	**24**	**57.79**	**59.00**	**18.86**
Republics	**141**	**39.24**	**36.00**	**16.94**

The mean Corruption Perceptions Index score is **57.79** among the world's monarchies and **39.24** among the world's republics. Thus, **monarchies score, on average, 18.55 points higher than republics; in other words, perceived corruption is lower among monarchies than among republics.**

The medians point in the same direction. The median score is **59** among monarchies and **36** among republics. Thus, the median monarchy scores **23 points** higher than the median republic. **This indicates that the observed advantage of monarchies is not confined to a few exceptional countries but is also evident across the broader distribution of the data.**

Dispersion is also relatively similar in the two groups. The standard deviation is **18.86** among monarchies and **16.94** among republics. The difference between the two group means therefore cannot be attributed simply to a difference in the dispersion of their scores.

Test of the Difference in Means

To assess whether the difference between the two group means is statistically significant, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**18.55**
Welch's t-statistic	**4.52**
Approximate degrees of freedom	**29.65**
Two-tailed *p*-value	**0.00009**
95% confidence interval	**10.16 to 26.94**
Hedges' g	**1.07**

Welch's test indicates that the difference in mean Corruption Perceptions Index scores between monarchies and republics worldwide is highly statistically significant (**$p < 0.001$**). The entire 95% confidence interval also lies above zero, supporting the observed higher mean score among monarchies.

Hedges' g is **1.07**, indicating a **large effect size**. Thus, **the observed difference is not only statistically significant but also substantial in magnitude**.

Mann–Whitney U Test

To ensure that the conclusion does not depend solely on the assumptions of the parametric test, the nonparametric **Mann–Whitney U test** was also conducted.

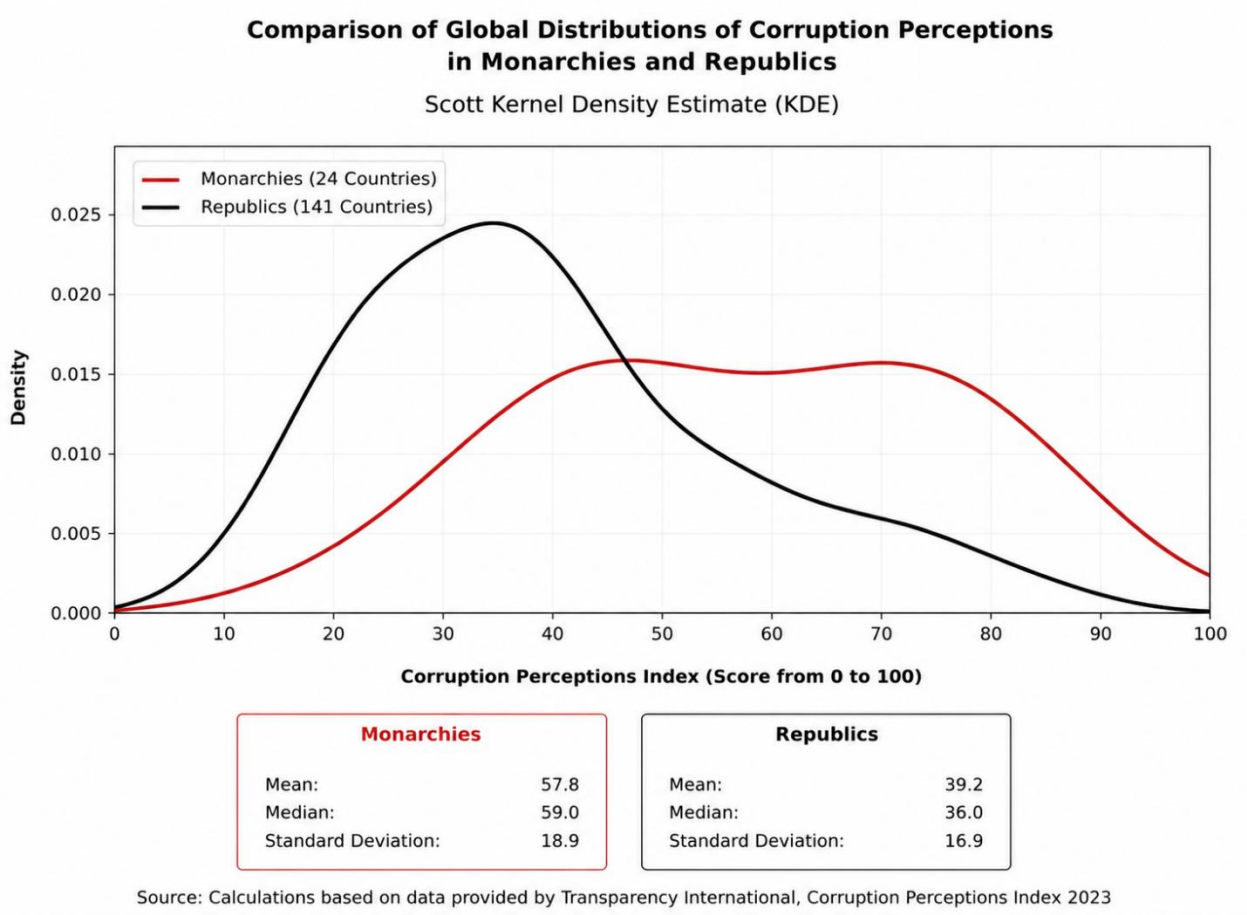
Statistical Measure	Value
Mann–Whitney U statistic	**2613.5**
Two-tailed *p*-value	**0.00002**
Rank-biserial correlation	**0.54**

The Mann–Whitney U test also confirms a difference between the two groups with very strong statistical evidence (**$p < 0.001$**). The rank-biserial correlation of **0.54** indicates a **large effect size**, showing that Corruption Perceptions Index scores among monarchies generally occupy higher ranks than those among republics.

Conclusion

In 2023, **the world's monarchies performed substantially better than the world's republics on the Corruption Perceptions Index. Both the mean and median scores were considerably higher among monarchies, and both Welch's t-test and the Mann–Whitney U test provide very strong statistical evidence of a difference between the two groups. The large effect sizes from both analyses further indicate that the observed difference is substantial in magnitude, rather than merely statistically significant.**

Overall, the data examined in this study show that monarchies had lower levels of **perceived public-sector corruption** than republics. As in the other chapters of this book, this finding represents an **observational association** and does not, **by itself**, establish a causal relationship between the form of government and the level of corruption.



The kernel density estimate (KDE) plot shows that the distribution of monarchies worldwide on the Corruption Perceptions Index is generally shifted toward higher scores, with most monarchies concentrated at relatively favorable levels of the index. In contrast, the distribution of republics is concentrated mainly at lower scores. Although republics exhibit a wide range of performance, their highest density occurs at levels below those of monarchies. Overall, this pattern indicates that

monarchies tend to perform more favorably than republics in terms of perceived public-sector corruption

Comparison of the Corruption Perceptions Index in Constitutional and Non-Constitutional Monarchies Worldwide

Descriptive Statistics

Type of Monarchy	Number of Countries	Mean	Median	Standard Deviation
Constitutional Monarchies	**14**	**63.00**	**72.00**	**20.33**
Semi-Constitutional and Absolute Monarchies	**10**	**50.50**	**49.00**	**12.19**

The mean Corruption Perceptions Index score is **63.00** for constitutional monarchies and **50.50** for semi-constitutional and absolute monarchies. Thus, **constitutional monarchies score, on average, 12.50 points higher on this index**.

The medians point in the same direction. The median score is **72** for constitutional monarchies and **49** for the second group, meaning that the median constitutional monarchy scores **23 points** higher than the median semi-constitutional or absolute monarchy.

The standard deviation is also higher among constitutional monarchies (**20.33**) than in the second group (**12.19**), indicating greater variation in performance among constitutional monarchies.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (constitutional minus semi-constitutional and absolute)	**12.50**
Welch's t-statistic	**1.80**
Approximate degrees of freedom	**21.60**
Two-tailed *p*-value	**0.086**
Hedges' *g*	**0.66**

Welch's test indicates that **the difference in means between the two types of monarchy is not statistically significant at the 5% level** (**p = 0.086**), although the *p*-value is relatively close to the conventional significance threshold.

Hedges' *g* is **0.66**, indicating a **moderate-to-large effect**. Thus, although the observed difference does not reach conventional statistical significance, its magnitude cannot be regarded as small.

Mann–Whitney Test

To compare the distributions of the two groups, the nonparametric **Mann–Whitney test** was also conducted.

Statistical Measure	Value
U statistic	**100**
Two-tailed *p*-value	**0.084**
Rank-biserial correlation	**0.43**

The Mann–Whitney test leads to a similar conclusion. The difference between the two groups is not statistically significant at the 5% level (**p = 0.084**), although the *p*-value is relatively close to the conventional threshold. The rank-biserial correlation of **0.43** indicates a **moderate effect** in favor of constitutional monarchies.

Conclusion

In 2023, constitutional monarchies performed better than semi-constitutional and absolute monarchies in terms of both the mean and median Corruption Perceptions Index scores. However, neither Welch's test nor the Mann–Whitney test found the difference statistically significant at the conventional 5% level, although both *p*-values were relatively close to that threshold and the effect sizes were moderate or larger.

Therefore, as with several other indicators, the main and clearest difference observed in this study on the Corruption Perceptions Index is between monarchies and republics, rather than necessarily between the two types of monarchy. This finding is also consistent with the broader pattern observed in the preceding chapters of the book.

Comparison of the Corruption Perceptions Index in Commonwealth Realms and Republics Worldwide

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Commonwealth Realms	**9**	**58.11**	**58.00**	**18.14**
Republics Worldwide	**141**	**39.24**	**36.00**	**16.94**

The mean Corruption Perceptions Index score is **58.11** among the Commonwealth realms and **39.24** among republics worldwide. Thus, **the Commonwealth realms score, on average, 18.87 points higher than republics**.

The medians point in the same direction. The median score is **58** among the Commonwealth realms and **36** among republics, meaning that the median country in the Commonwealth realm group scores **22 points** higher than the median republic.

The standard deviations of the two groups are also relatively similar. Therefore, the observed difference cannot simply be attributed to differences in the dispersion of the data.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Commonwealth realms minus republics)	**18.87**
Welch's t-statistic	**3.04**
Approximate degrees of freedom	**8.91**
Two-tailed *p*-value	**0.0142**
95% confidence interval	**4.79 to 32.95**
Hedges' *g*	**1.10**

Welch's test indicates that **the difference in mean Corruption Perceptions Index scores between the Commonwealth realms and republics worldwide is statistically significant at the 5% level** (**p = 0.014**). The 95% confidence interval lies entirely above zero, confirming that the direction of the difference favors the Commonwealth realms.

Hedges' *g* is **1.10**, indicating a **large effect**. Thus, the observed difference is not only statistically significant but also substantial in practical terms.

Mann–Whitney Test

To compare the overall distributions of the two groups, the nonparametric **Mann–Whitney test** was also conducted.

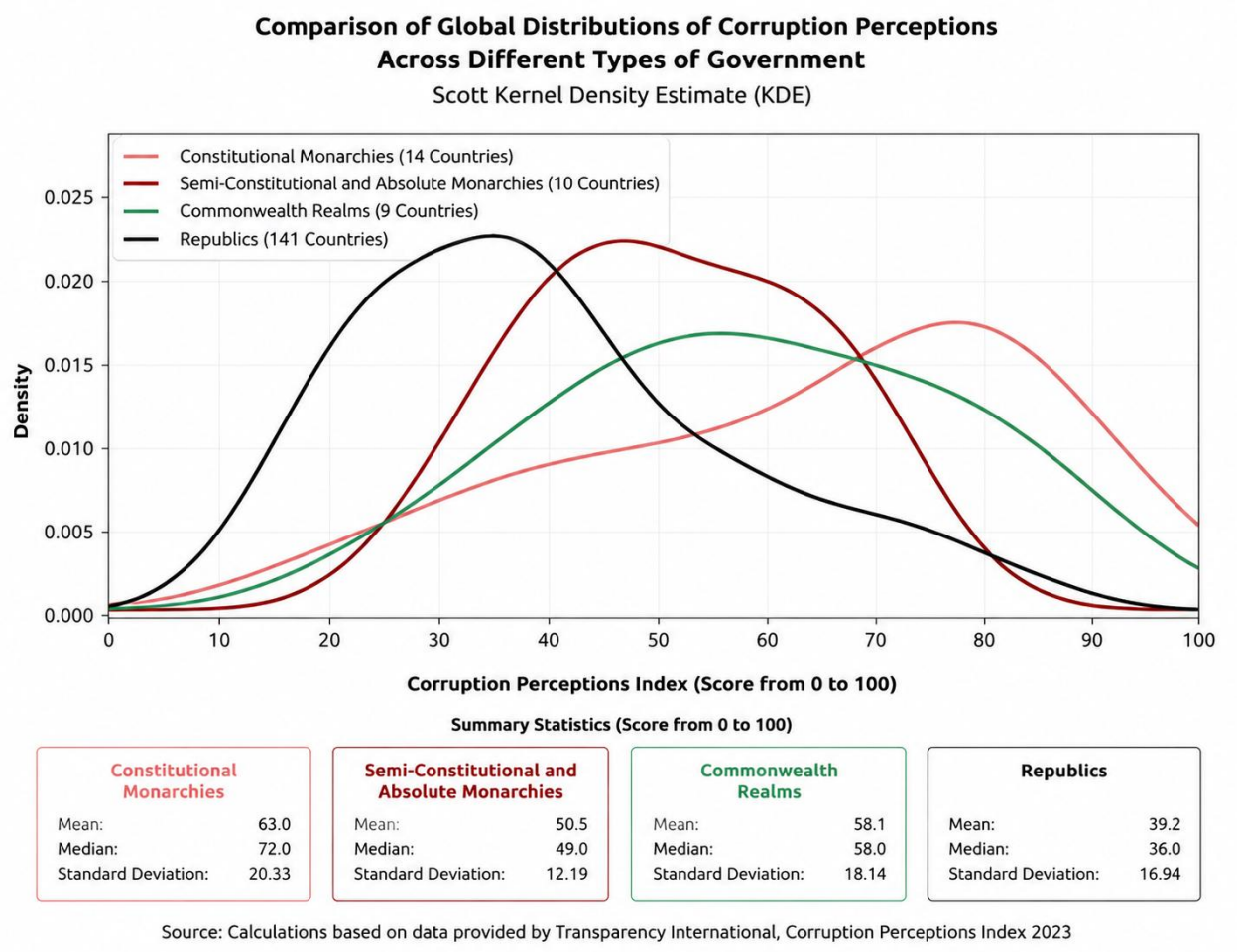
Statistical Measure	Value
U statistic	**1004**
Two-tailed *p*-value	**0.0035**
Rank-biserial correlation	**0.58**

The Mann–Whitney test also indicates a statistically significant difference between the two groups (**p < 0.01**). The rank-biserial effect size of **0.58** indicates a **large difference**, showing that Corruption Perceptions Index scores are generally higher among the Commonwealth realms than among republics worldwide.

Conclusion

In 2023, **the Commonwealth realms had higher Corruption Perceptions Index scores than republics worldwide in terms of both the mean and the median. Both Welch's test and the Mann–Whitney test confirm that the difference is statistically significant, and the effect sizes are large.**

Overall, the data indicate that perceived public-sector corruption was lower in the Commonwealth realms than in republics worldwide. As with the other comparisons in this book, however, this finding represents an **observational association** and does **not by itself establish a causal relationship** between the form of government and the level of corruption.



The kernel density estimate (KDE) plot shows that constitutional monarchies are most heavily concentrated at the higher end of the Corruption Perceptions Index, with their distribution clearly shifted toward higher scores. Commonwealth realms are also concentrated mainly at relatively high scores, followed by semi-constitutional and absolute monarchies, which occupy a more intermediate position. In contrast, the distribution of republics is concentrated primarily at lower scores and is more

widely dispersed, indicating considerable heterogeneity in their performance on the index. Overall, this pattern suggests that, among the four forms of government, constitutional monarchies have the most favorable performance on the Corruption Perceptions Index.

Comparison of the Corruption Perceptions Index in Monarchies and Republics Across Continents and the Middle East

Comparison of the Corruption Perceptions Index in Asian Monarchies and Republics

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Asian Monarchies	**13**	**51.00**	**50.00**	**14.39**
Asian Republics	**32**	**33.41**	**33.00**	**15.32**

The mean Corruption Perceptions Index score is **51.00** for Asian monarchies and **33.41** for Asian republics. Thus, **Asian monarchies score, on average, 17.59 points higher than the republics of the continent**.

The medians point in the same direction. The median score is **50** for monarchies and **33** for republics, meaning that the median country among Asian monarchies scores **17 points** higher than the median Asian republic.

The standard deviations of the two groups are also very similar (**14.39** and **15.32**, respectively). Thus, the observed difference cannot be attributed simply to differences in the dispersion of the data.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (monarchies minus republics)	**17.59**
Welch's t-statistic	**3.65**
Approximate degrees of freedom	**23.65**
Two-tailed *p*-value	**0.0013**
95% confidence interval	**7.63 to 27.56**
Hedges' *g*	**1.15**

Welch's test shows that the difference in mean Corruption Perceptions Index scores between Asian monarchies and republics is highly statistically significant (**p = 0.0013**). The 95% confidence interval also lies entirely above zero, confirming the higher scores of the monarchies.

Hedges' *g* is **1.15**, indicating a **large effect**. Thus, the observed difference is not only statistically significant but also substantial in practical terms.

Mann–Whitney Test

To ensure that the conclusion does not depend on the assumptions of the parametric test, the nonparametric **Mann–Whitney test** was also conducted.

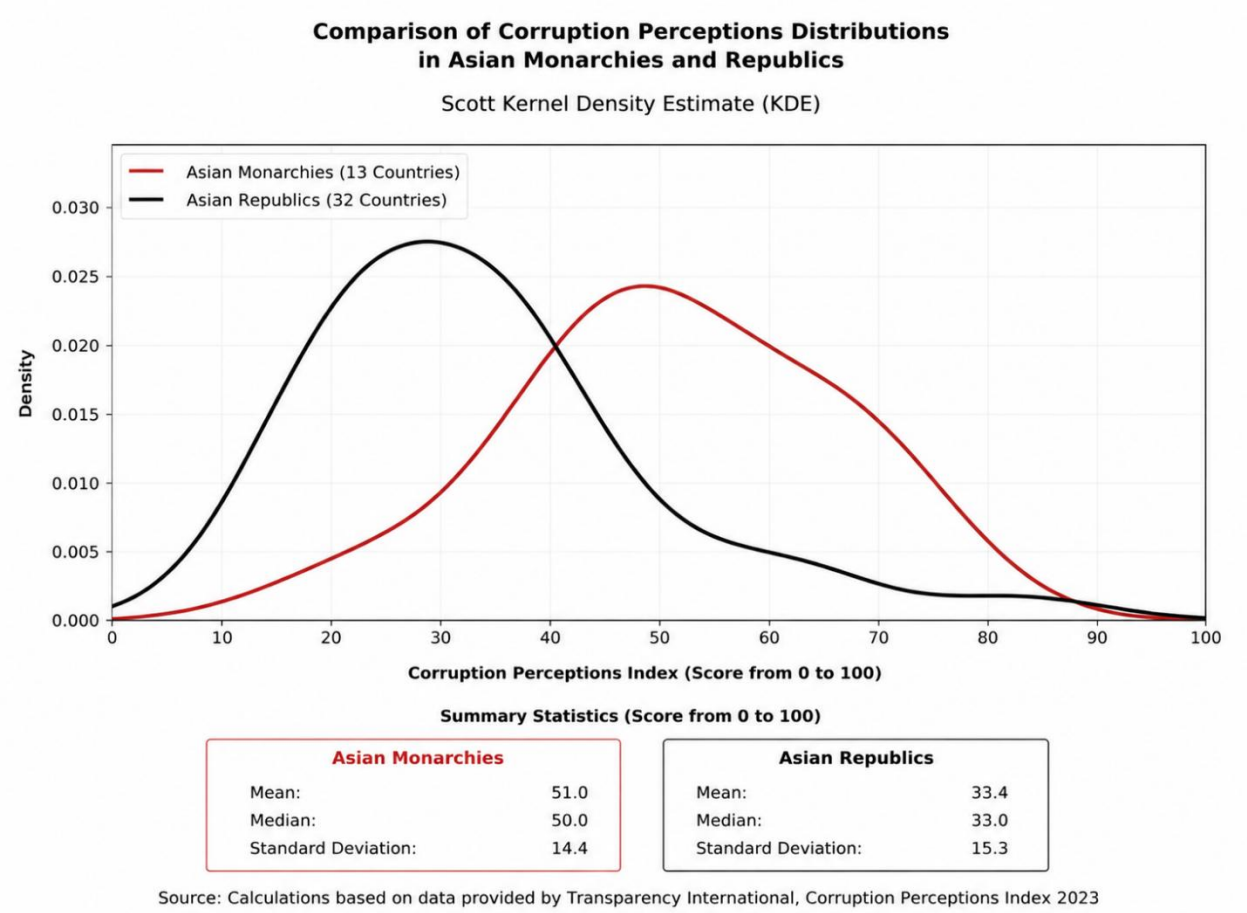
Statistical Measure	Value
U statistic	**344**
Two-tailed *p*-value	**0.00068**
Rank-biserial correlation	**0.65**

The Mann–Whitney test also confirms the difference between the two groups with very strong statistical evidence (**$p < 0.001$**). The rank-biserial correlation of **0.65** indicates a **large effect**, showing that Corruption Perceptions Index scores are generally substantially higher among Asian monarchies than among the republics of the continent.

Conclusion

In 2023, **Asian monarchies performed substantially better than Asian republics on the Corruption Perceptions Index. Both the mean and the median were considerably higher among monarchies, and both Welch's test and the Mann–Whitney test confirm the difference with very strong statistical evidence. The large effect sizes in both tests further indicate that the difference is substantial not only statistically but also in practical terms.**

Overall, the Asian data in this study indicate that monarchies on the continent had lower levels of perceived public-sector corruption than Asian republics. As in the other sections of the book, however, this finding represents an **observational association** and does **not by itself establish a causal relationship** between the form of government and the level of corruption.



The kernel density estimate (KDE) shows that the distribution of Asian monarchies on the Corruption Perceptions Index is generally shifted toward higher scores, with the greatest concentration in the middle-to-upper range of the index. In contrast, Asian republics are concentrated mainly at lower scores and also exhibit greater dispersion, indicating greater heterogeneity in their performance on perceived corruption. The separation between the two curves indicates that, on average, Asian monarchies perform more favorably on the Corruption Perceptions Index than Asian republics.

Comparison of the Corruption Perceptions Index Between Monarchies and Republics in Europe

Descriptive Statistics

Form of Government	Number of Countries	Mean	Median	Standard Deviation
European Monarchies	**8**	**77.13**	**78.50**	**9.17**
European Republics	**32**	**54.25**	**53.50**	**15.80**

The mean Corruption Perceptions Index score is **77.13** for European monarchies and **54.25** for European republics. Thus, **European monarchies score, on average, 22.88 points higher than European republics**.

The median confirms the same pattern. The median is **78.50** for monarchies and **53.50** for republics, meaning that the median country among European monarchies scores **25 points** higher than the median country among European republics. **This indicates that the advantage of monarchies is not limited to a few exceptional countries but is also evident across the overall distribution of the data.**

The standard deviation is also considerably lower among European monarchies (**9.17**) than among European republics (**15.80**). This suggests that European monarchies not only have lower levels of perceived corruption but also exhibit more consistent performance.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies minus Republics)	**22.88**
Welch's t-statistic	**5.35**
Approximate degrees of freedom	**18.89**
Two-tailed *p*-value	**0.000038**
95% confidence interval	**13.91 to 31.84**
Hedges' *g*	**1.51**

Welch's test indicates that the difference in mean Corruption Perceptions Index scores between European monarchies and republics is highly statistically significant (**$p < 0.001$**). The 95% confidence interval also lies entirely above zero, confirming the higher scores of the monarchies.

Hedges' *g* is **1.51**, indicating a **very large effect**. Thus, the observed difference is not only statistically significant but also substantial in practical terms.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the nonparametric **Mann–Whitney U test** was also conducted.

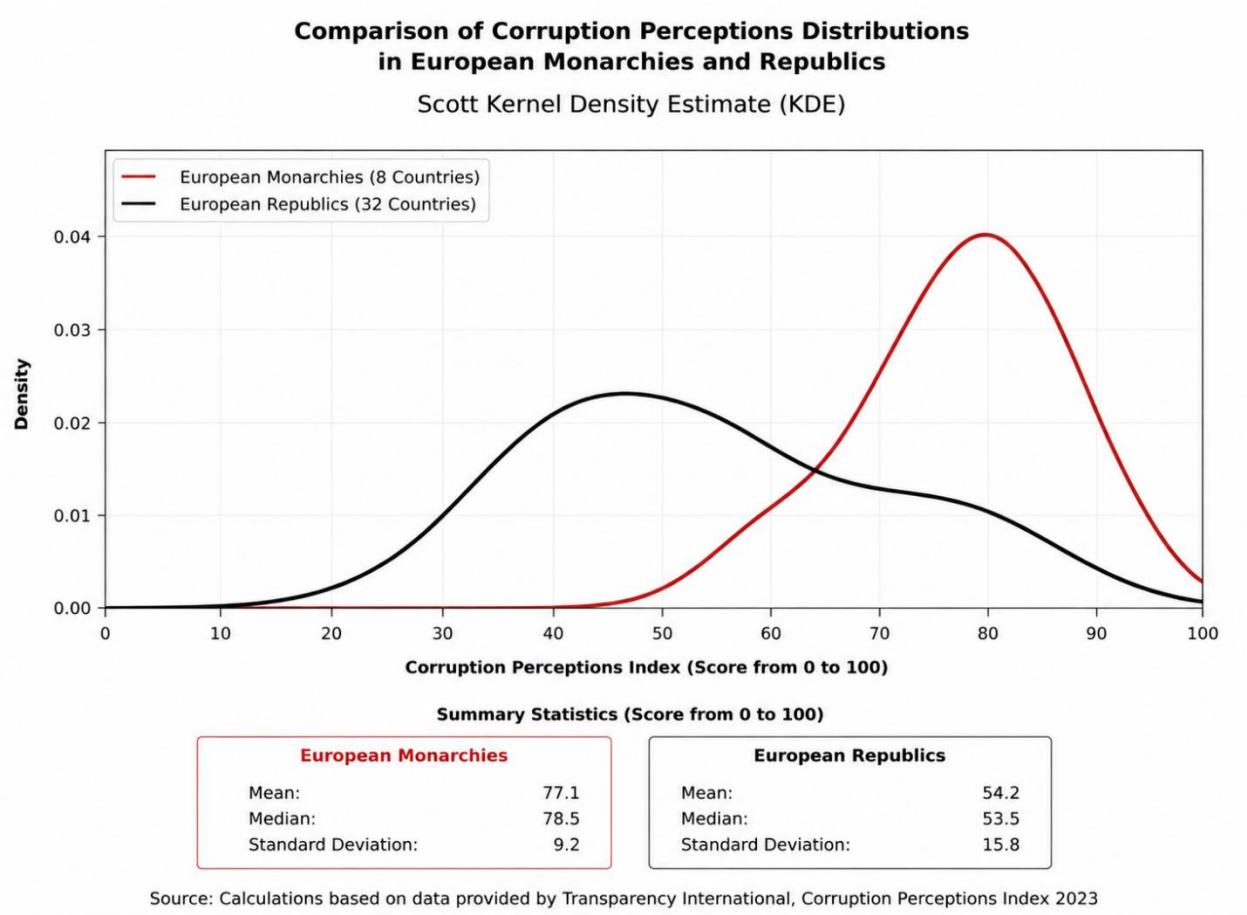
Statistical Measure	Value
U statistic	**226.5**
Two-tailed *p*-value	**0.00091**
Rank-biserial correlation	**0.77**

The Mann–Whitney test also confirms a highly statistically significant difference between the two groups (**$p < 0.001$**). The rank-biserial correlation of **0.77** indicates a **very large effect**, showing that Corruption Perceptions Index scores are generally substantially higher among European monarchies than among European republics.

Conclusion

In 2023, **European monarchies performed substantially better than European republics on the Corruption Perceptions Index. Both the mean and median scores were considerably higher among monarchies, and both Welch's t-test and the Mann–Whitney test confirmed the difference with very strong statistical evidence. Moreover, the very large effect sizes in both tests indicate that the difference is not only statistically significant but also substantial in practical terms.**

The lower standard deviation among European monarchies also indicates that these countries have achieved lower levels of perceived corruption with greater consistency. Overall, the European data show that monarchies performed better than republics on the Corruption Perceptions Index.



The kernel density estimate (KDE) shows that the distribution of European monarchies on the Corruption Perceptions Index is noticeably shifted toward higher scores, with most of these countries concentrated at favorable levels of the index. In contrast, although European republics include several countries with very strong performance, their scores are much more widely dispersed, with a substantial share concentrated at moderate and lower levels. This pattern indicates that, on average, European monarchies have both more favorable and more homogeneous outcomes on the Corruption Perceptions Index than European republics.

Comparison of the Corruption Perceptions Index Between European Monarchies and European Republics That Have Never Been Communist

Descriptive Statistics

Form of Government	Number of Countries	Mean	Median	Standard Deviation
European Monarchies	**8**	**77.13**	**78.50**	**9.17**
Non-Communist European Republics	**11**	**68.82**	**71.00**	**12.42**

The mean Corruption Perceptions Index score is **77.13** for European monarchies and **68.82** for European republics that have never been communist. Thus, **European monarchies score, on average, 8.31 points higher on this index**.

The median points in the same direction. The median is **78.50** for monarchies and **71.00** for non-communist republics, meaning that the median country among European monarchies scores **7.5 points** higher than the median country among non-communist European republics.

The standard deviation is also lower among monarchies (**9.17**) than among non-communist republics (**12.42**), indicating that European monarchies combine higher scores with greater consistency in their performance.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies minus Republics)	**8.31**
Welch's *t*-statistic	**1.68**
Approximate degrees of freedom	**16.98**
Two-tailed *p*-value	**0.112**

Welch's test indicates that **the difference in mean Corruption Perceptions Index scores between the two groups is not statistically significant** (**p = 0.112**). Given the available sample sizes, there is therefore insufficient evidence to draw a firm conclusion about a systematic difference between the two groups.

Mann–Whitney Test

The nonparametric **Mann–Whitney U test** was also conducted to compare the distributions of the two groups.

Statistical Measure	Value
U statistic	**63**
Two-tailed *p*-value	**0.126**

The Mann–Whitney test leads to the same conclusion, with the difference between the two groups not reaching statistical significance (**p = 0.126**).

Conclusion

Once European republics with a history of communist rule are excluded, the gap between monarchies and republics on the Corruption Perceptions Index narrows considerably. Although European monarchies still perform better in terms of both the mean and the median, this advantage does not reach statistical significance in either Welch's test or the Mann–Whitney test.

Nevertheless, this result remains noteworthy in the context of the book's central hypothesis. The republics included in this comparison are countries with long histories of democracy, competitive elections, and freedom of the press—factors commonly regarded in the political science literature as among the most important mechanisms for constraining corruption. These countries would therefore be expected to perform very strongly on the Corruption Perceptions Index. The data in this study confirm that these republics do indeed perform at a high level. At the same time, European monarchies still achieve higher mean and median scores, although the difference is not statistically significant given the available sample sizes. As in other sections of the book, this finding represents an observational relationship and does not, by itself, establish a causal relationship between the form of government and the level of corruption.

Comparison of the Corruption Perceptions Index in Monarchies and Republics in Africa, the Americas, and Oceania

Continent and Form of Government	Number of Countries	Mean	Median	Sample Standard Deviation
Africa — Monarchies	**3**	**35.67**	**38.00**	**4.93**
Africa — Republics	**49**	**32.94**	**31.00**	**12.75**
Americas — Commonwealth Realms	**5**	**58.20**	**58.00**	**11.71**
Americas — Republics	**26**	**39.00**	**35.50**	**16.57**
Oceania — Commonwealth Realms	**4**	**58.00**	**59.00**	**26.36**
Oceania — Republics	**2**	**50.00**	**50.00**	**2.83**

No data are available for **Tonga** in the dataset used for this indicator. Therefore, no independent monarchy remains for comparison in Oceania, and the analysis is limited to Commonwealth realms and republics for which data are available.

As with the preceding indicators, the limited number of monarchies and Commonwealth realms in Africa, the Americas, and Oceania makes inferential statistical testing of limited value. The results in these three regions are therefore reported at the descriptive level only. On the Corruption Perceptions Index, a higher score indicates a more favorable outcome and a lower level of perceived corruption.

Africa

The three monarchies with available data—**Lesotho, Morocco, and Eswatini**—have a mean score of **35.67**, compared with **32.94** among the continent's 49 republics. The median is also higher among monarchies, at **38**, compared with **31** among republics. Both measures of central tendency therefore point in the direction of better performance among the monarchies, although the difference in means is relatively modest.

There is also a notable difference in dispersion. The standard deviation is **4.93** among the monarchies and **12.75** among the republics. Thus, in addition to having higher mean and median scores, the three African monarchies have scores that are more closely clustered together. Given the very small number of monarchies, however, this pattern cannot support broad statistical conclusions.

The Americas

Because there are no independent monarchies in the Americas, the comparison is between **Commonwealth realms** and republics. The five Commonwealth realms with available data have

a mean score of **58.20**, compared with **39.00** among the continent's 26 republics. The **19.20-point difference** between the two groups is substantial on the scale of the Corruption Perceptions Index.

The medians point in the same direction. The median among Commonwealth realms is **58**, compared with **35.50** among republics. Thus, the observed difference is not merely the result of one or two exceptionally high-scoring countries; a clear gap is also visible in the center of the two distributions. The standard deviation is **11.71** among the Commonwealth realms and **16.57** among the republics.

The descriptive data for the Americas therefore present a relatively clear pattern: **Commonwealth realms have substantially higher mean and median Corruption Perceptions Index scores than the republics of the continent.**

Oceania

The available data are much more limited in Oceania. Tonga, the region's only independent monarchy, has no observation in the source used for this indicator. The comparison is therefore restricted to four Commonwealth realms and two republics with available data.

The mean score is **58** among the Commonwealth realms and **50** among the republics. The respective medians are **59** and **50**. Thus, the direction of the difference again favors the Commonwealth realms.

This result, however, must be interpreted with particular caution. Only **two republics** have data in this comparison, while the four Commonwealth realms display substantial dispersion, with a standard deviation of **26.36**. Oceania therefore provides a very limited descriptive comparison rather than a basis for meaningful statistical inference.

Overall, the descriptive statistics point in the same direction in all three comparisons that can be made: **African monarchies have higher mean and median Corruption Perceptions Index scores than African republics; Commonwealth realms in the Americas have higher scores than the republics of the Americas; and Commonwealth realms in Oceania have higher scores than the republics with available data in that region.** Nevertheless, particularly in Africa and Oceania, the small size of the comparison groups means that these findings should be treated strictly as **descriptive evidence rather than as independent statistical tests of the superiority of one form of government**.

Comparison of the Corruption Perceptions Index Between Monarchies and Republics in the Middle East

Descriptive Statistics

Form of Government	Number of Countries	Mean	Median	Standard Deviation
Middle Eastern Monarchies	**7**	**50.71**	**46.00**	**8.73**
Middle Eastern Republics	**9**	**31.56**	**24.00**	**15.56**

The mean Corruption Perceptions Index score is **50.71** for Middle Eastern monarchies and **31.56** for Middle Eastern republics. Thus, **monarchies in the region score, on average, 19.16 points higher than republics**.

The median confirms the same pattern. The median is **46** for monarchies and **24** for republics, meaning that the median country among Middle Eastern monarchies scores **22 points** higher than the median country among Middle Eastern republics. **This indicates that the advantage of monarchies is not confined to one or two particular countries but is also evident across the overall distribution of the data.**

The standard deviation is also considerably lower among monarchies (**8.73**) than among republics (**15.56**), indicating greater consistency in the performance of the region's monarchies on this index.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies minus Republics)	**19.16**
Welch's *t*-statistic	**2.923**
Approximate degrees of freedom	**13.06**
Two-tailed *p*-value	**0.0118**
Hedges' *g*	**1.30**

Welch's test indicates that **the difference in mean Corruption Perceptions Index scores between Middle Eastern monarchies and republics is statistically significant** (**p = 0.0118**). Hedges' *g* is **1.30**, indicating a **large effect**. Thus, the observed difference is not only statistically significant but also substantial in practical terms.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the nonparametric **Mann–Whitney U test** was also conducted.

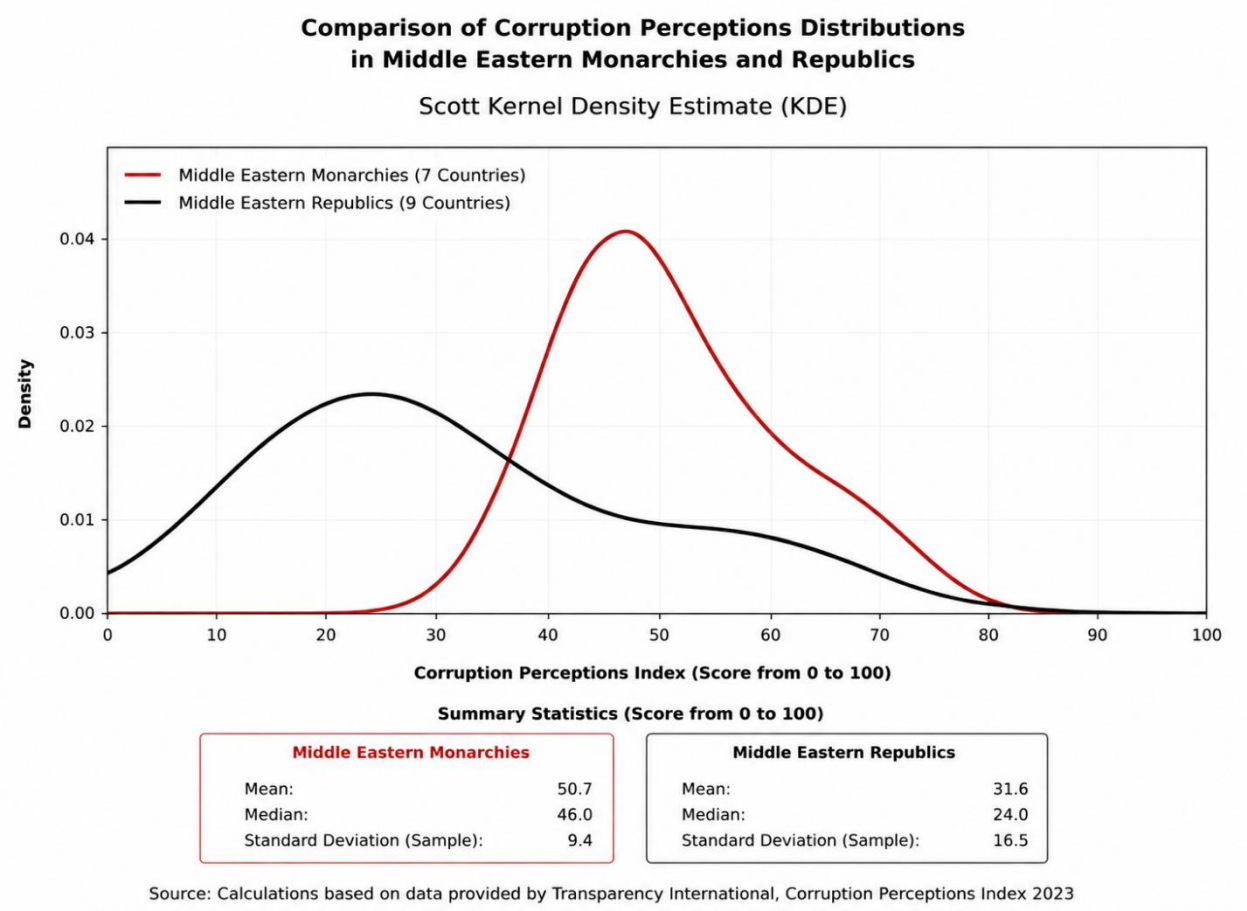
Statistical Measure	Value
U statistic	**52**
Two-tailed *p*-value	**0.0340**
Rank-biserial correlation	**0.65**

The Mann–Whitney test also indicates a statistically significant difference between the two groups (**p = 0.0340**). The rank-biserial correlation of **0.65** likewise indicates a **large effect**, showing a clear overall advantage for Middle Eastern monarchies on this index.

Conclusion

In 2023, **Middle Eastern monarchies performed better than the region's republics on the Corruption Perceptions Index. Both the mean and median scores were substantially higher among monarchies, and both Welch's t-test and the Mann–Whitney test confirmed that the difference was statistically significant. The large effect sizes also indicate that the difference is substantial in practical terms.**

Moreover, the lower dispersion of scores among Middle Eastern monarchies indicates that their stronger performance is not confined to one or two countries but is more consistent across the group. Overall, the data in this study show that Middle Eastern monarchies had lower levels of perceived public-sector corruption than the region's republics. As in other sections of the book, this finding represents an observational relationship and **does not, by itself, establish a causal relationship between the form of government and the level of corruption**.



The kernel density estimate shows that the distribution of Middle Eastern monarchies on the Corruption Perceptions Index is noticeably shifted toward higher scores, with the greatest concentration in the middle-to-upper range of the index. By contrast, Middle Eastern republics are concentrated mainly at lower scores and display greater dispersion, indicating more heterogeneous performance in terms of perceived corruption. The separation between the two curves suggests that, on average, the region's monarchies are in a more favorable position than its republics.

Status of the "New or Modern Monarchy" Hypothesis after the Tenth Indicator

With the addition of the Corruption Perceptions Index, the status of the hypothesis is updated as follows:

Test	Indicator	Result
✓	GDP per Capita	Consistent with the hypothesis
✓	Purchasing Power per Capita	Consistent with the hypothesis

Test	Indicator	Result
◐	Gini Coefficient	Mixed evidence; difference favors the hypothesis
✓	Political Stability and Absence of Violence	Consistent with the hypothesis
✓	Regulatory Quality	Consistent with the hypothesis
✓	Rule of Law	Consistent with the hypothesis
✓	Government Effectiveness	Consistent with the hypothesis
✓	Inflation	Consistent with the hypothesis
✓	Control of Corruption	Consistent with the hypothesis
✓	Corruption Perceptions Index	Consistent with the hypothesis

At this stage, the **“New or Modern Monarchy” hypothesis has been evaluated using ten independent indicators, and the results for all ten have been in the direction predicted by the hypothesis.** For the Corruption Perceptions Index as well, monarchies generally received higher scores than republics in both global and regional comparisons, particularly in Asia, Europe, and the Middle East. At the global level, the difference between monarchies and republics was both statistically significant and characterized by a large effect size.

This result does not constitute definitive proof of the hypothesis, nor does it establish a causal relationship between monarchy and lower levels of corruption. A more precise formulation is that **the empirical evidence gathered thus far is consistent with the predictions of the hypothesis, and none of the ten indicators examined has produced a result in the opposite direction.**

The Record on Liberty

Earlier, we noted that political philosophers have identified five principal aims of government. So far, we have examined the extent to which four of these aims—welfare, security, order, and justice—have been achieved in the world's monarchies and republics. The fifth aim of government is **liberty**.

One of the first things that may come to mind when the reader hears the word *liberty* is freedom of expression. Yet freedom of expression was not historically the first or most fundamental issue of liberty. Before people could speak freely, they first had to be freed from slavery, personal dependence, and the denial of the most basic rights. To understand the meaning of liberty, it is therefore useful to begin with the relationship between **rights, choice, and liberty**.

In earlier sections, we noted that for thousands of years land was the most important means of production, and that a large share of human needs was met directly or indirectly through it. In an age when industry and technology were limited, human labor was enormously important. Landowners and governments therefore sought to secure that labor at the lowest possible cost. As a result, many societies developed laws that recognized slavery, serfdom, and various forms of dependence on land or landlords.

Slaves were deprived of many of the rights that we now regard as fundamental human rights. Serfs, too, were often unable to choose freely where they lived, what occupation they pursued, or where they worked. **The more limited a person's legal rights are, the fewer choices that person has; and the narrower the range of available choices, the less liberty that person enjoys.**

If a serf is permitted to work only on the land of a particular landlord and cannot possess land of his own, he loses the ability to choose what to produce, where to work, or how to organize his life. A slave whose labor and wages are controlled by someone else cannot choose an occupation or decide how the fruits of that labor will be used. A girl denied the right to education not only loses the ability to choose her educational path in childhood; as an adult, her lack of knowledge and skills will also leave her with fewer options for building the life she wishes to lead.

In past societies, governments often assigned different rights to different classes, occupations, religions, and sexes according to their own interests and those of the groups that supported them. The social position into which a person was born largely determined his or her rights and course of life, while movement from one legal status to another was often extremely difficult or impossible.

Religion and custom sometimes reinforced and institutionalized these legal distinctions. Some restrictions took the form of religious rules, while others, even without direct legal or religious authority, were enforced through tradition and social pressure. In this way, **government,**

religion, and custom often restricted the range of choices otherwise available to individuals and thereby shaped the extent of their liberty.

With the emergence of natural-rights theory and modern political philosophy, the idea gained strength that human beings are naturally free and possess fundamental rights. From this perspective, the role of government is not merely to maintain order or carry out the commands of rulers. Government must also create conditions in which each person can choose the way of life he or she prefers and pursue happiness in his or her own way, so long as doing so does not violate the equal rights of others.

This responsibility presents government with a difficult problem. On the one hand, maintaining security, order, and the rights of others necessarily requires restrictions on certain forms of conduct. On the other hand, government must avoid unnecessary restrictions and preserve the widest possible range of lawful choices for its citizens. In the liberal conception of government, liberty does not mean the absence of all law; rather, it means that everyone should enjoy **the greatest liberty compatible with equal liberty for others**.

Within this framework, **the more extensive the legal rights enjoyed by citizens, the greater the range of choices open to them and the greater their practical liberty to shape their own lives**. We will now examine the extent to which monarchies and republics recognize and protect the rights and liberties of their citizens.

A Note on the Concept of Class Society

It is important here to distinguish between two different meanings of a "class society." In many historical societies, different classes did not enjoy equal rights under the law. Nobles, clergy, serfs, slaves, women, and members of different religious communities were subject to different rights and obligations, and movement from one legal class to another was often extremely difficult.

In the modern world, however, the term *class* is also sometimes used to describe differences in income and economic position. Economic inequality is not the same as legal inequality. Members of a society may have different levels of income and wealth while the law grants them equal civil rights and equal rights to property, education, employment, and social mobility. Such a society may be economically unequal, but in the historical and legal sense it is not necessarily a society based on hereditary classes with unequal rights.

This distinction does not imply that economic inequality or inequality of opportunity is unimportant. It simply means that differences in income should not automatically be equated with the legally stratified class systems of earlier societies. Historical experience has also shown that attempts to eliminate economic differences entirely, when accompanied by restrictions on property rights, occupational choice, and freedom of exchange, may themselves reduce both liberty and general welfare.

How Monarchies and Republics Have Approached Slavery

In ancient and premodern societies, slaves possessed the fewest rights and the least liberty. They had little or no control over where they lived, what work they performed, the fruits of their labor, and, in many cases, even whether and how they could form a family.

I initially intended to construct a historical indicator comparing the timing of the abolition of slavery in monarchies and republics. Such a comparison, however, faces serious methodological difficulties. Many present-day countries have changed their form of government during the past two centuries, and some republics inherited the prohibition of slavery from the monarchies that preceded them. Moreover, legal abolition did not always mean that slavery disappeared in practice. For these reasons, constructing a consistent and reliable indicator that would permit comparison across all countries is neither straightforward nor sufficiently robust.

Nevertheless, several historical examples are useful for illustrating the issue.

The United States declared independence from Britain in 1776 and adopted a republican form of government. Its Declaration of Independence proclaimed that human beings were created equal, yet in practice this principle did not extend to enslaved people or women and did not guarantee their political and legal equality.

Slavery was abolished throughout much of the British Empire in 1833, while it continued in the United States. The dispute over slavery was one of the principal causes of the American Civil War, which lasted from 1861 to 1865, and abolition was secured following the victory of the federal government. This experience illustrates that majority rule or the existence of republican institutions does not, by itself, guarantee recognition of the rights of minorities or groups lacking political power.

In Iran, too, although the Constitutional Revolution led to the establishment of the Majles in 1906, the legal prohibition of slavery came more than two decades later, during the reign of Reza Shah, through legislation passed by the National Consultative Assembly.

The Islamic Republic of Mauritania formally abolished slavery in 1981, but the prohibition long lacked sufficient enforcement mechanisms, and slavery was subsequently made a criminal offense. This example illustrates the important distinction between the formal recognition of a right in law and its effective enforcement in practice.

These cases alone are not sufficient to establish the superiority of monarchy or republic. They do, however, make one point clear: **a republican form of government does not necessarily guarantee the rights of minorities.** Even in an electoral system, a political majority may disregard the rights of a weaker group. This is the danger described in political philosophy as the **“tyranny of the majority.”**

The main method of this study, however, does not rest on a limited number of historical examples or case-by-case judgments. To reach more general conclusions, we need broad datasets and comparable indicators. Liberty is multidimensional and cannot be adequately measured by a single index. Accordingly, as in the preceding chapters, this chapter examines liberty from several different perspectives in order to determine whether any differences between forms of government recur consistently across multiple indicators.

We therefore begin in the next section with the **International Property Rights Index**, examining the extent to which monarchies and republics recognize and protect their citizens' property rights.

International Property Rights

As noted at the beginning of this chapter, liberty can be examined from several different perspectives. One of its most fundamental dimensions is the **right to property**, because property rights largely determine the extent to which individuals are free to make decisions about the fruits of their labor, their possessions, and their assets.

Full ownership of an asset means that, within the law, its owner is free to decide whether to use it or not, retain it, sell it, rent it out, give it away, leave it as an inheritance, or otherwise dispose of it. The broader these rights are, the greater the individual's freedom of action over that property.

For this reason, the owner of a house has far greater freedom in deciding what to do with it than the tenant of the same house, because the owner possesses a broader set of rights over that asset.

Throughout history, governments have been among the most important actors in defining the boundaries of property rights. In many societies, governments could determine who had the right to own land, who could hunt, fish, trade, or inherit, and even under what circumstances private property could be confiscated. Property rights have therefore long been one of the principal areas of tension between citizens and governments.

The more freely a government can interfere with private ownership without clear rules or constraints, the less control citizens have over their property, and the more their range of choices is reduced. Conversely, the more strongly a government protects property rights, the greater the freedom citizens have to make decisions about the fruits of their labor and their possessions.

Because this book also tests the **"New or Modern Monarchy" hypothesis**, we would expect, if the hypothesis is correct, modern monarchies to provide stronger protection for their citizens' property rights. According to this hypothesis, just as caring parents seek to give their children greater independence and autonomy, a king or queen would be expected to recognize broader rights for citizens, allowing them greater freedom to choose how they live and to develop their abilities.

To test this prediction, we use the **International Property Rights Index (IPRI)**. Published by the **Property Rights Alliance**, this index evaluates the extent to which countries protect both physical and intellectual property rights. The higher a country's score, the stronger the protection of property rights and the greater the freedom citizens have over their assets and achievements. Economic research has also found that stronger property-rights protection is generally associated with greater investment security, easier trade, and a higher capacity for economic growth.

The data used in this section are from **2024** and cover **123 countries** (Appendix RDS-K01).

Global Comparison of the International Property Rights Index in Monarchies and Republics

This section compares the performance of monarchies and republics worldwide on the **International Property Rights Index (IPRI)**. The data are from **2024**. On this index, **a higher score indicates stronger protection of property rights and therefore greater freedom for citizens to control and make use of their property and assets**.

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Monarchies Worldwide	**20**	**6.50**	**6.27**	**1.03**
Republics Worldwide	**99**	**4.83**	**4.63**	**1.28**

The mean International Property Rights Index score is **6.50** for monarchies and **4.83** for republics. Thus, **monarchies score, on average, 1.67 points higher than republics**.

The median confirms the same pattern. The median is **6.27** for monarchies and **4.63** for republics, meaning that the median monarchy scores **1.64 points** higher than the median republic. **This indicates that the observed advantage of monarchies is not limited to a few exceptional countries but is also visible in the broader distribution of the data.**

The standard deviation is also lower among monarchies (**1.03**) than among republics (**1.28**). Thus, monarchies not only have higher levels of property-rights protection but also exhibit somewhat more consistent performance.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies minus Republics)	**1.67**
Welch's *t*-statistic	**6.304**
Approximate degrees of freedom	**32.09**
Two-tailed *p*-value	**0.00000045**
95% confidence interval	**1.13 to 2.20**
Hedges' *g*	**1.33**

Welch's test shows that **the difference in mean property-rights scores between monarchies and republics worldwide is highly statistically significant** (**$p < 0.001$**). The 95% confidence interval also lies entirely above zero, supporting the observed higher mean among monarchies.

Hedges' *g* is **1.33**, indicating a **very large effect**. **Thus, the observed difference is not only statistically significant but also substantial in practical terms.**

Mann–Whitney U Test

To ensure that the conclusion does not depend solely on the assumptions of the parametric test, the nonparametric **Mann–Whitney U test** was also conducted.

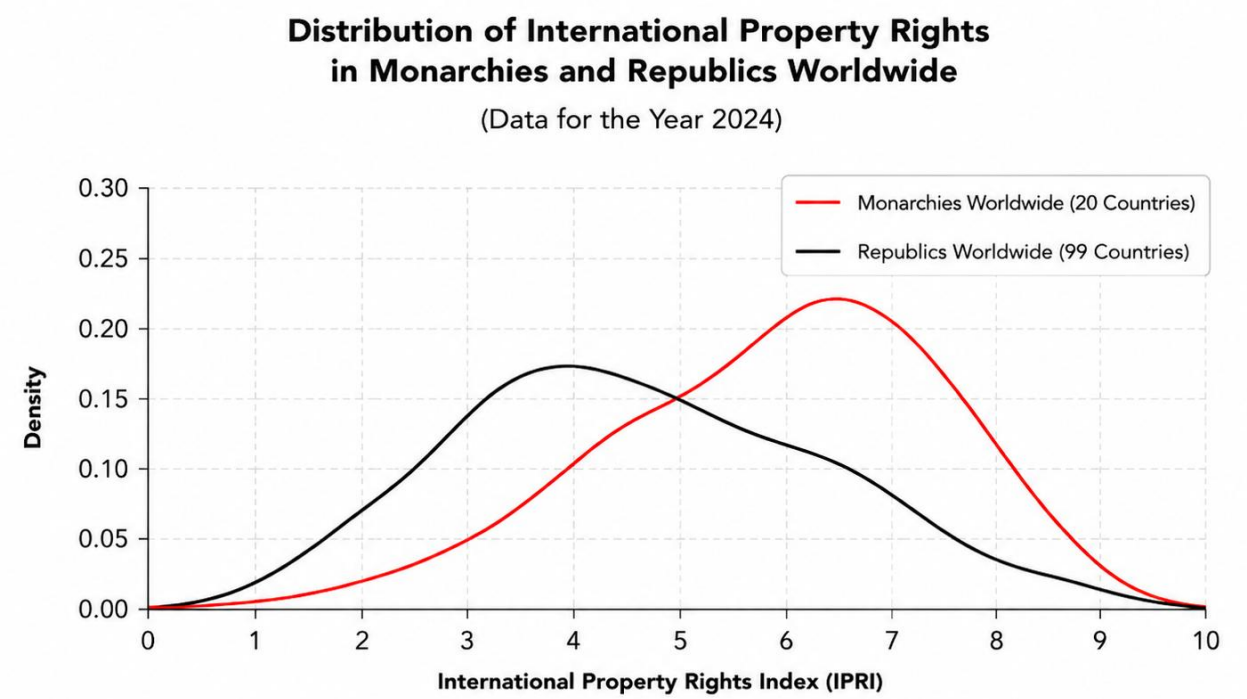
Statistical Measure	Value
U statistic	**1686.5**
Two-tailed *p*-value	**0.00000076**
Rank-biserial correlation	**0.704**

The Mann–Whitney test likewise confirms a difference between the two groups with very strong statistical evidence (**p < 0.001**). The rank-biserial correlation of **0.704** indicates a **very large effect**, showing that International Property Rights Index scores are generally substantially higher among monarchies than among republics.

Conclusion

In 2024, **the world's monarchies performed substantially better than the world's republics on the International Property Rights Index. Both the mean and median scores were considerably higher among monarchies, and both Welch's t-test and the Mann–Whitney U test confirmed the difference with very strong statistical evidence. The very large effect sizes in both tests further indicate that the observed difference is substantial not only statistically but also in practical terms.**

Overall, the data examined in this study show that **monarchies provide stronger protection for their citizens' property rights than republics do**. Since property rights are among the most fundamental dimensions of liberty, **this finding is also consistent with the predictions of the "New or Modern Monarchy" hypothesis**. As in the other chapters of this book, however, this result represents an **observational association** and does not, **by itself**, establish a causal relationship between the form of government and the degree of property-rights protection.



Monarchies Worldwide (20 Countries)	
Mean	6.50
Median	6.27
Standard Deviation (Sample)	1.03

Republics Worldwide (99 Countries)	
Mean	4.83
Median	4.63
Standard Deviation (Sample)	1.28

Source: Calculations based on data provided by Property Rights Alliance, International Property Rights Index (IPRI), 2024

The kernel density estimate (KDE) plot shows that the distribution of monarchies worldwide on the International Property Rights Index is noticeably shifted toward higher scores, with the greatest concentration occurring in the relatively high range of the index. By contrast, the distribution of republics is concentrated primarily at lower scores and is also more dispersed, indicating greater heterogeneity in their performance in protecting property rights. The separation between the two curves indicates that, on average, monarchies perform more favorably on the Property Rights Index.

Comparison of the International Property Rights Index between Constitutional and Non-Constitutional Monarchies Worldwide

Descriptive Statistics

Type of Monarchy	Number of Countries	Mean	Median	Standard Deviation
Constitutional monarchies	12	6.99	7.52	1.03

Type of Monarchy	Number of Countries	Mean	Median	Standard Deviation
Semi-constitutional and absolute monarchies	**8**	**5.76**	**5.78**	**0.43**

The mean International Property Rights Index score is **6.99** for constitutional monarchies and **5.76** for semi-constitutional and absolute monarchies. Thus, **constitutional monarchies score, on average, 1.23 points higher than the second group**.

The medians point in the same direction. The median is **7.52** for constitutional monarchies and **5.78** for semi-constitutional and absolute monarchies. In other words, the median country among constitutional monarchies scores approximately **1.74 points** higher than the median country in the second group.

The standard deviation among constitutional monarchies (**1.03**) is also more than twice that of the second group (**0.43**), indicating that although constitutional monarchies have a higher average score, their performance is also more dispersed.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (constitutional minus semi-constitutional and absolute)	**1.23**
Welch's *t* statistic	**3.69**
Approximate degrees of freedom	**15.90**
Two-tailed *p*-value	**0.0020**
95% confidence interval	**0.52 to 1.94**
Hedges' *g*	**1.39**

Welch's test shows that **the difference in mean Property Rights Index scores between the two types of monarchy is statistically significant** (**p = 0.002**). The 95% confidence interval also lies entirely above zero, **supporting the higher average score of constitutional monarchies**.

Hedges' *g* is **1.39**, indicating a **very large effect**. Thus, **the observed difference is not only statistically significant but also substantial in magnitude**.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the non-parametric **Mann–Whitney U test** was also conducted.

Statistical Measure	Value
U statistic	**81**
Two-tailed *p*-value	**0.0096**
Rank-biserial correlation	**0.69**

The Mann–Whitney test also indicates a statistically significant difference between the two groups (**$p < 0.01$**). The rank-biserial correlation of approximately **0.69** indicates a **large effect**, with constitutional monarchies generally receiving higher scores on this index.

Conclusion

In 2024, **constitutional monarchies performed better on the International Property Rights Index than semi-constitutional and absolute monarchies. Both the mean and median scores were noticeably higher among constitutional monarchies, and both Welch's t-test and the Mann–Whitney test confirmed that the difference was statistically significant. Moreover, the large effect sizes indicate that the difference is substantial in magnitude as well as statistically significant.**

This result suggests that, even among monarchies themselves, constitutional systems provide stronger protection for citizens' property rights. This is consistent with the theoretical expectation that more constrained government power and broader recognition of citizens' rights should be associated with stronger property-rights protection in constitutional monarchies.

Comparison of the International Property Rights Index between Commonwealth Realms and Republics Worldwide

It should be noted that, of the **16 Commonwealth realms**, only **4 countries** had data available in the 2024 edition of this index. This comparison is therefore based on those four countries.

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Commonwealth realms	**4**	**7.03**	**7.55**	**1.16**
Republics worldwide	**99**	**4.83**	**4.63**	**1.28**

The mean International Property Rights Index score is **7.03** for the Commonwealth realms and **4.83** for republics worldwide. Thus, **the Commonwealth realms score, on average, 2.19 points higher than the republics**.

The medians point in the same direction. The median score is **7.55** for the Commonwealth realms and **4.63** for the republics, meaning that the median country among the Commonwealth realms scores approximately **2.92 points** higher than the median republic.

The standard deviations of the two groups are also relatively similar. Thus, the observed difference primarily reflects a difference in the level of performance rather than merely a difference in the dispersion of scores.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Commonwealth realms minus republics)	**2.19**
Welch's *t* statistic	**3.68**
Approximate degrees of freedom	**3.30**
Two-tailed *p*-value	**0.0296**
95% confidence interval	**0.39 to 4.00**
Hedges' *g*	**1.70**

Welch's test indicates that **the difference in mean Property Rights Index scores between the Commonwealth realms and republics worldwide is statistically significant** (**p = 0.0296**). The 95% confidence interval lies entirely above zero, supporting the higher average score of the Commonwealth realms.

Hedges' *g* is **1.70**, indicating a **very large effect**. However, this result is based on **only four Commonwealth realms**. Therefore, although the observed difference is substantial, the very small sample size of this group should be kept in mind when interpreting the result.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the non-parametric **Mann–Whitney U test** was also conducted.

Statistical Measure	Value
U statistic	**358**
Two-tailed *p*-value	**0.0065**

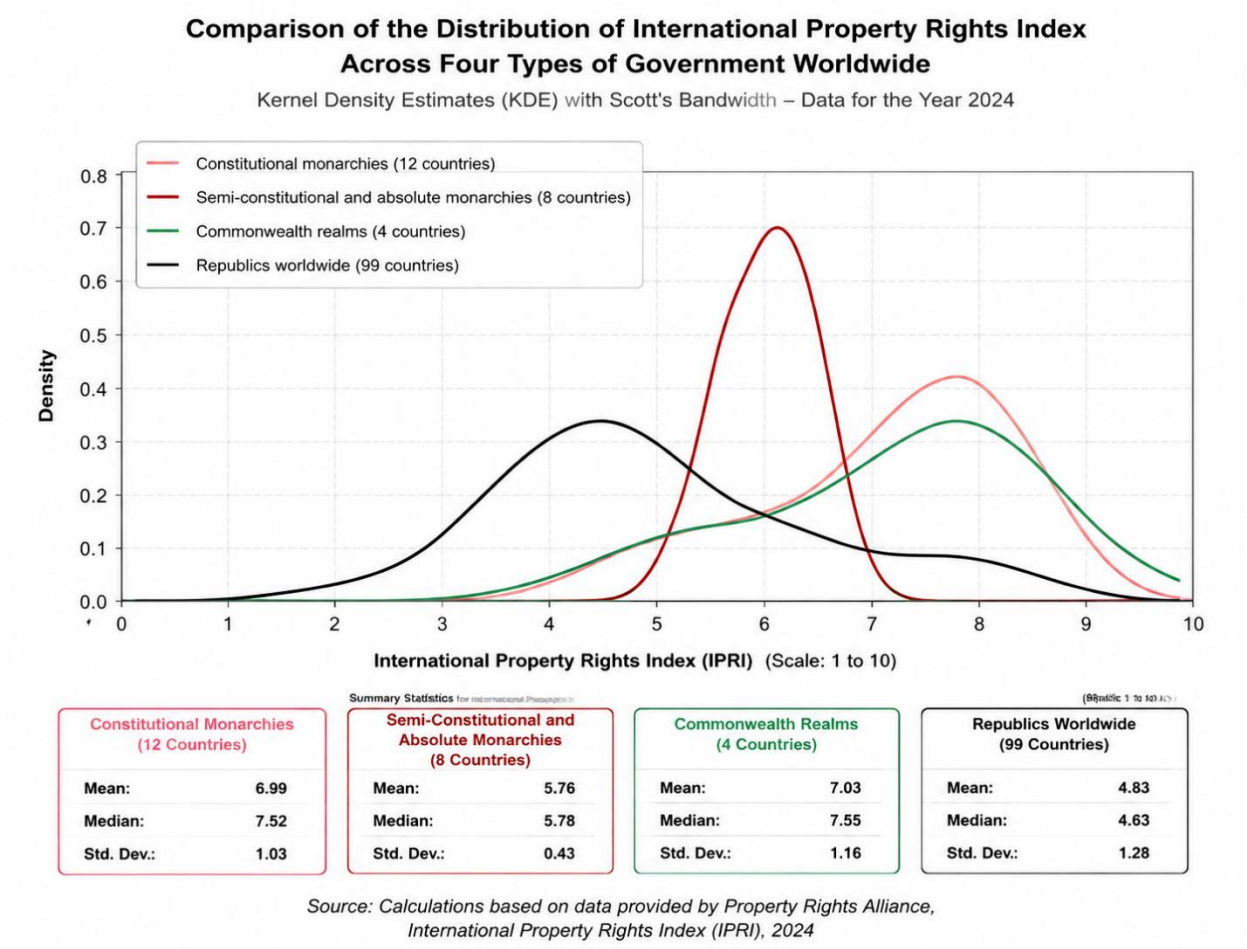
Statistical Measure	Value
Rank-biserial correlation	**0.81**

The Mann–Whitney test likewise indicates a statistically significant difference between the two groups (**$p < 0.01$**). The rank-biserial correlation of approximately **0.81** indicates a **very large effect**, with the Commonwealth realms generally receiving higher scores on this index.

Conclusion

In 2024, **the four Commonwealth realms for which International Property Rights Index data were available performed better, on average, than republics worldwide. Both their mean and median scores were substantially higher, and both Welch's t-test and the Mann–Whitney test indicated statistically significant differences between the groups.**

However, because this comparison is based on **only four Commonwealth realms**, the results should be interpreted with greater caution than the other comparisons in this book. Despite this limitation, the available evidence is consistent with the predictions of the **"New or Modern Monarchy" hypothesis**, indicating that this group of countries performs more favorably than republics worldwide in the protection of property rights.



The kernel density estimate (KDE) plot shows that the **Commonwealth realms and constitutional monarchies are concentrated primarily at the higher end of the International Property Rights Index**, with their greatest density occurring among relatively high scores. Semi-constitutional and absolute monarchies also perform better overall than republics, although their distribution is centered at lower levels than that of constitutional monarchies. By contrast, the distribution of republics worldwide is noticeably shifted toward lower scores and is also more dispersed. Overall, this pattern suggests that, at the global level, **monarchical systems—particularly constitutional monarchies and Commonwealth realms—tend to perform more favorably in the protection of property rights**.

Comparison of the International Property Rights Index between Monarchies and Republics across Continents and the Middle East

Comparison of the International Property Rights Index between Monarchies and Republics in Asia

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Asian monarchies	**11**	**5.90**	**5.84**	**0.73**
Asian republics	**20**	**4.75**	**4.67**	**1.22**

The mean International Property Rights Index score is **5.90** for Asian monarchies and **4.75** for Asian republics. Thus, **Asian monarchies score, on average, 1.15 points higher than the republics of the continent**.

The medians point in the same direction. The median is **5.84** for monarchies and **4.67** for republics, meaning that the median country among Asian monarchies scores approximately **1.16 points** higher than the median country among Asian republics.

The standard deviation is also substantially lower among monarchies (**0.73**) than among republics (**1.22**), indicating that Asian monarchies not only achieve higher scores but also exhibit more consistent performance.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (monarchies minus republics)	**1.15**
Welch's *t* statistic	**3.26**
Approximate degrees of freedom	**28.69**
Two-tailed *p*-value	**0.0028**
95% confidence interval	**0.43 to 1.87**
Hedges' *g*	**1.03**

Welch's test shows that the difference in mean Property Rights Index scores between monarchies and republics in Asia is statistically significant (**$p < 0.01$**). The 95% confidence interval lies entirely above zero, supporting the higher average score of the monarchies.

Hedges' *g* is **1.03**, indicating a **large effect**. Thus, the observed difference is not only statistically significant but also substantial in magnitude.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the non-parametric **Mann–Whitney U test** was also conducted.

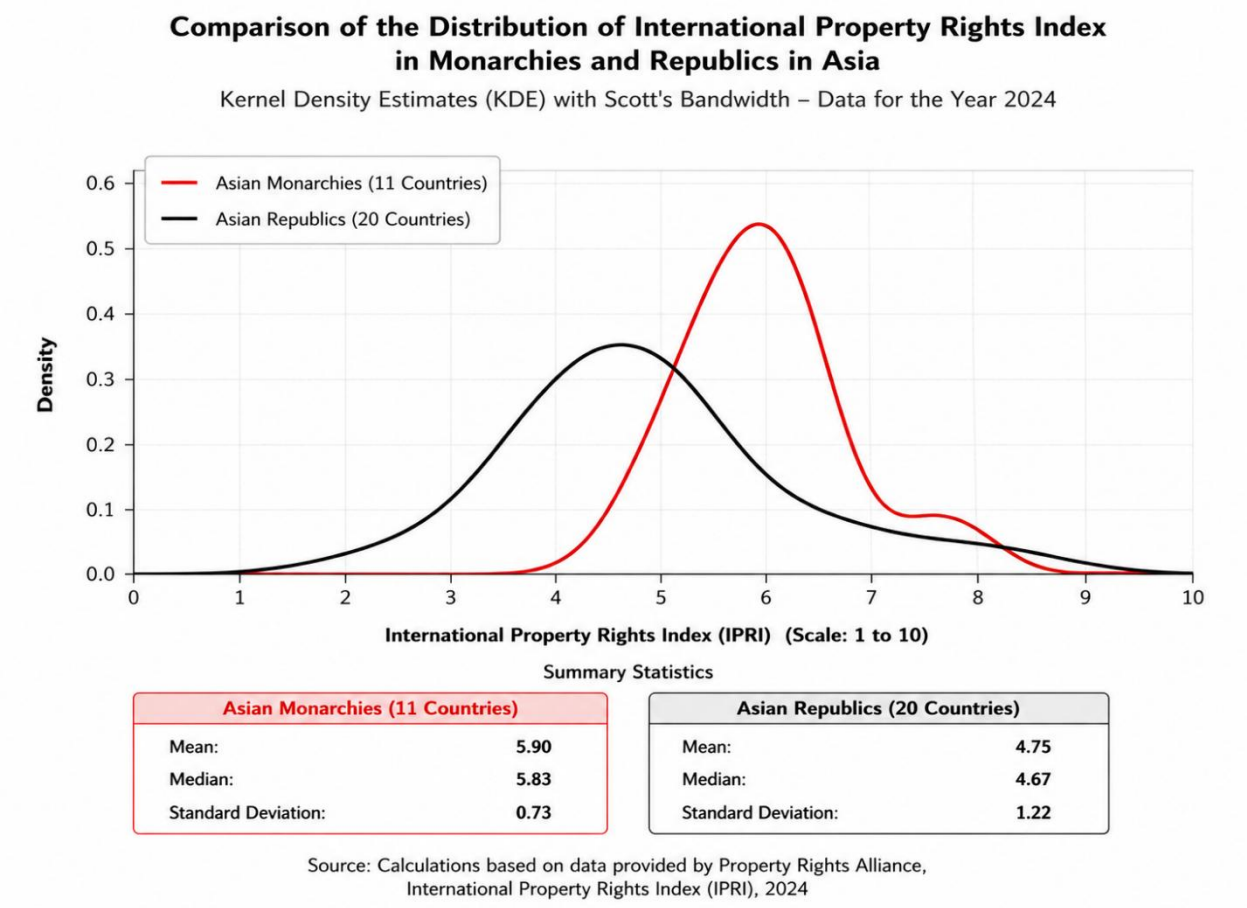
Statistical Measure	Value
U statistic	**182**
Two-tailed *p*-value	**0.0032**
Rank-biserial correlation	**0.65**

The Mann–Whitney test also indicates a statistically significant difference between the two groups (**$p < 0.01$**). The rank-biserial correlation of **0.65** indicates a **large effect**, with Asian monarchies generally receiving higher scores on this index.

Conclusion

In 2024, **Asian monarchies performed substantially better on the International Property Rights Index than Asian republics. Both the mean and median scores were considerably higher among monarchies, and both Welch's t-test and the Mann–Whitney test confirmed the difference with strong statistical evidence. The large effect sizes further indicate that the difference is substantial in magnitude as well as statistically significant.**

Overall, the Asian data indicate that monarchies in the region provide stronger protection for citizens' property rights than Asian republics, and **this finding is consistent with the predictions of the "New or Modern Monarchy" hypothesis**.



The kernel density estimate shows that the distribution of Asian monarchies on the International Property Rights Index is noticeably shifted toward higher scores, with the greatest concentration occurring at relatively high levels of the index. In contrast, Asian republics are concentrated mainly at lower scores and display greater dispersion, indicating more heterogeneous performance in the protection of property rights. The separation between the two curves suggests that, on average, Asian monarchies perform more favorably than the republics of the continent.

Comparison of the International Property Rights Index in European Monarchies and Republics

Descriptive Statistics

Form of Government	Number of Countries	Mean	Median	Standard Deviation
European Monarchies	**8**	**7.48**	**7.70**	**0.47**
European Republics	**30**	**5.82**	**5.89**	**1.19**

The mean International Property Rights Index score is **7.48** for European monarchies and **5.82** for European republics. Thus, **European monarchies score, on average, 1.66 points higher than European republics**.

The median confirms the same pattern. The median score is **7.70** for monarchies and **5.89** for republics, meaning that the median country among European monarchies scores approximately **1.82 points** higher than the median country among European republics.

The standard deviation is also substantially lower among European monarchies (**0.47**) than among European republics (**1.19**), indicating that European monarchies not only achieve higher scores but also exhibit greater consistency in their performance.

Test of Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**1.66**
Welch's *t* statistic	**6.07**
Approximate degrees of freedom	**30.22**
Two-tailed *p*-value	**0.0000011**
95% confidence interval	**1.10 to 2.22**
Hedges' *g*	**1.49**

Welch's test indicates that the difference in mean property-rights scores between European monarchies and republics is highly statistically significant (**p < 0.001**). The 95% confidence interval also lies entirely above zero, confirming that the difference favors the monarchies.

Hedges' *g* is **1.49**, indicating a **very large effect**. Thus, **the observed difference is not only statistically significant but also substantial in practical terms**.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the nonparametric **Mann–Whitney U test** was also conducted.

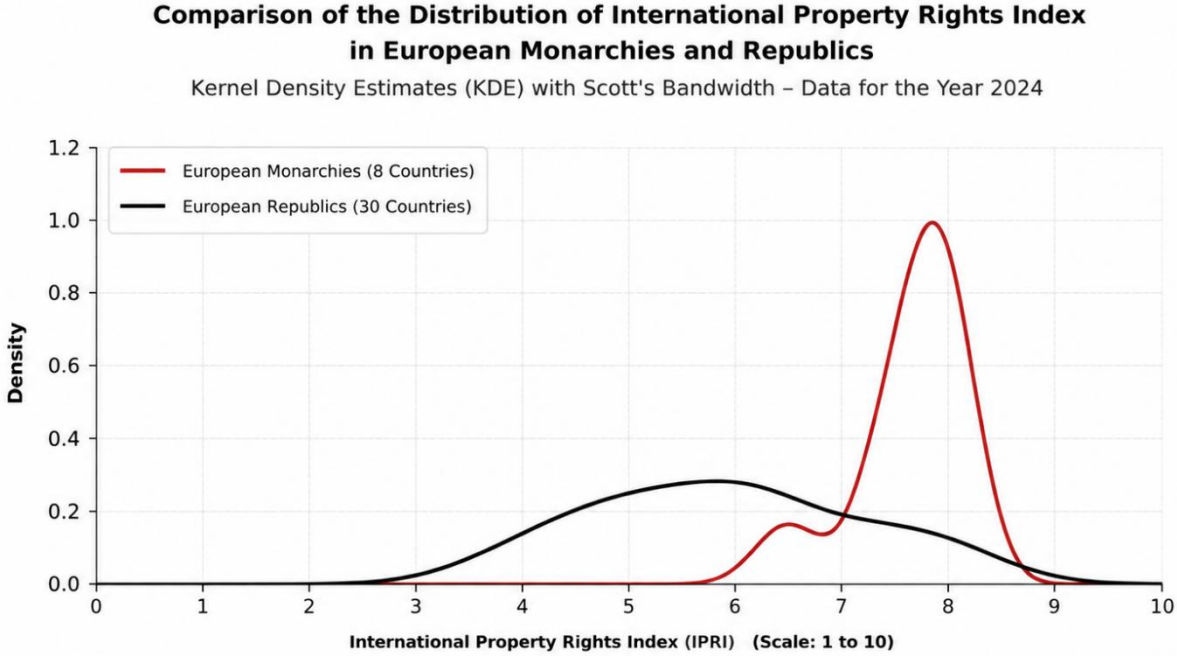
Statistical Measure	Value
U statistic	**218**
Two-tailed *p*-value	**0.00013**
Rank-biserial correlation	**0.82**

The Mann–Whitney test likewise indicates a highly statistically significant difference between the two groups (**$p < 0.001$**). The rank-biserial correlation of **0.82** represents a **very large effect**, indicating a pronounced advantage for European monarchies on this index.

Conclusion

In 2024, **European monarchies performed substantially better than European republics on the International Property Rights Index. Both the mean and median scores were considerably higher among monarchies, and both Welch's t-test and the Mann–Whitney test confirmed the difference with very strong statistical evidence. The very large effect sizes further indicate that the difference is substantial not only statistically but also in practical terms.**

Moreover, the lower dispersion of scores among European monarchies indicates that these countries have not only achieved a higher level of property-rights protection but have also done so with greater consistency. **This finding is also consistent with the prediction of the "New or Modern Monarchy" hypothesis.**



Comparison of the Distribution of International Property Rights Index in European Monarchies and Republics

Kernel Density Estimates (KDE) with Scott's Bandwidth – Data for the Year 2024

Summary Statistics

European Monarchies (8 Countries)	
Mean:	7.48
Median:	7.70
Standard Deviation:	0.47

European Republics (30 Countries)	
Mean:	7.82
Median:	5.89
Standard Deviation:	1.19

Source: Calculations based on data provided by Property Rights Alliance, International Property Rights Index (IPRI), 2024

The distribution of European monarchies on the International Property Rights Index is clearly shifted toward higher scores, with the greatest concentration at the very high end of the index. In contrast, although European republics generally perform well, their scores are more widely dispersed, with some countries falling in the middle range of the index. The separation between the two curves indicates that, on average, European monarchies perform more favorably than the republics of the continent in protecting property rights.

Comparison of the International Property Rights Index in European Monarchies and European Republics That Have Never Been Communist

Descriptive Statistics

Form of Government	Number of Countries	Mean	Median	Standard Deviation
European Monarchies	**8**	**7.48**	**7.70**	**0.47**
Never-Communist European Republics	**12**	**6.82**	**7.05**	**0.91**

The mean International Property Rights Index score is **7.48** for European monarchies and **6.82** for never-communist European republics. Thus, **European monarchies score, on average, 0.66 points higher than this group of republics**.

The median confirms the same pattern. The median score is **7.70** for monarchies and **7.05** for never-communist republics, meaning that the median country among European monarchies scores approximately **0.65 points** higher than the median country among never-communist republics.

The standard deviation among monarchies (**0.47**) is approximately half that of the never-communist republics (**0.91**), indicating that European monarchies not only achieve higher scores but also exhibit greater consistency in their performance.

Test of Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**0.66**
Welch's *t* statistic	**2.12**
Approximate degrees of freedom	**17.21**
Two-tailed *p*-value	**0.049**
95% confidence interval	**0.003 to 1.316**
Hedges' *g*	**0.82**

Welch's test indicates that **the difference in mean property-rights scores between European monarchies and never-communist European republics is statistically significant at the 5% level (p = 0.049)**. The 95% confidence interval also lies entirely above zero, although its lower bound is very close to zero. Thus, while statistically significant, this result is less robust than those of the preceding comparisons.

Hedges' *g* is **0.82**, indicating a **large effect**. The observed difference is therefore also substantial in practical terms.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the nonparametric **Mann–Whitney U test** was also conducted.

Statistical Measure	Value
U statistic	**75**
Two-tailed *p*-value	**0.0387**
Rank-biserial correlation	**0.56**

The Mann–Whitney test also indicates a statistically significant difference between the two groups (p = 0.039). The rank-biserial correlation of **0.56** indicates a **large effect**, favoring European monarchies on this index.

Conclusion

Even after excluding republics with a history of communist rule, European monarchies continue to perform better on the International Property Rights Index than European republics that have never been communist. Both the mean and median scores are higher among monarchies, and both Welch's t-test and the Mann–Whitney test confirm that the difference is statistically significant at the 5% level.

This finding is noteworthy because the never-communist European republics include some of the world's longest-established democratic systems and would therefore be expected to provide strong protection for property rights. Nevertheless, the 2024 data show that European monarchies, on average, still provide stronger protection for their citizens' property rights. As in other sections of this book, however, this finding represents an observational relationship and **does not, by itself, establish a causal relationship between form of government and the degree of property-rights protection**.

Comparison in Africa, the Americas, and Oceania

For this index, data were available for only a very limited number of monarchies and Commonwealth realms in Africa, the Americas, and Oceania. Consequently, comparisons with the republics of these continents would have insufficient analytical value and reliability because of the very small sample sizes. Such comparisons are therefore omitted from this section.

Comparison of the International Property Rights Index in Middle Eastern Monarchies and Republics

Descriptive Statistics

Form of Government	Number of Countries	Mean	Median	Standard Deviation
Middle Eastern Monarchies	7	**5.82**	**5.84**	**0.42**
Middle Eastern Republics	7	**4.34**	**4.26**	**1.38**

The mean International Property Rights Index score is **5.82** for Middle Eastern monarchies and **4.34** for Middle Eastern republics. **Thus, the region's monarchies score, on average, 1.48 points higher than its republics.**

The median confirms the same pattern. The median score is **5.84** for monarchies and **4.26** for republics, meaning that the median country among Middle Eastern monarchies scores approximately **1.58 points** higher than the median country among the region's republics.

The standard deviation is also substantially lower among monarchies (**0.42**) than among republics (**1.38**). This indicates that the region's monarchies not only achieve higher scores but also exhibit greater consistency in their performance.

Test of Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**1.48**
Welch's *t* statistic	**2.71**
Approximate degrees of freedom	**7.11**
Two-tailed *p*-value	**0.0297**
95% confidence interval	**0.19 to 2.76**
Hedges' *g*	**1.36**

Welch's test indicates that **the difference in mean property-rights scores between Middle Eastern monarchies and republics is statistically significant** (**p = 0.0297**). The 95% confidence interval also lies entirely above zero, supporting the higher scores observed among monarchies.

Hedges' *g* is **1.36**, indicating a **very large effect**. Thus, **the observed difference is not only statistically significant but also substantial in practical terms**.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the nonparametric **Mann–Whitney U test** was also conducted.

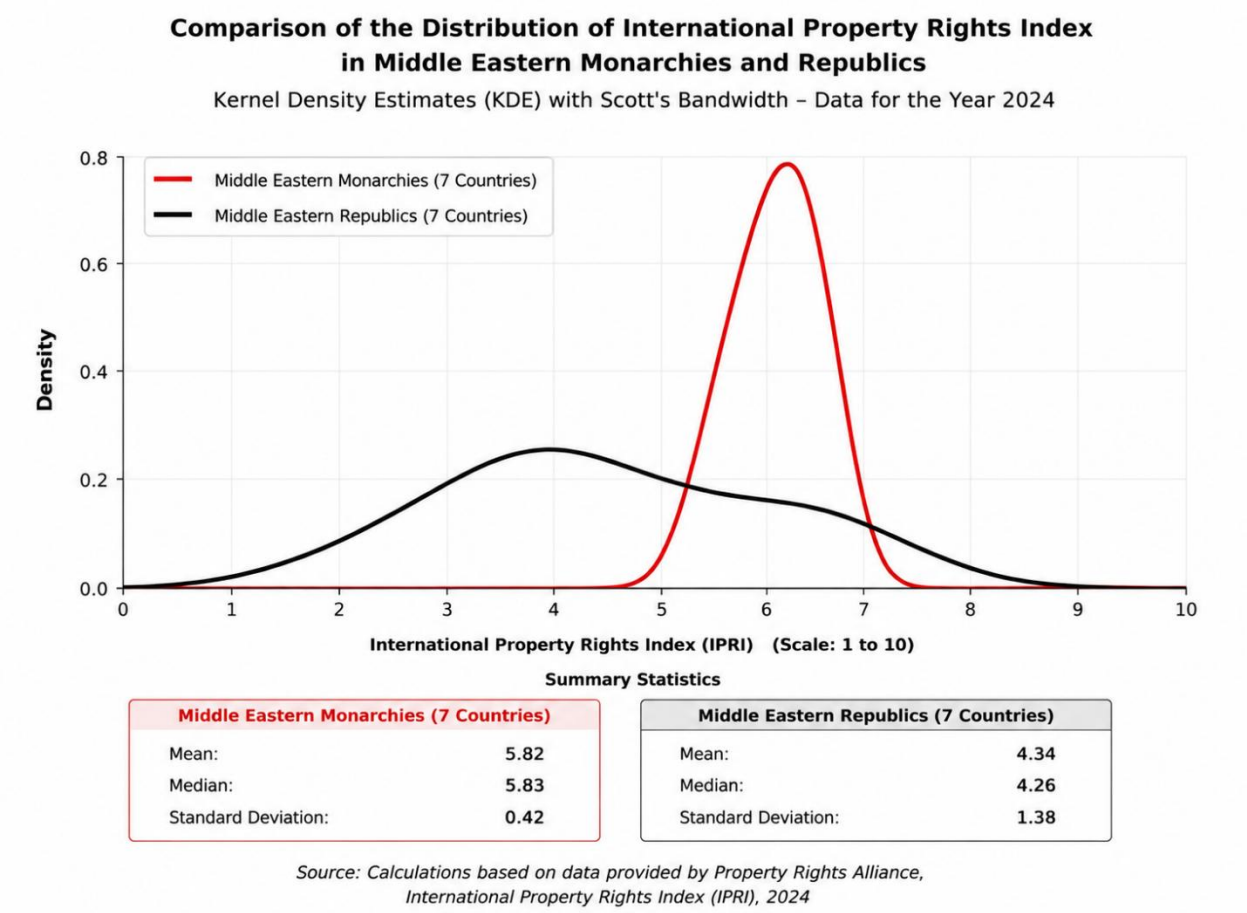
Statistical Measure	Value
U statistic	**39**
Two-tailed *p*-value	**0.0728**
Rank-biserial correlation	**0.59**

Unlike Welch's test, the Mann–Whitney test does not indicate a statistically significant difference at the 5% level (**p = 0.0728**), although the *p*-value is relatively close to the conventional significance threshold. The rank-biserial correlation of **0.59** nevertheless indicates a **large effect** favoring monarchies.

Conclusion

In 2024, **Middle Eastern monarchies performed better on average than the region's republics on the International Property Rights Index. Both the mean and median scores are higher among monarchies. Welch's test indicates a statistically significant difference, whereas the Mann–Whitney test does not confirm significance at the 5% level, likely reflecting the small sample size and some overlap in the groups' ranks; nevertheless, the effect size in the latter test is also large.**

Overall, the Middle Eastern data are consistent with the prediction of the "New or Modern Monarchy" hypothesis, indicating that the region's monarchies, on average, provide stronger protection for their citizens' property rights. Given the small number of countries and the differing results of the two statistical tests, however, this finding should be interpreted more cautiously than some of the global and European comparisons.



Status of the "New or Modern Monarchy" Hypothesis After the Eleventh Indicator

With the addition of the International Property Rights Index, the status of the "New or Modern Monarchy" hypothesis at this stage is as follows:

Test	Indicator	Result
✓	GDP per capita	Consistent with the hypothesis
✓	Purchasing power per capita	Consistent with the hypothesis
◐	Gini coefficient	**Mixed evidence; difference favors the hypothesis**
✓	Political Stability and Absence of Violence	Consistent with the hypothesis
✓	Regulatory Quality	Consistent with the hypothesis
✓	Rule of Law	Consistent with the hypothesis
✓	Government Effectiveness	Consistent with the hypothesis

Test	Indicator	Result
✓	Inflation	Consistent with the hypothesis
✓	Control of Corruption	Consistent with the hypothesis
✓	Corruption Perceptions	Consistent with the hypothesis
✓	Property Rights	Consistent with the hypothesis

At this stage, the "New or Modern Monarchy" hypothesis has been tested against **eleven independent indicators** covering a range of different domains. These indicators are not confined to a single dimension of governance; rather, they encompass various aspects of **welfare, security, order, justice, and liberty**, and their data have been drawn from a variety of international sources and institutions.

For the International Property Rights Index as well, monarchies achieved higher average scores in the global, Asian, European, and Middle Eastern comparisons. Even when European monarchies were compared with European republics that had never experienced communist rule, the difference remained in favor of the monarchies.

Thus, **none of the eleven indicators examined so far has produced a result running counter to the prediction of the hypothesis: ten indicators have provided clear evidence in its direction, while for the Gini coefficient, despite the mixed evidence, the direction of the difference favored the hypothesis.** As emphasized throughout this book, however, these findings do not establish a causal relationship between regime type and the quality of governance. A more precise conclusion is that **the empirical pattern emerging from the eleven indicators examined so far has not been inconsistent with the predictions of the "New or Modern Monarchy" hypothesis and, in most cases, has provided clear statistical evidence in its direction.**

Human Development

In the previous section, we noted that governments can expand citizens' liberty by recognizing their rights. However, simply possessing a right does not always mean that one has the practical ability to exercise it.

Suppose a girl has the legal right to education, but there is no school where she lives, or her family cannot afford to send her to school. In such a situation, although her right is legally recognized, she lacks the practical opportunity to exercise it, and her **liberty** to choose remains constrained.

Therefore, a government seeking to expand the liberty of its citizens should not merely recognize their rights; it should also help create the conditions and opportunities necessary for people to exercise those rights. **Real liberty** requires both **the existence of a right and the practical ability to exercise that right**.

One of the best-known measures relevant to this dimension of human well-being is the **Human Development Index (HDI)**, published by the United Nations Development Programme (UNDP).

The index combines three fundamental dimensions:

- Life expectancy (health)
- Education
- Standard of living

The United Nations Development Programme does not equate human development simply with increases in wealth and income. Rather, it relates human development to the expansion of people's capabilities and opportunities to lead lives they have reason to value. The Human Development Index measures three fundamental dimensions of this process: a long and healthy life, access to knowledge, and a decent standard of living. Since this book relates liberty to the expansion of people's range of real choices, the HDI can serve as an appropriate measure of **the conditions and capabilities that expand people's practical ability to exercise liberty**.

As we continue testing the "New or Modern Monarchy" hypothesis, we now examine whether monarchies and republics differ in their performance in providing citizens with the conditions and capabilities that enable them to pursue different paths in life.

The data in this section are for **2022** (Appendix RDS-L01). A higher HDI value indicates more favorable conditions in education, health, and standard of living.

Comparison of the Human Development Index in Monarchies and Republics Worldwide

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Monarchies	**27**	**0.835**	**0.875**	**0.124**
Republics	**150**	**0.698**	**0.727**	**0.153**

The mean Human Development Index is **0.835** for monarchies and **0.698** for republics. Thus, **the mean for monarchies is approximately 0.137 points higher than that for republics**.

The medians show the same pattern. The median HDI is **0.875** among monarchies and **0.727** among republics, meaning that the median country among monarchies scores approximately **0.149 points** higher than the median country among republics. The fact that both the mean and the median are higher indicates that the observed difference is not merely the result of a few exceptionally highly developed monarchies; it is also evident at the center of the distributions.

The standard deviation is also somewhat lower among monarchies than among republics. Thus, in addition to their higher overall level, the performance of monarchies on this index is somewhat more consistent.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchy − Republic)	**0.137**
Welch's *t* statistic	**5.07**
Approximate degrees of freedom	**41.72**
Two-tailed *p*-value	**0.0000087**
95% confidence interval	**0.082 to 0.191**
Hedges' *g*	**0.91**

Welch's test shows that **the difference in mean HDI between monarchies and republics worldwide is highly statistically significant** (**$p < 0.001$**). The 95% confidence interval lies entirely above zero, confirming that the direction of the difference favors monarchies.

Hedges' *g* is **0.91**, indicating a **large effect**. The observed difference is therefore not only statistically significant but also substantial in practical terms.

Mann–Whitney Test

To ensure that the result does not depend solely on a comparison of means or on the assumptions of a parametric test, the nonparametric **Mann–Whitney test** was also conducted.

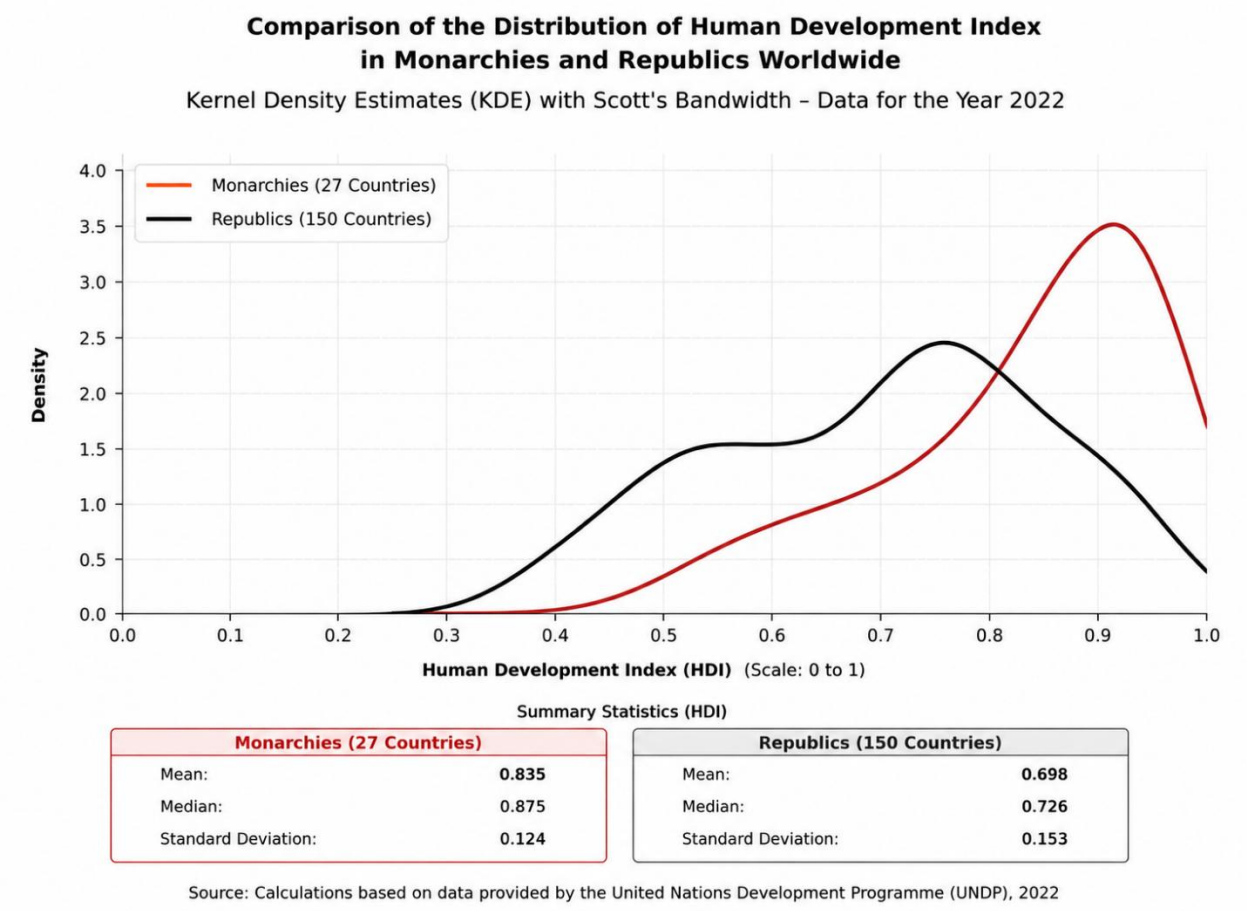
Statistical Measure	Value
U statistic	**3089.5**
Two-tailed *p*-value	**0.000014**
Rank-biserial correlation	**0.53**

The Mann–Whitney test likewise confirms a difference between the distributions of the two groups with very strong statistical evidence (**$p < 0.001$**). The rank-based effect size of **0.53** also indicates a large difference, with monarchies generally occupying higher ranks.

Conclusion

In 2022, **monarchies worldwide had a higher Human Development Index than republics in terms of both the mean and the median. Welch's t-test and the Mann–Whitney test both confirm this difference with very strong statistical evidence, and the effect sizes are also large.**

This result indicates that monarchies, on average, were in a more favorable position across the three fundamental dimensions of human development—health, education, and standard of living—dimensions that expand individuals' capacity to enjoy a wider range of real options in life. From this perspective, the Human Development Index also yields a result consistent with the prediction of the "New or Modern Monarchy" hypothesis. As with the other comparisons in this book, however, this finding alone does not establish a causal relationship between regime type and human development.



The kernel density estimate (KDE) shows that the distribution of monarchies worldwide on the Human Development Index is noticeably shifted toward higher scores, with the greatest concentration among countries with **high and very high levels of human development**. In contrast, the distribution of republics is not only concentrated at lower levels but also exhibits greater dispersion, spanning a wider range of HDI values.

This pattern is consistent with the descriptive statistics: the mean and median HDI scores for monarchies are **0.835** and **0.875**, respectively, compared with **0.698** and **0.726** for republics. Thus, the 2022 data indicate that, **on average, monarchies performed better than republics on the Human Development Index**

Comparison of the Human Development Index in Constitutional and Non-Constitutional Monarchies Worldwide

Descriptive Statistics

Type of Monarchy	Number of Countries	Mean	Median	Standard Deviation
Constitutional monarchies	**15**	**0.854**	**0.920**	**0.137**

Type of Monarchy	Number of Countries	Mean	Median	Standard Deviation
Semi-constitutional and absolute monarchies	**12**	**0.811**	**0.835**	**0.106**

The mean Human Development Index is **0.854** among constitutional monarchies and **0.811** among semi-constitutional and absolute monarchies. Thus, **constitutional monarchies score, on average, 0.043 points higher than the second group**.

The medians point in the same direction. The median is **0.920** among constitutional monarchies and **0.835** among semi-constitutional and absolute monarchies. In other words, the median country among constitutional monarchies scores **0.085 points** higher than the median country in the second group.

The standard deviations of the two groups are relatively similar (**0.137** and **0.106**, respectively), indicating no substantial difference in the dispersion of their scores.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Constitutional − Semi-constitutional and Absolute)	**0.043**
Welch's *t* statistic	**0.910**
Approximate degrees of freedom	**24.99**
Two-tailed *p*-value	**0.3713**

Welch's test shows that **the difference in mean HDI between the two types of monarchy is not statistically significant** (**p = 0.371**). Based on the available data, therefore, we cannot conclude that the type of monarchy alone is associated with a statistically significant difference in this index.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the nonparametric **Mann–Whitney test** was also conducted.

Statistical Measure	Value
U statistic	**120.5**
Two-tailed *p*-value	**0.1431**

The Mann–Whitney test leads to the same conclusion, finding no statistically significant difference between the two groups (p = 0.143).

Conclusion

In 2022, **constitutional monarchies had a higher average Human Development Index than semi-constitutional and absolute monarchies, and both the mean and the median were higher. However, neither Welch's test nor the Mann–Whitney test confirms that this difference is statistically significant.**

Thus, as with most of the previous indicators, the Human Development Index provides insufficient statistical evidence of a significant difference between the two types of monarchy. The results indicate that **both groups of monarchies, regardless of the extent to which the monarch's powers are constitutionally limited, have relatively high levels of human development**, and the observed difference is insufficient to support a definitive conclusion. As elsewhere in this book, this finding reports an observational relationship and should not be interpreted as evidence of causation.

Comparison of the Human Development Index in Commonwealth Realms and Republics Worldwide

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Commonwealth realms	**14**	**0.770**	**0.783**	**0.126**
Republics	**150**	**0.698**	**0.727**	**0.153**

The mean Human Development Index is **0.770** among Commonwealth realms and **0.698** among republics worldwide. Thus, **Commonwealth realms score, on average, 0.072 points higher than republics**.

The medians point in the same direction. The median is **0.783** among Commonwealth realms and **0.727** among republics, meaning that the median Commonwealth realm scores approximately **0.056 points** higher than the median republic.

The standard deviation is also slightly lower among Commonwealth realms (**0.126**) than among republics (**0.153**), although the dispersion of the two groups is relatively similar.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Commonwealth − Republic)	**0.072**
Welch's *t* statistic	**2.001**
Approximate degrees of freedom	**16.81**
Two-tailed *p*-value	**0.0618**
Hedges' *g*	**0.47**

Welch's test shows that **although the mean HDI is higher among Commonwealth realms than among republics worldwide, the difference is not statistically significant at the conventional 5% level** (**p = 0.0618**). Nevertheless, the *p*-value is relatively close to the significance threshold, and Hedges' *g* (**0.47**) indicates a **medium-sized effect**.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the nonparametric **Mann–Whitney test** was also conducted.

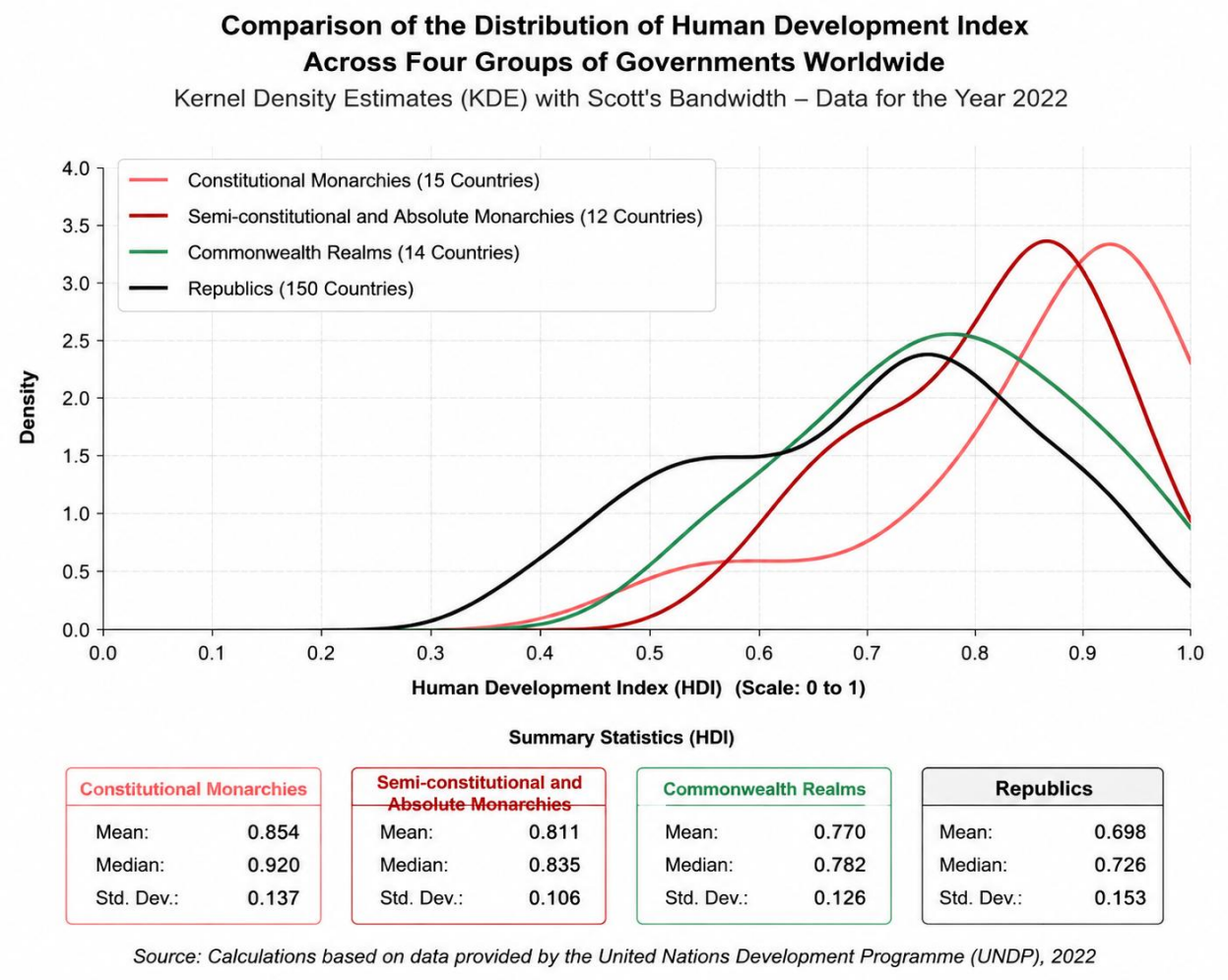
Statistical Measure	Value
U statistic	**1326.5**
Two-tailed *p*-value	**0.1043**
Rank-biserial correlation	**0.26**

The Mann–Whitney test likewise does not find the difference between the two groups statistically significant at the 5% level (**p = 0.1043**). The rank-biserial correlation of **0.26** indicates a **small-to-medium effect**.

Conclusion

In 2022, **Commonwealth realms had a higher average Human Development Index than republics worldwide, with the same advantage also appearing in the median. However, neither Welch's test nor the Mann–Whitney test confirms that the difference is statistically significant at the 5% level.**

Thus, although the direction of the difference is consistent with the prediction of the "New or Modern Monarchy" hypothesis, the statistical evidence in this comparison is insufficient for a definitive conclusion and should be interpreted more cautiously than the results of some of the previous indicators.



The kernel density estimate (KDE) shows that **constitutional monarchies have, overall, the highest level of human development**, with their distribution more heavily concentrated at very high HDI scores than those of the other groups. **Semi-constitutional and absolute monarchies**, although positioned at a somewhat lower level, also perform better on average than Commonwealth realms and republics. **Commonwealth realms occupy an intermediate position**, while the distribution of republics worldwide is shifted toward lower scores and exhibits greater dispersion.

Overall, this pattern indicates that **all three monarchy-related groups have higher average Human Development Index scores than republics**.

Comparison of the Human Development Index in Monarchies and Republics Across Continents and the Middle East

Comparison of the Human Development Index in Asian Monarchies and Republics

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Asian Monarchies	**13**	**0.816**	**0.823**	**0.096**
Asian Republics	**32**	**0.716**	**0.725**	**0.121**

The mean Human Development Index (HDI) score is **0.816** for Asian monarchies and **0.716** for Asian republics. Thus, **Asian monarchies score approximately 0.101 points higher on average than the republics of the continent**.

The median confirms the same pattern. The median HDI is **0.823** for monarchies and approximately **0.725** for republics, meaning that the median country among Asian monarchies scores about **0.099 points** higher than the median country among Asian republics. The fact that both the mean and the median are higher indicates that the observed difference is not merely the result of a few highly developed monarchies, but is also evident at the center of the distributions.

The standard deviation is also lower among monarchies than among republics. Thus, in addition to their higher average HDI, Asian monarchies exhibit somewhat more consistent performance on this index.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**0.101**
Welch's *t* statistic	**2.95**
Approximate degrees of freedom	**27.93**
Two-tailed *p*-value	**0.0063**
95% confidence interval	**0.031 to 0.170**
Hedges' *g*	**0.87**

Welch's test indicates that **the difference in mean HDI between Asian monarchies and republics is statistically significant** (**p = 0.0063**). The 95% confidence interval lies entirely above zero, confirming that the direction of the difference favors monarchies.

Hedges' *g* is **0.87**, indicating a **large effect**. Thus, the observed difference is not only statistically significant but also substantial in practical terms.

Mann–Whitney Test

To ensure that the result does not depend solely on a comparison of means or on the assumptions of the parametric test, the nonparametric **Mann–Whitney test** was also performed.

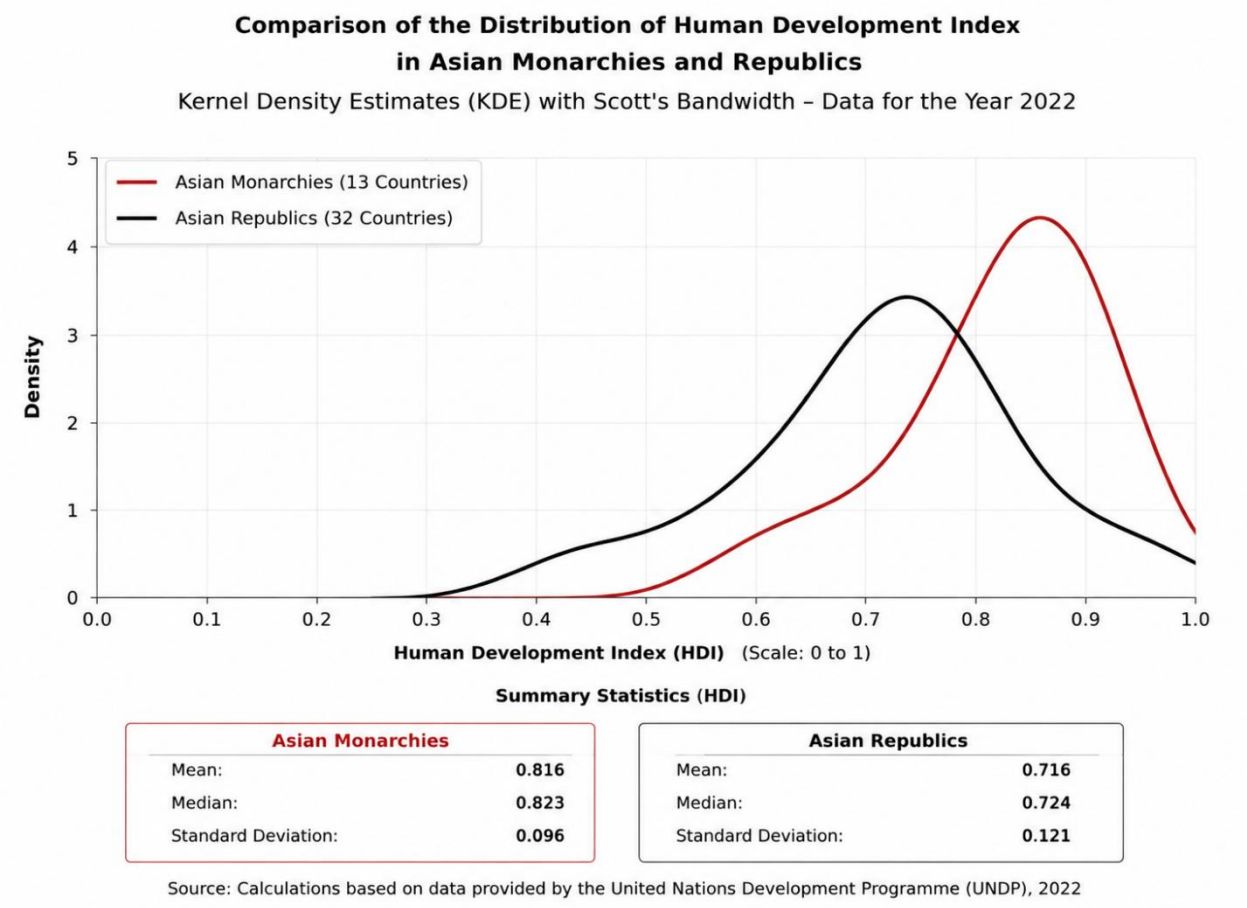
Statistical Measure	Value
U statistic	**319**
Two-tailed *p*-value	**0.0057**
Rank-biserial correlation	**0.53**

The Mann–Whitney test likewise indicates a statistically significant difference between the distributions of the two groups (**p = 0.0057**). The rank-biserial correlation of **0.53** also indicates a **large effect**, with monarchies generally occupying higher ranks.

Conclusion

In 2022, **Asian monarchies had a higher Human Development Index than Asian republics in terms of both the mean and the median. Welch's test and the Mann–Whitney test both confirm this difference with strong statistical evidence, and the effect size is also large.**

This finding indicates that, on average, citizens of Asian monarchies enjoyed more favorable conditions in **education, health, and standard of living**, and consequently greater opportunities and capabilities to choose and shape the course of their lives. Therefore, **the results of the Asian comparison are also consistent with the prediction of the "New or Modern Monarchy" hypothesis**; however, this comparison alone does not establish a causal relationship between type of government and human development.



The kernel density estimate (KDE) shows that the distribution of Asian monarchies on the Human Development Index is noticeably shifted toward higher scores, with the greatest concentration among countries with **high levels of human development**. In contrast, Asian republics not only have lower scores on average but also exhibit greater dispersion, spanning a wider range of human development levels.

This pattern is consistent with the descriptive statistics: the mean and median HDI scores for Asian monarchies are **0.816** and **0.823**, respectively, compared with **0.716** and **0.724** for Asian republics. Thus, the 2022 data indicate that, **on average, Asian monarchies have a higher level of human development than the republics of the continent**.

Comparison of the Human Development Index in European Monarchies and Republics

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
European Monarchies	**10**	**0.936**	**0.942**	**0.024**

Type of Government	Number of Countries	Mean	Median	Standard Deviation
European Republics	**33**	**0.868**	**0.879**	**0.063**

The mean Human Development Index (HDI) score is **0.936** for European monarchies and **0.868** for European republics. Thus, **European monarchies score 0.068 points higher on average than European republics**.

The median confirms the same pattern. The median HDI is **0.942** for monarchies and **0.879** for republics, meaning that the median country among European monarchies scores **0.063 points** higher than the median country among European republics.

The standard deviation is also substantially lower among monarchies (**0.024**) than among republics (**0.063**). This indicates that European monarchies not only have a very high level of human development but also exhibit more consistent performance.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**0.068**
Welch's *t* statistic	**5.182**
Approximate degrees of freedom	**38.76**
Two-tailed *p*-value	**0.000007**
Hedges' *g*	**1.19**

Welch's test indicates that **the difference in mean HDI between European monarchies and republics is highly statistically significant** (**$p < 0.001$**).

Hedges' *g* is **1.19**, indicating a **very large effect**. Thus, the observed difference is not only statistically significant but also substantial in practical terms.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the nonparametric **Mann–Whitney test** was also performed.

Statistical Measure	Value
U statistic	**280**
Two-tailed *p*-value	**0.0010**

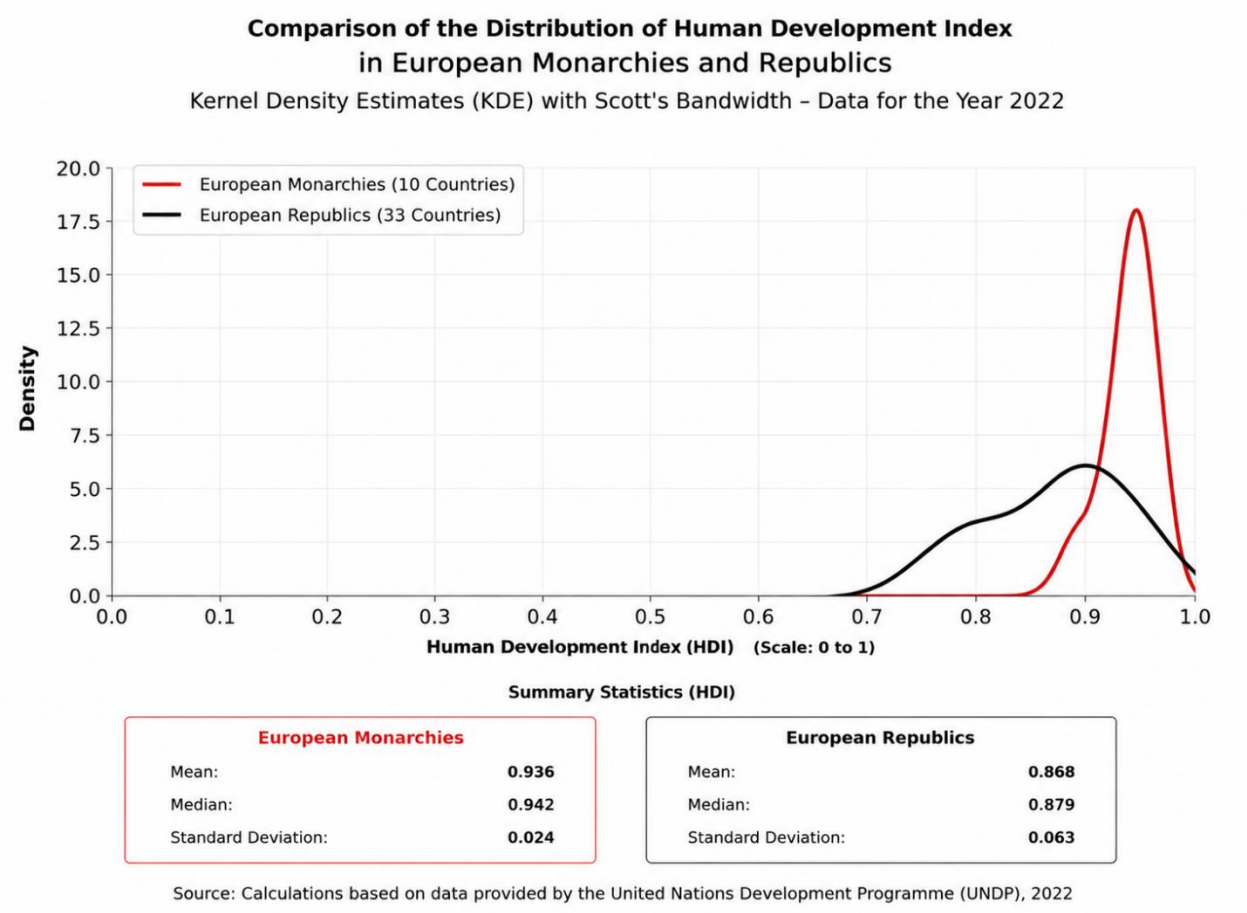
Statistical Measure	Value
Rank-biserial correlation	**0.697**

The Mann–Whitney test also indicates a highly statistically significant difference between the two groups (**p = 0.0010**). The rank-biserial correlation of **0.697** indicates a **large effect** and a clear tendency for European monarchies to occupy higher ranks on this index.

Conclusion

In 2022, **European monarchies performed better than European republics on the Human Development Index. Both the mean and median HDI scores are higher among monarchies, and both Welch's test and the Mann–Whitney test confirm the difference with very strong statistical evidence.**

This finding indicates that, on average, citizens of European monarchies enjoyed more favorable conditions in education, health, and standard of living, and that the observed advantage is substantial not only statistically but also in practical terms. Therefore, the European data for this index are also consistent with the prediction of the "New or Modern Monarchy" hypothesis.



The kernel density estimate (KDE) plot shows that the distribution of European monarchies on the Human Development Index is concentrated almost entirely at the very high end of the index, with relatively little dispersion. This indicates a comparatively consistent level of very high human development across these countries. European republics, by contrast, also perform well overall, but their distribution is broader: alongside countries with very high human development, a number of republics are found at lower levels of the index.

This pattern is consistent with the descriptive statistics. The mean and median HDI values for European monarchies are **0.936** and **0.942**, respectively, compared with **0.868** and **0.879** for European republics. Thus, the 2022 data indicate that **European monarchies, on average, have a higher level of human development than the republics of the continent**.

Comparison of the Human Development Index in European Monarchies and European Republics That Have Never Been Communist

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
European Monarchies	**10**	**0.936**	**0.942**	**0.024**
Never-Communist European Republics	**13**	**0.920**	**0.915**	**0.032**

The mean Human Development Index is **0.936** among European monarchies and **0.920** among European republics that have never experienced communist rule. Thus, **European monarchies score, on average, 0.016 points higher than this group of republics**.

The median points in the same direction. The median is **0.942** among monarchies and **0.915** among never-communist republics, meaning that the median country among European monarchies scores **0.027 points** higher than the median country among the never-communist republics.

The standard deviation is also lower among monarchies (**0.024**) than among the never-communist republics (**0.032**), indicating that European monarchies exhibit slightly more consistent performance on this index.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**0.016**
Welch's *t* statistic	**1.361**
Approximate degrees of freedom	**20.99**
Two-tailed *p*-value	**0.188**
Hedges' *g*	**0.53**

Welch's test indicates that **the difference in mean HDI between European monarchies and never-communist European republics is not statistically significant** (**p = 0.188**). Although the mean difference favors monarchies, the evidence is insufficient to reject the hypothesis of equal means.

Hedges' *g* is **0.53**, indicating a **medium-sized effect**.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the nonparametric **Mann–Whitney test** was also conducted.

Statistical Measure	Value
U statistic	**84**
Two-tailed *p*-value	**0.251**

The Mann–Whitney test likewise does not indicate a statistically significant difference between the two groups (**p = 0.251**), reinforcing the conclusion of Welch's test.

Conclusion

In 2022, **European monarchies had a higher average Human Development Index than European republics that had never experienced communist rule, and both the mean and median were higher among monarchies. However, neither Welch's test nor the Mann–Whitney test found the difference to be statistically significant.**

Comparison of the Human Development Index in Africa, the Americas, and Oceania

As with the previous indicators, this section compares monarchies and Commonwealth realms with republics in the same continent. Because the number of monarchies and Commonwealth realms is very small in some of these regions, the analysis is limited to descriptive statistics and no inferential statistical tests are conducted.

Africa

Type of Government	Number of Countries	Mean	Median	Standard Deviation
African Monarchies	**3**	**0.610**	**0.610**	**0.072**
African Republics	**50**	**0.552**	**0.537**	**0.110**

In Africa, the mean Human Development Index among the three monarchies (**0.610**) is higher than the mean among the continent's republics (**0.552**). The median points in the same direction.

The Americas

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Commonwealth Realms in the Americas	**9**	**0.791**	**0.793**	**0.071**

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Republics in the Americas	**26**	**0.751**	**0.761**	**0.081**

In the Americas, Commonwealth realms also have a higher average Human Development Index than the republics of the continent. The mean is **0.791** among Commonwealth realms and **0.751** among republics.

Oceania

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Oceanian Monarchy	**1**	**0.739**	**0.739**	—
Commonwealth Realms in Oceania	**5**	**0.734**	**0.653**	**0.174**
Oceanian Republics	**9**	**0.677**	**0.696**	**0.068**

In Oceania, the region's only independent monarchy, Tonga, also has an HDI score above the average for Oceanian republics. The mean among Commonwealth realms in Oceania (**0.734**) is likewise higher than the mean among the region's republics (**0.677**). However, because the number of monarchies and Commonwealth realms is very limited, these results are strictly descriptive, and no statistical tests are conducted for this section.

Comparison of the Human Development Index in Middle Eastern Monarchies and Republics

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Middle Eastern Monarchies	**7**	**0.854**	**0.875**	**0.063**
Middle Eastern Republics	**10**	**0.728**	**0.726**	**0.153**

The mean Human Development Index is **0.854** among Middle Eastern monarchies and **0.728** among the region's republics. Thus, **monarchies score approximately 0.126 points higher on average than republics**.

The median points in the same direction. The median is **0.875** among monarchies and approximately **0.726** among republics, meaning that the median country among Middle Eastern monarchies scores nearly **0.150 points** higher than the median country among the region's republics.

The standard deviation among republics is also more than twice that of monarchies. This indicates that human-development outcomes among Middle Eastern republics are considerably

more dispersed, whereas monarchies are concentrated at a higher and relatively more consistent level.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**0.126**
Welch's *t* statistic	**2.335**
Approximate degrees of freedom	**12.80**
Two-tailed *p*-value	**0.0365**
95% confidence interval	**0.009 to 0.243**
Hedges' *g*	**0.96**

Welch's test indicates that **the difference in mean HDI between Middle Eastern monarchies and republics is statistically significant at the 5% level** (**p = 0.037**). The 95% confidence interval lies entirely above zero, although its lower bound is close to zero.

Hedges' *g* is **0.96**, indicating a **large effect**. Thus, the observed difference is substantial in practical terms as well as statistically significant.

Mann–Whitney Test

To ensure that the result does not depend solely on the comparison of means or on the assumptions of the parametric test, the nonparametric **Mann–Whitney test** was also conducted.

Statistical Measure	Value
U statistic	**54**
Two-tailed *p*-value	**0.071**
Rank-biserial correlation	**0.54**

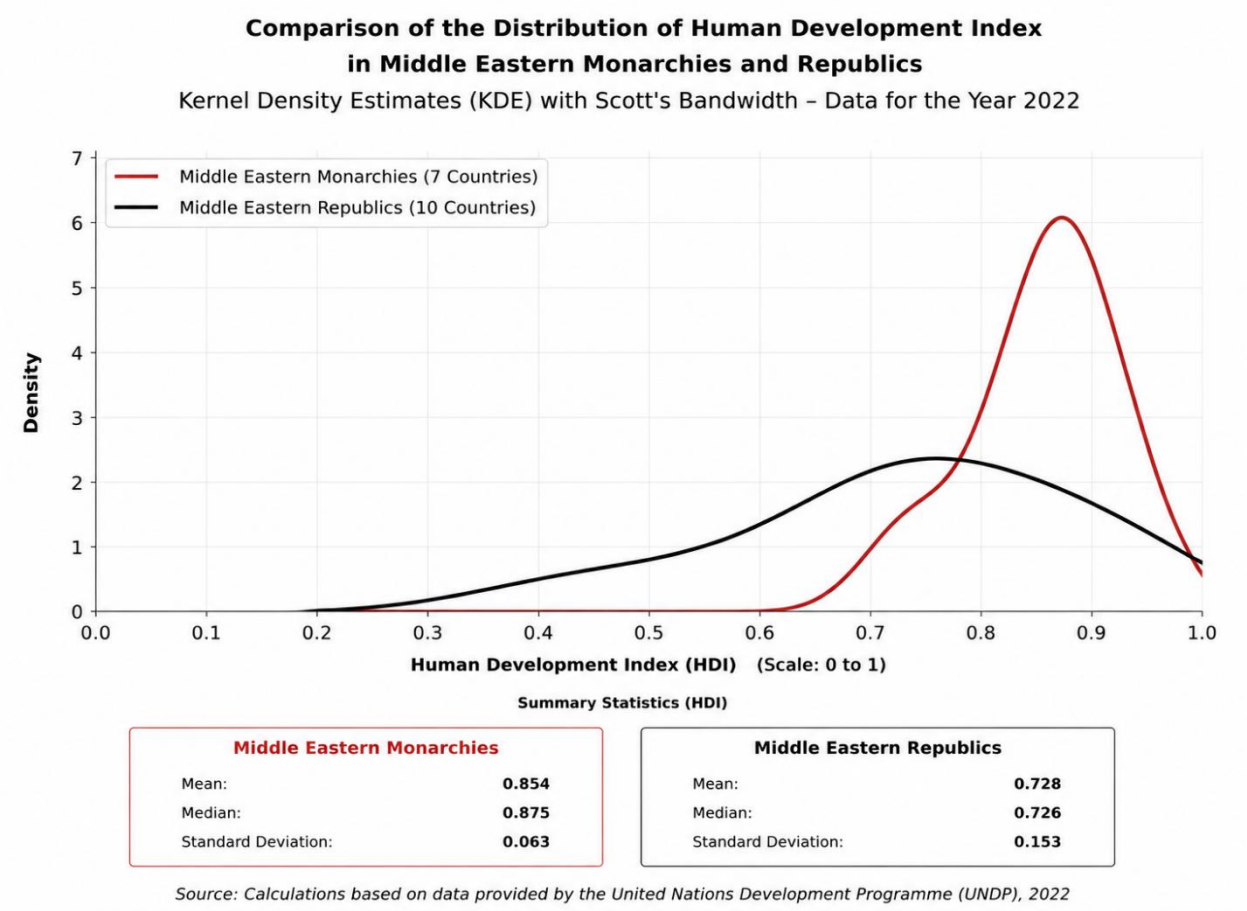
The Mann–Whitney test does not reach statistical significance at the 5% level (**p = 0.071**), although the *p*-value is relatively close to the conventional threshold. The rank-biserial correlation of **0.54** nevertheless indicates a **large effect** favoring monarchies.

Conclusion

In 2022, **Middle Eastern monarchies had a higher Human Development Index than the region's republics in terms of both the mean and the median. Welch's test indicates that the**

difference is statistically significant at the 5% level, with a large effect size. The Mann–Whitney test does not reach the 5% significance threshold, but both the direction of the result and its effect size still favor monarchies.

Overall, the Middle Eastern data indicate that monarchies in the region were, on average, in a more favorable position across the key dimensions of human development—health, education, and standard of living. From the perspective adopted in this chapter, this result is also consistent with the prediction of the "New or Modern Monarchy" hypothesis.



The kernel density estimate (KDE) plot shows that the distribution of Middle Eastern monarchies on the Human Development Index is noticeably shifted toward higher scores, with most of these countries falling within the **"very high human development"** category. Middle Eastern republics, by contrast, display considerably greater dispersion and span a much wider range of human development levels, from low to very high.

This pattern is also consistent with the descriptive statistics. The mean and median HDI values for Middle Eastern monarchies are **0.854** and **0.875**, respectively, compared with **0.728** and

0.726 for the region's republics. Thus, the 2022 data indicate that **Middle Eastern monarchies, on average, have a higher level of human development than the region's republics**.

Status of the "New or Modern Monarchy" Hypothesis After the Twelfth Indicator

With the addition of the Human Development Index, the status of the **"New or Modern Monarchy" hypothesis** at this stage is as follows:

Test	Indicator	Result
✓	GDP per Capita	Consistent with the hypothesis
✓	Purchasing Power per Capita	Consistent with the hypothesis
◐	Gini Coefficient	**Mixed evidence; difference favors the hypothesis**
✓	Political Stability and Absence of Violence	Consistent with the hypothesis
✓	Regulatory Quality	Consistent with the hypothesis
✓	Rule of Law	Consistent with the hypothesis
✓	Government Effectiveness	Consistent with the hypothesis
✓	Inflation	Consistent with the hypothesis
✓	Control of Corruption	Consistent with the hypothesis
✓	Corruption Perceptions	Consistent with the hypothesis
✓	Property Rights	Consistent with the hypothesis
✓	Human Development	Consistent with the hypothesis

At this stage, the **"New or Modern Monarchy" hypothesis has been tested against twelve independent indicators**. These indicators cover a range of the most important goals of government, from **welfare** and **order** to **justice** and now **liberty**. They have also been produced by reputable and independent international institutions, including the World Bank, the United Nations Development Programme (UNDP), Transparency International, and other international organizations.

For the Human Development Index, as with the Property Rights Index, monarchies have, on average, outperformed republics in the global, Asian, European, and Middle Eastern comparisons. Thus, **the second indicator associated with the goal of "liberty" also points in the direction predicted by the hypothesis.**

Overall, **none of the twelve indicators examined so far has produced a result inconsistent with the "New or Modern Monarchy" hypothesis.** As emphasized throughout this study,

however, these findings do not establish a causal relationship between form of government and quality of governance. Rather, they indicate that **the empirical evidence from all the tests conducted up to this point has been consistent with the predictions of the hypothesis.**

Voice and Accountability

So far, all the indicators examined have been consistent with the predictions of the **"New or Modern Monarchy" hypothesis**. The results have shown that monarchies, whether constitutional or non-constitutional, have on average performed better than republics in providing their citizens with the **rights** and **opportunities** necessary for liberty. Thus, for indicators that measure liberty in terms of the ability to choose and access rights and opportunities, the evidence so far has been consistent with the hypothesis.

Yet in the contemporary understanding, liberty is associated perhaps more than anything else with **freedom of expression and political participation**. For this reason, if the "New or Modern Monarchy" hypothesis is to face a serious test, one of the most appropriate indicators is the **Voice and Accountability** indicator.

Voice and Accountability is one of the World Bank's six Worldwide Governance Indicators. It measures the extent to which citizens are able to **express their views on public affairs, participate in selecting their government, enjoy freedom of expression and freedom of the media, and form and participate in associations**. The World Bank reports this indicator as a **percentile rank**, with higher scores indicating greater opportunities for citizens to express their views and participate in public life.

The data used in this section are from **2023** (Appendix RDS-M01). For brevity, from this point onward I refer to it simply as the **"Voice" indicator**.

We can now examine whether, in this dimension of liberty as well, the evidence remains consistent with the predictions of the "New or Modern Monarchy" hypothesis—or whether this will be the first indicator to challenge it.

Comparison of the "Voice and Accountability" Indicator Between Monarchies and Republics Worldwide

Descriptive Statistics

Form of Government	Number of Countries	Mean	Median	Standard Deviation
Monarchies Worldwide	**28**	**55.35**	**49.25**	**33.60**
Republics Worldwide	**151**	**45.47**	**44.60**	**27.66**

The mean percentile rank for Voice and Accountability is **55.35** among the world's monarchies and **45.47** among republics. **Thus, monarchies score approximately 9.88 percentile points higher on average than republics.**

The median points in the same direction. The median is **49.25** for monarchies and **44.60** for republics, meaning that the median country among monarchies ranks approximately **4.65 percentile points** higher than the median country among republics.

Both groups exhibit considerable dispersion. The standard deviation is **33.60** among monarchies and **27.66** among republics. This substantial variation indicates that both forms of government include countries with very high as well as very low scores.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference; monarchy minus republic	**9.88**
Welch's *t* statistic	**1.47**
Approximate degrees of freedom	**34.11**
Two-tailed *p*-value	**0.152**
95% confidence interval	**−3.81 to 23.57**
Hedges' *g*	**0.34**

Welch's test shows that **although the mean score is higher among monarchies, the difference is not statistically significant at the conventional 5 percent level** (**p = 0.152**). The 95 percent confidence interval also includes zero. Therefore, the available data do not provide sufficient evidence to conclude that there is a systematic difference between the mean scores of the two groups.

Hedges' *g* is **0.34**, indicating a **small-to-moderate effect** in favor of monarchies.

Mann–Whitney Test

To compare the overall distributions of the two groups without relying on the assumptions of a parametric test, the nonparametric **Mann–Whitney test** was also conducted.

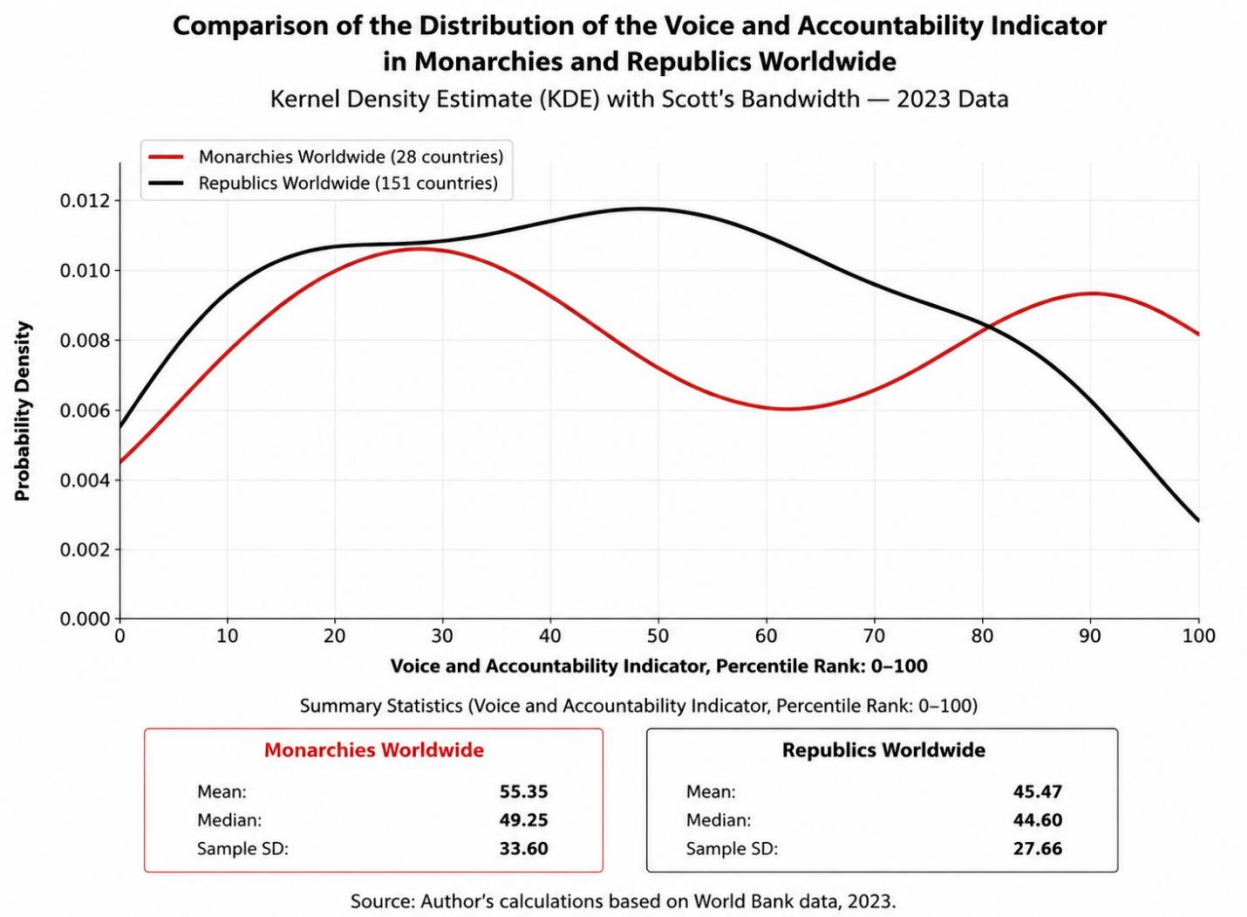
Statistical Measure	Value
U statistic	**2490**
Two-tailed *p*-value	**0.136**
Rank-biserial correlation	**0.18**

The Mann–Whitney test likewise does not find a statistically significant difference between the two groups at the 5 percent level (**p = 0.136**). The rank-based effect size of **0.18** is also small and favors monarchies.

Conclusion

In 2023, **monarchies worldwide had higher scores than republics on the Voice and Accountability indicator in terms of both the mean and the median; however, the observed difference did not reach statistical significance in either Welch's t-test or the Mann–Whitney test.**

The result is therefore not inconsistent with the "New or Modern Monarchy" hypothesis, since the observed difference is in the direction predicted by the hypothesis. Unlike most of the preceding indicators, however, the statistical evidence is insufficient to establish a systematic difference between the two groups. More precisely, this test neither contradicts the hypothesis nor provides strong statistical support for it.



The kernel density estimate shows that the distribution of monarchies worldwide on the Voice and Accountability indicator is generally shifted toward higher scores, although there is also considerable variation among them. In contrast, republics worldwide span a broader range of scores, with greater concentration in the middle and lower levels of the indicator. This pattern is also consistent with the descriptive statistics: the mean and median scores for monarchies are **55.35** and **49.25**, respectively,

compared with **45.47** and **44.60** for republics. Thus, the 2023 data show that monarchies, on average, performed better than republics on the Voice and Accountability indicator, although the substantial dispersion within both groups indicates considerable variation across countries.

Comparison of the "Voice and Accountability" Indicator Between Constitutional and Non-Constitutional Monarchies Worldwide

Descriptive Statistics

Type of Monarchy	Number of Countries	Mean	Median	Standard Deviation
Constitutional Monarchies	**15**	**73.26**	**87.70**	**29.87**
Semi-Constitutional and Absolute Monarchies	**13**	**34.68**	**28.90**	**25.20**

The mean Voice and Accountability score is **73.26** among constitutional monarchies and **34.68** among semi-constitutional and absolute monarchies. Thus, **constitutional monarchies score an average of 38.58 percentile points higher than the second group**.

The median reveals an even larger difference. The median is **87.70** among constitutional monarchies and **28.90** among semi-constitutional and absolute monarchies. In other words, the median country among constitutional monarchies scores **58.80 percentile points** higher than the median country in the second group.

Although both groups exhibit considerable dispersion, the difference between them is large enough to remain clearly visible despite this variation.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (constitutional minus semi-constitutional and absolute)	**38.58**
Welch's *t* statistic	**3.71**
Approximate degrees of freedom	**25.99**
Two-tailed *p*-value	**0.0010**

Welch's test shows that the difference in the mean Voice and Accountability score between the two types of monarchy is highly statistically significant (**p = 0.001**). The hypothesis of equal means can therefore be rejected with a high degree of confidence.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the nonparametric **Mann–Whitney test** was also conducted.

Statistical Measure	Value
U statistic	**163**
Two-tailed *p*-value	**0.0027**

The Mann–Whitney test likewise finds a **highly statistically significant** difference between the two groups (**p = 0.0027**), confirming the result of Welch's test.

Conclusion

In 2023, **constitutional monarchies performed substantially better than semi-constitutional and absolute monarchies on the Voice and Accountability indicator. Both the mean and the median are considerably higher among constitutional monarchies, and both Welch's t-test and the Mann–Whitney test confirm the difference with very strong statistical evidence.**

This finding indicates that the type of monarchy matters greatly for this indicator. Unlike many of the preceding indicators, for which the difference between the two types of monarchy was not statistically significant, constitutional monarchies perform significantly better than semi-constitutional and absolute monarchies in the area of political and civil liberties.

This result also helps explain why the overall comparison between monarchies and republics worldwide did not produce a statistically significant difference. Combining the two very different groups of constitutional and non-constitutional monarchies into a single category lowers the overall mean for monarchies. The comparison of "all monarchies" with republics therefore appears to provide an incomplete picture for this indicator.

In other words, the data show that **constitutional monarchies, which constitute a major group among modern monarchies, not only perform far better than non-constitutional monarchies on the Voice and Accountability indicator, but also score substantially above the worldwide average for republics.** Thus, **the findings in this section are not inconsistent with the "New or Modern Monarchy" hypothesis; rather, they suggest that, for this indicator, constitutional constraints are an important factor associated with the realization of political and civil liberties.**

Comparison of the "Voice and Accountability" Indicator Between Commonwealth Realms and Republics Worldwide

Descriptive Statistics

Form of Government	Number of Countries	Mean	Median	Standard Deviation
Commonwealth Realms	**14**	**74.89**	**72.55**	**15.20**
Republics Worldwide	**151**	**45.47**	**44.60**	**27.66**

The mean Voice and Accountability score is **74.89** among the Commonwealth realms and **45.47** among republics worldwide. Thus, **the Commonwealth realms score an average of 29.42 percentile points higher than republics**.

The median shows the same pattern. The median is **72.55** among the Commonwealth realms and **44.60** among republics, meaning that the median Commonwealth realm scores approximately **27.95 percentile points** higher than the median republic worldwide.

The standard deviation is also substantially lower among the Commonwealth realms (**15.20**) than among republics (**27.66**), indicating that the Commonwealth realms have not only achieved higher scores but have also displayed more consistent performance on this indicator.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Commonwealth realm minus republic)	**29.42**
Welch's *t* statistic	**6.34**
Approximate degrees of freedom	**22.02**
Two-tailed *p*-value	**0.0000022**

Welch's test shows that **the difference in the mean Voice and Accountability score between the Commonwealth realms and republics worldwide is highly statistically significant** (**$p < 0.001$**). The hypothesis of equal means can therefore be rejected with a very high degree of confidence.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the nonparametric **Mann–Whitney test** was also conducted.

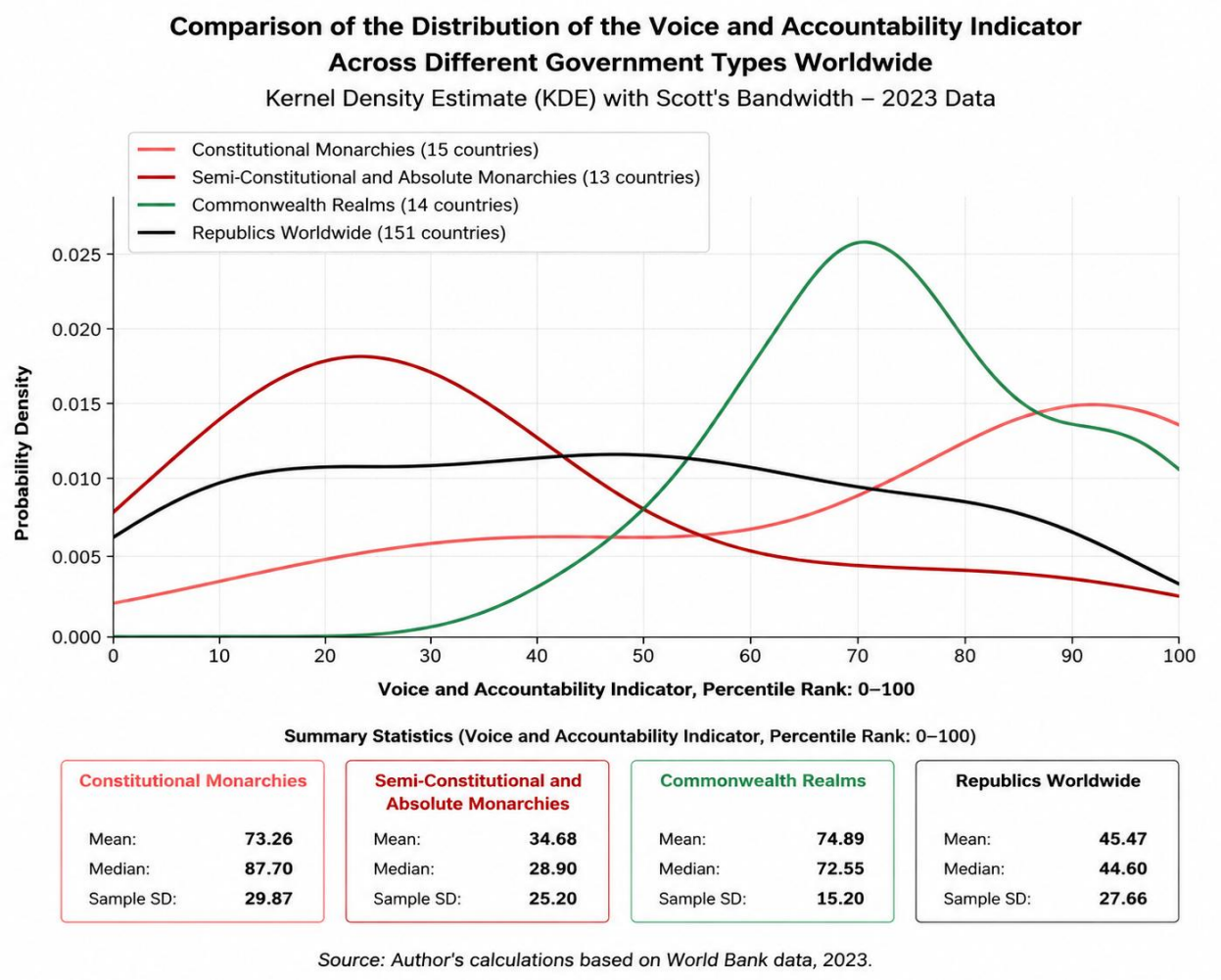
Statistical Measure	Value
U statistic	**1705**
Two-tailed *p*-value	**0.000153**

The Mann–Whitney test likewise finds a highly statistically significant difference between the two groups (**$p < 0.001$**), confirming the result of Welch's test.

Conclusion

In 2023, **the Commonwealth realms achieved substantially higher Voice and Accountability scores, on average, than republics worldwide. Both the mean and the median are considerably higher among the Commonwealth realms, and both Welch's t-test and the Mann–Whitney test confirm the difference with very strong statistical evidence.**

This result indicates that among countries that continue to retain the British monarch as their head of state, the level of political and civil liberties is, on average, substantially higher than among republics worldwide. Therefore, **unlike the comparison of "all monarchies" with republics, in which no statistically significant difference was found because constitutional and non-constitutional monarchies were combined, the comparison between the Commonwealth realms and republics provides substantially stronger evidence in the direction predicted by the "New or Modern Monarchy" hypothesis.**



A closer look at the distribution of the Voice and Accountability indicator, together with the statistical results, reveals an important pattern. As expected, the distributions of **constitutional monarchies** and **Commonwealth realms** are concentrated largely on the right side of the chart, corresponding to higher levels of this indicator. By contrast, **semi-constitutional and absolute monarchies** have lower scores on average, which helps explain why the difference between monarchies as a whole and republics did not reach statistical significance.

However, **these data do not allow us to conclude with confidence that the difference is caused simply by the non-constitutional character of these monarchies.** The relatively wide dispersion of scores among semi-constitutional and absolute monarchies suggests that, unlike constitutional monarchies and Commonwealth realms, this group consists of a heterogeneous set of countries and that other factors may also contribute to their lower average score.

In other words, rather than establishing a causal relationship, the chart points to the possible presence of one or more hidden variables—factors that may be associated both with the type of monarchy and with performance on the Voice and Accountability

indicator. The evidence in this section therefore shows only that **semi-constitutional and absolute monarchies, as a group, perform less well than constitutional monarchies**. Whether this difference is specifically attributable to the extent of the monarch's powers, or whether other factors are involved, is a question that this study alone cannot answer conclusively.

Comparison of the Voice and Accountability Indicator Between Monarchies and Republics Across Continents and in the Middle East

Comparison of the Voice and Accountability Indicator Between Monarchies and Republics in Asia

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Sample Standard Deviation
Asian Monarchies	**13**	**31.48**	**27.00**	**21.41**
Asian Republics	**33**	**27.03**	**23.50**	**21.43**

The mean Voice and Accountability score is **31.48** for Asian monarchies and **27.03** for Asian republics. **Thus, Asian monarchies score, on average, 4.45 percentile points higher than the republics of the continent.**

The median points in the same direction. The median is **27.00** for monarchies and **23.50** for republics, meaning that the median country among Asian monarchies scores **3.50 percentile points** higher than the median country among Asian republics.

At the same time, the standard deviations of the two groups are almost identical—approximately **21.4**—indicating a similar degree of variation in performance within both groups.

Test of the Difference Between Means

To assess the statistical significance of the difference between the two means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**4.46**
Welch's *t* statistic	**0.636**
Approximate degrees of freedom	**22.05**
Two-tailed *p*-value	**0.532**

Welch's test indicates that **the difference in the mean Voice and Accountability scores between Asian monarchies and republics is not statistically significant** (**p = 0.532**). Therefore, based on the available data, the hypothesis of equal means cannot be rejected.

Mann–Whitney Test

To ensure that the conclusion does not depend on the assumptions of the parametric test, the nonparametric **Mann–Whitney test** was also conducted.

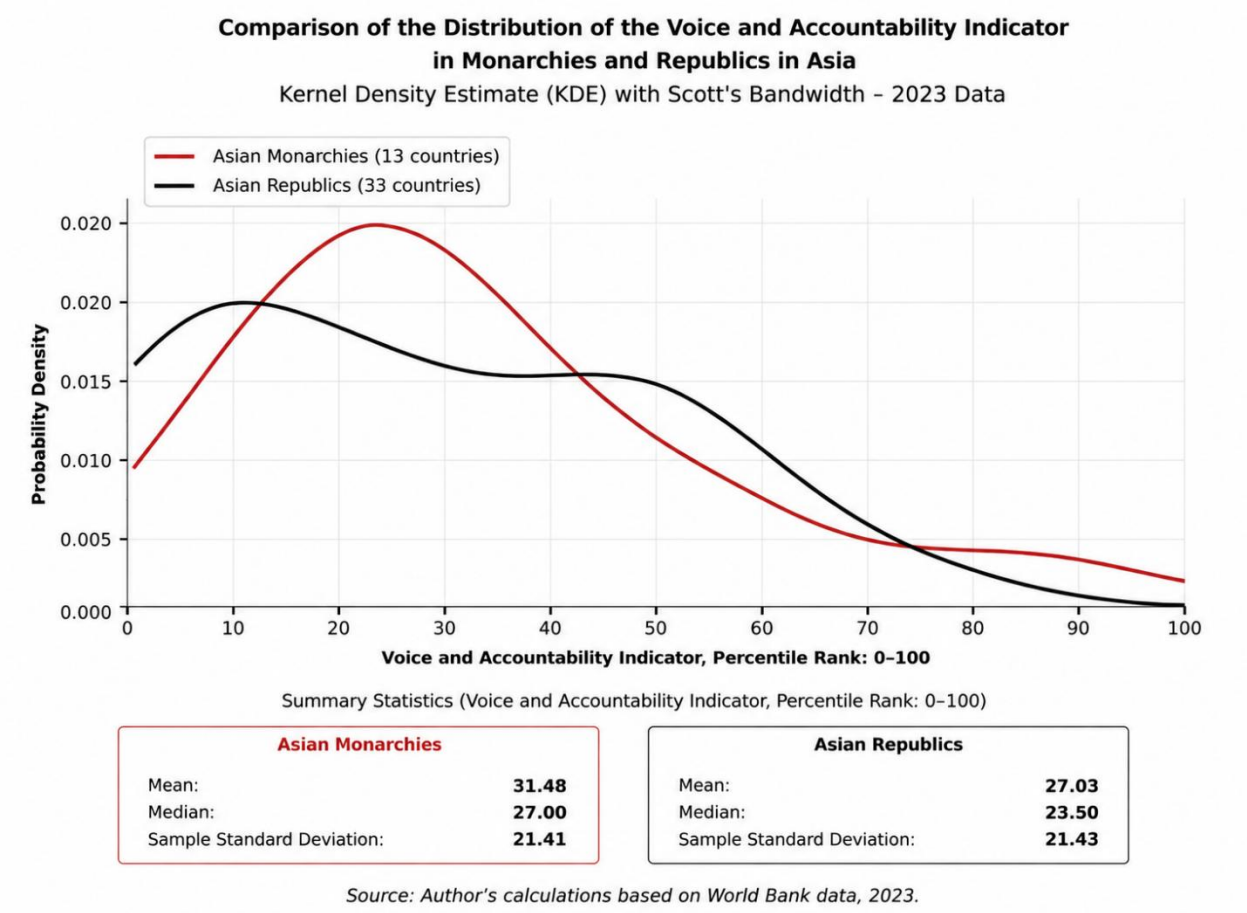
Statistical Measure	Value
U statistic	**247**
Two-tailed *p*-value	**0.435**

The Mann–Whitney test likewise finds no statistically significant difference between the two groups (**p = 0.435**), confirming the result of Welch's test.

Conclusion

In 2023, **Asian monarchies had a higher average Voice and Accountability score than Asian republics, with both the mean and the median favoring monarchies. However, the observed difference did not reach statistical significance in either Welch's test or the Mann–Whitney test.**

This result indicates that **in Asia, as in the global comparison, the data do not contradict the "New or Modern Monarchy" hypothesis, but neither do they provide sufficient statistical evidence to establish a systematic difference between the two groups.** The relatively small difference between the means, combined with substantial variation within both groups, means that the observed difference is not statistically significant.



The kernel density estimate shows that the distributions of **Asian monarchies and republics** on the Voice and Accountability indicator overlap substantially, although the monarchy curve is slightly shifted toward higher scores. This pattern is consistent with the descriptive statistics: the mean and median scores for Asian monarchies are **31.48** and **27.00**, respectively, compared with **27.03** and **23.50** for Asian republics. By contrast, the standard deviations of the two groups are nearly identical, indicating a similar degree of dispersion in scores among the monarchies and republics of the continent. Thus, the 2023 data suggest a **modest average advantage for Asian monarchies**, but the limited separation between the two distributions means that any firm conclusion should be based on the results of the statistical significance tests.

Comparison of the Voice and Accountability Indicator Between Monarchies and Republics in Europe

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Sample Standard Deviation
European Monarchies	11	87.91	93.10	18.79
European Republics	34	69.77	77.45	23.52

The mean Voice and Accountability score is **87.91** for European monarchies and **69.77** for European republics. Thus, **European monarchies score, on average, 18.14 percentile points higher than European republics**.

The median confirms the same pattern. The median is **93.10** for monarchies and **77.45** for republics, meaning that the median country among European monarchies scores **15.65 percentile points** higher than the median country among European republics.

The standard deviation is also lower among monarchies (**18.79**) than among republics (**23.52**), indicating that European monarchies have not only higher scores but also somewhat less variation in their performance.

Test of the Difference Between Means

To assess the statistical significance of the difference between the two means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**18.14**
Welch's *t* statistic	**2.61**
Approximate degrees of freedom	**21.07**
Two-tailed *p*-value	**0.0164**

Welch's test shows that **the difference in the mean Voice and Accountability scores between European monarchies and republics is statistically significant** (**p = 0.016**). Therefore, the hypothesis of equal means is rejected at the 5 percent significance level.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the nonparametric **Mann–Whitney test** was also conducted.

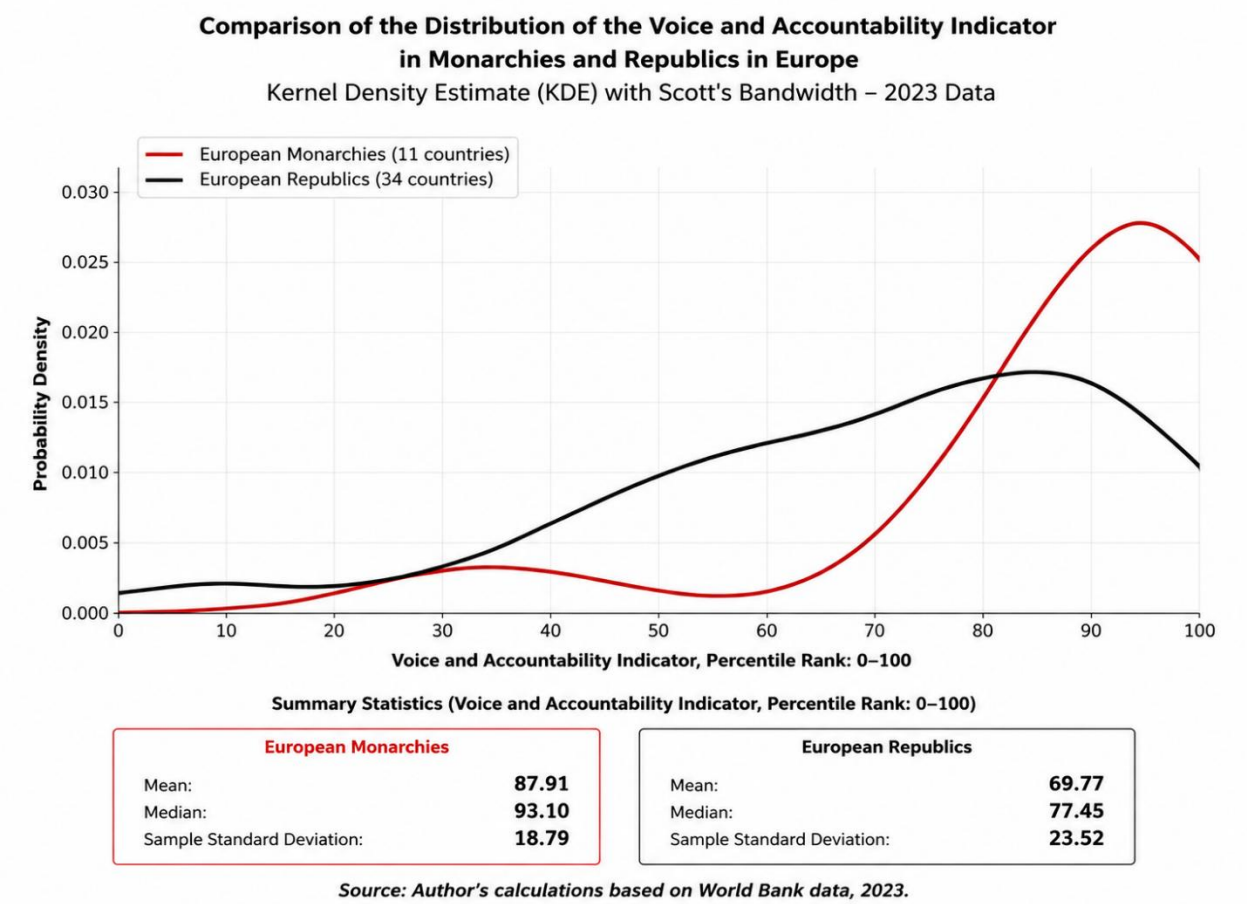
Statistical Measure	Value
U statistic	**294**
Two-tailed *p*-value	**0.0049**

The Mann–Whitney test likewise indicates a statistically significant difference between the two groups (**p = 0.0049**), confirming the result of Welch's test.

Conclusion

In 2023, **European monarchies had higher Voice and Accountability scores than European republics in terms of both the mean and the median. Both Welch's test and the Mann–Whitney test confirm that the difference is statistically significant.**

This finding indicates that, on average, citizens of European monarchies enjoyed greater freedom to express their views, participate in political life, operate through the media, and hold governments accountable. Therefore, the European evidence for this indicator is also consistent with the prediction of the "New or Modern Monarchy" hypothesis.



The kernel density estimate shows that the distribution of **European monarchies** on the Voice and Accountability indicator is generally concentrated at the higher end of the scale, with the greatest density occurring among very high scores. By contrast, although **European republics** include a substantial number of countries with high scores, their distribution is more dispersed, with some countries falling within the middle and lower ranges of the indicator. This pattern is also consistent with the descriptive statistics: the mean and median scores for European monarchies are **87.91** and **93.10**, respectively, compared with **69.77** and **77.45** for European republics. Thus, the 2023 data indicate that,

on average, European monarchies perform better than the republics of the continent on the Voice and Accountability indicator.

Comparison of the Voice and Accountability Indicator Between European Monarchies and European Republics That Have Never Been Communist

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Sample Standard Deviation
European Monarchies	**11**	**87.91**	**93.10**	**18.79**
Never-Communist European Republics	**13**	**89.17**	**89.70**	**7.47**

The mean Voice and Accountability score is **87.91** among European monarchies and **89.17** among European republics that have never experienced communist rule. Thus, **the never-communist republics score 1.26 percentile points higher on average than the European monarchies**.

The median, however, points slightly in the opposite direction. The median is **93.10** among monarchies and **89.70** among never-communist republics. The difference between the mean and the median suggests that one or more relatively low values among the European monarchies—such as Monaco on this indicator—pull down the group mean.

The standard deviation is also substantially higher among monarchies (**18.79**) than among the never-communist republics (**7.47**), indicating considerably greater variation among European monarchies on this indicator.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**−1.26**
Welch's *t* statistic	**−0.209**
Approximate degrees of freedom	**12.67**
Two-tailed *p*-value	**0.838**

Welch's test shows that **the difference in mean Voice and Accountability scores between European monarchies and never-communist European republics is not statistically significant** (**p = 0.838**). The hypothesis of equal means is therefore not rejected.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the nonparametric **Mann–Whitney test** was also conducted.

Statistical Measure	Value
U statistic	**87**
Two-tailed *p*-value	**0.385**

The Mann–Whitney test likewise finds **no statistically significant difference** between the two groups (**p = 0.385**), confirming the result of Welch's test.

Conclusion

Once European republics with a history of communist rule are excluded, the difference between monarchies and republics on the Voice and Accountability indicator effectively disappears. Although the median remains slightly higher among monarchies, the group means are extremely close, and neither Welch's test nor the Mann–Whitney test finds a statistically significant difference.

This result indicates that **among European countries with longer-established democratic institutions and civil liberties, regime type—monarchy or republic—is not associated with a statistically significant difference on this indicator.** Thus, unlike some previous indicators such as Control of Corruption or Property Rights, European monarchies do not show a statistically significant advantage over never-communist European republics in Voice and Accountability. As elsewhere in this book, this finding describes an observational relationship and is not by itself sufficient to establish causation.

Methodological Note

It should be emphasized that the comparison between European monarchies and European republics with no history of communist rule is only a **supplementary analysis** intended to explore the possible sources of observed differences; **it does not constitute the primary basis for the study's overall assessment of the relative performance of monarchy and republic**. From the outset, the hypothesis of this book concerns the comparison of the two forms of government in their broader categories, and the principal conclusions should therefore rest on comparisons between all monarchies and all republics.

Excluding republics with a communist past—just as excluding non-constitutional monarchies—may help clarify the role of historical, political, or cultural factors in shaping the results, but it does not by itself justify changing the basis of the main comparison. For this reason, such comparisons are used in this book only to **interpret the data more carefully and identify potentially relevant factors**, not to replace the principal analysis.

Comparison of the Voice and Accountability Indicator in Africa, the Americas, and Oceania

Because the number of monarchies in Africa and Oceania is small, and because the number of Commonwealth realms is also limited in some of these regions, this section is restricted to **descriptive statistics** and does not include inferential statistical tests.

Africa

Type of Government	Number of Countries	Mean	Median	Standard Deviation
African Monarchies	**3**	**32.83**	**34.30**	**15.45**
African Republics	**49**	**32.41**	**29.90**	**19.66**

In Africa, the mean Voice and Accountability scores of monarchies and republics are almost identical, and no substantial difference is evident between the two groups.

The Americas

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Commonwealth Realms in the Americas	**9**	**73.20**	**72.10**	**9.02**
Republics in the Americas	**26**	**51.64**	**54.15**	**23.53**

In the Americas, Commonwealth realms perform substantially better than republics in terms of both the mean and the median, while also displaying considerably less dispersion.

Oceania

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Oceanian Monarchy	**1**	**75.00**	**75.00**	—
Commonwealth Realms in Oceania	**5**	**77.94**	**89.20**	**23.86**
Oceanian Republics	**9**	**74.61**	**80.90**	**11.74**

Oceania has only **one independent monarchy**, Tonga, so a meaningful inferential comparison between monarchies and republics is not possible. Descriptively, however, both Tonga and the Commonwealth realms of Oceania perform at levels broadly similar to the region's republics, and the differences in group means are not large.

Comparison of the Voice and Accountability Indicator Between Monarchies and Republics in the Middle East

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Middle Eastern Monarchies	**7**	**20.10**	**21.10**	**8.06**
Middle Eastern Republics	**10**	**27.10**	**20.80**	**26.20**

The mean Voice and Accountability score is **20.10** among Middle Eastern monarchies and **27.10** among the region's republics. Thus, **the mean for republics is 7 percentile points higher than that for monarchies**.

The medians, however, are almost identical: 21.10 for monarchies and 20.80 for republics. This divergence between the mean and the median reflects the much greater dispersion among the republics. The standard deviation for republics is **26.20**, more than three times that of monarchies. Middle Eastern republics are therefore a highly heterogeneous group on this indicator, combining very low scores with two very high values.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**−7.00**
Welch's *t* statistic	**−0.793**
Approximate degrees of freedom	**11.29**
Two-tailed *p*-value	**0.444**
95% confidence interval	**−26.37 to 12.37**
Hedges' *g*	**−0.32**

Welch's test shows that the difference between the two group means is not statistically significant (**p = 0.444**). The 95% confidence interval includes zero, meaning that the available data are compatible with a positive difference, a negative difference, or no true difference between the groups.

Hedges' *g* is approximately **−0.32**, indicating a small effect in favor of republics. Given the limited sample size and wide confidence interval, however, this estimate is subject to substantial uncertainty.

Mann–Whitney Test

To compare the overall rankings of the two groups and reduce dependence on the means, the nonparametric **Mann–Whitney test** was also conducted.

Statistical Measure	Value
U statistic	**37**
Two-tailed *p*-value	**0.887**
Rank-biserial correlation	**0.06**

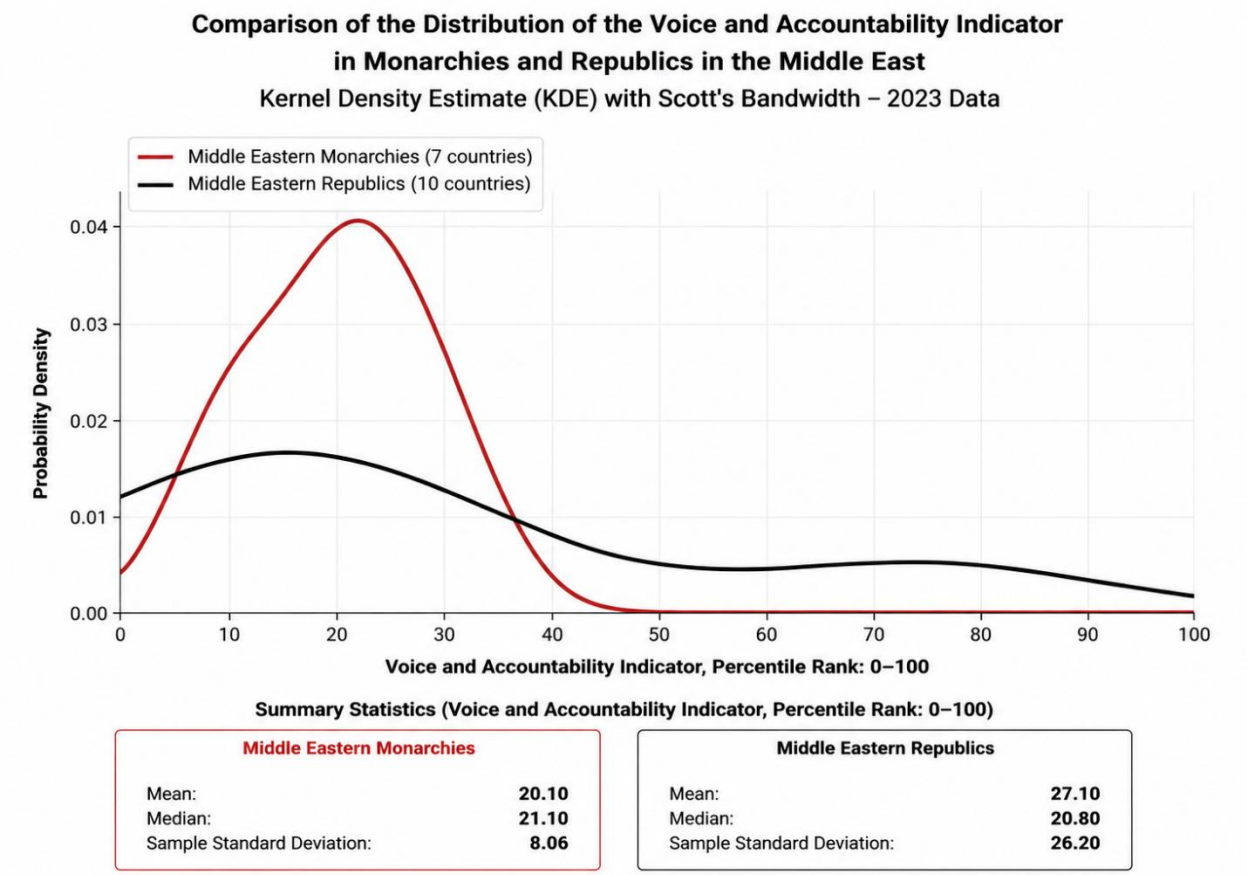
The Mann–Whitney test likewise finds no statistically significant difference between the two groups (**p = 0.887**). The rank-based effect size is also close to zero, indicating little overall difference in the relative positions of Middle Eastern monarchies and republics within the distribution.

Conclusion

In the Middle East, **the mean Voice and Accountability score is higher among republics than among monarchies, but the medians of the two groups are nearly identical, and neither Welch's test nor the Mann–Whitney test finds a statistically significant difference.**

The very high dispersion among republics indicates that their higher mean is strongly influenced by a small number of countries with exceptionally high scores, while most of the region's republics fall within a range similar to, or below, that of the monarchies. This does not, however, justify excluding those countries or disregarding the republican mean; they are genuine members of the group being studied.

Accordingly, the direction of the mean difference in this comparison runs contrary to the prediction of the "New or Modern Monarchy" hypothesis, but the statistical evidence does not indicate that this difference reflects a systematic distinction between the two forms of government. More precisely, the Middle Eastern result on this indicator provides neither clear support for the hypothesis nor sufficient evidence to reject it.



The kernel density estimate shows that the distributions of Middle Eastern monarchies and republics on the Voice and Accountability indicator overlap considerably, but the republics display substantially greater dispersion. This difference is also reflected in the descriptive statistics: the mean score for Middle Eastern republics (**27.10**) is somewhat higher than that for monarchies (**20.10**), while the standard deviation among republics (**26.20**) is more than three times that among monarchies (**8.06**). This pattern indicates that Middle Eastern republics constitute a highly heterogeneous group in terms of Voice and Accountability, with some countries receiving very high scores and others very low scores, whereas the region's monarchies are concentrated within a narrower and relatively similar range.

Why Does the Middle Eastern Result Differ from the Overall Pattern?

We appear to have reached the first result that differs from the dominant pattern observed throughout this study, making it one of the most interesting sections of the book.

In the Middle East, unlike the global pattern, the mean Voice and Accountability score is somewhat higher among republics than among monarchies, although the difference between the two groups is not statistically significant and their medians are nearly identical. Unlike many of

the preceding indicators, therefore, this result provides neither clear support for the "New or Modern Monarchy" hypothesis nor evidence sufficient to reject it.

This raises an important question.

If Voice and Accountability is so important to good governance, how is it possible that the same Middle Eastern monarchies that score relatively low on this indicator have performed better than the region's republics on nearly all the other indicators examined so far? These countries have performed favorably on indicators such as GDP per capita, purchasing power, the Gini coefficient, political stability, regulatory quality, rule of law, government effectiveness, control of corruption, property rights, and human development. If Voice and Accountability is a necessary condition for achieving these objectives, how can these two patterns coexist?

Earlier, in examining the Political Stability and Absence of Violence indicator, we also encountered a result in Oceania that ran contrary to the global pattern. There, the particular characteristics of the region's small island states appeared likely to have contributed to the result, and once data from other regions of the world were incorporated, the effect of that regional pattern disappeared from the overall result.

Something similar may be occurring in the Middle East: a regional factor, independent of the form of government, may be influencing Voice and Accountability scores.

One hypothesis worth examining concerns the role of religious fundamentalist movements in some countries of the region. In much of the Middle East, there are groups with substantial social support that openly advocate restricting a range of individual and civil liberties according to their particular interpretation of Sharia. If such groups gain power through elections or unrestricted political activity, they may themselves proceed to restrict the liberties of those same citizens, including the freedom of expression of their opponents.

This possibility raises an important question: **Could there, under certain circumstances, be a difficult trade-off between expanding Voice and Accountability and preserving other objectives of government? Conflicts may sometimes arise among different governmental objectives, and the best decision may not always be to maximize a single indicator. Can all political values be maximized simultaneously, or are choices among them sometimes unavoidable?**

In other words, could a government, under particular regional circumstances, restrict some political liberties in order to prevent the establishment of a regime that would abolish many fundamental liberties, while at the same time maintaining higher levels of security, stability, property rights, prosperity, and other dimensions of governance?

The contemporary history of Iran is an example often raised in this discussion. In the second half of the 1970s, the political environment became more open than before, and opponents of the government gained greater scope for political activity. Some analysts argue that this political

opening, alongside a range of other factors, contributed to the fall of the monarchy and the establishment of the Islamic Republic—a regime that, according to the data examined in this study, has performed more poorly on many governance indicators than many of the region's monarchies.

This example, however, **is not by itself sufficient to establish such a relationship.** The Iranian Revolution resulted from a combination of political, economic, social, religious, and international factors, and the contribution of each cannot be isolated using the data in this study. Nor can the available evidence establish that the relatively low Voice and Accountability scores of Middle Eastern monarchies necessarily result from attempts to preserve other liberties. This is merely one possible hypothesis for explaining the observed pattern, and evaluating it would require dedicated research beyond the scope of this study.

Nevertheless, the question itself is worth considering. **If greater Voice and Accountability always and under all circumstances led to better achievement of the objectives of government, one might expect Middle Eastern monarchies, with their lower scores on this indicator, also to perform more poorly on other governance indicators. Yet the evidence examined in this study shows precisely the opposite pattern.**

The evidence in this section therefore establishes neither that greater Voice and Accountability always leads to better achievement of other governmental objectives nor the reverse. What this study identifies is simply a **noteworthy coexistence of two patterns**: in the Middle East, monarchies that score lower on Voice and Accountability have nevertheless performed better than the region's republics on many other governance indicators. Whether this coexistence reflects an unavoidable trade-off among certain objectives of government under particular regional conditions, or instead results from other factors not examined here, is a question raised by this study but left for future research to answer.

One further point should also be emphasized. **Democracy is not itself an objective of government; it is one method of governing.** As discussed at the beginning of this book on the basis of the views of political philosophers, the objectives of government are to provide citizens with welfare, order, justice, liberty, and security. Democracy is valuable insofar as it succeeds in advancing these objectives. Therefore, if under particular circumstances a conflict emerges between certain democratic mechanisms and the achievement of other governmental objectives, that conflict should be examined empirically and on the basis of evidence rather than decided in advance solely on the basis of theoretical assumptions.

Status of the "New or Modern Monarchy" Hypothesis After the Thirteenth Indicator

So far, the "New or Modern Monarchy" hypothesis has been tested using **thirteen independent indicators**. Of these, **eleven indicators** have clearly been consistent with the hypothesis's prediction, with statistically significant differences between monarchies and republics.

For the **Gini coefficient**, the difference between monarchies and republics was in the direction predicted by the hypothesis but did not reach statistical significance. Thus, although the result was aligned with the hypothesis, this indicator alone did not provide sufficient evidence to support it statistically.

The **Voice and Accountability indicator** produced a different pattern. In the principal comparison—all monarchies worldwide versus all republics—**the mean score for monarchies was higher than that for republics, but the difference was not statistically significant**. **A similar result was found in Asia. In Europe, the difference favored monarchies and was statistically significant, whereas in the Middle East the direction of the mean difference ran contrary to the hypothesis's prediction, but this difference also failed to reach statistical significance.** On the basis of the principal comparison, therefore, this indicator provides insufficient evidence either to support or to reject the hypothesis.

At the same time, the supplementary analyses of this indicator produced noteworthy findings. Constitutional monarchies performed significantly better than both republics and semi-constitutional and absolute monarchies. **Commonwealth realms also scored significantly higher than republics worldwide.** Because these analyses concern subgroups, they are not incorporated into the study's principal conclusion, but they may be useful for interpreting the observed patterns and guiding future research.

Contrary to what might initially appear, this result is not a weakness of the study. Rather, it demonstrates that the hypothesis has been tested in a way that allows for divergent findings. If every result, regardless of its direction or statistical significance, were interpreted as supporting the hypothesis, the value of the empirical test would be diminished. In this study, results consistent with the hypothesis have been reported as such, while cases in which the evidence is insufficient for a definitive judgment have likewise been presented without exaggeration or post hoc justification.

Accordingly, at this stage of the study, eleven of the thirteen indicators examined have provided statistically significant evidence consistent with the "New or Modern Monarchy" hypothesis. The remaining two indicators did not reach statistical significance, but neither provided statistically significant evidence contrary to the hypothesis. Thus far, no indicator has produced statistically significant evidence against the "New or Modern Monarchy" hypothesis.

Test	Indicator	Result
✓	GDP per Capita	Consistent with the hypothesis
✓	Purchasing Power per Capita	Consistent with the hypothesis
◐	Gini Coefficient	**Mixed evidence; difference favors the hypothesis**
✓	Political Stability and Absence of Violence	Consistent with the hypothesis

Test	Indicator	Result
✓	Regulatory Quality	Consistent with the hypothesis
✓	Rule of Law	Consistent with the hypothesis
✓	Government Effectiveness	Consistent with the hypothesis
✓	Inflation	Consistent with the hypothesis
✓	Control of Corruption	Consistent with the hypothesis
✓	Corruption Perceptions	Consistent with the hypothesis
✓	Property Rights	Consistent with the hypothesis
✓	Human Development	Consistent with the hypothesis
◐	Voice and Accountability	**Mixed evidence; principal test not significant, but supplementary analyses favor the hypothesis**

A Composite Indicator

Passport Power

So far, we have examined all five principal goals of government, as identified at the beginning of this book on the basis of the views of political philosophers, across contemporary monarchies and republics. The results showed that on **eleven indicators**, monarchies performed significantly better than republics, while on the **two remaining indicators**, although the differences did not reach statistical significance, no statistically significant evidence contrary to the **"New or Modern Monarchy" hypothesis** was found.

To complete this picture, the final comparison concerns **passport power**. Although passport power is not itself one of the purposes for which governments are established, it may be regarded as a composite indicator reflecting **international standing, the confidence other countries place in a country's citizens, and the overall quality of governance**. The less citizens of a country are perceived as risks in terms of illegal residence, asylum claims, cross-border crime, or security concerns—and the greater the stability, order, and international credibility of their country—the fewer entry restrictions other governments are generally inclined to impose on them.

From this perspective, **passport power can be viewed as a practical summary of how other countries perceive the overall condition of a country and its citizens**—a perception shaped by a combination of economic, political, legal, security, and diplomatic factors rather than by any single variable.

The data used in this section are from **2024**[1] (Appendix RDS-N01), and passport power is measured by the **number of countries and territories** that citizens of each country can enter without obtaining a visa in advance or with a visa on arrival.

Comparison of Passport Power in Monarchies and Republics Worldwide

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Monarchies Worldwide	**28**	**137.46**	**168.50**	**54.83**
Republics Worldwide	**151**	**102.15**	**80.00**	**53.58**

The mean passport power score is **137.46** for monarchies worldwide and **102.15** for republics. Thus, **citizens of monarchies have access, on average, to 35.32 more destinations without obtaining a visa in advance or with a visa on arrival**.

[1] The Henley Passport Index

The median confirms the same pattern. The median passport power score is **168.50** among monarchies and **80.00** among republics. In other words, the median monarchy provides access to **88.50 more destinations** without a prior visa or with a visa on arrival than the median republic.

Test of the Difference in Means

To assess the statistical significance of the difference in means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**35.32**
Welch's *t* statistic	**3.142**
Approximate degrees of freedom	**37.20**
Two-tailed *p*-value	**0.0033**
95% confidence interval	**12.55 to 58.09**
Hedges' *g*	**0.65**

Welch's test shows that **the difference in mean passport power between monarchies and republics worldwide is statistically significant** (**p = 0.0033**). Hedges' *g* (**g = 0.65**) further indicates that the difference is also **moderate to large in practical terms**.

Mann–Whitney Test

To ensure that the result does not depend on the assumptions of the parametric test, the nonparametric **Mann–Whitney test** was also conducted.

Statistical Measure	Value
U statistic	**2963.5**
Two-tailed *p*-value	**0.00075**
Rank-biserial correlation	**0.40**

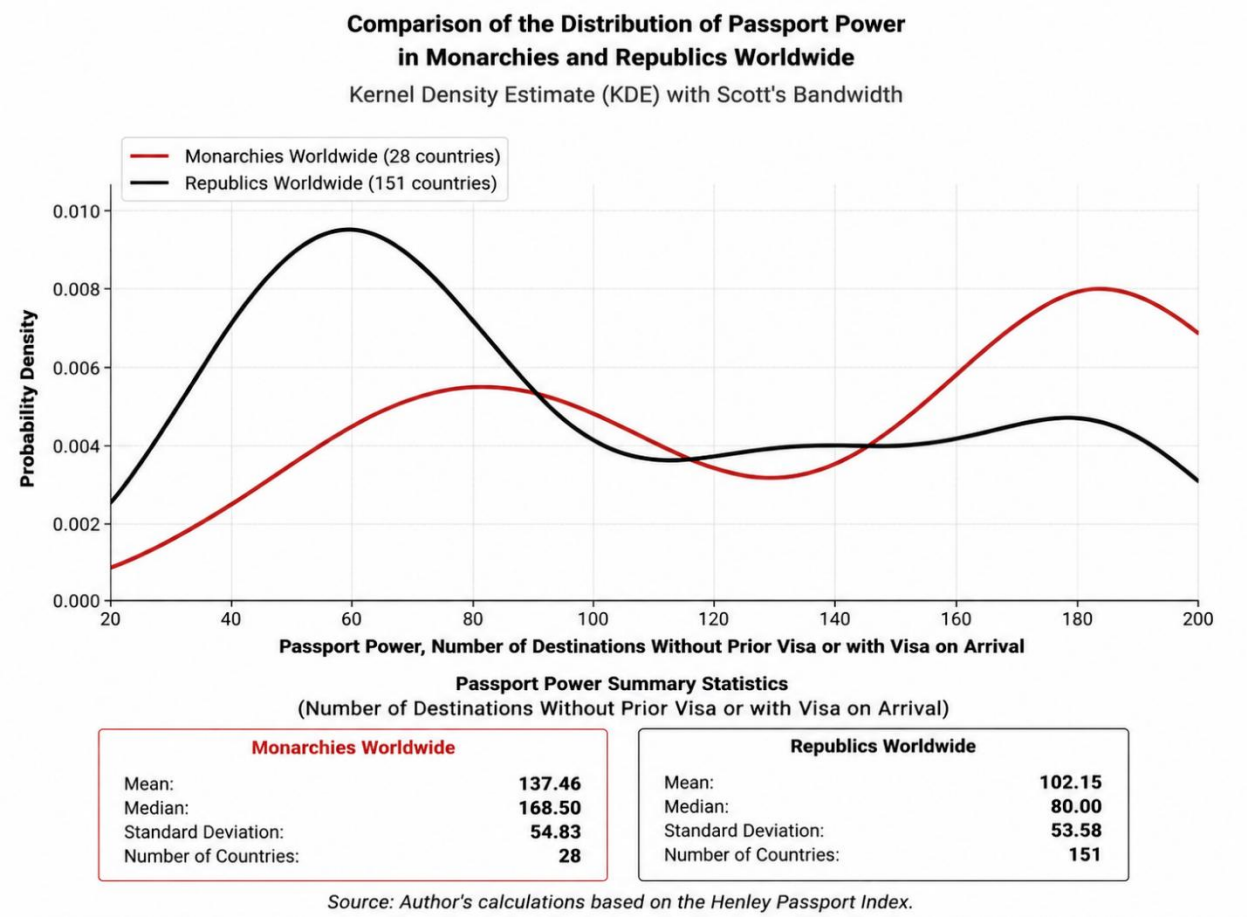
The Mann–Whitney test likewise indicates a statistically significant difference between the two groups (**p = 0.00075**). The rank-biserial correlation (**0.40**) indicates a **moderate effect** and confirms the result of Welch's test.

Conclusion

In **2024, monarchies worldwide performed better than republics worldwide in terms of passport power. Both the mean and the median favor monarchies, and both Welch's test and the Mann–Whitney test confirm that this advantage is statistically significant. In**

addition, the effect sizes indicate that the observed difference is not merely statistically significant but also meaningful in practical terms.

If passport power is understood, as proposed in the introduction to this section, as a composite indicator of international standing, the confidence other countries place in a country's citizens, and the overall quality of governance, this finding is consistent with the pattern observed across most of the preceding indicators and with the prediction of the "New or Modern Monarchy" hypothesis.



The kernel density estimate shows that the distribution of passport power among monarchies worldwide, based on 2024 data, is generally shifted toward higher values, with a substantial proportion of monarchies among the countries whose passports provide visa-free or visa-on-arrival access to a large number of destinations worldwide. In contrast, although republics span a very wide range of values, their distribution is more concentrated at the lower and middle levels of the index. This pattern is also consistent with the descriptive statistics: the mean and median passport power scores for monarchies are **137.46** and **168.50**, respectively, compared with **102.15** and **80.00** for republics. Overall, the available data indicate that monarchies, on average, have more powerful passports than republics.

Comparison of Passport Power between Constitutional and Non-Constitutional Monarchies Worldwide

Descriptive Statistics

Type of Monarchy	Number of Countries	Mean	Median	Standard Deviation
Constitutional Monarchies	**15**	**155.93**	**190.00**	**40.55**
Non-Constitutional Monarchies	**13**	**116.15**	**99.00**	**45.05**

The mean passport power is **155.93** in constitutional monarchies and **116.15** in non-constitutional monarchies. Thus, **citizens of constitutional monarchies have access, on average, to 39.78 more destinations without obtaining a visa in advance or with a visa on arrival.**

The median shows the same pattern. The median passport power is **190** among constitutional monarchies and **99** among non-constitutional monarchies. In other words, the median country among constitutional monarchies provides access to **91 more destinations** than the median country among non-constitutional monarchies.

Test of the Difference in Means

Welch's **t-test** was used to examine the difference between the group means.

Statistical Measure	Value
Mean Difference (Constitutional minus Non-Constitutional)	**39.78**
Welch's *t*	**2.049**
Approximate Degrees of Freedom	**25.97**
Two-tailed *p*-value	**0.0507**
95% Confidence Interval	**−0.14 to 79.70**
Effect Size (Hedges' *g*)	**0.74**

Welch's test shows that **the difference in mean passport power between the two groups is very close to the threshold of statistical significance** (**p ≈ 0.051**) and falls just short of the conventional 5% level. Nevertheless, the Hedges' *g* effect size (**g = 0.74**) indicates that the observed difference is **moderate to large** in practical terms.

Mann–Whitney Test

To compare the rankings of the two groups and reduce dependence on the assumption of normally distributed data, the non-parametric **Mann–Whitney test** was also conducted.

Statistical Measure	Value
U Statistic	**146.5**
Two-tailed *p*-value	**0.0252**
Rank-Biserial Correlation	**0.50**

The Mann–Whitney test indicates that the distributions of passport power in the two groups differ significantly (**p = 0.0252**). The rank-biserial correlation of approximately **0.50** also indicates a **moderate-to-large effect** in favor of constitutional monarchies.

Conclusion

In the within-monarchy comparison, **constitutional monarchies perform better in passport power than non-constitutional monarchies. The mean, median, and Mann–Whitney test all favor constitutional monarchies. Welch's test also indicates a substantial difference, although it falls just short of the conventional threshold for statistical significance.**

This finding suggests that, even among monarchies themselves, political structure may be associated with differences in international standing and the degree of trust extended to a country's citizens. As with the other within-group analyses, however, this finding should be regarded as **supplementary evidence** rather than a substitute for the primary comparison between monarchies and republics.

Comparison of Passport Power between Commonwealth Realms and Republics Worldwide

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Commonwealth Realms	**14**	**144.79**	**150.50**	**34.35**
Republics Worldwide	**151**	**102.15**	**80.00**	**53.58**

The mean passport power is **144.79** among Commonwealth realms and **102.15** among republics worldwide. Thus, **citizens of Commonwealth realms have access, on average, to approximately 42.64 more destinations without obtaining a visa in advance or with a visa on arrival.**

The median confirms the same pattern. The median passport power is **150.50** among Commonwealth realms and **80.00** among republics. In other words, the median Commonwealth realm provides access to approximately **70.50 more destinations** than the median republic.

The standard deviation is also lower among Commonwealth realms than among republics, indicating that passport power among Commonwealth realms is not only higher on average but also less dispersed.

Test of the Difference in Means

Welch's **t-test** was used to assess the statistical significance of the difference between the means.

Statistical Measure	Value
Mean Difference (Commonwealth minus Republic)	**42.64**
Welch's *t*	**4.195**
Approximate Degrees of Freedom	**19.44**
Two-tailed *p*-value	**0.00047**
95% Confidence Interval	**21.40 to 63.88**
Effect Size (Hedges' *g*)	**0.81**

Welch's test shows that **the difference in mean passport power between Commonwealth realms and republics worldwide is highly statistically significant** (**$p < 0.001$**). The 95% confidence interval lies entirely above zero, further confirming that the direction of the difference favors the Commonwealth realms.

Hedges' *g* is **0.81**, indicating a **large effect**. The observed difference is therefore not only statistically significant but also substantial in practical terms.

Mann–Whitney Test

To ensure that the result does not depend solely on a comparison of means or on the assumptions of the parametric test, the non-parametric **Mann–Whitney test** was also conducted.

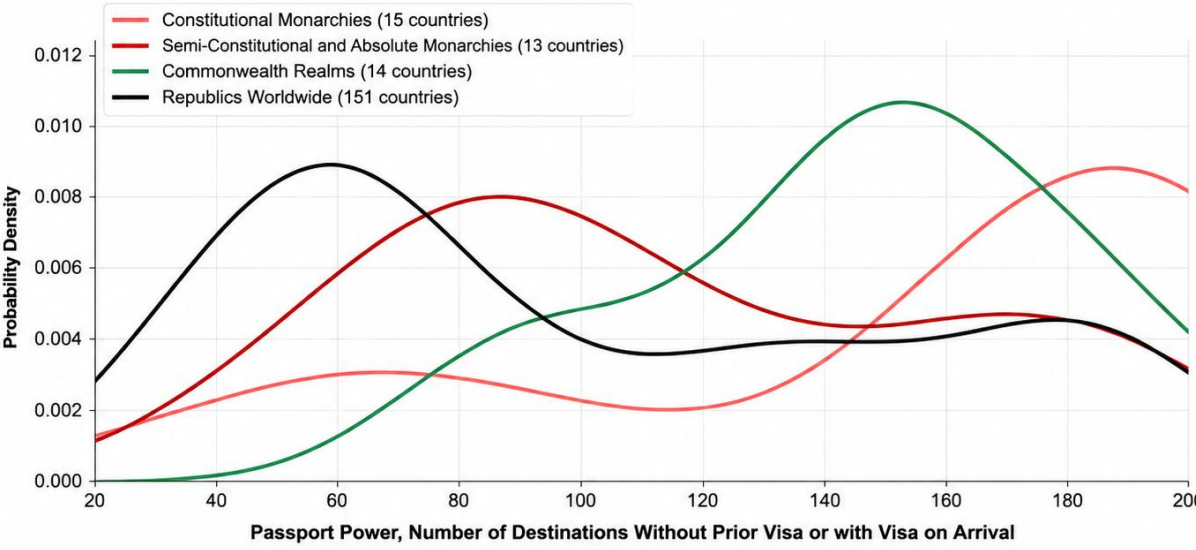
Statistical Measure	Value
U Statistic	**1564**
Two-tailed *p*-value	**0.0031**
Rank-Biserial Correlation	**0.48**

The Mann–Whitney test also shows a statistically significant difference between the two groups (**$p = 0.0031$**). The rank-biserial correlation of **0.48** indicates a **moderate-to-large effect** in favor of the Commonwealth realms.

Conclusion

In 2024, **Commonwealth realms had higher passport power than republics worldwide in terms of both the mean and the median. Both Welch's test and the Mann–Whitney test confirm this difference with strong statistical evidence, and the effect sizes are also substantial.**

Citizens of Commonwealth realms therefore enjoyed, on average, greater freedom of international travel than citizens of republics worldwide. **If passport power is regarded as a composite indicator of a country's international standing, the trust other governments place in its citizens, and the overall quality of governance, the result of this comparison is also consistent with the prediction of the "New or Modern Monarchy" hypothesis.**



Comparison of the Distribution of Passport Power Across Government Groups Worldwide

Kernel Density Estimate (KDE) with Scot's Bandwidth – Data for 2024

Passport Power Summary Statistics
(Number of Destinations Without Prior Visa or with Visa on Arrival)

Constitutional Monarchies (15 countries)		Semi-Constitutional and Absolute Monarchies (13 countries)		Commonwealth Realms (14 countries)		Republics Worldwide (151 countries)	
Mean:	155.93	Mean:	116.15	Mean:	144.79	Mean:	102.15
Median:	190.00	Median:	99.00	Median:	150.50	Median:	80.00
Standard Deviation:	40.55	Standard Deviation:	45.05	Standard Deviation:	34.35	Standard Deviation:	53.58
Number of Countries:	15	Number of Countries:	13	Number of Countries:	14	Number of Countries:	151

Source: Author's calculations based on the Henley Passport Index (2024).

The kernel density estimate shows substantial differences in the distribution of passport power across the four government groups worldwide. Constitutional monarchies exhibit the highest level of performance, with most of them among the countries whose citizens enjoy visa-free or visa-on-arrival access to the largest number of destinations worldwide. Commonwealth realms follow closely behind

constitutional monarchies, with a relatively concentrated distribution at the higher levels of the index. In contrast, although semi-constitutional and absolute monarchies have a higher mean than republics, their distribution is more dispersed, with a substantial proportion falling within the middle range of the index. Republics worldwide have the lowest mean among the four groups, and their distribution is more heavily concentrated at lower values. Overall, the 2024 data indicate that, as monarchical systems move toward the constitutional model, the passport power of their citizens tends, on average, to increase.

Comparison of Passport Power between Monarchies and Republics across Continents and in the Middle East

Comparison of Passport Power between Monarchies and Republics in Asia

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Asian Monarchies	**13**	**110.31**	**88.00**	**52.29**
Asian Republics	**33**	**68.97**	**58.00**	**43.94**

The mean passport power score is **110.31** for Asian monarchies and **68.97** for Asian republics. Thus, **citizens of Asian monarchies have access, on average, to approximately 41.34 more destinations without obtaining a visa in advance or with a visa on arrival.**

The median points in the same direction. The median passport power score is **88** for monarchies and **58** for republics, meaning that the median country among Asian monarchies provides access to **30 more destinations** than the median country among Asian republics.

The standard deviation is relatively high in both groups, although dispersion is slightly greater among monarchies than among republics. This indicates that both groups include countries with very strong as well as very weak passports.

Test of the Difference between Means

To assess the statistical significance of the difference between the two means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchy − Republic)	**41.34**
Welch's *t* statistic	**2.521**
Approximate degrees of freedom	**19.05**
Two-tailed *p*-value	**0.0208**
95% confidence interval	**7.03 to 75.65**
Hedges' *g*	**0.88**

Welch's test indicates that **the difference in mean passport power between Asian monarchies and republics is statistically significant at the 5% level** (**p = 0.0208**). The 95% confidence interval also lies entirely above zero, confirming that the direction of the difference favors monarchies.

Hedges' *g* is **0.88**, indicating a **large effect**. Thus, the observed difference is not only statistically significant but also substantial in practical terms.

Mann–Whitney Test

To ensure that the result does not depend solely on a comparison of means or on the assumptions of the parametric test, the nonparametric **Mann–Whitney test** was also conducted.

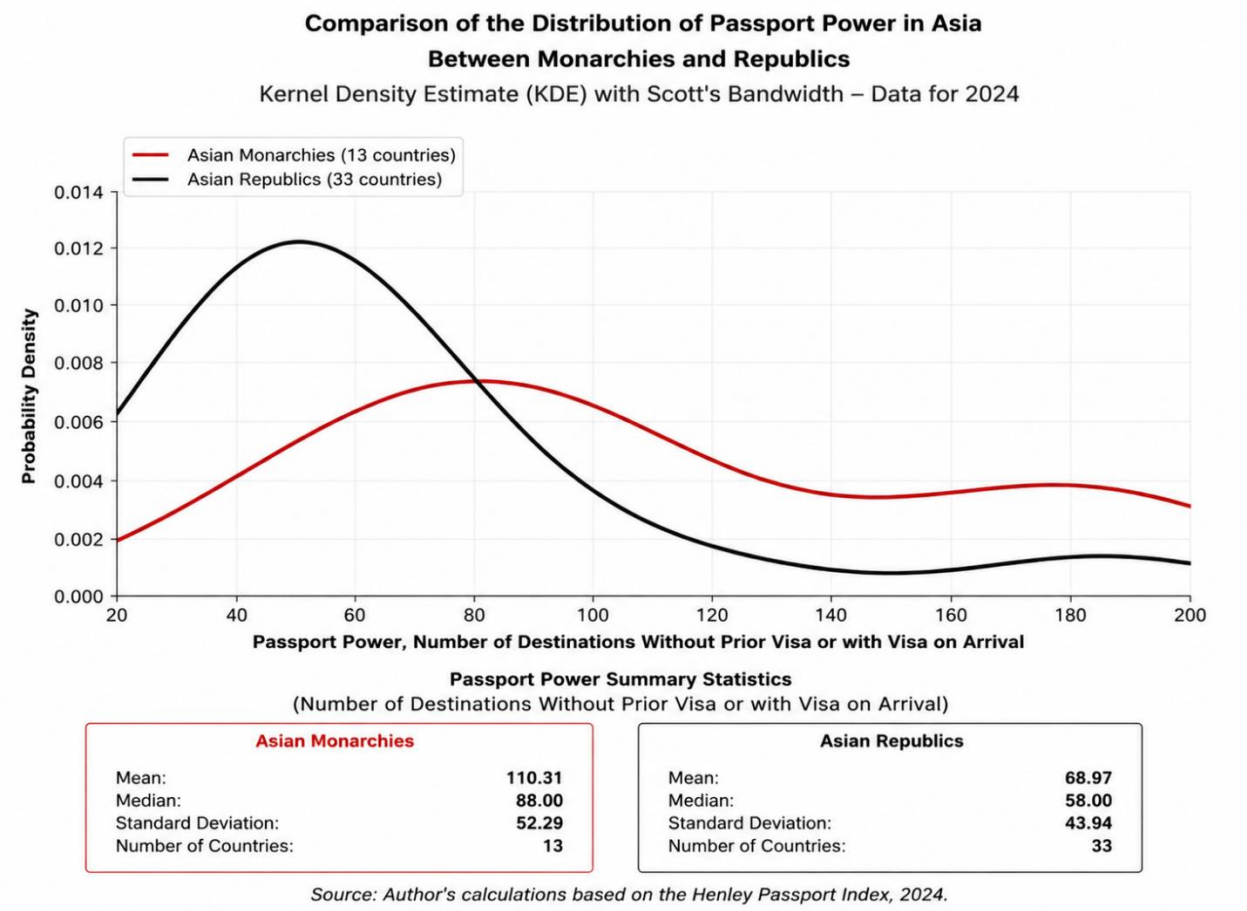
Statistical Measure	Value
U statistic	**335**
Two-tailed *p*-value	**0.0034**
Rank-biserial correlation	**0.56**

The Mann–Whitney test likewise indicates a statistically significant difference between the two groups (**p = 0.0034**). The rank-biserial correlation of **0.56** also represents a **large effect** in favor of Asian monarchies.

Conclusion

In 2024, **Asian monarchies had greater passport power than Asian republics in terms of both the mean and the median. Both Welch's t-test and the Mann–Whitney test confirm that the difference is statistically significant, and the effect sizes are large.**

Citizens of Asian monarchies therefore enjoyed, on average, greater freedom of international travel than citizens of Asian republics. **If passport power is regarded as a composite indicator of a country's international standing, the trust placed in its citizens by other governments, and the overall quality of governance, the result of this comparison is also consistent with the prediction of the "New or Modern Monarchy" hypothesis.**



The kernel density estimate shows that the distribution of passport power among Asian monarchies is generally concentrated at higher levels than that of the continent's republics. This advantage is also clearly reflected in the descriptive statistics: the mean and median passport power scores for Asian monarchies are **110.31** and **88**, respectively, compared with **68.97** and **58** for Asian republics. Both groups exhibit relatively high dispersion, but the greater concentration of monarchies at higher values indicates that their citizens, on average, enjoy greater freedom of international travel. Thus, the 2024 data indicate that Asian monarchies perform better than the continent's republics in terms of passport power.

Comparison of Passport Power Between Monarchies and Republics in Europe

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
European Monarchies	11	186.91	190.00	6.86
European Republics	34	166.06	184.00	33.41

The mean passport power score is **186.91** for European monarchies and **166.06** for European republics. Thus, **citizens of European monarchies have access, on average, to approximately 20.85 more destinations without obtaining a visa in advance or with a visa on arrival.**

The median points in the same direction. The median passport power score is **190** for monarchies and **184** for republics, meaning that the median country among European monarchies provides access to **6 more destinations** than the median country among European republics.

The standard deviation is only **6.86** among monarchies, compared with **33.41** among republics. European monarchies therefore not only have a higher average but also display much more consistent performance on this indicator. Among European republics, by contrast, there are countries with very powerful passports as well as countries whose passports provide substantially more limited access.

Test of the Difference Between Means

To assess the statistical significance of the difference between the two means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**20.85**
Welch's *t* statistic	**3.422**
Approximate degrees of freedom	**39.93**
Two-tailed *p*-value	**0.00145**
95% confidence interval	**8.54 to 33.16**
Effect size (Hedges' *g*)	**0.70**

Welch's test shows that **the difference in mean passport power between European monarchies and republics is statistically significant** (**p = 0.00145**). The 95% confidence interval also lies entirely above zero, confirming that the direction of the difference favors monarchies.

Hedges' *g* is **0.70**, indicating a **medium-to-large effect**. Thus, the observed difference is not only statistically significant but also substantial in practical terms.

Mann–Whitney Test

To ensure that the result does not depend solely on a comparison of means or on the assumptions of the parametric test, the nonparametric **Mann–Whitney test** was also conducted.

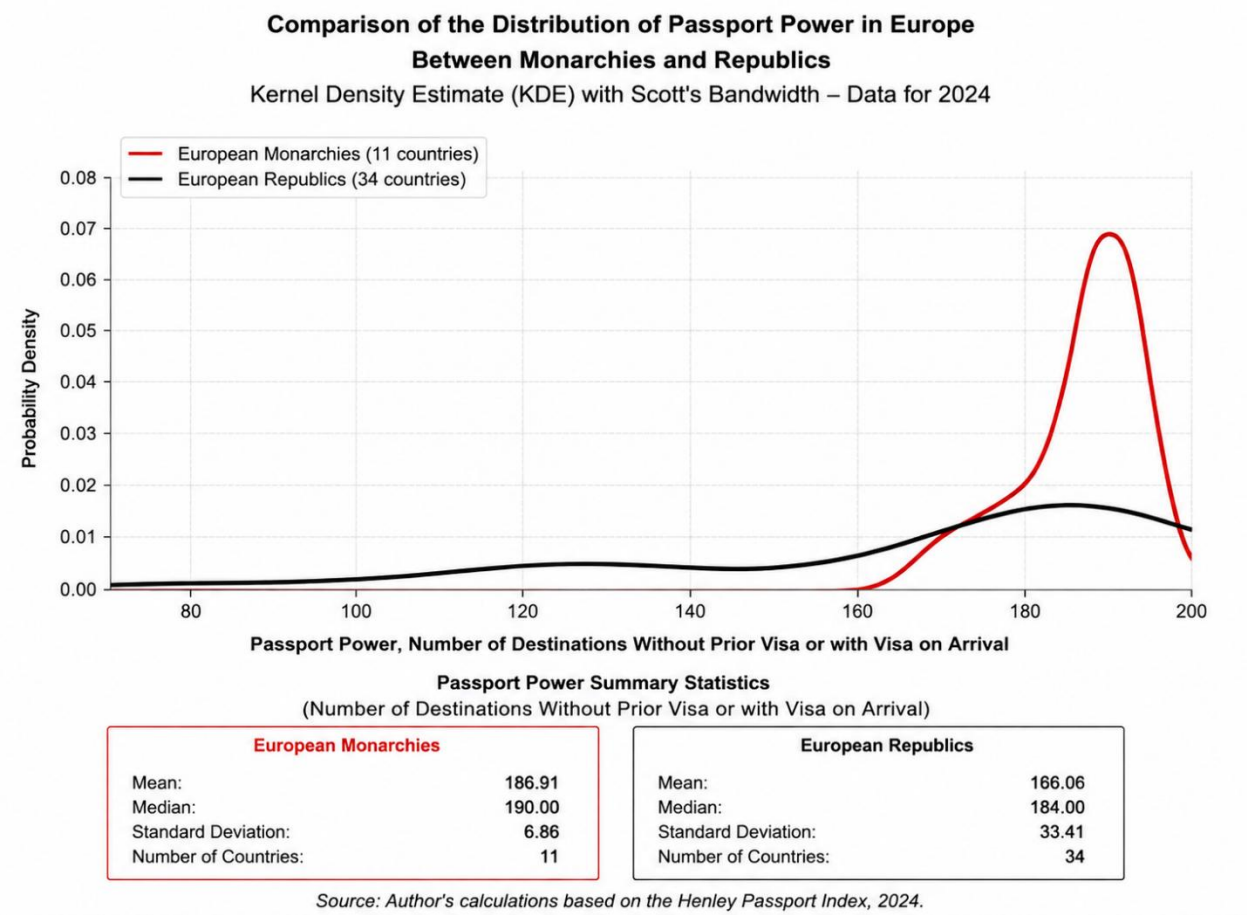
Statistical Measure	Value
U statistic	**268.5**
Two-tailed *p*-value	**0.0319**
Rank-biserial correlation	**0.44**

The Mann–Whitney test also indicates a statistically significant difference between the two groups at the 5% level (p = 0.0319). The rank-biserial correlation of **0.44** indicates a **medium effect** in favor of European monarchies.

Conclusion

In 2024, **European monarchies had higher passport power than European republics in terms of both the mean and the median. Welch's test and the Mann–Whitney test both confirm that this difference is statistically significant, while the effect sizes indicate that the observed advantage is also meaningful in practical terms.**

Moreover, the much lower dispersion among monarchies suggests that high passport power is a relatively widespread and consistent feature of this group rather than the result of a few exceptional countries. The European comparison is therefore also consistent with the predictions of the "New or Modern Monarchy" hypothesis.



The kernel density estimate shows that European monarchies are concentrated at a very high level of passport power, with little dispersion; nearly all of them rank among the countries with the world's most powerful passports. European republics, by contrast, also include many countries with very powerful passports, but their distribution is considerably wider, with a number of Eastern and Southeastern European republics recording lower scores and thereby reducing the group average.

This difference is also evident in the descriptive statistics: the mean and median passport power scores for European monarchies are **186.91** and **190**, respectively, compared with **166.06** and **184** for European republics. Thus, the 2024 data indicate that European monarchies, on average, have more powerful passports and a more homogeneous distribution than the continent's republics.

Comparison of Passport Power Between European Monarchies and European Republics That Have Never Been Communist

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
European Monarchies	**11**	**186.91**	**190.00**	**6.86**
Never-Communist European Republics	**13**	**187.46**	**190.00**	**6.12**

The mean passport power score is **186.91** for European monarchies and **187.46** for European republics that have never experienced communist rule. Thus, **the mean for this group of republics is only about 0.55 destinations higher than that of the monarchies.**

The median is exactly the same in both groups, at 190 destinations. The standard deviations are also very similar, indicating that passport power in both groups is very high and relatively consistent.

Test of the Difference Between Means

To assess the statistical significance of the difference between the two means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**−0.55**
Welch's *t* statistic	**−0.206**
Approximate degrees of freedom	**20.32**
Two-tailed *p*-value	**0.838**
95% confidence interval	**−6.13 to 5.02**
Hedges' *g*	**−0.08**

Welch's test shows that **the difference in mean passport power between the two groups is not statistically significant** (**p = 0.838**). The 95% confidence interval includes zero, and Hedges' *g* is also close to zero. Thus, there is no practically meaningful difference between the groups in terms of either their mean scores or effect size.

Mann–Whitney Test

To ensure that the conclusion does not depend solely on a comparison of means, the nonparametric **Mann–Whitney test** was also conducted.

Statistical Measure	Value
U statistic	**66.5**
Two-tailed *p*-value	**0.791**
Rank-biserial correlation	**−0.07**

The Mann–Whitney test likewise finds no statistically significant difference between the two groups (p = 0.791). The rank-based effect size is also very small and close to zero, confirming the result of Welch's test.

Conclusion

Once the comparison is restricted to European republics that have never experienced communist rule, the difference in passport power between monarchies and republics effectively disappears. The two groups have almost identical means and exactly the same median, and neither Welch's test nor the Mann–Whitney test finds a statistically significant difference.

This result suggests that the advantage of monarchies in the overall European comparison is associated, at least in part, with the presence of republics with lower passport power, many of which have a history of communist rule. However, as explained earlier in the methodological note, excluding republics with a communist past is only **a supplementary analysis intended to clarify the possible sources of the observed difference** and should not serve as the basis for the final judgment regarding the relative performance of monarchy and republic.

Comparison of Passport Power in Africa, the Americas, and Oceania

Because the number of monarchies and Commonwealth realms is limited in some of these regions, this section is restricted to descriptive statistics and no inferential statistical tests are conducted.

Africa

Type of Government	Number of Countries	Mean	Median	Standard Deviation
African Monarchies	**3**	**76.00**	**77.00**	**3.61**
African Republics	**49**	**63.51**	**59.00**	**22.62**

In Africa, both the mean and the median passport power scores are higher among monarchies than among the continent's republics. The dispersion of scores among the three African monarchies is also much lower than among republics.

The Americas

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Commonwealth Realms in the Americas	**9**	**144.56**	**153.00**	**30.35**
Republics in the Americas	**26**	**130.38**	**138.50**	**37.38**

In the Americas, both the mean and median passport power scores are higher among Commonwealth realms than among the continent's republics. Passport power is also somewhat less dispersed among the Commonwealth realms.

Oceania

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Oceanian Monarchy	**1**	**131.00**	**131.00**	—
Commonwealth Realms in Oceania	**5**	**145.20**	**134.00**	**44.64**
Oceanian Republics	**9**	**111.11**	**124.00**	**18.41**

In Oceania, the continent's only monarchy, Tonga, has a passport power score above the average of the region's republics. Both the mean and median passport power scores of the Commonwealth realms in Oceania are also higher than those of the republics, although dispersion is greater among the Commonwealth realms.

Comparison of Passport Power Between Monarchies and Republics in the Middle East

Descriptive Statistics

Type of Government	Number of Countries	Mean	Median	Standard Deviation
Middle Eastern Monarchies	**7**	**101.00**	**88.00**	**40.42**
Middle Eastern Republics	**10**	**73.60**	**44.00**	**58.58**

The mean passport power score is **101** among Middle Eastern monarchies and **73.60** among the region's republics. Thus, **citizens of Middle Eastern monarchies have access, on average, to 27.40 more destinations without obtaining a visa in advance or with a visa on arrival.**

The median reveals an even larger difference. The median passport power score is **88** among monarchies and **44** among republics, meaning that the median Middle Eastern monarchy provides access to **44 more destinations** than the median republic in the region.

The standard deviation is higher among republics (**58.58**) than among monarchies (**40.42**). This indicates that passport power is more widely dispersed among Middle Eastern republics and that the presence of a few republics with very powerful passports substantially raises the group mean.

Test of the Difference Between Means

To assess the statistical significance of the difference between the means, **Welch's t-test** was used.

Statistical Measure	Value
Mean difference (Monarchies − Republics)	**27.40**
Welch's *t* statistic	**1.141**
Approximate degrees of freedom	**15.00**
Two-tailed *p*-value	**0.272**
95% confidence interval	**−23.78 to 78.58**
Hedges' *g*	**0.50**

Welch's test shows that **the difference in mean passport power between Middle Eastern monarchies and republics is not statistically significant** (**p = 0.272**). The 95% confidence interval includes zero, so this test does not allow the hypothesis of equal means to be rejected.

Nevertheless, Hedges' *g* is **0.50**, indicating a **moderate effect** in favor of monarchies. This suggests that the observed difference is not trivial in practical terms, although the small sample sizes and substantial dispersion prevent it from reaching statistical significance.

Mann–Whitney Test

To ensure that the conclusion does not depend solely on a comparison of means or on the assumptions of the parametric test, the nonparametric **Mann–Whitney test** was also conducted.

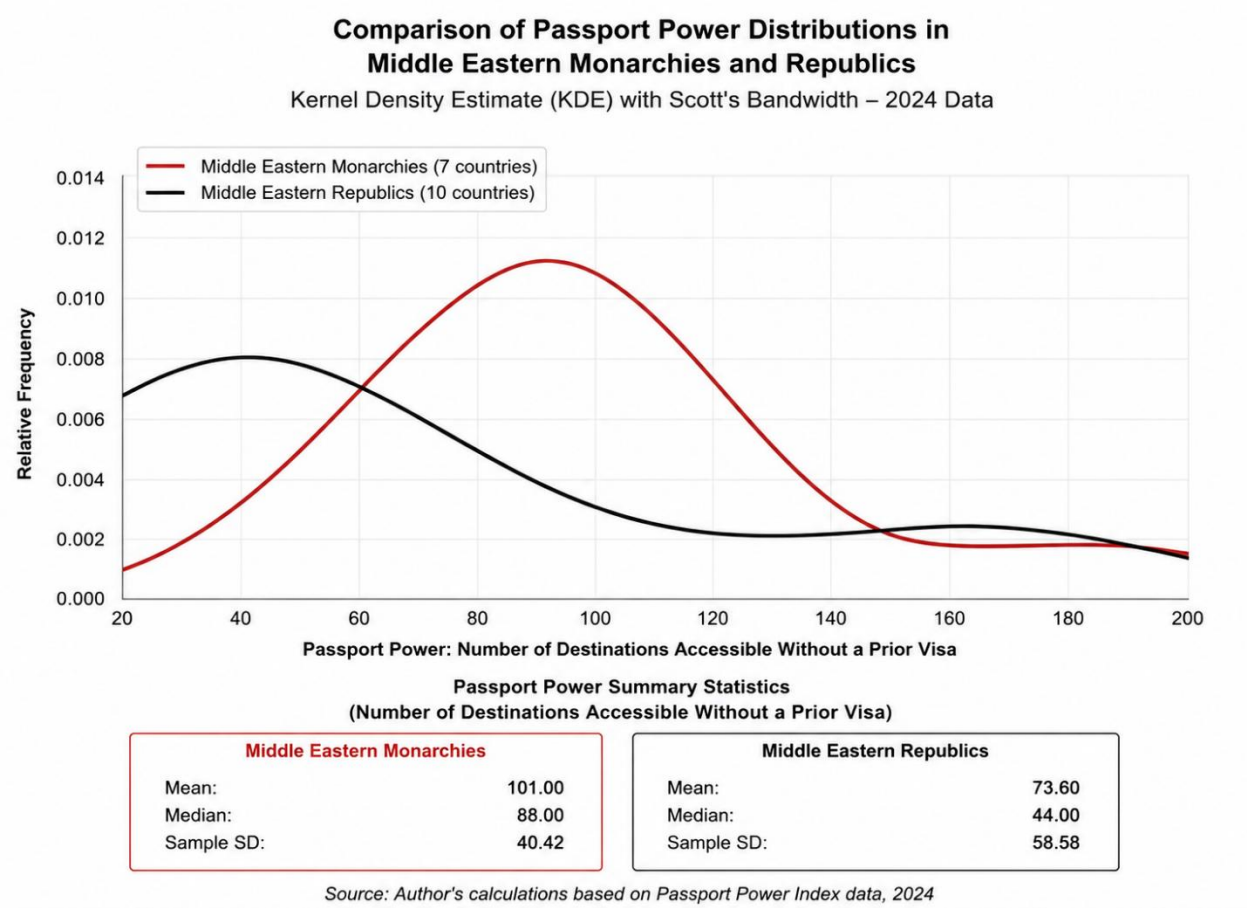
Statistical Measure	Value
U statistic	**52**
Two-tailed *p*-value	**0.109**
Rank-biserial correlation	**0.49**

The Mann–Whitney test likewise does not find a statistically significant difference between the two groups at the 5% level (**p = 0.109**). However, the rank-biserial correlation of **0.49** indicates a **moderate-to-large effect** in favor of monarchies.

Conclusion

In 2024, **Middle Eastern monarchies had higher passport power than the region's republics in terms of both the mean and the median. The difference between the group means is 27.40 destinations, while the difference between the medians is 44 destinations, and both effect-size measures indicate a substantial advantage for monarchies.**

However, neither Welch's test nor the Mann–Whitney test finds the difference statistically significant at the 5% level. Thus, the direction of the result is consistent with the prediction of the "New or Modern Monarchy" hypothesis, but the statistical evidence is not sufficient to establish a systematic difference between the two groups with confidence.



The kernel density estimate shows that the distribution of passport power among Middle Eastern monarchies is generally shifted toward higher values than that of the region's republics. This advantage is also evident in the descriptive statistics: the mean passport power score for Middle Eastern monarchies is **101**, with a median of **88**, compared with **73.60** and **44**, respectively, for Middle Eastern republics. However, the republics show considerably greater dispersion, as the group includes countries with relatively weak passports alongside countries such as Cyprus and

Israel with very strong passports. Middle Eastern monarchies, by contrast, are concentrated mainly in the middle and upper ranges of the index and display a more homogeneous distribution. Overall, the 2024 data indicate that Middle Eastern monarchies have, on average, greater passport power than the region's republics.

Status of the "New or Modern Monarchy" Hypothesis After the Fourteenth Indicator

With the addition of passport power, the "New or Modern Monarchy" hypothesis has now been tested against fourteen independent indicators. Of these, **twelve indicators have produced results clearly consistent with the hypothesis**.

In the main passport power comparison—that is, the comparison of all monarchies worldwide with all republics—the mean for monarchies was significantly higher. Both Welch's t-test and the Mann–Whitney test showed a statistically significant difference between the two groups, with an effect size ranging from moderate to large. In addition, **the supplementary comparisons in Asia and Europe also favored monarchies and were statistically significant. Commonwealth realms likewise had greater passport power than republics worldwide, with a statistically significant difference and a large effect size.**

In the comparison between the two types of monarchy, constitutional monarchies had greater passport power than semi-constitutional and absolute monarchies. The Mann–Whitney test found this difference statistically significant, while the Welch test, with a value very close to the five-percent threshold, fell just short of statistical significance. As supplementary evidence, this result points to a substantial difference between the two groups, but it does not form the basis of the main assessment of the hypothesis.

In the Middle East as well, both the mean and the median passport power scores favored monarchies, and the effect sizes indicated a moderate and meaningful difference. **However, because of the small number of countries and the substantial dispersion in the data, neither statistical test reached the conventional threshold for significance.** This regional result does not alter the overall direction of the findings, since the principal conclusion is based on the global comparison between the two forms of government.

The results for two other indicators were different. **For the Gini coefficient, the difference favored monarchies but was not statistically significant. For Voice and Accountability, the main global test likewise found no statistically significant difference, although constitutional monarchies and Commonwealth realms performed significantly better in the supplementary analyses.**

Thus, among the fourteen indicators examined:

- **Twelve indicators provide clear evidence consistent with the hypothesis;**

- **Two indicators either point in the direction predicted by the hypothesis or provide supplementary evidence in its favor, but their main tests did not reach statistical significance;**
- **No indicator has produced statistically significant evidence in the direction contrary to the "New or Modern Monarchy" hypothesis.**

Unlike the preceding indicators, passport power is not directly one of the five principal goals of government. Rather, it is a composite, externally manifested indicator of international standing, the confidence other governments place in a country's citizens, and the overall quality of governance. Its consistency with the predominant pattern found in this study may therefore be regarded as a form of external and independent confirmation of the pattern observed across indicators relating to welfare, security, order, justice, and liberty. As with the other findings, however, this consistency **does not by itself establish a causal relationship** between the form of government and national performance.

Test	Indicator	Result
✓	GDP per Capita	Consistent with the hypothesis
✓	Purchasing Power per Capita	Consistent with the hypothesis
◐	Gini Coefficient	**Mixed evidence; difference favors the hypothesis**
✓	Political Stability and Absence of Violence	Consistent with the hypothesis
✓	Regulatory Quality	Consistent with the hypothesis
✓	Rule of Law	Consistent with the hypothesis
✓	Government Effectiveness	Consistent with the hypothesis
✓	Inflation	Consistent with the hypothesis
✓	Control of Corruption	Consistent with the hypothesis
✓	Corruption Perceptions	Consistent with the hypothesis
✓	Property Rights	Consistent with the hypothesis
✓	Human Development	Consistent with the hypothesis
◐	Voice and Accountability	**Mixed evidence; main test not significant, but supplementary analyses favor the hypothesis**
✓	Passport Power	Consistent with the hypothesis

Final Comparison Table of Indicators

SYNTHESIS TABLE · FINAL COMPARISON

MONARCHIES AND REPUBLICS · COMPARATIVE RESULTS

Monarchy and Republic in the Laboratory of History · Third Edition · August 2026

LEGEND

Monarchy Republic

* Statistically significant mean difference

** 2023 or nearest available year

				More Successful Form of Government Based on the Mean of the Indicator Studied				
Government Goal	Indicator	Source	Indicator Year	Asia	Europe	European Republics without a Communist Past	Middle East	Global
Welfare	GDP per Capita (Current US$)	World Bank	2023**	Monarchy*	Monarchy*	Monarchy	Monarchy*	Monarchy*
	GDP per Capita, PPP (Current International $)	World Bank	2023**	Monarchy*	Monarchy	Monarchy	Monarchy*	Monarchy*
	Gini Index	World Bank	2023**	Monarchy	Monarchy	Monarchy	Monarchy	Monarchy
Security	Political Stability and Absence of Violence/Terrorism	World Bank	2023	Monarchy*	Monarchy*	Monarchy	Monarchy*	Monarchy*
Order	Regulatory Quality	World Bank	2023	Monarchy*	Monarchy*	Monarchy*	Monarchy*	Monarchy*
	Rule of Law	World Bank	2023	Monarchy*	Monarchy*	Monarchy	Monarchy*	Monarchy*
	Government Effectiveness	World Bank	2023	Monarchy*	Monarchy*	Monarchy*	Monarchy*	Monarchy*
	Inflation, 2019–2023	World Bank	2019–2023	Monarchy*	Monarchy*	Republic	Monarchy*	Monarchy*
Justice	Control of Corruption	World Bank	2023	Monarchy*	Monarchy*	Monarchy	Monarchy	Monarchy*
	Corruption Perceptions Index	Transparency International	2023	Monarchy*	Monarchy*	Monarchy	Monarchy*	Monarchy*
Liberty	International Property Rights Index	Property Rights Alliance	2024	Monarchy*	Monarchy*	Monarchy*	Monarchy*	Monarchy*
	Human Development Index	United Nations	2022	Monarchy*	Monarchy*	Monarchy	Monarchy*	Monarchy*
	Voice and Accountability	World Bank	2023	Monarchy	Monarchy*	Republic	Republic	Monarchy
Composite Index	Passport Power Index	Henley & Partners	2024	Monarchy*	Monarchy*	Republic	Monarchy	Monarchy*

** The statistical test indicates that the difference in the mean of the relevant indicator between the two forms of government is statistically significant.*

*** Data are from 2023 or the nearest available year to 2023.*

N. Fakhr

Monarchy and Republic in the Laboratory of History

Final Comparative Synthesis Table

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Republican Bias: Has Political Modernity Turned a Blind Eye to the Data?

In the modern era, republicanism has come to be regarded as a symbol of political modernity, liberty, and popular sovereignty. This view, rooted in major revolutions such as the French and American Revolutions, tends to regard the republic as inherently superior to monarchy. But do global data support this judgment? Is republicanism an empirically successful form of government, or merely an attractive political ideal? This chapter examines whether this prevailing perception is consistent with empirical evidence on government performance.

The Dominant Republican Discourse

In much of contemporary political discourse, republics are commonly associated with concepts such as democracy, liberty, equality, and political participation, while monarchy is often associated with traditional or authoritarian systems. This perception has meant that the success of contemporary monarchies—particularly constitutional monarchies—has received less attention or has been treated as an exception to a general rule.

What Do the Data Say?

In this study, the performance of contemporary monarchies and republics was compared using **fourteen internationally recognized indicators**, covering the five principal goals of government identified in this book: welfare, security, order, justice, and liberty.

The results showed that monarchies had **higher mean scores than republics across all fourteen indicators**. In **twelve indicators**, this advantage was also statistically significant. In the **two remaining indicators**—the Gini coefficient and Voice and Accountability—the main tests did not reach statistical significance, but neither produced statistically significant evidence contrary to the hypothesis.

This pattern was not confined to the global comparison. Similar results were observed in many regional analyses, including those for Europe and Asia. At the same time, some supplementary analyses—such as excluding European republics with a history of communist rule—showed that part of the observed differences may also be associated with historical factors.

Some Hypotheses About the Origins of This Bias

If the empirical findings reveal such a pattern, the question arises as to why republicanism continues to occupy a privileged position in contemporary political discourse. Providing a

definitive answer to this question lies beyond the scope of this study, but several possible hypotheses can at least be considered:

- The mental association of republicanism with concepts such as liberty, democracy, and opposition to hereditary rule.
- The legacy of republican revolutions, which have acquired a special place in the historical memory of many societies.
- The tendency to equate being a "republic" with being "democratic," even though many contemporary republics are, in practice, authoritarian or one-party states.
- The greater emphasis placed by contemporary political literature on the experience of republics, with comparatively less attention given to the performance of constitutional monarchies.

These hypotheses require independent investigation and are presented here only as possible explanations for the observed pattern.

Republicanism: Judged by Its Ideals or by Its Performance?

Republicanism is commonly presented in terms of ideals such as liberty, equality, and popular sovereignty. **But if the purpose is to compare systems of government, the central question is not what ideals each system proclaims, but how successfully it achieves in practice the goals expected of government.**

The data examined in this study show that many contemporary monarchies—particularly constitutional monarchies—have performed very well on indicators related to welfare, security, order, justice, and liberty. Being a republic or a monarchy, therefore, cannot by itself serve as an adequate criterion for judging the quality of governance.

Conclusion

The findings of this study show that, overall, contemporary monarchies have performed significantly better than republics on most of the indicators examined. This finding does not, by itself, establish the inherent superiority of either form of government. It does, however, suggest that the widespread assumption that republics are self-evidently superior to monarchies requires reconsideration, at least in light of the empirical evidence examined in this study.

If we are to judge between two forms of government, that judgment should be based less on historical assumptions or the appeal of political ideals and more on empirical evidence and their actual performance.

Are These Data Sufficient for Decision-Making?

It may be argued that this study, regardless of the breadth of its data and the statistical significance of many of its findings, still does not establish a causal relationship between the form of government and national performance. From a methodological perspective, this objection is largely valid. In empirical science—and especially in the social sciences—definitively establishing causation is **often impossible**, because no researcher can simultaneously turn the same country into both a republic and a monarchy while holding all other conditions perfectly constant.

But the more important question is this: **if absolute proof of causation is impossible, should empirical evidence therefore be ignored?**

All empirical sciences, from medicine to economics, operate according to this same logic. No drug is approved simply because its causal effect has been established with 100 percent certainty. A drug is accepted when a body of empirical evidence shows, with sufficiently high confidence, that patients are more likely to benefit from taking it than from not taking it.

If multiple studies show that patients receiving Drug A have **statistically significantly** better outcomes than patients receiving Drug B, no scientific institution concludes that, because the possibility of unknown factors cannot be reduced to zero, no decision should be made. On the contrary, those empirical findings become the basis for regulatory approval, clinical guidelines, and medical recommendations—while remaining open to revision if new evidence emerges. Just as modern medicine makes decisions on the basis of the best available evidence rather than absolute certainty—**Evidence-Based Medicine—a scientific choice among models of governance should likewise be based on the best available empirical evidence, not on ideals or ideological assumptions.**

If the choice of a form of government is to be approached scientifically, it should follow the same logic. If, over many decades, hundreds of countries, billions of people, and successive generations have unintentionally participated in a vast historical experiment, producing measurable outcomes, it would be irrational to disregard that evidence simply because causation cannot be demonstrated with absolute certainty.

In science, **the absence of absolute certainty does not mean the absence of a rational basis for decision-making**. Almost no important decision in medicine, engineering, economics, or public policy is made under conditions of complete certainty. **What decisions are based on is the best available evidence.**

It is equally important, however, not to draw conclusions that go beyond what the findings actually show. This study does not prove that every monarchy will succeed merely because it is a monarchy, or that every republic will fail merely because it is a republic. Nor does it prove that the form of government is the only factor determining a country's success.

But the study does provide one important conclusion: **if someone today wished to choose between two broad models of government solely on the basis of the available empirical evidence, without relying on ideological assumptions, the findings of this study would give that person reason at least to question the presumed self-evident superiority of the republic and to regard monarchy as an alternative for which the existing evidence provides stronger support in terms of observed performance.**

If someone still considers this evidence insufficient for decision-making, then that person must explain **what kind of evidence could realistically be broader, more costly, or more long-term than this for comparing two forms of government. If evidence on this scale is still considered insufficient, the critic must explain what superior form of empirical evidence could in principle be produced. What exactly are we waiting for before making a scientific judgment?**

Science does not place human beings between "certainty" and "ignorance." It requires us, at every moment, to make decisions on the basis of the best evidence available. If humanity had refused to act until absolute certainty was achieved, no medicine would ever have been approved, no bridge would have been built, no aircraft would have flown, and no public policy would ever have been implemented. What has advanced modern civilization has not been absolute certainty, but confidence in the best available empirical evidence, combined with a willingness to revise decisions in light of better evidence. "The absence of absolute certainty is not a license to ignore the best available evidence." If we set aside the best available evidence, it will not be replaced by better evidence; it will be replaced by assumptions, preferences, and ideologies. Science is not a substitute for ideology; it is a substitute for decision-making without evidence. Rejecting the best available evidence does not make a person neutral; it merely means that the decision will be based on something else—tradition, emotion, ideology, or habit.

No new experiment was conducted on human beings for this book. The experiment had already taken place, across decades and around the world, before this research began. Hundreds of countries, billions of people, and successive generations have, willingly or unwillingly, lived under two different broad models of government, and the consequences have been recorded in international indicators. This study has merely attempted to translate that enormous historical experience into the language of statistics. The researcher did not pay the cost of generating these data; history did. History has been the largest laboratory of political science, and we have simply read its results. **"History is the laboratory; statistics are the record book of that laboratory's results."**

Today, if someone asks me:

"Have you proved that monarchy causes success?"

My book answers:

“No.”

If they ask:

“Have you proved that republics fail?”

Again, the answer is:

“No.”

But if they ask:

“If we were to judge today between two broad models of governance solely on the basis of the best empirical evidence currently available, in which direction does that evidence point?”

Then my book can answer with confidence:

“The data in this study answer that question, not my assumptions.”

I am not asking you to trust me. I am asking you to do what science asks of us: examine the best available evidence and then judge for yourself.

So far, the performance of contemporary monarchies and republics has been examined using fourteen internationally recognized indicators. Across all fourteen, monarchies had higher mean values than republics, and in twelve cases the difference was statistically significant. In the remaining two indicators, although the main tests did not reach statistical significance, no statistically significant evidence was found in the direction contrary to the research hypothesis.

Even so, such findings should not be expected to transform prevailing political beliefs overnight. The history of science shows that people rarely abandon intellectual frameworks they have lived with for years simply and immediately. Max Planck, the distinguished German physicist, described this phenomenon in the following way:

“A new scientific truth does not triumph by convincing its opponents; rather, its opponents eventually disappear, and a new generation grows up familiar with it.”

If the findings of this study appear inconsistent with some widely held beliefs, such a reaction would hardly be surprising. The history of science repeatedly shows that paradigm shifts usually occur gradually rather than suddenly.

The book began with Aristarchus, who showed that evidence can prevail over intuition. It ends with Planck, who reminds us that even after the evidence has spoken, changing beliefs takes time.

But the central question remains: if a country today wished to make decisions about its model of governance solely on the basis of the **best available empirical evidence**, would it be rational to ignore that evidence merely because it does not provide absolute certainty?

Science has never required human beings to refrain from making decisions until certainty is complete. On the contrary, in medicine, engineering, economics, and other empirical disciplines, decisions are routinely made on the basis of the best available evidence, with the understanding that they can be revised if better evidence later emerges.

The purpose of this book, therefore, is not to defend a political ideology, but to defend a method of judgment: **if we are to choose between competing theories of government, that judgment should, as far as possible, be based on their measurable consequences rather than merely on the appeal of the ideals they promise to realize.**

History does not cast its vote through slogans; it casts its vote through consequences. This book is an attempt to read those consequences.

Throughout this study, I have tried not to act as an advocate for my own hypothesis, but simply as a translator of the evidence. I have tried not to fill gaps in the evidence with confident rhetoric, because I believe that transparent reasoning is stronger than emphatic assertion.

Aristarchus taught us that truth does not always emerge from prevailing beliefs; sometimes the evidence must be rearranged and examined anew. This book is likewise an attempt, when thinking about government, to ask history and its data before asking ideology.

Carl Sagan famously said: **“Extraordinary claims require extraordinary evidence.”** At this point in the book, I would like to borrow from him and offer a sentence that is both similar and different: **“Extraordinary claims do not require extraordinary rhetoric; they require extraordinarily transparent evidence.”**

The final judgment belongs to the reader. But if that judgment is to be scientific, it must be based on the best empirical evidence available, not on assumptions that disregard that evidence. So go as far as the data allow—and no farther. But when the data truly have spoken, do not be afraid to state what they say.

Does History Offer a Successful Model for a Transition from Republic to Monarchy?

If we take the findings of this study seriously, the next natural question is whether modern history provides any examples of a successful transition from a republic back to a monarchy.

Before answering this question, an important point should be considered. If the data examined in this study indicate that monarchies have, overall, performed better, one might ask: **why, then, are most countries in the world republics today?**

The answer is that political decisions are not always made on the basis of empirical evidence. Revolutions, wars, colonialism, coups, ideologies, international pressures, political traditions, and many other factors have played a role in creating or changing systems of government. Therefore, **the prevalence of a particular form of government is not, by itself, evidence of its empirical success—just as the popularity of a scientific theory or a medical treatment does not necessarily demonstrate its correctness or effectiveness.**

Modern history nevertheless provides several examples of movement in the opposite direction: a return from republican government to monarchy.

In **1649**, following the execution of Charles I, the monarchy was abolished in England, and Oliver Cromwell eventually became head of government. Although his official title was **Lord Protector**, what matters here is not the title of an office but the mechanism through which political authority is appointed, exercised, and removed. The new system was intended to operate through Parliament and according to republican principles. Cromwell, however, gradually purged Parliament, dissolved it on several occasions, and imposed extensive religious and political restrictions. After his death, an attempt was also made to transfer power to his son, but the regime did not endure, and in **1660 the monarchy was restored in England**.

In Cambodia, the Khmer Rouge period from **1975 to 1979**, following the earlier abolition of the monarchy and establishment of a republic, produced one of the most catastrophic regimes of the twentieth century, resulting in the deaths of roughly a quarter of the country's population. After that period ended, the monarchy was eventually restored in **1993**. Although Cambodia today still remains far behind developed countries, its condition across many indicators of governance and development is vastly better than during the Khmer Rouge era.

Spain provides another example. After the establishment of the **Second Republic in 1931**, the country experienced years of political instability, civil war, and subsequently the dictatorship of Francisco Franco. Following Franco's death, the monarchy was restored in **1975**, and with the adoption of a constitutional system in **1978**, Spain began its transition into one of Europe's successful democracies.

These examples are **by no means sufficient to establish the inherent superiority of monarchy**, nor are they presented as an exhaustive list of historical transitions from republic to monarchy. They simply demonstrate that, contrary to a common assumption, a return from republican government to monarchy is not impossible. Such transitions have occurred in modern history and, in some countries, have been followed by significant political and institutional developments.

From Self-Interest to the Long-Term Horizon of Government: A Hypothesis for Explaining the Findings

So far, our work has dealt primarily with data. We examined fourteen indicators from different angles, and in all fourteen cases, the overall direction of the evidence favored monarchies. But even if we fully accept this pattern, a more important question remains unanswered: **why?**

Statistics can show us that two groups differ; they can also measure the size of that difference, its dispersion, and the likelihood that it arose by chance. But observing a difference does not allow us to move directly to its cause. What follows, therefore, is not a conclusion established by the statistical tests in this book, but rather an **explanatory hypothesis** arising from the findings—one that could become the subject of independent future research.

Seeing Human Beings as They Are

From one perspective, the history of political and economic thought can be seen as the history of attempts to answer a simple question: **What should we do with human beings' individual motivations?**

Human beings are not purely altruistic or idealistic creatures. Self-interest, the desire to own property, the effort to improve one's own position and that of one's family, competition, and the desire to preserve and increase wealth are also parts of human behavior. Of course, the strength of these motivations varies across individuals and cultures, and human beings cannot be reduced to purely self-interested creatures. But ignoring such motivations when designing social institutions can have important consequences.

One of the problems raised in critiques of communist and centrally planned economic systems concerns precisely this issue of incentives. These systems did not face only the problems of economic calculation or the concentration of power; restricting private property and weakening the connection between individual effort, risk-taking, innovation, and economic reward could also alter people's incentives.

The market economy took a different path. Rather than expecting people to abandon their self-interest, it sought, within a framework of property, contracts, competition, and law, to channel that motivation into a productive force. In a competitive market, someone who wants to become wealthy generally has to produce something that someone else is willing to pay for. To earn greater profits, a producer tries to offer a better, cheaper, or different product; competitors do the same in their efforts to win customers, and consumers can benefit from this competition.

This mechanism is certainly not perfect. Without law, competition, and limits on economic power, it can also lead to monopoly, exploitation, and rent-seeking. The central point, however,

is different: **an institutional system can, instead of denying a human motivation, channel it in a direction in which, under appropriate conditions, private interest and the public interest become at least partly aligned.**

This brings us to a question about political systems.

Does the "Time Horizon" of Rulers Matter?

By the nature of the office, a president occupies the highest position in government for a limited period. The president knows that the office does not belong to him or her and that, at a specified point, it must be handed over to someone else. This temporary nature of power is one of the republic's most important mechanisms for preventing the permanent concentration of authority, and it can offer major advantages.

Yet every institutional mechanism may also create particular incentives alongside its benefits.

A politician whose horizon of power is four or eight years may be more influenced by the short-term consequences of decisions than a ruler with a longer time horizon. A policy that generates popularity today but imposes costs ten years later creates a different incentive structure for someone who knows that ten years from now he or she will no longer be in power. Elections, despite their fundamental benefits for government accountability, may also encourage politicians to focus more heavily on the electoral cycle.

This does not mean that presidents necessarily think in the short term. Many do not, and strong republican institutions can ensure continuity in long-term policy regardless of the individuals holding office. The point concerns a **potential incentive embedded in the structure of the office**, not a judgment about the character of the people who occupy it.

Now consider the same issue from the other side.

The Country as Intergenerational Capital

A hereditary monarch faces a different time horizon. Under normal circumstances, the monarch is not appointed for a four- or eight-year term, and the end of an electoral cycle does not end the royal family's relationship with the institution of government. The position held today may one day pass to a child or to a later generation of the same family.

Here, however, we must avoid a tempting but inaccurate formulation: **in a modern constitutional monarchy, the monarch is not the legal owner of the country.** The country is not the monarch's private property, to be sold or treated like a personal estate.

There may nevertheless be an institutional relationship that, in certain respects, produces consequences similar to a person's relationship with a long-term asset.

If the future of the royal family is tied to the stability, prestige, and success of the country, the monarch's horizon of interest may extend beyond his or her own political lifetime. The country passed on to the next generation is the same country upon which the status, prestige, and perhaps even the survival of the monarchy for that next generation will depend.

From this perspective, a difference may emerge between viewing political office as a temporary position and viewing it as part of an institution expected to pass from one generation to the next.

To understand this difference intuitively, we might use the analogy of an **owner and a tenant**, provided that we do not confuse the analogy with the legal reality of a country. Someone who expects a property to remain in his or her possession—and eventually in the possession of his or her children—for a long time will generally have different incentives to preserve its long-term value than someone who knows that he or she will leave it after a few years. This is not true of every owner or every tenant; it merely illustrates how a time horizon can influence behavior.

Perhaps a similar mechanism, though far more complex, exists in politics.

The "Time Horizon and Intergenerational Capital" Hypothesis

On this basis, we can formulate a testable hypothesis, which I will call the **"Time Horizon and Intergenerational Capital Hypothesis."**

According to this hypothesis, one possible difference between hereditary monarchy and the presidency lies in the time horizon of the incentives created by the two institutions. The more the survival of a ruler's position—and that of his or her successors—is tied to the country's long-term stability and success, the stronger the incentive may be to preserve the value of institutions, political stability, international standing, and the country's economic and social capital.

If such a mechanism actually exists, we do not need to assume that monarchs are more benevolent, wiser, or more ethical than presidents. In fact, the strength of this hypothesis lies precisely in the fact that it does **not** depend on the personal virtue of the ruler. The argument is not that a monarch takes care of the country out of altruism. Rather, the argument is that the institutional structure may align part of the monarch's personal and family interests with the preservation of the country's long-term success.

This is where the analogy with economic incentives becomes meaningful: just as an economic system can channel self-interest toward production and exchange without expecting human nature itself to change, political structures might also be examined in terms of the incentives they create and the extent to which they lengthen the time horizon of those who hold power.

But the same mechanism can also be dangerous.

If the argument ended here, the hypothesis would be one-sided.

The same hereditary connection that might create an incentive to preserve a country's long-term capital could, under different institutional conditions, produce exactly the opposite result. A ruler might view the country not as something whose interests should be aligned with those of the ruling family, but as a resource for enriching that family. Hereditary power can reduce accountability, generate rent-seeking, and, in the absence of effective legal constraints, even open the way to autocracy.

Likewise, the temporary nature of a presidency is not merely a possible source of short-term thinking. That same temporariness can remove an ineffective ruler from power, facilitate the circulation of elites, and increase the political cost of poor performance.

The scientific question, therefore, is not "Is an owner better than a tenant?" or "Is a monarch better than a president?" The more precise question is:

Under what institutional conditions does a long-term horizon of power create incentives to preserve and develop a country, and under what conditions does it lead to rent-seeking and the entrenchment of the ruling family's interests?

The answer probably lies in the interaction between the form of government and such factors as the rule of law, constraints on power, property rights, freedom of information, institutional independence, and mechanisms of accountability.

At this point, we need to return to the findings of this book.

These fourteen indicators should be regarded not as the end of the discussion, but as the beginning of a new question.

If monarchies repeatedly perform better across a broad range of indicators, **what mechanisms have produced this pattern?**

The Time Horizon and Intergenerational Capital Hypothesis is one possible answer—not the only answer, and not one that this study has been able to prove.

A Hypothesis That Can Be Tested

The advantage of formulating the argument in this way is that it moves beyond philosophical speculation and becomes an empirical hypothesis.

If the time-horizon hypothesis is correct, its predictions should be testable against data.

We can even look within monarchies themselves. If heredity alone is not sufficient, and what matters is the combination of a long-term horizon with legal constraints and effective institutions, then we should expect different patterns to emerge among different types of monarchy.

A Test Conducted Within This Very Book

The hypothesis of time horizon and intergenerational capital should not be regarded merely as a suggestion for future research. In fact, part of the test that can help evaluate this hypothesis has already been conducted within this very book.

In examining all fourteen indicators, monarchies were not compared with republics only as a single group. Constitutional monarchies and non-constitutional monarchies were also separated from one another, and the performance of both groups was examined. This distinction is particularly important for the argument presented in this chapter.

If the superior performance of monarchies were mainly the result of the severe limitation of monarchical power, the existence of parliamentary institutions, and the close resemblance of constitutional monarchies to democratic republics in their governing structures, then it would be reasonable to expect that moving from constitutional monarchies toward monarchies with broader royal authority would reduce a significant part of the observed advantage—or even reverse it.

However, the findings of this book do not show such a simple pattern. Across the majority of the indicators examined, non-constitutional monarchies also maintained their advantage over republics, and in some indicators their performance was even better than that of constitutional monarchies. Therefore, the observed advantage of monarchies in this dataset cannot be reduced simply to the explanation that "monarchies succeed because the monarch has no real power and government is actually carried out by laws and parliaments."

This finding is important for the hypothesis proposed in this chapter. The common feature that remains when moving from constitutional to non-constitutional monarchies is **the hereditary nature of the institution of monarchy, its continuity over time, and the connection between the fate of the ruling family and the future of the country**. By contrast, the degree of legal restriction and political power of the monarch differs substantially between these two types of systems.

Therefore, the continued advantage of non-constitutional monarchies—and their superior performance over constitutional monarchies in some indicators—is at least consistent with the possibility that part of the observed difference may be related to characteristics shared by different types of monarchies, rather than solely to the limitation of royal power.

In terms of the hypothesis discussed in this chapter, it is possible that **an intergenerational time horizon and the long-term alignment of the royal family's interests with the country's future** may have effects independent of the degree to which royal authority is legally restricted. It may even be suggested that, under certain conditions, when the long-term interests of a monarch are aligned with the preservation and development of the country, greater actual authority may allow this mechanism to operate more strongly.

However, here it is essential to distinguish between a **pattern consistent with a hypothesis** and **proof of that hypothesis**. The data in this book do not directly measure "a sense of ownership," "concern for the country," or the psychological time horizon of rulers. Nor do constitutional and non-constitutional monarchies differ only in the amount of power held by the monarch. Geography, natural resources, economic structures, population size, political history, and many other factors also differ among these countries.

Therefore, one cannot conclude from the stronger performance of some non-constitutional monarchies that reducing legal constraints on power, by itself, improves governance.

This distinction is crucial. The conclusion of this book **is not that "less law is better."**

However, when evaluating this risk, another methodological distinction is also necessary. The risk of a political system cannot be assessed only by imagining its best-case scenario; both the probability of failure and the consequences of failure must be considered. In simple terms:

Risk of a political system = Probability of failure × Severity of the consequences of failure

Estimating the probability of failure for each political system requires an independent historical study: How many constitutional monarchies throughout history have moved toward concentrated power and non-constitutional monarchy? How many electoral republics have transformed into authoritarian governments? This book does not attempt to answer these questions.

However, the second component of this equation—the **severity of the consequences of failure**—can be examined to some extent through the data presented in this study.

If the main concern is that a constitutional monarchy might, under certain circumstances, move away from constitutionalism, the relevant question is: Compared with the outcome of the failure of a democratic republic, how has the resulting situation actually performed in practice?

The findings of this book show that non-constitutional monarchies, at least across the set of indicators examined, have not performed worse than republics; rather, in many cases they have shown higher average values.

Therefore, the empirical evidence in this study does not support the assumption that merely moving from a constitutional monarchy to a non-constitutional monarchy necessarily leads to worse governance outcomes than those observed in republics.

This conclusion does not mean recommending fewer restrictions on power or ignoring the danger of authoritarian rule. It simply emphasizes that risk assessment must be balanced. If a failure scenario is considered for one political system, the failure scenarios of competing systems should be evaluated according to the same criteria.

The rule of law, the prevention of abuse of power, and the existence of accountable institutions are themselves fundamental elements of good governance. The more limited conclusion that can be drawn from the internal comparison among monarchies is that **the legal restriction of royal power alone does not appear sufficient to explain the observed advantage of monarchies over republics.**

This point makes the time-horizon hypothesis even more interesting. If the advantage had been observed only among constitutional monarchies, a simple explanation would have been available: perhaps what succeeds is not monarchy itself, but parliamentary democracy, rule of law, and limited government power.

However, when monarchies with greater executive authority also demonstrate favorable patterns across most indicators, other mechanisms must also be considered.

One such possible mechanism is what may be called the **"intergenerational capital effect."**

Under this hypothesis, the key issue is not legal ownership of the country—for even a powerful monarch does not legally own the country as private property—but rather the nature of the ruler's relationship with the future of the state.

A president knows that his or her position will end after a defined period of time. A royal family, by contrast, may perceive the success or failure of the country as directly connected to the reputation, security, and status of future generations of the dynasty.

From this perspective, the comparison between constitutional and non-constitutional monarchies conducted across the fourteen indicators in this book provides a kind of **initial within-group test** for this argument. The results do not show that increasing royal authority systematically eliminates the advantage of monarchies. In some cases, the opposite pattern is observed.

This finding does not prove the time-horizon hypothesis. However, it weakens a simple alternative explanation: the claim that the better performance of monarchies exists only because constitutional monarchs lack real power, or that the success of monarchies can easily be attributed to the merely "ceremonial" nature of the monarchy.

Perhaps We Have Been Asking the Wrong Question

Much of modern political thought has framed the debate between monarchy and republic around the question: **Who has the right to rule?** That remains a fundamental question. But perhaps another question deserves equal attention:

What incentives does a system of government create for the person who holds power?

Perhaps the best political system is not merely one in which an ideal human being would govern well. Perhaps it must also work with human beings as they actually are—with ambition, self-

interest, attachment to family, desire for prestige, fear of loss, and the wish to leave a legacy—and, as far as possible, channel these motivations toward the public good.

Modern economics owes an important part of its success to the insight that building an effective system does not require us first to create a different kind of human being. Institutions can be designed to channel some existing human motivations toward productive outcomes.

Perhaps the same question is worth taking seriously in the design of government.

If so, one of the questions that the findings of this book leave for future research is:

Can modern monarchy, alongside its other institutional features, align part of the self-interest of those who hold power with the country's long-term success by creating an intergenerational time horizon?

This book has not provided a definitive answer to that question.

But the fourteen indicators examined in the preceding pages provide, at the very least, a reason to take the question seriously

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This book was not written to defend any individual or political movement; it was written to defend a method of judgment. If its publication or translation can help broaden this discussion, I welcome it and would consider it an honor.

Knowledge is tested more thoroughly the more widely it is shared; and the more thoroughly it is tested, the closer it comes to the truth.

N. Fakhr

Part Two

Appendix I: Research Methodology and Statistical Analysis

1. General Research Design

The purpose of this study is to compare the performance of countries with monarchical and republican systems of government across a set of measurable indicators of governance, development, and welfare. The unit of analysis is the **country**, and comparisons are based, wherever possible, on data from the same year and from reputable international sources.

This study is **observational and comparative** in nature. Countries have not been randomly assigned to different political systems, and numerous historical, geographical, cultural, economic, and institutional factors may be associated both with the form of government and with the indicators examined. The results should therefore be understood primarily as observed empirical differences between groups of countries and should not, without additional evidence, be interpreted as proof of a causal relationship between the form of government and the outcomes under study.

The central principle of this methodology is to avoid beginning with the assumption that either form of government is inherently superior to the other. Instead, the actual performance of countries is examined using objective indicators, after which the magnitude of the differences between the groups is assessed.

2. Unit of Analysis and Classification of Countries

Countries are classified into two main groups according to their political structure:

- **Monarchies:** countries in which the head of state is a king, queen, emir, sultan, or holder of a comparable hereditary title.
- **Republics:** countries in which the office of head of state is not hereditary and the formal structure of government is republican.

For more detailed analysis, monarchies are divided in some comparisons into two subgroups: **constitutional monarchies** and **non-constitutional monarchies**. In this book, the latter category includes monarchies in which the monarch exercises greater political authority than in parliamentary constitutional monarchies, including absolute monarchies and certain semi-constitutional systems.

Commonwealth realms are also examined separately. These are independent states that recognize the British monarch as their head of state. Because of their distinctive institutional structure and shared monarch, they are treated as a separate group in the analyses rather than being combined with independent monarchies.

This distinction also prevents overlap between groups in the statistical comparisons.

3. Geographical Coverage

Comparisons are first conducted at the global level and then, where the number of countries permits meaningful analysis, at the continental level.

Regional analyses cover **Asia, Europe, Africa, the Americas, and Oceania**. The **Middle East** is also examined separately. It is treated as a distinct analytical region because its countries share certain relative similarities in history, geography, climate, culture, religion, and, for some indicators, economic structure and natural resources. From this perspective, the Middle East serves to some extent as a **"natural laboratory"** for comparing political systems, although this expression should not be understood to imply the controlled conditions of a true experiment.

Where the number of countries in a group is very small, inferential interpretation is avoided or the findings are reported with greater caution. For example, when a group contains only one country, within-group dispersion cannot be calculated and conventional tests of differences between means cannot be validly performed.

4. Data Sources and Missing Data

For each indicator, data are drawn from reputable international sources, and the source and year are specified in the corresponding section of the book. For indicators such as **GDP per capita** and **GDP per capita based on purchasing power parity**, World Bank data form the basis of the calculations.

The general principle of the study is to use the latest common and comparable dataset available for the countries under examination in the selected year.

If the primary source contains no data for a particular country, its value is neither estimated nor imputed from similar countries. Instead, that country is excluded from the calculation for that particular indicator. Consequently, the number of countries included in the analysis may vary from one indicator to another.

This point is especially important when interpreting the results, because missing data are not necessarily random, and the exclusion of particular countries may affect group means and other statistical measures.

5. Descriptive Statistics

For each group, at least three principal descriptive statistics are calculated:

- **Mean**
- **Median**

- **Sample standard deviation**

The mean provides a measure of the general level of an indicator, but in cross-country data it may be strongly affected by a small number of exceptionally high or low values. For this reason, the median is also reported alongside the mean. Considering the two measures together helps determine whether an observed difference is driven mainly by a few exceptional countries or is also evident near the center of the distribution.

The sample standard deviation is calculated to show the degree of dispersion of countries around the group mean. Where comparison of relative dispersion is important, the **coefficient of variation** is also used.

6. Tests of Differences Between Means

Differences between the means of two independent groups are tested using **Welch's t-test**.

Welch's test was chosen because, unlike the conventional Student's t-test under the assumption of equal variances, it does not require the two groups to have equal variances and performs well when group sizes differ. This is particularly important in the present study because republics generally far outnumber monarchies, and the dispersion of the indicators is not necessarily equal across the two groups.

The null hypothesis is that the means of the two groups are equal.

Where the research hypothesis specifies a direction in advance—for example, testing the hypothesis that the mean value of a particular indicator is higher among monarchies than among republics—a **one-tailed test** is reported. Where the question concerns simply whether the two groups differ, without a direction specified in advance, a **two-tailed test** is appropriate. The direction of the test must be determined before the result is interpreted and must not be changed in response to the observed outcome.

A conventional significance level of **5 percent ($\alpha = 0.05$)** is used as the primary criterion for statistical significance. However, exact *p*-values are also reported so that the results are not reduced merely to a binary distinction between "significant" and "not significant."

7. Effect Size

Statistical significance alone does not indicate the magnitude or practical importance of a difference. A difference may be statistically significant but practically small; conversely, a relatively large difference may be observed in a sample but fail to reach the conventional threshold of statistical significance because the number of countries is small.

For this reason, **Hedges' *g*** is calculated alongside Welch's test.

Hedges' *g* standardizes the difference between two group means relative to the dispersion of the data and, particularly in small samples, provides a more appropriate correction for sampling bias than Cohen's *d*.

In interpreting effect sizes, values close to zero indicate negligible differences, while larger absolute values of *g* indicate greater differences. These thresholds are treated as approximate interpretive guidelines rather than as fixed or natural boundaries.

The sign of the effect size indicates the direction of the difference. In the book's main comparisons, a positive sign means that the mean for monarchies is higher than the mean for republics.

8. Analysis of Data Distributions

Because comparisons of means alone cannot reveal the shape of the distribution of countries, **Kernel Density Estimation (KDE)** is used for appropriate indicators.

Unlike a histogram, a KDE plot does not depend on an arbitrary division of observations into bins. Instead, it provides a continuous estimate of the approximate distribution of the data.

For economic variables such as GDP per capita and GDP per capita based on purchasing power parity, whose distributions are strongly right-skewed and in which values for wealthy countries may be many times those of lower-income countries, KDE is calculated using the **base-10 logarithm** of the data. The horizontal axis is then converted back to the actual values of the indicator so that the graph remains easy for readers to interpret.

The same standardized rule for bandwidth selection is applied to all groups shown in a given chart. In the charts presented in this book, **Scott's rule** is used to determine bandwidth. The shapes of the curves are therefore direct products of the underlying data and a consistent computational method rather than curves drawn or adjusted manually.

9. Regional Analyses and Robustness Checks

Global comparisons may be influenced by historical, geographical, and economic differences across regions. For this reason, the analyses are repeated at the continental level and for the Middle East. The purpose of these comparisons is to **reduce—not eliminate—environmental heterogeneity** and to examine whether the global pattern is also observed across different regions.

For Europe, a **robustness check** is also conducted in which republics with a history of communist rule are excluded from the European republican group, and monarchies are compared with European republics that have never experienced communist rule.

This analysis should not be interpreted as implying that excluding countries with a communist past necessarily brings us closer to identifying the "pure effect" of republican government. A country's historical path into communism may itself have been associated with its political structure and institutional development. The robustness check therefore addresses a more limited question: **if republics with a history of communist rule are removed from the comparison, how much does the observed difference between monarchies and the remaining republics change?**

Accordingly, the robustness analysis does not replace the primary comparison; it complements it.

10. Limitations of Interpretation

The findings of this study should be interpreted in light of several important limitations.

First, the study is observational. Therefore, the existence of differences between monarchies and republics does not by itself establish a causal relationship between the form of government and the outcome being examined.

Second, the numbers of countries in the two groups are unequal, and in some regions the number of monarchies or Commonwealth realms is very small. This increases uncertainty in the estimates and reduces the statistical power of the tests.

Third, countries are not units that are entirely independent of one another historically or geographically. Geographic proximity, colonial history, wars, trade, natural resources, regional institutions, and the diffusion of political and economic models can create interdependence among countries, whereas conventional statistical tests generally assume independence of observations.

Fourth, any classification of countries necessarily simplifies some of the complexity of real political systems. Two countries may both be classified as "republics" or "monarchies" while differing substantially in institutional quality, degree of democracy, concentration of political power, and political history.

Fifth, missing data may affect the results. For example, if data for a particular indicator are unavailable for some exceptionally wealthy or poor countries, the estimated group mean may be lower or higher than it otherwise would have been. For this reason, important instances of missing data are explicitly reported in the text.

11. Governing Principle for Interpreting the Results

This study distinguishes among **three levels of evidence: descriptive differences, statistical significance, and effect size**.

A higher mean or median in one group does not by itself establish the existence of a persistent difference in the broader population of countries. Conversely, a *p*-value that does not cross the conventional threshold of 0.05 does not mean that there is "no difference," particularly when one of the groups contains only a small number of countries.

For this reason, final conclusions are based on the **totality of the evidence**: the direction and magnitude of differences between means, differences between medians, dispersion of the data, *p*-values, effect sizes, the shapes of the distributions, and the consistency of the observed pattern across regional comparisons.

The purpose of this approach is to allow the data to speak before theoretical judgments are made, while at the same time avoiding conclusions that go beyond what the research design can support.

Research Data Sheets

About the Research Data Sheets (RDS)

The **Research Data Sheets (RDS)** included in the appendices of this book contain the complete datasets on which the analyses and statistical tests in this study are based. These data are published to ensure the highest possible degree of research transparency, enable independent verification of the results, and allow interested readers to reproduce the calculations.

All data presented in these sheets have been collected from reputable international sources. For each indicator, the original data source, its official identifier, the year of the data, and a QR code linking to the original database are provided, allowing readers, if they wish, to compare and verify each value directly against the data published by the original source.

The purpose of these appendices is not merely to present the results, but to make the raw data available, enable independent examination of the calculations, and allow the conclusions of this study to be reassessed. The Research Data Sheets are therefore an integral part of the methodology of this research and reflect the principle that **scientific conclusions should, as far as possible, be based on data that are accessible, transparent, and open to scrutiny.**

In this book, the data are not merely evidence supporting the conclusions; **they are part of the argument itself.** Readers are therefore invited to approach the data as researchers themselves: to examine them, scrutinize them, and, if they wish, reproduce all calculations independently of the author.

APPENDICES

Research Data Sheets

Appendix A

GDP per Capita (Current US$)
(World Bank · 2023 Edition)

Appendix B

GDP per Capita, PPP (Current International $)
(World Bank · 2023 Edition)

Appendix C

Gini Index

Appendix D

Political Stability and Absence of Violence

Appendix E

Regulatory Quality

Appendix F

Rule of Law

Appendix G

Government Effectiveness

Appendix H

Inflation

Appendix I

Control of Corruption

Appendix J

Corruption Perceptions Index

Appendix K

Property Rights

Appendix L

Human Development Index

Appendix M

Voice and Accountability

Appendix N

Passport Power Index

RESEARCH DATA SHEET · RDS-A01

GDP PER CAPITA (CURRENT US$)

World Bank · NY.GDP.PCAP.CD · Dataset: 2023 Edition

SCAN SOURCE

LEGEND

Republic
Commonwealth realm *
Constitutional monarchy **
Semi-constitutional monarchy ***
Absolute monarchy ****

GLOBAL MONARCHIES

Countries 28 · Mean 50,800.0 · Median 34,653.5 · Population SD 59,254.8

Andorra**	46,812.4	Malaysia**	11,386.0
Bahrain***	29,290.1	Monaco***	256,799.9
Belgium**	55,244.7	Morocco***	3,813.7
Bhutan***	3,831.3	Netherlands**	63,515.6
Brunei****	32,880.5	Norway**	90,984.4
Cambodia**	2,429.7	Oman****	21,027.7
Denmark**	68,043.5	Qatar***	80,195.9
Eswatini****	3,755.5	Saudi Arabia****	36,156.9
Japan**	35,215.0	Spain**	33,493.2
Jordan**	4,906.2	Sweden**	54,950.3
Kuwait***	34,092.0	Thailand**	7,210.5
Lesotho**	921.4	Tonga***	5,651.6
Liechtenstein***	206,780.6	United Arab Emirates***	49,850.7
Luxembourg**	133,230.6	United Kingdom**	49,919.7

GLOBAL COMMONWEALTH

Countries 14 · Mean 22,328.7 · Median 12,515.5 · Population SD 20,223.7

Antigua and Barbuda*	22,012.3
Australia*	65,058.4
Belize*	7,424.8
Canada*	54,847.5
Grenada*	11,414.5
Jamaica*	7,542.4
New Zealand*	49,302.3
Papua New Guinea*	2,966.1
Saint Kitts and Nevis*	22,576.9
Saint Lucia*	13,616.6
Saint Vincent and the Grenadines*	10,582.4
Solomon Islands*	1,882.6
The Bahamas*	38,231.8
Tuvalu*	5,143.8

GLOBAL REPUBLICS

Countries 150 · Mean 12,987.7 · Median 5,446.2 · Population SD 19,482.2

Afghanistan	413.8	Ecuador	6,718.1	Libya	6,026.5	Sao Tome and Principe	2,990.7
Albania	9,740.7	Egypt	3,456.8	Lithuania	27,983.4	Senegal	1,698.0
Algeria	5,370.5	El Salvador	5,319.7	Madagascar	514.1	Serbia	12,281.7
Angola	2,885.5	Eritrea	688.7	Malawi	633.2	Seychelles	18,130.5
Argentina	14,261.8	Estonia	30,264.0	Maldives	12,587.6	Sierra Leone	758.3
Armenia	8,159.1	Ethiopia	1,055.8	Mali	1,042.4	Singapore	86,382.6
Austria	56,579.5	Fiji	5,926.2	Malta	40,932.5	Slovakia	24,614.9
Azerbaijan	7,133.0	Finland	52,864.7	Marshall Islands	6,799.4	Slovenia	32,660.5
Bangladesh	2,551.0	France	44,700.1	Mauritania	2,128.0	Somalia	596.9
Barbados	25,583.9	Gabon	7,802.8	Mauritius	11,270.0	South Africa	6,034.3
Belarus	7,896.8	Georgia	8,283.7	Mexico	13,830.8	South Korea	35,674.1
Benin	1,394.2	Germany	54,776.8	Micronesia, Federated States of	3,933.0	Sri Lanka	3,815.4
Bolivia	4,274.7	Ghana	2,383.9	Moldova	6,799.6	Sudan	797.3
Bosnia and Herzegovina	8,663.0	Greece	23,343.7	Mongolia	5,838.6	Sudan, South	1,080.1
Botswana	7,827.3	Guatemala	5,754.4	Montenegro	12,258.2	Suriname	5,522.0
Brazil	10,377.6	Guinea	1,555.5	Mozambique	622.2	Switzerland	104,449.8
Bulgaria	15,854.0	Guinea-Bissau	964.4	Myanmar	1,233.2	Syria	1,056.8
Burkina Faso	873.2	Guyana	20,473.7	Namibia	4,226.0	Tajikistan	1,178.5
Burundi	249.8	Haiti	1,681.8	Nauru	13,602.6	Tanzania	1,224.0
Cabo Verde	4,794.9	Honduras	3,227.5	Nepal	1,382.4	The Gambia	883.0
Cameroon	1,720.5	Hungary	22,208.7	Nicaragua	2,609.4	Togo	1,182.3
Central African Republic	496.0	Iceland	82,201.3	Niger	647.9	Trinidad and Tobago	18,308.5
Chad	950.0	India	2,434.4	Nigeria	2,138.8	Tunisia	3,951.1
Chile	17,081.5	Indonesia	4,876.3	North Macedonia	8,674.4	Turkey	13,375.1
China	12,951.2	Iran	5,049.3	Pakistan	1,360.3	Turkmenistan	5,298.5
Colombia	7,012.5	Iraq	5,965.3	Palau	15,578.3	Uganda	1,002.3
Comoros	1,723.9	Ireland	106,818.9	Palestine	3,607.4	Ukraine	5,139.6
Congo, Democratic Republic of the	660.2	Israel	52,126.4	Panama	18,797.2	United States	82,586.8
Congo, Republic of the	2,478.0	Italy	39,279.7	Paraguay	6,303.3	Uruguay	23,378.6
Costa Rica	17,140.8	Kazakhstan	12,879.4	Peru	7,919.0	Uzbekistan	3,016.0
Croatia	22,183.5	Kenya	1,942.6	Philippines	3,804.1	Vanuatu	3,708.2
Cuba	9,605.3	Kiribati	2,200.7	Poland	22,145.3	Venezuela	3,617.5
Cyprus	36,623.7	Kosovo	6,220.3	Portugal	27,634.6	Vietnam	4,322.8
Czech Republic	31,761.6	Kyrgyzstan	2,138.2	Romania	18,244.4	Yemen	633.9
Côte d'Ivoire	2,592.0	Laos	2,066.9	Russia	13,987.0	Zambia	1,330.7
Djibouti	3,381.3	Latvia	22,710.3	Rwanda	1,061.1	Zimbabwe	2,195.2
Dominica	9,913.0	Lebanon	3,477.7	Samoa	4,822.7		
Dominican Republic	10,630.4	Liberia	799.2	San Marino	59,871.3		

GDP PER CAPITA (CURRENT US$)

World Bank · NY.GDP.PCAP.CD · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Asia · Europe

LEGEND

Republic

Commonwealth realm *

Constitutional monarchy **

Semi-constitutional monarchy ***

Absolute monarchy ****

ASIA · MONARCHIES

Countries 13 · Mean 26,806.3 · Median 29,290.1 · Population SD 21,290.6

Bahrain***	29,290.1
Bhutan***	3,831.3
Brunei****	32,890.5
Cambodia**	2,429.7
Japan**	35,215.0
Jordan**	4,906.2
Kuwait***	34,092.0
Malaysia**	11,386.0
Oman****	21,027.7
Qatar***	80,195.9
Saudi Arabia****	36,156.9
Thailand**	7,210.5
United Arab Emirates***	49,850.7

ASIA · REPUBLICS

Countries 32 · Mean 9,846.0 · Median 4,069.1 · Population SD 17,198.4

Afghanistan	413.8	Mongolia	5,838.6
Armenia	8,159.1	Myanmar	1,233.2
Azerbaijan	7,133.0	Nepal	1,382.4
Bangladesh	2,551.0	Pakistan	1,360.3
China	12,951.2	Palestine	3,607.4
Georgia	8,283.7	Philippines	3,804.1
India	2,434.4	Singapore	86,382.6
Indonesia	4,876.3	South Korea	35,674.1
Iran	5,049.3	Sri Lanka	3,815.4
Iraq	5,965.3	Syria	1,056.8
Israel	52,126.4	Tajikistan	1,178.5
Kazakhstan	12,879.4	Turkey	13,375.1
Kyrgyzstan	2,138.2	Turkmenistan	5,298.5
Laos	2,066.9	Uzbekistan	3,016.0
Lebanon	3,477.7	Vietnam	4,322.8
Maldives	12,587.6	Yemen	633.9

EUROPE · MONARCHIES

Countries 11 · Mean 96,343.2 · Median 63,515.6 · Population SD 69,525.5

Andorra**	46,812.4
Belgium**	55,244.7
Denmark**	68,043.5
Liechtenstein***	206,780.6
Luxembourg**	133,230.6
Monaco***	256,799.9
Netherlands**	63,515.6
Norway**	90,984.4
Spain**	33,493.2
Sweden**	54,950.3
United Kingdom**	49,919.7

EUROPE · REPUBLICS

Countries 34 · Mean 32,128.5 · Median 23,979.3 · Population SD 25,610.2

Albania	9,740.7	Kosovo	6,220.3
Austria	56,579.5	Latvia	22,710.3
Belarus	7,896.8	Lithuania	27,983.4
Bosnia and Herzegovina	8,663.0	Malta	40,932.5
Bulgaria	15,854.0	Moldova	6,799.6
Croatia	22,183.5	Montenegro	12,258.2
Cyprus	36,623.7	North Macedonia	8,674.4
Czech Republic	31,761.6	Poland	22,145.3
Estonia	30,264.0	Portugal	27,634.6
Finland	52,864.7	Romania	18,244.4
France	44,700.1	Russia	13,987.0
Germany	54,776.8	San Marino	59,871.3
Greece	23,343.7	Serbia	12,281.7
Hungary	22,208.7	Slovakia	24,614.9
Iceland	82,201.3	Slovenia	32,660.5
Ireland	106,818.9	Switzerland	104,449.8
Italy	39,279.7	Ukraine	5,139.6

EUROPE · REPUBLICS WITHOUT A COMMUNIST PAST

Countries 13 · Mean 56,159.7 · Median 52,864.7 · Population SD 25,605.8

Austria	56,579.5
Cyprus	36,623.7
Finland	52,864.7
France	44,700.1
Germany	54,776.8
Greece	23,343.7
Iceland	82,201.3
Ireland	106,818.9
Italy	39,279.7
Malta	40,932.5
Portugal	27,634.6
San Marino	59,871.3
Switzerland	104,449.8

RESEARCH DATA SHEET · RDS-A01

GDP PER CAPITA (CURRENT US$)

World Bank · NY.GDP.PCAP.CD · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Africa · America
Oceania · Middle East

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy *****

AFRICA · MONARCHIES

Countries 3 · Mean 2,830.2

Eswatini*****	3,755.5
Lesotho**	921.4
Morocco***	3,813.7

AFRICA · REPUBLICS

Countries 50 · Mean 2,644.2

Algeria	5,370.5
Angola	2,885.5
Benin	1,394.2
Botswana	7,827.3
Burkina Faso	873.2
Burundi	249.8
Cabo Verde	4,794.9
Cameroon	1,720.5
Central African Republic	496.0
Chad	950.0
Comoros	1,723.9
Congo, Democratic Republic of the	660.2
Congo, Republic of the	2,478.0
Côte d'Ivoire	2,592.0
Djibouti	3,381.3
Egypt	3,456.8
Eritrea	688.7
Ethiopia	1,055.8
Gabon	7,802.8
Ghana	2,383.9
Guinea	1,555.5
Guinea-Bissau	964.4
Kenya	1,942.6
Liberia	799.2
Libya	6,026.5
Madagascar	514.1
Malawi	633.2
Mali	1,042.4
Mauritania	2,128.0
Mauritius	11,270.0
Mozambique	622.2
Namibia	4,226.0
Niger	647.9
Nigeria	2,138.8
Rwanda	1,061.1
Sao Tome and Principe	2,990.7
Senegal	1,698.0
Seychelles	18,130.5
Sierra Leone	758.3
Somalia	596.9
South Africa	6,034.3
Sudan	797.3
Sudan, South	1,080.1
Tanzania	1,224.0
The Gambia	883.0
Togo	1,182.3
Tunisia	3,951.1
Uganda	1,002.3
Zambia	1,330.7
Zimbabwe	2,195.2

AMERICA · COMMONWEALTH

Countries 9 · Mean 20,916.6

Antigua and Barbuda*	22,012.3
Belize*	7,424.8
Canada*	54,847.5
Grenada*	11,414.5
Jamaica*	7,542.4
Saint Kitts and Nevis*	22,576.9
Saint Lucia*	13,616.6
Saint Vincent and the Grenadines*	10,582.4
The Bahamas*	38,231.8

AMERICA · REPUBLICS

Countries 26 · Mean 13,535.8

Argentina	14,261.8
Barbados	25,583.9
Bolivia	4,274.7
Brazil	10,377.6
Chile	17,081.5
Colombia	7,012.5
Costa Rica	17,140.8
Cuba	9,605.3
Dominica	9,913.0
Dominican Republic	10,630.4
Ecuador	6,718.1
El Salvador	5,319.7
Guatemala	5,754.4
Guyana	20,473.7
Haiti	1,681.8
Honduras	3,227.5
Mexico	13,830.8
Nicaragua	2,609.4
Panama	18,797.2
Paraguay	6,303.3
Peru	7,919.0
Suriname	5,522.0
Trinidad and Tobago	18,308.5
United States	82,586.8
Uruguay	23,378.6
Venezuela	3,617.5

OCEANIA · MONARCHY

Countries 1 · Mean 5,651.6

Tonga***	5,651.6

OCEANIA · COMMONWEALTH

Countries 5 · Mean 24,870.6

Australia*	65,058.4
New Zealand*	49,302.3
Papua New Guinea*	2,966.1
Solomon Islands*	1,882.6
Tuvalu*	5,143.8

OCEANIA · REPUBLICS

Countries 8 · Mean 7,071.4

Fiji	5,926.2
Kiribati	2,200.7
Marshall Islands	6,799.4
Micronesia, Federated States of	3,933.0
Nauru	13,602.6
Palau	15,578.3
Samoa	4,822.7
Vanuatu	3,708.2

MIDDLE EAST · MONARCHIES

Countries 7 · Mean 36,502.8

Bahrain***	29,290.1
Jordan**	4,906.2
Kuwait***	34,092.0
Oman*****	21,027.7
Qatar***	80,195.9
Saudi Arabia*****	36,156.9
United Arab Emirates***	49,850.7

MIDDLE EAST · REPUBLICS

Countries 10 · Mean 12,537.2

Cyprus	36,623.7
Egypt	3,456.8
Iran	5,049.3
Iraq	5,965.3
Israel	52,126.4
Lebanon	3,477.7
Palestine	3,607.4
Syria	1,056.8
Turkey	13,375.1
Yemen	633.9

Data Source
World Bank Open Data
Indicator: NY.GDP.PCAP.CD
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-B01

GDP PER CAPITA, PPP (CURRENT INTERNATIONAL $)

World Bank · Indicator NY.GDP.PCAP.PP.CD · Dataset: 2023 Edition

SCAN SOURCE

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

GLOBAL MONARCHIES

Countries 26 · Mean 55,784.7 · Median 58,224 · Population SD 38,018.7

Andorra**	71,789.8	Malaysia**	36,496.9
Bahrain***	64,224	Morocco***	9,902.8
Belgium**	72,629.6	Netherlands**	80,391.6
Bhutan***	16,228.5	Norway**	107,768
Brunei****	85,102.7	Oman****	41,979.6
Cambodia**	7,436.9	Qatar***	129,024.8
Denmark**	77,425.4	Saudi Arabia****	71,623.8
Eswatini****	11,309.2	Spain**	55,661.3
Japan**	52,726.2	Sweden**	68,457.9
Jordan**	11,450.6	Thailand**	23,598.4
Kuwait***	53,083.4	Tonga***	7,810.1
Lesotho**	2,879.6	United Arab Emirates***	78,019.5
Luxembourg**	152,596.1	United Kingdom**	60,786.7

GLOBAL COMMONWEALTH

Countries 14 · Mean 28,704.4 · Median 23,045.1 · Population SD 21,526.9

Antigua and Barbuda*	31,892.6	Papua New Guinea*	4,672.3
Australia*	72,272.7	Saint Kitts and Nevis*	34,007.6
Belize*	13,741.2	Saint Lucia*	26,253.9
Canada*	64,962.2	Saint Vincent and the Grenadines*	19,836.2
Grenada*	18,986.4	Solomon Islands*	2,599
Jamaica*	12,661.3	The Bahamas*	39,122.3
New Zealand*	54,949.7	Tuvalu*	5,903.9

GLOBAL REPUBLICS

Countries 148 · Mean 23,435.4 · Median 15,320.3 · Population SD 24,756.6

Afghanistan	2,203.5	Ecuador	15,907.4	Lebanon	12,585.2	Samoa	8,202.6
Albania	24,847.5	Egypt	18,540.1	Liberia	1,796	San Marino	78,810.3
Algeria	16,838.4	El Salvador	12,667.5	Libya	13,858.5	Sao Tome and Principe	6,155.1
Angola	9,549.8	Eritrea	1,742.3	Lithuania	52,720.1	Senegal	4,781.6
Argentina	30,246.4	Estonia	48,848.1	Madagascar	1,810.1	Serbia	29,777
Armenia	21,551.8	Ethiopia	3,063.4	Malawi	1,832	Seychelles	31,894
Austria	74,261.2	Fiji	14,656.2	Maldives	24,806.6	Sierra Leone	3,371.1
Azerbaijan	23,671.1	Finland	63,466.8	Mali	3,178.4	Singapore	144,728.4
Bangladesh	9,155.3	France	60,927	Malta	67,101.1	Slovakia	45,864.6
Barbados	23,176	Gabon	20,772.8	Marshall Islands	7,525.2	Slovenia	56,150.7
Belarus	30,859.6	Georgia	25,092.6	Mauritania	6,971.7	Somalia	1,556.9
Benin	4,133.4	Germany	72,262.2	Mauritius	29,402.9	South Africa	15,212.3
Bolivia	12,902.8	Ghana	7,562.6	Mexico	25,188.3	Sri Lanka	14,512.1
Bosnia and Herzegovina	23,376.3	Greece	42,613.7	Micronesia, Federated States of	4,266.7	Sudan	2,422.7
Botswana	21,029.4	Guatemala	13,751.6	Moldova	17,750.6	Sudan, South	1,154.9
Brazil	21,193.1	Guinea	4,337.3	Mongolia	18,019.7	Suriname	21,047.2
Bulgaria	38,829.3	Guinea-Bissau	2,955.5	Montenegro	31,862.9	Switzerland	93,939.8
Burkina Faso	2,761.6	Guyana	54,774.1	Mozambique	1,680.2	Syria	4,905.2
Burundi	1,147.4	Haiti	3,264.7	Myanmar	5,958.3	Tajikistan	4,987.7
Cabo Verde	10,250.6	Honduras	7,184.3	Namibia	11,247.8	Tanzania	4,021.8
Cameroon	5,415.2	Hungary	46,376.2	Nauru	13,172	The Gambia	3,291.8
Central African Republic	1,258.1	Iceland	82,256.5	Nepal	5,399.7	Togo	3,572
Chad	2,703.1	India	9,853	Nicaragua	8,326.8	Trinidad and Tobago	34,651.9
Chile	33,173.5	Indonesia	15,428.4	Niger	1,902.7	Tunisia	14,052.2
China	25,200.1	Iran	18,932.7	Nigeria	8,711.7	Turkey	43,196.3
Colombia	21,281.7	Iraq	14,664.6	North Macedonia	25,354.1	Turkmenistan	19,644.7
Comoros	3,898.7	Ireland	133,440.4	Pakistan	6,018.9	Uganda	3,100.2
Congo, Democratic Republic of the	1,732	Israel	55,301.4	Palau	18,313.3	Ukraine	17,679.9
Congo, Republic of the	6,855.9	Italy	60,349.7	Panama	39,846.2	United States	82,586.8
Costa Rica	29,247.9	Kazakhstan	38,546.9	Paraguay	17,585.1	Uruguay	34,545.7
Croatia	47,759.6	Kenya	6,322.2	Peru	17,041.6	Uzbekistan	11,678.6
Cyprus	59,874.7	Kiribati	3,526.7	Philippines	10,994.9	Vanuatu	4,201.8
Czech Republic	56,062.4	Korea, South	57,429.8	Poland	48,473.1	Venezuela	21,282.6
Côte d'Ivoire	7,243.1	Kosovo	15,817.4	Portugal	49,800.3	Vietnam	15,033.9
Djibouti	7,231.8	Kyrgyzstan	7,304.1	Romania	45,981.6	Yemen	3,164.3
Dominica	20,302.7	Laos	9,299.4	Russia	44,298.7	Zambia	4,080.4
Dominican Republic	25,861.2	Latvia	42,783.8	Rwanda	3,518.9	Zimbabwe	5,796

Data Source
World Bank Open Data
Indicator: NY.GDP.PCAP.PP.CD
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-B01

GDP PER CAPITA, PPP (CURRENT INTERNATIONAL $)

World Bank · Indicator NY.GDP.PCAP.PP.CD · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Asia · Europe

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

ASIA · MONARCHIES

Countries 13 · Mean 51,615.0 · Median 52,726.2 · Population SD 33,162.8

Bahrain***	64,224
Bhutan***	16,228.5
Brunei****	85,102.7
Cambodia**	7,436.9
Japan**	52,726.2
Jordan**	11,450.6
Kuwait***	53,083.4
Malaysia**	36,496.9
Oman****	41,979.6
Qatar***	129,024.8
Saudi Arabia****	71,623.8
Thailand**	23,598.4
United Arab Emirates***	78,019.5

ASIA · REPUBLICS

Countries 31 · Mean 21,911.3 · Median 14,664.6 · Population SD 26,357.2

Afghanistan	2,203.5	Maldives	24,806.6
Armenia	21,551.8	Mongolia	18,019.7
Azerbaijan	23,671.1	Myanmar	5,958.3
Bangladesh	9,155.3	Nepal	5,399.7
China	25,200.1	Pakistan	6,018.9
Georgia	25,092.6	Philippines	10,994.9
India	9,853	Singapore	144,728.4
Indonesia	15,428.4	Sri Lanka	14,512.1
Iran	18,932.7	Syria	4,905.2
Iraq	14,664.6	Tajikistan	4,967.7
Israel	55,301.4	Turkey	43,196.3
Kazakhstan	38,546.9	Turkmenistan	19,644.7
Korea, South	57,429.8	Uzbekistan	11,678.6
Kyrgyzstan	7,304.1	Vietnam	15,033.9
Laos	9,299.4	Yemen	3,164.3
Lebanon	12,585.2		

EUROPE · MONARCHIES

Countries 9 · Mean 83,056.3 · Median 72,629.6 · Population SD 28,245.8

Andorra**	71,789.8
Belgium**	72,629.6
Denmark**	77,425.4
Luxembourg**	152,596.1
Netherlands**	80,391.6
Norway**	107,768
Spain**	55,661.3
Sweden**	68,457.9
United Kingdom**	60,786.7

EUROPE · REPUBLICS

Countries 34 · Mean 50,899.3 · Median 48,116.3 · Population SD 23,841.0

Albania	24,847.5	Kosovo	15,817.4
Austria	74,261.2	Latvia	42,783.8
Belarus	30,859.6	Lithuania	52,720.1
Bosnia and Herzegovina	23,376.3	Malta	67,101.1
Bulgaria	38,829.3	Moldova	17,750.6
Croatia	47,759.6	Montenegro	31,862.9
Cyprus	59,874.7	North Macedonia	25,354.1
Czech Republic	56,062.4	Poland	48,473.1
Estonia	48,848.1	Portugal	49,800.3
Finland	63,466.8	Romania	45,981.6
France	60,927	Russia	44,298.7
Germany	72,262.2	San Marino	78,810.3
Greece	42,613.7	Serbia	29,777
Hungary	46,376.2	Slovakia	45,864.6
Iceland	82,256.5	Slovenia	56,150.7
Ireland	133,440.4	Switzerland	93,939.8
Italy	60,349.7	Ukraine	17,679.9

EUROPE · REPUBLICS WITHOUT A COMMUNIST PAST

Countries 13 · Mean 72,238.7 · Median 67,101.1 · Population SD 21,963.0

Austria	74,261.2
Cyprus	59,874.7
Finland	63,466.8
France	60,927
Germany	72,262.2
Greece	42,613.7
Iceland	82,256.5
Ireland	133,440.4
Italy	60,349.7
Malta	67,101.1
Portugal	49,800.3
San Marino	78,810.3
Switzerland	93,939.8

Data Source
World Bank Open Data
Indicator: NY.GDP.PCAP.PP.CD
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-B01

GDP PER CAPITA, PPP (CURRENT INTERNATIONAL $)

World Bank · Indicator NY.GDP.PCAP.PP.CD · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Africa · America
Oceania · Middle East

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

AFRICA · MONARCHIES

Countries 3 · Mean 8,030.5 · Median 9,902.8 · Population SD 3,687.2

Eswatini****	11,309.2
Lesotho**	2,879.6
Morocco***	9,902.8

AFRICA · REPUBLICS

Countries 50 · Mean 7,154.3 · Median 4,106.9 · Population SD 7,047.3

Algeria	16,838.4	Madagascar	1,810.1
Angola	9,549.8	Malawi	1,832
Benin	4,133.4	Mali	3,178.4
Botswana	21,029.4	Mauritania	6,971.7
Burkina Faso	2,761.6	Mauritius	29,402.9
Burundi	1,147.4	Mozambique	1,680.2
Cabo Verde	10,250.6	Namibia	11,247.8
Cameroon	5,415.2	Niger	1,902.7
Central African Republic	1,258.1	Nigeria	8,711.7
Chad	2,703.1	Rwanda	3,518.9
Comoros	3,896.7	Sao Tome and Principe	6,155.1
Congo, Democratic Republic of the	1,732	Senegal	4,781.6
Congo, Republic of the	6,855.9	Seychelles	31,894
Côte d'Ivoire	7,243.1	Sierra Leone	3,371.1
Djibouti	7,231.8	Somalia	1,556.9
Egypt	18,540.1	South Africa	15,212.3
Eritrea	1,742.3	Sudan	2,422.7
Ethiopia	3,063.4	Sudan, South	1,154.9
Gabon	20,772.8	Tanzania	4,021.8
Ghana	7,562.6	The Gambia	3,291.8
Guinea	4,337.3	Togo	3,572
Guinea-Bissau	2,955.5	Tunisia	14,052.2
Kenya	6,322.2	Uganda	3,100.2
Liberia	1,796	Zambia	4,080.4
Libya	13,858.5	Zimbabwe	5,796

AMERICA · COMMONWEALTH

Countries 9 · Mean 29,051.5 · Median 26,253.9 · Population SD 15,350.5

Antigua and Barbuda*	31,892.6
Belize*	13,741.2
Canada*	64,962.2
Grenada*	18,986.4
Jamaica*	12,661.3
Saint Kitts and Nevis*	34,007.6
Saint Lucia*	26,253.9
Saint Vincent and the Grenadines*	19,836.2
The Bahamas*	39,122.3

AMERICA · REPUBLICS

Countries 25 · Mean 25,081.5 · Median 21,281.7 · Population SD 16,121.2

Argentina	30,246.4	Haiti	3,264.7
Barbados	23,176	Honduras	7,184.3
Bolivia	12,902.8	Mexico	25,188.3
Brazil	21,193.1	Nicaragua	8,326.8
Chile	33,173.5	Panama	39,846.2
Colombia	21,281.7	Paraguay	17,585.1
Costa Rica	29,247.9	Peru	17,041.6
Dominica	20,302.7	Suriname	21,047.2
Dominican Republic	25,861.2	Trinidad and Tobago	34,651.9
Ecuador	15,907.4	United States	82,586.8
El Salvador	12,667.5	Uruguay	34,545.7
Guatemala	13,751.6	Venezuela	21,282.6
Guyana	54,774.1		

MIDDLE EAST · MONARCHIES

Countries 7 · Mean 64,200.8

Bahrain****	64,224
Jordan**	11,450.6
Kuwait***	53,083.4
Oman****	41,979.6
Qatar***	129,024.8
Saudi Arabia*****	71,623.8
United Arab Emirates***	78,019.5

MIDDLE EAST · REPUBLICS

Countries 9 · Mean 25,684.9

Cyprus	59,874.7
Egypt	18,540.1
Iran	18,932.7
Iraq	14,664.6
Israel	55,301.4
Lebanon	12,585.2
Syria	4,905.2
Turkey	43,198.3
Yemen	3,164.3

OCEANIA

MONARCHY

Countries 1 · Mean 7,810.1

Tonga***	7,810.1

COMMONWEALTH

Countries 5 · Mean 28,079.5

Australia*	72,272.7
New Zealand*	54,949.7
Papua New Guinea*	4,672.3
Solomon Islands*	2,599
Tuvalu*	5,903.9

REPUBLICS

Countries 8 · Mean 9,233.1

Fiji	14,656.2
Kiribati	3,526.7
Marshall Islands	7,525.2
Micronesia, Federated States of	13,697
Nauru	13,172
Palau	18,313.3
Samoa	8,202.6
Vanuatu	4,201.8

N. Fakhr
Monarchy and Republic in the Laboratory of History
Appendix B · Research Data Sheet Series · RDS-B01
Third Edition · Version 1.0 · August 2026

Data Source
World Bank Open Data
Indicator: NY.GDP.PCAP.PP.CD
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-C01

GINI INDEX

World Bank · Indicator SI.POV.GINI · Dataset: 2024 Edition

SCAN SOURCE

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

GLOBAL MONARCHIES

Countries 19 · Mean 33.4 · Median 32.4 · Population SD 7.1

Belgium**	26.8	Netherlands**	25.7
Bhutan***	28.5	Norway**	26.5
Denmark**	29.9	Qatar***	35.1
Eswatini****	54.6	Spain**	33.4
Japan**	32.3	Sweden**	29.3
Jordan**	33.7	Thailand**	33.3
Lesotho**	44.9	Tonga***	27.1
Luxembourg**	33.6	United Arab Emirates***	26.4
Malaysia**	40.7	United Kingdom**	32.4
Morocco***	39.5		

GLOBAL COMMONWEALTH

Countries 9 · Mean 39.0 · Median 39.9 · Population SD 4.0

Australia*	33.8	Papua New Guinea*	41.9
Belize*	39.9	Saint Lucia*	43.7
Canada*	31.5	Solomon Islands*	37.1
Grenada*	43.8	Tuvalu*	39.1
Jamaica*	39.9		

GLOBAL REPUBLICS

Countries 143 · Mean 36.6 · Median 35.9 · Population SD 7.0

Albania	29.4	Egypt	28.5	Liberia	35.3	Samoa	38.7
Algeria	27.6	El Salvador	38.8	Lithuania	36.0	Sao Tome and Principe	40.7
Angola	51.3	Estonia	30.7	Madagascar	36.8	Senegal	36.2
Argentina	42.4	Ethiopia	31.1	Malawi	38.5	Serbia	32.8
Armenia	27.4	Fiji	30.7	Maldives	29.3	Seychelles	32.1
Austria	31.2	Finland	27.4	Mali	35.7	Sierra Leone	35.7
Azerbaijan	26.6	France	31.8	Malta	31.8	Slovakia	23.8
Bangladesh	30.9	Gabon	38.0	Marshall Islands	35.5	Slovenia	24.7
Barbados	34.1	Gambia	38.8	Mauritania	32.0	South Africa	54.1
Belarus	24.4	Georgia	33.9	Mauritius	36.8	South Korea	32.9
Benin	34.4	Germany	33.7	Mexico	42.6	South Sudan	44.0
Bolivia	40.9	Ghana	43.5	Micronesia	40.1	Sri Lanka	37.7
Bosnia and Herzegovina	30.3	Greece	33.4	Moldova	26.8	Sudan	34.2
Botswana	54.9	Guatemala	45.2	Mongolia	31.4	Suriname	39.2
Brazil	50.3	Guinea	29.6	Montenegro	34.3	Switzerland	33.8
Bulgaria	39.5	Guinea-Bissau	33.4	Mozambique	49.6	Syria	26.4
Burkina Faso	37.4	Guyana	45.1	Myanmar	30.7	Tajikistan	36.1
Burundi	37.5	Haiti	41.1	Namibia	59.1	Tanzania	40.5
Cabo Verde	42.4	Honduras	45.7	Nauru	32.4	Timor-Leste	28.7
Cameroon	42.2	Hungary	30.6	Nepal	30.0	Togo	37.9
Central African Republic	43.0	Iceland	26.8	Nicaragua	46.2	Trinidad and Tobago	40.2
Chad	37.4	India	25.5	Niger	32.9	Tunisia	33.7
Chile	43.0	Indonesia	38.3	Nigeria	33.9	Turkey	43.7
China	36.0	Iran	35.9	North Korea	36.7	Turkmenistan	40.8
Colombia	54.4	Iraq	29.8	North Macedonia	33.5	Uganda	42.7
Comoros	30.3	Ireland	34.3	Pakistan	33.5	Ukraine	25.6
Congo	48.9	Israel	38.3	Palestine	36.4	United States	41.8
Costa Rica	47.2	Italy	34.8	Panama	49.7	Uruguay	40.0
Croatia	30.1	Kazakhstan	29.2	Paraguay	44.2	Uzbekistan	31.2
Cyprus	31.8	Kenya	38.5	Peru	40.1	Vanuatu	32.3
Czechia	25.7	Kiribati	24.7	Philippines	39.3	Venezuela	44.7
Côte d'Ivoire	35.3	Kosovo	38.3	Poland	28.5	Vietnam	36.1
DR Congo	44.7	Kyrgyzstan	34.7	Portugal	33.9	Yemen	36.7
Djibouti	41.6	Laos	38.8	Romania	29.8	Zambia	51.5
Dominican Republic	39.0	Latvia	34.0	Russia	33.0	Zimbabwe	50.3
Ecuador	45.5	Lebanon	35.5	Rwanda	39.4		

N. Fakhr

Monarchy and Republic in the Laboratory of History
Appendix C · Research Data Sheet Series · RDS-C01
Third Edition · Version 1.0 · August 2026

Data Source
World Bank Open Data
Indicator: SI.POV.GINI
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-C01

GINI INDEX

World Bank · Indicator SI.POV.GINI · Dataset: 2024 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Asia · Europe

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy *****

ASIA · MONARCHIES

Countries 7 · Mean 32.9 · Median 33.3 · Population SD 4.3

Bhutan***	28.5
Japan**	32.3
Jordan**	33.7
Malaysia**	40.7
Qatar***	35.1
Thailand**	33.3
United Arab Emirates***	26.4

ASIA · REPUBLICS

Countries 31 · Mean 33.9 · Median 34.7 · Population SD 4.5

Armenia	27.4	Myanmar	30.7
Azerbaijan	26.6	Nepal	30.0
Bangladesh	30.9	North Korea	36.7
China	36.0	Pakistan	33.5
Georgia	33.9	Palestine	36.4
India	25.5	Philippines	39.3
Indonesia	38.3	South Korea	32.9
Iran	35.9	Sri Lanka	37.7
Iraq	29.8	Syria	26.4
Israel	38.3	Tajikistan	36.1
Kazakhstan	29.2	Turkey	43.7
Kyrgyzstan	34.7	Turkmenistan	40.8
Laos	38.8	Uzbekistan	31.2
Lebanon	35.5	Vietnam	36.1
Maldives	29.3	Yemen	36.7
Mongolia	31.4		

EUROPE · MONARCHIES

Countries 8 · Mean 29.7 · Median 29.6 · Population SD 3.0

Belgium**	26.8
Denmark**	29.9
Luxembourg**	33.6
Netherlands**	25.7
Norway**	26.5
Spain**	33.4
Sweden**	29.3
United Kingdom**	32.4

EUROPE · REPUBLICS

Countries 33 · Mean 31.1 · Median 31.8 · Population SD 3.9

Albania	29.4	Kosovo	38.3
Austria	31.2	Latvia	34.0
Belarus	24.4	Lithuania	36.0
Bosnia and Herzegovina	30.3	Malta	31.8
Bulgaria	39.5	Moldova	26.8
Croatia	30.1	Montenegro	34.3
Cyprus	31.8	North Macedonia	33.5
Czechia	25.7	Poland	28.5
Estonia	30.7	Portugal	33.9
Finland	27.4	Romania	29.8
France	31.8	Russia	33.0
Germany	33.7	Serbia	32.8
Greece	33.4	Slovakia	23.8
Hungary	30.6	Slovenia	24.7
Iceland	26.8	Switzerland	33.8
Ireland	34.3	Ukraine	25.6
Italy	34.8		

EUROPE · REPUBLICS WITHOUT A COMMUNIST PAST

Countries 12 · Mean 32.1 · Median 32.6 · Population SD 2.5

Austria	31.2
Cyprus	31.8
Finland	27.4
France	31.8
Germany	33.7
Greece	33.4
Iceland	26.8
Ireland	34.3
Italy	34.8
Malta	31.8
Portugal	33.9
Switzerland	33.8

Data Source
World Bank Open Data
Indicator: SI.POV.GINI
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-C01

GINI INDEX

World Bank · Indicator SI.POV.GINI · Dataset: 2024 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Africa · America
Oceania · Middle East

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

AFRICA · MONARCHIES

Countries 3 · Mean 46.3 · Median 44.9 · Population SD 6.2

Eswatini****	54.6
Lesotho**	44.9
Morocco***	39.5

AFRICA · REPUBLICS

Countries 47 · Mean 39.4 · Median 37.9 · Population SD 7.2

Algeria	27.6	Madagascar	36.8
Angola	51.3	Malawi	38.5
Benin	34.4	Mali	35.7
Botswana	54.9	Mauritania	32.0
Burkina Faso	37.4	Mauritius	36.8
Burundi	37.5	Mozambique	49.6
Cabo Verde	42.4	Namibia	59.1
Cameroon	42.2	Niger	32.9
Central African Republic	43.0	Nigeria	33.9
Chad	37.4	Rwanda	39.4
Comoros	30.3	Sao Tome and Principe	40.7
Congo	48.9	Senegal	36.2
Côte d'Ivoire	35.3	Seychelles	32.1
DR Congo	44.7	Sierra Leone	35.7
Djibouti	41.6	South Africa	54.1
Egypt	28.5	South Sudan	44.0
Ethiopia	31.1	Sudan	34.2
Gabon	38.0	Tanzania	40.5
Gambia	38.8	Togo	37.9
Ghana	43.5	Tunisia	33.7
Guinea	29.6	Uganda	42.7
Guinea-Bissau	33.4	Zambia	51.5
Kenya	38.5	Zimbabwe	50.3
Liberia	35.3		

AMERICA · COMMONWEALTH

Countries 5 · Mean 39.8 · Median 39.9 · Population SD 4.5

Belize*	39.9
Canada*	31.5
Grenada*	43.8
Jamaica*	39.9
Saint Lucia*	43.7

AMERICA · REPUBLICS

Countries 24 · Mean 43.4 · Median 42.8 · Population SD 4.3

Argentina	42.4	Haiti	41.1
Barbados	34.1	Honduras	45.7
Bolivia	40.9	Mexico	42.6
Brazil	50.3	Nicaragua	46.2
Chile	43.0	Panama	49.7
Colombia	54.4	Paraguay	44.2
Costa Rica	47.2	Peru	40.1
Dominican Republic	39.0	Suriname	39.2
Ecuador	45.5	Trinidad and Tobago	40.2
El Salvador	38.8	United States	41.8
Guatemala	45.2	Uruguay	40.0
Guyana	45.1	Venezuela	44.7

MIDDLE EAST · MONARCHIES

Countries 3 · Mean 31.7

Jordan**	33.7
Qatar***	35.1
United Arab Emirates***	26.4

MIDDLE EAST · REPUBLICS

Countries 10 · Mean 34.3

Cyprus	31.8
Egypt	28.5
Iran	35.9
Iraq	29.8
Israel	38.3
Lebanon	35.5
Palestine	36.4
Syria	26.4
Turkey	43.7
Yemen	36.7

OCEANIA

MONARCHY

Countries 1 · Mean 27.1

Tonga***	27.1

COMMONWEALTH

Countries 4 · Mean 38.0

Australia*	33.8
Papua New Guinea*	41.9
Solomon Islands*	37.1
Tuvalu*	39.1

REPUBLICS

Countries 8 · Mean 32.9

Fiji	30.7
Kiribati	24.7
Marshall Islands	35.5
Micronesia	40.1
Nauru	32.4
Samoa	38.7
Timor-Leste	28.7
Vanuatu	32.3

N. Fakhr
Monarchy and Republic in the Laboratory of History
Appendix C · Research Data Sheet Series · RDS-C01
Third Edition · Version 1.0 · August 2026

Data Source
World Bank Open Data
Indicator: SI.POV.GINI
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-D01

POLITICAL STABILITY AND ABSENCE OF VIOLENCE

World Bank · Indicator PV.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

GLOBAL MONARCHIES

Countries 28 · Mean 65.1 · Median 67.3 · Population SD 22.3

Andorra**	98.6	Malaysia**	50.7
Bahrain***	29.4	Monaco***	93.4
Belgium**	58.3	Morocco***	29.9
Bhutan***	83.4	Netherlands**	68.7
Brunei****	96.7	Norway**	77.7
Cambodia**	47.9	Oman****	65.9
Denmark**	76.9	Qatar***	84.4
Eswatini****	30.8	Saudi Arabia****	39.8
Japan**	81.5	Spain**	55.0
Jordan**	40.3	Sweden**	73.5
Kuwait***	58.8	Thailand**	36.0
Lesotho**	35.5	Tonga***	90.0
Liechtenstein***	99.1	United Arab Emirates***	70.1
Luxembourg**	87.7	United Kingdom**	62.1

GLOBAL COMMONWEALTH

Countries 14 · Mean 75.8 · Median 80.6 · Population SD 17.5

Antigua and Barbuda*	80.6
Australia*	79.6
Bahamas*	84.8
Belize*	65.4
Canada*	76.3
Grenada*	86.3
Jamaica*	57.3
New Zealand*	96.2
Papua New Guinea*	27.0
Saint Kitts and Nevis*	80.6
Saint Lucia*	86.3
Saint Vincent and the Grenadines*	86.3
Solomon Islands*	60.7
Tuvalu*	94.3

GLOBAL REPUBLICS

Countries 152 · Mean 41.6 · Median 39.1 · Population SD 27.4

Afghanistan	1.4	Ecuador	32.7	Lebanon	9.5	Samoa	90.5
Albania	51.7	Egypt	16.6	Liberia	42.7	San Marino	93.4
Algeria	23.2	El Salvador	46.4	Libya	4.3	Sao Tome and Principe	60.2
Angola	32.2	Eritrea	17.5	Lithuania	72.5	Senegal	41.2
Argentina	41.7	Estonia	89.2	Madagascar	19.9	Serbia	44.5
Armenia	17.1	Ethiopia	6.2	Malawi	37.9	Seychelles	73.9
Austria	71.6	Fiji	73.0	Maldives	62.5	Sierra Leone	37.4
Azerbaijan	18.0	Finland	71.1	Mali	0.5	Singapore	97.2
Bangladesh	15.6	France	55.9	Malta	77.3	Slovakia	64.5
Barbados	93.8	Gabon	33.6	Marshall Islands	93.4	Slovenia	75.8
Belarus	18.5	Georgia	33.2	Mauritania	26.1	Somalia	2.8
Benin	31.8	Germany	66.4	Mauritius	74.9	South Africa	20.9
Bolivia	35.1	Ghana	45.5	Mexico	22.7	Sri Lanka	25.6
Bosnia and Herzegovina	31.3	Greece	54.0	Micronesia, Federated States of	89.6	Sudan	1.9
Botswana	87.2	Guatemala	37.0	Moldova	20.4	Sudan, South	3.8
Brazil	28.4	Guinea	16.1	Mongolia	67.3	Suriname	59.7
Bulgaria	55.5	Guinea-Bissau	34.1	Montenegro	48.8	Switzerland	88.6
Burkina Faso	5.7	Guyana	43.6	Mozambique	11.8	Syria	0.0
Burundi	12.8	Haiti	10.4	Myanmar	4.7	Tajikistan	27.5
Cabo Verde	79.1	Honduras	28.0	Namibia	63.5	Tanzania	44.1
Cameroon	11.4	Hungary	72.0	Nauru	82.9	The Gambia	46.0
Central African Republic	3.3	Iceland	95.3	Nepal	39.3	Timor-Leste	54.5
Chad	9.0	India	21.3	Nicaragua	42.2	Togo	14.2
Chile	50.2	Indonesia	28.9	Niger	8.5	Trinidad and Tobago	57.8
China	25.1	Iran	8.1	Nigeria	7.6	Tunisia	22.3
Colombia	19.0	Iraq	2.4	North Macedonia	51.2	Turkey	13.7
Comoros	38.9	Ireland	78.7	Pakistan	6.6	Turkmenistan	43.1
Congo	46.9	Israel	10.0	Palau	89.6	Uganda	19.4
Costa Rica	83.9	Italy	64.9	Palestine	7.1	Ukraine	10.9
Croatia	67.8	Kazakhstan	36.5	Panama	52.6	United States	47.4
Cuba	56.9	Kenya	14.7	Paraguay	49.3	Uruguay	82.0
Cyprus	59.2	Kiribati	91.0	Peru	24.2	Uzbekistan	40.8
Czech Republic	82.5	Korea, North	30.3	Philippines	23.7	Vanuatu	81.0
Côte d'Ivoire	21.8	Korea, South	68.2	Poland	64.0	Venezuela	12.3
Djibouti	24.6	Kosovo	34.6	Portugal	70.6	Vietnam	45.0
Dominica	95.7	Kyrgyzstan	26.5	Romania	56.4	Yemen	0.9
Dominican Republic	53.6	Laos	75.4	Russia	13.3	Zambia	52.1
DR Congo	5.2	Latvia	66.8	Rwanda	49.8	Zimbabwe	15.2

RESEARCH DATA SHEET · RDS-D01

POLITICAL STABILITY AND ABSENCE OF VIOLENCE

World Bank · Indicator PV.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Asia · Europe

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

ASIA · MONARCHIES

Countries 13 · Mean 60.4 · Median 58.8 · Population SD 20.8

Bahrain***	29.4
Bhutan***	83.4
Brunei****	96.7
Cambodia**	47.9
Japan**	81.5
Jordan**	40.3
Kuwait***	58.8
Malaysia**	50.7
Oman****	65.9
Qatar***	84.4
Saudi Arabia****	39.8
Thailand**	36.0
United Arab Emirates***	70.1

ASIA · REPUBLICS

Countries 33 · Mean 28.3 · Median 25.1 · Population SD 23.6

Afghanistan	1.4	Maldives	62.5
Armenia	17.1	Mongolia	67.3
Azerbaijan	18.0	Myanmar	4.7
Bangladesh	15.6	Nepal	39.3
China	25.1	Pakistan	6.6
Georgia	33.2	Palestine	7.1
India	21.3	Philippines	23.7
Indonesia	28.9	Singapore	97.2
Iran	8.1	Sri Lanka	25.6
Iraq	2.4	Syria	0.0
Israel	10.0	Tajikistan	27.5
Kazakhstan	36.5	Turkey	13.7
Korea, North	30.3	Turkmenistan	43.1
Korea, South	68.2	Uzbekistan	40.8
Kyrgyzstan	26.5	Vietnam	45.0
Laos	75.4	Yemen	0.9
Lebanon	9.5		

EUROPE · MONARCHIES

Countries 11 · Mean 77.4 · Median 76.9 · Population SD 15.0

Andorra**	98.6
Belgium**	58.3
Denmark**	76.9
Liechtenstein***	99.1
Luxembourg**	87.7
Monaco***	93.4
Netherlands**	68.7
Norway**	77.7
Spain**	55.0
Sweden**	73.5
United Kingdom**	62.1

EUROPE · REPUBLICS

Countries 34 · Mean 59.4 · Median 64.7 · Population SD 21.3

Albania	51.7	Kosovo	34.6
Austria	71.6	Latvia	66.8
Belarus	18.5	Lithuania	72.5
Bosnia and Herzegovina	31.3	Malta	77.3
Bulgaria	55.5	Moldova	20.4
Croatia	67.8	Montenegro	48.8
Cyprus	59.2	North Macedonia	51.2
Czech Republic	82.5	Poland	64.0
Estonia	89.2	Portugal	70.6
Finland	71.1	Romania	56.4
France	55.9	Russia	13.3
Germany	66.4	San Marino	93.4
Greece	54.0	Serbia	44.5
Hungary	72.0	Slovakia	64.5
Iceland	95.3	Slovenia	75.8
Ireland	78.7	Switzerland	88.6
Italy	64.9	Ukraine	10.9

EUROPE · REPUBLICS WITHOUT A COMMUNIST PAST

Countries 13 · Mean 72.8 · Median 71.1 · Population SD 12.9

Austria	71.6
Cyprus	59.2
Finland	71.1
France	55.9
Germany	66.4
Greece	54.0
Iceland	95.3
Ireland	78.7
Italy	64.9
Malta	77.3
Portugal	70.6
San Marino	93.4
Switzerland	88.6

RESEARCH DATA SHEET · RDS-D01

POLITICAL STABILITY AND ABSENCE OF VIOLENCE

World Bank · Indicator PV.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Africa · America
Oceania · Middle East

LEGEND

Republic
Commonwealth realm *
Constitutional monarchy **
Semi-constitutional monarchy ***
Absolute monarchy ****

AFRICA · MONARCHIES

Countries 3 · Mean 32.1 · Median 30.8 · Population SD 2.5

Eswatini****	30.8
Lesotho**	35.5
Morocco***	29.9

AFRICA · REPUBLICS

Countries 50 · Mean 28.4 · Median 22.1 · Population SD 22.0

Algeria	23.2	Madagascar	19.9
Angola	32.2	Malawi	37.9
Benin	31.8	Mali	0.5
Botswana	87.2	Mauritania	26.1
Burkina Faso	5.7	Mauritius	74.9
Burundi	12.8	Mozambique	11.8
Cabo Verde	79.1	Namibia	63.5
Cameroon	11.4	Niger	8.5
Central African Republic	3.3	Nigeria	7.6
Chad	9.0	Rwanda	49.8
Comoros	38.9	Sao Tome and Principe	60.2
Congo	46.9	Senegal	41.2
Côte d'Ivoire	21.8	Seychelles	73.9
Djibouti	24.6	Sierra Leone	37.4
DR Congo	5.2	Somalia	2.8
Egypt	16.6	South Africa	20.9
Eritrea	17.5	Sudan	1.9
Ethiopia	6.2	Sudan, South	3.8
Gabon	33.6	Tanzania	44.1
Ghana	45.5	The Gambia	46.0
Guinea	16.1	Togo	14.2
Guinea-Bissau	34.1	Tunisia	22.3
Kenya	14.7	Uganda	19.4
Liberia	42.7	Zambia	52.1
Libya	4.3	Zimbabwe	15.2

AMERICA · COMMONWEALTH

Countries 9 · Mean 78.2 · Median 80.6 · Population SD 9.8

Antigua and Barbuda*	80.6
Bahamas*	84.8
Belize*	65.4
Canada*	76.3
Grenada*	86.3
Jamaica*	57.3
Saint Kitts and Nevis*	80.6
Saint Lucia*	86.3
Saint Vincent and the Grenadines*	86.3

AMERICA · REPUBLICS

Countries 26 · Mean 46.4 · Median 45.0 · Population SD 22.6

Argentina	41.7	Guyana	43.6
Barbados	93.8	Haiti	10.4
Bolivia	35.1	Honduras	28.0
Brazil	28.4	Mexico	22.7
Chile	50.2	Nicaragua	42.2
Colombia	19.0	Panama	52.6
Costa Rica	83.9	Paraguay	49.3
Cuba	56.9	Peru	24.2
Dominica	95.7	Suriname	59.7
Dominican Republic	53.6	Trinidad and Tobago	57.8
Ecuador	32.7	United States	47.4
El Salvador	46.4	Uruguay	82.0
Guatemala	37.0	Venezuela	12.3

OCEANIA

MIDDLE EAST · MONARCHIES

Countries 7 · Mean 55.5

Bahrain***	29.4
Jordan**	40.3
Kuwait***	58.8
Oman****	65.9
Qatar***	84.4
Saudi Arabia*****	39.8
United Arab Emirates***	70.1

MIDDLE EAST · REPUBLICS

Countries 10 · Mean 12.8

Cyprus	59.2
Egypt	16.6
Iran	8.1
Iraq	2.4
Israel	10.0
Lebanon	9.5
Palestine	7.1
Syria	0.0
Turkey	13.7
Yemen	0.9

MONARCHY

Countries 1 · Mean 90.0

Tonga***	90.0

COMMONWEALTH

Countries 5 · Mean 71.6

Australia*	79.6
New Zealand*	96.2
Papua New Guinea*	27.0
Solomon Islands*	60.7
Tuvalu*	94.3

REPUBLICS

Countries 9 · Mean 82.8

Fiji	73.0
Kiribati	91.0
Marshall Islands	93.4
Micronesia, Federated States of	84.8
Nauru	82.9
Palau	89.6
Samoa	90.5
Timor-Leste	54.5
Vanuatu	81.0

Data Source
World Bank Open Data
Indicator: PV.PER.RNK
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-E01

REGULATORY QUALITY

World Bank · Indicator RQ.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

LEGEND

Republic
Commonwealth realm *
Constitutional monarchy **
Semi-constitutional monarchy ***
Absolute monarchy ****

GLOBAL MONARCHIES

Countries 28 · Mean 72.6 · Median 80.9 · Population SD 22.7

Andorra**	89.6	Malaysia**	73.1
Bahrain***	84.0	Monaco***	89.6
Belgium**	85.8	Morocco***	50.9
Bhutan***	36.8	Netherlands**	97.2
Brunei****	80.7	Norway**	93.4
Cambodia**	35.5	Oman****	66.5
Denmark**	98.1	Qatar***	81.1
Eswatini****	24.1	Saudi Arabia****	68.9
Japan**	92.5	Spain**	74.5
Jordan**	59.4	Sweden**	95.3
Kuwait***	63.7	Thailand**	57.1
Lesotho**	29.7	Tonga***	37.3
Liechtenstein***	94.3	United Arab Emirates***	82.1
Luxembourg**	99.1	United Kingdom**	92.9

GLOBAL COMMONWEALTH

Countries 14 · Mean 61.2 · Median 62.0 · Population SD 23.7

Antigua and Barbuda*	70.3
Australia*	99.5
Bahamas*	51.9
Belize*	35.8
Canada*	94.8
Grenada*	61.8
Jamaica*	53.8
New Zealand*	98.6
Papua New Guinea*	27.8
Saint Kitts and Nevis*	69.3
Saint Lucia*	64.6
Saint Vincent and the Grenadines*	62.3
Solomon Islands*	21.7
Tuvalu*	44.8

GLOBAL REPUBLICS

Countries 152 · Mean 41.8 · Median 39.9 · Population SD 27.0

Afghanistan	9.4	Ecuador	23.6	Lebanon	14.6	Samoa	42.9
Albania	58.5	Egypt	26.9	Liberia	18.4	San Marino	79.2
Algeria	16.0	El Salvador	39.6	Libya	2.4	Sao Tome and Principe	16.5
Angola	22.6	Eritrea	0.5	Lithuania	88.2	Senegal	39.2
Argentina	35.4	Estonia	91.0	Madagascar	21.2	Serbia	56.6
Armenia	52.8	Ethiopia	14.2	Malawi	22.2	Seychelles	59.0
Austria	90.1	Fiji	50.0	Maldives	26.4	Sierra Leone	13.7
Azerbaijan	49.1	Finland	98.7	Mali	25.9	Singapore	100.0
Bangladesh	18.9	France	85.4	Malta	74.1	Slovakia	71.2
Barbados	67.5	Gabon	25.0	Marshall Islands	36.3	Slovenia	75.0
Belarus	7.5	Georgia	79.7	Mauritania	15.1	Somalia	2.8
Benin	40.6	Germany	92.0	Mauritius	82.5	South Africa	44.3
Bolivia	11.8	Ghana	45.3	Mexico	46.2	Sri Lanka	33.0
Bosnia and Herzegovina	46.7	Greece	70.8	Micronesia, Federated States of	30.7	Sudan	4.7
Botswana	67.0	Guatemala	41.5	Moldova	54.7	Sudan, South	0.9
Brazil	40.1	Guinea	13.2	Mongolia	45.8	Suriname	23.1
Bulgaria	65.6	Guinea-Bissau	9.9	Montenegro	64.2	Switzerland	95.8
Burkina Faso	34.9	Guyana	31.6	Mozambique	24.5	Syria	3.8
Burundi	15.6	Haiti	8.5	Myanmar	7.1	Tajikistan	11.3
Cabo Verde	59.9	Honduras	32.1	Namibia	51.4	Tanzania	30.2
Cameroon	19.3	Hungary	62.7	Nauru	52.4	The Gambia	28.3
Central African Republic	6.1	Iceland	87.3	Nepal	27.4	Timor-Leste	34.4
Chad	12.3	India	47.2	Nicaragua	20.3	Togo	34.0
Chile	76.9	Indonesia	60.8	Niger	20.8	Trinidad and Tobago	47.6
China	38.7	Iran	4.2	Nigeria	17.0	Tunisia	28.8
Colombia	54.2	Iraq	6.6	North Macedonia	66.0	Turkey	43.9
Comoros	10.8	Ireland	96.2	Pakistan	19.8	Turkmenistan	1.4
Congo	10.4	Israel	84.4	Palau	65.1	Uganda	32.5
Costa Rica	69.8	Italy	72.6	Palestine	42.0	Ukraine	43.4
Croatia	72.2	Kazakhstan	53.3	Panama	55.7	United States	90.6
Cuba	5.2	Kenya	37.7	Paraguay	50.5	Uruguay	73.6
Cyprus	75.9	Kiribati	42.5	Peru	60.4	Uzbekistan	31.1
Czech Republic	87.7	Korea, North	0.0	Philippines	57.5	Vanuatu	48.1
Côte d'Ivoire	48.6	Korea, South	84.9	Poland	76.4	Venezuela	1.9
Djibouti	17.5	Kosovo	41.0	Portugal	75.5	Vietnam	38.2
Dominica	61.3	Kyrgyzstan	29.2	Romania	63.2	Yemen	3.3
Dominican Republic	58.0	Laos	17.9	Russia	12.7	Zambia	33.5
DR Congo	8.0	Latvia	86.3	Rwanda	56.1	Zimbabwe	9.0

RESEARCH DATA SHEET · RDS-E01

REGULATORY QUALITY

World Bank · Indicator RQ.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Asia · Europe

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

ASIA · MONARCHIES

Countries 13 · Mean 67.8 · Median 68.9 · Population SD 16.8

Bahrain***	84.0
Bhutan***	36.8
Brunei****	80.7
Cambodia**	35.5
Japan**	92.5
Jordan**	59.4
Kuwait***	63.7
Malaysia**	73.1
Oman****	66.5
Qatar***	81.1
Saudi Arabia****	68.9
Thailand**	57.1
United Arab Emirates***	82.1

ASIA · REPUBLICS

Countries 33 · Mean 34.7 · Median 31.1 · Population SD 26.4

Afghanistan	9.4	Maldives	26.4
Armenia	52.8	Mongolia	45.8
Azerbaijan	49.1	Myanmar	7.1
Bangladesh	18.9	Nepal	27.4
China	38.7	Pakistan	19.8
Georgia	79.7	Palestine	42.0
India	47.2	Philippines	57.5
Indonesia	60.8	Singapore	100.0
Iran	4.2	Sri Lanka	33.0
Iraq	6.6	Syria	3.8
Israel	84.4	Tajikistan	11.3
Kazakhstan	53.3	Turkey	43.9
Korea, North	0.0	Turkmenistan	1.4
Korea, South	84.9	Uzbekistan	31.1
Kyrgyzstan	29.2	Vietnam	38.2
Laos	17.9	Yemen	3.3
Lebanon	14.6		

EUROPE · MONARCHIES

Countries 11 · Mean 91.8 · Median 93.4 · Population SD 6.7

Andorra**	89.6
Belgium**	85.8
Denmark**	98.1
Liechtenstein***	94.3
Luxembourg**	99.1
Monaco***	89.6
Netherlands**	97.2
Norway**	93.4
Spain**	74.5
Sweden**	95.3
United Kingdom**	92.9

EUROPE · REPUBLICS

Countries 34 · Mean 70.1 · Median 73.3 · Population SD 21.1

Albania	58.5	Kosovo	41.0
Austria	90.1	Latvia	86.3
Belarus	7.5	Lithuania	88.2
Bosnia and Herzegovina	46.7	Malta	74.1
Bulgaria	65.6	Moldova	54.7
Croatia	72.2	Montenegro	64.2
Cyprus	75.9	North Macedonia	66.0
Czech Republic	87.7	Poland	76.4
Estonia	91.0	Portugal	75.5
Finland	96.7	Romania	63.2
France	85.4	Russia	12.7
Germany	92.0	San Marino	79.2
Greece	70.8	Serbia	56.6
Hungary	62.7	Slovakia	71.2
Iceland	87.3	Slovenia	75.0
Ireland	96.2	Switzerland	95.8
Italy	72.6	Ukraine	43.4

EUROPE · REPUBLICS WITHOUT A COMMUNIST PAST

Countries 13 · Mean 84.0 · Median 85.4 · Population SD 9.3

Austria	90.1
Cyprus	75.9
Finland	96.7
France	85.4
Germany	92.0
Greece	70.8
Iceland	87.3
Ireland	96.2
Italy	72.6
Malta	74.1
Portugal	75.5
San Marino	79.2
Switzerland	95.8

Data Source
World Bank Open Data
Indicator: RQ.PER.RNK
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-E01

REGULATORY QUALITY

World Bank · Indicator RQ.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Africa · America
Oceania · Middle East

LEGEND

Republic
Commonwealth realm *
Constitutional monarchy **
Semi-constitutional monarchy ***
Absolute monarchy ****

AFRICA · MONARCHIES

Countries 3 · Mean 34.9 · Median 29.7 · Population SD 11.5

Eswatini****	24.1
Lesotho**	29.7
Morocco***	50.9

AFRICA · REPUBLICS

Countries 50 · Mean 25.9 · Median 21.7 · Population SD 18.2

Algeria	16.0	Madagascar	21.2
Angola	22.6	Malawi	22.2
Benin	40.6	Mali	25.9
Botswana	67.0	Mauritania	15.1
Burkina Faso	34.9	Mauritius	82.5
Burundi	15.6	Mozambique	24.5
Cabo Verde	59.9	Namibia	51.4
Cameroon	19.3	Niger	20.8
Central African Republic	6.1	Nigeria	17.0
Chad	12.3	Rwanda	56.1
Comoros	10.8	Sao Tome and Principe	16.5
Congo	10.4	Senegal	39.2
Côte d'Ivoire	48.6	Seychelles	59.0
Djibouti	17.5	Sierra Leone	13.7
DR Congo	8.0	Somalia	2.8
Egypt	26.9	South Africa	44.3
Eritrea	0.5	Sudan	4.7
Ethiopia	14.2	Sudan, South	0.9
Gabon	25.0	Tanzania	30.2
Ghana	45.3	The Gambia	28.3
Guinea	13.2	Togo	34.0
Guinea-Bissau	9.9	Tunisia	28.8
Kenya	37.7	Uganda	32.5
Liberia	18.4	Zambia	33.5
Libya	2.4	Zimbabwe	9.0

AMERICA · COMMONWEALTH

Countries 9 · Mean 62.7 · Median 62.3 · Population SD 15.1

Antigua and Barbuda*	70.3
Bahamas*	51.9
Belize*	35.8
Canada*	94.8
Grenada*	61.8
Jamaica*	53.8
Saint Kitts and Nevis*	69.3
Saint Lucia*	64.6
Saint Vincent and the Grenadines*	62.3

AMERICA · REPUBLICS

Countries 26 · Mean 43.3 · Median 43.9 · Population SD 23.0

Argentina	35.4	Guyana	31.6
Barbados	67.5	Haiti	8.5
Bolivia	11.8	Honduras	32.1
Brazil	40.1	Mexico	46.2
Chile	76.9	Nicaragua	20.3
Colombia	54.2	Panama	55.7
Costa Rica	69.8	Paraguay	50.5
Cuba	5.2	Peru	60.4
Dominica	61.3	Suriname	23.1
Dominican Republic	58.0	Trinidad and Tobago	47.6
Ecuador	23.6	United States	90.6
El Salvador	39.6	Uruguay	73.6
Guatemala	41.5	Venezuela	1.9

OCEANIA

MIDDLE EAST · MONARCHIES

Countries 7 · Mean 72.2

Bahrain***	84.0
Jordan**	59.4
Kuwait***	63.7
Oman****	66.5
Qatar***	81.1
Saudi Arabia*****	68.9
United Arab Emirates***	82.1

MIDDLE EAST · REPUBLICS

Countries 10 · Mean 30.6

Cyprus	75.9
Egypt	26.9
Iran	4.2
Iraq	6.6
Israel	84.4
Lebanon	14.6
Palestine	42.0
Syria	3.8
Turkey	43.9
Yemen	3.3

MONARCHY

Countries 1 · Mean 37.3

Tonga***	37.3

COMMONWEALTH

Countries 5 · Mean 58.5

Australia*	99.5
New Zealand*	98.6
Papua New Guinea*	27.8
Solomon Islands*	21.7
Tuvalu*	44.8

REPUBLICS

Countries 9 · Mean 44.7

Fiji	50.0
Kiribati	42.5
Marshall Islands	36.3
Micronesia, Federated States of	31.0
Nauru	52.4
Palau	65.1
Samoa	42.9
Timor-Leste	34.4
Vanuatu	48.1

Data Source
World Bank Open Data
Indicator: RQ.PER.RNK
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-F01

RULE OF LAW

World Bank · Indicator RL.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

GLOBAL MONARCHIES

Countries 28 · Mean 73.7 · Median 78.5 · Population SD 21.2

Andorra**	91.5	Malaysia**	67.5
Bahrain***	65.6	Monaco***	91.5
Belgium**	88.2	Morocco***	48.6
Bhutan***	72.6	Netherlands**	94.3
Brunei****	79.2	Norway**	99.1
Cambodia**	23.6	Oman****	70.3
Denmark**	99.5	Qatar***	80.2
Eswatini****	28.3	Saudi Arabia****	62.7
Japan**	92.5	Spain**	78.3
Jordan**	58.0	Sweden**	93.4
Kuwait***	59.0	Thailand**	57.5
Lesotho**	34.0	Tonga***	67.0
Liechtenstein***	96.2	United Arab Emirates***	78.8
Luxembourg**	97.6	United Kingdom**	89.2

GLOBAL COMMONWEALTH

Countries 14 · Mean 63.9 · Median 67.0 · Population SD 21.1

Antigua and Barbuda*	59.9
Australia*	92.0
Bahamas*	55.7
Belize*	27.8
Canada*	90.1
Grenada*	69.3
Jamaica*	44.8
New Zealand*	95.8
Papua New Guinea*	29.7
Saint Kitts and Nevis*	66.0
Saint Lucia*	69.8
Saint Vincent and the Grenadines*	67.9
Solomon Islands*	43.4
Tuvalu*	82.1

GLOBAL REPUBLICS

Countries 152 · Mean 41.7 · Median 39.4 · Population SD 27.3

Afghanistan	5.2	Ecuador	19.3	Lebanon	13.2	Samoa	79.7
Albania	45.8	Egypt	44.3	Liberia	17.0	San Marino	91.5
Algeria	26.4	El Salvador	39.2	Libya	2.4	Sao Tome and Principe	25.5
Angola	14.6	Eritrea	2.8	Lithuania	86.8	Senegal	42.9
Argentina	37.7	Estonia	89.6	Madagascar	18.9	Serbia	51.4
Armenia	49.1	Ethiopia	26.9	Malawi	46.2	Seychelles	63.7
Austria	97.2	Fiji	58.5	Maldives	50.9	Sierra Leone	22.2
Azerbaijan	29.2	Finland	100.0	Mali	18.4	Singapore	98.1
Bangladesh	33.5	France	84.9	Malta	75.5	Slovakia	68.9
Barbados	62.3	Gabon	20.8	Marshall Islands	73.1	Slovenia	82.5
Belarus	11.3	Georgia	55.2	Mauritania	27.4	Somalia	0.0
Benin	33.0	Germany	92.9	Mauritius	77.8	South Africa	54.2
Bolivia	11.8	Ghana	49.5	Mexico	24.1	Sri Lanka	50.0
Bosnia and Herzegovina	39.6	Greece	57.1	Micronesia, Federated States of	72.2	Sudan	4.2
Botswana	61.3	Guatemala	15.6	Moldova	47.2	Sudan, South	0.9
Brazil	42.0	Guinea	14.2	Mongolia	43.9	Suriname	51.9
Bulgaria	53.8	Guinea-Bissau	7.1	Montenegro	52.4	Switzerland	98.6
Burkina Faso	24.5	Guyana	84.0	Mozambique	17.5	Syria	1.4
Burundi	10.8	Haiti	7.5	Myanmar	6.1	Tajikistan	9.9
Cabo Verde	61.8	Honduras	15.1	Namibia	64.6	Tanzania	38.7
Cameroon	16.5	Hungary	43.2	Nauru	53.3	The Gambia	38.2
Central African Republic	3.3	Iceland	98.7	Nepal	34.9	Timor-Leste	21.7
Chad	8.0	India	56.1	Nicaragua	9.4	Togo	32.1
Chile	71.2	Indonesia	46.7	Niger	25.0	Trinidad and Tobago	42.5
China	52.8	Iran	16.0	Nigeria	20.3	Tunisia	47.6
Colombia	35.8	Iraq	3.8	North Macedonia	45.3	Turkey	32.5
Comoros	8.5	Ireland	93.9	Pakistan	21.2	Turkmenistan	6.6
Congo	17.9	Israel	77.4	Palau	80.7	Uganda	34.4
Costa Rica	60.4	Italy	60.8	Palestine	25.9	Ukraine	19.8
Croatia	59.4	Kazakhstan	36.8	Panama	40.1	United States	88.7
Cuba	36.3	Kenya	41.0	Paraguay	28.8	Uruguay	75.9
Cyprus	71.7	Kiribati	68.4	Peru	30.2	Uzbekistan	23.1
Czech Republic	84.4	Korea, North	5.7	Philippines	37.3	Vanuatu	54.7
Côte d'Ivoire	35.4	Korea, South	85.8	Poland	65.1	Venezuela	0.5
Djibouti	13.7	Kosovo	41.5	Portugal	83.5	Vietnam	50.5
Dominica	70.8	Kyrgyzstan	12.7	Romania	64.2	Yemen	1.9
Dominican Republic	48.1	Laos	22.6	Russia	12.3	Zambia	31.6
DR Congo	4.7	Latvia	83.0	Rwanda	56.6	Zimbabwe	10.4

RESEARCH DATA SHEET · RDS-F01

RULE OF LAW

World Bank · Indicator RL.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Asia · Europe

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

ASIA · MONARCHIES

Countries 13 · Mean 66.7 · Median 67.5 · Population SD 15.9

Bahrain***	65.6
Bhutan***	72.6
Brunei****	79.2
Cambodia**	23.6
Japan**	92.5
Jordan**	58.0
Kuwait***	59.0
Malaysia**	67.5
Oman****	70.3
Qatar***	80.2
Saudi Arabia****	62.7
Thailand**	57.5
United Arab Emirates***	78.8

ASIA · REPUBLICS

Countries 33 · Mean 33.2 · Median 32.5 · Population SD 24.4

Afghanistan	5.2	Maldives	50.9
Armenia	49.1	Mongolia	43.9
Azerbaijan	29.2	Myanmar	6.1
Bangladesh	33.5	Nepal	34.9
China	52.8	Pakistan	21.2
Georgia	55.2	Palestine	25.9
India	56.1	Philippines	37.3
Indonesia	46.7	Singapore	98.1
Iran	16.0	Sri Lanka	50.0
Iraq	3.8	Syria	1.4
Israel	77.4	Tajikistan	9.9
Kazakhstan	36.8	Turkey	32.5
Korea, North	5.7	Turkmenistan	6.6
Korea, South	85.8	Uzbekistan	23.1
Kyrgyzstan	12.7	Vietnam	50.5
Laos	22.6	Yemen	1.9
Lebanon	13.2		

EUROPE · MONARCHIES

Countries 11 · Mean 92.6 · Median 93.4 · Population SD 5.8

Andorra**	91.5
Belgium**	88.2
Denmark**	99.5
Liechtenstein***	96.2
Luxembourg**	97.6
Monaco***	91.5
Netherlands**	94.3
Norway**	99.1
Spain**	78.3
Sweden**	93.4
United Kingdom**	89.2

EUROPE · REPUBLICS

Countries 34 · Mean 66.2 · Median 67.0 · Population SD 24.7

Albania	45.8	Kosovo	41.5
Austria	97.2	Latvia	83.0
Belarus	11.3	Lithuania	86.8
Bosnia and Herzegovina	39.6	Malta	75.5
Bulgaria	53.8	Moldova	47.2
Croatia	59.4	Montenegro	52.4
Cyprus	71.7	North Macedonia	45.3
Czech Republic	84.4	Poland	65.1
Estonia	89.6	Portugal	83.5
Finland	100.0	Romania	64.2
France	84.9	Russia	12.3
Germany	92.9	San Marino	91.5
Greece	57.1	Serbia	51.4
Hungary	43.2	Slovakia	68.9
Iceland	96.7	Slovenia	82.5
Ireland	93.9	Switzerland	98.6
Italy	60.8	Ukraine	19.8

EUROPE · REPUBLICS WITHOUT A COMMUNIST PAST

Countries 13 · Mean 84.9 · Median 91.5 · Population SD 13.9

Austria	97.2
Cyprus	71.7
Finland	100.0
France	84.9
Germany	92.9
Greece	57.1
Iceland	96.7
Ireland	93.9
Italy	60.8
Malta	75.5
Portugal	83.5
San Marino	91.5
Switzerland	98.6

RESEARCH DATA SHEET · RDS-F01

RULE OF LAW

World Bank · Indicator RL.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Africa · America
Oceania · Middle East

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

AFRICA · MONARCHIES

Countries 3 · Mean 37.0 · Median 34.0 · Population SD 8.5

Eswatini****	28.3
Lesotho**	34.0
Morocco***	48.6

AFRICA · REPUBLICS

Countries 50 · Mean 27.7 · Median 24.8 · Population SD 19.4

Algeria	26.4	Madagascar	18.9
Angola	14.6	Malawi	46.2
Benin	33.0	Mali	18.4
Botswana	61.3	Mauritania	27.4
Burkina Faso	24.5	Mauritius	77.8
Burundi	10.8	Mozambique	17.5
Cabo Verde	61.8	Namibia	64.6
Cameroon	16.5	Niger	25.0
Central African Republic	3.3	Nigeria	20.3
Chad	8.0	Rwanda	56.6
Comoros	8.5	Sao Tome and Principe	25.5
Congo	17.9	Senegal	42.9
Côte d'Ivoire	35.4	Seychelles	63.7
Djibouti	13.7	Sierra Leone	22.2
DR Congo	4.7	Somalia	0.0
Egypt	44.3	South Africa	54.2
Eritrea	2.8	Sudan	4.2
Ethiopia	26.9	Sudan, South	0.9
Gabon	20.8	Tanzania	38.7
Ghana	49.5	The Gambia	38.2
Guinea	14.2	Togo	32.1
Guinea-Bissau	7.1	Tunisia	47.6
Kenya	41.0	Uganda	34.4
Liberia	17.0	Zambia	31.6
Libya	2.4	Zimbabwe	10.4

AMERICA · COMMONWEALTH

Countries 9 · Mean 61.3 · Median 66.0 · Population SD 16.5

Antigua and Barbuda*	59.9
Bahamas*	55.7
Belize*	27.8
Canada*	90.1
Grenada*	69.3
Jamaica*	44.8
Saint Kitts and Nevis*	66.0
Saint Lucia*	69.8
Saint Vincent and the Grenadines*	67.9

AMERICA · REPUBLICS

Countries 26 · Mean 40.4 · Median 38.5 · Population SD 24.1

Argentina	37.7	Guyana	84.0
Barbados	62.3	Haiti	7.5
Bolivia	11.8	Honduras	15.1
Brazil	42.0	Mexico	24.1
Chile	71.2	Nicaragua	9.4
Colombia	35.8	Panama	40.1
Costa Rica	60.4	Paraguay	28.8
Cuba	36.3	Peru	30.2
Dominica	70.8	Suriname	51.9
Dominican Republic	48.1	Trinidad and Tobago	42.5
Ecuador	19.3	United States	88.7
El Salvador	39.2	Uruguay	75.9
Guatemala	15.6	Venezuela	0.5

OCEANIA

MIDDLE EAST · MONARCHIES

Countries 7 · Mean 67.8

Bahrain***	65.6
Jordan**	58.0
Kuwait***	59.0
Oman****	70.3
Qatar***	80.2
Saudi Arabia****	62.7
United Arab Emirates***	78.8

MIDDLE EAST · REPUBLICS

Countries 10 · Mean 28.8

Cyprus	71.7
Egypt	44.3
Iran	16.0
Iraq	3.8
Israel	77.4
Lebanon	13.2
Palestine	25.9
Syria	1.4
Turkey	32.5
Yemen	1.9

MONARCHY

Countries 1 · Mean 67.0

Tonga***	67.0

COMMONWEALTH

Countries 5 · Mean 68.6

Australia*	92.0
New Zealand*	95.8
Papua New Guinea*	29.7
Solomon Islands*	43.4
Tuvalu*	82.1

REPUBLICS

Countries 9 · Mean 62.5

Fiji	58.5
Kiribati	68.4
Marshall Islands	73.1
Micronesia, Federated States of	62.0
Nauru	53.3
Palau	80.7
Samoa	79.7
Timor-Leste	21.7
Vanuatu	54.7

RESEARCH DATA SHEET · RDS-G01

GOVERNMENT EFFECTIVENESS

World Bank · Indicator GE.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

GLOBAL MONARCHIES

Countries 28 · Mean 74.6 · Median 80.2 · Population SD 23.6

Andorra**	91.0	Malaysia**	79.7
Bahrain***	74.5	Monaco***	99.1
Belgium**	80.7	Morocco***	50.0
Bhutan***	69.3	Netherlands**	95.8
Brunei****	90.1	Norway**	97.6
Cambodia**	38.2	Oman****	62.3
Denmark**	98.6	Qatar***	85.5
Eswatini****	19.3	Saudi Arabia****	78.8
Japan**	96.7	Spain**	76.9
Jordan**	64.6	Sweden**	94.8
Kuwait***	51.4	Thailand**	58.5
Lesotho**	16.5	Tonga***	45.8
Liechtenstein***	96.2	United Arab Emirates***	95.3
Luxembourg**	98.1	United Kingdom**	84.4

GLOBAL COMMONWEALTH

Countries 14 · Mean 58.2 · Median 58.8 · Population SD 22.6

Antigua and Barbuda*	63.7
Australia*	93.9
Bahamas*	61.8
Belize*	36.8
Canada*	92.0
Grenada*	50.9
Jamaica*	66.0
New Zealand*	92.5
Papua New Guinea*	22.2
Saint Kitts and Nevis*	63.7
Saint Lucia*	55.7
Saint Vincent and the Grenadines*	55.7
Solomon Islands*	25.0
Tuvalu*	35.4

GLOBAL REPUBLICS

Countries 151 · Mean 41.9 · Median 40.1 · Population SD 26.9

Afghanistan	1.4	Ecuador	32.5	Lebanon	6.1	Samoa	68.9
Albania	60.4	Egypt	42.0	Liberia	9.0	San Marino	87.3
Algeria	27.4	El Salvador	53.8	Libya	5.2	Sao Tome and Principe	17.9
Angola	14.6	Eritrea	2.8	Lithuania	81.6	Senegal	54.2
Argentina	36.3	Estonia	88.7	Madagascar	13.7	Serbia	51.9
Armenia	43.9	Ethiopia	24.1	Malawi	19.8	Seychelles	70.8
Austria	89.6	Fiji	68.4	Maldives	45.3	Sierra Leone	12.3
Azerbaijan	50.5	Finland	97.2	Mali	12.7	Singapore	100.0
Bangladesh	26.9	France	83.0	Malta	65.6	Slovakia	59.0
Barbados	64.2	Gabon	23.1	Marshall Islands	54.7	Slovenia	81.1
Belarus	15.1	Georgia	77.8	Mauritania	24.5	Somalia	1.9
Benin	42.5	Germany	85.4	Mauritius	75.0	South Africa	40.6
Bolivia	29.2	Ghana	47.2	Mexico	43.4	Sri Lanka	41.0
Bosnia and Herzegovina	15.6	Greece	57.5	Micronesia, Federated States of	59.4	Sudan	2.4
Botswana	67.0	Guatemala	18.4	Moldova	44.3	Sudan, South	0.0
Brazil	32.1	Guinea	17.0	Mongolia	34.0	Suriname	16.0
Bulgaria	53.3	Guinea-Bissau	8.0	Montenegro	59.9	Switzerland	99.5
Burkina Faso	21.2	Guyana	39.6	Mozambique	25.9	Syria	3.3
Burundi	11.3	Haiti	0.9	Myanmar	3.8	Tajikistan	23.6
Cabo Verde	52.4	Honduras	22.6	Namibia	52.8	Tanzania	34.9
Cameroon	17.5	Hungary	62.7	Nauru	48.6	The Gambia	27.9
Central African Republic	4.2	Iceland	93.4	Nepal	21.7	Togo	30.2
Chad	7.5	India	67.9	Nicaragua	13.2	Trinidad and Tobago	49.5
Chile	75.9	Indonesia	69.8	Niger	28.8	Tunisia	39.2
China	73.6	Iran	14.2	Nigeria	20.3	Turkey	41.5
Colombia	48.1	Iraq	8.5	North Macedonia	49.1	Turkmenistan	10.8
Comoros	6.6	Ireland	94.3	Pakistan	30.7	Uganda	31.6
Congo	9.4	Israel	83.5	Palau	67.5	Ukraine	37.7
Costa Rica	60.8	Italy	70.3	Palestine	10.4	United States	87.7
Croatia	75.5	Kazakhstan	57.1	Panama	42.9	Uruguay	79.2
Cuba	33.5	Kenya	38.7	Paraguay	34.4	Uzbekistan	40.1
Cyprus	76.4	Kiribati	46.2	Peru	33.0	Vanuatu	31.1
Czech Republic	82.5	Korea, North	7.1	Philippines	58.0	Venezuela	5.7
Côte d'Ivoire	37.3	Korea, South	90.6	Poland	66.5	Vietnam	56.1
Djibouti	25.5	Kosovo	47.6	Portugal	80.2	Yemen	0.5
Dominica	61.3	Kyrgyzstan	18.9	Romania	46.7	Zambia	27.8
Dominican Republic	56.6	Laos	28.3	Russia	26.4	Zimbabwe	11.7
DR Congo	4.7	Latvia	74.1	Rwanda	65.1		

RESEARCH DATA SHEET · RDS-G01

GOVERNMENT EFFECTIVENESS

World Bank · Indicator GE.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Asia · Europe

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

ASIA · MONARCHIES

Countries 13 · Mean 72.7 · Median 74.5 · Population SD 16.8

Bahrain***	74.5
Bhutan***	69.3
Brunei****	90.1
Cambodia**	38.2
Japan**	96.7
Jordan**	64.6
Kuwait***	51.4
Malaysia**	79.7
Oman****	62.3
Qatar***	85.5
Saudi Arabia****	78.8
Thailand**	58.5
United Arab Emirates***	95.3

ASIA · REPUBLICS

Countries 33 · Mean 37.8 · Median 34.0 · Population SD 28.0

Afghanistan	1.4	Maldives	45.3
Armenia	43.9	Mongolia	34.0
Azerbaijan	50.5	Myanmar	3.8
Bangladesh	26.9	Nepal	21.7
China	73.6	Pakistan	30.7
Georgia	77.8	Palestine	10.4
India	67.9	Philippines	58.0
Indonesia	69.8	Singapore	100.0
Iran	14.2	Sri Lanka	41.0
Iraq	8.5	Syria	3.3
Israel	83.5	Tajikistan	23.6
Kazakhstan	57.1	Turkey	41.5
Korea, North	7.1	Turkmenistan	10.8
Korea, South	90.6	Uzbekistan	40.1
Kyrgyzstan	18.9	Vietnam	56.1
Laos	28.3	Yemen	0.5
Lebanon	6.1		

EUROPE · MONARCHIES

Countries 11 · Mean 92.1 · Median 95.8 · Population SD 7.5

Andorra**	91.0
Belgium**	80.7
Denmark**	98.6
Liechtenstein***	96.2
Luxembourg**	98.1
Monaco***	99.1
Netherlands**	95.8
Norway**	97.6
Spain**	76.9
Sweden**	94.8
United Kingdom**	84.4

EUROPE · REPUBLICS

Countries 34 · Mean 66.5 · Median 68.4 · Population SD 22.1

Albania	60.4	Kosovo	47.6
Austria	89.6	Latvia	74.1
Belarus	15.1	Lithuania	81.6
Bosnia and Herzegovina	15.6	Malta	65.6
Bulgaria	53.3	Moldova	44.3
Croatia	75.5	Montenegro	59.9
Cyprus	76.4	North Macedonia	49.1
Czech Republic	82.5	Poland	66.5
Estonia	88.7	Portugal	80.2
Finland	97.2	Romania	48.7
France	83.0	Russia	26.4
Germany	85.4	San Marino	87.3
Greece	57.5	Serbia	51.9
Hungary	62.7	Slovakia	59.0
Iceland	93.4	Slovenia	81.1
Ireland	94.3	Switzerland	99.5
Italy	70.3	Ukraine	37.7

EUROPE · REPUBLICS WITHOUT A COMMUNIST PAST

Countries 13 · Mean 83.1 · Median 85.4 · Population SD 12.2

Austria	89.6
Cyprus	76.4
Finland	97.2
France	83.0
Germany	85.4
Greece	57.5
Iceland	93.4
Ireland	94.3
Italy	70.3
Malta	65.6
Portugal	80.2
San Marino	87.3
Switzerland	99.5

RESEARCH DATA SHEET · RDS-G01

GOVERNMENT EFFECTIVENESS

World Bank · Indicator GE.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Africa · America
Oceania · Middle East

LEGEND

Republic	Constitutional monarchy **
Commonwealth realm *	Semi-constitutional monarchy ***
	Absolute monarchy ****

AFRICA · MONARCHIES

Countries 3 · Mean 28.6 · Median 19.3 · Population SD 15.2

Eswatini****	19.3
Lesotho**	16.5
Morocco***	50.0

AFRICA · REPUBLICS

Countries 50 · Mean 26.2 · Median 23.6 · Population SD 19.2

Algeria	27.4	Madagascar	13.7
Angola	14.6	Malawi	19.8
Benin	42.5	Mali	12.7
Botswana	67.0	Mauritania	24.5
Burkina Faso	21.2	Mauritius	75.0
Burundi	11.3	Mozambique	25.9
Cabo Verde	52.4	Namibia	52.8
Cameroon	17.5	Niger	28.8
Central African Republic	4.2	Nigeria	20.3
Chad	7.5	Rwanda	65.1
Comoros	6.6	Sao Tome and Principe	17.9
Congo	9.4	Senegal	54.2
Côte d'Ivoire	37.3	Seychelles	70.8
Djibouti	25.5	Sierra Leone	12.3
DR Congo	4.7	Somalia	1.9
Egypt	42.0	South Africa	40.6
Eritrea	2.8	Sudan	2.4
Ethiopia	24.1	Sudan, South	0.0
Gabon	23.1	Tanzania	34.9
Ghana	47.2	The Gambia	27.9
Guinea	17.0	Togo	30.2
Guinea-Bissau	8.0	Tunisia	39.2
Kenya	38.7	Uganda	31.6
Liberia	9.0	Zambia	27.8
Libya	5.2	Zimbabwe	11.7

AMERICA · COMMONWEALTH

Countries 9 · Mean 60.7 · Median 61.8 · Population SD 13.9

Antigua and Barbuda*	63.7
Bahamas*	61.8
Belize*	36.8
Canada*	92.0
Grenada*	50.9
Jamaica*	66.0
Saint Kitts and Nevis*	63.7
Saint Lucia*	55.7
Saint Vincent and the Grenadines*	55.7

AMERICA · REPUBLICS

Countries 26 · Mean 41.2 · Median 38.0 · Population SD 21.8

Argentina	36.3	Guyana	39.6
Barbados	64.2	Haiti	0.9
Bolivia	29.2	Honduras	22.6
Brazil	32.1	Mexico	43.4
Chile	75.9	Nicaragua	13.2
Colombia	48.1	Panama	42.9
Costa Rica	60.8	Paraguay	34.4
Cuba	33.5	Peru	33.0
Dominica	61.3	Suriname	16.0
Dominican Republic	56.6	Trinidad and Tobago	49.5
Ecuador	32.5	United States	87.7
El Salvador	53.8	Uruguay	79.2
Guatemala	18.4	Venezuela	5.7

OCEANIA

MIDDLE EAST · MONARCHIES

Countries 7 · Mean 73.2

Bahrain***	74.5
Jordan**	64.6
Kuwait***	51.4
Oman****	62.3
Qatar***	85.5
Saudi Arabia*****	78.8
United Arab Emirates***	95.3

MIDDLE EAST · REPUBLICS

Countries 10 · Mean 28.6

Cyprus	76.4
Egypt	42.0
Iran	14.2
Iraq	8.5
Israel	83.5
Lebanon	6.1
Palestine	10.4
Syria	3.3
Turkey	41.5
Yemen	0.5

MONARCHY

Countries 1 · Mean 45.8

Tonga***	45.8

COMMONWEALTH

Countries 5 · Mean 53.8

Australia*	93.9
New Zealand*	92.5
Papua New Guinea*	22.2
Solomon Islands*	25.0
Tuvalu*	35.4

REPUBLICS

Countries 8 · Mean 55.6

Fiji	68.4
Kiribati	46.2
Marshall Islands	54.7
Micronesia, Federated States of	58.7
Nauru	48.6
Palau	67.5
Samoa	68.9
Vanuatu	31.1

Data Source
World Bank Open Data
Indicator: GE.PER.RNK
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-H01

INFLATION · 2019–2023 AVERAGE

World Bank · Indicator FP.CPI.TOTL.ZG · Dataset: 2019–2023 Average

SCAN SOURCE

LEGEND

Republic
Commonwealth realm *
Constitutional monarchy **
Semi-constitutional monarchy ***
Absolute monarchy ****

GLOBAL MONARCHIES

Countries 25 · Mean 2.8 · Median 3.0 · Population SD 1.4

Bahrain***	0.4	Morocco***	3.0
Belgium**	3.6	Netherlands**	4.1
Bhutan***	5.0	Norway**	3.6
Brunei****	1.5	Oman****	0.9
Cambodia**	3.1	Qatar***	1.4
Denmark**	2.8	Saudi Arabia****	1.8
Japan**	1.2	Spain**	3.1
Jordan**	1.7	Sweden**	4.2
Kuwait***	2.6	Thailand**	2.2
Lesotho**	6.2	Tonga***	4.5
Liechtenstein***	3.0	United Arab Emirates***	0.5
Luxembourg**	3.0	United Kingdom**	4.0
Malaysia**	1.6		

GLOBAL COMMONWEALTH

Countries 14 · Mean 3.2 · Median 3.1 · Population SD 1.3

Antigua and Barbuda*	3.4
Australia*	3.5
Bahamas*	2.7
Belize*	2.9
Canada*	3.3
Grenada*	1.3
Jamaica*	6.4
New Zealand*	4.0
Papua New Guinea*	3.7
Saint Kitts and Nevis*	1.2
Saint Lucia*	2.7
Saint Vincent and the Grenadines*	2.4
Solomon Islands*	2.9
Tuvalu*	4.9

GLOBAL REPUBLICS

Countries 148 · Mean 66.9 · Median 5.3 · Population SD 492.5

Afghanistan	5.9	Ecuador	1.1	Lebanon	130.7	Rwanda	9.9
Albania	3.5	Egypt	14.5	Liberia	11.4	Samoa	3.3
Algeria	6.0	El Salvador	2.9	Libya	7.3	San Marino	3.0
Angola	19.9	Eritrea	2.3	Lithuania	7.4	Sao Tome and Principe	13.2
Argentina	69.0	Estonia	7.1	Madagascar	6.7	Senegal	4.6
Armenia	4.1	Ethiopia	25.9	Malawi	15.4	Serbia	6.4
Austria	4.4	Fiji	1.0	Maldives	1.2	Seychelles	3.4
Azerbaijan	7.0	Finland	3.4	Mali	2.7	Sierra Leone	23.0
Bangladesh	6.8	France	2.7	Malta	3.0	Singapore	2.7
Barbados	6.1	Gabon	2.5	Marshall Islands	3.5	Slovakia	6.2
Belarus	8.2	Georgia	6.8	Mauritania	4.6	Slovenia	4.0
Benin	2.8	Germany	3.6	Mauritius	5.0	South Africa	5.0
Bolivia	1.6	Ghana	19.3	Mexico	5.2	Sri Lanka	15.8
Bosnia and Herzegovina	4.3	Greece	2.7	Micronesia, Federated States of	3.2	Sudan	3561.6
Botswana	5.7	Guatemala	4.9	Moldova	11.9	Sudan, South	21.4
Brazil	5.8	Guinea	10.2	Mongolia	8.7	Suriname	44.7
Bulgaria	6.3	Guinea-Bissau	4.9	Montenegro	4.8	Switzerland	1.0
Burkina Faso	3.5	Guyana	3.0	Mozambique	5.9	Tajikistan	7.1
Burundi	12.1	Haiti	26.7	Myanmar	12.6	Tanzania	3.8
Cabo Verde	3.2	Honduras	5.6	Namibia	4.3	The Gambia	9.8
Cameroon	4.2	Hungary	8.7	Nauru	3.0	Timor-Leste	4.1
Central African Republic	3.6	Iceland	5.5	Nepal	5.6	Togo	4.0
Chad	3.4	India	5.5	Nicaragua	6.8	Trinidad and Tobago	3.0
Chile	5.8	Indonesia	2.8	Niger	2.4	Tunisia	7.1
China	1.7	Iran	40.3	Nigeria	16.7	Turkey	34.6
Colombia	6.3	Iraq	3.2	North Macedonia	5.8	Turkmenistan	7.5
Comoros	5.4	Ireland	3.4	Pakistan	15.3	Uganda	4.3
Congo	2.8	Israel	2.1	Palau	5.3	Ukraine	10.6
Costa Rica	2.7	Italy	3.2	Palestine	2.3	United States	4.0
Croatia	4.5	Kazakhstan	9.8	Panama	0.8	Uruguay	8.1
Cyprus	2.5	Kenya	6.4	Paraguay	4.7	Uzbekistan	11.9
Czech Republic	7.1	Kiribati	3.6	Peru	4.6	Vanuatu	5.1
Côte d'Ivoire	3.4	Korea, South	2.4	Philippines	4.2	Venezuela	4877.6
Djibouti	2.9	Kosovo	4.6	Poland	7.3	Vietnam	2.9
Dominica	2.9	Kyrgyzstan	8.8	Portugal	2.7	Yemen	21.9
Dominican Republic	5.5	Laos	13.3	Romania	7.1	Zambia	13.9
DR Congo	10.8	Latvia	6.5	Russia	6.4	Zimbabwe	224.2

Data Source
World Bank DataBank
Indicator: FP.CPI.TOTL.ZG
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-H01

INFLATION · 2019–2023 AVERAGE

World Bank · Indicator FP.CPI.TOTL.ZG · Dataset: 2019–2023 Average

SCAN SOURCE

REGIONAL COMPARISONS

Asia · Europe

LEGEND

Republic
Commonwealth realm *
Constitutional monarchy **
Semi-constitutional monarchy ***
Absolute monarchy ****

ASIA · MONARCHIES

Countries 13 · Mean 1.9 · Median 1.6 · Population SD 1.2

Bahrain***	0.4
Bhutan***	5.0
Brunei****	1.5
Cambodia**	3.1
Japan**	1.2
Jordan**	1.7
Kuwait***	2.6
Malaysia**	1.6
Oman****	0.9
Qatar***	1.4
Saudi Arabia****	1.8
Thailand**	2.2
United Arab Emirates***	0.5

ASIA · REPUBLICS

Countries 31 · Mean 13.1 · Median 6.8 · Population SD 23.2

Afghanistan	5.9	Maldives	1.2
Armenia	4.1	Mongolia	8.7
Azerbaijan	7.0	Myanmar	12.6
Bangladesh	6.8	Nepal	5.6
China	1.7	Pakistan	15.3
Georgia	6.8	Palestine	2.3
India	5.5	Philippines	4.2
Indonesia	2.8	Singapore	2.7
Iran	40.3	Sri Lanka	15.8
Iraq	3.2	Tajikistan	7.1
Israel	2.1	Turkey	34.6
Kazakhstan	9.8	Turkmenistan	7.5
Korea, South	2.4	Uzbekistan	11.9
Kyrgyzstan	8.8	Vietnam	2.9
Laos	13.3	Yemen	21.9
Lebanon	130.7		

EUROPE · MONARCHIES

Countries 9 · Mean 3.5 · Median 3.6 · Population SD 0.5

Belgium**	3.6
Denmark**	2.8
Liechtenstein***	3.0
Luxembourg**	3.0
Netherlands**	4.1
Norway**	3.6
Spain**	3.1
Sweden**	4.2
United Kingdom**	4.0

EUROPE · REPUBLICS

Countries 34 · Mean 5.3 · Median 4.7 · Population SD 2.4

Albania	3.5	Kosovo	4.6
Austria	4.4	Latvia	6.5
Belarus	8.2	Lithuania	7.4
Bosnia and Herzegovina	4.3	Malta	3.0
Bulgaria	6.3	Moldova	11.9
Croatia	4.5	Montenegro	4.8
Cyprus	2.5	North Macedonia	5.8
Czech Republic	7.1	Poland	7.3
Estonia	7.1	Portugal	2.7
Finland	3.4	Romania	7.1
France	2.7	Russia	6.4
Germany	3.6	San Marino	3.0
Greece	2.7	Serbia	6.4
Hungary	8.7	Slovakia	6.2
Iceland	5.5	Slovenia	4.0
Ireland	3.4	Switzerland	1.0
Italy	3.2	Ukraine	10.6

EUROPE · REPUBLICS WITHOUT A COMMUNIST PAST

Countries 13 · Mean 3.2 · Median 3.0 · Population SD 1.0

Austria	4.4
Cyprus	2.5
Finland	3.4
France	2.7
Germany	3.6
Greece	2.7
Iceland	5.5
Ireland	3.4
Italy	3.2
Malta	3.0
Portugal	2.7
San Marino	3.0
Switzerland	1.0

RESEARCH DATA SHEET · RDS-H01

INFLATION · 2019–2023 AVERAGE

World Bank · Indicator FP.CPI.TOTL.ZG · Dataset: 2019–2023 Average

SCAN SOURCE

REGIONAL COMPARISONS

Africa · America
Oceania · Middle East

LEGEND

Republic
Commonwealth realm *
Constitutional monarchy **
Semi-constitutional monarchy ***
Absolute monarchy ****

AFRICA · MONARCHIES

Countries 2 · Mean 4.6 · Median 4.6 · Population SD 1.6

Lesotho**	6.2
Morocco***	3.0

AFRICA · REPUBLICS

Countries 49 · Mean 85.0 · Median 5.7 · Population SD 502.8

Algeria	6.0	Madagascar	6.7
Angola	19.9	Malawi	15.4
Benin	2.8	Mali	2.7
Botswana	5.7	Mauritania	4.6
Burkina Faso	3.5	Mauritius	5.0
Burundi	12.1	Mozambique	5.9
Cabo Verde	3.2	Namibia	4.3
Cameroon	4.2	Niger	2.4
Central African Republic	3.6	Nigeria	16.7
Chad	3.4	Rwanda	9.9
Comoros	5.4	Sao Tome and Principe	13.2
Congo	2.8	Senegal	4.6
Côte d'Ivoire	3.4	Seychelles	3.4
Djibouti	2.9	Sierra Leone	23.0
DR Congo	10.8	South Africa	5.0
Egypt	14.5	Sudan	3561.6
Eritrea	2.3	Sudan, South	21.4
Ethiopia	25.9	Tanzania	3.8
Gabon	2.5	The Gambia	9.8
Ghana	19.3	Togo	4.0
Guinea	10.2	Tunisia	7.1
Guinea-Bissau	4.9	Uganda	4.3
Kenya	6.4	Zambia	13.9
Liberia	11.4	Zimbabwe	224.2
Libya	7.3		

AMERICA · COMMONWEALTH

Countries 9 · Mean 2.9 · Median 2.7 · Population SD 1.4

Antigua and Barbuda*	3.4
Bahamas*	2.7
Belize*	2.9
Canada*	3.3
Grenada*	1.3
Jamaica*	6.4
Saint Kitts and Nevis*	1.2
Saint Lucia*	2.7
Saint Vincent and the Grenadines*	2.4

AMERICA · REPUBLICS

Countries 25 · Mean 204.4 · Median 5.2 · Population SD 954.0

Argentina	69.0	Haiti	26.7
Barbados	6.1	Honduras	5.6
Bolivia	1.6	Mexico	5.2
Brazil	5.8	Nicaragua	6.8
Chile	5.8	Panama	0.8
Colombia	6.3	Paraguay	4.7
Costa Rica	2.7	Peru	4.6
Dominica	2.9	Suriname	44.7
Dominican Republic	5.5	Trinidad and Tobago	3.0
Ecuador	1.1	United States	4.0
El Salvador	2.9	Uruguay	8.1
Guatemala	4.9	Venezuela	4877.6
Guyana	3.0		

MIDDLE EAST · MONARCHIES

Countries 7 · Mean 1.4

Bahrain***	0.4
Jordan**	1.7
Kuwait***	2.6
Oman****	0.9
Qatar***	1.4
Saudi Arabia*****	1.8
United Arab Emirates***	0.5

MIDDLE EAST · REPUBLICS

Countries 9 · Mean 28.0

Cyprus	2.5
Egypt	14.5
Iran	40.3
Iraq	3.2
Israel	2.1
Lebanon	130.7
Palestine	2.3
Turkey	34.6
Yemen	21.9

OCEANIA

MONARCHY

Countries 1 · Mean 4.5

Tonga***	4.5

COMMONWEALTH

Countries 5 · Mean 3.8

Australia*	3.5
New Zealand*	4.0
Papua New Guinea*	3.7
Solomon Islands*	2.9
Tuvalu*	4.9

REPUBLICS

Countries 9 · Mean 3.6

Fiji	1.0
Kiribati	3.6
Marshall Islands	3.5
Micronesia, Federated States of	3.7
Nauru	3.0
Palau	5.3
Samoa	3.3
Timor-Leste	4.1
Vanuatu	5.1

RESEARCH DATA SHEET · RDS-I01

CONTROL OF CORRUPTION

World Bank · Indicator CC.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

LEGEND

Republic
Commonwealth realm *
Constitutional monarchy **
Semi-constitutional monarchy ***
Absolute monarchy ****

GLOBAL MONARCHIES

Countries 28 · Mean 72.0 · Median 85.1 · Population SD 25.5

Andorra**	87.7	Malaysia**	61.8
Bahrain***	88.2	Monaco***	87.7
Belgium**	89.2	Morocco***	33.0
Bhutan***	91.0	Netherlands**	96.2
Brunei****	88.7	Norway**	99.1
Cambodia**	9.4	Oman****	60.8
Denmark**	100.0	Qatar***	75.0
Eswatini****	25.0	Saudi Arabia****	66.5
Japan**	90.1	Spain**	71.7
Jordan**	57.5	Sweden**	97.6
Kuwait***	60.4	Thailand**	35.8
Lesotho**	33.5	Tonga***	45.3
Liechtenstein***	95.3	United Arab Emirates***	82.5
Luxembourg**	96.7	United Kingdom**	90.6

GLOBAL COMMONWEALTH

Countries 14 · Mean 68.9 · Median 67.7 · Population SD 20.0

Antigua and Barbuda*	62.3
Australia*	95.8
Bahamas*	88.2
Belize*	48.1
Canada*	94.8
Grenada*	68.4
Jamaica*	50.9
New Zealand*	98.6
Papua New Guinea*	27.4
Saint Kitts and Nevis*	64.6
Saint Lucia*	67.0
Saint Vincent and the Grenadines*	76.4
Solomon Islands*	50.5
Tuvalu*	72.2

GLOBAL REPUBLICS

Countries 152 · Mean 41.8 · Median 39.0 · Population SD 27.2

Afghanistan	13.7	Ecuador	27.8	Lebanon	10.8	Samoa	75.9
Albania	43.4	Egypt	24.5	Liberia	20.8	San Marino	87.7
Algeria	30.2	El Salvador	32.1	Libya	3.8	Sao Tome and Principe	63.2
Angola	29.2	Eritrea	5.7	Lithuania	77.8	Senegal	56.6
Argentina	42.0	Estonia	91.5	Madagascar	18.4	Serbia	38.2
Armenia	57.1	Ethiopia	37.3	Malawi	29.7	Seychelles	93.9
Austria	83.5	Fiji	66.0	Maldives	39.6	Sierra Leone	31.1
Azerbaijan	12.3	Finland	99.5	Mali	21.2	Singapore	98.1
Bangladesh	14.6	France	84.0	Malta	58.5	Slovakia	61.3
Barbados	89.6	Gabon	17.9	Marshall Islands	64.2	Slovenia	77.4
Belarus	26.9	Georgia	71.2	Mauritania	22.6	Somalia	0.9
Benin	52.8	Germany	94.3	Mauritius	65.6	South Africa	45.8
Bolivia	21.7	Ghana	51.4	Mexico	17.5	Sri Lanka	40.1
Bosnia and Herzegovina	30.7	Greece	58.0	Micronesia, Federated States of	78.3	Sudan	4.2
Botswana	73.6	Guatemala	14.2	Moldova	46.7	Sudan, South	0.0
Brazil	34.7	Guinea	20.3	Mongolia	35.4	Suriname	39.2
Bulgaria	50.0	Guinea-Bissau	12.7	Montenegro	51.9	Switzerland	97.2
Burkina Faso	49.5	Guyana	40.6	Mozambique	22.2	Syria	0.5
Burundi	3.3	Haiti	6.1	Myanmar	11.8	Tajikistan	7.5
Cabo Verde	81.1	Honduras	15.1	Namibia	59.0	Tanzania	43.9
Cameroon	13.2	Hungary	54.7	Nauru	69.8	The Gambia	46.2
Central African Republic	9.0	Iceland	92.0	Nepal	34.0	Timor-Leste	47.6
Chad	4.7	India	41.5	Nicaragua	7.1	Togo	28.8
Chile	80.7	Indonesia	36.3	Niger	31.6	Trinidad and Tobago	41.0
China	54.2	Iran	10.4	Nigeria	17.0	Tunisia	42.9
Colombia	44.8	Iraq	8.5	North Macedonia	42.5	Turkey	34.9
Comoros	19.3	Ireland	92.9	Pakistan	18.9	Turkmenistan	6.6
Congo	8.0	Israel	78.8	Palau	69.8	Uganda	16.5
Costa Rica	72.6	Italy	67.9	Palestine	26.4	Ukraine	25.9
Croatia	59.9	Kazakhstan	47.2	Panama	28.3	United States	83.0
Cuba	52.4	Kenya	24.1	Paraguay	16.0	Uruguay	92.5
Cyprus	62.7	Kiribati	64.2	Peru	25.5	Uzbekistan	23.1
Czech Republic	76.9	Korea, North	2.4	Philippines	32.5	Vanuatu	53.8
Côte d'Ivoire	44.3	Korea, South	79.7	Poland	68.9	Venezuela	1.4
Djibouti	23.6	Kosovo	48.6	Portugal	74.1	Vietnam	38.7
Dominica	67.5	Kyrgyzstan	11.3	Romania	56.1	Yemen	1.9
Dominican Republic	37.7	Laos	19.8	Russia	15.6	Zambia	36.8
DR Congo	5.2	Latvia	74.5	Rwanda	73.1	Zimbabwe	9.9

RESEARCH DATA SHEET · RDS-I01

CONTROL OF CORRUPTION

World Bank · Indicator CC.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Asia · Europe

LEGEND

Republic
Commonwealth realm *
Constitutional monarchy **
Semi-constitutional monarchy ***
Absolute monarchy ****

ASIA · MONARCHIES

Countries 13 · Mean 66.7 · Median 66.5 · Population SD 22.9

Bahrain***	88.2
Bhutan***	91.0
Brunei****	88.7
Cambodia**	9.4
Japan**	90.1
Jordan**	57.5
Kuwait***	60.4
Malaysia**	61.8
Oman****	60.8
Qatar***	75.0
Saudi Arabia****	66.5
Thailand**	35.8
United Arab Emirates***	82.5

ASIA · REPUBLICS

Countries 33 · Mean 30.9 · Median 26.4 · Population SD 24.5

Afghanistan	13.7	Maldives	39.6
Armenia	57.1	Mongolia	35.4
Azerbaijan	12.3	Myanmar	11.8
Bangladesh	14.6	Nepal	34.0
China	54.2	Pakistan	18.9
Georgia	71.2	Palestine	26.4
India	41.5	Philippines	32.5
Indonesia	36.3	Singapore	98.1
Iran	10.4	Sri Lanka	40.1
Iraq	8.5	Syria	0.5
Israel	78.8	Tajikistan	7.5
Kazakhstan	47.2	Turkey	34.9
Korea, North	2.4	Turkmenistan	6.6
Korea, South	79.7	Uzbekistan	23.1
Kyrgyzstan	11.3	Vietnam	38.7
Laos	19.8	Yemen	1.9
Lebanon	10.8		

EUROPE · MONARCHIES

Countries 11 · Mean 92.0 · Median 95.3 · Population SD 7.7

Andorra**	87.7
Belgium**	89.2
Denmark**	100.0
Liechtenstein***	95.3
Luxembourg**	96.7
Monaco***	87.7
Netherlands**	96.2
Norway**	99.1
Spain**	71.7
Sweden**	97.6
United Kingdom**	90.6

EUROPE · REPUBLICS

Countries 34 · Mean 63.9 · Median 62.0 · Population SD 22.2

Albania	43.4	Kosovo	48.6
Austria	83.5	Latvia	74.5
Belarus	26.9	Lithuania	77.8
Bosnia and Herzegovina	30.7	Malta	58.5
Bulgaria	50.0	Moldova	46.7
Croatia	59.9	Montenegro	51.9
Cyprus	62.7	North Macedonia	42.5
Czech Republic	76.9	Poland	68.9
Estonia	91.5	Portugal	74.1
Finland	99.5	Romania	56.1
France	84.0	Russia	15.6
Germany	94.3	San Marino	87.7
Greece	58.0	Serbia	38.2
Hungary	54.7	Slovakia	61.3
Iceland	92.0	Slovenia	77.4
Ireland	92.9	Switzerland	97.2
Italy	67.9	Ukraine	25.9

EUROPE · REPUBLICS WITHOUT A COMMUNIST PAST

Countries 13 · Mean 80.9 · Median 84.0 · Population SD 14.4

Austria	83.5
Cyprus	62.7
Finland	99.5
France	84.0
Germany	94.3
Greece	58.0
Iceland	92.0
Ireland	92.9
Italy	67.9
Malta	58.5
Portugal	74.1
San Marino	87.7
Switzerland	97.2

RESEARCH DATA SHEET · RDS-I01

CONTROL OF CORRUPTION

World Bank · Indicator CC.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Africa · America
Oceania · Middle East

LEGEND

Republic
Commonwealth realm *
Constitutional monarchy **
Semi-constitutional monarchy ***
Absolute monarchy ****

AFRICA · MONARCHIES

Countries 3 · Mean 30.5 · Median 33.0 · Population SD 3.9

Eswatini****	25.0
Lesotho**	33.5
Morocco***	33.0

AFRICA · REPUBLICS

Countries 50 · Mean 30.9 · Median 24.3 · Population SD 22.7

Algeria	30.2	Madagascar	18.4
Angola	29.2	Malawi	29.7
Benin	52.8	Mali	21.2
Botswana	73.6	Mauritania	22.6
Burkina Faso	49.5	Mauritius	65.6
Burundi	3.3	Mozambique	22.2
Cabo Verde	81.1	Namibia	59.0
Cameroon	13.2	Niger	31.6
Central African Republic	9.0	Nigeria	17.0
Chad	4.7	Rwanda	73.1
Comoros	19.3	Sao Tome and Principe	63.2
Congo	8.0	Senegal	56.6
Côte d'Ivoire	44.3	Seychelles	93.9
Djibouti	23.6	Sierra Leone	31.1
DR Congo	5.2	Somalia	0.9
Egypt	24.5	South Africa	45.8
Eritrea	5.7	Sudan	4.2
Ethiopia	37.3	Sudan, South	0.0
Gabon	17.9	Tanzania	43.9
Ghana	51.4	The Gambia	46.2
Guinea	20.3	Togo	28.8
Guinea-Bissau	12.7	Tunisia	42.9
Kenya	24.1	Uganda	16.5
Liberia	20.8	Zambia	36.8
Libya	3.8	Zimbabwe	9.9

AMERICA · COMMONWEALTH

Countries 9 · Mean 69.0 · Median 67.0 · Population SD 14.6

Antigua and Barbuda*	62.3
Bahamas*	88.2
Belize*	48.1
Canada*	94.8
Grenada*	68.4
Jamaica*	50.9
Saint Kitts and Nevis*	64.6
Saint Lucia*	67.0
Saint Vincent and the Grenadines*	76.4

AMERICA · REPUBLICS

Countries 26 · Mean 39.7 · Median 36.2 · Population SD 26.1

Argentina	42.0	Guyana	40.6
Barbados	89.6	Haiti	6.1
Bolivia	21.7	Honduras	15.1
Brazil	34.7	Mexico	17.5
Chile	80.7	Nicaragua	7.1
Colombia	44.8	Panama	28.3
Costa Rica	72.6	Paraguay	16.0
Cuba	52.4	Peru	25.5
Dominica	67.5	Suriname	39.2
Dominican Republic	37.7	Trinidad and Tobago	41.0
Ecuador	27.8	United States	83.0
El Salvador	32.1	Uruguay	92.5
Guatemala	14.2	Venezuela	1.4

OCEANIA

MIDDLE EAST · MONARCHIES

Countries 7 · Mean 70.1

Bahrain***	88.2
Jordan**	57.5
Kuwait***	60.4
Oman****	60.8
Qatar***	75.0
Saudi Arabia*****	66.5
United Arab Emirates***	82.5

MIDDLE EAST · REPUBLICS

Countries 10 · Mean 25.9

Cyprus	62.7
Egypt	24.5
Iran	10.4
Iraq	8.5
Israel	78.8
Lebanon	10.8
Palestine	26.4
Syria	0.5
Turkey	34.9
Yemen	1.9

MONARCHY

Countries 1 · Mean 45.3

Tonga***	45.3

COMMONWEALTH

Countries 5 · Mean 68.9

Australia*	95.8
New Zealand*	98.6
Papua New Guinea*	27.4
Solomon Islands*	50.5
Tuvalu*	72.2

REPUBLICS

Countries 9 · Mean 65.5

Fiji	66.0
Kiribati	64.2
Marshall Islands	64.2
Micronesia, Federated States of	61.7
Nauru	69.8
Palau	69.8
Samoa	75.9
Timor-Leste	47.6
Vanuatu	53.8

RESEARCH DATA SHEET · RDS-J01

CORRUPTION PERCEPTIONS INDEX

Transparency International · CPI Score (0–100) · Dataset: 2023 Edition

SCAN SOURCE

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

GLOBAL MONARCHIES

Countries 24 · Mean 57.8 · Median 59.0 · Population SD 18.5

Bahrain***	42.0	Malaysia**	50.0
Belgium**	73.0	Morocco***	38.0
Bhutan***	68.0	Netherlands**	79.0
Brunei****	60.0	Norway**	84.0
Cambodia**	22.0	Oman****	43.0
Denmark**	90.0	Qatar***	58.0
Eswatini****	30.0	Saudi Arabia****	52.0
Japan**	73.0	Spain**	60.0
Jordan**	46.0	Sweden**	82.0
Kuwait***	46.0	Thailand**	35.0
Lesotho**	39.0	United Arab Emirates***	68.0
Luxembourg**	78.0	United Kingdom**	71.0

GLOBAL COMMONWEALTH

Countries 9 · Mean 58.1 · Median 58.0 · Population SD 17.1

Australia*	75.0
Canada*	76.0
Grenada*	53.0
Jamaica*	44.0
New Zealand*	85.0
Papua New Guinea*	29.0
Saint Lucia*	58.0
Saint Vincent and the Grenadines*	60.0
Solomon Islands*	43.0

GLOBAL REPUBLICS

Countries 141 · Mean 39.2 · Median 36.0 · Population SD 16.9

Afghanistan	20.0	Dominican Republic	35.0	Kyrgyzstan	26.0	Senegal	43.0
Albania	37.0	DR Congo	20.0	Laos	28.0	Serbia	36.0
Algeria	36.0	Ecuador	34.0	Latvia	60.0	Seychelles	71.0
Angola	33.0	Egypt	35.0	Lebanon	24.0	Sierra Leone	35.0
Argentina	37.0	El Salvador	31.0	Liberia	25.0	Singapore	83.0
Armenia	46.0	Eritrea	21.0	Libya	18.0	Slovakia	54.0
Austria	71.0	Estonia	76.0	Lithuania	61.0	Slovenia	56.0
Azerbaijan	23.0	Ethiopia	37.0	Madagascar	25.0	Somalia	11.0
Bangladesh	24.0	Fiji	52.0	Malawi	34.0	South Africa	41.0
Barbados	69.0	Finland	87.0	Maldives	39.0	Sri Lanka	34.0
Belarus	37.0	France	71.0	Mali	28.0	Sudan	20.0
Benin	43.0	Gabon	28.0	Mauritania	30.0	Sudan, South	13.0
Bolivia	29.0	Georgia	53.0	Mauritius	51.0	Suriname	40.0
Bosnia and Herzegovina	35.0	Germany	78.0	Mexico	31.0	Switzerland	82.0
Botswana	59.0	Ghana	43.0	Moldova	42.0	Syria	13.0
Brazil	36.0	Greece	49.0	Mongolia	33.0	Tajikistan	20.0
Bulgaria	45.0	Guatemala	23.0	Montenegro	46.0	Tanzania	40.0
Burkina Faso	41.0	Guinea	26.0	Mozambique	25.0	Togo	31.0
Burundi	20.0	Guinea-Bissau	22.0	Myanmar	20.0	Trinidad and Tobago	42.0
Cabo Verde	64.0	Guyana	40.0	Namibia	49.0	Tunisia	40.0
Cameroon	27.0	Haiti	17.0	Nepal	35.0	Turkey	34.0
Central African Republic	24.0	Honduras	23.0	Nicaragua	17.0	Turkmenistan	18.0
Chad	20.0	Hungary	42.0	Niger	32.0	Uganda	26.0
Chile	66.0	Iceland	72.0	Nigeria	25.0	Ukraine	36.0
China	42.0	India	39.0	North Macedonia	42.0	United States	69.0
Colombia	40.0	Indonesia	34.0	Pakistan	29.0	Uruguay	73.0
Comoros	20.0	Iran	24.0	Panama	35.0	Uzbekistan	33.0
Congo	22.0	Iraq	23.0	Paraguay	28.0	Vanuatu	48.0
Costa Rica	55.0	Ireland	77.0	Peru	33.0	Venezuela	13.0
Croatia	50.0	Israel	62.0	Philippines	34.0	Vietnam	41.0
Cuba	42.0	Italy	56.0	Poland	54.0	Yemen	16.0
Cyprus	53.0	Kazakhstan	39.0	Portugal	61.0	Zambia	37.0
Czech Republic	57.0	Kenya	31.0	Romania	46.0	Zimbabwe	24.0
Côte d'Ivoire	40.0	Korea, North	17.0	Russia	26.0		
Djibouti	30.0	Korea, South	63.0	Rwanda	53.0		
Dominica	56.0	Kosovo	41.0	Sao Tome and Principe	45.0		

Data Source
Transparency International · CPI 2023
Indicator: CPI Score (0–100)
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-J01

CORRUPTION PERCEPTIONS INDEX

Transparency International · CPI Score (0–100) · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Asia · Europe

LEGEND

Republic
Commonwealth realm *
Constitutional monarchy **
Semi-constitutional monarchy ***
Absolute monarchy ****

ASIA · MONARCHIES

Countries 13 · Mean 51.0 · Median 50.0 · Population SD 13.8

Bahrain***	42.0
Bhutan***	68.0
Brunei****	60.0
Cambodia**	22.0
Japan**	73.0
Jordan**	46.0
Kuwait***	46.0
Malaysia**	50.0
Oman****	43.0
Qatar***	58.0
Saudi Arabia****	52.0
Thailand**	35.0
United Arab Emirates***	68.0

ASIA · REPUBLICS

Countries 32 · Mean 33.4 · Median 33.0 · Population SD 15.1

Afghanistan	20.0	Lebanon	24.0
Armenia	46.0	Maldives	39.0
Azerbaijan	23.0	Mongolia	33.0
Bangladesh	24.0	Myanmar	20.0
China	42.0	Nepal	35.0
Georgia	53.0	Pakistan	29.0
India	39.0	Philippines	34.0
Indonesia	34.0	Singapore	83.0
Iran	24.0	Sri Lanka	34.0
Iraq	23.0	Syria	13.0
Israel	62.0	Tajikistan	20.0
Kazakhstan	39.0	Turkey	34.0
Korea, North	17.0	Turkmenistan	18.0
Korea, South	63.0	Uzbekistan	33.0
Kyrgyzstan	26.0	Vietnam	41.0
Laos	28.0	Yemen	16.0

EUROPE · MONARCHIES

Countries 8 · Mean 77.1 · Median 78.5 · Population SD 8.6

Belgium**	73.0
Denmark**	90.0
Luxembourg**	78.0
Netherlands**	79.0
Norway**	84.0
Spain**	60.0
Sweden**	82.0
United Kingdom**	71.0

EUROPE · REPUBLICS

Countries 32 · Mean 54.2 · Median 53.5 · Population SD 15.5

Albania	37.0	Italy	56.0
Austria	71.0	Kosovo	41.0
Belarus	37.0	Latvia	60.0
Bosnia and Herzegovina	35.0	Lithuania	61.0
Bulgaria	45.0	Moldova	42.0
Croatia	50.0	Montenegro	46.0
Cyprus	53.0	North Macedonia	42.0
Czech Republic	57.0	Poland	54.0
Estonia	76.0	Portugal	61.0
Finland	87.0	Romania	46.0
France	71.0	Russia	26.0
Germany	78.0	Serbia	36.0
Greece	49.0	Slovakia	54.0
Hungary	42.0	Slovenia	56.0
Iceland	72.0	Switzerland	82.0
Ireland	77.0	Ukraine	36.0

EUROPE · REPUBLICS WITHOUT A COMMUNIST PAST

Countries 11 · Mean 68.8 · Median 71.0 · Population SD 11.8

Austria	71.0
Cyprus	53.0
Finland	87.0
France	71.0
Germany	78.0
Greece	49.0
Iceland	72.0
Ireland	77.0
Italy	56.0
Portugal	61.0
Switzerland	82.0

Data Source
Transparency International · CPI 2023
Indicator: CPI Score (0–100)
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-J01

CORRUPTION PERCEPTIONS INDEX

Transparency International · CPI Score (0–100) · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Africa · America
Oceania · Middle East

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

AFRICA · MONARCHIES

Countries 3 · Mean 35.7 · Median 38.0 · Population SD 4.0

Eswatini****	30.0
Lesotho**	39.0
Morocco***	38.0

AFRICA · REPUBLICS

Countries 49 · Mean 32.9 · Median 31.0 · Population SD 12.6

Algeria	36.0	Madagascar	25.0
Angola	33.0	Malawi	34.0
Benin	43.0	Mali	28.0
Botswana	59.0	Mauritania	30.0
Burkina Faso	41.0	Mauritius	51.0
Burundi	20.0	Mozambique	25.0
Cabo Verde	64.0	Namibia	49.0
Cameroon	27.0	Niger	32.0
Central African Republic	24.0	Nigeria	25.0
Chad	20.0	Rwanda	53.0
Comoros	20.0	Sao Tome and Principe	45.0
Congo	22.0	Senegal	43.0
Côte d'Ivoire	40.0	Seychelles	71.0
Djibouti	30.0	Sierra Leone	35.0
DR Congo	20.0	Somalia	11.0
Egypt	35.0	South Africa	41.0
Eritrea	21.0	Sudan	20.0
Ethiopia	37.0	Sudan, South	13.0
Gabon	28.0	Tanzania	40.0
Ghana	43.0	Togo	31.0
Guinea	26.0	Tunisia	40.0
Guinea-Bissau	22.0	Uganda	26.0
Kenya	31.0	Zambia	37.0
Liberia	25.0	Zimbabwe	24.0
Libya	18.0		

AMERICA · COMMONWEALTH

Countries 5 · Mean 58.2 · Median 58.0 · Population SD 10.5

Canada*	76.0
Grenada*	53.0
Jamaica*	44.0
Saint Lucia*	58.0
Saint Vincent and the Grenadines*	60.0

AMERICA · REPUBLICS

Countries 26 · Mean 39.0 · Median 35.5 · Population SD 16.2

Argentina	37.0	Guyana	40.0
Barbados	69.0	Haiti	17.0
Bolivia	29.0	Honduras	23.0
Brazil	36.0	Mexico	31.0
Chile	66.0	Nicaragua	17.0
Colombia	40.0	Panama	35.0
Costa Rica	55.0	Paraguay	28.0
Cuba	42.0	Peru	33.0
Dominica	56.0	Suriname	40.0
Dominican Republic	35.0	Trinidad and Tobago	42.0
Ecuador	34.0	United States	69.0
El Salvador	31.0	Uruguay	73.0
Guatemala	23.0	Venezuela	13.0

OCEANIA

MIDDLE EAST · MONARCHIES

Countries 7 · Mean 50.7

Bahrain***	42.0
Jordan**	46.0
Kuwait***	46.0
Oman****	43.0
Qatar***	58.0
Saudi Arabia*****	52.0
United Arab Emirates***	68.0

MIDDLE EAST · REPUBLICS

Countries 9 · Mean 31.6

Cyprus	53.0
Egypt	35.0
Iran	24.0
Iraq	23.0
Israel	62.0
Lebanon	24.0
Syria	13.0
Turkey	34.0
Yemen	16.0

MONARCHY

Countries 0 · With data 0

COMMONWEALTH

Countries 4 · Mean 58.0

Australia*	75.0
New Zealand*	85.0
Papua New Guinea*	29.0
Solomon Islands*	43.0

REPUBLICS

Countries 2 · Mean 50.0

Fiji	52.0
Vanuatu	48.0

RESEARCH DATA SHEET · RDS-K01

PROPERTY RIGHTS

Property Rights Alliance · IPRI Score (0–10) · Dataset: 2024 Edition

SCAN SOURCE

LEGEND

Republic
Commonwealth realm *
Constitutional monarchy **
Semi-constitutional monarchy ***
Absolute monarchy ****

GLOBAL MONARCHIES

Countries 20 · Mean 6.5 · Median 6.3 · Population SD 1.0

Bahrain***	5.719	Netherlands**	7.773
Belgium**	7.297	Norway**	7.684
Brunei****	5.539	Oman****	6.092
Denmark**	7.774	Qatar***	6.352
Japan**	7.588	Saudi Arabia****	5.835
Jordan**	5.452	Spain**	6.402
Kuwait***	5.151	Sweden**	7.725
Luxembourg**	7.751	Thailand**	4.807
Malaysia**	6.191	United Arab Emirates***	6.157
Morocco***	5.232	United Kingdom**	7.452

GLOBAL COMMONWEALTH

Countries 4 · Mean 7.0 · Median 7.5 · Population SD 1.0

Australia*	7.689
Canada*	7.406
Jamaica*	5.294
New Zealand*	7.721

GLOBAL REPUBLICS

Countries 99 · Mean 4.8 · Median 4.6 · Population SD 1.3

Albania	4.509	Czech Republic	6.542	Kenya	4.646	Russia	3.797
Algeria	4.142	Côte d'Ivoire	4.071	Latvia	6.149	Rwanda	5.326
Angola	3.221	Dominican Republic	4.707	Lebanon	3.63	Senegal	4.712
Argentina	4.181	Ecuador	4.142	Lithuania	6.333	Serbia	4.63
Armenia	5.216	Egypt	4.33	Madagascar	3.597	Singapore	7.942
Austria	7.56	El Salvador	4.272	Malawi	4.508	Slovakia	5.829
Azerbaijan	5.008	Ethiopia	3.173	Mali	3.534	Slovenia	6.064
Bangladesh	3.657	Finland	8.081	Malta	6.065	South Africa	5.083
Benin	4.257	France	7.002	Mauritania	3.242	South Korea	6.684
Bolivia	3.364	Gabon	3.763	Mexico	4.591	Sri Lanka	4.702
Bosnia and Herzegovina	4.143	Georgia	4.931	Moldova	4.689	Switzerland	7.613
Botswana	5.558	Germany	7.681	Montenegro	4.861	Tanzania	4.543
Brazil	4.546	Ghana	4.945	Mozambique	3.754	Trinidad and Tobago	4.971
Bulgaria	5.132	Greece	5.111	Nepal	4.148	Tunisia	4.657
Burkina Faso	4.089	Haiti	2.568	Nicaragua	3.612	Turkey	4.258
Burundi	3.735	Honduras	4.002	Niger	3.52	Uganda	4.265
Cameroon	3.64	Hungary	5.302	North Macedonia	4.575	Ukraine	3.82
Chad	3.023	Iceland	7.106	Pakistan	3.745	United States	7.494
Chile	5.927	India	5.09	Panama	5.034	Uruguay	6.366
China	5.285	Indonesia	5	Paraguay	4.349	Venezuela	1.927
Colombia	4.57	Iran	3.558	Peru	4.316	Vietnam	4.407
Congo, Democratic Republic of the	2.992	Ireland	7.404	Philippines	4.422	Yemen	2.382
Costa Rica	5.835	Israel	6.308	Poland	5.42	Zambia	4.316
Croatia	5.197	Italy	5.99	Portugal	6.314	Zimbabwe	3.44
Cyprus	5.948	Kazakhstan	4.644	Romania	5.762		

Data Source
Property Rights Alliance · International Property Rights Index
Indicator: IPRI Score (0–10)
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-K01

PROPERTY RIGHTS

Property Rights Alliance · IPRI Score (0–10) · Dataset: 2024 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Asia · Europe

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy *****

ASIA · MONARCHIES

Countries 11 · Mean 5.9 · Median 5.8 · Population SD 0.7

Bahrain***	5.719
Brunei*****	5.539
Japan**	7.588
Jordan**	5.452
Kuwait***	5.151
Malaysia**	6.191
Oman*****	6.092
Qatar***	6.352
Saudi Arabia*****	5.835
Thailand**	4.807
United Arab Emirates***	6.157

ASIA · REPUBLICS

Countries 20 · Mean 4.8 · Median 4.7 · Population SD 1.2

Armenia	5.216	Lebanon	3.63
Azerbaijan	5.008	Nepal	4.148
Bangladesh	3.657	Pakistan	3.745
China	5.285	Philippines	4.422
Georgia	4.931	Singapore	7.942
India	5.09	South Korea	6.684
Indonesia	5	Sri Lanka	4.702
Iran	3.558	Turkey	4.258
Israel	6.308	Vietnam	4.407
Kazakhstan	4.644	Yemen	2.382

EUROPE · MONARCHIES

Countries 8 · Mean 7.5 · Median 7.7 · Population SD 0.4

Belgium**	7.297
Denmark**	7.774
Luxembourg**	7.751
Netherlands**	7.773
Norway**	7.684
Spain**	6.402
Sweden**	7.725
United Kingdom**	7.452

EUROPE · REPUBLICS

Countries 30 · Mean 5.8 · Median 5.9 · Population SD 1.2

Albania	4.509	Latvia	6.149
Austria	7.56	Lithuania	6.333
Bosnia and Herzegovina	4.143	Malta	6.065
Bulgaria	5.132	Moldova	4.689
Croatia	5.197	Montenegro	4.861
Cyprus	5.948	North Macedonia	4.575
Czech Republic	6.542	Poland	5.42
Finland	8.081	Portugal	6.314
France	7.002	Romania	5.762
Germany	7.681	Russia	3.797
Greece	5.111	Serbia	4.63
Hungary	5.302	Slovakia	5.829
Iceland	7.106	Slovenia	6.064
Ireland	7.404	Switzerland	7.613
Italy	5.99	Ukraine	3.82

EUROPE · REPUBLICS WITHOUT A COMMUNIST PAST

Countries 12 · Mean 6.8 · Median 7.1 · Population SD 0.9

Austria	7.56
Cyprus	5.948
Finland	8.081
France	7.002
Germany	7.681
Greece	5.111
Iceland	7.106
Ireland	7.404
Italy	5.99
Malta	6.065
Portugal	6.314
Switzerland	7.613

Data Source
Property Rights Alliance · International Property Rights Index
Indicator: IPRI Score (0–10)
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-K01

PROPERTY RIGHTS

Property Rights Alliance · IPRI Score (0–10) · Dataset: 2024 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Africa · America
Oceania · Middle East

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy *****

AFRICA · MONARCHIES

Countries 0 · With data 0

AFRICA · REPUBLICS

Countries 0 · With data 0

AMERICA · COMMONWEALTH

Countries 0 · With data 0

AMERICA · REPUBLICS

Countries 0 · With data 0

OCEANIA

MIDDLE EAST · MONARCHIES

Countries 7 · Mean 5.8

Bahrain***	5.719
Jordan**	5.452
Kuwait***	5.151
Oman****	6.092
Qatar***	6.352
Saudi Arabia*****	5.835
United Arab Emirates***	6.157

MIDDLE EAST · REPUBLICS

Countries 7 · Mean 4.3

Cyprus	5.948
Egypt	4.33
Iran	3.558
Israel	6.308
Lebanon	3.63
Turkey	4.258
Yemen	2.382

MONARCHY

Countries 0 · With data 0

COMMONWEALTH

Countries 0 · With data 0

REPUBLICS

Countries 0 · With data 0

RESEARCH DATA SHEET · RDS-L01

HUMAN DEVELOPMENT INDEX

UNDP Human Development Reports · HDI Value (0–1) · Dataset: 2022 Edition

SCAN SOURCE

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

GLOBAL MONARCHIES

Countries 27 · Mean 0.835 · Median 0.875 · Population SD 0.122

Andorra**	0.884	Malaysia**	0.807
Bahrain***	0.888	Morocco***	0.698
Belgium**	0.942	Netherlands**	0.946
Bhutan***	0.681	Norway**	0.966
Brunei****	0.823	Oman****	0.819
Cambodia**	0.600	Qatar***	0.875
Denmark**	0.952	Saudi Arabia****	0.875
Eswatini****	0.610	Spain**	0.911
Japan**	0.920	Sweden**	0.952
Jordan**	0.736	Thailand**	0.803
Kuwait***	0.847	Tonga***	0.739
Lesotho**	0.521	United Arab Emirates***	0.937
Liechtenstein***	0.942	United Kingdom**	0.940
Luxembourg**	0.927		

GLOBAL COMMONWEALTH

Countries 14 · Mean 0.770 · Median 0.782 · Population SD 0.121

Antigua and Barbuda*	0.826
Australia*	0.946
Bahamas*	0.820
Belize*	0.700
Canada*	0.935
Grenada*	0.793
Jamaica*	0.706
New Zealand*	0.939
Papua New Guinea*	0.568
Saint Kitts and Nevis*	0.838
Saint Lucia*	0.725
Saint Vincent and the Grenadines*	0.772
Solomon Islands*	0.562
Tuvalu*	0.653

GLOBAL REPUBLICS

Countries 150 · Mean 0.698 · Median 0.726 · Population SD 0.153

Afghanistan	0.462	Egypt	0.728	Lithuania	0.879	Sao Tome and Principe	0.613
Albania	0.789	El Salvador	0.674	Madagascar	0.487	Senegal	0.517
Algeria	0.745	Eritrea	0.493	Malawi	0.508	Serbia	0.805
Angola	0.591	Estonia	0.899	Maldives	0.762	Seychelles	0.802
Argentina	0.849	Ethiopia	0.492	Mali	0.410	Sierra Leone	0.458
Armenia	0.786	Fiji	0.729	Malta	0.915	Singapore	0.949
Austria	0.926	Finland	0.942	Marshall Islands	0.731	Slovakia	0.855
Azerbaijan	0.760	France	0.910	Mauritania	0.540	Slovenia	0.926
Bangladesh	0.670	Gabon	0.693	Mauritius	0.796	Somalia	0.380
Barbados	0.809	Gambia	0.495	Mexico	0.781	South Africa	0.717
Belarus	0.801	Georgia	0.814	Micronesia	0.634	South Korea	0.929
Benin	0.504	Germany	0.950	Moldova	0.763	South Sudan	0.381
Bolivia	0.698	Ghana	0.602	Mongolia	0.741	Sri Lanka	0.780
Bosnia and Herzegovina	0.779	Greece	0.893	Montenegro	0.844	Sudan	0.516
Botswana	0.708	Guatemala	0.629	Mozambique	0.461	Suriname	0.690
Brazil	0.760	Guinea	0.471	Myanmar	0.608	Switzerland	0.967
Bulgaria	0.799	Guinea-Bissau	0.483	Namibia	0.610	Syria	0.557
Burkina Faso	0.438	Guyana	0.742	Nauru	0.696	Tajikistan	0.679
Burundi	0.420	Haiti	0.552	Nepal	0.601	Tanzania (United Republic of)	0.532
Cabo Verde	0.661	Honduras	0.624	Nicaragua	0.669	Timor-Leste	0.566
Cameroon	0.587	Hungary	0.851	Niger	0.394	Togo	0.547
Central African Republic	0.387	Iceland	0.959	Nigeria	0.548	Trinidad and Tobago	0.814
Chad	0.394	India	0.644	North Macedonia	0.765	Tunisia	0.732
Chile	0.860	Indonesia	0.713	Pakistan	0.540	Turkey	0.855
China	0.788	Iran	0.780	Palau	0.797	Turkmenistan	0.744
Colombia	0.758	Iraq	0.673	Palestine, State of	0.716	Uganda	0.550
Comoros	0.586	Ireland	0.950	Panama	0.820	Ukraine	0.734
Costa Rica	0.806	Israel	0.915	Paraguay	0.731	United States	0.927
Croatia	0.878	Italy	0.906	Peru	0.762	Uruguay	0.830
Cuba	0.764	Kazakhstan	0.802	Philippines	0.710	Uzbekistan	0.727
Cyprus	0.907	Kenya	0.601	Poland	0.881	Vanuatu	0.614
Czechia	0.895	Kiribati	0.628	Portugal	0.874	Venezuela	0.699
Côte d'Ivoire	0.534	Kyrgyzstan	0.701	Republic of the Congo	0.593	Vietnam	0.726
Djibouti	0.515	Laos	0.620	Romania	0.827	Yemen	0.424
Dominica	0.740	Latvia	0.879	Russia	0.821	Zambia	0.569
Dominican Republic	0.766	Lebanon	0.723	Rwanda	0.548	Zimbabwe	0.550
DR Congo	0.481	Liberia	0.487	Samoa	0.702		
Ecuador	0.765	Libya	0.746	San Marino	0.867		

RESEARCH DATA SHEET · RDS-L01

HUMAN DEVELOPMENT INDEX

UNDP Human Development Reports · HDI Value (0–1) · Dataset: 2022 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Asia · Europe

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

ASIA · MONARCHIES

Countries 13 · Mean 0.816 · Median 0.823 · Population SD 0.092

Bahrain***	0.888
Bhutan***	0.681
Brunei****	0.823
Cambodia**	0.600
Japan**	0.920
Jordan**	0.736
Kuwait***	0.847
Malaysia**	0.807
Oman****	0.819
Qatar***	0.875
Saudi Arabia****	0.875
Thailand**	0.803
United Arab Emirates***	0.937

ASIA · REPUBLICS

Countries 32 · Mean 0.716 · Median 0.724 · Population SD 0.119

Afghanistan	0.462	Mongolia	0.741
Armenia	0.786	Myanmar	0.608
Azerbaijan	0.760	Nepal	0.601
Bangladesh	0.670	Pakistan	0.540
China	0.788	Palestine, State of	0.716
Georgia	0.814	Philippines	0.710
India	0.644	Singapore	0.949
Indonesia	0.713	South Korea	0.929
Iran	0.780	Sri Lanka	0.780
Iraq	0.673	Syria	0.557
Israel	0.915	Tajikistan	0.679
Kazakhstan	0.802	Turkey	0.855
Kyrgyzstan	0.701	Turkmenistan	0.744
Laos	0.620	Uzbekistan	0.727
Lebanon	0.723	Vietnam	0.726
Maldives	0.762	Yemen	0.424

EUROPE · MONARCHIES

Countries 10 · Mean 0.936 · Median 0.942 · Population SD 0.022

Andorra**	0.884
Belgium**	0.942
Denmark**	0.952
Liechtenstein***	0.942
Luxembourg**	0.927
Netherlands**	0.946
Norway**	0.966
Spain**	0.911
Sweden**	0.952
United Kingdom**	0.940

EUROPE · REPUBLICS

Countries 33 · Mean 0.868 · Median 0.879 · Population SD 0.062

Albania	0.789	Latvia	0.879
Austria	0.926	Lithuania	0.879
Belarus	0.801	Malta	0.915
Bosnia and Herzegovina	0.779	Moldova	0.763
Bulgaria	0.799	Montenegro	0.844
Croatia	0.878	North Macedonia	0.765
Cyprus	0.907	Poland	0.881
Czechia	0.895	Portugal	0.874
Estonia	0.899	Romania	0.827
Finland	0.942	Russia	0.821
France	0.910	San Marino	0.867
Germany	0.950	Serbia	0.805
Greece	0.893	Slovakia	0.855
Hungary	0.851	Slovenia	0.926
Iceland	0.959	Switzerland	0.967
Ireland	0.950	Ukraine	0.734
Italy	0.906		

EUROPE · REPUBLICS WITHOUT A COMMUNIST PAST

Countries 13 · Mean 0.920 · Median 0.915 · Population SD 0.031

Austria	0.926
Cyprus	0.907
Finland	0.942
France	0.910
Germany	0.950
Greece	0.893
Iceland	0.959
Ireland	0.950
Italy	0.906
Malta	0.915
Portugal	0.874
San Marino	0.867
Switzerland	0.967

RESEARCH DATA SHEET · RDS-L01

HUMAN DEVELOPMENT INDEX

UNDP Human Development Reports · HDI Value (0–1) · Dataset: 2022 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Africa · America
Oceania · Middle East

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

AFRICA · MONARCHIES

Countries 3 · Mean 0.610

Eswatini****	0.610
Lesotho**	0.521
Morocco***	0.698

AFRICA · REPUBLICS

Countries 50 · Mean 0.552

Algeria	0.745
Angola	0.591
Benin	0.504
Botswana	0.708
Burkina Faso	0.438
Burundi	0.420
Cabo Verde	0.661
Cameroon	0.587
Central African Republic	0.387
Chad	0.394
Comoros	0.586
Côte d'Ivoire	0.534
Djibouti	0.515
DR Congo	0.481
Egypt	0.728
Eritrea	0.493
Ethiopia	0.492
Gabon	0.693
Gambia	0.495
Ghana	0.602
Guinea	0.471
Guinea-Bissau	0.483
Kenya	0.601
Liberia	0.487
Libya	0.746
Madagascar	0.487
Malawi	0.508
Mali	0.410
Mauritania	0.540
Mauritius	0.796
Mozambique	0.461
Namibia	0.610
Niger	0.394
Nigeria	0.548
Republic of the Congo	0.593
Rwanda	0.548
Sao Tome and Principe	0.613
Senegal	0.517
Seychelles	0.802
Sierra Leone	0.458
Somalia	0.380
South Africa	0.717
South Sudan	0.381
Sudan	0.516
Tanzania (United Republic of)	0.532
Togo	0.547
Tunisia	0.732
Uganda	0.550
Zambia	0.569
Zimbabwe	0.550

AMERICA · COMMONWEALTH

Countries 9 · Mean 0.791

Antigua and Barbuda*	0.826
Bahamas*	0.820
Belize*	0.700
Canada*	0.935
Grenada*	0.793
Jamaica*	0.706
Saint Kitts and Nevis*	0.838
Saint Lucia*	0.725
Saint Vincent and the Grenadines*	0.772

AMERICA · REPUBLICS

Countries 26 · Mean 0.751

Argentina	0.849
Barbados	0.809
Bolivia	0.698
Brazil	0.760
Chile	0.860
Colombia	0.758
Costa Rica	0.806
Cuba	0.764
Dominica	0.740
Dominican Republic	0.766
Ecuador	0.765
El Salvador	0.674
Guatemala	0.629
Guyana	0.742
Haiti	0.552
Honduras	0.624
Mexico	0.781
Nicaragua	0.669
Panama	0.820
Paraguay	0.731
Peru	0.762
Suriname	0.690
Trinidad and Tobago	0.814
United States	0.927
Uruguay	0.830
Venezuela	0.699

MIDDLE EAST · MONARCHIES

Countries 7 · Mean 0.854

Bahrain***	0.888
Jordan**	0.736
Kuwait***	0.847
Oman****	0.819
Qatar***	0.875
Saudi Arabia****	0.875
United Arab Emirates***	0.937

MIDDLE EAST · REPUBLICS

Countries 10 · Mean 0.728

Cyprus	0.907
Egypt	0.728
Iran	0.780
Iraq	0.673
Israel	0.915
Lebanon	0.723
Palestine, State of	0.716
Syria	0.557
Turkey	0.855
Yemen	0.424

OCEANIA · MONARCHY

Countries 1 · Mean 0.739

Tonga***	0.739

OCEANIA · COMMONWEALTH

Countries 5 · Mean 0.734

Australia*	0.946
New Zealand*	0.939
Papua New Guinea*	0.568
Solomon Islands*	0.562
Tuvalu*	0.653

OCEANIA · REPUBLICS

Countries 9 · Mean 0.677

Fiji	0.729
Kiribati	0.628
Marshall Islands	0.731
Micronesia	0.634
Nauru	0.696
Palau	0.797
Samoa	0.702
Timor-Leste	0.566
Vanuatu	0.614

RESEARCH DATA SHEET · RDS-M01

VOICE AND ACCOUNTABILITY

World Bank · VA.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

LEGEND

Republic
Commonwealth realm *
Constitutional monarchy **
Semi-constitutional monarchy ***
Absolute monarchy ****

GLOBAL MONARCHIES

Countries 28 · Mean 55.4 · Median 49.2 · Population SD 33.0

Andorra**	78.9	Malaysia**	51.0
Bahrain***	11.3	Monaco***	34.3
Belgium**	92.6	Morocco***	34.3
Bhutan***	54.9	Netherlands**	96.6
Brunei****	28.9	Norway**	100.0
Cambodia**	12.7	Oman****	21.1
Denmark**	98.5	Qatar***	22.5
Eswatini****	16.7	Saudi Arabia****	8.8
Japan**	85.3	Spain**	87.7
Jordan**	27.0	Sweden**	97.1
Kuwait***	31.4	Thailand**	35.8
Lesotho**	47.5	Tonga***	75.0
Liechtenstein***	93.1	United Arab Emirates***	18.6
Luxembourg**	98.0	United Kingdom**	90.2

GLOBAL COMMONWEALTH

Countries 14 · Mean 74.9 · Median 72.5 · Population SD 14.6

Antigua and Barbuda*	70.1
Australia*	96.1
Belize*	63.7
Canada*	95.1
Grenada*	71.1
Jamaica*	65.2
New Zealand*	99.5
Papua New Guinea*	47.1
Saint Kitts and Nevis*	72.1
Saint Lucia*	74.5
Saint Vincent and the Grenadines*	73.0
Solomon Islands*	57.8
The Bahamas*	74.0
Tuvalu*	89.2

GLOBAL REPUBLICS

Countries 151 · Mean 45.5 · Median 44.6 · Population SD 27.6

Afghanistan	2.0	Ecuador	44.1	Liberia	49.0	San Marino	89.7
Albania	53.9	Egypt	10.8	Libya	9.8	Sao Tome and Principe	56.9
Algeria	22.1	El Salvador	34.8	Lithuania	81.9	Senegal	52.0
Argentina	63.2	Eritrea	1.0	Madagascar	37.7	Serbia	44.6
Armenia	50.5	Estonia	88.7	Malawi	50.0	Seychelles	64.7
Austria	93.6	Ethiopia	20.1	Maldives	41.2	Sierra Leone	38.7
Azerbaijan	9.3	Fiji	54.4	Mali	20.6	Singapore	43.6
Bangladesh	27.5	Finland	97.5	Malta	77.0	Slovakia	76.5
Barbados	87.3	France	86.8	Marshall Islands	86.3	Slovenia	82.8
Belarus	3.9	Gabon	24.0	Mauritania	30.4	Somalia	4.4
Benin	40.2	Georgia	48.0	Mauritius	66.7	South Africa	70.6
Bolivia	39.2	Germany	94.6	Mexico	42.2	South Korea	73.5
Bosnia and Herzegovina	38.2	Ghana	60.8	Micronesia, Federated States of	83.8	Sri Lanka	39.7
Botswana	62.7	Greece	79.4	Moldova	55.9	Sudan	5.9
Brazil	59.8	Guatemala	33.3	Mongolia	56.4	Sudan, South	2.9
Bulgaria	60.3	Guinea	17.6	Montenegro	58.3	Suriname	58.8
Burkina Faso	25.5	Guinea-Bissau	36.3	Mozambique	31.9	Switzerland	99.0
Burundi	11.8	Guyana	52.9	Myanmar	2.5	Syria	1.5
Cabo Verde	77.5	Haiti	16.2	Namibia	66.2	Tajikistan	4.9
Cameroon	19.6	Honduras	35.3	Nauru	68.6	Tanzania	32.8
Central African Republic	14.2	Hungary	59.3	Nepal	46.1	The Gambia	45.6
Chad	8.3	Iceland	94.1	Nicaragua	10.3	Togo	28.5
Chile	79.9	India	51.5	Niger	29.9	Trinidad and Tobago	67.2
China	7.4	Indonesia	52.5	Nigeria	32.4	Tunisia	40.7
Colombia	55.4	Iran	7.8	North Macedonia	53.4	Turkey	25.0
Comoros	26.0	Iraq	23.5	Pakistan	23.0	Turkmenistan	0.5
Congo, Democratic Republic of the	15.7	Ireland	95.6	Palau	83.8	Uganda	29.4
Congo, Republic of the	17.2	Israel	69.1	palestine	18.1	Ukraine	42.6
Costa Rica	81.4	Italy	85.8	Panama	64.2	United States	75.5
Croatia	65.7	Kazakhstan	21.6	Paraguay	48.5	Uruguay	92.2
Cuba	14.7	Kenya	41.7	Peru	49.5	Uzbekistan	12.3
Cyprus	77.9	Kiribati	84.3	Philippines	46.6	Vanuatu	68.1
Czech Republic	82.4	Korea, North	0.0	Poland	67.6	Venezuela	6.9
Côte d'Ivoire	36.8	Kosovo	43.1	Portugal	88.2	Vietnam	15.2
Djibouti	13.7	Kyrgyzstan	28.4	Romania	62.3	Yemen	6.4
Dominica	72.5	Laos	5.4	Russia	13.2	Zambia	45.1
Dominican Republic	57.4	Latvia	78.4	Rwanda	24.5	Zimbabwe	19.1
East Timor	61.3	Lebanon	30.9	Samoa	80.9		

RESEARCH DATA SHEET · RDS-M01

VOICE AND ACCOUNTABILITY

World Bank · VA.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Asia · Europe

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

ASIA · MONARCHIES

Countries 13 · Mean 31.5 · Median 27.0 · Population SD 20.6

Bahrain***	11.3
Bhutan***	54.9
Brunei****	28.9
Cambodia**	12.7
Japan**	85.3
Jordan**	27.0
Kuwait***	31.4
Malaysia**	51.0
Oman****	21.1
Qatar***	22.5
Saudi Arabia****	8.8
Thailand**	35.8
United Arab Emirates***	18.6

ASIA · REPUBLICS

Countries 33 · Mean 27.0 · Median 23.5 · Population SD 21.1

Afghanistan	2.0	Mongolia	56.4
Armenia	50.5	Myanmar	2.5
Azerbaijan	9.3	Nepal	46.1
Bangladesh	27.5	Pakistan	23.0
China	7.4	palestine	18.1
Georgia	48.0	Philippines	46.6
India	51.5	Singapore	43.6
Indonesia	52.5	South Korea	73.5
Iran	7.8	Sri Lanka	39.7
Iraq	23.5	Syria	1.5
Israel	69.1	Tajikistan	4.9
Kazakhstan	21.6	Turkey	25.0
Korea, North	0.0	Turkmenistan	0.5
Kyrgyzstan	28.4	Uzbekistan	12.3
Laos	5.4	Vietnam	15.2
Lebanon	30.9	Yemen	6.4
Maldives	41.2		

EUROPE · MONARCHIES

Countries 11 · Mean 87.9 · Median 93.1 · Population SD 17.9

Andorra**	78.9
Belgium**	92.6
Denmark**	98.5
Liechtenstein***	93.1
Luxembourg**	98.0
Monaco***	34.3
Netherlands**	96.6
Norway**	100.0
Spain**	87.7
Sweden**	97.1
United Kingdom**	90.2

EUROPE · REPUBLICS

Countries 34 · Mean 69.8 · Median 77.5 · Population SD 23.2

Albania	53.9	Kosovo	43.1
Austria	93.6	Latvia	78.4
Belarus	3.9	Lithuania	81.9
Bosnia and Herzegovina	38.2	Malta	77.0
Bulgaria	60.3	Moldova	55.9
Croatia	65.7	Montenegro	58.3
Cyprus	77.9	North Macedonia	53.4
Czech Republic	82.4	Poland	67.6
Estonia	88.7	Portugal	88.2
Finland	97.5	Romania	62.3
France	86.8	Russia	13.2
Germany	94.6	San Marino	89.7
Greece	79.4	Serbia	44.6
Hungary	59.3	Slovakia	76.5
Iceland	94.1	Slovenia	82.8
Ireland	95.6	Switzerland	99.0
Italy	85.8	Ukraine	42.6

EUROPE · REPUBLICS WITHOUT A COMMUNIST PAST

Countries 13 · Mean 89.2 · Median 89.7 · Population SD 7.2

Austria	93.6
Cyprus	77.9
Finland	97.5
France	86.8
Germany	94.6
Greece	79.4
Iceland	94.1
Ireland	95.6
Italy	85.8
Malta	77.0
Portugal	88.2
San Marino	89.7
Switzerland	99.0

Data Source
World Bank · Worldwide Governance Indicators
Indicator: VA.PER.RNK
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-M01

VOICE AND ACCOUNTABILITY

World Bank · VA.PER.RNK · Dataset: 2023 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Africa · America
Oceania · Middle East

LEGEND

Republic
Commonwealth realm *
Constitutional monarchy **
Semi-constitutional monarchy ***
Absolute monarchy ****

AFRICA · MONARCHIES

Countries 3 · Mean 32.8

Eswatini****	16.7
Lesotho**	47.5
Morocco***	34.3

AFRICA · REPUBLICS

Countries 49 · Mean 32.4

Algeria	22.1
Benin	40.2
Botswana	62.7
Burkina Faso	25.5
Burundi	11.8
Cabo Verde	77.5
Cameroon	19.6
Central African Republic	14.2
Chad	8.3
Comoros	26.0
Congo, Democratic Republic of the	15.7
Congo, Republic of the	17.2
Côte d'Ivoire	36.8
Djibouti	13.7
Egypt	10.8
Eritrea	1.0
Ethiopia	20.1
Gabon	24.0
Ghana	60.8
Guinea	17.6
Guinea-Bissau	36.3
Kenya	41.7
Liberia	49.0
Libya	9.8
Madagascar	37.7
Malawi	50.0
Mali	20.6
Mauritania	30.4
Mauritius	66.7
Mozambique	31.9
Namibia	66.2
Niger	29.9
Nigeria	32.4
Rwanda	24.5
Sao Tome and Principe	56.9
Senegal	52.0
Seychelles	64.7
Sierra Leone	38.7
Somalia	4.4
South Africa	70.6
Sudan	5.9
Sudan, South	2.9
Tanzania	32.8
The Gambia	45.6
Togo	26.5
Tunisia	40.7
Uganda	29.4
Zambia	45.1
Zimbabwe	19.1

AMERICA · COMMONWEALTH

Countries 9 · Mean 73.2

Antigua and Barbuda*	70.1
Belize*	63.7
Canada*	95.1
Grenada*	71.1
Jamaica*	65.2
Saint Kitts and Nevis*	72.1
Saint Lucia*	74.5
Saint Vincent and the Grenadines*	73.0
The Bahamas*	74.0

AMERICA · REPUBLICS

Countries 26 · Mean 51.6

Argentina	63.2
Barbados	87.3
Bolivia	39.2
Brazil	59.8
Chile	79.9
Colombia	55.4
Costa Rica	81.4
Cuba	14.7
Dominica	72.5
Dominican Republic	57.4
Ecuador	44.1
El Salvador	34.8
Guatemala	33.3
Guyana	52.9
Haiti	16.2
Honduras	35.3
Mexico	42.2
Nicaragua	10.3
Panama	64.2
Paraguay	48.5
Peru	49.5
Suriname	58.8
Trinidad and Tobago	67.2
United States	75.5
Uruguay	92.2
Venezuela	6.9

MIDDLE EAST · MONARCHIES

Countries 7 · Mean 20.1

Bahrain***	11.3
Jordan**	27.0
Kuwait***	31.4
Oman****	21.1
Qatar***	22.5
Saudi Arabia****	8.8
United Arab Emirates***	18.6

MIDDLE EAST · REPUBLICS

Countries 10 · Mean 27.1

Cyprus	77.9
Egypt	10.8
Iran	7.8
Iraq	23.5
Israel	69.1
Lebanon	30.9
palestine	18.1
Syria	1.5
Turkey	25.0
Yemen	6.4

OCEANIA · MONARCHY

Countries 1 · Mean 75.0

Tonga***	75.0

OCEANIA · COMMONWEALTH

Countries 5 · Mean 77.9

Australia*	96.1
New Zealand*	99.5
Papua New Guinea*	47.1
Solomon Islands*	57.8
Tuvalu*	89.2

OCEANIA · REPUBLICS

Countries 9 · Mean 74.6

East Timor	61.3
Fiji	54.4
Kiribati	84.3
Marshall Islands	86.3
Micronesia, Federated States of	83.8
Nauru	68.6
Palau	83.8
Samoa	80.9
Vanuatu	68.1

Data Source
World Bank · Worldwide Governance Indicators
Indicator: VA.PER.RNK
Alphabetical order within each group

RESEARCH DATA SHEET · RDS-N01

PASSPORT POWER INDEX

Henley & Partners · Henley Passport Index Score · Dataset: 2024 Edition

SCAN SOURCE

LEGEND

- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

GLOBAL MONARCHIES

Countries 28 · Mean 137.5 · Median 168.5 · Population SD 53.8

Andorra**	171.0	Malaysia**	182.0
Bahrain***	87.0	Monaco***	178.0
Belgium**	190.0	Morocco***	72.0
Bhutan***	52.0	Netherlands**	191.0
Brunei****	166.0	Norway**	190.0
Cambodia**	53.0	Oman****	86.0
Denmark**	190.0	Qatar***	107.0
Eswatini****	77.0	Saudi Arabia****	88.0
Japan**	192.0	Spain**	192.0
Jordan**	55.0	Sweden**	191.0
Kuwait***	99.0	Thailand**	82.0
Lesotho**	79.0	Tonga***	131.0
Liechtenstein***	182.0	United Arab Emirates***	185.0
Luxembourg**	191.0	United Kingdom**	190.0

GLOBAL COMMONWEALTH

Countries 14 · Mean 144.8 · Median 150.5 · Population SD 33.1

Antigua and Barbuda*	153.0
Australia*	189.0
Bahamas*	161.0
Belize*	102.0
Canada*	187.0
Grenada*	147.0
Jamaica*	89.0
New Zealand*	190.0
Papua New Guinea*	85.0
Saint Kitts and Nevis*	157.0
Saint Lucia*	148.0
Saint Vincent and the Grenadines*	157.0
Solomon Islands*	134.0
Tuvalu*	128.0

GLOBAL REPUBLICS

Countries 151 · Mean 102.1 · Median 80.0 · Population SD 53.4

Afghanistan	26.0	Egypt	52.0	Liberia	51.0	San Marino	172.0
Albania	123.0	El Salvador	136.0	Libya	39.0	Sao Tome and Principe	63.0
Algeria	55.0	Eritrea	42.0	Lithuania	185.0	Senegal	58.0
Angola	52.0	Estonia	185.0	Madagascar	60.0	Serbia	140.0
Argentina	172.0	Ethiopia	46.0	Malawi	75.0	Seychelles	156.0
Armenia	68.0	Fiji	90.0	Maldives	94.0	Sierra Leone	66.0
Austria	191.0	Finland	191.0	Mali	55.0	Singapore	195.0
Azerbaijan	71.0	France	192.0	Malta	187.0	Slovakia	184.0
Bangladesh	40.0	Gabon	60.0	Marshall Islands	129.0	Slovenia	184.0
Barbados	165.0	Georgia	122.0	Mauritania	57.0	Somalia	35.0
Belarus	81.0	Germany	192.0	Mauritius	150.0	South Africa	106.0
Benin	64.0	Ghana	68.0	Mexico	159.0	Sri Lanka	44.0
Bolivia	80.0	Greece	188.0	Micronesia, Federated States of	124.0	Sudan	43.0
Bosnia and Herzegovina	123.0	Guatemala	135.0	Moldova	122.0	Sudan, South	44.0
Botswana	88.0	Guinea	59.0	Mongolia	64.0	Suriname	79.0
Brazil	171.0	Guinea-Bissau	55.0	Montenegro	128.0	Switzerland	190.0
Bulgaria	177.0	Guyana	89.0	Mozambique	64.0	Syria	28.0
Burkina Faso	60.0	Haiti	53.0	Myanmar	45.0	Tajikistan	58.0
Burundi	50.0	Honduras	133.0	Namibia	81.0	Tanzania	73.0
Cabo Verde	69.0	Hungary	187.0	Nauru	89.0	The Gambia	71.0
Cameroon	50.0	Iceland	184.0	Nepal	39.0	Timor-Leste	97.0
Central African Republic	53.0	India	58.0	Nicaragua	128.0	Togo	61.0
Chad	53.0	Indonesia	76.0	Niger	57.0	Trinidad and Tobago	151.0
Chile	175.0	Iran	43.0	Nigeria	45.0	Tunisia	69.0
China	85.0	Iraq	31.0	North Macedonia	128.0	Turkey	116.0
Colombia	134.0	Ireland	191.0	Pakistan	33.0	Turkmenistan	50.0
Comoros	54.0	Israel	170.0	Palau	124.0	Uganda	70.0
Congo	50.0	Italy	192.0	Palestine	40.0	Ukraine	148.0
Costa Rica	151.0	Kazakhstan	77.0	Panama	149.0	United States	186.0
Croatia	183.0	Kenya	75.0	Paraguay	147.0	Uruguay	157.0
Cuba	62.0	Kiribati	124.0	Peru	141.0	Uzbekistan	62.0
Cyprus	178.0	Korea, North	41.0	Philippines	67.0	Vanuatu	92.0
Czech Republic	187.0	Korea, South	191.0	Poland	188.0	Venezuela	124.0
Côte d'Ivoire	59.0	Kosovo	79.0	Portugal	189.0	Vietnam	51.0
Djibouti	49.0	Kyrgyzstan	64.0	Romania	177.0	Yemen	33.0
Dominica	143.0	Laos	49.0	Russia	116.0	Zambia	70.0
Dominican Republic	75.0	Latvia	184.0	Rwanda	65.0	Zimbabwe	65.0
Ecuador	95.0	Lebanon	45.0	Samoa	131.0		

RESEARCH DATA SHEET · RDS-N01

PASSPORT POWER INDEX

Henley & Partners · Henley Passport Index Score · Dataset: 2024 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Asia · Europe

LEGEND

Republic	Constitutional monarchy **
Commonwealth realm *	Semi-constitutional monarchy ***
	Absolute monarchy ****

ASIA · MONARCHIES

Countries 13 · Mean 110.3 · Median 88.0 · Population SD 50.2

Bahrain***	87.0
Bhutan***	52.0
Brunei****	166.0
Cambodia**	53.0
Japan**	192.0
Jordan**	55.0
Kuwait***	99.0
Malaysia**	182.0
Oman****	86.0
Qatar***	107.0
Saudi Arabia****	88.0
Thailand**	82.0
United Arab Emirates***	185.0

ASIA · REPUBLICS

Countries 33 · Mean 69.0 · Median 58.0 · Population SD 43.3

Afghanistan	26.0	Maldives	94.0
Armenia	68.0	Mongolia	64.0
Azerbaijan	71.0	Myanmar	45.0
Bangladesh	40.0	Nepal	39.0
China	85.0	Pakistan	33.0
Georgia	122.0	Palestine	40.0
India	58.0	Philippines	67.0
Indonesia	76.0	Singapore	195.0
Iran	43.0	Sri Lanka	44.0
Iraq	31.0	Syria	28.0
Israel	170.0	Tajikistan	58.0
Kazakhstan	77.0	Turkey	116.0
Korea, North	41.0	Turkmenistan	50.0
Korea, South	191.0	Uzbekistan	62.0
Kyrgyzstan	64.0	Vietnam	51.0
Laos	49.0	Yemen	33.0
Lebanon	45.0		

EUROPE · MONARCHIES

Countries 11 · Mean 186.9 · Median 190.0 · Population SD 6.5

Andorra**	171.0
Belgium**	190.0
Denmark**	190.0
Liechtenstein***	182.0
Luxembourg**	191.0
Monaco***	178.0
Netherlands**	191.0
Norway**	190.0
Spain**	192.0
Sweden**	191.0
United Kingdom**	190.0

EUROPE · REPUBLICS

Countries 34 · Mean 166.1 · Median 184.0 · Population SD 32.9

Albania	123.0	Kosovo	79.0
Austria	191.0	Latvia	184.0
Belarus	81.0	Lithuania	185.0
Bosnia and Herzegovina	123.0	Malta	187.0
Bulgaria	177.0	Moldova	122.0
Croatia	183.0	Montenegro	128.0
Cyprus	178.0	North Macedonia	128.0
Czech Republic	187.0	Poland	188.0
Estonia	185.0	Portugal	189.0
Finland	191.0	Romania	177.0
France	192.0	Russia	116.0
Germany	192.0	San Marino	172.0
Greece	188.0	Serbia	140.0
Hungary	187.0	Slovakia	184.0
Iceland	184.0	Slovenia	184.0
Ireland	191.0	Switzerland	190.0
Italy	192.0	Ukraine	148.0

EUROPE · REPUBLICS WITHOUT A COMMUNIST PAST

Countries 13 · Mean 187.5 · Median 190.0 · Population SD 5.9

Austria	191.0
Cyprus	178.0
Finland	191.0
France	192.0
Germany	192.0
Greece	188.0
Iceland	184.0
Ireland	191.0
Italy	192.0
Malta	187.0
Portugal	189.0
San Marino	172.0
Switzerland	190.0

RESEARCH DATA SHEET · RDS-N01

PASSPORT POWER INDEX

Henley & Partners · Henley Passport Index Score · Dataset: 2024 Edition

SCAN SOURCE

REGIONAL COMPARISONS

Africa · America
Oceania · Middle East

LEGEND

Republic
Commonwealth realm *
Constitutional monarchy **
Semi-constitutional monarchy ***
Absolute monarchy ****

AFRICA · MONARCHIES

Countries 3 · Mean 76.0 · Median 77.0 · Population SD 2.9

Eswatini****	77.0
Lesotho**	79.0
Morocco***	72.0

AFRICA · REPUBLICS

Countries 49 · Mean 63.5 · Median 59.0 · Population SD 22.4

Algeria	55.0	Malawi	75.0
Angola	52.0	Mali	55.0
Benin	64.0	Mauritania	57.0
Botswana	88.0	Mauritius	150.0
Burkina Faso	60.0	Mozambique	64.0
Burundi	50.0	Namibia	81.0
Cabo Verde	69.0	Niger	57.0
Cameroon	50.0	Nigeria	45.0
Central African Republic	53.0	Rwanda	65.0
Chad	53.0	Sao Tome and Principe	63.0
Comoros	54.0	Senegal	58.0
Congo	50.0	Seychelles	156.0
Côte d'Ivoire	59.0	Sierra Leone	66.0
Djibouti	49.0	Somalia	35.0
Egypt	52.0	South Africa	106.0
Eritrea	42.0	Sudan	43.0
Ethiopia	46.0	Sudan, South	44.0
Gabon	60.0	Tanzania	73.0
Ghana	68.0	The Gambia	71.0
Guinea	59.0	Togo	61.0
Guinea-Bissau	55.0	Tunisia	69.0
Kenya	75.0	Uganda	70.0
Liberia	51.0	Zambia	70.0
Libya	39.0	Zimbabwe	65.0
Madagascar	60.0		

AMERICA · COMMONWEALTH

Countries 9 · Mean 144.6 · Median 153.0 · Population SD 28.6

Antigua and Barbuda*	153.0
Bahamas*	161.0
Belize*	102.0
Canada*	187.0
Grenada*	147.0
Jamaica*	89.0
Saint Kitts and Nevis*	157.0
Saint Lucia*	148.0
Saint Vincent and the Grenadines*	157.0

AMERICA · REPUBLICS

Countries 26 · Mean 130.4 · Median 138.5 · Population SD 36.7

Argentina	172.0	Guyana	89.0
Barbados	165.0	Haiti	53.0
Bolivia	80.0	Honduras	133.0
Brazil	171.0	Mexico	159.0
Chile	175.0	Nicaragua	128.0
Colombia	134.0	Panama	149.0
Costa Rica	151.0	Paraguay	147.0
Cuba	62.0	Peru	141.0
Dominica	143.0	Suriname	79.0
Dominican Republic	75.0	Trinidad and Tobago	151.0
Ecuador	95.0	United States	186.0
El Salvador	136.0	Uruguay	157.0
Guatemala	135.0	Venezuela	124.0

OCEANIA

MIDDLE EAST · MONARCHIES

Countries 7 · Mean 101.0

Bahrain***	87.0
Jordan**	55.0
Kuwait***	99.0
Oman****	86.0
Qatar***	107.0
Saudi Arabia****	88.0
United Arab Emirates***	185.0

MIDDLE EAST · REPUBLICS

Countries 10 · Mean 73.6

Cyprus	178.0
Egypt	52.0
Iran	43.0
Iraq	31.0
Israel	170.0
Lebanon	45.0
Palestine	40.0
Syria	28.0
Turkey	116.0
Yemen	33.0

MONARCHY

Countries 1 · Mean 131.0

Tonga***	131.0

COMMONWEALTH

Countries 5 · Mean 145.2

Australia*	189.0
New Zealand*	190.0
Papua New Guinea*	85.0
Solomon Islands*	134.0
Tuvalu*	128.0

REPUBLICS

Countries 9 · Mean 111.1

Fiji	90.0
Kiribati	124.0
Marshall Islands	129.0
Micronesia, Federated States of	89.0
Nauru	89.0
Palau	124.0
Samoa	131.0
Timor-Leste	97.0
Vanuatu	92.0

RESEARCH DATA SHEET · RDS-O01

CRUDE OIL RESERVES BY COUNTRY

Worldometer · Oil Reserves · Dataset: 2023 Edition · Unit: Million Barrels

SCAN SOURCE

GLOBAL RANKING

Highest to Lowest Oil Reserves

Source-file order preserved; no alphabetical reordering

LEGEND

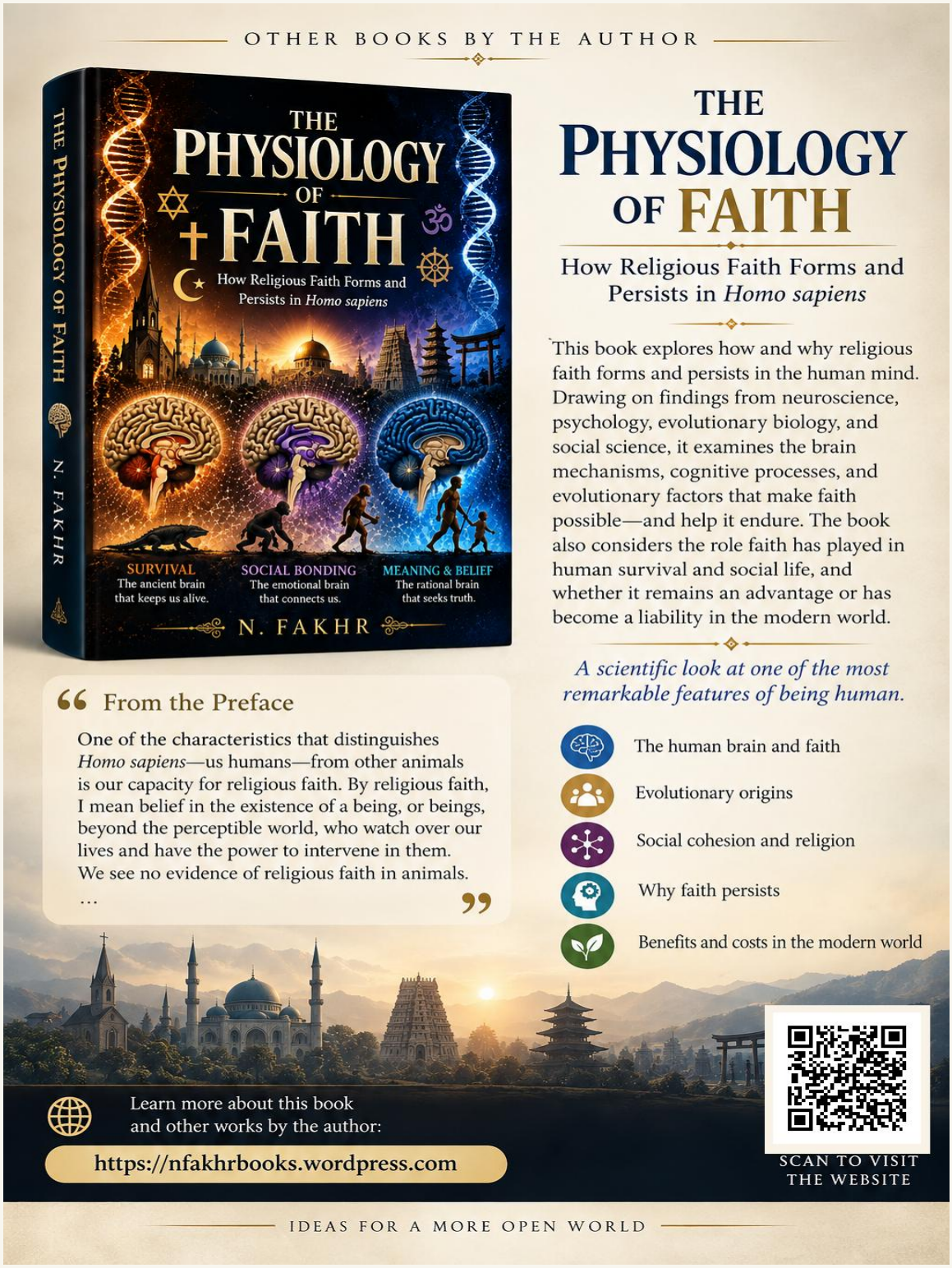
- Republic
- Commonwealth realm *
- Constitutional monarchy **
- Semi-constitutional monarchy ***
- Absolute monarchy ****

WORLD RANKING · PROVEN OIL RESERVES

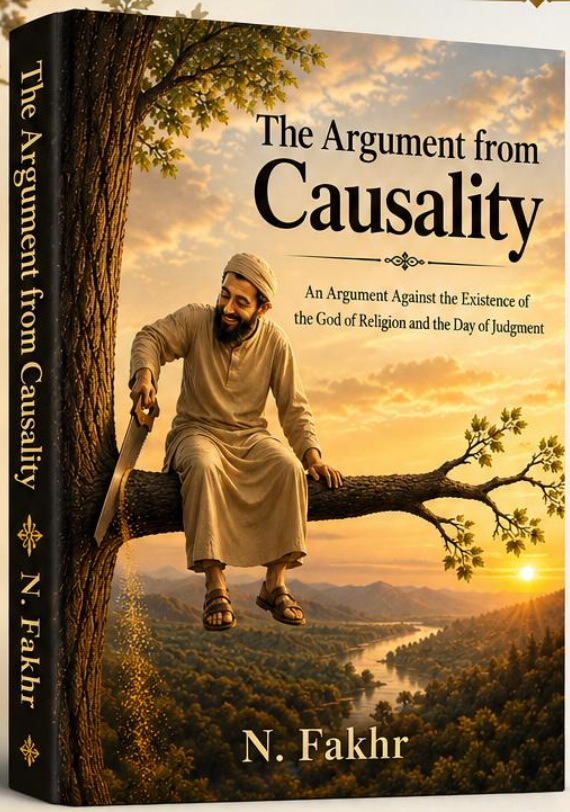
Countries: 85 · Ranking follows the spreadsheet from largest to smallest reserves

RANK	COUNTRY	MILLION BARRELS	RANK	COUNTRY	MILLION BARRELS	RANK	COUNTRY	MILLION BARRELS
1	Venezuela	304,000	30	Congo, Republic of the	2,900	59	Cameroon	200
2	Saudi Arabia****	259,000	31	Argentina	2,500	60	Chile	200
3	Iran	209,000	32	Indonesia	2,500	61	DR Congo	200
4	Canada*	170,000	33	Syria	2,500	62	Niger	200
5	Iraq	145,000	34	Uganda	2,500	63	Papua New Guinea*	200
6	Kuwait***	102,000	35	United Kingdom**	2,500	64	Spain**	200
7	United Arab Emirates***	98,000	36	Australia*	2,400	65	Trinidad and Tobago	200
8	Russia	80,000	37	Colombia	2,000	66	Côte d'Ivoire	100
9	United States	48,300	38	Gabon	2,000	67	Cuba	100
10	Libya	48,000	39	Chad	1,500	68	Germany	100
11	Nigeria	37,000	40	Turkey	1,400	69	Myanmar	100
12	Kazakhstan	30,000	41	Sudan	1,300	70	Netherlands**	100
13	China	26,000	42	Brunei****	1,100	71	Philippines	100
14	Qatar***	25,000	43	Guinea	1,100	72	Poland	100
15	Brazil	13,000	44	Peru	900	73	Guatemala	86
16	Algeria	12,000	45	Ghana	700	74	Serbia	77
17	Guyana	11,000	46	Romania	600	75	Croatia	71
18	Ecuador	8,300	47	Turkmenistan	600	76	France	61
19	Norway**	8,100	48	Uzbekistan	600	77	Japan**	44
20	Angola	7,800	49	Italy	500	78	Kyrgyzstan	40
21	Azerbaijan	7,000	50	Pakistan	500	79	New Zealand*	40
22	Mexico	5,800	51	Denmark**	400	80	Israel	12
23	Oman****	5,400	52	Tunisia	400	81	Jordan**	1
24	India	4,600	53	Ukraine	400	82	Somalia	0
25	Vietnam	4,400	54	Thailand**	300	83	Cyprus	0
26	Sudan, South	3,800	55	Albania	200	84	Lebanon	0
27	Malaysia**	3,600	56	Bahrain***	200	85	Palestine	0
28	Egypt	3,300	57	Belarus	200			
29	Yemen	3,000	58	Bolivia	200			

OTHER BOOKS BY THE AUTHOR



OTHER BOOKS BY THE AUTHOR



The Argument from Causality

An Argument Against the Existence of the God of Religion and the Day of Judgment

This book takes the reader, through a series of engaging dialogues, on a careful and step-by-step examination of the Argument from Causality. It shows why this famous philosophical argument does not support the God of religion, and why it also undermines the idea of the Day of Judgment. The book also discusses the Argument from Design, the concept of the soul, free will and determinism, and the problems with viewing the Quran as both a miracle and something sacred. Using modern concepts and clear language, it invites readers to examine the evidence and decide for themselves.

From the Introduction

The story of the Argument from Causality may be one of the most remarkable stories in the history of philosophy. From the very beginning, this argument should have been used against the existence of the God of religion and the Day of Judgment. Instead, it was presented to followers of religion as an argument for the existence of a personal God.

...

A rational conversation about life's biggest questions.

- Philosophy and reason
- The Argument from Causality
- The Argument from Design
- The soul, free will, and determinism
- The Quran: miracle or sacred text?
- A fair and open-minded discussion



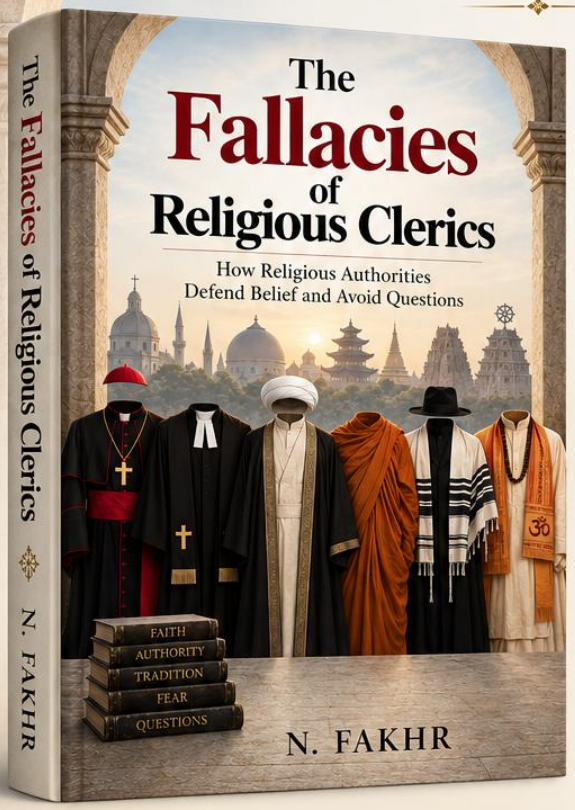
Learn more about this book and other works by the author:

https://nfakhrbooks.wordpress.com

SCAN TO VISIT THE WEBSITE

IDEAS FOR A MORE OPEN WORLD

OTHER BOOKS BY THE AUTHOR



The Fallacies of Religious Clerics

How Religious Authorities Defend Belief and Avoid Questions

This book examines the common logical fallacies and psychological strategies used by religious clerics to defend their beliefs and avoid difficult questions. It is not about attacking individuals, but about understanding how human beings—especially those in religious authority—can sincerely believe and present weak arguments as if they were strong. Drawing on examples from different religions, the book helps readers recognize flawed reasoning, examine claims more critically, and think more clearly and independently about faith, religion, and life's biggest questions.

"From the Introduction

This book is not about clerics being bad people.
It is about human beings.
We all have beliefs that we care about. Some are beliefs we grew up with. Some have become part of our identity. Some give meaning to our lives or make us feel safe. When such a belief faces a serious question, our minds do not always act like impartial judges. Sometimes, before asking, "What is the truth?" we begin looking for an answer to a different question: "How can I defend what I already believe?"

...

Better questions. A more open mind.

- Common logical fallacies
- Psychology of belief and bias
- Authority, tradition, and sacredness
- Faith, fear, and social pressure
- Science, philosophy, and religion
- A call for critical thinking

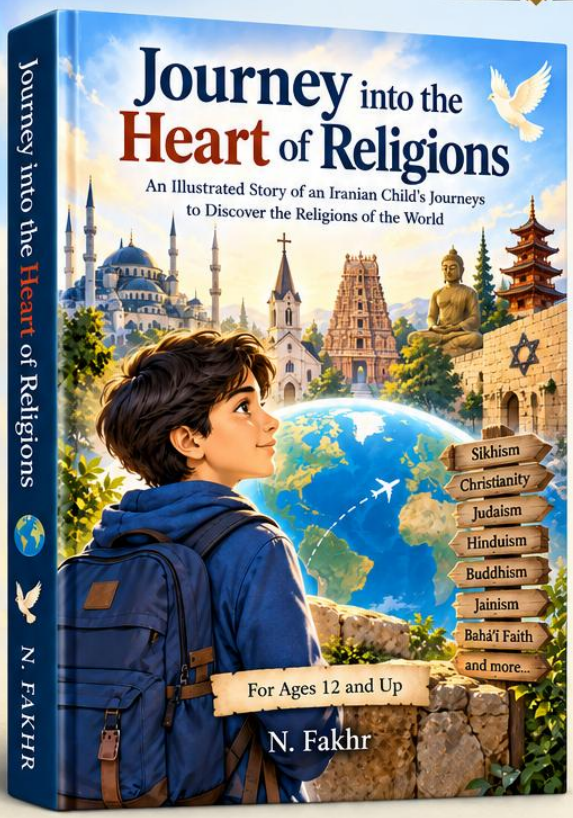
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SCAN TO VISIT THE WEBSITE

IDEAS FOR A MORE OPEN WORLD

OTHER BOOKS BY THE AUTHOR



Journey into the Heart of Religions

An Illustrated Story of an Iranian Child's Journeys to Discover the Religions of the World

Follow the journey of a curious 12-year-old Iranian boy as he and his father travel to different countries, explore the world's major religions, meet new people, and ask big questions. Through engaging stories and vivid illustrations, this book introduces young readers to different faiths, cultures, and ways of thinking—encouraging curiosity, respect, and independent thought.

Written in simple and enjoyable language for readers aged 12 and up, this book also offers valuable insights for readers of all ages.

Different paths. A more open world.

- Explore world religions
- Real places, real people
- Ask questions, seek the truth
- Respect, understanding, and tolerance
- For curious minds, ages 12 and up

From the Introduction

Since then, I have felt that I should try to find a way to introduce children and teenagers to religion—a way that does not silence their questions, but instead helps them search for the truth and learn to think freely. *Journey into the Heart of Religions* is one possible way of teaching children about religion. ...

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